

Lecture Notes on General Relativity

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1 Einstein Equivalence Principle: from Gravity to Geometry

1.1 Motivation: The Einstein Equivalence Principle

The highly successful Newtonian theory of gravity can be succinctly summarised by two sets of differential equations, one describing the dynamics (motion) of particles in a given gravitational field (described by a potential ϕ), and the other describing the dynamics of the gravitational field itself, namely how ϕ is to be determined from a given mass configuration. The former takes the standard Newtonian form

$$m\ddot{\vec{x}} = \vec{F}_g = -m\vec{\nabla}\phi \tag{1.1}$$

(but we will come back in some detail below to the question if/why the same mass parameter m appears on both sides of this equation, so as to incorporate the observation, going back to Galileo, that "all bodies fall at the same rate in a a gravitational field"). The latter is the Poisson equation

$$\Delta\phi = 4\pi G_N \mu = (4\pi G_N/c^2)\rho \ , \tag{1.2}$$

with G_N denoting, here and throughout, Newton's constant, i.e. the gravitational coupling constant, and where μ is the mass density, and $\rho = \mu c^2$ the associated rest mass energy density - I will set $c = 1$ in the following and use ρ.

Let us start with the field equation. It is immediately evident that this cannot be the final story. Not only is this equation not Lorentz invariant. Because of the absence of time-derivatives in (1.2), it actually describes an "action at a distance" and an instantaneous propagation of the gravitational field to every point in space (if you wiggle your mass distribution here now, this will immediately effect the gravitational potential arbitrarily far away). This is something that Einstein had just successfully exorcised from other aspects of physics, and clearly Newtonian gravity had to be revised as well.

It is then also immediately clear that what would have to replace Newton's theory is something rather more complicated. The reason for this is that, according to Special Relativity, mass is just another form of energy. Then, since gravity couples to masses, in a relativistically invariant theory gravity will also have to couple to energy. In particular, therefore, gravity would have to couple to gravitational energy, i.e. to itself. As a consequence, the new gravitational field equations will, unlike Newton's, have to be non-linear: the field of the sum of two masses cannot equal the sum of the gravitational fields of the two masses because it should also take into account the gravitational energy of the two-body system.

Now, having realised that Newton's theory cannot be the final word on the issue, how does one go about finding a better theory?

I will first very briefly discuss (and then dismiss) what at first sight may appear to be the most natural and naive approach to formulating a relativistic theory of gravity,

namely the simple replacement of Newton's field equation (1.2) by its relativistically covariant version

$$\Delta\phi = 4\pi G_N \rho \quad \longrightarrow \quad \Box\phi = 4\pi G_N \rho \ , \tag{1.3}$$

where $\Box$ is the Lorentz invariant d'Alembert or wave operator. While this looks promising, something can't be quite right about this equation. We already know (from Special Relativity) that ρ is not a scalar but rather the 00-component of a tensor, the energy-momentum tensor T_{ab}, so if actually ρ appears on the right-hand side, ϕ cannot be a scalar, while if ϕ is a scalar something needs to be done to fix the right-hand side.

Turning first to the latter possibility, one option that suggests itself is to replace ρ by the trace $T = T^a_{\ a}$ of the energy-momentum tensor. This is by definition / construction a scalar, and it will agree with ρ in the non-relativistic limit (where rest mass dominates over other contributions). With the sign conventions that we will use, one has $T_{00} = \rho \Rightarrow T = -\rho + \ldots$, and thus a first attempt at fixing the above equation might look like

$$\Box\phi = -4\pi G_N T \ . \tag{1.4}$$

This is certainly an attractive equation, but it definitely has the drawback that it is too linear. Recall from the discussion above that the universality of gravity (coupling to all forms of matter) and the equivalence of mass and energy lead to the conclusion that gravity should couple to gravitational energy, invariably predicting non-linear (self-interacting) equations for the gravitational field. However, the left hand side could be such that it only reduces to $\Box$ or Δ of the Newtonian potential in the Newtonian limit of weak time-independent fields. Thus a second attempt at fixing the above equation might look like

$$\Box\Phi(\phi) = -4\pi G_N T \ , \tag{1.5}$$

where $\Phi(\phi) \approx \phi$ for weak fields.

Such scalar relativistic theories of gravity (or rather some minor variants thereof) were proposed and studied among others by Abraham, Mie, and Nordstrøm. As it stands, the field equation appears to be perfectly consistent (and it may be interesting to discuss if/how the Einstein equivalence principle, which will put us on our route towards metrics and space-time curvature is realised in such a theory).

However, regardless of this, this theory is incorrect simply because it is ruled out experimentally. The easiest way to see this (with hindsight) is to note that the energy-momentum tensor of Maxwell theory (7.47) is traceless (7.121), and thus the above equation would predict no coupling of gravity to the electro-magnetic field, in particular to light. Hence in such a theory there would be no deflection of light by the sun etc.[1]

[1] For more on the history and properties of scalar theories of gravity see the review by D. Giulini, *What is (not) wrong with scalar gravity?*, `arXiv:gr-qc/0611100`.

The other possibility to render (1.3) consistent is the, a priori perhaps much less compelling, option to think of ϕ and $\Delta\phi$ or $\Box\phi$ not as scalars but as (00)-components of some tensor, in which case one could try to salvage (1.3) by promoting it to a tensorial equation

$$\{\text{Some tensor generalising } \Delta\phi\}_{ab} \sim 4\pi G_N T_{ab} \ . \tag{1.6}$$

This appears to require not just one potential, but actually 10 of them,

$$\phi \to \phi_{ab} = \phi_{ba} \tag{1.7}$$

and this seems to be a rather crazy thing to do at this stage (in particular, without any insight into the nature of these potentials). However, this is indeed the form of the field equations for gravity (the Einstein equations) we will ultimately be led to (see section 19.4), but Einstein arrived at this in a completely different, and much more insightful, way.

Let us now, very briefly and in a streamlined way, try to retrace (one aspect of) Einstein's thoughts, namely on the relation between inertial and gravitational mass, which, as we will see, will lead us rather quickly to the geometric picture of gravity sketched in the Introduction.

To that end we return to the Newtonian equation of motion (1.1). Recall that in this Newtonian theory, there are two *a priori* completely independent concepts of mass:

- inertial mass m_i (or acceleration mass), which accounts for the resistance of a body or particle against acceleration and appears universally on the left-hand-side of the Newtonian equation of motion

  $$m_i \vec{a} = \vec{F} \tag{1.8}$$

  in conjunction with the acceleration $\vec{a}$;

- gravitational mass m_g which is the mass the gravitational field couples to, i.e. it is the gravitational charge of a particle,

  $$\vec{F}_g = -m_g \vec{\nabla}\phi \ . \tag{1.9}$$

Now it is an important empirical fact that the inertial mass of a body is equal to its gravitational mass. This realisation, at least with this clarity, is usually attributed to Newton, although it goes back to experiments and observations by Galileo usually paraphrased as "all bodies fall at the same rate in a gravitational field". (It is not true, though, that Galileo dropped objects from the leaning tower of Pisa to test this - he used an inclined plane, a water clock and a pendulum).

These experiments were later on improved, in various forms, by Huygens, Newton, Bessel and others and reached unprecedented accuracy with the work of Baron von

Eötvös (starting in 1889), who was able to show that inertial and gravitational mass of different materials (like wood and platinum) agree to one part in 10^9. In the 1950/60's, this was still further improved by R. Dicke to something like one part in 10^{11}. More recently, rumours of a 'fifth force', based on a reanalysis of Eötvös' data (but buried in the meantime) motivated experiments with even higher accuracy and no difference between m_i and m_g was found.

Newton's theory would in principle be perfectly consistent with $m_i \neq m_g$, just as the formally analogous equation for an electrically charged particle with charge q_e in an electrostatic field $\vec{E} = -\vec{\nabla}\phi$,

$$m_i \ddot{\vec{x}} = -q_e \vec{\nabla}\phi \quad , \tag{1.10}$$

is perfectly acceptable for any ratio q_e/m_i, and Einstein was very impressed with the observed equality of m_i and m_g. This should, he reasoned, not be a mere coincidence but is probably trying to tell us something rather deep about the nature of gravity.

To see what this could be, let us recall that there is a very common class of "forces" for which the equality between the inertial mass and the coupling constant is actually built in and automatic. These are the "pseudo-forces" or "fictitious forces" $\vec{P}$ (like centrifugal forces) which arise when one transforms the Newtonian equations of motion via a non-linear coordinate transformation to accelerated (or other non-Cartesian) coordinates,

$$x^i \to z^m = z^m(x^i) \tag{1.11}$$

(like spherical coordinates). These "forces" arise from the non-trivial transformation behaviour of the acceleration $\ddot{\vec{x}}$ under such non-linear coordinate transformations, and are therefore inevitably and automatically proportional to m_i,

$$m_i \ddot{\vec{x}} = \vec{F} \quad \Rightarrow \quad m_i \ddot{\vec{z}} = \vec{F} + \vec{P} \quad \text{with} \quad \vec{P} \sim m_i \quad . \tag{1.12}$$

To be explicit, note that if we perform such a time-independent coordinate transformation, the velocity and acceleration of a particle with trajectory $x^i(t)$ transform as

$$\begin{aligned} \dot{z}^m &= \frac{\partial z^m}{\partial x^i} \dot{x}^i \\ \ddot{z}^m &= \frac{\partial z^m}{\partial x^i} \ddot{x}^i + \frac{\partial^2 z^m}{\partial x^i \partial x^j} \dot{x}^i \dot{x}^j \end{aligned} \tag{1.13}$$

(cf. section 1.5 for an analogous calculation in relativistic mechanics in Minkowski space). The first line expresses the fact that velocities transform linearly (with the Jacobi matrix) under arbitrary coordinate transformations (and are thus the protyotype of what we will call vectors or tensors later on). In the coordinates z^m, the equations of motion thus take the form

$$m_i \ddot{z}^m = \frac{\partial z^m}{\partial x^i} F^i + P^m \equiv F^m + P^m \quad , \tag{1.14}$$

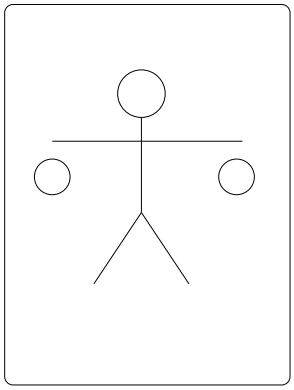


Figure 1: Experimenter and his two stones freely floating somewhere in outer space, i.e. in the absence of forces.

where it is the second term in the second line of (1.13) that gives rise to the (centrifugal etc.) pseudo-force

$$P^m = m_i \frac{\partial^2 z^m}{\partial x^i \partial x^j} \dot{x}^i \dot{x}^j \quad , \tag{1.15}$$

manifestly proportional to m_i. If one considers 3-dimensional coordinate transformations that depend explicitly on time, $z^m = z^m(x^i, t)$ (such as a transformation to a rotating reference system), then one finds additional contributions to the pseudo-force, like Coriolis forces etc. Conversely, any such pseudo-forces can be eliminated by transforming the equations of motion to a suitable (inertial) coordinate system or reference system.

With his unequalled talent for discovering profound truths in such simple observations, he concluded (calling this "der glücklichste Gedanke meines Lebens" (the happiest thought of my life)) that the equality of inertial and gravitational mass suggests a close relation between inertia and gravity itself, suggests, in fact, that locally effects of gravity and acceleration (or non-linear transformations of the reference system) are indistinguishable,

gravitational mass = inertial mass <u>because</u> (locally) GRAVITY = ACCELERATION

He substantiated this with some classical thought experiments, *Gedankenexperimente*, as he called them, which in the meantime have morphed into and have come to be known as the *elevator thought experiments*, which we will now discuss.

1. Consider somebody in a small sealed box (elevator) somewhere in outer space. In the absence of any forces, this person will float. Likewise, two stones he has just dropped (see Figure 1) will float with him.

2. Now assume (Figure 2) that somebody on the outside suddenly pulls the box up with a constant acceleration. Then of course, our friend will be pressed to the

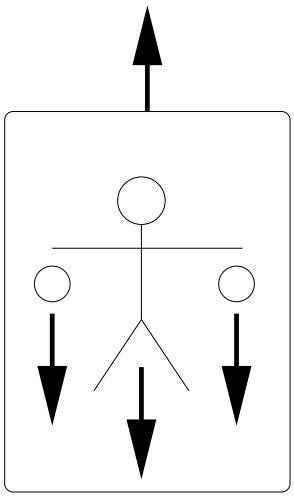


Figure 2: Constant acceleration upwards mimics the effect of a gravitational field: experimenter and stones drop to the bottom of the box.

bottom of the elevator with a constant force and he will also see his stones drop to the floor.

3. Now consider (Figure 3) this same box brought into a constant gravitational field. Then again, the experimenter will be pressed to the bottom of the elevator with a constant force and will see the stones drop to the floor. There is no experiment inside the elevator that permits him to decide if this is actually due to a gravitational field or due to the fact that somebody is pulling the elevator upwards.

   Thus our first lesson is that, indeed, locally the effects of acceleration and gravity are indistinguishable.

4. Now consider somebody cutting the cable of the elevator (Figure 4). Then the elevator will fall freely downwards but, as in Figure 1, our experimenter and his stones will float as in the absence of gravity.

   Thus lesson number two is that, locally the effect of gravity can be eliminated by going to a freely falling reference frame (or coordinate system). This should not come as a surprise, since this is of course built into the Newtonian theory: free fall in a constant gravitational field (in the vertial z-direction, say) is described by the equation

$$\ddot{z} = -g \ \ (+ \ \text{other forces}) \ . \tag{1.16}$$

   In the accelerated (freely falling) coordinate system

$$Z(z,t) = z + gt^2/2 \tag{1.17}$$

   the same physics is described by the equation

$$\ddot{Z} = 0 \ \ (+ \ \text{other forces}) \ , \tag{1.18}$$

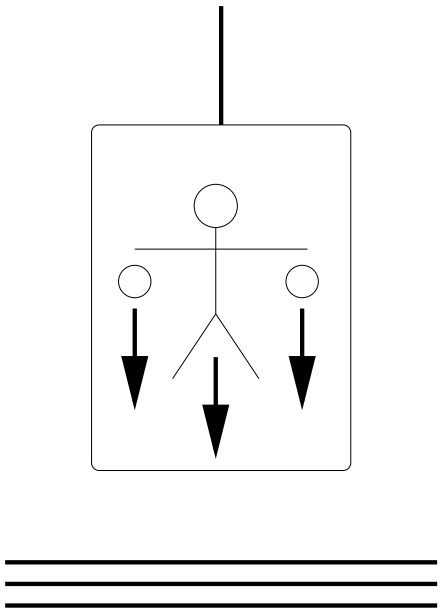


Figure 3: Effect of a constant gravitational field: indistinguishable for our experimenter from that of a constant acceleration in Figure 2.

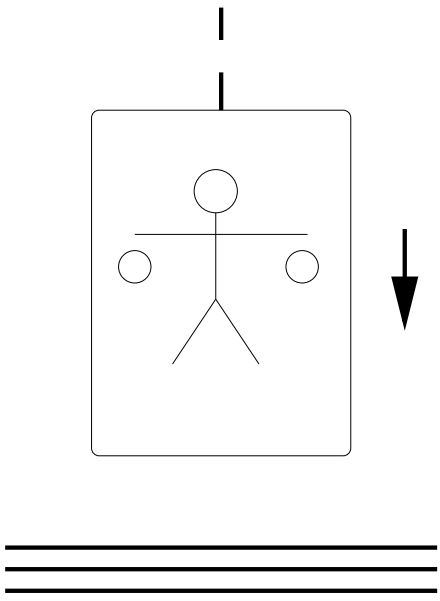


Figure 4: Free fall in a gravitational field has the same effect as no gravitational field (Figure 1): experimenter and stones float.

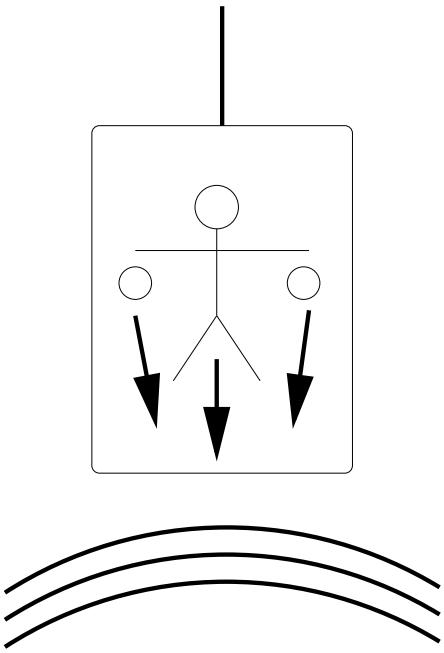


Figure 5: Experimenter and his stones in a non-uniform gravitational field: the stones will approach each other slightly as they fall to the bottom of the elevator.

and the effect of gravity has been eliminated completely by going to the freely falling reference system described by Z. The crucial point here is that in such a reference frame not only our observer will float freely, but because of the equality of inertial and gravitational mass he will also observe all other objects obeying the usual laws of motion in the absence of gravity.

5. In the above discussion, I have put the emphasis on constant accelerations and on 'locally'. To see the significance of this, consider our experimenter with his elevator in the gravitational field of the earth (Figure 5). This gravitational field is not constant but spherically symmetric, pointing towards the center of the earth. Therefore the stones will slightly approach each other as they fall towards the bottom of the elevator, in the direction of the center of the gravitational field.

6. Thus, if somebody cuts the cable now and the elevator is again in free fall (Figure 6), our experimenter will float again, so will the stones, but our experimenter will also notice that the stones move closer together for some reason. He will have to conclude that there is some force responsible for this.

   This is lesson number three: in a non-uniform gravitational field the effects of gravity cannot be eliminated by going to a freely falling coordinate system. This is only possible locally, on such scales on which the gravitational field is essentially constant.

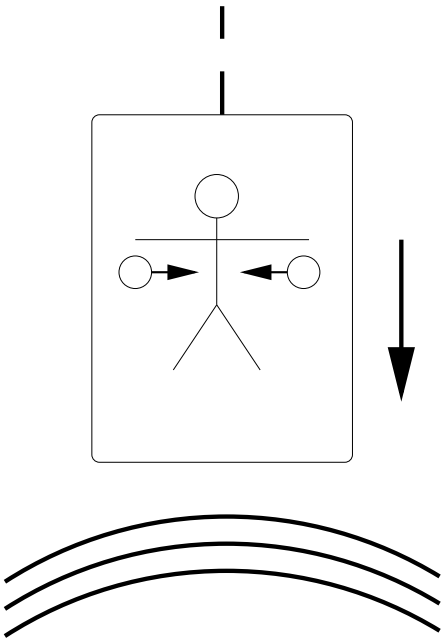


Figure 6: Experimentator and stones freely falling in a non-uniform gravitational field. The experimenter floats, so do the stones, but they move closer together, indicating the presence of some external force.

Einstein formalised the outcome of these thought experiments in what is now known as the *Einstein Equivalence Principle* which roughly states that physics in a freely falling frame in a gravitational field is the same as physics in an inertial frame in Minkowski space in the absence of gravitation. Two formulations are

> At every space-time point in an arbitrary gravitational field it is possible to choose a *locally inertial* (or *freely falling*) *coordinate system* such that, within a sufficiently small region of this point, the laws of nature take the same form as in unaccelerated Cartesian coordinate systems in the absence of gravitation.[2]

and

> Experiments in a sufficiently small freely falling laboratory, over a sufficiently short time, give results that are indistinguishable from those of the same experiments in an inertial frame in empty space.[3]

There are different versions of this principle depending on what precisely one means by 'the laws of nature'. If one just means the laws of Newtonian (or relativistic) mechanics,

[2] S. Weinberg, *Gravitation and Cosmology.*
[3] J. Hartle, *Gravity. An Introduction to Einstein's General Relativity.*

then this principle essentially reduces to the statement that inertial and gravitational mass are equal. Usually, however, this statement is taken to imply also Maxwell's theory, quantum mechanics etc.[4] What it pragmatically asserts in one of its stronger forms is that

> [...] there is no experiment that can distinguish a uniform acceleration from a uniform gravitational field. (J. Hartle, loc. cit.)

The power of the above principle, which we will regard as a heuristic guideline, rather than trying to (prematurely) give it a mathematically precise formulation, lies in the fact that we can combine it with our understanding of physics in accelerated reference systems to gain insight into the physics in a gravitational field. Two immediate consequences of this (which cannot be derived on the basis of Newtonian physics or Special Relativity alone) are

- light is deflected by a gravitational field just like material objects;
- clocks run slower in a gravitational field than in the absence of gravity.

To see the inevitability of the first assertion, imagine a lightray entering the rocket / elevator in Figure 1 horizontally through a window on the left hand side and exiting again at the same height through a window on the right. Now imagine, as in Figure 2, accelerating the elevator upwards. Then clearly the lightray that enters on the left will exit at a lower point of the elevator on the right because the elevator is accelerating upwards. By the equivalence principle one should observe exactly the same thing in a constant gravitational field (Figure 3). It follows that in a gravitational field the lightray is bent downwards, i.e. it experiences a downward acceleration with the (locally constant) gravitational acceleration g.

To understand the second assertion, one can e.g. simply appeal to the so-called "twin-paradox" of Special Relativity: the accelerated twin is younger than his unaccelerated inertial sibling. Hence accelerated clocks run slower than inertial clocks. Hence, by the equivalence principle, clocks in a gravitational field run slower than clocks in the absence of gravity.

Alternatively, one can imagine two observers at the top and bottom of the elevator, having identical clocks and sending light signals to each other at regular intervals as determined by their clocks. Once the elevator accelerates upwards, the observer at the bottom will receive the signals at a higher rate than he emits them (because he is accelerating towards the signals he receives), and he will interpret this as his clock

[4] For a discussion of different formulations of the equivalence principle and the logical relations among them, see E. di Casola, S. Liberati, S. Sonego, *Nonequivalence of equivalence principles*, `arXiv:1310.7426 [gr-qc]`.

running more slowly than that of the observer at the top. By the equivalence principle, the same conclusion now applies to two observers at different heights in a gravitational field. This can also be interpreted in terms of a gravitational redshift or blueshift (photons losing or gaining energy by climbing or falling in a gravitational field), and we will return to a more quantitative discussion of this effect in section 3.5.

1.2 Lorentz-Covariant Formulation of Special Relativity (Review)

What the equivalence principle tells us is that we can expect to learn something about the effects of gravitation by transforming the laws of nature (equations of motion) from an inertial Cartesian coordinate system to other (accelerated, curvilinear) coordinates. As a first step, we will, in section 1.3 below, discuss the above example of an observer undergoing constant acceleration in the context of special relativity.

As a preparation for this, and the remainder of the course, this section will provide a lightning review of the Lorentz-covariant formulation of special relativity, mainly to set the notation and conventions that will be used throughout, and only to the extent that it will be used in the following.

1. Minkowski space(-time)

   (a) The arena of special relativity is Minkowski space-time [henceforth *Minkowski space* for short, the union of space and time is implied by uttering the word "Minkowski"]. It is the space of events, labelled by 3 Cartesian spatial coordinates x^k and a time-coordinate t or, more usefully, by the coordinates

   $$(\xi^a) = (\xi^0 = ct, \xi^k = x^k) \ , \tag{1.19}$$

   where c is the speed of light. Typically in these notes ξ^a will indicate such a (locally) inertial coordinate system, whereas generic coordinates will be called x^μ etc. We will almost always work in units in which $c = 1$.

   (b) Minkowski space is equipped with a prescription for measuring distances, encoded in a line-element which, in these coordinates, takes the form

   $$ds^2 = -(d\xi^0)^2 + \sum_k (d\xi^k)^2 \ . \tag{1.20}$$

   (c) This can be written as

   $$ds^2 = -(d\xi^0)^2 + \sum_k (d\xi^k)^2 \equiv \eta_{ab}\, d\xi^a d\xi^b \ . \tag{1.21}$$

   with metric $(\eta_{ab}) = \mathrm{diag}(-1, +1, +1, +1)$ or, more explicitly,

   $$\eta_{ab} = \begin{pmatrix} -1 & 0 & 0 & 0 \\ 0 & +1 & 0 & 0 \\ 0 & 0 & +1 & 0 \\ 0 & 0 & 0 & +1 \end{pmatrix} \tag{1.22}$$

(thus we are using the "mostly plus" convention).

2. Lorentz Transformations

   (a) Lorentz transformations are by definition those linear transformations

   $$\xi^a \mapsto \bar{\xi}^a = L^a_{\ b}\xi^b \tag{1.23}$$

   that leave the Minkowski line-element invariant,

   $$d\bar{s}^2 \equiv \eta_{ab}d\bar{\xi}^a d\bar{\xi}^b = \eta_{ab}d\xi^a d\xi^b = ds^2 \quad \Leftrightarrow \quad \eta_{ab}L^a_{\ c}L^b_{\ d} = \eta_{cd} \ . \tag{1.24}$$

   In matrix notation this can also be written as

   $$\bar{\xi} = L\xi \quad , \quad L^t\eta L = \eta \tag{1.25}$$

   where L^t is the transpose of L. This is the defining condition for Lorentz transformations, and the Lorentz signature analogue of the condition $R^t\mathbb{1}R = \mathbb{1}$ for an orthogonal transformation (rotation or reflection) in Euclidean space, with metric $\eta_{ab} \to \mathbb{1}_{ik} = \delta_{ik}$.
   Alternative notation:

   $$\bar{\xi}^{\bar{a}} = L^{\bar{a}}_{\ b}\xi^b \quad \text{or} \quad \xi^{\bar{a}} = L^{\bar{a}}_{\ b}\xi^b \tag{1.26}$$

   Strictly speaking $\bar{\xi}^a$ and $\xi^{\bar{a}}$ may be considered to refer to two different quantities, to the coordinates of the new point $\bar{\xi}$ in the old coordinate system versus the coordinates of the old point ξ in the new coordinate system. However, for many elementary purposes this difference between what is known as the active (moving points) versus the passive (relabelling points) point of view is not crucial, and one should not be hung-up on notation: coordinates (and in particular indices referring to them) are fundamentally just bookkeeping devices so use whatever is convenient for current bookkeeping or other purposes.

   (b) Infinitesimal Lorentz rotations, i.e. Lorentz transformations with L of the form $L = \mathbb{1} + \omega$, ω infinitesimal, are characterised by

   $$(\mathbb{1} + \omega)^t\eta(\mathbb{1} + \omega) = \eta \quad \Rightarrow \quad (\eta\omega) + (\eta\omega)^t = 0 \ . \tag{1.27}$$

   Thus the matrix $\eta\omega$ is anti-symmetric. In components, an infinitesimal Lorentz transformation therefore has the form

   $$\delta\xi^a = \omega^a_{\ b}\xi^b \quad \text{with} \quad \omega_{ab} \equiv \eta_{ac}\omega^c_{\ b} = -\omega_{ba} \ . \tag{1.28}$$

   (c) Poincaré transformations are those affine transformations that leave the Minkowski line-element invariant. They are composed of Lorentz transformations and arbitrary constant translations and thus have the form

   $$\xi^a \mapsto \bar{\xi}^a = L^a_{\ b}\xi^b + \zeta^a \ , \tag{1.29}$$

infinitesimally

$$\delta\xi^a = \omega^a_{\ b}\xi^b + \epsilon^a \quad . \tag{1.30}$$

Any two inertial systems in the sense of the equivalence principle of special relativity are related by a Poincaré transformation.

3. Distance & Causal Structure

   (a) The Minkowski metric defines the Lorentz (and Poincaré) invariant distance

   $$(\Delta\xi)^2 = \eta_{ab}(\xi^a_P - \xi^a_Q)(\xi^b_P - \xi^b_Q) \tag{1.31}$$

   betwen two events P and Q with coordinates ξ^a_P and ξ^a_Q respectively.

   (b) Depending on the sign of $(\Delta\xi)^2$, the two events P, Q are called, spacelike, lightlike (null) or timelike separated,

   $$(\Delta\xi)^2 = \begin{cases} > 0 & \text{spacelike} \quad \text{separated} \\ = 0 & \text{lightlike} \quad \text{separated} \\ < 0 & \text{timelike} \quad \text{separated} \end{cases} \tag{1.32}$$

   (c) The set of events that are lightlike separated from P define the lightcone at P. It consists of two components (joined at P), the future and the past lightcone, distinguished by the sign of $\xi^0_Q - \xi^0_P$ (positive for Q on the future lightcone, $\xi^0_Q > \xi^0_P$, negative for Q on the past lightcone).

4. Curves and Tangent Vectors

   (a) A parametrised curve is given by a map $\lambda \mapsto \xi^a(\lambda)$. The tangent vector to the curve at the point $\xi(\lambda_0)$ has components

   $$\xi'^a(\lambda_0) = \frac{d}{d\lambda}\xi^a(\lambda)|_{\lambda=\lambda_0} \quad . \tag{1.33}$$

   It is called spacelike, lightlike (null) or timelike, depending on the sign of $\eta_{ab}\xi'^a\xi'^b$,

   $$\eta_{ab}\xi'^a\xi'^b \quad \begin{cases} > 0 & \text{spacelike} \\ = 0 & \text{lightlike} \\ < 0 & \text{timelike} \end{cases} \tag{1.34}$$

   This sign (and hence this classification) depends only on the image of the curve, not its parametrisation.

   (b) A curve whose tangent vector is everywhere timelike is called a timelike curve (and likewise for lightlike and spacelike curves). A curve whose tangent vector is everywhere timelike or null (i.e. non-spacelike) is called a causal curve. Worldlines of massive particles are timelike curves, those of massless particles (light) are null curves.

(c) A natural Lorentz-invariant parametrisation of timelike curves is provided by the Lorentz-invariant proper time τ along the curves,

$$\xi^a = \xi^a(\tau) \ , \tag{1.35}$$

with

$$\begin{aligned} & c d\tau = \sqrt{-ds^2} = \sqrt{-\eta_{ab} d\xi^a d\xi^b} = \sqrt{-\eta_{ab} \xi'^a \xi'^b} d\lambda \\ \Rightarrow \quad & \eta_{ab} \frac{d\xi^a(\tau)}{d\tau} \frac{d\xi^b(\tau)}{d\tau} = -c^2 \ . \end{aligned} \tag{1.36}$$

Likewise spacelike curves are naturally parametrised by proper distance ds. The derivative with respect to proper time will be denoted by an overdot,

$$\dot{\xi}^a(\tau) = \frac{d}{d\tau} \xi^a(\tau) \ . \tag{1.37}$$

Because τ is Lorentz-invariant, $\bar{\tau} = \tau$, tangent vectors $\dot{\xi}^a$ of τ-parametrised curves transform linearly under Lorentz transformations,

$$\dot{\bar{\xi}}^a(\tau) = \frac{d}{d\tau} \bar{\xi}^a(\tau) = \frac{\partial \bar{\xi}^a}{\partial \xi^b} \frac{d}{d\tau} \xi^b(\tau) = L^a_{\ b} \dot{\xi}^b(\tau) \ . \tag{1.38}$$

These are the prototypes of what are called Lorentz vectors or, more generally, Lorentz tensors.

5. Lorentz Vectors

(a) Lorentz vectors (or 4-vectors) are objects with components v^a which transform under Lorentz transformations with the matrix $L^a_{\ b}$ (to be thought of as the Jacobian of the transformation relating $\bar{\xi}^a$ and ξ^a),

$$\bar{v}^a = L^a_{\ b} v^b \ . \tag{1.39}$$

(b) η_{ab} can be regarded as defining an indefinite scalar product on the space of Lorentz vectors, and the Minkowski norm $\eta_{ab} v^a v^b$ and the Minkowski scalar product $\eta_{ab} v^a w^b$ of Lorentz vectors are invariant under Lorentz transformations,

$$\eta_{ab} \bar{v}^a \bar{v}^b = \eta_{ab} v^a v^b \quad , \quad \eta_{ab} \bar{v}^a \bar{w}^b = \eta_{ab} v^a w^b \ . \tag{1.40}$$

A vector is called, spacelike, lightlike (null) or timelike depending on the sign of its Minkowski norm.

(c) One can identify Minkowski space with its "tangent space", i.e. with the vector space $\mathbb{V} = \mathbb{R}^{1,3}$ of 4-vectors equipped with the quadratic form or scalar product η_{ab} with signature (1,3).

6. Other Lorentz Tensors

(a) Lorentz scalars are objects that are invariant under Lorentz transformations. Examples are scalar products and norms of Lorentz vectors.

(b) Lorentz covectors are objects u_a that transform under Lorentz transformations with the dual (or contragredient = transpose inverse) representation

$$\Lambda = (L^t)^{-1} = \eta L \eta^{-1} \ , \tag{1.41}$$

i.e.

$$\bar{u}_a = \Lambda_a{}^b u_b \quad , \quad \Lambda_a{}^b = \eta_{ac} L^c{}_d \eta^{db} \ , \tag{1.42}$$

where η^{ab} denotes the components of the inverse metric η^{-1}. In terms of Λ, the condition that L is a Lorentz transformation (i.e. preserves η_{ab}) can evidently be written as

$$L^t \eta L = \eta \quad \Leftrightarrow \quad \Lambda \eta \Lambda^t = \eta \quad \Leftrightarrow \quad \Lambda_a^c \Lambda_b{}^d \eta_{cd} = \eta_{ab} \ . \tag{1.43}$$

Covectors can be regarded as elements of the dual $\mathbb{V}^*$ of the space $\mathbb{V}$ of 4-vectors, with u_a defining the Lorentz-invariant linear mapping

$$u : \ v \in \mathbb{V} \mapsto u(v) = u_a v^a \in \mathbb{R} \ . \tag{1.44}$$

Examples are $u_a = \eta_{ab} v^b \equiv v_a$ with v^a a Lorentz vector, the scalar product η_{ab} defining the isomorphism $\mathbb{V}^* \cong \mathbb{V}$.

(c) Lorentz (p,q)-tensors are objects that transform under Lorentz transformations like a product of p vectors and q covectors,

$$T^{a_1 \ldots a_p}_{c_1 \ldots c_q} \to \bar{T}^{a_1 \ldots a_p}_{c_1 \ldots c_q} = L^{a_1}_{\ b_1} \ldots L^{a_p}_{\ b_p} \Lambda_{c_1}^{\ d_1} \ldots \Lambda_{c_q}^{\ d_q} \ T^{b_1 \ldots b_p}_{d_1 \ldots d_q} \ . \tag{1.45}$$

In particular, direct products of vectors and covectors like $V^a W^b U_c$ are tensors. A special case is η_{ab}, which is a Lorentz-invariant $(0,2)$-tensor by definition,

$$\bar{\eta}_{ab} = \Lambda_a^c \Lambda_b{}^d \eta_{cd} = \eta_{ab} \ . \tag{1.46}$$

Linear combinations of (p,q)-tensors are again (p,q)-tensors. Arbitrary products and contractions of Lorentz tensors are again Lorentz tensors (and the tensor type can be read off from the number and position of the "free" indices).

7. Tensor Fields

(a) Lorentz tensor fields are assignments of Lorentz tensors to each point of Minkowski space,

$$T : \xi \mapsto T^{a_1 \ldots a_p}_{c_1 \ldots c_q}(\xi) \ . \tag{1.47}$$

(b) Given a vector field $V^a(\xi)$, $\eta_{ab}V^a(\xi)V^b(\xi)$ is an example of a scalar field, and given a scalar field $f(\xi)$, its partial derivatives give a covector field

$$U_a(\xi) = \partial_{\xi^a} f(\xi) \equiv \partial_a f(\xi) \tag{1.48}$$

(providing the justification for abbreviating $\partial_{\xi^a} = \partial_a$). More generally, the partial derivatives of the components of a (p,q)-tensor,

$$T^{a_1\ldots a_p}_{c_1\ldots c_q}(\xi) \quad \to \quad \partial_a T^{a_1\ldots a_p}_{c_1\ldots c_q}(\xi) \tag{1.49}$$

are the components of a $(p, q+1)$-tensor, and the wave operator

$$\Box = \eta^{ab}\partial_a\partial_b \tag{1.50}$$

is a Lorentz-invariant differential operator mapping (p,q) tensor fields to (p,q) tensor fields.

(c) Tensorial equations of the form

$$T^{a_1\ldots a_p}_{c_1\ldots c_q}(\xi) = 0 \tag{1.51}$$

are Lorentz invariant in the sense that they are satisfied in one inertial system iff they are satisfied in all inertial systems. (Here and throughout these notes "iff" is the usual mathematicians' shorthand for "if and only if".)

8. Worldlines of Massive Particles

(a) In the covariant formulation, the timelike worldline of a massive particle is parametrised by proper time, $\xi^a = \xi^a(\tau)$. The velocity (tangent) vector

$$u^a \equiv \dot{\xi}^a(\tau) \tag{1.52}$$

is a Lorentz vector, normalised as

$$u^a u_a \equiv \eta_{ab}u^a u^b = -c^2 \ . \tag{1.53}$$

(b) The Lorentz-covariant acceleration is the 4-vector

$$a^c = \frac{d}{d\tau}u^c = \frac{d^2}{d\tau^2}\xi^c \ , \tag{1.54}$$

and the equation of motion of a massive free particle is

$$a^c = \frac{d^2}{d\tau^2}\xi^c(\tau) = 0 \ . \tag{1.55}$$

We will study this equation further (in any arbitrary coordinate system) in section 1.5. For observers with non-zero acceleration it follows from (1.53) by differentiation that a^c is orthogonal to u^b,

$$a^c u_c \equiv \eta_{cb}a^c u^b = 0 \ , \tag{1.56}$$

and therefore spacelike,

$$\eta_{cb}a^c a^b \equiv \mathsf{a}^2 > 0 \ . \tag{1.57}$$

Observers with constant acceleration will be the subject of section 1.3.

(c) The action for a free massive particle with worldline $\xi^a(\tau)$ is essentially the total proper time along the path,

$$S[\xi] = -mc^2 \int d\tau = -mc \int \sqrt{-\eta_{ab} d\xi^a d\xi^b} \ , \tag{1.58}$$

worldlines of free massive particles extremising (maximising) the proper time. In terms of an arbitrary parametrisation $\xi^a = \xi^a(\lambda)$ of the path, this action can be written as

$$S[\xi] = \int d\lambda \, L_\lambda \quad , \quad L_\lambda = -mc \left(-\eta_{ab} \frac{d\xi^a}{d\lambda} \frac{d\xi^b}{d\lambda} \right)^{1/2} \ . \tag{1.59}$$

A special choice is $\lambda = t$, for which

$$L_t = -mc^2 \sqrt{1 - \vec{v}^2/c^2} \qquad \vec{v} = d\vec{\xi}/dt = d\vec{x}/dt \ . \tag{1.60}$$

9. Energy-Momentum 4-Vector

(a) The covariant momenta p_a are defined by

$$p_a = \frac{\partial L_\lambda}{\partial (d\xi^a/d\lambda)} = m \eta_{ab} u^b \quad \Rightarrow \quad p^a = m u^a = m(d\xi^a/d\tau) \tag{1.61}$$

(independently of the choice of λ).

(b) Its components are

$$p^0 = E/c \quad , \quad p^k = p^{(c)k} \tag{1.62}$$

where $p^{(c)k}$ are the canonical momenta associated to the Lagrangian L_t,

$$p_k^{(c)} = \frac{\partial L_t}{\partial v^k} = m\gamma(v) v_k \ , \tag{1.63}$$

with $\gamma(v) = (1 - \vec{v}^2/c^2)^{-1/2}$, and $E = H$ is the corresponding energy or Hamiltonian

$$H = p_k^{(c)} v^k - L_t = m\gamma(v) c^2 \ . \tag{1.64}$$

(c) The p^a are the components of a Lorentz vector, the energy-momentum 4-vector. It satisfies the mass-shell relation

$$\eta_{ab} p^a p^b = -m^2 c^2 \quad \Leftrightarrow \quad E^2 = m^2 c^4 + \vec{p}^2 c^2 \ . \tag{1.65}$$

1.3 Accelerated Observers in Minkowski Space and the Rindler Metric

We return to the issue discussed in the context of the Einstein equivalence principle in section 1.1, namely physics as experienced by an observer undergoing constant acceleration (as a precursor to studying this observer in a genuine gravitational field), now specifically within the framework of special relativity.

Specialising (1.56) to an observer accelerating in the ξ^1-direction (so that in the momentary restframe of this observer one has $u^a = (1,0,0,0), a^a = (0,\mathsf{a},0,0)$), we will say that the observer undergoes constant acceleration if a is time-independent. To determine the worldline of such an observer, we note that the general solution to (1.53) with $u^2 = u^3 = 0$,

$$\eta_{ab}u^a u^b = -(u^0)^2 + (u^1)^2 = -1 \quad , \tag{1.66}$$

is

$$u^0 = \cosh F(\tau) \quad , \quad u^1 = \sinh F(\tau) \tag{1.67}$$

for some function $F(\tau)$. Thus the acceleration is

$$a^a = \dot{F}(\tau)(\sinh F(\tau), \cosh F(\tau), 0, 0) \quad , \tag{1.68}$$

with norm

$$\eta_{ab}a^a a^b = \mathsf{a}^2 = \dot{F}^2 \quad , \tag{1.69}$$

and an observer with constant acceleration is characterised by $F(\tau) = \mathsf{a}\tau$,

$$u^a(\tau) = (\cosh \mathsf{a}\tau, \sinh \mathsf{a}\tau, 0, 0) \quad . \tag{1.70}$$

This can now be integrated, and in particular

$$\xi^a(\tau) = (\mathsf{a}^{-1}\sinh \mathsf{a}\tau, \mathsf{a}^{-1}\cosh \mathsf{a}\tau, 0, 0) \tag{1.71}$$

is the worldline of an observer with constant acceleration a and initial condition $\xi^a(\tau = 0) = (0, \mathsf{a}^{-1}, 0, 0)$. The worldlines of this observer is the hyperbola

$$\eta_{ab}\xi^a\xi^b = -(\xi^0)^2 + (\xi^1)^2 = \mathsf{a}^{-2} \tag{1.72}$$

in the quadrant $\xi^1 > |\xi^0|$ of Minkowski space-time.

We can now ask the question what the Minkowski metric or line-element looks like in the restframe of such an observer. Note that one cannot expect this to be again the constant Minkowski metric η_{ab}: the transformation to an accelerated reference system, while certainly allowed in special relativity, is not a Lorentz transformation, while η_{ab} is, by definition, invariant under Lorentz-transformations.

We are thus looking for coordinates that are adapted to these accelerated observers in the same way that the inertial coordinates are adapted to static observers (ξ^0 is proper time, and the spatial components ξ^i remain constant). In other words, we seek a coordinate transformation $(\xi^0, \xi^1) \to (\eta, \rho)$ such that the worldlines of these accelerated observers are characterised by $\rho =$ constant (this is what we mean by restframe, the observer stays at a fixed value of ρ) and ideally such that then η is proportional to the proper time of the observer.

Comparison with (1.71) suggests the coordinate transformation

$$\xi^0(\eta, \rho) = \rho \sinh \eta \qquad \xi^1(\eta, \rho) = \rho \cosh \eta \quad . \tag{1.73}$$

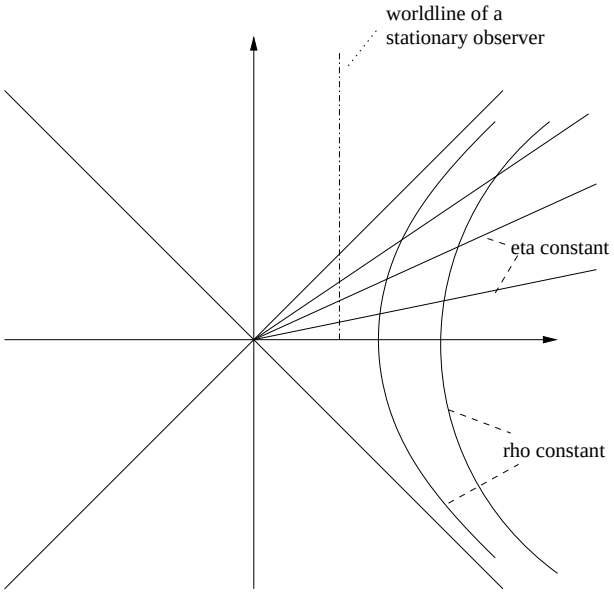


Figure 7: Rindler metric: Rindler coordinates (η, ρ) cover the first quadrant $\xi^1 > |\xi^0|$. Indicated are lines of constant ρ (hyperbolas, worldlines of constantly accelerating observers) and lines of constant η (straight lines through the origin). The quadrant is bounded by the lightlike lines $\xi^0 = \pm\xi^1 \Leftrightarrow \eta = \pm\infty$. An inertial observer reaches and crosses the line $\eta = \infty$ in finite proper time $\tau = \xi^0$.

It is now easy to see that in terms of these new coordinates the 2-dimensional Minkowski metric $ds^2 = -(d\xi^0)^2 + (d\xi^1)^2$ (we are now suppressing, here and in the remainder of this subsection, the transverse spectator dimensions 2 and 3) takes the form

$$ds^2 = -\rho^2 d\eta^2 + d\rho^2 \quad . \tag{1.74}$$

This is the so-called *Rindler metric*.

Let us try to gain a better understanding of these Rindler coordinates (illustrated in Figure 7 - see also Figure 25 in section 28.4 for a *Penrose Diagram* illustration).

- The Rindler coordinates ρ and η are obvisouly in some sense hyperbolic (Lorentzian) analogues of polar coordinates ($x = r\cos\phi, y = r\sin\phi, ds^2 = dx^2 + dy^2 = dr^2 + r^2 d\phi^2$). In particular, since

  $$(\xi^1)^2 - (\xi^0)^2 = \rho^2 \qquad , \qquad \frac{\xi^0}{\xi^1} = \tanh\eta \quad , \tag{1.75}$$

  by construction the lines of constant ρ, $\rho = \rho_0$, are hyperbolas, $(\xi^1)^2 - (\xi^0)^2 = \rho_0^2$, while the lines of constant $\eta = \eta_0$ are straight lines through the origin, $\xi^0 = (\tanh\eta_0)\xi^1$.

- The null lines $\xi^0 = \pm\xi^1$ correspond to $\eta = \pm\infty$. Thus the Rindler coordinates cover the first quadrant $\xi^1 > |\xi^0|$ of Minkowski space and can be used as coordinates there.

- The metric in these new coordinates is time-independent, where time means η, and time-independent means that the coefficients of the metric or line-element in (1.74) do not depend on η. This is due to the fact that the generator ∂_η of η-"time evolution" is actually the generator of a Lorentz boost in the (ξ^0, ξ^1)-plane in Minkowski space,

$$\partial_\eta = (\partial_\eta \xi^0)\partial_{\xi^0} + (\partial_\eta \xi^1)\partial_{\xi^1} = \xi^1 \partial_{\xi^0} + \xi^0 \partial_{\xi^1} \quad . \tag{1.76}$$

  Since a Lorentz boost leaves the Minkowski metric invariant, the latter has to be invariant under translations in η, i.e. it has to be η-independent, as is indeed the case.

- Along the worldline of an observer with constant ρ one has $d\tau = \rho_0 d\eta$, so that his proper time parametrised path is

$$\xi^0(\tau) = \rho_0 \sinh \tau/\rho_0 \qquad \xi^1(\tau) = \rho_0 \cosh \tau/\rho_0 \quad , \tag{1.77}$$

  and his 4-velocity is given by

$$u^0 = \frac{d}{d\tau}\xi^0(\tau) = \cosh \tau/\rho_0 \qquad u^1 = \frac{d}{d\tau}\xi^1(\tau) = \sinh \tau/\rho_0 \quad . \tag{1.78}$$

  These satisfy $-(u^0)^2+(u^1)^2 = -1$ (as they should), and comparison with (1.70,1.71) shows that the observer's (constant) acceleration is $\mathsf{a} = 1/\rho_0$.

Even though (1.74) is just the metric of Minkowski space-time, written in accelerated coordinates, this metric exhibits a number of interesting features that are prototypical of more general metrics that one encounters in general relativity:

1. First of all, we notice that the coefficients of the line element (metric) in (1.74) are no longer constant (space-time independent). Since in the case of constant acceleration we are just describing a "fake" gravitational field, this dependence on the coordinates is such that it can be completely and globally eliminated by passing to appropriate new coordinates (namely inertial Minkowski coordinates). Since, by the equivalence principle, locally an observer cannot distinguish between a fake and a "true" gravitational field, this now suggests that a "true" gravitational field can be described in terms of a space-time coordinate dependent line-element

$$ds^2 = g_{\alpha\beta}(x) dx^\alpha dx^\beta \quad , \tag{1.79}$$

   where the coordinate dependence on the x^α is now such that it cannot be eliminated globally by a suitable choice of coordinates.

2. We observe that (1.74) appears to be ill-defined at $\rho = 0$. However, in this case we already know that this is a mere *coordinate singularity* at $\rho = 0$ (akin to the

coordinate singularity at the origin of standard polar coordinates in the Cartesian plane). More generally, whenever a metric written in some coordinate system appears to exhibit some singular behaviour, one needs to investigate whether this is just a coordinate singularity or a true singularity of the gravitational field itself.

3. The above coordinates do not just fail at $\rho = 0$, they actually fail to cover large parts of Minkowski space. Thus the next lesson is that, given a metric in some coordinate system, one has to investigate if the space-time described in this way needs to be extended beyond the range of the original coordinates. One way to analyse this question (which we will make extensive use of in sections 26 and 27 when trying to understand and come to terms with black holes) is to study lightrays or the worldlines of freely falling (inertial) observers.

   In the present case, an example of an inertial observer is a static observer in Minkowski space, i.e. an observer at a fixed value of ξ^1, say, with $\xi^0 = \tau$ his proper time. In Rindler coordinates this is described by the condition that $\xi^1 = \rho \cosh \eta$ is a constant, so this is most certainly not a straight line in an (η, ρ)-diagram.

   Such an observer will of course "discover" that $\eta = +\infty$ is not the end of the world (indeed, he crosses this line at finite proper time $\tau = \xi^1$) and that Minkowski space continues (at the very least) into the quadrant $\xi^0 > |\xi^1|$ (see Figure 7 for an illustration of this).

4. Related to this is the behaviour of lightcones when expressed in terms of the coordinates (η, ρ) or when drawn in the (η, ρ)-plane (do this!). These lightcones satisfy $ds^2 = 0$, i.e.

$$\rho^2 d\eta^2 = d\rho^2 \quad \Rightarrow \quad d\eta = \pm \rho^{-1} d\rho \ . \tag{1.80}$$

   describing outgoing (ρ grows with η) respectively ingoing (ρ decreases with increasing η) lightrays. These lightcones have the familiar Minkowskian shape at $\rho = 1$, but the lightcones open up for $\rho > 1$ and become more and more narrow for $\rho \to 0$, once again exactly as we will find for the Schwarzschild black hole metric (see Figure 16 in section 26).

5. It follows from (1.76) that the Minkowski norm of ∂_η is

$$|\partial_\eta|^2 = (\xi^0)^2 - (\xi^1)^2 \ . \tag{1.81}$$

   Thus this generator of Rindler time-translations really is timelike in the region $\xi^1 > |\xi^0|$ covered by the Rindler coordinates, but it actually becomes lightlike on the lightlike boundary $\xi^1 = |\xi^0|$ of that region. As we will discuss in section 27.10, such a *Killing horizon* also happens to be one of the characteristic properties of a black hole.

6. Finally we note that there is a large region of Minkowski space that is "invisible" to the constantly accelerated observers. While a static observer will eventually receive information from any event anywhere in space-time (his past lightcone will eventually cover all of Minkowski space ...), the past lightcone of one of the Rindler accelerated observers (whose worldlines asymptote to the lightcone direction $\xi^0 = \xi^1$) will asymptotically only cover one half of Minkowski space, namely the region $\xi^0 < \xi^1$. Thus any event above the line $\xi^0 = \xi^1$ will forever be invisible to this class of observers. Such an observer-dependent horizon has some similarities with the *event horizon* characterising a black hole (see section 27.5 for a first encounter with such an object, and section 32 for a detailed discussion).

For more on Rindler coordinates, see sections 3.4 and 7.8.

1.4 General Coordinate Transformations in Minkowski Space I: Metric

In order to move away from constant accelerations (as models of observers in constant gravitational fields only), we now consider the effect of arbitrary (general) coordinate transformations on the laws of special relativity and the geometry of Minkowski space. This may look like a somewhat exaggerated move at this point (should we perhaps not just look at coordinate transformations to coordinates that somehow correspond to adapted coordinates for some arbitrary accelerated observer?), but

- it is actually easier to just do this than to understand what is meant precisely by this parenthetical remark and how to implement it;
- there are many useful things that one can learn from doing this;
- and we will see later (when discussing the relation between the *Einstein Equivalence Principle* and the *Principle of General Covariance* in section 4.1), that the relation between the description of physics in an arbitrary gravitational field and the behaviour of this description under arbitrary coordinate transformations is much closer and more far-reaching than we perhaps have the right to expect at the moment.

Let us see what things look like when written in some other (non-inertial, accelerating) coordinate system. It is extremely useful for bookkeeping purposes and for avoiding algebraic errors to use different kinds of indices for different coordinate systems. Thus we will usually call the new coordinates $x^\mu(\xi^b)$ or $x^\alpha(\xi^a)$, and not, say, $x^a(\xi^b)$ (although there would be nothing wrong with that).

We start with the definition of proper time, as described in inertial coordinates by the Minkowski line element,

$$d\tau^2 = -\eta_{ab} d\xi^a d\xi^b \tag{1.82}$$

First of all, this proper time should not depend on which coordinates we use to describe the motion of the particle, but only on the world line of the particle itself. After all, the particle could not care less what coordinates we experimenters or observers use to describe the particle's proper time.

[By the way: this is the best way to resolve the so-called "twin-paradox", which should really be referred to (and presented) as the "twin-non-paradox" or simply the "twin-fact" - everything else is deliberate and unhelpful obfuscation! It does not matter which reference system you use - the accelerating twin in the rocket will always be younger than her brother when they meet again.]

Thus all we need to know is how the same proper time τ is expressed in terms of the new coordinates, which simply follows from

$$d\tau^2 = -\eta_{ab} d\xi^a d\xi^b = -\eta_{ab} \frac{\partial \xi^a}{\partial x^\mu} \frac{\partial \xi^b}{\partial x^\nu} dx^\mu dx^\nu \quad . \tag{1.83}$$

Here

$$J^a_\mu(x) = \frac{\partial \xi^a}{\partial x^\mu} \tag{1.84}$$

is the Jacobi matrix associated to the coordinate transformation $\xi^a = \xi^a(x^\mu)$, and we will make the assumption that (locally) this matrix is non-degenerate, thus has an inverse $J^\mu_a(x)$ or $J^\mu_a(\xi)$ which is the Jacobi matrix associated to the inverse coordinate transformation $x^\mu = x^\mu(\xi^a)$,

$$J^a_\mu J^\mu_b = \delta^a_b \qquad J^\mu_a J^a_\nu = \delta^\mu_\nu \quad . \tag{1.85}$$

We see that in the new coordinates, proper time and distance are no longer measured by the Minkowski metric in its standard form (the constant matrix η_{ab}), but by

$$d\tau^2 = -\eta_{\mu\nu}(x) dx^\mu dx^\nu \quad , \tag{1.86}$$

where the *metric tensor* (or *metric* for short) $\eta_{\mu\nu}(x)$, the Minkowski metric in the coordinates x^μ, is

$$\eta_{\mu\nu}(x) = \eta_{ab} \frac{\partial \xi^a}{\partial x^\mu} \frac{\partial \xi^b}{\partial x^\nu} \quad . \tag{1.87}$$

Remarks:

1. The fact that the Minkowski metric written in the coordinates x^μ in general depends on x should not come as a surprise - after all, this also happens when one writes the Euclidean metric in spherical coordinates, $ds^2 = dx^2 + dy^2 = dr^2 + r^2 d\phi^2$ etc.

2. Even though the components of the metric are not those of the Minkowski metric in inertial coordinates, this metric or line element (with its associated presciption for time- and space-measurements) still describes exactly the same Minkowskian

geometry as the standard Minkowski metric η_{ab}: just as passing from Cartesian to spherical coordinates in $\mathbb{R}^3$ does not change the Euclidean geometry of the space, using some other than inertial coordinates does not change the Minkowskian geometry of Minkowski space.

3. It is easy to check, using (1.85), that the inverse metric, which we will denote by $\eta^{\mu\nu}$,

$$\eta^{\mu\nu}(x)\eta_{\nu\lambda}(x) = \delta^{\mu}_{\lambda} \ , \tag{1.88}$$

is given by

$$\eta^{\mu\nu}(x) = \eta^{ab}\frac{\partial x^{\mu}}{\partial \xi^{a}}\frac{\partial x^{\nu}}{\partial \xi^{b}} \ . \tag{1.89}$$

In terms of Jacobi matrices we have

$$\eta_{\mu\nu} = J^{a}_{\mu}J^{b}_{\nu}\eta_{ab} \quad , \quad \eta^{\mu\nu} = J^{\mu}_{a}J^{\nu}_{b}\eta^{ab} \ . \tag{1.90}$$

The rationale for using a different positioning of the indices for the metric and its inverse is that, as is apparent form the above equations, they transform in a *different* way under coordinate transformations: one with the Jacobi matrices, the other with their inverses. Thus geometrically they are different kinds of objects, and (as in the tensor algebra for Lorentz tensors of special relativity) the positioning of the indices is used to indicate and keep track of this fact in a simple and intuitive manner.

We will have much more to say about the metric below and, indeed, throughout this course.

1.5 General Coordinate Transformations in Minkowski Space II: Free Particle

We now turn to the equation of motion of a free particle, given in inertial coordinates by

$$\frac{d^2}{d\tau^2}\xi^{a}(\tau) = 0 \ . \tag{1.91}$$

The usual rules for a change of variables give

$$\frac{d}{d\tau}\xi^{a} = \frac{\partial \xi^{a}}{\partial x^{\mu}}\frac{dx^{\mu}}{d\tau} \ , \tag{1.92}$$

where $J^{a}_{\mu} = \frac{\partial \xi^{a}}{\partial x^{\mu}}$ is the invertible Jacobi matrix. This shows that, as usual, velocities transform in a particularly simple (linear, vectorial) way under arbitrary coordinate transformations, namely just with the Jacobi matrix,

$$\dot{\xi}^{a} = J^{a}_{\mu}\dot{x}^{\mu} \ . \tag{1.93}$$

Differentiating once more, one finds

$$\begin{aligned}\frac{d^2}{d\tau^2}\xi^a &= \frac{\partial\xi^a}{\partial x^\mu}\frac{d^2x^\mu}{d\tau^2} + \frac{\partial^2\xi^a}{\partial x^\nu\partial x^\lambda}\frac{dx^\nu}{d\tau}\frac{dx^\lambda}{d\tau} \\ &= \frac{\partial\xi^a}{\partial x^\mu}\frac{d^2x^\mu}{d\tau^2} + \delta^a_{\ b}\frac{\partial^2\xi^b}{\partial x^\nu\partial x^\lambda}\frac{dx^\nu}{d\tau}\frac{dx^\lambda}{d\tau} \\ &= \frac{\partial\xi^a}{\partial x^\mu}\left[\frac{d^2x^\mu}{d\tau^2} + \frac{\partial x^\mu}{\partial\xi^b}\frac{\partial^2\xi^b}{\partial x^\nu\partial x^\lambda}\frac{dx^\nu}{d\tau}\frac{dx^\lambda}{d\tau}\right] \ . \end{aligned} \tag{1.94}$$

Thus, since the matrix appearing outside the square bracket is invertible, in terms of the coordinates x^μ the equation of motion, or the equation for a straight (and, in the case at hand, timelike) line in Minkowski space, becomes

$$\frac{d^2x^\mu}{d\tau^2} + \frac{\partial x^\mu}{\partial\xi^a}\frac{\partial^2\xi^a}{\partial x^\nu\partial x^\lambda}\frac{dx^\nu}{d\tau}\frac{dx^\lambda}{d\tau} = 0 \ . \tag{1.95}$$

We will write this as

$$\frac{d^2x^\mu}{d\tau^2} + \gamma^\mu_{\ \nu\lambda}\frac{dx^\nu}{d\tau}\frac{dx^\lambda}{d\tau} = 0 \ , \tag{1.96}$$

or just

$$\ddot{x}^\mu + \gamma^\mu_{\ \nu\lambda}\dot{x}^\nu\dot{x}^\lambda = 0 \ , \tag{1.97}$$

where

$$\gamma^\mu_{\ \nu\lambda} = \frac{\partial x^\mu}{\partial\xi^a}\frac{\partial^2\xi^a}{\partial x^\nu\partial x^\lambda} \ . \tag{1.98}$$

REMARKS:

1. Of course the statement (1.93) regarding the linear transformation rule of velocities under coordinate transformations remains true for transformations between arbitary coordinate systems in Minkowski space, $\{x^\mu\}$ and $\{y^\alpha\}$, say, i.e. one has

$$\dot{y}^\alpha = \frac{\partial y^\alpha}{\partial x^\mu}\dot{x}^\mu \equiv J^\alpha_\mu\dot{x}^\mu \ . \tag{1.99}$$

In general this simply follows from the chain rule. It can also be deduced (in a somewhat unnecessarily long way included here only for later reference purposes) from what we have already done, namely by simply repeating the calculation leading to (1.93), but now for the coordinates y^α,

$$\dot{\xi}^a = J^a_\alpha\dot{y}^\alpha \ . \tag{1.100}$$

This now implies

$$\dot{y}^\alpha = J^\alpha_a\dot{\xi}^a = J^\alpha_a J^a_\mu\dot{x}^\mu = J^\alpha_\mu\dot{x}^\mu \tag{1.101}$$

where

$$J^\alpha_\mu = \frac{\partial y^\alpha}{\partial\xi^a}\frac{\partial\xi^a}{\partial x^\mu} = \frac{\partial y^\alpha}{\partial x^\mu} \tag{1.102}$$

and in the last step the chain rule (or multiplicativity of the Jacobi matrix under consecutive coordinate transformations) was used.

2. While (1.97) looks a bit complicated and unattractive, it is simply the general variant of a calculation that you have probably done numerous times before in various specific contexts. In particular, the second term in this equation, is just the general expression for a pseudo-force or fictitious gravitational force (like a centrifugal force or the Coriolis force) that arises whenever one describes inertial motion in non-inertial coordinates.

3. More compactly, this pseudo-force term can be written as

$$\gamma^{\mu}_{\nu\lambda} = J^{\mu}_{a}\partial_{\nu}J^{a}_{\lambda} = J^{\mu}_{a}\partial_{\lambda}J^{a}_{\nu} \equiv J^{\mu}_{a}J^{a}_{\nu\lambda} \ . \tag{1.103}$$

It is absent precisely for linear coordinate transformations $\xi^{a}(x^{\mu}) = M^{a}_{\mu}x^{\mu}$,

$$\gamma^{\mu}_{\nu\lambda} = 0 \quad (\forall\, ^{\mu}, _{\nu}, _{\lambda}) \quad \Leftrightarrow \quad \xi^{a} = M^{a}_{\mu}x^{\mu} \tag{1.104}$$

for some constant matrix M^{a}_{μ}. In particular, this means that the equation of motion for a free particle is invariant under Lorentz transformations, as it should be.

4. By the same reasoning, the quantity $\gamma^{\mu}_{\nu\lambda}$ is independent of the choice of reference inertial coordinate system ξ^{a}. I.e. if $\zeta^{a} = L^{a}_{b}\xi^{b}$ for some Lorentz (or more general linear) transformation matrix L^{a}_{b}, then

$$\frac{\partial x^{\mu}}{\partial \xi^{a}}\frac{\partial^{2}\xi^{a}}{\partial x^{\nu}\partial x^{\lambda}} = \frac{\partial x^{\mu}}{\partial \zeta^{a}}\frac{\partial^{2}\zeta^{a}}{\partial x^{\nu}\partial x^{\lambda}} \tag{1.105}$$

5. In the same way that the equation of motion for a free particle in inertial coordinates follows from the extremisation of the proper time (written in inertial coordinates),

$$\delta\int d\tau = \delta\int\sqrt{-\eta_{ab}d\xi^{a}d\xi^{b}} = 0 \quad \Rightarrow \quad \ddot{\xi}^{a} = 0 \ , \tag{1.106}$$

the equation of motion for a free particle in noninertial coordinates follows from the extremisation of the proper time (written in these noninertial coordinates),

$$\delta\int d\tau = \delta\int\sqrt{-\eta_{\mu\nu}dx^{\mu}dx^{\nu}} = 0 \quad \Rightarrow \quad \ddot{x}^{\mu} + \gamma^{\mu}_{\nu\lambda}\dot{x}^{\nu}\dot{x}^{\lambda} = 0 \ . \tag{1.107}$$

It is a straightforward exercise to establish this, and simply reflects the well-known covariance of the Euler-Lagrange equations under coordinate transformations. The proof will not be given here also because we will straightaway establish a more general statement (for a space equipped with an arbitrary metric) in section 2.3 below.

6. It is immediately apparent from this action principle, that the (pseudo-)force terms in the equation of motion will arise from the fact that in general the components of the metric $\eta_{\mu\nu}(x)$ are not constant in these cooordinates. In particular, therefore, it must be possible to write the pseudo-force terms $\gamma^{\mu}_{\nu\lambda}$ in terms of the first partial derivatives of the metric $\eta_{\mu\nu}$. We will come back to this in section 1.6 below.

Even though the resulting equations look a bit uninviting at the moment, that is just what you get when you do write things in arbitrary coordinates. Moreover, there are at least two very useful things that we can extract or anticipate from this, namely

1. candidates for the appropriate generalisation of the Newtonian gravitational potential
2. the prototypical *general covariance* of physical equations

in *any* theory of gravity satisfying the Einstein equivalence principle. Let us now discuss these features in turn (relegating some uninspiring calculational details to the end of this subsection):

1. the Metric as a Candidate for the Gravitational Potential

   Recall that in non-inertial coordinates the metric takes the form

   $$\eta_{\mu\nu} = J^a_\mu J^b_\nu \eta_{ab} \tag{1.108}$$

   and the equation of motion of a free particle is (1.97) with the pseudo-force term (1.103)

   $$\gamma^\mu_{\ \nu\lambda} = J^\mu_a \partial_\nu J^a_\lambda \ . \tag{1.109}$$

   It turns out that this term can be expressed in terms of the partial derivatives of the metric (as anticipated from the action principle above). Indeed, given any symmetric and invertible objectg $g_{\mu\nu}(x)$ (a "metric"), with inverse $g^{\mu\nu}$, define

   $$\begin{aligned} \Gamma^\mu_{\ \nu\lambda} &= g^{\mu\rho}\Gamma_{\rho\nu\lambda} \\ \Gamma_{\rho\nu\lambda} &= \tfrac{1}{2}(g_{\rho\nu,\lambda} + g_{\rho\lambda,\nu} - g_{\nu\lambda,\rho}) \end{aligned} \tag{1.110}$$

   (this definition is such that it remains valid for an arbitary metric). Then one finds

   $$g_{\mu\nu} = \eta_{\mu\nu} \equiv J^a_\mu J^b_\nu \eta_{ab} \quad \Rightarrow \quad \Gamma^\mu_{\ \nu\lambda} = \gamma^\mu_{\ \nu\lambda} \ . \tag{1.111}$$

   It is an elementary but nevertheless useful exercise to check this (see below - but do try this yourself as well).

   This shows that the components $\eta_{\mu\nu}$ of the metric appear to play the role of "potentials" for the gravitational pseudo-force. In particular, since in principle all components of the metric can contribute to $\gamma_{\rho\nu\lambda}$, we learn the interesting fact that in order to achieve this a single scalar potential is completely insufficient (and one could have discovered the possibility or necessity of a multitude of potentials simply by the study of pseudo-forces in non-gravitational Newtonian mechanics).

It is enormously pleasing to note that the "number" of potentials that we seem to have discovered, namely 10 (for the symmetric (4×4)-matrix $g_{\mu\nu}(x)$), agrees with the number of potentials anticipated in our discussion of section 1.1 when we contemplated a tensorial generalisations (1.6) of the Poisson equation (with source T_{ab}).

If the metric indeed plays the role of the gravitational potential, as suggested by these considerations, then it will play the role of the fundamental dynamical variable of gravity. Since the metric encodes what one usually refers to as the geometry of a space(-time), as we will discuss in much more detail below, namely the information required to determine distances, areas, volumes etc., this means that we are being led to the conclusion that any theory of gravity based on the equivalence principle is a theory of dynamical geometry. Wow ...

2. the General Covariance of the Equation of Motion

   The equation of motion (1.97) has one other fundamental redeeming and attractive feature which will also make it the prototype of the kind of equations that we will be looking for in general. This feature is its *covariance* under general coordinate transformations, i.e. its *general covariance*, which means that the equation takes the same form in any coordinate system. Equivalently, but somewhat more to the point, as discussed in more detail in section 4.1 below, this is the statement that the equation is satisified in one coordinate system if and only if it is satisfied in all coordinate systems.

   This covariance is in some sense tautologically true since the coordinate system $\{x^\mu\}$ that we have chosen is indeed arbitrary. However, it is instructive to see how this comes about by explicitly transforming (1.97) from one coordinate system x^μ to another, say y^α.

   If one does this (cf. below for a proof), one finds that the equations of motion (1.97) in the coordinates x^μ and y^α are related by

$$\frac{d^2y^\alpha}{d\tau^2} + \gamma^\alpha_{\beta\gamma}\frac{dy^\beta}{d\tau}\frac{dy^\gamma}{d\tau} = \frac{\partial y^\alpha}{\partial x^\mu}\left[\frac{d^2x^\mu}{d\tau^2} + \gamma^\mu_{\nu\lambda}\frac{dx^\nu}{d\tau}\frac{dx^\lambda}{d\tau}\right] \tag{1.112}$$

   Thus the geodesic equation transforms in the simplest possible non-trivial way under coordinate transformations $x \to y$, namely with the Jacobi matrix

$$J^\alpha_\mu = \frac{\partial y^\alpha}{\partial x^\mu} \ . \tag{1.113}$$

   We will see later that this transformation behaviour characterises/defines tensors, in this particular case a vector (or contravariant tensor of rank 1).

   In particular, since this matrix is assumed to be invertible, we reach the conclusion that the left hand side of (1.112) is zero if and only if the term in square brackets

on the right hand side is zero,

$$\frac{d^2y^\alpha}{d\tau^2} + \gamma^\alpha_{\beta\gamma}\frac{dy^\beta}{d\tau}\frac{dy^\gamma}{d\tau} = 0 \quad \Leftrightarrow \quad \frac{d^2x^\mu}{d\tau^2} + \gamma^\mu_{\nu\lambda}\frac{dx^\nu}{d\tau}\frac{dx^\lambda}{d\tau} = 0 \tag{1.114}$$

This is what is meant by the statement that the equation takes the same form in any coordinate system, and is therefore satisfied in one coordinate system if and only if it is satisfied in all coordinate systems. We see that in this case this is achieved by having the equation transform in a particularly simple way under coordinate transformations, namely as a tensor. We will discuss this in more generality in section 4.1.

---

1. Proof of (1.111):

   - From
   $$\eta_{\mu\nu} = \eta_{ab}J^a_\mu J^b_\nu \tag{1.115}$$
   one deduces
   $$\eta_{\mu\nu,\lambda} = \eta_{ab}(J^a_{\mu\lambda}J^b_\nu + J^a_\mu J^b_{\nu\lambda}) \tag{1.116}$$
   where
   $$J^a_{\mu\lambda} = \partial_\lambda J^a_\mu = \frac{\partial^2\xi^a}{\partial x^\mu \partial x^\lambda} = J^a_{\lambda\mu} \ . \tag{1.117}$$
   - Therefore, form the definition (1.110) of the Γ-symbols, one has (for $g_{\mu\nu} = \eta_{\mu\nu}$)
   $$\begin{aligned}\Gamma_{\mu\nu\lambda} &= \tfrac{1}{2}(\eta_{\mu\nu,\lambda} + \eta_{\mu\lambda,\nu} - \eta_{\nu\lambda,\mu}) \\ &= \tfrac{1}{2}\eta_{ab}(J^a_{\mu\lambda}J^b_\nu + J^a_\mu J^b_{\nu\lambda} + J^a_{\mu\nu}J^b_\lambda + J^a_\mu J^b_{\lambda\nu} - J^a_{\nu\mu}J^b_\lambda - J^a_\nu J^b_{\lambda\mu}) \\ &= \eta_{ab}J^a_\mu J^b_{\nu\lambda} \ ,\end{aligned} \tag{1.118}$$
   where the cancellations in passing to the last line arise from the symmetries $\eta_{ab} = \eta_{ba}$, $J^b_{\lambda\mu} = J^b_{\mu\lambda}$ etc.
   - Thus, finally (and writing out everything in detail for once),
   $$\begin{aligned}\Gamma^\mu_{\nu\lambda} &= \eta^{\mu\rho}\Gamma_{\rho\nu\lambda} = \eta^{cd}J^\mu_c J^\rho_d \eta_{ab}J^a_\rho J^b_{\nu\lambda} = \eta^{cd}J^\mu_c \delta^a_d \eta_{ab}J^b_{\nu\lambda} \\ &= \eta^{ca}J^\mu_c \eta_{ab}J^b_{\nu\lambda} = \delta^c_b J^\mu_c J^b_{\nu\lambda} = J^\mu_b J^b_{\nu\lambda} \ ,\end{aligned} \tag{1.119}$$
   as was to be shown.

2. Proof of (1.112):

   - We proceed as in the proof of (1.99). Thus consider transforming the free particle equation of motion in inertial coordinates (1.55) not to the coordinate systsem x^μ, as we did before, but to another coordinate system $\{y^\alpha\}$.

Following the same steps as above, one arrives at the y-version of (1.94), namely

$$\frac{d^2}{d\tau^2}\xi^a = \frac{\partial \xi^a}{\partial y^\alpha}\left[\frac{d^2 y^\alpha}{d\tau^2} + \frac{\partial y^\alpha}{\partial \xi^b}\frac{\partial^2 \xi^b}{\partial y^\beta \partial y^\gamma}\frac{dy^\beta}{d\tau}\frac{dy^\gamma}{d\tau}\right] \quad . \tag{1.120}$$

- Equating this result to (1.94) and using the chain rule for partial derivatives

$$\frac{\partial y^\alpha}{\partial x^\mu} = \frac{\partial y^\alpha}{\partial \xi^a}\frac{\partial \xi^a}{\partial x^\mu} \quad , \tag{1.121}$$

one finds

$$\frac{d^2 y^\alpha}{d\tau^2} + \gamma^\alpha_{\beta\gamma}\frac{dy^\beta}{d\tau}\frac{dy^\gamma}{d\tau} = \frac{\partial y^\alpha}{\partial x^\mu}\left[\frac{d^2 x^\mu}{d\tau^2} + \gamma^\mu_{\nu\lambda}\frac{dx^\nu}{d\tau}\frac{dx^\lambda}{d\tau}\right] \tag{1.122}$$

as claimed.

2 Metrics, Geometry and Geodesics

Above we saw that the motion of free particles in Minkowski space in curvilinear coordinates is described in terms of a modified metric, $g_{\mu\nu}$, and a force term $\gamma^{\mu}_{\nu\lambda}$ representing the "pseudo-force" on the particle. Thus the Einstein Equivalence Principle suggests that an appropriate description of true gravitational fields is in terms of a metric tensor $g_{\mu\nu}(x)$ (and its associated Γ-symbols) which can only locally be related to the Minkowski metric via a suitable coordinate transformation (to locally inertial coordinates). We adopt this as our working hypothesis.

2.1 Metrics and Geometry I: Definition and Examples

Thus our starting point will now be a space-time equipped with some metric $g_{\mu\nu}(x)$, which (by analogy with the Euclidean and Minkowski metrics) we will assume to be symmetric and non-degenerate, i.e.

$$g_{\mu\nu}(x) = g_{\nu\mu}(x) \qquad \mathsf{det}(g_{\mu\nu}(x)) \neq 0 \ . \tag{2.1}$$

The metric encodes the information how to measure (spatial and temporal) distances, as well as areas, volumes etc., via the associated line element

$$g_{\mu\nu}(x) \quad \Rightarrow \quad ds^2 = g_{\mu\nu}(x) dx^{\mu} dx^{\nu} \ . \tag{2.2}$$

As an example, the most general 2-dimensional line element (on a space with local coordinates $(x^i) = (x^1, x^2)$) has the form

$$ds^2 = g_{ij} dx^i dx^j = g_{11}(dx^1)^2 + (g_{22})(dx^2)^2 + 2g_{12} dx^1 dx^2 \tag{2.3}$$

(which is non-degenerate if $g_{11}g_{22} - (g_{12})^2 \neq 0$).

A metric determines a geometry (in the literal sense of a prescription for measuring distances etc.), but different metrics may well determine the same geometry, namely those metrics which are just related by coordinate transformations. In particular, distances should not depend on which coordinate system is used. Hence, changing coordinates from the $\{x^{\mu}\}$ to new coordinates $\{y^{\alpha}(x^{\mu})\}$ and demanding that

$$g_{\mu\nu}(x) dx^{\mu} dx^{\nu} = g_{\alpha\beta}(y) dy^{\alpha} dy^{\beta} \ , \tag{2.4}$$

one finds that under a coordinate transformation a metric transforms as

$$g_{\alpha\beta}(y) = g_{\mu\nu}(x) \frac{\partial x^{\mu}}{\partial y^{\alpha}} \frac{\partial x^{\nu}}{\partial y^{\beta}} \equiv J^{\mu}_{\alpha} J^{\nu}_{\beta} \ g_{\mu\nu} \ . \tag{2.5}$$

Objects which transform in such a nice and simple way under coordinate transformations are known as *tensors* - the metric is an example of what is known as (and we will get to

know as) a covariant symmetric rank two tensor. We will study tensors in much more detail and generality later, starting in section 4.

REMARKS:

1. Here I have denoted the components of the metric in the new coordinates y^α simply by $g_{\alpha\beta}$. Occasionally it is more convenient to use a more elaborate notation, such as
$$x^\mu \to x'^\alpha = y^\alpha \quad \Rightarrow \quad g_{\mu\nu} \to g'_{\alpha\beta} = J^\mu_\alpha J^\nu_\beta g_{\mu\nu} \ , \tag{2.6}$$
which allows one to distinguish notationally specific components of the metric in 2 different coordinate systems, such as g'_{11} (the (11)-component of the metric in the y-coordinates) from g_{11} (the (11)-component of the metric in the x-coordinates). As mentioned before, indices and other decorations are primarily bookkeeping devices; therefore I will usually not be overly-pedantic about these things in the following and will use whatever notation is more convenient in the case at hand.

2. As a consequence of the non-degeneracy condition, pointwise $g_{\mu\nu}(x)$ possesses an inverse, whose components we will denote by $g^{\mu\nu}(x)$, i.e.
$$g^{\mu\nu}(x)g_{\nu\lambda}(x) = \delta^\mu_\lambda \quad , \quad g_{\mu\nu}(x)g^{\nu\lambda}(x) = \delta^\lambda_\mu \ . \tag{2.7}$$
Clearly, the inverse metric then transforms inversely, i.e. with the inverse Jacobi matrices J^α_μ, and this is now nicely compatible with the convention to denote the inverse metric by upper indices,
$$g^{\alpha\beta} = J^\alpha_\mu J^\beta_\nu g^{\mu\nu} \ . \tag{2.8}$$
This is also the rationale for writing the inverse metric with "upper" indices: the positioning of indices is used to indicate how an object transforms under coordinate transformations (and we will formalise this in the discussion of section 4 on tensor algebra).

3. A space-time equipped with a metric tensor $g_{\mu\nu}(x)$ is called a metric space-time or (pseudo-)Riemannian space-time. Here "Riemannian" usually refers to a space equipped with a positive-definite metric (all eigenvalues positive), while pseudo-Riemannian (or Lorentzian) refers to a space-time with a metric with one negative and 3 (or 27, or whatever) positive eigenvalues.

4. One point to note about the tensorial transformation behaviour is that pointwise it is a similarity transformation in the sense of linear algebra, in matrix notation
$$g \mapsto J^t g J \ . \tag{2.9}$$
In particular, therefore, if in one coordinate system the space-time metric tensor has one negative and three positive eigenvalues (as in a locally inertial coordinate

system), then the same will be true in any other coordinate system (even though the eigenvalues themselves will in general be different) - this statement should be familiar from linear algebra (e.g. as Sylvester's law, but it also goes under various other names).

Here are some examples of Riemannian metrics that you may already be familiar with.

EXAMPLES:

1. The Euclidean metrics or line-elements on $\mathbb{R}^2$ or $\mathbb{R}^3$, but written in polar or spherical coordinates,

$$\begin{aligned} ds^2(\mathbb{R}^2) &= dx^2 + dy^2 = dr^2 + r^2 d\phi^2 \\ ds^2(\mathbb{R}^3) &= dx^2 + dy^2 + dz^2 = dr^2 + r^2(d\theta^2 + \sin^2\theta d\phi^2) \ . \end{aligned} \tag{2.10}$$

   E.g. for the latter case one has

$$(x, y, z) = (r\sin\theta\cos\phi, r\sin\theta\sin\phi, r\cos\theta) \ , \tag{2.11}$$

   and plugging this into the Euclidean line-element $dx^2 + dy^2 + dz^2$, one finds the above result.

   Denoting the Cartesian coordinates by x^α and the spherical coordinates by y^α, with $(y^1 = r, y^2 = \theta, y^3 = \phi)$, the non-vanishing components of the metric in the two coordinate systems are thus (using the prime notation (2.6))

$$g_{11} = g_{22} = g_{33} = 1 \quad , \quad g'_{11} = 1 \ , \ g'_{22} = r^2 \ , \ g'_{33} = r^2\sin^2\theta \ . \tag{2.12}$$

   Alternatively, it is often more informative (and very common) to use the coordinates themselves, rather than indices, as the labels of the components of the metric tensor. In this case one can dispense with the prime notation and simply write the components of the metric in spherical coordinates as

$$g_{rr} = 1 \ , \ g_{\theta\theta} = r^2 \ , \ g_{\phi\phi} = r^2\sin^2\theta \ . \tag{2.13}$$

2. Restricting the first example above to constant radius $r = R$, this gives us the line-element on the circle S^1_R of radius R,

$$ds^2(S^1_R) = R^2 d\phi^2 \ . \tag{2.14}$$

   Restricting the second to the 2-sphere S^2_R of radius R,

$$x^2 + y^2 + z^2 = r^2 = R^2 \quad \text{or} \quad r = R \ , \tag{2.15}$$

   one finds the line-element

$$ds^2(S^2_R) = R^2(d\theta^2 + \sin^2\theta d\phi^2) \equiv R^2 d\Omega^2 \ . \tag{2.16}$$

Here

$$d\Omega^2 = d\theta^2 + \sin^2\theta d\phi^2 \tag{2.17}$$

is usually called the solid angle, and we can now interpret it as the line element on the unit 2-sphere. We will use the notation / abbrevation $d\Omega^2$ for this line element throughout the notes.

This example provides a nice illustration of the fact that by drawing the coordinate grid / infinitesimal parallelograms determined by the metric tensor, one can get a feeling for the geometry and can in particular convince oneself that in general a metric space or space-time need not or cannot be flat, i.e. is not the flat Euclidean space of Euclidean geometry.

Indeed, the coordinate grid of the metric $d\theta^2 + \sin^2\theta d\phi^2$ cannot be drawn in flat space because the infinitesimal parallelograms described by ds^2 degenerate to triangles not just at $\theta = 0$ (as would also be the case for the flat metric $ds^2 = dr^2 + r^2 d\phi^2$ in polar coordinates at $r = 0$), but also at $\theta = \pi$. This coordinate grid can, on the other hand, of course be drawn on the 2-sphere.

3. This line-element on the unit 2-sphere generalises to the line-element on a unit 3-sphere,

$$ds^2(S^3) = d\psi^2 + \sin^2\psi(d\theta^2 + \sin^2\theta d\phi^2) \ . \tag{2.18}$$

This can be obtained by simply generalising the construction of spherical coordinates from $\mathbb{R}^3$ to $\mathbb{R}^4$, and (if required) this can be continued iteratively to yet higher-dimensional spheres.

Alternatively, by thinking of the 3-sphere as the locus

$$x^2 + y^2 + z^2 + w^2 = 1 \tag{2.19}$$

in $\mathbb{R}^4$, and "solving" this equation by first setting

$$x^2 + y^2 = \sin^2\alpha \quad , \quad z^2 + w^2 = \cos^2\alpha \ , \tag{2.20}$$

and then refining this to

$$x = \sin\alpha\cos\beta \quad , \quad y = \sin\alpha\sin\beta \quad , \quad z = \cos\alpha\cos\gamma \quad , \quad w = \cos\alpha\sin\gamma \ , \tag{2.21}$$

one finds that the standard Euclidean line-element

$$ds^2 = dx^2 + dy^2 + dz^2 + dw^2 \tag{2.22}$$

induces the line-element

$$ds^2(S^3) = d\alpha^2 + \sin^2\alpha d\beta^2 + \cos^2\alpha d\gamma^2 \tag{2.23}$$

on the sphere. This is the same metric on S^3 as above (2.18), namely the one induced from the Euclidean metric on $\mathbb{R}^4$, but written in different coordinates.

In particular, both are invariant under 4-dimensional rotations, i.e. under $SO(4)$-transformations.

However, we can obtain genuinely different metrics on the 3-sphere e.g. by starting with different metrics on $\mathbb{R}^4$. One of the simplest possibilities is to replace (2.22) by

$$d\tilde{s}^2 = a^2(dx^2 + dy^2) + b^2(dz^2 + dw^2) \ , \tag{2.24}$$

with a, b real non-zero parameters. Then the induced metric on the 3-sphere $x^2 + y^2 + z^2 + w^2 = 1$ is

$$d\tilde{s}^2(S^3) = (a^2\cos^2\alpha + b^2\sin^2\alpha)d\alpha^2 + a^2\sin^2\alpha d\beta^2 + b^2\cos^2\alpha d\gamma^2 \ . \tag{2.25}$$

For $a^2 \neq b^2$, this metric is not invariant under full 4-dimensional rotations, but only under rotations in the (x, y) and (z, w) planes, i.e. under $SO(2) \times SO(2)$ transformations. Thus this equips the 3-sphere with a genuinely different geometry (and is an example of what is sometimes referred to as a "squashed 3-sphere geometry").

4. If instead of the unit 2-sphere one considers the "unit" hyperboloid H^2, defined by

$$x^2 + y^2 + z^2 = +1 \quad \longrightarrow \quad x^2 + y^2 - z^2 = -1 \ , \tag{2.26}$$

then this is naturally thought of as being embedded not in $\mathbb{R}^3$ but in $\mathbb{R}^{1,2}$, i.e. into the 3-dimensional vector space with line-element

$$ds^2 = dx^2 + dy^2 - dz^2 \ . \tag{2.27}$$

The hyperbolic analogues (r, σ, ϕ) of the spherical coordinates, defined by

$$(x, y, z) = (r\sinh\sigma\cos\phi, r\sinh\sigma\sin\phi, r\cosh\sigma) \ , \tag{2.28}$$

are naturally adapted to this situation, because

$$x^2 + y^2 - z^2 = -r^2 \tag{2.29}$$

so that the unit hyperboloid is evidently just the surface $r = 1$. In these coordinates, the metric (2.27) takes the form

$$ds^2 = -dr^2 + r^2(d\sigma^2 + \sinh^2\sigma\ d\phi^2) \ , \tag{2.30}$$

and therefore the induced metric on the unit hyperboloid $r = 1$ is

$$ds^2(H^2) = d\sigma^2 + \sinh^2\sigma\ d\phi^2 \ . \tag{2.31}$$

2.2 Metrics and Geometry II: Lorentzian (Pseudo-Riemannian) Metrics

We now turn to Lorentzian (pseudo-Riemannian) metrics and geometries. These will of course occupy and accompany us throughout these notes, so this section is meant to just provide a first brief encounter with these objects.

For a metric with Lorentzian signature, and with coordinates $x^\alpha = (x^0 = t, x^k)$, say, the metric has components $g_{00}, g_{0k} = g_{k0}$ and $g_{ik} = g_{ki}$, and the corresponding line element has the form

$$ds^2 = g_{00}dt^2 + 2g_{0k}dt\,dx^k + g_{ik}dx^i dx^k \tag{2.32}$$

Without any further conditions on the coefficients (except those ensuring non-degeneracy), this could a priori be a metric of any signature, and the signature of the metric may not always be readily apparent even when the coefficients of the metric are given explicitly.

Before looking at this in somewhat more detail, here are some simple examples, where the Lorentzian signature of the metrics is reasonably manifest:

Examples:

1. Of course any of the Riemannian metrics of the previous section can be promoted to space-time metrics by simply adding a $(-dt^2)$ (i.e. by taking the direct product with the time-axis). Thus the Minkowski metric in spatial spherical coordinates has the form
$$ds^2(\mathbb{R}^{1,3}) = -dt^2 + dr^2 + r^2(d\theta^2 + \sin^2\theta d\phi^2) \ . \tag{2.33}$$

2. A generalisation of this is provided by the so-called *ultrastatic* metrics, i.e. metrics that are just a product of the standard metric $-dt^2$ along the time-direction and a spatial metric $\tilde{g}_{ij}(x)$
$$ds^2 = -dt^2 + \tilde{g}_{ij}(x)dx^i dx^j \tag{2.34}$$
(i.e. the components depend only on the spatial coordinates x^i, not on t).

3. Somewhat more generally, the spatial components of the metric can depend non-trivially on time. For example, a space-time metric describing a spatially spherical universe with a time-dependent radius (expansion of the universe!) might be described by the line element
$$ds^2 = -dt^2 + a(t)^2\left(d\psi^2 + \sin^2\psi(d\theta^2 + \sin^2\theta d\phi^2)\right) \ , \tag{2.35}$$
and more generally one can consider the corresponding generalisation of (2.34), namely metrics of the form
$$ds^2 = -dt^2 + a(t)^2\tilde{g}_{ij}(x)dx^i dx^j \ . \tag{2.36}$$
This describes a space-time with spatial metric $\tilde{g}_{ij}(x)dx^i dx^j$ and a time-dependent overall scale factor $a(t)$; in particular, such a space-time metric can describe an

expanding universe in cosmology. We will discuss such metrics in detail later on in the context of cosmology, sections 33-38.

4. The (time-time)-component of the metric can of course in general depend non-trivially on the spatial coordinates. We already encountered this in the example of the Rindler metric (1.74), which has the form

$$ds^2 = -\rho^2 d\eta^2 + d\rho^2 \quad . \tag{2.37}$$

5. A particularly prominent example is the *Schwarzschild Metric*

$$ds^2 = -(1-2m/r)dt^2 + (1-2m/r)^{-1}dr^2 + r^2(d\theta^2 + \sin^2\theta d\phi^2) \quad . \tag{2.38}$$

   It is of fundamental importance for General Relativity, and perhaps the most important exact solution of the Einstein field equations for the gravitational field, as it describes the gravitational field outside a spherical star (as well as black holes, as it turns out ...). We will discuss this metric in great detail in sections 24-27.

The characteristic feature of metrics with Lorentzian signature is of course the presence of timelike and null (lightlike) directions, and thus in a pseudo-Riemannian space-time one has the same distinction between spacelike, timelike and lightlike separations as in Minkowski space(-time). Infinitesimal

- spacelike distances correspond to $ds^2 > 0$,
- timelike distances to $d\tau^2 = -ds^2 > 0$,
- and null or lightlike distances to $ds^2 = d\tau^2 = 0$.

Likewise, a vector $V^\mu(x)$ at a point x is called

- spacelike if $g_{\mu\nu}(x)V^\mu(x)V^\nu(x) > 0$,
- timelike if $g_{\mu\nu}(x)V^\mu(x)V^\nu(x) < 0$,
- and null or lightlike if $g_{\mu\nu}(x)V^\mu(x)V^\nu(x) = 0$,

and a curve $x^\mu(\lambda)$ is called spacelike if its tangent vector is everywhere spacelike etc.

Using the definition of a vector in general relativity (to be introduced in section 4), namely an object that transforms in the obvious way, with the Jacobi matrix, under coordinate transformations, one sees that $g_{\mu\nu}(x)V^\mu(x)V^\nu(x)$ is a scalar, i.e. invariant under coordinate transformations, and hence the statement that a vector is, say, spacelike is a coordinate-independent statement, as it should be.

REMARKS:

1. When the metric (2.32) is (time-space) block-diagonal, i.e. when the mixed components $g_{0k} = 0$ (as in all of the above examples), then the timelike and spacelike directions are easy to distinguish by inspection. Typically then the "spatial" metric g_{ik} is positive definite, and thus necessarily $g_{00} < 0$.

2. When some of the g_{0k} are non-zero, on the other hand, one has a more intricate mixing of time- and space-directions. For example, consider the simple 2-dimensional metric

$$ds^2 = dx^2 + dy^2 + 2a\, dx\, dy \tag{2.39}$$

for some real parameter a. For any a, the coordinate lines of x and y are spacelike curves. However, this does not imply all by itself that the metric is Euclidean (just like the sum of two spacelike vectors in Minkowski space is not necessarily spacelike: it may be spacelike, timelike or null: e.g. if $v = (0,1)$, choose $w = (0,1), (1/2,-1), (1,-2)$ respectively).

Indeed, by calculating the determinant of the metric,

$$\det \begin{pmatrix} 1 & a \\ a & 1 \end{pmatrix} = 1 - a^2 \ , \tag{2.40}$$

one sees that

- when $a^2 < 1$, the metric has Euclidean signature
  (it is actually related to the standard 2-dimensional Euclidean metric by a linear transformation)
- when $a^2 = 1$, the metric is degenerate
  (in this case, the line element can be written as $ds^2 = d(x \pm y)^2$, which is a 1-dimensional metric for the single coordinate $x \pm y$)
- when $a^2 > 1$, the metric has Lorentzian signature
  (it is actually related to the standard 2-dimensional Minkowski metric by a linear transformation)

3. This mixing of time- and space-directions for a metric which is not block-diagonal can also be seen from the components of the inverse metric. Indeed, from (2.7), one finds

$$g_{0\nu} g^{\nu k} = g_{00} g^{0k} + g_{0i} g^{ik} = \delta_0^k = 0 \ , \tag{2.41}$$

and thus (for $g_{00} \neq 0$)

$$g^{0k} = -\frac{1}{g_{00}} g_{0i} g^{ik} \ . \tag{2.42}$$

Likewise from (2.7) one deduces

$$g_{i\nu} g^{\nu k} = g_{i0} g^{0k} + g_{ij} g^{jk} = \delta_i^k \ . \tag{2.43}$$

In particular, this shows that in general (i.e. unless the off-diagonal components g_{0k} are all zero), the spatial components g^{ik} of the inverse metric are not the inverse of the spatial components g_{ij} of the metric. Rather, using (2.42) one has

$$\left(g_{ij} - \frac{1}{g_{00}} g_{i0} g_{j0}\right) g^{jk} = \delta_i^k \quad . \tag{2.44}$$

Outlook:

This ends our first brief encounter with metrics and geometries. At this point the question naturally arises how one can tell whether a given (perhaps complicated looking) metric is just the "flat" (Euclidean or Minkowski) metric written in other coordinates or whether it describes a genuinely new geometry.

We will see later that there is an object, the *Riemann curvature tensor*, constructed from the metric and its 1st and 2nd derivatives, which has the property that all of its components vanish if and only if the metric is a coordinate transform of the flat space Minkowski metric. Thus, given a metric, by calculating its curvature tensor one can decide if the metric is just the flat metric in disguise or not. The curvature tensor will be introduced in section 8, and the above statement will be established in section 11.2.

2.3 Geodesic Equation from the Extremisation of Proper Time

We have seen that the equation for a straight line in Minkowski space, written in arbitrary coordinates, is

$$\frac{d^2 x^\mu}{d\tau^2} + \gamma^\mu_{\nu\lambda} \frac{dx^\nu}{d\tau} \frac{dx^\lambda}{d\tau} = 0 \quad , \tag{2.45}$$

where the pseudo-force term $\gamma^\mu_{\nu\lambda}$ is given by (1.98). We have also seen in (1.111) (provided you checked this) that $\gamma^\mu_{\nu\lambda}$ can be expressed in terms of the metric (1.87) as

$$\gamma^\mu_{\nu\lambda} = \tfrac{1}{2} \eta^{\mu\rho} (\eta_{\rho\nu,\lambda} + \eta_{\rho\lambda,\nu} - \eta_{\nu\lambda,\rho}) \quad . \tag{2.46}$$

This gravitational force term is fictitious since it can globally be transformed away by going to the global inertial coordinates ξ^a. The equivalence principle suggests, however, that in general the equation for the worldline of a massive particle, i.e. a path that extremises proper time, in a true gravitational field described by a non-trivial metric $g_{\mu\nu}(x)$ (not related to the Minkowski metric by a coordinate transformation) is also of the above form.

We will now confirm this by deriving the equations for a timelike path that extremises proper time from a variational principle. These paths will be referred to as (timelike) *geodesics*. We will briefly return below to the (delicate) issue to which extent these can be regarded as world lines of actual massive particles.

Recall first of all from special relativity that the Lorentz-covariant description of the dynamics of a massive particle is based on describing the timelike worldline of the particle in the parametric form

$$\xi^a = \xi^a(\tau) \tag{2.47}$$

where τ is the proper time along the worldline,

$$d\tau^2 = -\eta_{ab} d\xi^a d\xi^b \ . \tag{2.48}$$

In particular, the 4-velocity

$$u^a = \frac{d\xi^a(\tau)}{d\tau} \tag{2.49}$$

is normalised as

$$\eta_{ab} u^a u^b = -1 \ . \tag{2.50}$$

The Lorentz-invariant action for a free massive particle with mass m is

$$S_0 = -m \int d\tau \ . \tag{2.51}$$

We can adopt the same set-up and action in the present setting. Thus we parametrise the worldlines by

$$x^\mu = x^\mu(\tau) \ , \tag{2.52}$$

with τ the proper time

$$d\tau^2 = -g_{\mu\nu}(x) dx^\mu dx^\nu \ , \tag{2.53}$$

invariant under general coordinate transformations (provided that one transforms the metric appropriately). The corresponding 4-velocity

$$u^\mu = \frac{dx^\mu}{d\tau} \tag{2.54}$$

is again normalised as

$$g_{\mu\nu} u^\mu u^\nu = -1 \ , \tag{2.55}$$

and we are led to consider the coordinate-invariant Lagrangian

$$S_0[x] = -m \int d\tau = -m \int \sqrt{-g_{\mu\nu}(x) dx^\mu dx^\nu} \ . \tag{2.56}$$

Of course m drops out of the variational equations (as it should by the equivalence principle) and we will therefore ignore m in the following.

In order to perform the variation, it is useful to introduce an arbitrary auxiliary parameter λ in the initial stages of the calculation via

$$d\tau = (-g_{\mu\nu} \tfrac{dx^\mu}{d\lambda} \tfrac{dx^\nu}{d\lambda})^{1/2} d\lambda \ , \tag{2.57}$$

and to write

$$\int d\tau = \int (d\tau/d\lambda) d\lambda = \int (-g_{\mu\nu} \tfrac{dx^\mu}{d\lambda} \tfrac{dx^\nu}{d\lambda})^{1/2} d\lambda \ . \tag{2.58}$$

We are varying the paths

$$x^\mu(\tau) \to x^\mu(\tau) + \delta x^\mu(\tau) \tag{2.59}$$

keeping the end-points fixed, and will denote the τ-derivatives by $\dot{x}^\mu(\tau)$. Under this variation, the metric $g_{\mu\nu}(x)$ varies as

$$\delta g_{\mu\nu} = g_{\mu\nu,\lambda}\, \delta x^\lambda \quad . \tag{2.60}$$

By the standard variational procedure one then finds, first of all,

$$\delta \int d\tau = \tfrac{1}{2} \int (-g_{\mu\nu} \tfrac{dx^\mu}{d\lambda} \tfrac{dx^\nu}{d\lambda})^{-1/2} d\lambda \left[-\delta g_{\mu\nu} \frac{dx^\mu}{d\lambda} \frac{dx^\nu}{d\lambda} - 2 g_{\mu\nu} \frac{d\delta x^\mu}{d\lambda} \frac{dx^\nu}{d\lambda} \right] \quad . \tag{2.61}$$

Already at this stage we can revert from λ to τ, and the expression simplifies to

$$\delta \int d\tau = \tfrac{1}{2} \int d\tau \left[-(\delta g_{\mu\nu}) \dot{x}^\mu \dot{x}^\nu - 2 g_{\mu\nu} \frac{d\delta x^\mu}{d\tau} \dot{x}^\nu \right] \quad . \tag{2.62}$$

Integration by parts of the 2nd term (in order to eliminate the derivative of the variation) and use of (2.60) then leads to

$$\begin{aligned} \delta \int d\tau &= \frac{1}{2} \int d\tau \left[-g_{\mu\nu,\lambda}\, \dot{x}^\mu \dot{x}^\nu \delta x^\lambda + 2 g_{\mu\nu} \ddot{x}^\nu \delta x^\mu + 2 g_{\mu\nu,\lambda}\, \dot{x}^\lambda \dot{x}^\nu \delta x^\mu \right] \\ &= \int d\tau \left[g_{\mu\nu} \ddot{x}^\nu + \tfrac{1}{2} (g_{\mu\nu,\lambda} + g_{\mu\lambda,\nu} - g_{\nu\lambda,\mu}) \dot{x}^\nu \dot{x}^\lambda \right] \delta x^\mu \end{aligned} \tag{2.63}$$

after a suitable relabelling of the indices.

If we now adopt the definition (2.46) for an arbitrary metric,

$$\Gamma^\mu_{\ \nu\lambda} = g^{\mu\rho} \Gamma_{\rho\nu\lambda} = \tfrac{1}{2} g^{\mu\rho} (g_{\rho\nu,\lambda} + g_{\rho\lambda,\nu} - g_{\nu\lambda,\rho}) \quad , \tag{2.64}$$

we can write the result as

$$\delta \int d\tau = \int d\tau g_{\mu\nu} (\ddot{x}^\nu + \Gamma^\nu_{\ \rho\lambda} \dot{x}^\rho \dot{x}^\lambda) \delta x^\mu \quad . \tag{2.65}$$

Thus we see that indeed the equations for a timelike geodesic in an arbitrary gravitational field are

$$\frac{d^2 x^\mu}{d\tau^2} + \Gamma^\mu_{\ \nu\lambda} \frac{dx^\nu}{d\tau} \frac{dx^\lambda}{d\tau} = 0 \quad . \tag{2.66}$$

REMARKS:

1. Given a metric $g_{\mu\nu}$, the quantities $\Gamma_{\rho\nu\lambda}$ and $\Gamma^\mu_{\ \nu\lambda}$ defined in (2.64) are known as the *Christoffel Symbols* of the 1st kind and 2nd kind.

2. The Christoffel symbols (2.64) play the role of the gravitational force term, and thus in this sense the components of the metric play the role of the gravitational potential. These Christoffel symbols play an important role not just in the geodesic equation but, as we will see later on, more generally in the definition of a covariant derivative operator and the construction of the curvature tensor, and thus ultimately also in the generally covariant description of the dynamics of the gravitational field itself.

3. Two elementary important properties of the Christoffel symbols are that they are symmetric in the second and third indices,

$$\Gamma_{\mu\nu\lambda} = \Gamma_{\mu\lambda\nu} \quad , \quad \Gamma^{\mu}_{\ \nu\lambda} = \Gamma^{\mu}_{\ \lambda\nu} \tag{2.67}$$

(this follows simply from the definition), and that symmetrising $\Gamma_{\mu\nu\lambda}$ over the first pair of indices one finds

$$\Gamma_{\mu\nu\lambda} + \Gamma_{\nu\mu\lambda} = g_{\mu\nu,\lambda} \tag{2.68}$$

(and this follows from noting that 4 of the 6 partial derivative terms of the metric cancel in this linear combination while 2 add up)

4. One can also consider spacelike paths that extremise (minimise) proper distance, by using the action

$$S_0 \sim \int ds \tag{2.69}$$

where

$$ds^2 = g_{\mu\nu}(x) dx^\mu dx^\nu \tag{2.70}$$

is the proper distance (or arc-length in the traditional terminology of the differnetial geometry of curves).

One should also consider massless particles, whose worldlines will be null (or lightlike) paths. However, in that case one can evidently not use proper time or proper distance, since these are by definition zero along a null path, $ds^2 = 0$. We will come back to this special case, and a unified description of the massive and massless case, below (section 2.5). In all cases, we will refer to the resulting paths as *geodesics*. If required, we add the qualifier "timelike", "spacelike" or "null", and this is meaningful and unambiguous since, as we will see below, a geodesic that is initially timelike will always remain timelike etc.

5. By definition, massive test particles are those particles that satisfy the above geodesic equation, i.e. that follow timelike geodesics in space-time. However, it needs to be borne in mind that this notion of a test particle is a fiction, in particular as it neglects the backreaction, i.e. the change in the background gravitational field due to the mass of the particle. Moreover, real particles either have a finite extent (in which case this finite size should play a role in their equations of motion) or are considered to be point-like. However, the notion of a point-like particle is extremely dangerous and delicate in general relativity: as we will see later, if a given total mass is concentrated in a sufficiently small region of space-time (and "point-like" certainly qualifies as "sufficiently small"), then one will end up with a black hole rather than with the description of a particle. The correct description of point particles in general relativity is a complicated issue and an active area of research.[5]

[5] See e.g. E. Poisson, A. Pound, I. Vega, *The Motion of Point Particles in Curved Spacetime*,

2.4 Geodesic Equation and Coordinate Transformations

As we have seen, in a coordinate system x^μ, with metric $g_{\mu\nu}$ and Christoffel symbols $\Gamma^\mu_{\ \nu\lambda}$ the geodesic equation takes the form (2.66)

$$\frac{d^2x^\mu}{d\tau^2} + \Gamma^\mu_{\ \nu\lambda}\frac{dx^\nu}{d\tau}\frac{dx^\lambda}{d\tau} = 0 \ . \tag{2.71}$$

Since the coordinate system x^μ was arbitrary, evidently this equation is true for an arbitrary coordinate system. Nevertheless, when one changes from coordinates x^μ to y^α, the components of the metric and the Christoffel symbols will change (the latter in a somewhat complicated way). It is therefore of interest to find out explicitly how it comes about that the validity of (2.71) is independent of the coordinate system. It turns out that the geodesic equations in the two coordinate systems are simply related by the Jacobi matrix,

$$\frac{d^2y^\alpha}{d\tau^2} + \Gamma^\alpha_{\ \beta\gamma}\frac{dy^\beta}{d\tau}\frac{dy^\gamma}{d\tau} = \frac{\partial y^\alpha}{\partial x^\mu}\left[\frac{d^2x^\mu}{d\tau^2} + \Gamma^\mu_{\ \nu\lambda}\frac{dx^\nu}{d\tau}\frac{dx^\lambda}{d\tau}\right] \ . \tag{2.72}$$

This generalises the result (1.112) that we had obtained before regarding coordinate transformations in Minkowski space, and the same remarks about covariance and tensors etc. apply. In particular, the invertibility of the Jacobi matrix allows us to conclude that

$$\frac{d^2y^\alpha}{d\tau^2} + \Gamma^\alpha_{\ \beta\gamma}\frac{dy^\beta}{d\tau}\frac{dy^\gamma}{d\tau} = 0 \quad \Leftrightarrow \quad \frac{d^2x^\mu}{d\tau^2} + \Gamma^\mu_{\ \nu\lambda}\frac{dx^\nu}{d\tau}\frac{dx^\lambda}{d\tau} = 0 \ , \tag{2.73}$$

as anticipated. I will now give 3 arguments to establish (2.72).

1. General Argument at the Level of the Variation of the Action

   That the geodesic equation transforms in this simple way (namely as a vector) should not come as a surprise. We obtained this equation as a variational equation. The Lagrangian itself is a scalar (invariant under coordinate transformations),

   $$d\tau = \sqrt{-g_{\alpha\beta}dy^\alpha dy^\beta} = \sqrt{-g_{\mu\nu}dx^\mu dx^\nu} \ . \tag{2.74}$$

   Moreover, the variations of y^α and x^μ are related by

   $$\delta y^\alpha = \frac{\partial y^\alpha}{\partial x^\mu}\delta x^\mu = J^\alpha_\mu \delta x^\mu \ . \tag{2.75}$$

   Putting these pieces together, one finds the desired result (2.72).

2. General Argument at the Level of the Euler-Lagrange Equations

   A general feature of the Lagrange formalism and the Euler-Lagrange equations is their covariance under what are known as "point-transformations" in classical

`arXiv:1102.0529 [gr-qc]` for a detailed discussion and many references (but you will need to acquire a solid understanding of tensor analysis first).

mechanics, arbitrary (possibly even time-dependent) coordinate transformations of the configuration space variables. In the case at hand, this reduces to the statement that the Euler-Lagrange equations with respect to x^μ and y^α respectively are related by the Jacobi matrix,

$$\frac{d}{d\tau}\frac{\partial L}{\partial x'^\mu} - \frac{\partial L}{\partial x^\mu} = J^\alpha_\mu \left[\frac{d}{d\tau}\frac{\partial L}{\partial y'^\alpha} - \frac{\partial L}{\partial y^\alpha}\right] \tag{2.76}$$

where $x'^\mu = dx^\mu/d\lambda$ etc. This also implies (2.72).

3. Specific Argument at the Level of the Geodesic Equation

   Knowing how the metric transforms under coordinate transformations,

$$g_{\alpha\beta} = J^\mu_\alpha J^\nu_\beta g_{\mu\nu} \ , \tag{2.77}$$

   we can now also determine how the Christoffel symbols (2.64) and the geodesic equation transform. A straightforward but not particularly inspiring calculation (which you should nevertheless do) shows that under $x^\mu \to y^\alpha$ the Christoffel symbols are related by

$$\Gamma^\alpha_{\beta\gamma} = \Gamma^\mu_{\nu\lambda}\frac{\partial y^\alpha}{\partial x^\mu}\frac{\partial x^\nu}{\partial y^\beta}\frac{\partial x^\lambda}{\partial y^\gamma} + \frac{\partial y^\alpha}{\partial x^\mu}\frac{\partial^2 x^\mu}{\partial y^\beta \partial y^\gamma} \ , \tag{2.78}$$

   or

$$\Gamma^\alpha_{\beta\gamma} = J^\alpha_\mu J^\nu_\beta J^\lambda_\gamma \Gamma^\mu_{\nu\lambda} + J^\alpha_\mu \partial_\beta J^\mu_\gamma \ . \tag{2.79}$$

   Thus, $\Gamma^\mu_{\nu\lambda}$ transforms inhomogenously under coordinate transformations. If only the first term on the right hand side were present, then $\Gamma^\mu_{\nu\lambda}$ would be a tensor. However, the second term is there precisely to compensate for the fact that $\ddot{x}^\mu$ is also not a tensor - the combined geodesic equation turns out to transform in a nice way under coordinate transformations.

   Namely, after another not terribly inspiring calculation (which you should nevertheless also do at least once in your life) , one finds precisely (2.72),

$$\frac{d^2 y^\alpha}{d\tau^2} + \Gamma^\alpha_{\beta\gamma}\frac{dy^\beta}{d\tau}\frac{dy^\gamma}{d\tau} = \frac{\partial y^\alpha}{\partial x^\mu}\left[\frac{d^2 x^\mu}{d\tau^2} + \Gamma^\mu_{\nu\lambda}\frac{dx^\nu}{d\tau}\frac{dx^\lambda}{d\tau}\right] \ . \tag{2.80}$$

   An explicit proof of (2.79) and (2.80) is given at the end of this subsection. A more general result along these lines will be established in section 5.1 below, when we introduce the covariant derivative of a vector field.

REMARKS:

1. Argument 3 in particular highlights the importance and significance of the Christoffel symbols: they are not a nuisance but a blessing, because they allow one to modify the definition of the acceleration $\ddot{x}^\mu = (d/d\tau)\dot{x}^\mu$ in such a way that it

behaves nicely under coordinate transformations, namely just like the velocity $\dot{x}^\mu$ itself. We will see later on, in section 5, that the Christoffel symbols allow us to accomplish this much more generally. Namely, with their aid we can define a (generally covariant) notion of a derivative that generalises the ordinary (or partial) derivative and maps tensors to tensors.

2. General covariance, i.e. invariance of the equations of motion under general coordinate transformations, as exhibited e.g. by the geodesic equation, is of course a desirable feature regardless of whether or not one is attempting to describe gravity. After all, the particle could not care less which coordinates we use to describe its motion, and therefore we should also formulate the equations of motion for a particle in a way that does not single out some preferred coordinate system or class of coordinate systems. This is precisely what is achieved by general covariance.

   However, here general covariance seems to have arisen somewhat coincidentally and spontaneously, and the relation between general covariance and gravity, or general covariance and the equivalence principle, may still appear to be somewhat mysterious at this point. The precise relation between the two concepts will be explained in section 4.1.

3. There is of course a very good physical reason for why the force term in the geodesic equation (quadratic in the 4-velocities) is not tensorial. This simply reflects the equivalence principle that locally, at a point (or in a sufficiently small neighbourhood of a point) you can eliminate the gravitational force by going to a freely falling (inertial) coordinate system. This would not be possible if the gravitational force term in the equation of motion for a particle were tensorial.

---

1. Proof of (2.79)

   For partial derivatives one has the chain rule $\partial_\gamma = J^\lambda_\gamma \partial_\lambda$ ("∂_λ is a covector"). Therefore for the partial derivatives of the metric one has

$$g_{\alpha\beta,\gamma} = (J^\mu_\alpha J^\nu_\beta g_{\mu\nu})_{,\gamma} = g_{\mu\nu,\lambda} J^\mu_\alpha J^\nu_\beta J^\lambda_\gamma + (J^\mu_{\alpha\gamma} J^\nu_\beta + J^\mu_\alpha J^\nu_{\beta\gamma}) g_{\mu\nu} \quad . \tag{2.81}$$

   Adding up the 3 terms comprising the Christoffel symbol $\Gamma_{\alpha\beta\gamma}$, one obtains

$$\begin{aligned} 2\Gamma_{\alpha\beta\gamma} &= g_{\alpha\beta,\gamma} + g_{\alpha\gamma,\beta} - g_{\beta\gamma,\alpha} \\ &= 2 J^\mu_\alpha J^\nu_\beta J^\lambda_\gamma \Gamma_{\mu\nu\lambda} \\ &\quad + (J^\mu_{\alpha\gamma} J^\nu_\beta + J^\mu_\alpha J^\nu_{\beta\gamma} + J^\mu_{\alpha\beta} J^\nu_\gamma + J^\mu_\alpha J^\nu_{\gamma\beta} - J^\mu_{\beta\alpha} J^\nu_\gamma - J^\mu_\beta J^\nu_{\gamma\alpha}) g_{\mu\nu} \quad . \end{aligned} \tag{2.82}$$

   In the last line, the 3rd term cancels against the 5th (because $J^\mu_{\alpha\beta}$ is symmetric), the 1st term cancels against the 6th (because $J^\mu_{\alpha\gamma}$ and $g_{\mu\nu}$ are symmetric), while

the 2nd and 4th term add up, so that one finds

$$\Gamma_{\alpha\beta\gamma} = J^\mu_\alpha J^\nu_\beta J^\lambda_\gamma \Gamma_{\mu\nu\lambda} + J^\mu_\alpha J^\nu_{\beta\gamma} g_{\mu\nu} \quad . \tag{2.83}$$

Now the hard work has been done. Raising the 1st index of the Christoffel symbol, using the inverse metric

$$g^{\alpha\delta} = g^{\sigma\rho} J^\alpha_\sigma J^\delta_\rho \quad , \tag{2.84}$$

it is now simple to see that one obtains the claimed result (4),

$$\Gamma^\alpha_{\beta\gamma} = g^{\alpha\delta}\Gamma_{\delta\beta\gamma} = J^\alpha_\mu J^\nu_\beta J^\lambda_\gamma \Gamma^\mu{}_{\nu\lambda} + J^\alpha_\mu J^\mu_{\beta\gamma} \quad . \tag{2.85}$$

For example, for the 2nd term one has (just using properties of inverse Jacobi matrices and metrics)

$$\begin{aligned} g^{\alpha\delta} J^\mu_\delta J^\nu_{\beta\gamma} g_{\mu\nu} &= g^{\sigma\rho} J^\alpha_\sigma J^\delta_\rho J^\mu_\delta J^\nu_{\beta\gamma} g_{\mu\nu} = g^{\sigma\rho} J^\alpha_\sigma \delta^\mu_\rho J^\nu_{\beta\gamma} g_{\mu\nu} \\ &= g^{\sigma\mu} J^\alpha_\sigma J^\nu_{\beta\gamma} g_{\mu\nu} = \delta^\sigma_\nu J^\alpha_\sigma J^\nu_{\beta\gamma} = J^\alpha_\nu J^\nu_{\beta\gamma} \end{aligned} \tag{2.86}$$

2. Proof of (2.72)

   The 4-velocities transform as vectors (the chain rule again), $\dot{y}^\alpha = J^\alpha_\mu \dot{x}^\mu$. Therefore for the acceleration one has

$$\ddot{y}^\alpha = J^\alpha_\mu \ddot{x}^\mu + J^\alpha_{\mu\nu} \dot{x}^\mu \dot{x}^\nu \quad . \tag{2.87}$$

   Therefore

$$\begin{aligned} \ddot{y}^\alpha + \Gamma^\alpha_{\beta\gamma} \dot{y}^\beta \dot{y}^\gamma &= J^\alpha_\mu (\ddot{x}^\mu + \Gamma^\mu{}_{\nu\lambda} J^\nu_\beta J^\lambda_\gamma \dot{y}^\beta \dot{y}^\gamma) + J^\alpha_\mu J^\mu_{\beta\gamma} \dot{y}^\beta \dot{y}^\gamma + J^\alpha_{\mu\nu} \dot{x}^\mu \dot{x}^\nu \\ &= J^\alpha_\mu (\ddot{x}^\mu + \Gamma^\mu{}_{\nu\lambda} \dot{x}^\nu \dot{x}^\lambda) + (J^\alpha_\mu J^\mu_{\beta\gamma} + J^\alpha_{\mu\nu} J^\mu_\beta J^\nu_\gamma) \dot{y}^\beta \dot{y}^\gamma \end{aligned} \tag{2.88}$$

   The 1st term will give us the desired result, and cooperatively the 2nd term is identically zero because (use $\partial_\gamma = J^\nu_\gamma \partial_\nu$ again)

$$0 = (\delta^\alpha_\beta)_{,\gamma} = (J^\alpha_\mu J^\mu_\beta)_{,\gamma} = J^\alpha_{\mu\nu} J^\nu_\gamma J^\mu_\beta + J^\alpha_\mu J^\mu_{\beta\gamma} \quad . \tag{2.89}$$

Apology and Outlook:

You may feel that, after a promising start, some of the things that we have done subsequently (in particular in this subsection) look terribly messy. I agree, indeed they are! However, I can assure you that this is by far the messiest part of the entire lecture notes and that things will improve dramatically rather quickly.

Indeed, the main purpose and benefit of developing tensor calculus in the next couple of sections is to develop a formalism in the framework of which (among other things)

- one can avoid having to deal explicitly with objects that transform in complicated ways under coordinate transformations
- the transformation behaviour of any object is manifest (and does not have to be checked)
- it is straightforward to write down equations that are *generally covariant*, i.e. independent of the coordinate system in the sense that they are satisfied in all coordinate systems if and only if they are satisfied in one.

This tensor calculus formalism is simple, elegant and efficient and will then allow us to make rapid progress towards describing the dynamics in a (and subsequently of the) gravitational field in a way compatible with the Einstein equivalence principle.

2.5 Alternative Action Principles for Geodesics

As we have already noted in section 2.3, there is a problem with the action $S \sim m\int d\tau$ (2.56) for massless particles (null geodesics). For this reason and many other practical purposes (the square root in the action is awkward) it is much more convenient to use, instead of the action

$$S_0[x] = -m\int d\tau = -m\int d\lambda\sqrt{-g_{\alpha\beta}\frac{dx^\alpha}{d\lambda}\frac{dx^\beta}{d\lambda}} \equiv \int d\lambda\ \mathcal{L}_0^\lambda \tag{2.90}$$

the simpler Lagrangian

$$\mathcal{L} = \tfrac{1}{2}g_{\alpha\beta}\frac{dx^\alpha}{d\lambda}\frac{dx^\beta}{d\lambda} \tag{2.91}$$

and action

$$S_1[x] = \int d\lambda\ \mathcal{L}\ . \tag{2.92}$$

Let us first verify that S_1 really leads to the same equations of motion as S_0. Either by direct variation of the action, or by using the Euler-Lagrange equations

$$\frac{d}{d\lambda}\frac{\partial\mathcal{L}}{\partial(dx^\gamma/d\lambda)} - \frac{\partial\mathcal{L}}{\partial x^\gamma} = 0\ , \tag{2.93}$$

one finds that the action is extremised by solutions to the equation

$$\frac{d}{d\lambda}\left(g_{\gamma\beta}\frac{dx^\beta}{d\lambda}\right) = \tfrac{1}{2}g_{\alpha\beta,\gamma}\frac{dx^\alpha}{d\lambda}\frac{dx^\beta}{d\lambda}\ . \tag{2.94}$$

The terms involving first derivatives of the metric cooperatively combine into the Christoffel symbols,

$$\left(\frac{d}{d\lambda}g_{\gamma\beta}\right)\frac{dx^\beta}{d\lambda} - \tfrac{1}{2}g_{\alpha\beta,\gamma}\frac{dx^\alpha}{d\lambda}\frac{dx^\beta}{d\lambda} = g_{\gamma\beta,\alpha}\frac{dx^\alpha}{d\lambda}\frac{dx^\beta}{d\lambda} - \tfrac{1}{2}g_{\alpha\beta,\gamma}\frac{dx^\alpha}{d\lambda}\frac{dx^\beta}{d\lambda} = \Gamma_{\gamma\alpha\beta}\frac{dx^\alpha}{d\lambda}\frac{dx^\beta}{d\lambda}\ . \tag{2.95}$$

Here we have used the fact that we can write

$$g_{\gamma\beta,\alpha}\frac{dx^\alpha}{d\lambda}\frac{dx^\beta}{d\lambda} = \tfrac{1}{2}\left(g_{\gamma\beta,\alpha} + g_{\gamma\alpha,\beta}\right)\frac{dx^\alpha}{d\lambda}\frac{dx^\beta}{d\lambda} \tag{2.96}$$

because

$$\frac{dx^\alpha}{d\lambda}\frac{dx^\beta}{d\lambda} = \frac{dx^\beta}{d\lambda}\frac{dx^\alpha}{d\lambda} \tag{2.97}$$

is symmetric. Therefore one has

$$g_{\gamma\beta}\frac{d^2x^\beta}{d\lambda^2} + \Gamma_{\gamma\alpha\beta}\frac{dx^\alpha}{d\lambda}\frac{dx^\beta}{d\lambda} = 0 \ . \tag{2.98}$$

By raising the index (or multiplying with the inverse metric) one can write this as

$$\frac{d^2x^\gamma}{d\lambda^2} + \Gamma^\gamma_{\ \alpha\beta}\frac{dx^\alpha}{d\lambda}\frac{dx^\beta}{d\lambda} = 0 \ . \tag{2.99}$$

This is identical to the geodesic equation derived from S_0 (with $\lambda \to \tau$, the proper time).

We will make extensive use of this simpler Lagrangian for geodesics throughout these notes. In particular, in practice the version (2.94) of the geodesic action is much more efficient and user-friendly than the standard form, because everything is expressed directly in terms of the metric and its first derivatives (neither does one need the inverse metric, nor does one have to assemble the derivatives of the metric into Christoffel symbols first).

Moreover, as will be explained in section 3.1 below, (2.94) actually also provides one with a fairly efficient method to determine (essentially read off) the Christoffel symbols, simply by comparing (2.94) with (2.98) or (2.99).

One important consequence of (2.99) is that the quantity $\mathcal{L}$ is a constant of motion, i.e. constant along the geodesic,

$$\frac{d^2x^\gamma}{d\lambda^2} + \Gamma^\gamma_{\ \alpha\beta}\frac{dx^\alpha}{d\lambda}\frac{dx^\beta}{d\lambda} = 0 \quad \Rightarrow \quad \frac{d}{d\lambda}\left(g_{\alpha\beta}\frac{dx^\alpha}{d\lambda}\frac{dx^\beta}{d\lambda}\right) = 0 \ . \tag{2.100}$$

This useful result can be understood and derived in a variety of ways:

- The least insightful way is just direct calculation. Nevertheless, this is straightforward and a good exercise in Γ-ology (and as such is left as an exercise).
- Alternatively, noting that $\mathcal{L}$ does not depend explicitly on λ, this result can be derived (as the corresponding conserved "energy") from Noether's theorem (cf. section 2.6 below for this argument).
- Yet another derivation will be given in section 5.8, using the concept of "parallel transport".

One obvious consequence of (2.100) is that, if one imposes the initial condition

$$g_{\mu\nu}\frac{dx^\mu}{d\lambda}\frac{dx^\nu}{d\lambda}\bigg|_{\lambda=0} = \epsilon \ , \tag{2.101}$$

then this condition will be satisfied for all λ. This is as it should be. After all, something that starts off as a massless particle will remain a massless particle etc. In particular, therefore, even though with this choice of Lagrangian λ is a priori unrelated to proper time, say, this shows that one can choose $\epsilon = \mp 1$ for timelike (spacelike) geodesics, and λ can then be identified with proper time (proper distance), while the choice $\epsilon = 0$ sets the initial conditions appropriate to massless particles (for which λ is then not related to proper time or proper distance).

Moreover, the constancy of $\mathcal{L}$ for solutions of the Euler-Lagrange equations is the reason (or one explanation for) why $\mathcal{L}$ and $\sqrt{\pm\mathcal{L}}$ (more generally any monotonic function $f(\mathcal{L})$ of $\mathcal{L}$) give rise to equivalent equations of motion.

Indeed, this can be seen by simply comparing the Euler-Lagrange equations for $f(\mathcal{L})$ with those for $\mathcal{L}$. Denoting the Euler-Lagrange equations for a Lagrangian $\mathcal{L}$ by

$$\mathcal{E}^{\mathcal{L}}_\gamma = \frac{d}{d\lambda}\frac{\partial\mathcal{L}}{\partial(dx^\gamma/d\lambda)} - \frac{\partial\mathcal{L}}{\partial x^\gamma} \ , \tag{2.102}$$

for any $\mathcal{L}$ and any f one has

$$\mathcal{E}^{f(\mathcal{L})}_\gamma = f'(\mathcal{L})\mathcal{E}^{\mathcal{L}}_\gamma + f''(\mathcal{L})\left(\frac{d}{d\lambda}\mathcal{L}\right)\frac{\partial\mathcal{L}}{\partial(dx^\gamma/d\lambda)} \ . \tag{2.103}$$

Thus, if

$$\mathcal{E}^{\mathcal{L}}_\gamma = 0 \quad \Rightarrow \quad \frac{d}{d\lambda}\mathcal{L} = 0 \ , \tag{2.104}$$

then one has

$$\mathcal{E}^{\mathcal{L}}_\gamma = 0 \quad \Rightarrow \quad \mathcal{E}^{f(\mathcal{L})}_\gamma = 0 \ , \tag{2.105}$$

and if f is monotonic ($f' \neq 0$ everywhere), then one also has the converse,

$$f' \neq 0 \quad \Rightarrow \quad \left(\mathcal{E}^{\mathcal{L}}_\gamma = 0 \quad \Leftrightarrow \quad \mathcal{E}^{f(\mathcal{L})}_\gamma = 0\right) \ . \tag{2.106}$$

2.6 On the Relation between the two Action Principles

Even though not strictly required in the following, it is nevertheless quite instructive in its own right to try to understand and establish the precise relation between the two actions S_0 and S_1, and this is the subject of this subsection.

The first thing to notice is that S_0 is manifestly parametrisation-invariant, i.e. independent of how one parametrises the path. The reason for this is that

$$d\tau = (d\tau/d\lambda)d\lambda \tag{2.107}$$

is evidently independent of λ. This is not the case for S_1, which changes under parametrisations or, put more positively, singles out a preferred parametrisation (more precisely, as we will see below, a special class of parametrisations).

Thus, what is the relation (if any) between the two actions? In order to explain this, it will be useful to introduce an additional field $e(\lambda)$ (i.e. in addition to the $x^\alpha(\lambda)$), and a "master action" (or parent action) S which we can relate to both S_0 and S_1. Consider the action

$$S[x,e] = \tfrac{1}{2}\int d\lambda \left(e(\lambda)^{-1} g_{\alpha\beta} \frac{dx^\alpha}{d\lambda}\frac{dx^\beta}{d\lambda} - m^2 e(\lambda)\right) = \int d\lambda \left(e(\lambda)^{-1}\mathcal{L} - \tfrac{1}{2}m^2 e(\lambda)\right) \tag{2.108}$$

The crucial property of this action is that it is parametrisation invariant provided that one declares $e(\lambda)$ to transform appropriately. It is easy to see that under a transformation $\lambda \to \bar\lambda = f(\lambda)$, with

$$\bar x^\alpha(\bar\lambda) = x^\alpha(\lambda) \qquad d\bar\lambda = f'(\lambda) d\lambda \tag{2.109}$$

the action $S[x,e]$ is invariant provided that $e(\lambda)$ transforms such that $e(\lambda)d\lambda$ is invariant, i.e.

$$\bar e(\bar\lambda) d\bar\lambda \stackrel{!}{=} e(\lambda) d\lambda \quad \Rightarrow \quad \bar e(\bar\lambda) = e(\lambda)/f'(\lambda) \ . \tag{2.110}$$

Indeed, this is evident when one writes the action (2.108) in the form

$$S[x,e] = \tfrac{1}{2}\int e(\lambda) d\lambda \left(e(\lambda)^{-2} g_{\alpha\beta} \frac{dx^\alpha}{d\lambda}\frac{dx^\beta}{d\lambda} - m^2\right) \tag{2.111}$$

and notes that $d\lambda$ and $e(\lambda)$ only appear in the combinations $e(\lambda)d\lambda$ and $e(\lambda)^{-1}(d/d\lambda)$.

Now what is the relation between the action $S[x,e]$ and the two "standard" actions $S_0[x]$ and $S_1[x]$?

- Courtesy of this parametrisation invariance, we can always choose a "gauge" in which $e(\lambda) = 1$. With this choice, the action $S[x,e]$ manifestly reduces to the action $S_1[x]$ modulo an irrelevant field-independent constant,

  $$S[x, e=1] = \int d\lambda\ \mathcal{L} - \tfrac{1}{2}m^2 \int d\lambda = S_1[x] + \text{const.} \ . \tag{2.112}$$

  Thus we can regard S_1 as a gauge-fixed version of S (no wonder it is not parametrisation invariant ...). We will come back to the small residual gauge invariance (reparametrisations that preserve the gauge condition $e(\lambda) = 1$) below.

- Alternatively, instead of fixing the gauge, we can try to eliminate $e(\lambda)$ (which appears purely algebraically, i.e. without derivatives, in the action) by its equation of motion. Varying $S[x,e]$ with respect to $e(\lambda)$, one finds the constraint

  $$g_{\alpha\beta}\frac{dx^\alpha}{d\lambda}\frac{dx^\beta}{d\lambda} + m^2 e(\lambda)^2 = 0 \ . \tag{2.113}$$

This is just the usual mass-shell condition in disguise. It suggests that a better gauge fixing than $e(\lambda) = 1$ would have been $e(\lambda) = m^{-1}$. However, the sole effect of this would have been to replace $\mathcal{L}$ in (2.112) by $m\mathcal{L}$,

$$e(\lambda) = 1 \to e(\lambda) = m^{-1} \quad \Rightarrow \quad \mathcal{L} \to m\mathcal{L} \ . \tag{2.114}$$

In any case, for a massive particle, $m^2 \neq 0$, one can alternatively solve (2.113) for $e(\lambda)$,

$$e(\lambda) = m^{-1}\sqrt{-g_{\alpha\beta}\frac{dx^\alpha}{d\lambda}\frac{dx^\beta}{d\lambda}} \ . \tag{2.115}$$

Using this to eliminate $e(\lambda)$ from the action, one finds

$$S[x, e = m^{-1}\sqrt{\dots}] = -m\int d\lambda\sqrt{-g_{\alpha\beta}\frac{dx^\alpha}{d\lambda}\frac{dx^\beta}{d\lambda}} = -m\int d\tau = S_0[x] \ . \tag{2.116}$$

Thus for $m^2 \neq 0$ we find exactly the original action (integral of the proper time) $S_0[x]$ (and since we have not touched or fixed the parametrisation invariance, no wonder that S_0 is parametrisation invariant).

Thus we have elucidated the common origin of the actions S_0 and S_1 for a massive particle.

The perspective provided by the parent action $S[x, e]$ also gives some further insights. For example, an added benefit of the parent action $S[x, e]$ is that it also makes perfect sense for a massless particle. For $m^2 = 0$, the mass shell condition

$$g_{\alpha\beta}\frac{dx^\alpha}{d\lambda}\frac{dx^\beta}{d\lambda} = 0 \tag{2.117}$$

says that these particles move along null lines, and the action reduces to

$$S[x, e] = \tfrac{1}{2}\int d\lambda\ e(\lambda)^{-1} g_{\alpha\beta}\frac{dx^\alpha}{d\lambda}\frac{dx^\beta}{d\lambda} \tag{2.118}$$

which is parametrisation invariant but can (as in the massive case) be fixed to $e(\lambda) = 1$, upon which the action reduces to $S_1[x]$. Thus we see that $S_1[x]$ indeed provides a simple and unified action for both massive and massless particles, and in both cases the resulting equation of motion is the (affinely parametrised) geodesic equation (2.99),

$$\frac{d^2x^\alpha}{d\lambda^2} + \Gamma^\alpha_{\beta\gamma}\frac{dx^\beta}{d\lambda}\frac{dx^\gamma}{d\lambda} = 0 \ . \tag{2.119}$$

REMARKS:

1. The infinitesimal form of the invariance of the action $S[x, e]$ under (2.109) and (2.110) is obtained by considering the infinitesimal transformation of $x^\alpha(\lambda)$ and

$e(\lambda)$ induced by an infinitesimal transformation $\bar{\lambda} = \lambda + \epsilon(\lambda)$,

$$\begin{aligned} \delta\lambda = \epsilon(\lambda) \quad \Rightarrow \quad \delta x^\alpha(\lambda) &= \epsilon(\lambda)\frac{dx^\alpha(\lambda)}{d\lambda} \\ \delta e(\lambda) &= \frac{d\epsilon(\lambda)}{d\lambda}e(\lambda) + \epsilon(\lambda)\frac{de(\lambda)}{d\lambda} \quad . \end{aligned} \tag{2.120}$$

Here the (at first perhaps somewhat peculiar looking) transformation behaviour of the auxiliary field $e(\lambda)$ arises from the transformation behaviour (2.110) by setting

$$\delta e(\lambda) = e(\lambda) - \bar{e}(\lambda) \tag{2.121}$$

and calculating (keeping at most linear terms in $\epsilon(\lambda)$)

$$\begin{aligned} e(\lambda) &= f'(\lambda)\bar{e}(\bar{\lambda}) = (1 + \epsilon'(\lambda))\ \bar{e}(\lambda + \epsilon(\lambda)) \\ &= (1 + \epsilon'(\lambda))\ (\bar{e}(\lambda) + \epsilon(\lambda)\bar{e}'(\lambda)) \\ &= \bar{e}(\lambda) + \epsilon'(\lambda)\bar{e}(\lambda) + \epsilon(\lambda)\bar{e}'(\lambda) \quad . \end{aligned} \tag{2.122}$$

Under this infinitesimal transformation, the Lagrangian

$$\mathcal{L}_e = e(\lambda)^{-1}\mathcal{L} - \tfrac{1}{2}m^2 e(\lambda) \tag{2.123}$$

of the action $S[x, e]$ (2.108) transforms as

$$\delta\mathcal{L}_e = \frac{d}{d\lambda}(\epsilon(\lambda)\mathcal{L}_e) \quad , \tag{2.124}$$

implying the invariance of the action.

2. We saw in (2.100), that the Lagrangian $\mathcal{L}$ itself is a constant of motion,

$$\frac{d}{d\lambda}\left(g_{\alpha\beta}\frac{dx^\alpha}{d\lambda}\frac{dx^\beta}{d\lambda}\right) = 0 \tag{2.125}$$

for a solution to the geodesic equation. From the present (action-based) perspective it is most useful to think of this as the conserved quantity associated (via Noether's theorem) to the invariance of the action $S_1[x]$ under translations in λ. Note that evidently $S_1[x]$ has this invariance (as there is no explicit dependence on λ) and that this invariance is precisely the residual parametrisation invariance $f(\lambda) = \lambda + a$, $f'(\lambda) = 1$, that leaves invariant the "gauge" condition $e(\lambda) = 1$. For an infinitesimal constant λ-translation one has $\delta x^\alpha(\lambda) = dx^\alpha/d\lambda$ etc., so that

$$\frac{\partial}{\partial\lambda}\mathcal{L} = 0 \quad \Rightarrow \quad \delta\mathcal{L} = \frac{d}{d\lambda}\mathcal{L} = \frac{d}{d\lambda}\left(\frac{\partial\mathcal{L}}{\partial(dx^\alpha/d\lambda)}\frac{dx^\alpha}{d\lambda}\right) + \quad \text{Euler} - \text{Lagrange} \quad . \tag{2.126}$$

Thus via Noether's theorem the associated conserved charge for a solution to the Euler-Lagrange equations is the Legendre transform

$$\mathcal{H} = \left(\frac{\partial\mathcal{L}}{\partial(dx^\alpha/d\lambda)}\frac{dx^\alpha}{d\lambda}\right) - \mathcal{L} \tag{2.127}$$

of the Lagrangian (also known as the Hamiltonian, once expressed in terms of the momenta). In the case at hand, with the Lagrangian $\mathcal{L}$ consisting of a purely quadratic term in the velocities (the $dx^\alpha/d\lambda$), the Hamiltonian is equal to the Lagrangian, and hence the Lagrangian $\mathcal{L}$ itself is conserved,

$$\mathcal{H} = \mathcal{L} \quad , \quad \frac{d}{d\lambda}\mathcal{L}|_{\text{solution}} = 0 \ . \tag{2.128}$$

3. The above non-trivial Hamiltonian associated to the Lagrangian $\mathcal{L}$ should be contrasted with the Hamiltonian associated to the action S_0 with Lagrangian (2.90)

$$\mathcal{L}_0^\lambda = -m\sqrt{-g_{\alpha\beta}\frac{dx^\alpha}{d\lambda}\frac{dx^\beta}{d\lambda}} \ , \tag{2.129}$$

which turns out to be zero identically. Indeed, the canonical momenta are

$$p_\alpha = mg_{\alpha\beta}(dx^\beta/d\lambda)/\sqrt{-g_{\alpha\beta}\frac{dx^\alpha}{d\lambda}\frac{dx^\beta}{d\lambda}} = mg_{\alpha\beta}(dx^\beta/d\tau) \ , \tag{2.130}$$

which evidently satisfy the mass shell condition

$$p_\alpha p^\alpha + m^2 = 0 \tag{2.131}$$

and lead to the Hamiltonian

$$\mathcal{H}_0 = p_\alpha(dx^\alpha/d\lambda) - \mathcal{L}_0^\lambda = 0 \ . \tag{2.132}$$

This vanishing of the Hamiltonian is strictly related to the reparametrisation invariance of the action S_0.

2.7 Affine and Non-affine Parametrisations

To understand the significance of how one parametrises the geodesic, observe that the geodesic equation itself,

$$\ddot{x}^\mu + \Gamma^\mu_{\ \nu\lambda}\dot{x}^\nu\dot{x}^\lambda = 0 \ , \tag{2.133}$$

is not reparametrisation invariant. Indeed, consider a change of parametrisation $\tau \to \sigma = f(\tau)$. Then

$$\frac{dx^\mu}{d\tau} = \frac{df}{d\tau}\frac{dx^\mu}{d\sigma} \ , \tag{2.134}$$

and therefore the geodesic equation written in terms of σ reads

$$\frac{d^2x^\mu}{d\sigma^2} + \Gamma^\mu_{\ \nu\lambda}\frac{dx^\nu}{d\sigma}\frac{dx^\lambda}{d\sigma} = -\frac{\ddot{f}}{\dot{f}^2}\frac{dx^\mu}{d\sigma} \ . \tag{2.135}$$

Thus the geodesic equation retains its form only under affine changes of the proper time parameter τ, $f(\tau) = a\tau + b$, and parameters $\sigma = f(\tau)$ related to τ by such an affine transformation are known as affine parameters.

From the first variational principle, based on S_0, the term on the right hand side arises in the calculation of (2.63) from the integration by parts if one does not switch back from λ to the affine parameter τ. The second variational principle, based on S_1 and the Lagrangian $\mathcal{L}$, on the other hand, always and automatically yields the geodesic equation in affine form.

Conversely, if we find a curve that satisfies

$$\frac{d^2x^\mu}{d\sigma^2} + \Gamma^\mu_{\nu\lambda}\frac{dx^\nu}{d\sigma}\frac{dx^\lambda}{d\sigma} = \kappa(\sigma)\frac{dx^\mu}{d\sigma} \quad , \tag{2.136}$$

for some function $\kappa(\sigma)$ (the *inaffinity*), we can deduce that this curve is the trajectory of a geodesic, but that it is simply not parametrised by an affine parameter (like proper time in the case of a timelike curve). Comparison of (2.135) and (2.136) shows that, given $\kappa(\sigma)$, an affine parameter τ is determined by

$$\kappa(f(\tau)) = -\frac{\ddot{f}}{\dot{f}^2} \quad \Leftrightarrow \quad \kappa(\sigma) = \frac{d}{d\sigma}\ln\frac{d\tau}{d\sigma} \tag{2.137}$$

or

$$\frac{d\tau}{d\sigma} = \mathrm{e}^{\int^\sigma ds\, \kappa(s)} \quad . \tag{2.138}$$

The two integration constants, the first hidden in the lower limit of integration in the exponent and the second in the additive constant arising from integrating $d\tau/d\sigma$, are precisely the two constants a, b that parametrise the freedom in the choice of affine parameter, $\tau \to a\tau + b$.

In the following, whenever we talk about geodesics we will practically always have in mind the variational principle based on S_1 leading to the geodesic equation (2.119) in affinely parametrised form.

However, it should be kept in mind that sometimes non-affine parameters appear naturally. For instance, it is occasionally convenient to parametrise timelike geodesics in a geometry with coordinates $x^\alpha = (x^0 = t, x^k)$ not by $x^\alpha = x^\alpha(\tau)$, where τ is the proper time along the geodesic, but rather as $x^k = x^k(t)$. This is the same curve, but described with respect to coordinate time (which could for instance agree with the proper time of some other, perhaps static, observer). The curve $t \to (t, x^k(t))$ will not be an affinely parametrised curve unless t itself satisfies the geodesic equation

$$\ddot{t} = 0 \quad \Leftrightarrow \quad t = a\tau + b \ . \tag{2.139}$$

One occasion where this will play a role (and from where I have borrowed the symbol κ for the "inaffinity") is in our discussion, much later, of the horizon of a black hole, where the lack of a certain coordinate to be an affine parameter is directly related to the physical properties of black holes (see section 27.10). In this context κ is known as the surface gravity of a black hole.

2.8 Example: Geodesics in $\mathbb{R}^2$ in Polar Coordinates

It is high time to consider an example. We will consider the simplest non-trivial metric, namely the standard Euclidean metric on $\mathbb{R}^2$ in polar coordinates. Thus the line element is

$$ds^2 = dx^2 + dy^2 = dr^2 + r^2 d\phi^2 \tag{2.140}$$

and the non-zero components of the metric are

$$g_{xx} = g_{yy} = 1 \tag{2.141}$$

and

$$g_{rr} = 1 \quad , \quad g_{\phi\phi} = r^2 \ . \tag{2.142}$$

respectively. Since this metric is diagonal, the non-zero components of the inverse metric $g^{\mu\nu}$ are

$$g^{xx} = g^{yy} = 1 \tag{2.143}$$

and

$$g^{rr} = 1 \quad , \quad g^{\phi\phi} = r^{-2} \tag{2.144}$$

respectively.

A reminder on notation (cf. the discussion leading to (2.13)): since μ, ν in $g_{\mu\nu}$ are coordinate indices, we should really have called $x^1 = r$, $x^2 = \phi$, say, and written $g_{11} = 1, g_{22} = r^2$, etc. However, writing g_{rr} etc. is more informative and useful since one then knows that this is the (rr)-component of the metric without having to remember if one called $r = x^1$ or $r = x^2$. In the following we will frequently use this kind of notation when dealing with a specific coordinate system, while we retain the index notation $g_{\mu\nu}$ etc. for general purposes.

Let us now look at the geodesic equations for this metric, first in the Cartesian coordinates (x, y) and then in the polar coordinates (r, ϕ).

1. Cartesian coordinates

   Since the metric in Cartesian coordinates is the constant Euclidean metric $g_{\mu\nu} = \delta_{\mu\nu}$, all the partial derivatives of the metric are zero, and therefore also all the Christoffel symbols are zero. The geodesic equations thus take the form

   $$\ddot{x} = \ddot{y} = 0 \ . \tag{2.145}$$

   These equations could also have been obtained as the Euler-Lagrange equations of the Lagrangian

   $$\mathcal{L} = \tfrac{1}{2}(\dot{x}^2 + \dot{y}^2) \ . \tag{2.146}$$

   The general solution is

   $$x(s) = as + b \quad , \quad y(s) = cs + d \ . \tag{2.147}$$

Combining these two, one finds the standard representation

$$y = kx + e \tag{2.148}$$

for a straight line in $\mathbb{R}^2$, with slope k and intercept e.

2. Polar coordinates

Now let us consider the same problem in polar coordinates. The crucial point here is that in these coordinates the geodesic equations will not simply be $\ddot{r} = \ddot{\phi} = 0$, but that there are additional terms arising

- either from the non-linear coordinate transformation between Cartesian and polar coordinates
- or equivalently from the fact that the coefficients of the metric are not constant in polar coordinates.

Taking the latter point of view, the Christoffel symbols of this metric are to be calculated from

$$\Gamma_{\mu\nu\lambda} = \tfrac{1}{2}(g_{\mu\nu,\lambda} + g_{\mu\lambda,\nu} - g_{\nu\lambda,\mu}) \ . \tag{2.149}$$

Since the only non-trivial derivative of the metric is $g_{\phi\phi,r} = 2r$, only Christoffel symbols with exactly two ϕ's and one r are non-zero,

$$\begin{aligned} \Gamma_{r\phi\phi} &= \tfrac{1}{2}(g_{r\phi,\phi} + g_{r\phi,\phi} - g_{\phi\phi,r}) = -r \\ \Gamma_{\phi\phi r} &= \Gamma_{\phi r\phi} = r \ . \end{aligned} \tag{2.150}$$

Thus, since the metric is diagonal, the non-zero $\Gamma^{\mu}_{\ \nu\lambda}$ are

$$\begin{aligned} \Gamma^{r}_{\ \phi\phi} &= g^{r\mu}\Gamma_{\mu\phi\phi} = g^{rr}\Gamma_{r\phi\phi} = -r \\ \Gamma^{\phi}_{\ r\phi} &= \Gamma^{\phi}_{\ \phi r} = g^{\phi\mu}\Gamma_{\mu r\phi} = g^{\phi\phi}\Gamma_{\phi r\phi} = \frac{1}{r} \ . \end{aligned} \tag{2.151}$$

Note that here it was even convenient to use a hybrid notation, as in $g^{r\mu}$, where r is a coordinate and μ is a coordinate index. Once again, it is very convenient to permit oneself to use such a mixed notation.

In any case, having assembled all the Christoffel symbols, we can now write down the geodesic equations (once again in the convenient hybrid notation). For r one has

$$\ddot{r} + \Gamma^{r}_{\ \mu\nu}\dot{x}^{\mu}\dot{x}^{\nu} = 0 \ , \tag{2.152}$$

which, since the only non-zero $\Gamma^{r}_{\ \mu\nu}$ is $\Gamma^{r}_{\ \phi\phi}$, reduces to

$$\ddot{r} - r\dot{\phi}^2 = 0 \ . \tag{2.153}$$

Likewise for ϕ one finds

$$\ddot{\phi} + \frac{2}{r}\dot{\phi}\dot{r} = 0 \ . \tag{2.154}$$

Here the factor of 2 arises because both $\Gamma^{\phi}_{r\phi}$ and $\Gamma^{\phi}_{\phi r} = \Gamma^{\phi}_{r\phi}$ contribute.

REMARKS:

(a) This equation is supposed to describe geodesics in $\mathbb{R}^2$, i.e. straight lines. This can be verified in general (but, in general, polar coordinates are of course not particularly well suited to describe straight lines). However, it is easy to find a special class of solutions to the above equations, namely curves with $\dot{\phi} = \ddot{r} = 0$. These correspond to paths of the form

$$(r(s), \phi(s)) = (s, \phi_0) \ , \tag{2.155}$$

which are a special case of straight lines, namely straight lines through the origin.

(b) The geodesic equations can of course also be derived as the Euler-Lagrange equations of the Lagrangian

$$\mathcal{L} = \tfrac{1}{2}(\dot{r}^2 + r^2\dot{\phi}^2) \ . \tag{2.156}$$

Indeed, one has

$$\begin{aligned} \frac{d}{ds}\frac{\partial \mathcal{L}}{\partial \dot{r}} - \frac{\partial \mathcal{L}}{\partial r} &= \ddot{r} - r\dot{\phi}^2 = 0 \\ \frac{d}{ds}\frac{\partial \mathcal{L}}{\partial \dot{\phi}} - \frac{\partial \mathcal{L}}{\partial \phi} &= r^2\ddot{\phi} + 2r\dot{r}\dot{\phi} = 0 \ , \end{aligned} \tag{2.157}$$

which are obviously identical to the equations derived above.

(c) You may have the impression that getting the geodesic equation in this way, rather than via calculation of the Christoffel symbols first, is much simpler. I agree wholeheartedly. Not only is the Lagrangian approach the method of choice to determine the geodesic equations. It is also frequently the most efficient method to determine the Christoffel symbols. This will be described in section 3.1.

(d) Another advantage of the Lagrangian formulation is, as in classical mechanics, that it makes it much easier to detect and exploit symmetries. Indeed, you may have already noticed that the above second-order equation for ϕ is overkill. Since the Lagrangian does not depend on ϕ (i.e. it is invariant under rotations), one has

$$\frac{d}{ds}\frac{\partial \mathcal{L}}{\partial \dot{\phi}} = 0 \ , \tag{2.158}$$

which means that $\partial\mathcal{L}/\partial\dot{\phi}$ is a constant of motion, the angular momentum L,

$$\frac{\partial \mathcal{L}}{\partial \dot{\phi}} = r^2\dot{\phi} = L \ . \tag{2.159}$$

This equation is a first integral of the second-order equation for ϕ. We will come back to this in somewhat more generality below.

The next simplest example to discuss would be the two-sphere with its standard metric $d\theta^2 + \sin^2\theta d\phi^2$. It will appear, in bits and pieces, in section 3.1 to illustrate the general remarks.

2.9 Example: Geodesics for Ultrastatic and Direct Product Metrics

As another example, let us consider the ultrastatic metrics introduced in (2.34) with coordinates $x^\mu = (t, x^k)$ and line-element

$$ds^2 = -dt^2 + \tilde{g}_{ik}(x)dx^i dx^k \ . \tag{2.160}$$

Because $g_{00} = -1$, $g_{0k} = 0$, and the $g_{ik} = \tilde{g}_{ik}$ are time-independent, all Christoffel symbols with at least one x^0- or t-index are zero,

$$\Gamma_{0\mu\nu} = \Gamma_{\mu 0\nu} = \Gamma_{\mu\nu 0} = 0 \ , \tag{2.161}$$

and the purely spatial components of the Christoffel symbols agree with those of the spatial metric,

$$\Gamma^i_{\ jk} = \tilde{\Gamma}^i_{\ jk} \ . \tag{2.162}$$

Therefore the geodesic equations read, for the t-component,

$$\ddot{t} = 0 \ , \tag{2.163}$$

and for the spatial components

$$\ddot{x}^i + \tilde{\Gamma}^i_{\ jk}\dot{x}^j\dot{x}^k = 0 \tag{2.164}$$

where the dot denotes a derivative with respect to the affine parameter τ. The first equation tells us that

$$\ddot{t} = 0 \quad \Leftrightarrow \quad t(\tau) = a\tau + t_0 \ . \tag{2.165}$$

Thus provided that $a \neq 0$ we can use t instead of τ to parametrise the paths (and in the present case t is then also an affine parameter, cf. the discussion in section 2.7 in connection with (2.139)), and then one can rewrite the spatial equations as equations for $x^i = x^i(t)$,

$$\frac{d^2x^i}{dt^2} + \tilde{\Gamma}^i_{\ jk}\frac{dx^j}{dt}\frac{dx^k}{dt} = 0 \ . \tag{2.166}$$

Therefore the solutions to the space-time geodesic equations have the form

$$x^\alpha(t) = (t, x^i(t)) \tag{2.167}$$

where $x^i(t)$ is an affinely parametrised geodesic for the metric $\tilde{g}_{ij}$. When $a = 0$, one cannot change variables from τ to t because $t = t_0$ is fixed. One is then necessarily dealing with spacelike geodesics in space-time and the solutions have the form

$$x^\alpha(\tau) = (t_0, x^i(\tau)) \tag{2.168}$$

where $x^i(\tau)$ is again an affinely parametrised geodesic for the metric $\tilde{g}_{ij}$.

These sorts of considerations evidently generalise to more general metrics of this direct product form,

$$ds^2 = g_{ab}(y)dy^a dy^b + g_{ik}(x)dx^i dx^k \quad , \tag{2.169}$$

with the conclusion that geodesics in such space-times have the form $(y^a(\tau), x^i(\tau))$ with $y^a(\tau)$ and $x^i(\tau)$ individually solutions of the geodesic equations for the metric $g_{ab}(y)$ respectively $g_{ik}(x)$.

3 Geodesics and Motion in a Gravitational Field

3.1 Consequences and Uses of the Euler-Lagrange Equations

Recall from above that the geodesic equation for a metric $g_{\mu\nu}$ can be derived from the Lagrangian $\mathcal{L} = (1/2)g_{\mu\nu}\dot{x}^\mu\dot{x}^\nu$

$$\frac{d}{d\tau}\frac{\partial\mathcal{L}}{\partial\dot{x}^\mu} - \frac{\partial\mathcal{L}}{\partial x^\mu} = 0 \ . \tag{3.1}$$

This has several immediate consequences which are useful for the determination of Christoffel symbols and geodesics in practice.

1. Conserved charges / first integrals of the geodesic equation

   Just as in classical mechanics, a coordinate the Lagrangian does not depend on explicitly (a cyclic coordinate) leads to a conserved quantity, associated with the translation invariance of the system in that direction. In the present context this means that if, say, $\partial\mathcal{L}/\partial x^1 = 0$ (this means that the coeffcients of the metric do not depend on x^1), then the corresponding momentum

   $$p_1 = \partial\mathcal{L}/\partial\dot{x}^1 = g_{1\nu}\dot{x}^\nu \tag{3.2}$$

   is conserved along the geodesic.

   Remarks:

   (a) One might perhaps have wanted to argue that the definition (and interpretation) of conserved momenta should be based on the physical Lagrangian (2.90)

   $$\mathcal{L}_0^\lambda = -m\sqrt{-g_{\alpha\beta}\frac{dx^\alpha}{d\lambda}\frac{dx^\beta}{d\lambda}} \tag{3.3}$$

   with action $S = -m\int d\tau$, but this makes no difference since the two momenta are essentially equal: one has

   $$\frac{\partial\mathcal{L}_0^\lambda}{\partial(dx^1/d\lambda)} = mp_1 \tag{3.4}$$

   with p_1 as defined in (3.2), so that this just supplies us with the additional information that the momenta obtained from the Lagrangian $\mathcal{L}$ should (for a massive particle) be interpreted as momenta per unit mass. This discrepancy could have been avoided by working with the Lagrangian $m\mathcal{L}$ (alternatively: fixing the gauge $e(\lambda) = m^{-1}$ in section 2.5, see (2.114)), but unless or until one starts coupling the particle to fields other than the gravitational field it is unnecessary (and a nuisance) to carry m around all the time.

(b) For example, on the two-sphere the Lagrangian reads

$$\mathcal{L} = \tfrac{1}{2}(\dot{\theta}^2 + \sin^2\theta\dot{\phi}^2) \ . \tag{3.5}$$

The angle ϕ is a cyclic variable and the angular momentum (actually angular momentum per unit mass for a massive particle)

$$p_\phi = \frac{\partial \mathcal{L}}{\partial \dot{\phi}} = \sin^2\theta\dot{\phi} \tag{3.6}$$

is a conserved quantity. This generalises to conservation of angular momentum for a particle moving in an arbitrary spherically symmetric gravitational field.

(c) Likewise, if the metric is independent of the time coordinate $x^0 = t$, the corresponding conserved quantity

$$p_0 = g_{0\nu}\dot{x}^\nu \equiv -E \tag{3.7}$$

has the interpretation as minus the energy (per unit mass) of the particle, "minus" because, with our sign conventions, $p_0 = -E$ in special relativity. We will discuss the relation between this notion of energy and the notion of energy familiar from special relativity (this requires an asymptotically Minkowski-like metric) in more detail in section 25.3.

(d) We will discuss in more detail in section 3.2 (and then again in sections 9 and 10) how to detect and describe symmetries and conserved charges in coordinate systems in which the symmetries are not as manifest (via cyclic variables) as above.

2. Reading off (some) geodesics directly from the metric

Another immediate consequence is the following: consider a space or space-time with coordinates $\{y, x^\mu\}$ and a metric of the form

$$ds^2 = \pm dy^2 + g_{\mu\nu}(x, y)dx^\mu dx^\nu \ . \tag{3.8}$$

Then the coordinate lines of y are geodesics.

The quickest way to see this is (as usual) from the Lagrangian point of view. Indeed, since the Lagrangian is

$$\mathcal{L} = \tfrac{1}{2}(\pm\dot{y}^2 + g_{\mu\nu}(y, x)\dot{x}^\mu\dot{x}^\nu) \ , \tag{3.9}$$

the Euler-Lagrange equations are of the form

$$\begin{aligned} &\pm\ddot{y} - \tfrac{1}{2}g_{\mu\nu,y}\,\dot{x}^\mu\dot{x}^\nu = 0 \\ &\ddot{x}^\mu + \text{terms proportional to } \ \dot{x} = 0 \ . \end{aligned} \tag{3.10}$$

Therefore $\dot{x}^\mu = 0$, $\ddot{y} = 0$ is a solution of the geodesic equation, and it describes motion along the coordinate lines of y.

Alternatively, this special form of the metric implies that any Christoffel symbol with at least two y-indices is zero, and the conclusion then follows in the same way as above.

REMARKS:

(a) In the case of the two-sphere, with its metric $ds^2 = d\theta^2 + \sin^2\theta d\phi^2$, this translates into the familiar statement that the great circles, the coordinate lines of $y = \theta$, are geodesics.

(b) The result is also valid when y is a timelike coordinate. For example, consider a space-time with coordinates (t, x^i) and metric (2.36)

$$ds^2 = -dt^2 + a(t)^2 \tilde{g}_{ij}(x) dx^i dx^j \quad . \tag{3.11}$$

In such a cosmological space-time, there is, according to the above result, a privileged class of freely falling (i.e. geodesic) observers, namely those that stay at fixed values of the spatial coordinates x^i. For such *comoving observers*, the coordinate-time t coincides with their proper time τ.

(c) In general, these preferred geodesics are orthogonal to the hypersurfaces of constant y, and coordinates in which the metric (locally) takes such a form in the neighbourhood of some timelike or spacelike hypersurface are occasionally called *Gaussian normal coordinates*.

3. Using the Euler-Lagrange equations to determine the Christoffel symbols

Finally, as mentioned and observed above, the Euler-Lagrange form of the geodesic equations frequently provides the most direct way of calculating Christoffel symbols - by comparing the Euler-Lagrange equations with the expected form of the geodesic equation in terms of Christoffel symbols. More precisely, by rewriting the Euler-Lagrange equations (2.94)

$$\frac{d}{d\lambda}\left(g_{\gamma\beta}\frac{dx^\beta}{d\lambda}\right) = \tfrac{1}{2} g_{\alpha\beta,\gamma}\frac{dx^\alpha}{d\lambda}\frac{dx^\beta}{d\lambda} \quad . \tag{3.12}$$

in the form

$$\ddot{x}^\gamma + \text{terms proportional to } \dot{x}\dot{x} = 0 \quad , \tag{3.13}$$

and comparing with the geodesic equation

$$\frac{d^2x^\gamma}{d\lambda^2} + \Gamma^\gamma_{\alpha\beta}\frac{dx^\alpha}{d\lambda}\frac{dx^\beta}{d\lambda} = 0 \quad , \tag{3.14}$$

one can read off the $\Gamma^\gamma_{\alpha\beta}$.

REMARKS:

(a) Careful - in this and similar calculations beware of factors of 2:

$$\Gamma^{\mu}_{\ \nu\lambda}\dot{x}^{\nu}\dot{x}^{\lambda} = \Gamma^{\mu}_{\ 11}(\dot{x}^1)^2 + 2\Gamma^{\mu}_{\ 12}\dot{x}^1\dot{x}^2 + \ldots \tag{3.15}$$

(b) For example, once again in the case of the two-sphere, for the θ-equation one has

$$\frac{d}{d\tau}\frac{\partial \mathcal{L}}{\partial \dot{\theta}} = \ddot{\theta} \quad , \quad \frac{\partial \mathcal{L}}{\partial \theta} = \sin\theta\cos\theta\dot{\phi}^2 \ . \tag{3.16}$$

Comparing the resulting Euler-Lagrange equation

$$\ddot{\theta} - \sin\theta\cos\theta\dot{\phi}^2 = 0 \tag{3.17}$$

with the geodesic equation

$$\ddot{\theta} + \Gamma^{\theta}_{\ \theta\theta}\dot{\theta}^2 + 2\Gamma^{\theta}_{\ \theta\phi}\dot{\theta}\dot{\phi} + \Gamma^{\theta}_{\ \phi\phi}\dot{\phi}^2 = 0 \ , \tag{3.18}$$

one can immediately read off that

$$\Gamma^{\theta}_{\ \phi\phi} = -\sin\theta\cos\theta \quad , \quad \Gamma^{\theta}_{\ \theta\theta} = \Gamma^{\theta}_{\ \theta\phi} = 0 \ . \tag{3.19}$$

Likewise, from

$$\frac{d}{d\tau}\frac{\partial \mathcal{L}}{\partial \dot{\phi}} - \frac{\partial \mathcal{L}}{\partial \phi} = 0 \quad \Leftrightarrow \quad \sin^2\theta(\ddot{\phi} + 2\cot\theta\dot{\theta}\dot{\phi}) = 0 \tag{3.20}$$

one deduces that

$$\Gamma^{\phi}_{\ \theta\phi} = \Gamma^{\phi}_{\ \phi\theta} = \cot\theta \quad , \quad \Gamma^{\phi}_{\ \theta\theta} = \Gamma^{\phi}_{\ \phi\phi} = 0 \ . \tag{3.21}$$

(c) As another example, which will turn out to be of considerable importance later on, consider a space-time metric of the form

$$ds^2 = -A(r)dt^2 + B(r)dr^2 + r^2 d\Omega^2 \ . \tag{3.22}$$

As will be discussed in section 24.2, this is the general form of a static spherically symmetric metric, and as such will provide us with the starting point for describing the gravitational field of a star. The corresponding Lagrangian is

$$\mathcal{L} = \tfrac{1}{2}\left(-A(r)\dot{t}^2 + B(r)\dot{r}^2 + r^2(\dot{\theta}^2 + \sin^2\theta\dot{\phi}^2)\right) \ , \tag{3.23}$$

and therefore the Euler-Lagrange equations for t (a cyclic variable) are

$$0 = \frac{d}{d\tau}(-A(r)\dot{t}) = -A(r)\left(\ddot{t} + \frac{A'(r)}{A(r)}\dot{r}\dot{t}\right) \tag{3.24}$$

(a prime denoting an r-derivative), from which one can immediately read off

$$\Gamma^{t}_{\ rt} = \Gamma^{t}_{\ tr} = \frac{A'}{2A} \quad , \quad \Gamma^{t}_{\ \mu\nu} = 0 \quad \text{otherwise} \ . \tag{3.25}$$

Likewise, the equation for r takes the form

$$\ddot{r} + \frac{B'}{2B}\dot{r}^2 + \frac{A'}{2B}\dot{t}^2 + \cdots = 0 \ , \tag{3.26}$$

and from this one can read off that

$$\Gamma^r{}_{rr} = \frac{B'}{2B} \quad , \quad \Gamma^r{}_{tt} = \frac{A'}{2B} \quad , \quad \ldots \tag{3.27}$$

As we will need them anyway in section 24.3, it is a good exercise to determine all the Christoffel symbols in this way.

3.2 Conserved Charges and (a first encounter with) Killing Vectors

In the previous section we have seen that cyclic coordinates, i.e. coordinates the metric does not depend on, lead to conserved charges, as in (3.2). As nice and useful as this may be (and it is nice and useful), it is obvioulsy somewhat unsatisfactory because it is an explicitly coordinate-dependent statement: the metric may well be independent of one coordinate in some coordinate system, but if one now performs a coordinate transformation which depends on that coordinate, then in the new coordinate system the metric will typically depend on all the new coordinates. Nevertheless,

- the statement that a metric has a certain symmetry (a translational symmetry in the first coordinate system) should be coordinate-independent, and
- thus there should be a corresponding first integral of the geodesic equation in any coordinate system.

To see how this works, let us reconsider the situation discussed in the previous section, namely a metric which in some coordinate system, we will now call it $\{y^\mu\}$, has components $g_{\mu\nu}$ which are independent of y^1, say. Translation invariance of the geodesic Lagrangian is the statement that the Lagrangian is invariant under the infinitesimal variation $\delta y^1 = \epsilon$, $\delta y^\mu = 0$ otherwise, and via Noether's theorem this leads to a conserved charge $g_{1\mu}\dot{y}^\mu$, as in (3.2).

Now we ask ourselves what this statement corresponds to in another coordinate system. Note that in the y-coordinates, invariance is the statement that the metric is invariant under the (infinitesimal) coordinate transformation $y^1 \to y^1 + \epsilon$ or $\delta y^1 = \epsilon$, $\delta y^\mu = 0$ otherwise,

$$\delta g_{\mu\nu} \equiv \epsilon \partial_{y^1} g_{\mu\nu} = 0 \ . \tag{3.28}$$

It is then clear that in another coordinate system, infinitesimal y^1-translations must also correspond to some infinitesimal coordinate transformation (but not necessarily just a translation),

$$\delta x^\alpha = \epsilon V^\alpha(x) \ . \tag{3.29}$$

In particular, if (as in the above example) in y-coordinates V^μ has the components $V^1 = 1$, $V^\mu = 0$ otherwise, then in any other coordinate system one has

$$\delta x^\alpha = (\partial x^\alpha / \partial y^\mu)\delta y^\mu = \epsilon(\partial x^\alpha / \partial y^1) \tag{3.30}$$

so that

$$V^\alpha = J_1^\alpha \tag{3.31}$$

is just the corresponding column of the Jacobi matrix.

In order to determine how to characterise the translational symmetry (3.28) of the metric in an arbitrary coordinate system, we will now proceed in two (as it turns out ultimately equivalent) ways.

1. We can investigate directly, under which conditions on the V^α the transformation (3.29) leads to an invariance of the Lagrangian (2.91). Using

   $$\delta\dot{x}^\alpha = \epsilon\dot{x}^\gamma\partial_\gamma V^\alpha \tag{3.32}$$

   one straightforwardly finds

   $$\delta\left(g_{\alpha\beta}\dot{x}^\alpha\dot{x}^\beta\right) = \epsilon\left(\delta_V g_{\alpha\beta}\right)\dot{x}^\alpha\dot{x}^\beta \tag{3.33}$$

   where

   $$\delta_V g_{\alpha\beta} = V^\gamma\partial_\gamma g_{\alpha\beta} + (\partial_\alpha V^\gamma)g_{\gamma\beta} + (\partial_\beta V^\gamma)g_{\alpha\gamma} \tag{3.34}$$

   Thus the condition for the infinitesimal transformation (3.29) to leave the Lagrangian invariant is

   $$\delta_V g_{\alpha\beta} = 0 \ . \tag{3.35}$$

   Noether's theorem then leads to the corresponding conserved charge

   $$Q_V = p_\alpha V^\alpha = g_{\alpha\beta}V^\alpha\dot{x}^\beta \ . \tag{3.36}$$

   Note that for constant components V^α, (3.35) is simply the statement that the metric is constant in the direction V, $V^\gamma\partial_\gamma g_{\alpha\beta} = 0$.

2. Alternatively, we can determine the variation $\delta_V g_{\alpha\beta}$ of the components $g_{\alpha\beta}$ of the metric in x-coordinates from the variation (3.28) of the components $g_{\mu\nu}$ of the metric in y-coordinates by demanding that under a coordinate transformation the variation (3.28) of the metric transforms like the metric. Since we know how the metric transforms (2.5), and we also know how ∂_{y^1} transforms,

   $$g_{\mu\nu} = J_\mu^\alpha J_\nu^\beta g_{\alpha\beta} \quad , \quad \partial_{y^1} = (\partial_{y^1}x^\alpha)\partial_\alpha \equiv J_1^\alpha\partial_\alpha \equiv V^\alpha\partial_\alpha \tag{3.37}$$

   we find the condition

   $$\begin{aligned} &\partial_{y^1}g_{\mu\nu} = J_1^\gamma\partial_\gamma(J_\mu^\alpha J_\nu^\beta g_{\alpha\beta}) \stackrel{!}{=} J_\mu^\alpha J_\nu^\beta\delta_V g_{\alpha\beta} \\ \Leftrightarrow\quad &\delta_V g_{\alpha\beta} = J_\alpha^\mu J_\beta^\nu J_1^\gamma\partial_\gamma(J_\mu^\delta J_\nu^\epsilon g_{\delta\epsilon}) \ . \end{aligned} \tag{3.38}$$

In order to disentangle this, one can make use of identities such as

$$J_1^\gamma \partial_\gamma J_\mu^\delta = \partial_1 J_\mu^\delta = \partial_\mu J_1^\delta = \partial_\mu V^\delta = J_\mu^\alpha \partial_\alpha V^\delta \tag{3.39}$$

to show that this expression for $\delta_V g_{\alpha\beta}$ is identical to that given in (3.34).

All of this may seem a bit ham-handed at this point, and indeed it is. However, we will see later how these results can be written and understood in a much more pleasing and covariant way. In particular, we will see in section 5.5 how to write (3.34) in a way that makes it completely manifest that it transforms like the metric under coordinate transformations. Moreover, we will discover in section 9 that (3.34) is a special case of the *Lie derivative* of a tensor field along a vector field V, denoted by L_V. Continuous symmetries of a metric correspond to vector fields along which the Lie derivative of the metric vanishes. Such vectors are known as *Killing vectors*, and are thus vectors V^α satisfying the *Killing equation* (3.35),

$$L_V g_{\alpha\beta} \equiv \delta_V g_{\alpha\beta} = 0 \ . \tag{3.40}$$

3.3 Newtonian Limit of the Geodesic Equation

We saw that the 10 components of the metric $g_{\mu\nu}$ appear to play the role of potentials for the gravitational force. In order to substantiate this, and to show that in an appropriate limit this setting is able to reproduce the Newtonian results, we now want to find the relation of these potentials to the Newtonian potential, and the relation between the geodesic equation and the Newtonian equation of motion for a particle moving in a gravitational field.

First let us determine the conditions under which we might expect the general relativistic equation of motion (namely the non-linear coupled set of partial differential geodesic equations) to reduce to the linear equation of motion

$$\frac{d^2}{dt^2}\vec{x} = -\vec{\nabla}\phi \tag{3.41}$$

of Newtonian mechanics, with ϕ the gravitational potential, e.g.

$$\phi = -\frac{G_N M}{r} \ . \tag{3.42}$$

Thus we are trying to characterise the circumstances in which we know and can trust the validity of Newton's equations, such as those provided e.g. by the gravitational field of the earth or the sun, the gravitational fields in which Newton's laws were discovered and tested. Two of these are fairly obvious:

1. Weak Fields: our first plausible assumption is that the gravitational field is in a suitable sense sufficiently weak. We will need to make more precise by what we mean by this, and we will come back to this below.

2. Slow Motion: our second, equally reasonable and plausible, assumption is that the test particle moves at speeds at which we can neglect special relativistic effects, so "slow" should be taken to mean that its velocity is small compared to the velocity of light.

Interestingly, it turns out that one more condition is required. Note that the gravitational fields we have access to are not only quite weak but also only very slowly varying in time, and we will add this condition,

3. Stationary Fields: we will assume that the gravitational field does not vary significantly in time (over the time scale probed by our test particle).

The very fact that we have to add this condition in order to find Newton's equations (as will be borne out by the calculations below) is interesting in its own right, because it also shows that general relativity predicts phenomena deviating from the Newtonian picture even for weak fields, provided that they vary sufficiently rapidly (e.g. quickly oscillating fields), and one such phenomenon is that of gravitational waves (see section 23).

Now, having formulated in words the conditions that we wish to impose, we need to translate these conditions into equations that we can then use in conjunction with the geodesic equation.

1. In order to define a notion of weak fields, we need to keep in mind that this is not a coordinate-independent statement since we can simulate arbitrarily strong gravitational fields even in Minkowski space by going to suitably accelerated coordinates, and therefore a "weak field" condition will be a condition not only on the metric but also on the choice of coordinates. Thus we assume that we can choose coordinates $\{x^\mu\} = \{t, x^i\}$ in such a way that *in these coordinates* the metric differs from the standard constant Minkowski metric $\eta_{\mu\nu}$ only by a small amount,
$$g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu} \tag{3.43}$$
where we will implement "by a small amount" in the calculations below by dropping all terms that are at least quadratic in $h_{\mu\nu}$ (and/or its derivatives).

2. The second condition is obviously (with the coordinates chosen above) $dx^i/dt \ll 1$ or, expressed in terms of proper time,
$$\frac{dx^i}{d\tau} \ll \frac{dt}{d\tau} \ . \tag{3.44}$$

3. The third condition of stationarity we implement simply by considering time-independent fields,
$$g_{\alpha\beta,0} = 0 \quad \Rightarrow \quad h_{\alpha\beta,0} = 0 \tag{3.45}$$

(for a discussion and explanation of the difference betwen the term "stationary" used here and the term "static" used e.g. to describe the metric (3.22), see section 16.4 - it is not crucial here).

Before embarking on the calculation, we note that for the inverse $g^{\alpha\beta}$ of the metric $g_{\alpha\beta} = \eta_{\alpha\beta} + h_{\alpha\beta}$, with $h_{\alpha\beta}$ "small", one evidently has

$$g_{\alpha\beta} = \eta_{\alpha\beta} + h_{\alpha\beta} \quad \Rightarrow \quad g^{\alpha\beta} = \eta^{\alpha\beta} + \mathcal{O}(h) \ , \tag{3.46}$$

where $\eta^{\alpha\beta}$ is just the inverse Minkowski metric. The explicit expression of the order h term (which we will not need) is given in (3.65) below.

Now we look at the geodesic equation

$$\ddot{x}^\mu + \Gamma^\mu_{\nu\lambda}\dot{x}^\nu\dot{x}^\lambda = 0 \ . \tag{3.47}$$

From the decomposition $g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu}$ we see that $\Gamma^\mu_{\nu\lambda}$ is at least linear in $h_{\mu\nu}$, and by the weak field condition (condition 1) we will only retain the terms linear in $h_{\mu\nu}$. Then the condition of slow motion (condition 2) implies that among the quadratic terms $\dot{x}^\nu\dot{x}^\lambda$ we need to only retain the leading term, namely $\dot{t}\dot{t}$. Thus the geodesic equation can be approximated by

$$\ddot{x}^\mu + \Gamma^\mu_{00}\dot{t}^2 = 0 \ . \tag{3.48}$$

Thus we need to determine

$$\Gamma^\mu_{00} = \tfrac{1}{2}g^{\mu\nu}(g_{\nu 0,0} + g_{\nu 0,0} - g_{00,\nu}) \ . \tag{3.49}$$

Stationarity (condition 3) tells us that the first two terms are zero, and

$$\Gamma^\mu_{00} = -\tfrac{1}{2}g^{\mu\nu}\partial_\nu g_{00} = -\tfrac{1}{2}g^{\mu i}\partial_i g_{00} \ . \tag{3.50}$$

Now $\partial_i g_{00} = \partial_i h_{00}$ is already of order h. Therefore, by the weak field condition, working to linear order in h we can can replace the inverse metric $g^{\mu i}$ by the inverse Minkowski metric $\eta^{\mu i}$, so that in this approximation

$$\Gamma^\mu_{00} = -\tfrac{1}{2}\eta^{\mu i}\partial_i h_{00} \ . \tag{3.51}$$

Thus the relevant Christoffel symbols are

$$\Gamma^0_{00} = 0 \ , \qquad \Gamma^i_{00} = -\tfrac{1}{2}\partial^i h_{00} \ , \tag{3.52}$$

and the geodesic equation splits into

$$\begin{aligned} \ddot{t} &= 0 \\ \ddot{x}^i &= \tfrac{1}{2}\partial_i h_{00}\dot{t}^2 \ . \end{aligned} \tag{3.53}$$

The first of these just says that $\dot{t}$ is constant, or that t is also an affine parameter,

$$t(\tau) = a\tau + b \ . \tag{3.54}$$

In other words, in the Newtonian limit there is essentially (up to a choice of scale/units) no difference between coordinate time and proper time. We can use this in the second equation to convert the τ-derivatives into derivatives with respect to the coordinate time t,

$$\ddot{t} = 0 \quad \Rightarrow \quad \frac{1}{\dot{t}^2}\frac{d^2}{d\tau^2} = \frac{1}{\dot{t}}\frac{d}{d\tau}\frac{1}{\dot{t}}\frac{d}{d\tau} = \frac{d^2}{dt^2} \ . \tag{3.55}$$

Hence we obtain

$$\frac{d^2x^i}{dt^2} = \tfrac{1}{2}h_{00,i} \tag{3.56}$$

(the spatial index i in this expression is raised or lowered with the Kronecker symbol, $\eta^{ik} = \delta^{ik}$). Comparing this with the Newtonian equation (3.41),

$$\frac{d^2x^i}{dt^2} = -\phi_{,i} \tag{3.57}$$

leads us (with the constant of integration absorbed into an arbitrary constant term in the gravitational potential) to the key identification

$$h_{00} = -2\phi \tag{3.58}$$

between the Newtonian gravitational potential and the (00)-component of the deviation of the space-time metric from the Minkowski metric. By relating this back to $g_{\alpha\beta}$,

$$g_{00} = -(1 + 2\phi) \ . \tag{3.59}$$

we find the sought-for relation between the Newtonian potential and the space-time metric. Thus Newtonian gravity can be captured or described by a space-time metric of the form

$$ds^2 = -(1 + 2\phi(\vec{x}))dt^2 + d\vec{x}^2 \ . \tag{3.60}$$

For a radial gravitational field, with $\phi = \phi(r)$, it is also natural to write this in terms of spatial spherical coordinates as

$$ds^2 = -(1 + 2\phi(r))dt^2 + dr^2 + r^2 d\Omega_2^2 \ . \tag{3.61}$$

Remarks:

1. With the speed of light not set equal to $c = 1$, the dimensionally correct form of this identification is (recall that kinetic and potential energy have the same dimension so that the dimension of ϕ, the energy per unit mass, is that of a velocity-squared; thus ϕ/c^2 is dimensionless)

$$g_{00} = -(1 + 2\phi/c^2) \ . \tag{3.62}$$

2. For the gravitational field of isolated systems, it makes sense to choose the integration constant in such a way that the potential goes to zero at infinity, and this choice also ensures that the metric approaches the flat Minkowski metric at infinity.

3. Restoring the appropriate units, in particular the above factor of c^2, one finds that the dimensionless factor $\phi/c^2 \sim 10^{-9}$ on the surface of the earth, 10^{-6} on the surface of the sun (see section 24.4 for some more details), so that the distortion in the space-time geometry produced by gravitation is in general quite small (justifying our approximations).

4. Just for the record, here is the explicit expression for the inverse $g^{\alpha\beta}$ of a metric of the form $g_{\alpha\beta} = \eta_{\alpha\beta} + h_{\alpha\beta}$, with $h_{\alpha\beta}$ "small". In analogy with the series expansion

$$(1+x)^{-1} = 1 - x + x^2 \mp \dots \tag{3.63}$$

for a real number x with $|x| < 1$, the exact result for $g^{\alpha\beta}$ can be written as an infinite power series in $h_{\alpha\beta}$. We will not need the exact result here, but only the result to linear order in $h_{\alpha\beta}$.

In linear algebra notation, if I is an invertible matrix and A is sufficiently small so that $I + A$ is still invertible, one has (as a matrix generalisation of the power series (3.63) for $(1+x)^{-1}$)

$$(I+A)^{-1} = I^{-1} - I^{-1}AI^{-1} + \mathcal{O}(A^2) \ . \tag{3.64}$$

In the case at hand (with $I \to \eta, A \to h$), this is

$$g_{\alpha\beta} = \eta_{\alpha\beta} + h_{\alpha\beta} \quad \Rightarrow \quad g^{\alpha\beta} = \eta^{\alpha\beta} - h^{\alpha\beta} + \mathcal{O}(h^2) \ , \tag{3.65}$$

where

$$h^{\alpha\beta} = \eta^{\alpha\gamma}\eta^{\beta\delta}h_{\gamma\delta} \ . \tag{3.66}$$

Indeed it is now easily verified that this satisfies

$$g^{\alpha\beta}g_{\beta\gamma} = \delta^\alpha_\gamma + \mathcal{O}(h^2) \ . \tag{3.67}$$

5. Within this Newtonian approximation, we cannot distinguish the above result $g_{00} = -(1+2\phi)$ from $g_{00} = -(1+\phi)^2$, say (or a host of other possibilities).

6. Likewise, in this approximation it does not make sense to inquire about the other subleading components of the metric. As we have seen, a slowly moving particle in a weak static gravitational field is not sensitive to them, and hence can also not be used to probe or determine these components.

7. In this approximation, the modification of the space-time geometry can equivalently be described as, or attributed to, a space-time dependent speed of light in Minkowski space, along the lines of

$$ds^2 = -c(x)^2dt^2 + d\vec{x}^2 \ , \tag{3.68}$$

with

$$c^2(x) = (1 + 2\phi(x)/c^2)c^2 \ . \tag{3.69}$$

Einstein realised fairly early on (1911) in his search for a relativistic theory of gravity that this would have to be part of the story. However, this interpretation is neither useful nor tenable when considering gravitational fields beyond the static Newtonian approximation (which requires one to go beyond a theory with a single scalar potential).

8. Later on, we will determine the exact solution of the Einstein equations (the field equations for the gravitational field, i.e. for the metric) for the gravitational field outside a spherically symmetric mass distribution with mass M (the Schwarzschild metric). The metric turns out to have the simple form (24.31)

$$ds^2 = -\left(1 - \frac{2G_N M}{c^2 r}\right) c^2 dt^2 + \left(1 - \frac{2G_N M}{c^2 r}\right)^{-1} dr^2 + r^2 d\Omega^2 \quad . \tag{3.70}$$

From this expression one can read off that the leading correction to the flat metric indeed arises from the 00-component of the metric,

$$\begin{aligned} ds^2 &\approx -c^2 dt^2 + dr^2 + r^2 d\Omega^2 + \frac{2G_N M}{r} dt^2 + \dots \\ &= \eta_{\alpha\beta} dx^\alpha dx^\beta + \frac{2G_N M}{r c^2} (dx^0)^2 + \dots \quad . \end{aligned} \tag{3.71}$$

This is indeed precisely of the above Newtonian form, with the standard Newtonian potential

$$\phi(r) = -\frac{G_N M}{r} \quad . \tag{3.72}$$

One can then also determine the subleading (known as "post-Newtonian") corrections to the general relativistic gravitational field, which are evidently suppressed by additional inverse powers of c^2.

9. The key relation (3.58) can also be obtained at the level of the action. Starting with the action S_0 (the integral of the proper time), and using the time-coordinate t as the parameter, using the same approximations as above one finds that the action can be written as (keeping c explicit for a change, so that $x^0 = ct$)

$$\begin{aligned} S_0[x] &= -mc \int dt \sqrt{-g_{\mu\nu}(dx^\mu/dt)(dx^\nu/dt)} \\ &= -mc \int dt \sqrt{-\eta_{\mu\nu}(dx^\mu/dt)(dx^\nu/dt) - h_{\mu\nu}(dx^\mu/dt)(dx^\nu/dt)} \\ &= -mc \int dt \sqrt{c^2 - \eta_{ik}(dx^i/dt)(dx^k/dt) - h_{00} c^2} \\ &= -mc^2 \int dt \sqrt{1 - \vec{v}^2/c^2 - h_{00}} \quad . \end{aligned} \tag{3.73}$$

Expanding the square root and dropping the first (irrelevant) term, one finds that in this limit the action reduces to

$$S_0[x] \quad \to \quad \int dt \, \left(\frac{m}{2}\vec{v}^2 + \frac{mc^2}{2} h_{00}\right) \tag{3.74}$$

which is precisely the Newtonian action for a particle in a gravitational potential ϕ,

$$S_N[x] = \int dt \ \left(\frac{m}{2}\vec{v}^2 - m\phi\right) \tag{3.75}$$

provided that one makes the identification (3.58),

$$h_{00} = -2\phi/c^2 \ . \tag{3.76}$$

In this compact (but slightly dubious) derivation of this relation, the significance of the stationarity condition is not manifest: it enters through the condition of the equivalence of the 4-dimensional and 3-dimensional variational principles (with respect to the fields $x^\mu(\tau)$ and $x^k(t)$ respectively), guaranteed by the affine relation between t and τ implied by requiring in addition stationarity.

3.4 Rindler Coordinates Revisited

In section 1.3 we had discussed the Minkowski metric in Rindler coordinates, i.e. in coordinates adapted to a constantly accelerating observer. For an observer accelerating in the x^1-direction, the metric took the form (1.74),

$$ds^2 = -\rho^2 d\eta^2 + d\rho^2 + d\vec{y}^2 \ , \tag{3.77}$$

with $\vec{y} = (x^2, x^3)$ denoting the transverse spectator coordinates (which will again be suppressed in the following).

What is the relation, if any, between this metric and the metric describing a weak gravitational field, as derived above (after all, small accelerations should mimic weak gravitational fields)? At first sight, the only thing they appear to have in common is that the departure from what would be the Minkowski-metric in these coordinates is encoded in the time-time component of the metric, ρ^2 in one case, $(1 + 2\phi)$ in the other, but apart from that ρ^2 and $(1 + 2\phi)$ look quite different. This difference is, however, again a coordinate artefact and the Rindler metric can be made to look like the weak-field metric with the help of a suitable further redefinition of the coordinates.

For starters, it will be convenient, for this purpose and for a generalisation which we will discuss below, to introduce the acceleration a explicitly into the coordinates by redefining the coordinate transformation (1.73) to (I will now also call the Minkowski coordinates $\xi^0 = t$ and $\xi^1 = x$)

$$t(\eta, \rho) = (\rho/a)\sinh a\eta \qquad x(\eta, \rho) = (\rho/a)\cosh a\eta \tag{3.78}$$

(so this differs by $\rho \to \rho/a, \eta \to a\eta$ from the transformation given in (1.73)). Thus, now it is the observer at $\rho = 1$ who has acceleration a and whose proper time is $\tau = \eta$. The Rindler metric now has the form

$$ds^2 = -dt^2 + dx^2 = -\rho^2 d\eta^2 + a^{-2} d\rho^2 \ . \tag{3.79}$$

Now the transformation $\rho = 1 + a\hat{x}$ (reminding us that we are talking about acceleration in the $x = x^1$ direction), leads to

$$ds^2 = -(1 + a\hat{x})^2 d\eta^2 + d\hat{x}^2 \tag{3.80}$$

and it is the observer at $\hat{x} = 0$ who has proper time $\tau = \eta$ and constant acceleration a. If one now assumes that the acceleration a is sufficiently small, one can approximate

$$(1 + a\hat{x})^2 \approx 1 + 2a\hat{x} \equiv 1 + 2\phi(\hat{x}) \quad , \tag{3.81}$$

and in $\phi = a\hat{x}$ we recognise precisely the Newtonian potential for a constant force a in the $\hat{x}$-direction. Thus the Rindler and weak field form of the metric agree in this case.

Remarks:

1. Remarkably, this same form of the metric remains valid for an arbitrary time-dependent acceleration $a = a(\tau)$, and thus is capable of reproducing the weak field form of the metric for general potentials. To see this, consider the worldline $(t(\tau), x(\tau))$ with general 4-velocity (actually 2-velocity in this case)

$$u^0 = \dot{t}(\tau) = \cosh v(\tau) \quad , \quad u^1 = \dot{x}(\tau) = \sinh v(\tau) \quad , \tag{3.82}$$

which satisfies

$$(u^0)^2 - (u^1)^2 = 1 \quad , \tag{3.83}$$

as it should, and has the time-dependent acceleration $a(\tau) = \dot{v}(\tau)$,

$$(\dot{u}^1)^2 - (\dot{u}^0)^2 = \ddot{x}^2 - \ddot{t}^2 = \dot{v}(\tau)^2 \equiv a(\tau)^2 \quad . \tag{3.84}$$

We can pass to adapted coordinates $(\eta, \hat{x})$, as above, by setting

$$(t(\eta, \hat{x}), x(\eta, \hat{x})) = (t(\eta) + \hat{x} \sinh v(\eta), x(\eta) + \hat{x} \cosh v(\eta)) \quad , \tag{3.85}$$

leading to the metric

$$ds^2 = -dt^2 + dx^2 = -(1 + a(\eta)\hat{x})^2 d\eta^2 + d\hat{x}^2 \quad . \tag{3.86}$$

This is indeed the direct generalisation of (3.80) to arbitrary accelerations, the original worldline manifestly corresponding to the observer at fixed $\hat{x} = 0$, with $\eta = \tau$, and the same remarks regarding the weak field limit / small accelerations apply.

2. Another useful alternative coordinate transformation for constant a is $\rho = \exp a\xi$, leading to

$$ds^2 = \mathrm{e}^{2a\xi}(-d\eta^2 + d\xi^2) \quad . \tag{3.87}$$

Note that in these coordinates the metric is *conformally flat*, i.e. it differs from the flat Minkowski metric $-d\eta^2 + d\xi^2$ in these coordinates only by an overall factor.

Moreover, for small values of a the time-component of the metric again reduces to something like the Newtonian limit expression (3.81), namely

$$\mathrm{e}^{2a\xi} \approx 1 + 2a\xi \quad . \tag{3.88}$$

For the record, and for later use, we note that the complete coordinate transformation between the Minkowski coordinates (t, x) and the conformally flat Rindler coordinates (η, ξ) is

$$(t, x) = (a^{-1}\mathrm{e}^{a\xi} \sinh a\eta, a^{-1}\mathrm{e}^{a\xi} \cosh a\eta) \quad . \tag{3.89}$$

The generator of Rindler time evolution in these coordinates is

$$\partial_\eta = a(t\partial_x + x\partial_t) \quad . \tag{3.90}$$

This is a boost in the (t, x)-plane, but the limit $a \to 0$ appears to be singular.

3. A simple and useful way to rectify this is to introduce a further constant shift of x, $x \to x - 1/a$, into the 1-parameter family (3.89) of coordinate transformations,

$$(t, x) = (a^{-1}\mathrm{e}^{a\xi} \sinh a\eta, a^{-1}\mathrm{e}^{a\xi} \cosh a\eta - a^{-1}) \quad . \tag{3.91}$$

This transformation now has the desirable property that as $a \to 0$ it continuously connects the Rindler and Minkowski coordinates,

$$a \to 0 \quad \Rightarrow \quad t \to \eta \quad , \quad x \to \xi \quad . \tag{3.92}$$

As a consequence, also the Rindler time evolution generator

$$\partial_\eta = \partial_t + a(t\partial_x + x\partial_t) \tag{3.93}$$

now has a non-singular limit as $a \to 0$, namely the Minkowski time generator ∂_t.

4. In terms of the Minkowski null (or advanced and retarded time) coordinates

$$u_M = t - x \quad , \quad v_M = t + x \quad \Rightarrow \quad ds^2 = -du_M\, dv_M \quad , \tag{3.94}$$

and their Rindler counterparts

$$u_R = \eta - \xi \quad , \quad v_R = \eta + \xi \quad \Rightarrow \quad ds^2 = -\mathrm{e}^{a(v_R - u_R)} du_R\, dv_R \quad , \tag{3.95}$$

the transformation (3.89) takes the form

$$t \mp x = a^{-1}\mathrm{e}^{a\xi} \left(\sinh a\eta \mp \cosh a\eta\right) = \mp a^{-1}\mathrm{e}^{a(\xi \mp \eta)} \tag{3.96}$$

or, compactly,

$$u_M = -a^{-1}\mathrm{e}^{-au_R} \quad , \quad v_M = +a^{-1}\mathrm{e}^{+av_R} \quad . \tag{3.97}$$

Note that the range of the coordinates is $-\infty < \eta, \xi < +\infty$ or $-\infty < u_R, v_R < +\infty$, and that the coordinates (η, ξ) or (u_R, v_R) cover (and can be used in) the right-hand quadrant $x > |t|$ of Minkowski space-time, corresponding to $-\infty < u_M = t - x < 0$ and $0 < v_M = t + x < +\infty$, the so called (right) *Rindler wedge*.

As we will see in section 7.8, these null Rindler coordinates are particularly useful for studying the solutions of the scalar wave equation in the Rindler wedge.

Let me close this section with some comments on other versions of $(3+1)$-dimensional Rindler space. First of all, instead of looking at acceleration in the x^1-direction, say, one can consider radial accelerations. To that end one first writes the metric in spatial spherical coordinates,

$$ds^2 = -dt^2 + dr^2 + r^2 d\Omega_2^2 \ , \tag{3.98}$$

and introduces Rindler coordinates (ρ, τ), say, via

$$(t, r) = (\rho \sinh \tau, \rho \cosh \tau) \ , \tag{3.99}$$

leading to the $(3+1)$-dimensional spherical Rindler metric

$$ds^2 = -\rho^2 d\tau^2 + d\rho^2 + \rho^2 \cosh^2 \tau \, d\Omega_2^2 \ . \tag{3.100}$$

This form of the metric is adapted to the hyperboloids

$$r^2 - t^2 = \rho^2 \tag{3.101}$$

i.e. to a family of constantly radially accelerating observers whose worldlines asymptotically approach the lightcone through the origin $(t = 0, r = 0)$. These coordinates cover precisely the region of Minkowski space outside this lightcone, i.e. the region of events at spacelike distance from the origin while the region that is not covered is the past and future of the origin.

The "complementary" metric (adapted to the hyperboloids with $r^2 - t^2 < 0$ and covering precisely the interior of the lightcone) is the so-called *Milne metric* to be discussed in section 37.1.

A non-trivial variant of this metric[6] is obtained by shifting $r \to r - r_0$ in the above coordinate transformation,

$$(t, r) = (\rho \sinh \tau, r_0 + \rho \cosh \tau) \ , \tag{3.102}$$

leading to the metric

$$ds^2 = -\rho^2 d\tau^2 + d\rho^2 + (r_0 + \rho \cosh \tau)^2 d\Omega_2^2 \tag{3.103}$$

[6] V. Balasubramanian, B. Czech, B. Chowdhury, J. de Boer, *The entropy of a hole in spacetime*, `arXiv:1305.0856 [hep-th]`, V. Balasubramanian, B. Chowdhury, B. Czech, J. de Boer, M. Heller, *A hole-ographic spacetime*, `arXiv:1310.4204 [hep-th]`.

(an analogous shift $x \to x - x_0$ for acceleration in the x-direction would have had no effect on the metric since such a translation is a symmetry of the Minkowski metric, whereas a translation in the radial direction is not). This form of the metric is adapted to the hyperboloids

$$(r - r_0)^2 - t^2 = \rho^2 \tag{3.104}$$

and now describes radially accelerating observers, each one asymptotically approaching the radial lightray emanating from a distance r_0 from the origin (and correspondingly the region of space-time covered by these coordinates is the complement of the past and future of the 2-sphere of radius r_0 at the origin, a "hole" in space-time).

3.5 Gravitational Redshift

Following Einstein, the gravitational redshift (i.e. the fact that photons appear to lose or gain energy when rising or falling in a gravitational field) is usually presented as a direct consequence of the Einstein Equivalence Principle (and is therefore also said to provide an experimental test of the Einstein Equivalence Principle itself). It (or, better, its Newtonian weak field limit) can indeed be derived in this way (see Remark 1 at the end of this section for one such argument, albeit not the original one). However, here we will derive this effect without any approximation within the framework that we have already adopted, inspired by the equivalence principle, namely in terms of the description of the gravitational field by a metric.

This has several advantages. It allows us to further familiarise ourselves with the formalism and to illustrate how to extract physical effects from our description of lightrays as null geodesics (much as we employed timelike geodesics above to study the Newtonian limit). Moreover, it allows us to derive formulae for this effect in quite some generality and I will actually give 3 different derivations in increasing order of generality. In conjunction with the Newtonian approximation to the gravitational field these then reduce to the result in the form in which it is usually presented, e.g. as in (3.130) or (3.131) (and as then rederived on the basis of the equivalence principle in (3.137)).

To set the stage, note that it is manifest from the expression

$$d\tau^2 = -g_{\alpha\beta}(x)dx^\alpha dx^\beta \tag{3.105}$$

for the proper time that e.g. the rate of clocks is affected by where one is in a gravitational field. However, as by the universality of gravity everything is (and in particular all ideal clocks are) affected in the same way by gravity, it is impossible to measure this effect locally, at a fixed point in a gravitational field. In order to find an observable effect, one needs to compare data from two different points in a gravitational potential.

The situation we could consider is that of two observers A and B moving on worldlines (paths) γ_A and γ_B, A sending light signals to B. In general the frequency, measured

in the observers rest-frame at A (or in a locally inertial coordinate system there) will differ from the frequency measured by B upon receiving the signal.

In order to separate out Doppler-like effects due to relative velocities, we consider two observers A and B at rest radially to each other, at radii r_A and r_B, in a static spherically symmetric gravitational field. This means that the metric depends only on a radial coordinate r and we can choose it to be of the form

$$ds^2 = g_{00}(r)dt^2 + g_{rr}(r)dr^2 + r^2 d\Omega^2 \quad , \tag{3.106}$$

where $d\Omega^2$ is the standard volume element on the two-sphere (see section 24 for a more detailed justification of this ansatz for the metric).

Observer A sends out light of a given frequency ν_A, say n pulses per proper time unit $\Delta\tau_A$. Observer B receives these n pulses in his proper time $\Delta\tau_B$ and interprets this as a frequency ν_B. Thus the relation between the frequency ν_A emitted at A and the frequency ν_B observed at B is

$$\frac{\nu_A}{\nu_B} = \frac{\Delta\tau_B}{\Delta\tau_A} \quad . \tag{3.107}$$

I will now give two arguments to show that this ratio depends on the metric (i.e. the gravitational field) at r_A and r_B through

$$\frac{\nu_A}{\nu_B} = \frac{(-g_{00}(r_B))^{1/2}}{(-g_{00}(r_A))^{1/2}} \quad , \tag{3.108}$$

and then a 3rd argument establishing a slightly more general result.

1. The first argument is essentially one based on geometric optics (and is best accompanied by drawing a (1+1)-dimensional space-time diagram of the lightrays and worldlines of the observers).

   The geometry of the situation dictates that the coordinate time intervals recorded at A and B are equal, $\Delta t_A = \Delta t_B$ as nothing in the metric actually depends on t. In equations, this can be seen as follows. First of all, the equation for a radial lightray is

   $$-g_{00}(r)dt^2 = g_{rr}(r)dr^2 \quad , \tag{3.109}$$

   or

   $$\frac{dt}{dr} = \pm\left(\frac{g_{rr}(r)}{-g_{00}(r)}\right)^{1/2} \quad . \tag{3.110}$$

   From this we can calculate the coordinate time for the lightray to go from A to B. Say that the first light pulse is emitted at point A at time $t(A)_1$ and received at B at coordinate time $t(B)_1$. Then

   $$t(B)_1 - t(A)_1 = \int_{r_A}^{r_B} dr(-g_{rr}(r)/g_{00}(r))^{1/2} \tag{3.111}$$

The right hand side obviously does not depend on t, so we also have

$$t(B)_2 - t(A)_2 = \int_{r_A}^{r_B} dr(-g_{rr}(r)/g_{00}(r))^{1/2} \tag{3.112}$$

where t_2 denotes the coordinate time for the arrival of the n-th pulse. Therefore,

$$t(B)_1 - t(A)_1 = t(B)_2 - t(A)_2 \ , \tag{3.113}$$

or

$$t(A)_2 - t(A)_1 = t(B)_2 - t(B)_1 \ , \tag{3.114}$$

as claimed. Thus the coordinate time intervals recorded at A and B between the first and last pulse are equal. However, to convert this to proper time, we have to multiply the coordinate time intervals by an r-dependent function,

$$\Delta\tau_{A,B} = (-g_{\alpha\beta}(r_{A,B})\frac{dx^\alpha}{dt}\frac{dx^\beta}{dt})^{1/2}\Delta t_{A,B} \ , \tag{3.115}$$

and therefore the proper time intervals will not be equal. For observers at rest, $dx^i/dt = 0$, one has

$$\Delta\tau_{A,B} = (-g_{00}(r_{A,B}))^{1/2}\Delta t_{A,B} \ . \tag{3.116}$$

Since $\Delta t_A = \Delta t_B$, (3.108) now follows from (3.107).

2. The second argument uses the null geodesic equation, in particular the conserved quantity associated to time-translations (recall that we have assumed that the metric (3.106) is time-independent), as well as a somewhat more covariant looking, but equivalent, notion of frequency.

   First of all, let the lightray be described by the wave vector k^α. In special relativity, we would parametrise this as $k^\alpha = (\omega, \vec{k})$ with $\omega = 2\pi\nu$ the frequency. This is the frequency observed by an inertial observer at rest, with 4-velocity $u^\alpha = (1, 0, 0, 0)$. A Lorentz-invariant, and in our context now coordinate-independent, notion of the frequency as measured by an observer with velocity u^α is thus

   $$\omega = -u^\alpha k_\alpha \ . \tag{3.117}$$

   This includes as special cases

   - the standard (special) relativistic Doppler effect (where one compares ω with $\bar{\omega} = -\bar{u}^\alpha k_\alpha$, $\bar{u}^\alpha$ the 4-velocity of a boosted observer),
   - and the gravitational redshift between static observers we want to discuss here,

   but more generally also the redshift for observers with arbitrary 4-velocity u^α. And indeed we will employ this method in section 26.5 to look at

- the redshift between a static and a freely falling observer in the Schwarzschild geometry.

Returning to the case at hand, a static observer in the spherically-symmetric and static gravitational field (3.106) is described by the 4-velocity

$$u^\alpha = (u^0, 0, 0, 0) \qquad g_{\alpha\beta} u^\alpha u^\beta = g_{00} (u^0)^2 = -1 \quad . \tag{3.118}$$

Thus for the static observer at $r = r_A$, say, one has

$$u^0_A = (-g_{00}(r_A))^{-1/2} \tag{3.119}$$

(and likewise for the observer at $r = r_B$). The wave vector k^μ is a null tangent vector, $k^\mu k_\mu = 0$, to a null geodesic corresponding to the Lagrangian

$$\mathcal{L} = \tfrac{1}{2} g_{\alpha\beta} \dot{x}^\alpha \dot{x}^\beta = \tfrac{1}{2} g_{00}(r) \dot{t}^2 + \dots \tag{3.120}$$

Since the metric is time-independent, there is (cf. the discussion in section 3.1) the corresponding conserved quantity

$$E = -\frac{\partial \mathcal{L}}{\partial \dot{t}} = -g_{00}(r) \dot{t} \tag{3.121}$$

(the minus sign serving only to make this quantity positive for $\dot{t} > 0$). Then one finds that the frequency measured by the static observer at $r = r_A$ is

$$\begin{aligned} \omega_A = -u^\alpha_A k_\alpha &= -\left(-g_{00}(r_A)\right)^{-1/2} k_0 = -\left(-g_{00}(r_A)\right)^{-1/2} g_{0\alpha}(r_A) \dot{x}^\alpha \\ &= -\left(-g_{00}(r_A)\right)^{-1/2} g_{00}(r_A) \dot{t} = E \left(-g_{00}(r_A)\right)^{-1/2} \end{aligned} \tag{3.122}$$

Since E is a conserved quantity, i.e. the same for the lightray at $r = r_A$ or $r = r_B$, one sees that $\omega_A / \omega_B = \nu_A / \nu_B$ is given by (3.108), as claimed.

Note that this derivation shows that the relation between ω and E is exactly like the relation (3.116) between $(\Delta\tau)^{-1}$ and $(\Delta t)^{-1}$, which provides us with an interpretation of the conserved quantity E for a massless particle / photon: it is the frequency measured with respect to coordinate time (as the momentum conjugate to the time-coordinate t this should not be too surprising).

3. Even with the restriction to static observers in static gravitational fields, the above derivation is not completely general, and still not completely covariant, because we used the explicit form of the metric (which is the general form of a metric with a time-translation invariance in spherical symmetry, but not in general). We can improve this somewhat by using the more general characterisation of time-translation invariance in terms of Killing vectors (section 3.2) and the associated conserved charge (3.36).

   Thus assume that we have a timelike Killing vector V^α. Then by definition a static observer is one whose 4-velocity u^α is proportional to V^α,

$$u^\alpha \sim V^\alpha \quad . \tag{3.123}$$

For $V = \partial_t$ this evidently reduces to the statement that only t changes along the worldline, i.e. that the observer remains at fixed values of the spatial coordinates, and this is the sense in which we have informally used the term "static observer" so far. Denoting the norm of V by

$$\mathbb{V} = (-V^\alpha V_\alpha)^{1/2} \ , \tag{3.124}$$

the normalisation condition $u^\alpha u_\alpha = -1$ fixes the proportionality factor between u^α and V^α to be

$$u^\alpha = V^\alpha / \mathbb{V} \ . \tag{3.125}$$

Given the null wave vector k^α, we have the conserved energy (3.36),

$$E = -k_\alpha V^\alpha \ . \tag{3.126}$$

Therefore, adopting the definition (3.117), the frequency observed by a static observer is

$$\omega = -u^\alpha k_\alpha = -k_\alpha V^\alpha / \mathbb{V} = E/\mathbb{V} \ . \tag{3.127}$$

Since E is constant along the lightray, frequencies observed by two different static observers are related by

$$\frac{\omega_A}{\omega_B} = \frac{\mathbb{V}_B}{\mathbb{V}_A} \ . \tag{3.128}$$

For this reason, the norm $\mathbb{V}$ is also known as the *redshift factor* associated with a timelike Killing vector.

Note that this result reduces to (3.108) if the metric has the form (3.106) and $V = \partial_t$ since then

$$V = \partial_t \quad \Rightarrow \quad \mathbb{V}(r) = (-g_{00}(r))^{1/2} \ . \tag{3.129}$$

Having derived (3.108) in 3 different ways, let us now look at what the result tells us in specific situations of interest. Since on earth and in the solar system we only have access to gravitational fields that are to a reasonably high degree of precision well described by Newtonian gravity, we can use the Newtonian approximation (3.59). Then (3.108) becomes

$$g_{00} = -(1 + 2\phi) \quad \Rightarrow \quad \frac{\nu_A}{\nu_B} \sim 1 + \phi(r_B) - \phi(r_A) \ , \tag{3.130}$$

or, with $\phi(r) = -G_N M/r$,

$$\frac{\nu_A - \nu_B}{\nu_B} = \frac{G_N M (r_B - r_A)}{r_A r_B} \tag{3.131}$$

Thus for $r_B > r_A$ one has

$$r_B > r_A \quad \Rightarrow \quad \nu_B < \nu_A \ , \tag{3.132}$$

so that the frequency of a lightray appears to be redshifted when rising (moving upwards) in a gravitational field, and one has a gravitational blueshift effect

$$r_B < r_A \quad \Rightarrow \quad \nu_B > \nu_A \ . \tag{3.133}$$

for photons falling (moving downwards) in a gravitational field.

So far these are the objective facts, just the results of the equation (3.108) that we have derived. Let me now turn to the more subtle question of how to interpret this result, or what words to use to best describe this result.

It is tempting (and unfortunately quite common) to interpret these results as saying that "as expected, a photon loses energy when rising in (and against the pull of) a gravitational field". This kind of non-relativistic Newtonian reasoning, wildly extrapolated to include the energy of lightrays, can to a certain extent be justified by appealing to the Einstein Equivalence Principle (see Remark 1 below), but I believe that this is misleading and provides the wrong intuition for the physics of the situation, and I will subsequently provide a different explanation that (to me) appears to be much more to the point (Remark 4).

REMARKS:

1. The Newtonian limit (3.130) of the exact result (3.108) can to some extent also be deduced from energy conservation applied to Newtonian gravity.[7] By the Einstein Equivalence Principle a local observer at the emitter A will see a change in the internal mass of the emitter $\Delta m_A = -h\nu_A$ when a photon of frequency of ν_A is emitted. Likewise, the absorber at point B will experience an increase in inertial mass by $\Delta m_B = h\nu_B$, but the total internal plus gravitational potential energy

$$mc^2 + m\phi \equiv m + m\phi = m(1+\phi) \tag{3.134}$$

must be conserved, i.e.

$$m_A(1+\phi(r_A))+m_B(1+\phi(r_B)) = (m_A+\Delta m_A)(1+\phi(r_A))+(m_B+\Delta m_B)(1+\phi(r_B)) \ . \tag{3.135}$$

Thus

$$0 = \Delta m_A(1+\phi(r_A)) + \Delta m_B(1+\phi(r_B)) \ , \tag{3.136}$$

leading to (3.130),

$$\frac{\nu_A}{\nu_B} = \frac{1+\phi(r_B)}{1+\phi(r_A)} \sim 1 + \phi(r_B) - \phi(r_A) \ . \tag{3.137}$$

2. This "derivation" (in quotes, because we are wildly mixing Newtonian gravity, special relativity and quantum mechanics - do take this "derivation" with an

[7] See S. Weinberg, *Gravitation and Cosmology* section 3.5 for a careful discussion.

appropriately sized grain of salt, please) shows that gravitational redshift experiments test the Einstein Equivalence Principle in its strong form, in which the term 'laws of nature' is not restricted to mechanics (inertial = gravitational mass), but also includes quantum mechanics in the sense that it tests if in an inertial frame the relation between photon energy and frequency is unaffected by the presence of a gravitational field.

3. One significant shortcoming of the above "explanation" is that it relies on the Newtonian concept of a gravitational potential energy, and that it therefore applies at best to the Newtonian limit (3.130) of the exact result (3.108). It also seems to suggest that something is happening to the photon / lightray as it travels from A to B (namely that it "loses energy"), and it is in this sense that the Newtonian intuition is misleading.

   Indeed, note that the simple general result (3.108) depends only on the value of the gravitational field at the points r_A and r_B, not on the gravitational field inbetween. This is at it should be. After all, this gravitational time delay between clocks at r_A and r_B is all that we have used is the derivation (and not some new law of physics that causes photons to lose energy in some unexplained way). This strongly suggest the interpretation, that the gravitational redshift is only due to the different rate of clocks / proper time at the positions r_A and r_B, and not due to the fact that "something happens to the lightray as it travels through a gravitational field" (which should lead to a cumulative effect depending also on the intermediate gravitational field).

4. Indeed, let us reanalyse the gravitational redshift (without any approximation) from this point of view. We note that the observer A at r_A uses his (or her) proper time τ_A to define a notion of frequency ν_A, and hence a notion of energy $E_A = h\nu_A$. Likewise the observer B defines an energy $E_B = h\nu_B$ based on his (or her) proper time τ_B. The fact that $E_B \neq E_A$ has nothing to do with the photon or lightray having lost or gained energy, it is simply due to the fact that the observers A and B define energy in a different way!

   Moreover, there is a rigorous and well-defined sense in which the energy of a photon moving in a time-independent gravitational field is conserved. Indeed, recall that in such a case there is a conserved quantity $p_0 = -E$ (3.7). In the case at hand it takes the form (3.121),

$$E = -g_{00}(r)\dot{t} \tag{3.138}$$

   and (as the momentum conjugate to the time coordinate t) has the interpretation as (Planck's constant times) the frequency ν_t of the lightray with respect to this (observer-independent) coordinate time t,

$$E = E_t = h\nu_t \quad . \tag{3.139}$$

Putting all these facts together, one sees that the gravitational redshift can and should be attributed entirely to the gravitational time-delay between observers A and B, and thus to the different conventions for measuring frequencies and energies, and not to some kind of "loss of energy of a photon in a gravitational field".

Let me close this section with a brief comment on the observational tests of the redshift prediction (3.131) (but you should consult other sources for a more detailed discussion of experimental tests of General Relativity). While difficult to observe directly (by looking at light from the sun), the prediction (3.131) has been verified in the laboratory, first by Pound and Rebka (1960), and subsequently, with one percent accuracy, by Pound and Snider in 1964 (using the Mössbauer effect).

Here is a rough estimate of the expected effect. We first consider light reaching us (B) from the sun (A). In this case, we have $r_B \gg r_A$, where r_A is the radius of the sun, and (also inserting a so far suppressed factor of c^2) we obtain

$$\frac{\nu_A - \nu_B}{\nu_B} = \frac{G_N M(r_B - r_A)}{c^2 r_A r_B} \simeq \frac{G_N M}{c^2 r_A} \quad . \tag{3.140}$$

Using the approximate values

$$\begin{aligned} r_A &\simeq 0.7 \times 10^6 \ \mathsf{km} \\ M_{\rm sun} &\simeq 2 \times 10^{33} \ \mathsf{g} \\ G_N &\simeq 7 \times 10^{-8} \ \mathsf{g}^{-1}\mathsf{cm}^3\mathsf{s}^{-2} \\ G_N c^{-2} &\simeq 7 \times 10^{-29} \ \mathsf{g}^{-1}\mathsf{cm} = 7 \times 10^{-34} \ \mathsf{g}^{-1}\mathsf{km} \ , \end{aligned} \tag{3.141}$$

one finds

$$\frac{\Delta\nu}{\nu} \simeq 2 \times 10^{-6} \quad . \tag{3.142}$$

In principle, such a frequency shift should be observable. In practice, however, the spectral lines of light emitted by the sun are strongly effected e.g. by convection in the atmosphere of the sun (Doppler effect), and this makes it difficult to measure this effect with the required precision.

In the Pound-Snider experiment, the actual value of $\Delta\nu/\nu$ is much smaller. In the original set-up one has $r_B - r_A \simeq 20\mathsf{m}$ (the distance from floor to ceiling of the laboratory), and $r_A = r_{\rm earth} \simeq 6.4 \times 10^6 \mathsf{m}$, leading to

$$\frac{\Delta\nu}{\nu} \simeq 2.5 \times 10^{-15} \quad . \tag{3.143}$$

However, here the experiment is much better controlled, and the gravitational redshift was verified with 1% accuracy.

3.6 Equivalence Principle Revisited: Existence of Locally Inertial Coordinates

Central to our initial discussion of gravity was the Einstein Equivalence Principle which postulates the existence of locally inertial (or freely falling) coordinate systems in which locally at (or around) a point the effects of gravity are absent. Now that we have decided that the arena of gravity is a general metric space-time, we should establish that such coordinate systems indeed exist. Looking at the geodesic equation, it is clear that at least in this context "absence of gravitational effects" is tantamount to the existence of a coordinate system $\{\xi^a\}$ in which at a given point p the metric is the Minkowski metric, $g_{ab}(p) = \eta_{ab}$ and the Christoffel symbols are zero, $\Gamma^a_{\ bc}(p) = 0$,

$$g_{ab}(p) = \eta_{ab} \quad , \quad \Gamma^a_{\ bc}(p) = 0 \ . \tag{3.144}$$

Owing to the identity

$$g_{\mu\nu\,,\lambda} = \Gamma_{\mu\nu\lambda} + \Gamma_{\nu\mu\lambda} \ , \tag{3.145}$$

the latter condition is equivalent to $g_{ab\,,\,c}(p) = 0$. Below (after the Remarks) I will sketch three arguments establishing the existence of such coordinate systems, each one having its own virtues and providing its own insights into the issue.

Remarks:

1. Actually it is physically plausible (and fortuitously moreover true) that one can always find coordinates which embody the equivalence principle in the stronger sense that the metric is the flat metric η_{ab} and the Christoffel symbols are zero not just at a point but along the entire worldline of an inertial (freely falling) observer, i.e. along a geodesic γ,

$$g_{ab}|_\gamma = \eta_{ab} \quad , \quad \Gamma^a_{\ bc}|_\gamma = 0 \ . \tag{3.146}$$

   Such coordinates, based on a geodesic rather than on a point, are known as Fermi normal coordinates. The construction is similar to that of Riemann normal coordinates (based at a point) to be discussed below.[8]

2. In this mathematically idealised realisation of the equivalence principle, nothing is said about the metric and the Christoffel symbols in a neighbourhood of that point (or of the geodesic), and nothing is said about the 2nd and higher derivatives of the metric at that point.

[8] Most discussions of Fermi coordinates in the literature follow the presentation given in F. Menasse, C. Misner, *Fermi normal coordinates and some basic concepts in differential geometry*, J. Math. Phys. 4 (1963) 735-745; for a geometrically transparent treatment see also section 1.11 of E. Poisson, *A Relativist's Toolkit*; Fermi coordinates for *null* geodesics are constructed in M. Blau, D. Frank, S. Weiss, *Fermi Coordinates and Penrose Limits*, `arXiv:hep-th/0603109`.

3. In particular, thinking of the 1st derivatives of the metric as encoding the gravitational force, the 2nd derivatives of the metric must then correspond to gravitational *tidal forces*. (We will see this in more detail in section 8.4, where these tidal forces are related to components of the Riemann curvature tensor, a tensor that involves up to 2nd derivatives of the metric.) Such tidal forces are objective, i.e. physically real, as they lead to stresses in (or deformations of) extended bodies. One can and should therefore not expect to be able to eliminate such tidal forces by a suitable choice of reference system.

4. Therefore, physically what the equivalence principle says (or should say) is that in a gravitational field there is locally a reference system in which the effects of gravity are absent, provided that you choose the spacetime region to be sufficently small so that you can neglect the effect of gravitational tidal forces.

Here is a sketch of 3 arguments establishing the existence of locally inertial coordinate systems:

1. Direct Construction

   We know that given a coordinate system $\{\xi^a\}$ that is inertial at a point p, the metric and Christoffel symbols at p in a new coordinate system $\{x^\mu\}$ are determined by (1.87,1.98). Conversely, we will now see that knowledge of the metric and Christoffel symbols at a point p is sufficient to construct a locally inertial coordinate system at p.

   We will construct this coordinate system $\xi^a = \xi^a(x)$ locally around the point p (with coordinates x_0^α, say, in the original coordinate system) by a Taylor series expansion,

   $$\xi^a(x) = d^a + (x - x_0)^\alpha e^a_\alpha + \tfrac{1}{2}(x - x_0)^\beta (x - x_0)^\gamma f^a_{\beta\gamma} + \dots \quad . \tag{3.147}$$

   Here

   $$d^a = \xi^a(x_0) \equiv \xi^a_0 \tag{3.148}$$

   are the (arbitrary) coordinate values of the point p in the new coordinates ξ^a,

   $$e^a_\alpha = \frac{\partial \xi^a}{\partial x^\alpha}(x_0) \tag{3.149}$$

   is the Jacobi matrix of the coordinate transformation at $x = x_0$, and

   $$f^a_{\beta\gamma} = \frac{\partial^2 \xi^a}{\partial x^\beta \partial x^\gamma}(x_0) \tag{3.150}$$

   is its 1st derivative at x_0.

   Form the tensorial transformation behaviour of the metric we know that

   $$g_{\alpha\beta}(x_0) = g_{ab}(\xi_0) e^a_\alpha e^b_\beta \quad . \tag{3.151}$$

Requiring that $g_{ab}(\xi_0) = \eta_{ab}$ leads to the condition

$$g_{ab}(\xi_0) = \eta_{ab} \quad \Rightarrow \quad g_{\alpha\beta}(x_0) = \eta_{ab} e^a_\alpha e^b_\beta \ . \tag{3.152}$$

This shows that the e^a_α, thought of as a matrix, are an invertible (4×4)-matrix in $\mathsf{GL}(4, \mathbb{R})$. Denoting its inverse by e^α_a, with

$$e^\alpha_a e^a_\beta = \delta^\alpha_\beta \quad , \quad e^\alpha_a e^b_\alpha = \delta^a_b \ , \tag{3.153}$$

we see that the inverse matrix diagonalises (and scales) the metric at the point p in such a way that

$$g_{\alpha\beta}(x_0) e^\alpha_a e^\beta_b = \eta_{ab} \ . \tag{3.154}$$

Since $g_{\alpha\beta}(x_0)$ is a symmetric non-degenerate matrix, such matrices always exist (and are unique up to similarity transformations that leave η_{ab} invariant, i.e. up to Lorentz transformations). The notation e^a_α and e^α_a reflects the fact that these matrices are the components of an orthonormal vierbein (or vielbein) at the point p, which are traditionally denoted this way (cf. the discussion in section 4.8 below).

Taking stock, we see that the condition $g_{ab}(p) = \eta_{ab}$ determines the coordinate system to 1st order in a Taylor series expansion, up to translations (the choice of d^a) and Lorentz transformations, i.e. up to Poincaré transformation.

We now turn to the 2nd condition characterising a locally inertial coordinate system, namely $\Gamma^a_{\ bc}(p) = 0$. We can write the inhomogeneous transformation behaviour of the Christoffel symbols as

$$\frac{\partial \xi^a}{\partial x^\alpha} \Gamma^\alpha_{\ \beta\gamma} = \Gamma^a_{\ bc} \frac{\partial \xi^b}{\partial x^\beta} \frac{\partial \xi^c}{\partial x^\gamma} + \frac{\partial^2 \xi^a}{\partial x^\beta \partial x^\gamma} \ . \tag{3.155}$$

Thus at the point p we have

$$e^a_\alpha \Gamma^\alpha_{\ \beta\gamma}(x_0) = \Gamma^a_{\ bc}(\xi_0) e^b_\beta e^c_\gamma + f^a_{\beta\gamma} \ . \tag{3.156}$$

Requiring $\Gamma^a_{\ bc}(p) = 0$ now uniquely determines the 2nd order Taylor coefficients,

$$\Gamma^a_{\ bc}(\xi_0) = 0 \quad \Rightarrow \quad f^a_{\beta\gamma} = e^a_\alpha \Gamma^\alpha_{\ \beta\gamma}(x_0) \ . \tag{3.157}$$

Thus to 2nd order in a Taylor series expansion, the transformation from arbitrary coordinates x^α to inertial coordinates ξ^a at the point p is given by

$$\xi^a(x) = \xi^a_0 + (x - x_0)^\alpha e^a_\alpha + \tfrac{1}{2}(x - x_0)^\beta (x - x_0)^\gamma e^a_\alpha \Gamma^\alpha_{\ \beta\gamma}(x_0) + \ldots \ . \tag{3.158}$$

To this order we can also write the inverse coordinate transformation as

$$x^\alpha(\xi) = x^\alpha_0 + (\xi - \xi_0)^a e^\alpha_a - \tfrac{1}{2}(\xi - \xi_0)^b (\xi - \xi_0)^c \Gamma^\alpha_{\ \beta\gamma}(x_0) e^\beta_b e^\gamma_c + \ldots \ . \tag{3.159}$$

We have therefore established that for an arbitrary point p in an arbitrary gravitational field one can always introduce local coordinates which are inertial at that

point, and that up to 2nd order in a Taylor series expansion such a coordinate system is unique up to Poincaré transformations.

Since this leaves the infinite number of higher-order terms of the Taylor expansion undetermined, this shows that inertial coordinate systems are highly non-unique, and raises the following questions:

- Can one continue in this vein and choose the (so far undetermined) higher-order terms in the Taylor expansion such that also e.g. the 2nd derivatives of the metric at p are equal to zero,

$$\exists? \quad \xi^a(x): \quad g_{ab,\,cd}(p) = 0 \ ? \tag{3.160}$$

  The answer to this is a resounding "no", as the 3rd (numerological) argument below will show. In fact, as we will see (and study in detail) later on, the 2nd derivatives of the metric contain important coordinate-invariant information about the curvature of the metric.

- Are there nevertheless preferred inertial coordinate systems, i.e. preferred choices for the higher-order terms in the Taylor expansion? The answer to this is "yes". One such preferred and geometrically natural class of inertial coordinate systems are e.g. Riemann normal coordinates, based on geodesics at the point p, and briefly discussed below.

2. Geodesic (or Riemann Normal) Coordinates

   A slightly more insightful way of constructing a locally inertial coordinate system, rather than by directly solving the relevant differential equation, makes use of geodesics at p. Recall that in Minkowski space the metric takes the simplest possible form in coordinates whose coordinate lines are (orthogonal) geodesics. One might thus suspect that in a general metric space-time the metric will also (locally) look particularly simple when expressed in terms of such *geodesic coordinates*.

   Roughly speaking (I will give a more detailed argument below), since locally around p we can solve the geodesic equation with four linearly independent initial conditions, we can assume the existence of a coordinate system $\{\xi^a\}$ in which the coordinate lines are geodesics $\xi^a(\tau) = \xi^a\tau$. This means that in these coordinates geodesics satisfy $\ddot{\xi}^a = 0$. Comparing with the full geodesic equation in these coordinates, one sees that this implies that

$$\Gamma^a_{\ bc}\dot{\xi}^b\dot{\xi}^c = 0 \ . \tag{3.161}$$

   As at p the $\dot{\xi}^a$ were chosen to be linearly independent, this implies $\Gamma^a_{\ bc}(p) = 0$, as desired. It is easy to see that the coordinates ξ^a can also be chosen in such a way that $g_{ab}(p) = \eta_{ab}$ (by choosing the four directions at p to be orthonormal unit vectors).

Before turning to the more detailed construction, let us look at an example. Consider the standard metric $ds^2 = d\theta^2 + \sin^2\theta d\phi^2$ on the two-sphere. Any point is as good as any other point, and one can construct an inertial coordinate system at the north pole $\theta = 0$ in terms of geodesics shot off from the north pole into the $\phi = 0$ (ξ^1) and $\phi = \pi/2$ (ξ^2) directions. The affine parameter along a great circle (geodesic) connecting the north pole to a point (θ, ϕ) is θ, and thus θ is also the geodesic distance, and the coordinates of the point (θ, ϕ) are $(\xi^1 = \theta\cos\phi, \xi^2 = \theta\sin\phi)$. In particular, the north pole is the origin $\xi^1 = \xi^2 = 0$. Note that one could have guessed these coordinates from the fact that near $\theta = 0$ the metric is $d\theta^2 + \theta^2 d\phi^2$, which is the Euclidean metric in polar coordinates $(\theta\cos\phi, \theta\sin\phi)$.

Calculating the metric in these new components, using

$$(\xi^1 = \theta\cos\phi, \xi^2 = \theta\sin\phi) \quad \Rightarrow \quad ((\xi^1)^2 + (\xi^2)^2 = \theta^2, \xi^2/\xi^1 = \tan\phi) \tag{3.162}$$

and thus

$$d\theta = \frac{\xi^1 d\xi^1 + \xi^2 d\xi^2}{\sqrt{(\xi^1)^2 + (\xi^2)^2}} \quad , \quad d\phi = \frac{\xi^1 d\xi^2 - \xi^2 d\xi^1}{(\xi^1)^2 + (\xi^2)^2} \ , \tag{3.163}$$

one finds

$$d\theta^2 + \sin^2\theta d\phi^2 = (d\xi^1)^2 + (d\xi^2)^2 + \mathcal{O}(\xi^2 d\xi^2) \ , \tag{3.164}$$

i.e.

$$g_{ab}(\xi) = \delta_{ab} + \mathcal{O}(\xi^2) \ . \tag{3.165}$$

Therefore

$$g_{ab}(\xi = 0) = \delta_{ab} \quad , \quad g_{ab,c}(\xi = 0) = 0 \ , \tag{3.166}$$

as required.

We now (re)turn to the general construction of such coordinates, starting with the geodesic equation

$$\ddot{x}^\alpha + \Gamma^\alpha_{\beta\gamma}\dot{x}^\beta\dot{x}^\gamma = 0 \ . \tag{3.167}$$

We consider geodesics passing through (or emanating from) the point p with coordinates x_0^α at $\tau = 0$, and with initial 4-velocity u_0^α,

$$x^\alpha(\tau = 0) = x_0^\alpha \quad , \quad \dot{x}^\alpha(\tau = 0) = u_0^\alpha \ . \tag{3.168}$$

It then follows that the 2nd derivative at $\tau = 0$ is given by

$$\ddot{x}^\alpha(\tau = 0) = -\Gamma^\alpha_{\beta\gamma}(x_0)u_0^\beta u_0^\gamma \ . \tag{3.169}$$

Hence in a Taylor expansion around $\tau = 0$ we can write the solution to the geodesic equation as

$$x^\alpha(\tau) = x_0^\alpha + \tau u_0^\alpha - \tfrac{1}{2}\tau^2\Gamma^\alpha_{\beta\gamma}(x_0)u_0^\beta u_0^\gamma + \dots \ . \tag{3.170}$$

We can expand the (arbitrary) initial 4-velocity u_0^α in terms of 4 linearly independent (and orthonormal, say) vectors at p as

$$u_0^\alpha = \lambda^a e_a^\alpha \quad , \quad g_{\alpha\beta}(x_0) e_a^\alpha e_b^\beta = \eta_{ab} \ . \tag{3.171}$$

We can then think of the Taylor expansion (3.170) as defining a coordinate transformation

$$x^\alpha(\xi) = x_0^\alpha + (\xi - \xi_0)^a e_a^\alpha - \tfrac{1}{2}(\xi - \xi_0)^b (\xi - \xi_0)^c \Gamma^\alpha_{\ \beta\gamma}(x_0) e_b^\beta e_c^\gamma + \dots \quad , \tag{3.172}$$

which has the following properties:

(a) First of all, for

$$\xi^a(\tau) = \xi_0^a + \tau \lambda^a \tag{3.173}$$

this reduces to the Taylor-expanded solution (3.170) of the geodesic equation with $u_0^\alpha = \lambda^a e_a^\alpha$. Thus in this coordinate system in particular the 4 coordinate lines

$$\xi^a_{(b)}(\tau) = \xi_0^a + \tau \delta_b^a \quad , \quad b = 0, 1, 2, 3 \ , \tag{3.174}$$

are (affinely parametrised) geodesics, as desired.

(b) Moreover, up to quadratic order in the Taylor expansion (3.172) is identical to the coordinate transformation (3.159). In particular, this establishes that the geodesic coordinates constructed here are a special class of inertial coordinates, with

$$g_{ab}(\xi_0) = \eta_{ab} \quad , \quad \Gamma^a_{\ bc}(\xi_0) = 0 \ . \tag{3.175}$$

(c) From the present point of view, the 2nd condition arises from the fact (mentioned above) that in these coordinates the geodesic equation for the above geodesics reduces to

$$\ddot{\xi}^a + \Gamma^a_{\ bc} \dot{\xi}^b \dot{\xi}^c = 0 \quad \Rightarrow \quad \Gamma^a_{\ bc}(\xi_0^a + \tau\lambda^a)\lambda^b\lambda^c = 0 \ . \tag{3.176}$$

At ξ_0, i.e. for $\tau = 0$, the Christoffel symbols are independent of the λ^a, and therefore

$$\Gamma^a_{\ bc}(\xi_0)\lambda^b\lambda^c = 0 \quad \forall \lambda^a \quad \Rightarrow \Gamma^a_{\ bc}(\xi_0) = 0 \ , \tag{3.177}$$

as claimed.

(d) In contrast to the previous construction leading to (3.159), here the higher-order terms in the Taylor expansion of the coordinate transformation are now determined by the higher-order terms in the Taylor expansion of the solution (3.170) of the geodesic equation. These higher-order terms will depend on 2nd and higher derivatives of the metric $g_{\alpha\beta}(x)$ at x_0, and these in turn will

also determine the quadratic and higher terms of the Taylor expansion of the metric in these coordinates,

$$\begin{aligned} g_{ab}(\xi) &= g_{ab}(\xi_0) + (\xi-\xi_0)^c g_{ab,\,c}(\xi_0) + \tfrac{1}{2}(\xi-\xi_0)^c(\xi-\xi_0)^d g_{ab,\,cd}(\xi_0) + \dots \\ &= \eta_{ab} + \tfrac{1}{2}(\xi-\xi_0)^c(\xi-\xi_0)^d g_{ab,\,cd}(\xi_0) + \dots \quad . \end{aligned} \tag{3.178}$$

We will determine the quadratic term in this expansion (expressed in terms of the Riemann curvature tensor) in section 8.9.

3. A Numerological Argument

This is my favourite argument because it requires no calculations and at the same time provides additional insight into the nature of curved space-times.

Assuming that the local existence of solutions to differential equations is guaranteed by some mathematical theorems, it is frequently sufficient to check that one has enough degrees of freedom to satisfy the desired initial conditions (one may also need to check integrability conditions). Here we are looking at something even more elementary, namely the functional freedom contained in the coordinate transformations to impose certain conditions at one point. In the present context, this argument is useful because it also reveals some information about the 'true' curvature hidden in the second derivatives of the metric. It works as follows:

(a) Zero'th Derivatives:

Consider a Taylor expansion of the metric around p in the sought-for new coordinates. Then the metric at p will transform with the matrix $(\partial x^\mu/\partial\xi^a)(p)$. This matrix has $(4\times 4) = 16$ independent components, precisely enough to impose the 10 conditions $g_{ab}(p) = \eta_{ab}$ up to Lorentz transformations.

(b) First Derivatives:

The derivative of the metric at p, $g_{ab,c}\,(p)$, will appear in conjunction with the second derivative $\partial^2 x^\mu/\partial\xi^a\partial\xi^b$. The $4\times(4\times 5)/2 = 40$ coefficients are precisely sufficient to impose the 40 conditions $g_{ab,c}\,(p) = 0$.

(c) Second Derivatives:

Now let us look at the second derivatives of the metric. $g_{ab,cd}$ has $(10\times 10) = 100$ independent components, while the third derivative of $x^\mu(\xi)$ at p, $\partial^3 x^\mu/\partial\xi^a\partial\xi^b\partial\xi^c$ has $4\times(4\times 5\times 6)/(2\times 3) = 80$ components. Thus 20 linear combinations of the second derivatives of the metric at p cannot in general be set to zero by a coordinate transformation. Thus these encode the information about the real curvature at p. This agrees nicely with the fact that the Riemann curvature tensor we will construct later turns out to have precisely 20 independent components.

Repeating this argument in space-time dimension $D = d + 1$, one finds that the number of 2nd derivatives of the metric modulo coordinate transformations is

$$\left(\frac{D(D+1)}{2}\right)^2 - D\frac{D(D+1)(D+2)}{6} = \frac{1}{12}D^2(D^2-1) \ . \tag{3.179}$$

Again this turns out to agree with the number of independent components (8.27) of the curvature tensor in D dimensions.

Note:

At this point in the course I find it useful to develop in parallel (and suggest to read in parallel)

- the more formal material on tensor analysis in sections 4, 5, 6, 7, 8 and 11, say (and then moving on to the Einstein equations themselves)
- and a detailed discussion of the basic properties of the Schwarzschild metric (sections 25 - 27),

since much of the latter (in particular geodesics, solar system tests of general relativity, even the issues that arise in connection with the Schwarzschild radius) can be understood just on the basis of what has been done so far (if, for the time being, one accepts on faith that the Schwarzschild metric is the unique spherically symmetric vacuum solution of the Einstein field equations).

Not only is this an interesting and physically relevant application of the machinery developed so far, it also provides an appropriate balance between physics and formalism in the lectures. More advanced material in the intervening sections can then be covered and dealt with if and when needed or desired (or, ideally, both).

4 Tensor Algebra

4.1 Principle of General Covariance

The Einstein Equivalence Principle tells us that the laws of nature (including the effects of gravity) should be such that in an inertial frame they reduce to the laws of Special Relativity.

A prime example of this is the geodesic equation

$$\ddot{x}^\alpha + \Gamma^\alpha_{\beta\gamma}\dot{x}^\beta\dot{x}^\gamma = 0 \ , \tag{4.1}$$

describing the motion of a particle in a gravitational field. Using the Einstein Equivalence Principle, and the *General Covariance* of the geodesic equation established in section 1.6, one can argue as follows to establish that this is the "correct" equation at an aribitrary point p in an arbitrary gravitational field described by a metric $g_{\alpha\beta}(x)$:

- Choose an inertial coordinate system z^a centered at p, i.e. with $g_{ab}(p) = \eta_{ab}$ and $\Gamma^a_{bc}(p) = 0$. Then one has

$$\left(\ddot{x}^\alpha + \Gamma^\alpha_{\beta\gamma}\dot{x}^\beta\dot{x}^\gamma\right)(p) = \left(\frac{\partial x^\alpha}{\partial z^a}\right)\left(\ddot{z}^a + \Gamma^a_{bc}\dot{z}^b\dot{z}^c\right)(p) = \left(\frac{\partial x^\alpha}{\partial z^a}\right)\ddot{z}^a(p) \ . \tag{4.2}$$

- By the Equivalence Principle, $\ddot{z}^a = 0$ is the correct equation for a particle in a freely falling reference system (such as at the origin p of an inertial coordinate system). Therefore, by the above identity, at the point p the geodesic equation is also satisfied in arbitrary coordinates x^α.

- Both the point p and the metric $g_{\alpha\beta}(x)$ are arbitrary, and therefore the geodesic equation holds at an arbitrary point of an arbitrary gravitational field.

This argument generalises and shows that the Einstein Equivalence Principle provides us with a general link between equations describing physics in an arbitrary gravitational field and general covariance. In other words, general covariance provides us with a concrete way of implementing the Einstein Equivalence Principle. One classical (but not entirely satisfactory - see the discussion below) textbook formulation of this *Principle of General Covariance* is[9]

Principle of General Covariance

> By virtue of the Einstein Equivalence Principle, a physical equation holds in an arbitrary gravitational field if

[9] S. Weinberg, *Gravitation and Cosmology* section 4.1

1. the equation holds in the absence of gravity, i.e. when $g_{\mu\nu} = \eta_{\mu\nu}$, $\Gamma^{\mu}_{\nu\lambda} = 0$, and
2. the equation is *generally covariant*, i.e. preserves its form under a general coordinate transformation.

We will turn momentarily to a proof of (a slightly modified version of) this statement. First, however, I would like to add a *caveat* to the 1st and a clarification to the 2nd condition (starting with the latter):

ad 2: We first need to clarify (and then reformulate slightly) the 2nd condition, as the statement "preserves its form under a general coordinate transformation" is neither completely unambiguous (without further explanation or definitions) nor totally to the point.

Concretely, the 2nd condition means the following: assume that you have some physical equation that in some coordinate system takes the form $T = 0$, where $T = 0$ could be some multi-component (thus T is adorned with various indices) differential equation. Now perform a coordinate transformation $x \to y(x)$, and assume that the new object T' has the form

$$T' = (\ldots)T + \text{junk} \tag{4.3}$$

where the term in brackets is some invertible matrix or operator. Then clearly the presence of the junk-terms means that the equation $T' = 0$ is not equivalent to the equation $T = 0$. An example of an object that transform in this way is, as we have seen, the Christoffel symbols. On the other hand, if these junk terms are absent, so that we have

$$T' = (\ldots)T \tag{4.4}$$

then one might like to say that the equation has preserved its form under a general coordinate transformation.

As a consequence of this, one also has $T' = 0$ if and only if $T = 0$, i.e. the equation is satisfied in one coordinate system if and only if it is satisfied in any other (or all) coordinate systems. An example of this is the geodesic equation which, as we have seen, transform precisely in such a way, with the term $(\ldots)$ in brackets being the Jacobi matrix.

However, if this is what one desires (and it is), then one may as well say this directly. Thus, to be more concrete, we can replace the 2nd condition above by

2' the equation is *generally covariant*, i.e. it is satisfied in one coordinate system iff it is satisfied in all coordinate systems.

ad 1: The argument below will invoke general covariance in order to be able to look at a given equation at the origin of an inertial (freely falling) coordinate system. As we have seen in various ways in section 3.6, at that point p, the effects of gravity are absent to the extent that the metric at that point is the Minkowski metric, $g_{ab}(p) = \eta_{ab}$, and that the derivatives of the metric (or Christoffel symbols) at that point are zero, $g_{ab,c}(p) = 0$. However, as also noted there, higher derivatives of the metric can in general not also be chosen to be zero at that point (and indeed the second derivatives of the metric at that point will turn out to contain coordinate independent information about the curvature of the space-time).

In that sense, looking at an equation at the origin of an inertial coordinate system is not strictly identical to looking at that same equation in the absence of gravity (i.e. in Minkowski space), where all these higher derivatives of the metric are also zero (in inertial coordinates). For the purposes of the argument below we will ignore this difference, and thus allow for the possibility that the Einstein Equivalence Principle holds only to first order in derivatives of the metric. Indeed, as discussed in more detail in section 8.4, and as could be anticipated from our identification of the metric with the potential of the gravitational field, second derivatives of the metric encode tidal gravitational forces (which one cannot expect to be able to eliminate by passing to a freely falling reference system), so this is a plausible relaxation of some stricter interpretation of the Einstein Equivalence Principle. We will look at the implications of this in the remarks below.

With these remarks in mind, let us now establish the above statement, namely that the Einstein equivalence principle implies that an equation that satisfies the conditions 1 and 2 (or 2') is valid in an arbitrary gravitational field:

- consider some equation that satisfies these conditions, and assume that we are in an arbitrary gravitational field;
- condition 2' implies that this equation is true (or satisfied) in all coordinate systems if it is satisfied just in one coordinate system;
- now we know that we can always (locally) construct a freely falling coordinate system in which the effects of gravity are absent;
- the Einstein Equivalence Principle now posits that in such a reference system the physics is that of Minkowski space-time;
- condition 1 means that the equation is true (satisfied) there;
- thus it is valid in all coordinate systems;
- since we started off by considering an arbitrary gravitational field, it follows that the equation is now valid in an arbitrary gravitational field, as claimed in the Principle of General Covariance.

Remarks:

1. Note that general covariance alone is an empty statement since any equation (whether correct or not) can be made generally covariant simply by writing it in an arbitrary coordinate system (cf. also the discussion in section 6.4). It develops its power only when used in conjunction with the Einstein Equivalence Principle as a statement about physics in a gravitational field, namely that by virtue of its general covariance an equation will be true in a gravitational field if it is true in the absence of gravitation.

2. As alluded to above, the principle of general covariance does not fix the equations uniquely because there are generally covariant objects that one can construct e.g. from the (second) derivatives of the metric (via the Riemann curvature tensor to be introduced in section 8) that can therefore be added to an equation and which vanish for Minkowski space, i.e. in the absence of gravitation.

3. In section 6.1 we will introduce a recipe / algorithm, the *principle of minimal coupling*, that allows us to produce generally covariant equations from those of special relativity. However, as we will discuss in section 8.10, also this description is ambiguous. The upshot is that there is no unique way of implementing the principle of general covariance, but this was probably too much to hope for anyway.

4.2 Tensors and Tensor Fields

In order to construct generally covariant equations, we need objects that transform in a simple way under coordinate transformations

$$x^\alpha \to y^\mu(x) = \bar{x}^\mu(x) \tag{4.5}$$

(it will be convenient to be able to switch between denoting the new coordinates by either y^μ or $\bar{x}^\mu$, depending on what is easier to read or write in a given situation). The prime examples of such objects are *tensors*, which transform multi-linearly with the Jacobi matrices

$$J^\mu_\alpha = \frac{\partial y^\mu}{\partial x^\alpha} \quad , \quad J^\alpha_\mu = \frac{\partial x^\alpha}{\partial y^\mu} \quad , \tag{4.6}$$

satisfying

$$J^\alpha_\mu J^\mu_\beta = \delta^\alpha_\beta \quad , \quad J^\mu_\alpha J^\alpha_\nu = \delta^\mu_\nu \quad . \tag{4.7}$$

If you are already familiar with Lorentz tensors from special relativity (as briefly recalled in section 1.2, these are objects which transform in a particularly simple multi-linear way under Lorentz transformations), then hardly anything in this or the subsequent section 4.3 should be new or unexpected (but interesting new features will arise in particular when we move on from tensor algebra to tensor analysis in section 5).

1. Scalars

   The simplest example of a tensor is a function (or scalar) f which under a coordinate transformation $x^\alpha \to \bar{x}^\mu(x)$ simply transforms as

   $$\bar{f}(\bar{x}(x)) = f(x) \ . \tag{4.8}$$

   One frequently suppresses the argument, and thus writes simply $\bar{f} = f$, expressing the fact that, up to the obvious and familiar change of argument, functions are invariant under coordinate transformations.

   REMARKS:

   (a) Just in case this transformation behaviour is not as familiar as it should be, consider the transformation from Cartesian (x^1, x^2) to polar $(y^1 = r, y^2 = \phi)$ coordinates, and the function $f(x^1, x^2) = (x^1)^2 + (x^2)^2$. In terms of polar coordinates, this function can be written as $\bar{f}(r, \phi) = r^2$. While f and $\bar{f}$ are evidently different functions of 2 variables, they are equal up to the change of coordinates, and this is how functions or scalars transform in general.

   (b) Note that the coordinates themselves are functions (that assign coordinates to a point p). Thus the x-coordinates can be regarded as 4 functions $f^{(\alpha)}$ defined in the coordinate system x by $f^{(\alpha)}(x) = x^\alpha$. Likewise, the new coordinates y^μ can be regarded as being defined by 4 in general genuinely *different* functions of x, say $g^{(\mu)}(x) = y^\mu(x)$.

   Under changes of coordinates, these functions transform as they should, i.e. one has $\bar{f}^{(\alpha)}(y) = f^{(\alpha)}(x) = x^\alpha$, and likewise for the functions $g^{(\mu)}(x)$. In the example above, the former would simply be the Cartesian coordinates expressed in terms of polar coordinates, $\bar{f}^{(1)}(r, \phi) = r\cos\phi = x^1 = f^{(1)}(x^1, x^2)$. On the other hand, $g^{(1)}(r, \phi) = r$, which is quite distinct from $\bar{f}^{(1)}(r, \phi) = r\cos\phi$. In that sense, allowing oneself to also denote the new coordinates y^μ by $\bar{x}^\mu$ may initially be a source of confusion (but is frequently so useful that we will continue to do this anyway ...).

   For a slightly more sophisticated perspective on these things, which disentangles functions from their representatives in terms of local coordinates, see the discussion of manifolds and local charts in section 5.11.

2. Vectors

   The next simplest case are objects that transform under coordinate transformations with the Jacobi matrix J^μ_α. Such objects are called vectors and have components $V^\alpha(x)$ transforming as

   $$\bar{V}^\mu(y(x)) = J^\mu_\alpha(x) V^\alpha(x) \ . \tag{4.9}$$

These are the prototypes of tensors and indeed all the other tensorial objects that we will encounter in the following are rather straightforward generalisations of such vectors.

REMARKS:

(a) One comment on notation: since already the index μ on $\bar{V}^\mu$ indicates that these are the components of a vector with resepct to the coordinates $y^\mu = \bar{x}^\mu$, it is not strictly speaking necessary to indicate this as well by putting a bar on the vector. Therefore one could (and commonly just does) write simply

$$V^\mu(y) \equiv \bar{V}^\mu(y) \quad . \tag{4.10}$$

(b) One comment on terminology: it is sometimes useful to distinguish *vectors* from *vector fields* and, likewise, *tensors* from *tensor fields*. A vector is then just a vector $V^\alpha(x)$ at some point x of space-time whereas a vector field is something that assigns a vector to each point of space-time,

$$\text{vector field}: \quad x \mapsto V^\alpha(x) \tag{4.11}$$

and likewise for scalars and scalar fields, and more general tensors and tensor fields.

(c) An intermediate situation is that of a vector or tensor defined neither just at a single point nor in the entire space-time but for example along some curve (representing e.g. the worldline of a particle). A prime example is the tangent vector $\dot{x}^\alpha$ to a curve, which indeed transforms as a vector,

$$\dot{x}^\alpha \to \dot{y}^\mu = \frac{\partial y^\mu}{\partial x^\alpha}\dot{x}^\alpha = J^\mu_a \dot{x}^\alpha \quad . \tag{4.12}$$

And while the naive acceleration $\ddot{x}^\alpha$ is *not* a vector, as we have seen on several occasions before, we can define a vectorial acceleration by

$$a^\alpha = \ddot{x}^\alpha + \Gamma^\alpha_{\beta\gamma}\dot{x}^\beta\dot{x}^\gamma \quad . \tag{4.13}$$

We will rederive this expression for a covariant acceleration from a different perspective in section 5.7.

(d) One way of thinking about vector fields is as tangent vector fields to families of curves on a space or space-time which arise as the solutions to the differential equation

$$\frac{d}{d\sigma}x^\alpha(\sigma) = V^\alpha(x(\sigma)) \tag{4.14}$$

(and we take local existence and uniqueness of these solutions under suitable regularity and differentiability conditions for granted). These curves $x^\alpha(s)$ are the *integral curves* (or *orbits*) of the vector field V^α, and by by construction they are characterised by the fact that at any point x the tangent

vector to the curve passing through that point is the vector $V^\alpha(x)$ at that point. Thus vector fields also generate a *flow* on the space(-time), namely the motion of points along these integral curves, $x^\alpha(\sigma) \mapsto x^\alpha(\sigma+s)$ for $s \in \mathbb{R}$.

(e) An extremely useful related way of thinking about vectors (vector fields) is as first order differential operators, via the correspondence

$$V^\alpha \Leftrightarrow V := V^\alpha \partial_\alpha \quad . \tag{4.15}$$

This operator gives the directional derivative along the flow of the vector field V, as in

$$\frac{d}{d\sigma} f(x(\sigma)) = V^\alpha \partial_\alpha f(x(\sigma)) \tag{4.16}$$

(by the chain rule). One of the advantages of this point of view is that the object V is completely invariant under coordinate transformations as the *components* V^α of V transform inversely to the *basis vectors* ∂_α. For more on this see sections 4.6 and 4.8 on the coordinate-independent interpretation of tensors below.

3. Covectors

A covector (field) is an object $U_\alpha(x)$ which under a coordinate transformation transforms inversely to a vector, i.e. as

$$\bar{U}_\mu(y(x)) \equiv U_\mu(y(x)) = J^\alpha_\mu(y(x)) U_\alpha(x) \quad . \tag{4.17}$$

A familiar example of a covector is the derivative $U_\alpha = \partial_\alpha f$ of a function (scalar) which of course transforms as

$$\partial_\mu \bar{f}(y(x)) = J^\alpha_\mu \partial_\alpha f(x) \quad , \tag{4.18}$$

because that is how partial derivatives transform.

REMARKS:

(a) As in the case of covectors of special relativity (1.44), one should think of covectors pointwise as elements of the dual vector space $\mathbb{V}^*$ to the space of vectors $\mathbb{V}$, i.e. as linear functionals on the space of vectors, given by

$$U(x): \quad V^\alpha(x) \;\mapsto\; U_\alpha(x) V^\alpha(x) \;\in\; \mathbb{R} \quad . \tag{4.19}$$

The transformation properties of U_α and V^α guarantee that the result is a scalar (function) under coordinate transformations,

$$U_\alpha V^\alpha = \bar{U}_\mu \bar{V}^\mu \quad . \tag{4.20}$$

(b) Just as it was useful to think of a vector field $V^\alpha(x)$ in a more coordinate independent way as the components of the coordinate-independent object $V = V^\alpha \partial_\alpha$ with respect to the basis ∂_α, one can think of covector fields U_α as the components of an object

$$U = U_\alpha dx^\alpha \tag{4.21}$$

(which is invariant under coordinate transformations) with respect to the basis dx^α. The prime example is again the *differential*

$$df = \partial_\alpha f(x) dx^\alpha = \partial_\mu \bar{f}(y) dy^\mu \tag{4.22}$$

of a scalar.

(c) Combining the two points of view in the remarks above, one can thus think of df as the linear functional on vector fields that assigns to a vector field V the scalar which is the derivative of f along V,

$$\begin{aligned} df: &\quad \text{vector fields} \longrightarrow \text{scalar fields} \\ &\quad df(V) = V^\alpha \partial_\alpha f \equiv Vf \ . \end{aligned} \tag{4.23}$$

4. Covariant 2-Tensors

Clearly, given the above objects, we can construct more general objects which transform in a nice way under coordinate transformations by taking products of them. Tensors in general are objects which transform like (but need not be equal to) products of vectors and covectors.

In particular, a covariant 2-tensor, or (0,2)-tensor, is an object $A_{\alpha\beta}$ that transforms under coordinate transformations like the product of two covectors, i.e.

$$\bar{A}_{\mu\nu}(y(x)) \equiv A_{\mu\nu}(y(x)) = J^\alpha_\mu(y(x)) J^\beta_\nu(y(x)) A_{\alpha\beta}(x) \ . \tag{4.24}$$

We already know one example of such a tensor, namely the metric tensor $g_{\alpha\beta}$ (which happens to be a symmetric tensor).

In order to avoid unnecessary clutter and to make tensorial equations more readable, I will from now on typically use a shorthand notation in which I drop not only the bars, overlines or other decorations on the transformed objects but in which I also omit the argument x or $y(x)$. In this notation, the above equation would then simply become

$$A_{\mu\nu} = J^\alpha_\mu J^\beta_\nu A_{\alpha\beta} \ , \tag{4.25}$$

which is more user-friendly.

5. Contravariant 2-Tensors

Likewise we define a contravariant 2-tensor (or a (2,0)-tensor) to be an object $B^{\alpha\beta}$ that transforms like the product of two vectors,

$$B^{\mu\nu} = J^{\mu}_{\alpha} J^{\nu}_{\beta} B^{\alpha\beta} \quad . \tag{4.26}$$

An example is the inverse metric tensor $g^{\alpha\beta}$.

6. (p, q)-Tensors

   It should now be clear how to define a general (p, q)-tensor - namely as an object with p contravariant (upper) and q covariant (lower) indices which under a coordinate transformation transforms like a product of p vectors and q covectors,

$$T^{\mu_1 \cdots \mu_p}_{\nu_1 \ldots \nu_q} = J^{\mu_1}_{\alpha_1} \ldots J^{\mu_p}_{\alpha_p} \; J^{\beta_1}_{\nu_1} \ldots J^{\beta_q}_{\nu_q} \; T^{\alpha_1 \cdots \alpha_p}_{\beta_1 \cdots \beta_q} \quad . \tag{4.27}$$

Remarks:

1. Note that, in particular, a tensor is zero (at a point) in one coordinate system if and only if the tensor is zero (at the same point) in another coordinate system.

   Thus, any law of nature (field equation, equation of motion) expressed in terms of tensors, say in the form $T^{\alpha_1 \cdots \alpha_p}_{\beta_1 \ldots \beta_q} = 0$, preserves its form under coordinate trasformations and is therefore automatically generally covariant,

$$T^{\alpha_1 \cdots \alpha_p}_{\beta_1 \cdots \beta_q}(x) = 0 \quad \forall x \quad \Leftrightarrow \quad T^{\mu_1 \cdots \mu_p}_{\nu_1 \ldots \nu_q}(y) = 0 \quad \forall y \quad . \tag{4.28}$$

2. An important special example of a tensor is the Kronecker tensor δ^{α}_{β}. That this is indeed a tensor follows from the fact that it can e.g. be written as a contraction of a product of tensors, namely the metric tensor and its inverse,

$$\delta^{\alpha}_{\beta} = g^{\alpha\gamma} g_{\gamma\beta} \quad , \tag{4.29}$$

   and the calculational rules of tensor algebra to be established in section 4.3 below. Calculating explicitly the components with respect to the new coordinates $y^{\mu} = \bar{x}^{\mu}$, one finds

$$\bar{\delta}^{\mu}_{\nu} = J^{\mu}_{\alpha} J^{\beta}_{\nu} \; \delta^{\alpha}_{\beta} = J^{\mu}_{\alpha} J^{\alpha}_{\nu} = \delta^{\mu}_{\nu} \quad . \tag{4.30}$$

   Thus the Kronecker tensor has the same components in all coordinate systems. This is reassuring but should not be too surprising. Together with scalars and products of scalars and Kronecker tensors it is the only tensor whose components are the same in all coordinate systems.

3. A covariant 2-tensor $T_{\alpha\beta}$, say, is said to be symmetric if $T_{\alpha\beta} = T_{\beta\alpha}$ and anti-symmetric if $T_{\alpha\beta} = -T_{\beta\alpha}$. This is well-defined because it is a generally covariant notion: a tensor is symmetric in all coordinate system iff it is symmetric in one coordinate system, etc.

This definition can be extended to any or all pairs of covariant indices or pairs of contravariant indices. Thus e.g. a tensor $T^{\alpha_1\ldots\alpha_p}$ is called totally symmetric (or totally anti-symmetric) if it is symmetric (anti-symmetric) under the exchange of any pair of indices.

On the other hand, it is not meaningful to talk of the symmetry of a (1,1)-tensor, say, as an equation like $T^{\alpha}_{\beta} = T^{\beta}_{\alpha}$ does not make any sense.

Symmetrisation and anti-symmetrisation of tensors will be discussed in section 4.3 below.

4. The number of independent components of a general (p, q)-tensor is 4^{p+q}. The number of independent components is reduced if the tensor has some symmetry properties. Thus a symmetric (0,2)- or (2,0)-tensor has $4 \times 5/2 = 10$ independent components, an anti-symmetric (0,2)- or (2,0)-tensor has $4 \times 3/2 = 6$ independent components, and a totally anti-symmetric $(0, 4)$-tensor $T_{\beta_1\ldots\beta_4}$ has only got one independent component, namely T_{0123} (all the others being determined by anti-symmetry).

5. Important examples of non-tensors are the Christoffel symbols. Another important example is the the ordinary partial derivative of a (p, q)-tensor, $\partial_\gamma T^{\alpha_1\ldots\alpha_p}_{\beta_1\ldots\beta_q}$ which is *not* a $(p, q+1)$-tensor unless $p = q = 0$. This failure of the partial derivative to map tensors to tensors will motivate us below to introduce a *covariant derivative* which generalises the usual notion of a partial derivative and has the added virtue of mapping tensors to tensors.

4.3 Tensor Algebra

Tensors can be added, multiplied and contracted in certain obvious ways. The basic algebraic operations are the following:

1. Linear Combinations

   Given two (p, q)-tensors $A^{\alpha_1\ldots\alpha_p}_{\beta_1\ldots\beta_q}$ and $B^{\alpha_1\ldots\alpha_p}_{\beta_1\ldots\beta_q}$, their sum

   $$C^{\alpha_1\ldots\alpha_p}_{\beta_1\ldots\beta_q} = A^{\alpha_1\ldots\alpha_p}_{\beta_1\ldots\beta_q} + B^{\alpha_1\ldots\alpha_p}_{\beta_1\ldots\beta_q} \tag{4.31}$$

   is also a (p, q)-tensor.

2. Direct Products

   Given a (p, q)-tensor $A^{\alpha_1\ldots\alpha_p}_{\beta_1\ldots\beta_q}$ and a (p', q')-tensor $B^{\gamma_1\ldots\gamma_{p'}}_{\delta_1\ldots\delta_{q'}}$, their direct product

   $$A^{\alpha_1\ldots\alpha_p}_{\beta_1\ldots\beta_q} B^{\gamma_1\ldots\gamma_{p'}}_{\delta_1\ldots\delta_{q'}} \tag{4.32}$$

   is a $(p + p', q + q')$-tensor,

3. Contractions

   Given a (p,q)-tensor with p and q non-zero, one can associate to it a $(p-1,q-1)$-tensor via contraction of (i.e. summation over) one covariant and one contravariant index, for example

$$A^{\alpha_1\ldots\alpha_p}_{\beta_1\ldots\beta_q} \quad \to \quad B^{\alpha_1\ldots\alpha_{p-1}}_{\beta_1\ldots\beta_{q-1}} = A^{\alpha_1\ldots\alpha_{p-1}\gamma}_{\beta_1\ldots\beta_{q-1}\gamma} \ . \tag{4.33}$$

   This operation of contraction generalises the trace. In general, there are in total $(p \times q)$ distinct possibilities to do this, by contracting the 1st oder 2nd or ... or p'th upper index with the 1st or 2nd or ... or q'th lower index.

   That this indeed leads to a $(p-1,q-1)$-tensor follows from the fact that the Jacobi matrices corresponding to the two contracted indices are inverses to each other, and are multiplied (by the contraction), resulting in the identity matrix. This then leaves only $(p-1)$ Jacobi matrices and $(q-1)$ inverse Jacobi matrices, corresponding to the transformation behaviour of a $(p-1,q-1)$-tensor.

   For example, for a $(1,1)$-tensor T^α_β one has the unique contraction

$$T^\alpha_\beta \to T^\alpha_\alpha \ . \tag{4.34}$$

   This is just the standard trace of the linear tranformation represented by the $(1,1)$-tensor T^α_β. Under the coordinate transformation $x^\alpha \to y^\mu$ this trace transforms as

$$T^\mu_\mu = J^\mu_\alpha J^\beta_\mu T^\alpha_\beta = \delta^\beta_\alpha T^\alpha_\beta = T^\alpha_\alpha \ . \tag{4.35}$$

   Thus the trace is invariant (i.e. a scalar or $(0,0)$-tensor). Clearly this cancellation between the two Jacobi matrices corresponding to two contracted indices also takes place when the tensor T carries other non-contracted indices, and therefore the contraction of a (p,q)-tensor is a $(p-1,q-1)$-tensor, as claimed.

4. Raising and Lowering of Indices

   These operations can of course be combined in various ways. A particular important operation is, given a metric tensor, the lowering of indices with the metric, and the raising of indices with the inverse metric.

   In particular, by direct product with the metric and subsequent contraction, we can construct a covector from a vector by

$$V^\alpha \to g_{\alpha\beta} V^\beta \ . \tag{4.36}$$

   When it is clear from the context which metric is being used, it is common and convenient to simply call this covector V_α,

$$V_\alpha \equiv g_{\alpha\beta} V^\beta \ . \tag{4.37}$$

Likewise, to a covector one can associate a vector by using the inverse metric,

$$A_\alpha \to g^{\alpha\beta}A_\beta \equiv A^\alpha \quad . \tag{4.38}$$

REMARKS:

(a) Interpreted in terms of covectors as linear functions on vectors, this perhaps somewhat obscure convention and notation has a perfectly natural interpretation. Namely, recall that even though a vector space $\mathbb{V}$ and its dual $\mathbb{V}^*$ are isomorphic (in finite dimensions), there is no natural isomorphism between them, i.e. no natural identification of vectors in $\mathbb{V}$ with covectors in $\mathbb{V}^*$. However, if one has a scalar product (metric) $< v, w >$ on $\mathbb{V}$, then this provides an identification of $\mathbb{V}$ and $\mathbb{V}^*$ through

$$v \in \mathbb{V} \mapsto \alpha_v \in \mathbb{V}^* : \quad \alpha_v(w) =< v, w > \quad . \tag{4.39}$$

Thus a metric allows one to associate a covector to a vector, and in the notation of tensor algebra favoured by physicists this is conveniently just written as $V^\alpha \mapsto V_\alpha$.

(b) This convention of using the metric and its inverse to lower and raise indices on a tensor can of course be extended to higher rank tensors, but the result will in general depend on which index is lowered or raised. One then needs to make sure that the notation is sufficiently unambiguous to keep track of this. For example, the two (in general distinct) (1,1)-tensors one obtains from the lowering of one index of a (2,0)-tensor might be denoted amd distinguished by

$$g_{\gamma\alpha}T^{\alpha\beta} = T_\gamma{}^\beta \quad , \quad g_{\gamma\beta}T^{\alpha\beta} = T^\alpha{}_\gamma \quad . \tag{4.40}$$

(c) Finally note that this notation of raising and lowering indices with the metric is consistent with denoting the inverse metric by raised indices, i.e. it is indeed true that

$$g^{\alpha\beta} = g^{\alpha\gamma}g^{\beta\delta}g_{\gamma\delta} \tag{4.41}$$

(both indices of $g_{\gamma\delta}$ raised with two inverse metrics). That this is indeed correct follows from

$$g^{\alpha\gamma}g^{\beta\delta}g_{\gamma\delta} = g^{\alpha\gamma}\delta^\beta_\gamma = g^{\alpha\beta} \quad . \tag{4.42}$$

5. Symmetrisation and anti-Symmetrisation

Given any $(0,2)$-tensor $T_{\alpha\beta}$, say, one can decompose it into its symmetric and anti-symmetric parts as

$$T_{\alpha\beta} = \tfrac{1}{2}(T_{\alpha\beta} + T_{\beta\alpha}) + \tfrac{1}{2}(T_{\alpha\beta} - T_{\beta\alpha}) \equiv T_{(\alpha\beta)} + T_{[\alpha\beta]} \quad . \tag{4.43}$$

The decomposition into symmetric and anti-symmetric parts is invariant under coordinate transformations. In particular, when $T_{\alpha\beta}$ is a tensor, also $T_{(\alpha\beta)}$ and

$T_{[\alpha\beta]}$ are tensors, and thus (anti-)symmetrisation is yet another linear operation that one can perform on tensors.

The factor $\frac{1}{2}$ is chosen such that the symmetrisation of a symmetric tensor is the same as the original tensor,

$$T_{\alpha\beta} = T_{\beta\alpha} \quad \Rightarrow \quad T_{(\alpha\beta)} = T_{\alpha\beta} \quad , \quad T_{[\alpha\beta]} = 0 \tag{4.44}$$

(and likewise for the anti-symmetrisation of anti-symmetric tensors).

This can be generalised to the (anti-)symmetrisation of any pair of (contravariant or covariant) indices; e.g.

$$T_{(\alpha\beta)\gamma} = \tfrac{1}{2}(T_{\alpha\beta\gamma} + T_{\beta\alpha\gamma}) \tag{4.45}$$

is the symmetrisation of $T_{\alpha\beta\gamma}$ in its first and second index. It can also be generalised to the total (anti-)symmetrisation of a higher-rank tensor; e.g.

$$T_{(\alpha\beta\gamma)} \equiv \tfrac{1}{3!}(T_{\alpha\beta\gamma} + T_{\beta\alpha\gamma} + T_{\gamma\beta\alpha} + T_{\beta\gamma\alpha} + T_{\alpha\gamma\beta} + T_{\gamma\alpha\beta}) \tag{4.46}$$

is totally symmetric, i.e. symmetric under the exchange of any pair of indices, and

$$T_{[\alpha\beta\gamma]} \equiv \tfrac{1}{3!}(T_{\alpha\beta\gamma} - T_{\beta\alpha\gamma} - T_{\gamma\beta\alpha} + T_{\beta\gamma\alpha} - T_{\alpha\gamma\beta} + T_{\gamma\alpha\beta}) \tag{4.47}$$

is totally anti-symmetric. The prefactor $\frac{1}{6}$ is again there to ensure that the total symmetrisation of a totally symmetric tensor is the original tensor (and likewise for the total anti-symmetrisation of totally anti-symmetric tensors). This generalises in an evident way to higher rank p tensors, with the combinatorial prefactor $1/p!$.

An observation we will frequently make use of to recognise when some object is a tensor is the following (occasionally known as the *quotient theorem* or quotient lemma):

For example, say that in an equation of the form

$$A_\alpha = B_{\alpha\beta} C^\beta \tag{4.48}$$

you know that A_α transforms as a covector for any vector C^β. Then it follows that $B_{\alpha\beta}$ has to be a tensor. Likewise for higher rank tensors and more contractions, as in

$$A_{\alpha\beta} = B_{\alpha\beta\gamma\delta} C^{\gamma\delta} \quad . \tag{4.49}$$

Also in that case one has the statement that if A transforms as a tensor for every tensor C, then B itself has to be a tensor.

An elementary and ham-handed proof of these kinds of statements can be obtained by contradiction: assume that B does not transform as a tensor and write its transformation, as in (4.3), as

$$B' = (\ldots)B + \text{junk} \quad . \tag{4.50}$$

If "junk" $\neq 0$, then there will be some C such that "junk" contributes to the contraction $B'C'$. That means that "junk" contributes to A', the transformed A, contradicting the premise that A is a tensor.

4.4 Generally Covariant Integration and Volume Elements

While tensors are the objects which, in a sense, transform in the nicest and simplest possible way under coordinate transformations, they are not the only relevant objects. An important class of non-tensors (but "almost" tensors) are so-called *tensor densities*. They will play a crucial role for us in order to have a generally-covariant notion of integration at our disposal, and thus ultimately also a way of writing down generally covariant action principles for fields etc.

In this section we will address the issue of generally covariant integration in a space-time equipped with a metric. This will be accomplished with the help of a particular tensor density constructed from the metric. Having thus established that tensor densities are objects of legitimate interest in their own right, we will then discuss their properties in more generality in section 4.5 below.

To set the stage, consider once again first the situation in special relativity. In that case, the integral of a Lorentz scalar $f(\xi)$ with respect to the volume element $d^4\xi$ (or $d^{27}\xi$ ...) is itself a Lorentz scalar, i.e. independent of the inertial reference frame in which the integral is evaluated,

$$\int d^4\xi \; f(\xi) = \int d^4\bar{\xi} \; \bar{f}(\bar{\xi}) \qquad (\bar{\xi}^a = L^a_b \xi^b) \tag{4.51}$$

The reasons for this are that

1. f is a scalar by assumption,
$$\bar{f}(\bar{\xi}) = f(\xi) \;\; , \tag{4.52}$$

2. by the fundamental theorem of integral calculus, under an arbitrary coordinate transformation $\xi \to \bar{\xi} = \bar{\xi}(\xi)$ the volume element transforms with the *Jacobian*, the (absolute value of the) determinant of the Jacobi matrix,
$$d^4\bar{\xi} = \left| \det \frac{\partial \bar{\xi}}{\partial \xi} \right| d^4\xi \;\; , \tag{4.53}$$

3. for a Lorentz transformation one has
$$\left| \det \frac{\partial \bar{\xi}}{\partial \xi} \right| = |\det L| = 1 \;\; . \tag{4.54}$$
and thus the volume element is invariant under Lorentz transformations,
$$d^4\bar{\xi} = d^4\xi \;\; . \tag{4.55}$$

Turning now to general relativity and general covariance, it is immediately apparent that the integral of a scalar $\int d^4x \; f(x)$ will *not* be generally covariant, i.e. under a general coordinate transformation $x \to y = y(x)$ generically one has

$$\int d^4x \; f(x) \neq \int d^4y \; \bar{f}(y)(y) \tag{4.56}$$

because of the non-trivial Jacobian,

$$d^4y = \left|\det\left(\frac{\partial y}{\partial x}\right)\right| d^4x \ . \tag{4.57}$$

One way out would be to abandon the idea that one should integrate scalars and to require that the integrand $\rho(x)$ should transform in such a way that it cancels the Jacobian arising from the measure, namely as

$$\bar{\rho}(y) = \left|\det\left(\frac{\partial y}{\partial x}\right)\right|^{-1} \rho(x) \ . \tag{4.58}$$

This is indeed an option, and we will return to this below (see remark 1 in section 4.5), but at this stage this is rather unintuitive and not particularly useful, in particular because it is not clear how one should go about finding or constructing such objects in the first place.

A first simplification arises from the fact that (4.58) implies that

- if $\rho(x)$ satisfies (4.58), then so will $f(x)\rho(x)$ for any scalar $f(x)$;
- and conversely, if $\rho_1(x)$ and $\rho_2(x)$ satisfy (4.58), then $\rho_2(x)/\rho_1(x) \equiv f(x)$ is a scalar.

Thus, in order to find all objects that satisfy (4.58), we just need to find a single one, and with respect to that one as our measure we can then integrate any scalar function $f(x)$ in a generally covariant way,

$$\bar{\rho}(y) = \left|\det\left(\frac{\partial y}{\partial x}\right)\right|^{-1} \rho(x) \quad \Rightarrow \quad \int d^4y\ \bar{\rho}(y)\bar{f}(y) = \int d^4x\ \rho(x)f(x) \tag{4.59}$$

In order to find one candidate $\rho(x)$, let us approach this question in a different way. Integrals are used to calculate or measure volumes (or areas, or lenghts, or ...). Such integrals should have a coordinate-independent meaning, but they should depend on the prescription one uses for *measuring* volumes, areas, lenghts, ... These prescriptions are concisely encoded in the *metric*. Thus it is plausible that in order to define a generally covariant notion of integration one may need to specify the metric, but that this is all that one should need to know (while the Jacobian between two coordinate systems should fundamentally be irrelevant and be considered to be a red herring).

With this in mind, let us recall the standard tensorial transformation behaviour of the metric under coordinate transformations,

$$\bar{g}_{\alpha\beta}(y) = \frac{\partial x^\mu}{\partial y^\alpha}\frac{\partial x^\nu}{\partial y^\beta} g_{\mu\nu}(x) \ . \tag{4.60}$$

It follows from this that the *absolute value of the determinant of the metric*

$$g := |\det(g_{\mu\nu}(x))| \tag{4.61}$$

does not transform like a scalar or some other tensor at all, but instead transforms as

$$\bar{g} = \left|\det\left(\frac{\partial x}{\partial y}\right)^2\right| g = \left|\det\left(\frac{\partial y}{\partial x}\right)\right|^{-2} g \ . \tag{4.62}$$

In particular, its square-root $\sqrt{g}$ transforms as

$$\sqrt{\bar{g}} = \left|\det\left(\frac{\partial y}{\partial x}\right)\right|^{-1} \sqrt{g} \ . \tag{4.63}$$

Thus it satisfies (4.58) and provides us with our required reference object

$$\rho(x) = \sqrt{g(x)} \ . \tag{4.64}$$

In particular, the combined expression $\sqrt{g}d^4x$ is *invariant* under general coordinate transformations,

$$\sqrt{\bar{g}}d^4y = \sqrt{g}d^4x \ , \tag{4.65}$$

and can therefore be used to define integrals of scalars in a generally covariant (but metric-dependent) way,

$$\int \sqrt{\bar{g}}d^4y \ \bar{f}(y) = \int \sqrt{g}d^4x \ f(x) \ . \tag{4.66}$$

This will of course be important in order to formulate action principles etc. in a space-time equipped with a metric in a generally covariant way.

REMARKS:

1. This is also frequently the quickest way to determine the volume element in non-Cartesian coordinates in Euclidean space. In Cartesian coordinates one has $\sqrt{g} = 1$, so the correct integration measure is just the familiar

$$\sqrt{g}d^3x = d^3x \ . \tag{4.67}$$

To now determine what is the volume element in spherical coordinates $\{y^k\} = (r, \theta, \phi)$, say, instead of laboriously determining the Jacobi matrix for the coordinate transformation, and then (equally laboriously) calculating its determinant (which would be the standard uninspiring and uninspired procedure), all one needs to know is the metric in these coordinates (which one usually needs to determine anyway) to deduce

$$ds^2 = dr^2 + r^2(d\theta^2 + \sin^2\theta d\phi^2) \quad \Rightarrow \quad \bar{g} = r^4 \sin^2\theta \tag{4.68}$$

and therefore

$$d^3x = \sqrt{\bar{g}} \ d^3y = r^2 \sin\theta \ dr \ d\theta \ d\phi \ . \tag{4.69}$$

This is of course the standard result.

2. As a variation of this theme, it is now also easy to construct a spherical (i.e. radial + angular) coordinate system z^k in which $\sqrt{g} = 1$, i.e. which is such that simply $d^3z = d^3x$ (without any Jacobian factors). To that end it suffices to let (4.69) suggest to introduce a new radial variable $\rho = \rho(r)$ by

$$d\rho = r^2 dr \quad \Rightarrow \quad \rho = r^3/3 \in (0, \infty) \tag{4.70}$$

and a new angular variable $\psi = \psi(\theta)$ by

$$d\psi = \sin\theta d\theta \quad \Rightarrow \quad \psi = -\cos\theta \in [-1, 1) \ . \tag{4.71}$$

In these coordinates, the Euclidean line element takes the form

$$\begin{aligned} ds^2 &= r(\rho)^{-4} d\rho^2 + r(\rho)^2 (d\psi^2/\sin^2\theta(\psi) + \sin^2\theta(\psi) d\phi^2) \\ &= (3\rho)^{-4/3} d\rho^2 + (3\rho)^{2/3} \left(\frac{d\psi^2}{1-\psi^2} + (1-\psi^2) d\phi^2 \right) \end{aligned} \tag{4.72}$$

It is now manifest that in these coordinates $\{z^k\} = (\rho, \psi, \phi)$ one has

$$\sqrt{g} = 1 \quad \Rightarrow \quad d^3x = d^3z = d\rho\, d\psi\, d\phi \ . \tag{4.73}$$

The Euclidean metric in these coordinates will make a brief appearance in the discussion of the derivation of the Schwarzschild metric in section 24.3 (cf. the discussion leading to (24.33)).

3. More generally, given any metric $g_{\alpha\beta}$, one can find a coordinate system in which $g = |\det(g_{\alpha\beta})| = 1$. Such a ("*unimodular*") coordinate system is highly non-unique. In fact, given any such coordinate system x, any other coordinate system y with unit Jacobi matrix $|\det(\partial y/\partial x)| = 1$ will also satisfy this condition. Such coordinate transformations form an infinite-dimensional subgroup of the group of all coordinate transformations, known as the group of *volume-preserving coordinate transformations*.

   In particular, the above transformation from Cartesian coordinates x^k to the coordinates z^k is a non-trivial example of a volume-preserving coordinate transformation.

4.5 Tensor Densities and Volume Elements

In the previous section we have encountered certain not strictly tensorial objects which nevertheless turned out to be useful. Having thus established the basic credentials of such objects, we will now formalise this somewhat.

Thus the prime example of what we will call a tensor density is the (absolute value of the) determinant $g := |\det g_{\mu\nu}|$ of the metric tensor, which, as we have seen, transforms as

$$\bar{g} = \left| \det\left(\frac{\partial y}{\partial x} \right) \right|^{-2} g \ . \tag{4.74}$$

An object which transforms in such a way under coordinate transformations is called a *scalar tensor density of weight* $w = +2$, and the square root of the determinant $\sqrt{g}$ transforms as, and hence is, a tensor density of weight $w = +1$.

In general, a tensor density of weight w is an object that transforms as a tensor, with an additional factor of $|\det(\partial y/\partial x)|^{-w}$. It thus transforms like a tensor multiplied by $g^{w/2}$, and one has

$$\tilde{T} \quad \text{a tensor density of weight } w \quad \Leftrightarrow \quad \tilde{T} = g^{w/2} T \quad \text{with } T \text{ a tensor} \ . \tag{4.75}$$

The algebraic rules for tensor densities are strictly analogous to those for tensors. Thus, for example, the sum of two (p, q) tensor densities of weight w (let us call this a $(p, q; w)$ tensor) is again a $(p, q; w)$ tensor, and the direct product of a $(p_1, q_1; w_1)$ and a $(p_2, q_2; w_2)$ tensor is a $(p_1 + p_2, q_1 + q_2; w_1 + w_2)$ tensor. Contractions and the raising and lowering of indices of tensor densities can also be defined just as for ordinary tensors.

Remarks:

1. Generalising the argument in section 4.4, we now learn that if ρ is any scalar density of weight $w = +1$, then its integral is well-defined and coordinate independent,

$$\int d^4x \ \rho(x) = \int d^4y \ \bar{\rho}(y) \ . \tag{4.76}$$

   See remark 4 at the end of this section for one way of constructing such objects without taking recourse to a metric.

2. There is one important tensor density which - like the Kronecker tensor - has the same components in all coordinate systems. This is the totally anti-symmetric Levi-Civita symbol $\in_{\alpha\beta\gamma\delta}$ (taking the values $0, \pm 1$) which is a tensor density of weight $w = -1$. Then $\sqrt{g} \in_{\alpha\beta\gamma\delta}$ is a tensor (strictly speaking it is a pseudo-tensor because of its behaviour under reversal of orientation - see below).

   To see this, recall first of all the definition of the Levi-Civita symbol: it is totally anti-symmetric,

$$\in_{\alpha\beta\gamma\delta} = \in_{[\alpha\beta\gamma\delta]} \ , \tag{4.77}$$

   and has therefore only got one independent component which we will normalise to be

$$\in_{0123} = +1 \ . \tag{4.78}$$

   Thus $\in_{\alpha\beta\gamma\delta} = +1$ if the indices $(\alpha\beta\gamma\delta)$ are an even permutation of (0123), $\in_{\alpha\beta\gamma\delta} = -1$ if the indices $(\alpha\beta\gamma\delta)$ are an odd permutation of (0123), and $\in_{\alpha\beta\gamma\delta} = 0$ iff any two indices are equal. This definition makes no reference to any coordinate system whatsoever, and thus tautologically the purely combinatorial object $\in_{\alpha\beta\gamma\delta}$ has the same components in all coordinate systems.

This evidently extends to other dimensions, and we will define the $D = (d+1)$-dimensional Levi-Civita symbol in the same way,

$$\in_{\alpha_1 \ldots \alpha_D} = \in_{[\alpha_1 \ldots \alpha_D]} \quad , \quad \in_{01\ldots d} = +1 \ . \tag{4.79}$$

Next, recall one possible definition of the determinant $\det M$ of a $(D \times D)$-matrix $M^\mu_{\ \nu}$, namely as the coefficient (proportionality factor) on the right-hand side of

$$\in_{\alpha_1 \ldots \alpha_D} \ M^{\alpha_1}_{\ \beta_1} \ldots M^{\alpha_D}_{\ \beta_D} = (\det M) \ \in_{\beta_1 \ldots \beta_D} \ . \tag{4.80}$$

Now choose M to be the Jacobi matrix $J^\mu_\alpha = (\partial y^\mu / \partial x^\alpha)$. Then the above equation can be rearranged to

$$\in_{\mu_1 \ldots \mu_D} = \det \left(\frac{\partial y}{\partial x} \right) J^{\alpha_1}_{\mu_1} \ldots J^{\alpha_D}_{\mu_D} \in_{\alpha_1 \ldots \alpha_D} \ . \tag{4.81}$$

This shows that $\in_{\alpha_1 \ldots \alpha_D}$ transforms as a tensor density of weight $w = -1$, provided that $\det(\partial y / \partial x) > 0$. The latter condition means that the coordinate transformation preserves the orientation. Thus, $\in_{\alpha_1 \ldots \alpha_D}$ transforms as a tensor density under orientation-preserving coordinate transformations but picks up a sign when the orientation is reversed. Strictly speaking $\in_{\alpha_1 \ldots \alpha_D}$ is then not a tensor density but something that is called a *pseudo-tensor density*.

Going back to 4 dimensions, it follows that

$$\epsilon_{\alpha\beta\gamma\delta} \equiv \sqrt{g} \in_{\alpha\beta\gamma\delta} \tag{4.82}$$

is a totally anti-symmetric $(0,4)$ (pseudo-)tensor. Likewise, the totally anti-symmetric symbol $\in^{\alpha\beta\gamma\delta}$ is a tensor density of weight $w = +1$ and

$$\epsilon^{\alpha\beta\gamma\delta} \equiv \frac{1}{\sqrt{g}} \in^{\alpha\beta\gamma\delta} \tag{4.83}$$

is a totally anti-symmetric $(4,0)$ (pseudo-)tensor. Here, as usual, we have raised the indices of the tensor on the left-hand side with the metric, and $\in^{\alpha\beta\gamma\delta}$ is totally anti-symmetric, with $\in^{0123} = -1$. The minus-sign arises because the contraction with the metrics on the left-hand side produces a factor of

$$\sqrt{g} \det(g^{\alpha\beta}) = \sqrt{g} (\det(g_{\alpha\beta}))^{-1} = -\frac{1}{\sqrt{g}} \ . \tag{4.84}$$

We could have chosen to not absorb the minus sign into the definition of $\in^{\alpha\beta\gamma\delta}$, at the expense of an explicit minus sign on the right-hand side of (4.83). The convention we have adopted is more convenient, however, in particular since it is compatible with the standard practice in special relativity to (tacitly) identify $\epsilon_{abcd} = \in_{abcd}$, the minus sign arising from raising the indices on $\in_{abcd}$ with the Minkowski metric η^{ab} with $\eta^{00} = -1$, so that $\in^{0123} = - \in_{0123}$.

3. There is an intimate relation between the preceding observations regarding the Levi-Civita symbol (remark 2) and those in section 4.4 above regarding invariant volume elements. Namely, the usual coordinate volume element d^4x can be written as

$$d^4x = \frac{1}{4!} \in_{\alpha\beta\gamma\delta} dx^\alpha dx^\beta dx^\gamma dx^\delta \quad . \tag{4.85}$$

This is not a tensor but transforms like a scalar density. On the other hand, if one works instead with the tensor $\epsilon_{\lambda\mu\nu\rho}$ one obtains a scalar, and this scalar is precisely the invariant volume element (4.65),

$$\frac{1}{4!}\epsilon_{\alpha\beta\gamma\delta} dx^\alpha dx^\beta dx^\gamma dx^\delta = \sqrt{g} d^4x \quad . \tag{4.86}$$

4. More generally (and without invoking a metric to provide the weight $w = +1$ density required for an invariant integration) this can be phrased in the following way: Let $A_{\alpha\beta\gamma\delta}$ be a totally anti-symmetric $(0,4)$-tensor. Thus it will be proportional to the metric dependent Levi-Civita tensor $\epsilon_{\alpha\beta\gamma\delta}$, but we will now not make use of this fact (which would return us to the setting of the previous remark). Rather, we consider its contraction with the contravariant Levi-Civita symbol $\in^{\alpha\beta\gamma\delta}$ (which exists independently of any additional structure like a metric),

$$A_{\alpha\beta\gamma\delta} \to \tilde{A} = \in^{\alpha\beta\gamma\delta} A_{\alpha\beta\gamma\delta} \quad . \tag{4.87}$$

This is now clearly a scalar density of weight $w = +1$. As a consequence, its integral (4.76) is well-defined and coordinate independent, without reference to any metric. Thus totally anti-symmetric $(0,4)$-forms provide natural 4-dimensional volume elements (and likewise for totally anti-symmetric $(0,p)$ tensors and p-dimensional volume elements).

4.6 Towards a Coordinate-Independent Interpretation of Tensors

There is a more invariant and coordinate-independent way of looking at tensors than we have developed so far. The purpose of this section (and the subsequent section 4.7) is to briefly explain this point of view, even though it is not indispensable for an understanding of the remainder of the course.

Consider first of all the differential

$$df = \partial_\alpha f(x) dx^\alpha \tag{4.88}$$

of a function (scalar field) $f = f(x)$. This is clearly a coordinate-independent object, because partial deriviatives and the differentials dx^α transform inversely to each other,

$$dy^\mu = J^\mu_\alpha dx^\alpha \quad , \quad \partial_\mu = J^\alpha_\mu \partial_\alpha \quad \Rightarrow \quad \partial_\alpha f(x) dx^\alpha = \partial_\mu \bar{f}(y) dy^\mu \quad . \tag{4.89}$$

This suggests that it is useful to regard the quantities $\partial_\alpha f$ as the coefficients of the coordinate independent object df in a particular coordinate system, namely when df is expanded in the basis $\{dx^\alpha\}$.

We can do the same thing for any covector A_α. If A_α is a covector (i.e. transforms like one under coordinate transformations), then $A := A_\alpha(x)dx^\alpha$ is coordinate-independent, and it is useful to think of the A_α as the coefficients of the covector A when expanded in a coordinate basis, $A = A_\alpha dx^\alpha$. Linear combinations of dx^α built in this way from covectors are known as *1-forms*.

From this point of view, we interpret the $\{A_\alpha\}$ simply as the (coordinate dependent) components of the (coordinate independent) 1-form A when expressed with respect to the (coordinate dependent) differentials $\{dx^\alpha\}$, considered as a basis of the space of covectors.

Something similar can be done for vector fields. Just as covectors transform inversely to coordinate differentials, vectors V^α transform inversely to partial derivatives ∂_α. Thus

$$V := V^\alpha(x)\frac{\partial}{\partial x^\alpha} \tag{4.90}$$

is coordinate-independent - a coordinate-independent linear first-order differential operator. One can thus always think of a vector field as a 1st order differential operator and this is a very fruitful point of view.

Acting on a function (scalar) f, V produces the derivative of f along V,

$$Vf = V^\alpha \partial_\alpha f \quad . \tag{4.91}$$

This is also a coordinate independent object, a scalar, arising from the contraction of a vector and a covector. And this is as it should be because, after all, both a function and a vector field can be specified on a space-time without having to introduce coordinates (e.g. by simply drawing the vector field and the profile of the function). Therefore also the change of the function along a vector field should be coordinate independent and, as we have seen, it is.

So far we have only discussed vectors and covectors. All this can, in principle, be extended to higher rank tensors, but at this point it would be very useful to introduce the notion (or at least the notation) of tensor products. I will briefly describe this in section 4.7 below.

For those who do not want to delve into this (and it is not required for the following): fact of the matter is that any (p, q)-tensor $T^{\alpha_1\ldots\alpha_p}_{\beta_1\ldots\beta_q}$ can be thought of as the collection of components of a coordinate independent object T when expanded in a particular coordinate basis in terms of the dx^α and $(\partial/\partial x^\alpha)$.

Any choice of coordinate system $\{x^\alpha\}$ gives rise to such a basis $\{dx^\alpha\}$, and such bases are known as *coordinate bases* or *natural bases*. This is not the only possible choice of basis, however, and we will return to this issue in section 4.8.

4.7 Multilinear Algebra and Tensors

In (multi-)linear algebra, the tensor product is used to describe multilinear maps. Let V be a vector space, and V^* its dual, consisting of the linear maps $V \to \mathbb{R}$, and denote the action of $a \in V^*$ on $v \in V$ by

$$a \in V^*, v \in V \quad \to \quad a(v) \in \mathbb{R} \ . \tag{4.92}$$

In components, with respect to a basis E_k in V and its dual basis e^k in V^*,

$$e^i(E_k) = \delta^i_k \ , \tag{4.93}$$

this would be written as

$$(a_i e^i)(v^k E_k) = a_i v^k e^i(E_k) = a_i v^k \delta^i_k = a_k v^k \ . \tag{4.94}$$

Then a bilinear map on the Cartesian product $V \times V$ can be considered as an element of $V^* \otimes V^*$, the tensor product being defined by

$$(a \otimes b)(v, w) = a(v)\ b(w) \ . \tag{4.95}$$

Note that this is not symmetric, i.e.

$$a \otimes b \neq b \otimes a \ . \tag{4.96}$$

Again in components this would read

$$(a \otimes b)(v, w) = a_i b_k v^i w^k \ , \tag{4.97}$$

and a general element of $V^* \otimes V^*$ can be written as an object with components a_{ik} (a covariant tensor),

$$a = a_{ik} e^i \otimes e^k \ , \tag{4.98}$$

acting as

$$a(v, w) = a_{ik} v^i w^k \ . \tag{4.99}$$

From these definitions it follows that the tensor product is evidently linear,

$$a \otimes (b + c) = a \otimes b + a \otimes c \tag{4.100}$$

(and likewise for the first factor), and $\mathbb{R}$-linear, i.e. for $r \in \mathbb{R}$ one has

$$r(a \otimes b) = (ra) \otimes b = a \otimes (rb) \ . \tag{4.101}$$

The $\mathbb{R}$-linearity is in a sense the characteristic feature of the tensor product $V \otimes W$ of two vector spaces (here $V^* \otimes V^*$) that sets it apart from the direct (Cartesian) product $V \times W$ of vector spaces (here $V^* \times V^*$), which consists of the pairs (v, w), and for which there is obviously no identification between (rv, w) and (v, rw) since these are just distinct points of the Cartesian product.

This can be straightforwardly extended to a description of general multilinear maps on vector spaces:

- Using the canonical isomorphism $(V^*)^* \cong V$ for finite-dimensional vector spaces,

$$v \in V \to \hat{v} \in (V^*)^* : \quad \hat{v}(a) = a(v) \ , \tag{4.102}$$

one can also in the same way define the tensor product $V \otimes V$ as the space of bilinear functions on $V^* \times V^*$,

$$(v \otimes w)(a, b) = a(v)\ b(w) \ , \tag{4.103}$$

and a general elelement of $V \otimes V$ can be represented in terms of its components T^{ik},

$$T = T^{ik} E_i \otimes E_k \ , \tag{4.104}$$

acting as

$$T(a, b) = T^{ik} a_i b_k \ . \tag{4.105}$$

- By the same token, the tensor product $V \otimes W$ is the space of bilinear maps on $V^* \otimes W^*$.

- Multilinear maps from $V \times \ldots \times V$ to $\mathbb{R}$ are elements of

$$\otimes^p V^* = \underbrace{V^* \otimes \ldots \otimes V^*}_{p \text{ times}} \tag{4.106}$$

(and can be represented as covariant tensors of rank p), etc.

- These multilinear maps can be added and multipled and thus form an algebra, the *tensor algebra* of V^*, denoted by $T(V^*)$. As a vector space, it consists of the sums of all the p-linear maps,

$$T(V^*) = \oplus_{p=0} \otimes^p V^* \ . \tag{4.107}$$

The tensor product can also be used to describe multilinear maps between vector spaces:

- An element $a \otimes v$ of $V^* \otimes V$ can be regarded as a linear map from V to itself via

$$(a \otimes v_1)(v_2) = a(v_2) v_1 \tag{4.108}$$

and a general element of $V^* \otimes V$ (a "matrix" $M^i_{\ k}$) can be written as a linear combination of such maps,

$$M = M^i_{\ k} E_i \otimes e^k : \quad M(v) = M^i_{\ k} e^k(v) E_i = (M^i_{\ k} v^k) E_i \ . \tag{4.109}$$

- Likewise a linear map from V to some other vector space W can be regarded as an element of $V^* \otimes W$.

- Multilinear maps from V to W ("W-valued multilinear maps") are elements of $V^* \otimes \ldots \otimes V^* \otimes W$, etc.

Clearly, in general, given a basis of V and a dual basis of V^*, the tensor product can be used to construct a basis

$$(E_{i_1} \otimes \ldots \otimes E_{i_p}) \otimes (e^{k_1} \otimes \ldots e^{k_q}) \tag{4.110}$$

in the space

$$T^{p,q} = \underbrace{(V \otimes \ldots \otimes V)}_{p \text{ times}} \otimes \underbrace{(V^* \otimes \ldots \otimes V^*)}_{q \text{ times}} \tag{4.111}$$

of (p,q)-tensors,

$$T \in T^{p,q} : T = T^{i_1 \ldots i_p}_{k_1 \ldots k_q} \ (E_{i_1} \otimes \ldots \otimes E_{i_p}) \otimes (e^{k_1} \otimes \ldots \otimes e^{k_q}) \ . \tag{4.112}$$

This is the way we will use the tensor product notation below, as a multilinear operation providing us with a basis for higher rank tensor fields.

Now, as we have seen above, in the standard component/index formulation of general relativity, say, a (p,q)-tensor is defined as an object with components $T^{\mu_1 \ldots \mu_p}_{\nu_1 \ldots \nu_q}$ which transforms multi-linearly with the Jacobi matrix under coordinate transformations, i.e. under $x^\mu \to y^\alpha$ one has

$$T^{\alpha_1 \ldots \alpha_p}_{\beta_1 \ldots \beta_q}(y) = J^{\alpha_1}_{\mu_1} \ldots J^{\alpha_p}_{\mu_p} \ J^{\nu_1}_{\beta_1} \ldots J^{\nu_q}_{\beta_q} \ T^{\mu_1 \ldots \mu_p}_{\nu_1 \ldots \nu_q}(x(y)) \tag{4.113}$$

where $J^\alpha_\mu = \partial y^\alpha / \partial x^\mu$ is the Jacobi matrix and $J^\mu_\alpha = \partial x^\mu / \partial y^\alpha$ is its inverse.

The reason for introducing and working with tensors, defined in this way, is that tensorial equations have the virtue that they are *generally covariant*, i.e. that they are satisfied in all coordinate system if and only if they are satisfied in one coordinate system. The emphasis in this formulation is thus not on tensors as multilinear maps but on how they transform under coordinate transformations. This seems to be somewhat at odds with the definition of tensors in multilinear algebra, but as we will see below this is simply due to the choice of a particular class of bases (coordinate bases), with respect to which multilinear maps indeed transform in this way under changes of the coordinate basis, i.e. under changes of coordinates.

We had already noted above, that there is a more coordinate independent way of looking at covector fields and vector fields, by associating to them the objects

$$A_\mu(x) \to A(x) = A_\mu(x) dx^\mu \quad , \quad V^\alpha(x) \to V(x) = V^\alpha(x) \partial_\alpha \ . \tag{4.114}$$

which are completely invariant under coordinate transformations, with the dx^μ and the ∂_α providing a basis for the space of covector and vector fields respectively.

This perspective can now be extended to higher-rank and mixed tensors. In particular, associated with the metric $g_{\mu\nu}(x)$ we have the coordinate independent line element

$$ds^2 = g_{\mu\nu} dx^\mu dx^\nu \ . \tag{4.115}$$

which we can now also think of as the tensor

$$g = g_{\mu\nu} dx^\mu \otimes dx^\nu \quad . \tag{4.116}$$

where $\otimes$ is now again the *tensor product*.

Since we are now dealing with tensor fields rather than just with tensors (multilinear maps at a given point), the tensor product in this context is required to be multilinear not just over $\mathbb{R}$, but over functions (scalars) so that e.g.

$$dx^\mu \otimes (f(x) dx^\nu) = (f(x) dx^\mu) \otimes dx^\nu \quad . \tag{4.117}$$

Now let us return to (4.116). If one wants to emphasise that the metric is a symmetric (0,2)-tensor, one can also expand it with respect to the symmetrised basis as

$$g = g_{\mu\nu} \, (dx^\mu \otimes dx^\nu + dx^\nu \otimes dx^\mu)/2 \quad , \tag{4.118}$$

but for the metric the tensor-product is often omitted and one simply writes it as the line element (4.115).

If one has a non-symmetric $(0,2)$-tensor $T_{\mu\nu}$, say, then one can also group these coefficients into the components of a coordinate-invariant object, but now the tensor product notation

$$T_{\mu\nu} \to T = T_{\mu\nu} dx^\mu \otimes dx^\nu \tag{4.119}$$

is more useful than just writing $T_{\mu\nu} dx^\mu dx^\nu$, simply to emphasise the fact that all components of $T_{\mu\nu}$, not just the symmetric part of $T_{\mu\nu}$, contribute to T because $dx^\mu \otimes dx^\nu$ is not symmetric,

$$dx^\mu \otimes dx^\nu \neq dx^\nu \otimes dx^\mu \quad , \tag{4.120}$$

(whereas just writing $dx^\mu dx^\nu$ might lead one to believe that dx^μ and dx^ν commute).

More generally, to a $(0,p)$-tensor we can associate the object

$$T = T_{\mu_1 \dots \mu_p} dx^{\mu_1} \otimes \dots \otimes dx^{\mu_p} \quad . \tag{4.121}$$

If $T_{\mu_1 \cdots \mu_p}$ is totally anti-symmetric, the resulting object is also referred to as a *p-form*. As we have already seen above, such p-forms provide natural and invariant p-dimensional volume elements. In particular, applying this to the Levi-Civita tensor discussed in section 4.5, we reproduce the statement (4.65) that (4.86)

$$\frac{1}{4!} \epsilon_{\alpha\beta\gamma\delta} dx^\alpha dx^\beta dx^\gamma dx^\delta = \sqrt{g} d^4x \tag{4.122}$$

is a space-time volume element that is invariant under coordinate transformations.

The tensor product notation is also useful for higher-rank contravariant or mixed tensors. Given a $(2,0)$-tensor with components $T^{\mu\nu}$, say, one really does not want to write the corresponding coordinate-invariant object as $T^{\mu\nu} \partial_\mu \partial_\nu$, say, because this may be

interpreted as a second order differential operator whereas what one really means is a bilinear first order differential operator, which one writes as

$$T = T^{\mu\nu}\, \partial_\mu \otimes \partial_\nu \ , \tag{4.123}$$

and whose components with respect to the basis $\partial_\mu \otimes \partial_\nu$ are the $T^{\mu\nu}$.

In general, we can thus think of a (p, q)-tensor field, as given in (4.113), as the components of a coordinate-independent object

$$T = T^{\mu_1 \dots \mu_p}_{\nu_1 \dots \nu_q}(x)\ (\partial_{\mu_1} \otimes \dots \partial_{\mu_p}) \otimes (dx^{\nu_1} \otimes \dots \otimes dx^{\nu_q}) \ , \tag{4.124}$$

when expanded with respect to the coordinate basis in the space of tensor fields generated by dx^μ and $\partial_\mu = \partial_{x^\mu}$.

4.8 Vielbeins and Orthonormal Frames

As we saw in section 4.6, a choice of coordinates provides one with a choice of basis for vectors, covectors and other tensors, and a quantity like V^μ is then interpreted as the collection of components of an object $V = V^\mu \partial_\mu$ with respect to the coordinate basis ∂_μ. In classical tensor calculus one always works in such a basis, and with the components of tensors with respect to such a basis. This is very convenient and natural, but this is now clearly not the only choice.

Indeed, the above point of view suggests a reformulation and generalisation that is extremely natural and useful (but that I will nevertheless hardly ever make use of in these notes).

Namley, let $\{e^m_\mu(x)\}$ be such that it is an invertible matrix for every point x. Then another possible choice of basis for the space of covectors are the linear combinations

$$e^m := e^m_\mu dx^\mu \ . \tag{4.125}$$

A general such basis is called a *vielbein*, which is German for *multileg*, quite appropriate actually, as one should visualise this as a bunch of linearly independent (co-)vectors at every point of space-time.

In two, three, and four dimensions these are also known more specifically as *zweibeins, dreibeins* and *vierbeins* respectively. In four dimensions, the Greek word *tetrads* is also commonly used. The e^m are sometimes also referred to as *frame fields*, mostly in the context of *orthonormal frames* (see below).

In general, this new basis is *not* a coordinate basis, i.e. there does not exist a coordinate system $\{y^m\}$ such that $e^m = dy^m$. If such a coordinate system does exist, then one has

$$\begin{aligned} e^m = dy^m \quad &\Rightarrow \quad e^m_\mu = \frac{\partial y^m}{\partial x^\mu} \\ &\Rightarrow \quad \partial_\nu e^m_\mu = \partial_\mu e^m_\nu \ , \end{aligned} \tag{4.126}$$

and locally also the converse is true. In particular, if

$$\partial_\nu e^m_\mu - \partial_\mu e^m_\nu \neq 0 \quad \Rightarrow \quad e^m \quad \text{is not a coordinate basis} \ . \tag{4.127}$$

For many purposes, bases other than coordinate bases can also be extremely useful and natural, in particular the orthonormal bases we will introduce below.

The inverse relation to (4.125) is

$$dx^\mu = e^\mu_{\ m} e^m \ , \tag{4.128}$$

where $e^\mu_{\ m}(x)$ is (pointwise) the inverse matrix of $e^m_\mu(x)$,

$$e^m_\mu e^\mu_n = \delta^m_n \qquad e^\mu_m e^m_\nu = \delta^\mu_\nu \ . \tag{4.129}$$

With respect to this basis, one can expand a covector A as

$$A = A_\mu dx^\mu = A_\mu e^\mu_{\ m} e^m \equiv A_m e^m \ , \tag{4.130}$$

so that the components of A with respect to the new basis $\{e^m\}$ are

$$A_m = A_\mu e^\mu_{\ m} \ . \tag{4.131}$$

Likewise, the vielbeins allow us to pass from a natural (or coordinate) basis for vector fields, the $\{\partial_\mu\}$, to another basis

$$E_m = e^\mu_{\ m} \partial_\mu \ , \tag{4.132}$$

allowing us to write the coordinate independent vector field

$$V = V^\mu(x)\partial_\mu = V^m E_m \tag{4.133}$$

with

$$V^m = e^m_\mu V^\mu \ . \tag{4.134}$$

Note that, unlike the ∂_μ, the E_m do not commute in general, i.e.

$$[E_m, E_n] \neq 0 \ . \tag{4.135}$$

In fact, a 'dual' characterization of a coordinate basis is that the corresponding E_m do commute. This is clearly a necessary condition and, as above, locally it is also sufficient to ensure that there is a coordinate system y^m such that $E_m = \partial/\partial y^m$.

We can apply the same reasoning to any other tensor field, e.g. to the metric tensor itself. We can write the invariant line element as

$$ds^2 = g_{\mu\nu} dx^\mu dx^\mu = g_{\mu\nu} e^\mu_{\ m} e^\nu_{\ n} e^m e^n \equiv g_{mn} e^m e^n \ , \tag{4.136}$$

so that the components of the metric with respect to the new basis are

$$g_{mn} = g_{\mu\nu} e^\mu_{\ m} e^\nu_{\ n} \ . \tag{4.137}$$

Given a metric, there is a preferred class of bases $\{e^a\}$ which are such that the corresponding matrices $e^a_{\ \mu}(x)$ diagonalise (and normalise) the metric at every point x, i.e. which are such that $g_{ab} = \eta_{ab}$ or

$$g_{ab} = \eta_{ab} \quad \Leftrightarrow \quad g_{\mu\nu}(x) e^\mu_{\ a}(x) e^\nu_{\ b}(x) = \eta_{ab} \ . \tag{4.138}$$

Such a basis e^a, with respect to which the components of the metric are the Minkowski metric η_{ab}, is known as an *orthonormal basis* or *orthonormal frame*.

In the more mathematical literature, the e^a are also referred to as *soldering forms* because they identify (solder, glue) an abstract space of (co-)vectors at each point x, labelled by $a, b, \ldots$ with the concrete space of (co-)vectors tangent to the space-time at the point x, labelled e.g. by the indices $\mu, \nu, \ldots$.

For a general metric, a basis which achieves this cannot be a coordinate basis (because this would mean that the metric is equivalent to the Minkowski metric by a coordinate transformation). However, clearly there is no obstacle to finding a more general basis which will do this: for every point x we can find a matrix $e^a_{\ \mu}(x)$ which achieves (4.138) As the metric varies smoothly with x, we can also choose the matrices $e^a_{\ \mu}(x)$ to vary smoothly with x, and hence we can put them together to define the smooth matrix-valued function $e^a_{\ \mu}(x)$ for all x. [I am ignoring some global (topological) issues here. We will not need to worry about them here.]

The reason why I referred to a "class of bases" above is that, clearly, such an orthonormal basis is not unique. At every point x it is determined up to a Lorentz transformation

$$\begin{aligned} & e^a(x) \to \Lambda^a_{\ b}(x) e^b(x) \\ & \Lambda^a_{\ b}(x) \Lambda^c_{\ d}(x) \eta_{ac} = \eta_{bd} \ . \end{aligned} \tag{4.139}$$

Thus a given metric does not determine a unique orthonormal basis, but only an orthonormal basis up to Lorentz transformations

$$e^a(x) \to \Lambda^a_{\ b}(x) e^b(x) \ . \tag{4.140}$$

Conversely, however, an orthonormal basis uniquely determines a metric via

$$ds^2 = \eta_{ab} e^a(x) e^b(x) \ . \tag{4.141}$$

If one wants the components of the metric in a given coordinate system $\{x^\mu\}$, one expands the orthonormal basis e^a in terms of the natural basis dx^μ as above as

$$e^a(x) = e^a_{\ \mu}(x) dx^\mu \ , \tag{4.142}$$

to find, as above,

$$g_{\mu\nu}(x) = e^a_{\ \mu}(x) e^b_{\ \nu}(x) \eta_{ab} \ . \tag{4.143}$$

Thus instead of the metric one can choose orthonormal vielbeins as the basic variables of General Relativity. In that case one has to demand not only general covariance but

also invariance under local Lorentz transformations (acting on the orthonormal indices $a, b, \ldots$). [One could also allow for general vielbeins, in which case one would have to replace Lorentz transformations by the larger group of general linear transformations.]

EXAMPLES:

Here are a few examples to illustrate that orthonormal frames are not something mysterious but can usually be read off very easily from the metric in a coordinate basis.

1. The 2-Sphere Metric (2.16)

   The standard metric on a sphere of radius R is

   $$ds^2 = R^2(d\theta^2 + \sin^2\theta d\phi^2) \ . \tag{4.144}$$

   Now define

   $$e^1 = Rd\theta \quad , \quad e^2 = R\sin\theta d\phi \ , \tag{4.145}$$

   i.e.

   $$e^a = e^a{}_\alpha dx^\alpha \tag{4.146}$$

   with

   $$e^1_\theta = R \ , \ e^1_\phi = 0 \ , \ e^2_\theta = 0 \ , \ e^2_\phi = R\sin\theta \ . \tag{4.147}$$

   Then the metric can be written as

   $$ds^2 = e^1e^1 + e^2e^2 = \delta_{ab}e^ae^b \ , \tag{4.148}$$

   so the e^a are an orthonormal basis. They are obviously not a coordinate basis because (4.127)

   $$\partial_\theta e^2_\phi = R\cos\theta \neq \partial_\phi e^2_\theta = 0 \ . \tag{4.149}$$

   Likewise, we can introduce an orthonormal basis

   $$E_a = E^\alpha_a \partial_\alpha \tag{4.150}$$

   for vectors. A simple choice is

   $$E_1 = R^{-1}\partial_\theta \quad , \quad E_2 = (R\sin\theta)^{-1}\partial_\phi \ , \tag{4.151}$$

   which satisfies

   $$g_{\alpha\beta}E^\alpha_a E^\beta_b = \delta_{ab} \ . \tag{4.152}$$

   That this is not a coordinate basis is reflected in the fact that the commutator $[E_1, E_2] \neq 0$,

   $$[E_1, E_2] = R^{-2}(\partial_\theta(\sin\theta)^{-1})\partial_\phi = -R^{-1}\cot\theta \ E_2 \ . \tag{4.153}$$

2. The Schwarzschild Metric (2.38)

   The metric is

$$ds^2 = -(1-2m/r)dt^2 + (1-2m/r)^{-1}dr^2 + r^2(d\theta^2 + \sin^2\theta d\phi^2) \ . \tag{4.154}$$

   With

$$(e^0, e^1, e^2, e^3) = \left((1-\frac{2m}{r})^{1/2}dt, (1-\frac{2m}{r})^{-1/2}dr, rd\theta, r\sin\theta d\phi\right) \tag{4.155}$$

   the metric can be written as

$$ds^2 = \eta_{ab}e^a e^b \ , \tag{4.156}$$

   so the e^a are an orthonormal basis for the Schwarzschild metric.

3. The Kaluza-Klein Metric (section 44)

   Here is an example of a non-diagonal metric. The five-dimensional Kaluza-Klein metric is

$$d\widehat{s}^2_{KK} = g_{\mu\nu}dx^\mu dx^\nu + (dx^5 + A_\mu dx^\mu)^2 \ . \tag{4.157}$$

   Let $e^a_{\ \mu}$, $a = 0, 1, 2, 3$, be vielbeins (tetrads) for the four-dimensional space-time metric $g_{\mu\nu}$. Then an orthonormal frame $\widehat{e}^A$ for the Kaluza-Klein metric is

$$\widehat{e}^a = e^a_{\ \mu}dx^\mu \quad , \quad \widehat{e}^5 = dx^5 + A_\mu dx^\mu \ . \tag{4.158}$$

Remarks:

1. Consider the timelike trajectory $x^\mu = x^\nu(\tau)$ of an observer, parametrised by proper time τ. Then his 4-velocity $u^\mu = dx^\mu/d\tau$ satisfies (2.55)

$$g_{\mu\nu}u^\mu u^\nu = -1 \ . \tag{4.159}$$

   Recalling the defining relation for an orthonormal frame,

$$g_{\mu\nu}e^\mu_{\ a}e^\nu_{\ b} = \eta_{ab} \quad \Rightarrow \quad g_{\mu\nu}e^\mu_{\ a=0}e^\nu_{\ b=0} = \eta_{00} = -1 \ , \tag{4.160}$$

   we see that the 4-velocity u^μ can be interpreted as the timelike component $e^\mu_{\ a=0}$ of an orthonormal frame along the worldline,

$$u^\mu = e^\mu_{\ a=0} \ , \tag{4.161}$$

   with the spacelike components providing an orthonormal laboratory reference system. At this point there is still considerable freeedom in the choice of the spatial components of the orthonormal frame. This freedom can be significantly reduced (to rigid τ-independent rotations) by adopting a particular "parallel transport"

condition of these vectors along the worldline, such as the Fermi-Walker parallel transport to be discussed in section 5.10.

In any case, however the laboratory system is defined, the frame components

$$V^a = e^a_{\ \mu} V^\mu \tag{4.162}$$

of a vector now acquire the physical interpretation as the components of V as measured with respect to the observer's proper time and his laboratory frame. Note that this generalises the fact, already used in our second derivation of the gravitational redshift in section 3.5, that the frequency of a wave with wave vector k^μ as measured by an observer with 4-velocity u^μ is (3.117)

$$\omega = -u^\mu k_\mu = -e^\mu_{\ a=0} k_\mu = e^{a=0}_{\ \mu} k^\mu \equiv k^{a=0} \quad . \tag{4.163}$$

2. The $e^a_{\ \mu}$ can in some sense be regarded as the square-root of the metric. In particular denoting the determinant of the matrix $e^a_{\ \mu}$ by

$$e(x) := \det(e^a_{\ \mu}(x)) \quad , \tag{4.164}$$

(4.138) implies

$$g(x) := |\det(g_{\alpha\beta}(x))| = e(x)^2 \quad \Leftrightarrow \quad |e(x)| = \sqrt{g(x)} \quad . \tag{4.165}$$

3. Coordinate indices can, as usual, be raised and lowered with the space-time metric $g_{\mu\nu}$ and its inverse, and Minkowski (tangent space) indices with the Minkowski metric η_{ab} and its inverse.

Note that this is consistent with the notation for $e^a_{\ \mu}$ and its inverse $e^\mu_{\ a}$ because

$$e^\mu_{\ a} = g^{\mu\nu} \eta_{ab} e^b_{\ \nu} \quad . \tag{4.166}$$

One also has other fairly evident relations like

$$g^{\mu\nu} = \eta^{ab} e^\mu_{\ a} e^\nu_{\ b} \quad , \tag{4.167}$$

etc. The reason why I have called the basis of vector fields in a general frame E_m rather than e_m is that e^m and E_m are of course not related just by lowering or raising the indices of the metric, $E_m \neq g_{mn} e^n$. The former are linear combinations of the dx^μ, the latter linear combinations of the ∂_μ, so they are very different objects.

One could now go ahead and develop the entire machinery of tensor calculus (covariant derivatives, curvature, ...) that we are about to develop in the following sections in terms of vielbeins as the basic variables instead of the metric. This is rather straightforward. For example, given the expression for the Christoffel symbols in terms of the

metric, and for the metric in terms of the vielbeins, one can express the Christoffel symbols (and hence covariant derivatives and curvatures) in terms of vielbeins, but the resulting expressions are rather unenlightning and not of much use in practice.

The real power of the vielbein formalism emerges when one combines it with the formalism of *differential forms*. And in practice the most useful and efficient alternative to working in components in a coordinate basis is working with differential forms in an orthonormal basis.

I do most of my (curvature) calculations in the latter framework (and e.g. only then translate them into coordinate components for the purposes of inserting them into these notes), but this is (for the time being) not something I will develop further here.[10]

4.9 Epilogue: Indices? Indices!

Having reached this point, you may have the impression that the notation we have introduced for tensors, $T^{\mu_1\ldots\mu_p}_{\nu_1\ldots\nu_q}$ say, and which, as you might have noticed by looking ahead, we will continue to use in these notes, with its morass of indices, is somewhat cumbersome and unelegant. And perhaps you might prefer to at the very least see everything written in terms of the index-free coordinate-invariant objects like $V = V^\mu\partial_\mu$ or $A = A_\mu dx^\mu$ introduced in section 4.6.

I cannot disagree with the sentiment that using all these indices does not appear to be particularly elegant. Mathematicians abhor it. Physicists, however, are pragmatists by nature - they will use whatever turns out to be useful or efficient for what they want to achieve, regardless of whether or not it is considered or perceived to be beautiful or elegant according to some external criteria.

In particular, in the case at hand, the index-laden notation would not be that commonly used and widespread if it did not have some distinct advantages over other options. Indeed, this notation is an extremely useful and informative bookkeeping device that conveys a lot of information in a very compact way. In particular, as we have seen, the index notation allows one to reliably read off what kind of tensor one is dealing with, along the lines of "if it has p upper and q lower indices, it transform like, hence is, a (p, q)-tensor". Moreover, as we will see below, it provides one with a much more concise and informative way of describing and performing algebraic manipulations of tensors than some index-free notation is capable of.

Let me first make clear what the issue is and what it is not when one writes something like V^μ or $V^\mu(x)$, as this can be interpreted in (at least) 2 distinct ways:

[10] See e.g. W. Thirring, *Classical Mathematical Physics* for a presentation of general relativity entirely in the coordinate-independent formalism of differential forms, and N. Straumann, *General Relativity*, where differential forms are used whenever it is convenient or useful (and also occasionally when it is not ...).

1. On the one hand, V^μ may refer to the numerical values of the components of a specific vector V in a specific coordinate system.

2. On the other hand, the notation V^μ may be used to indicate that the object V transforms like a vector.

The first use of V^μ is completely uncontentious: if one wants to write down the components of some object with respect to some basis, one has to write down the components of that object with respect to that basis, there is no way around that.

It is mainly the second use and interpretation of the notation that is at stake, and it is also mainly in this sense that the index notation is used for tensor algebra and tensor calculus in general and in these notes in particular.

To a somewhat lesser extent the fact that the notation itself does not indicate whether one has in mind the first or the second interpretation is also an issue (even though this is usually clear from the context). It is actually not so much an issue (if desired this is something that can easily be remedied - I will come back to this at the end of this section) as possibly the source of a major misunderstanding between mathematicians and physicists - namely that a dislike of the index notation arises from the (false!) belief that it means that one is always writing down objects with respect to a particular basis. If this were the case, this would indeed be clumsy and silly, and quite contrary to the spirit of general covariance. However, as interpretation 2 indicates, this is absolutely not what is meant.

Returning to the use of indices as a way to indicate tensorial type and tensorial operations (like contractions), let us consider the alternatives. If one wants to indicate in symbols that some object V is a vector field, then as a mathematician one might write something like $V \in \Gamma(TM)$, stating that V is a section of the tangent bundle of the space or space-time (manifold) M. This is fine, but if the space M is clear from the context, why not declare once and for all that writing V^μ means the same thing? And perhaps use different kinds of indices to refer to tensors on different spaces?

If this were all then this would hardly be an issue and even physicists could be convinced to write "$V \in \Gamma(TM)$", at least when talking to mathematicians. Where the index notation really pays off, however, is when it comes to algebraic manipulations such as those discussed in section 4.3 (and even more so when it comes to tensor analysis, which is the subject of section 5, but tensor algebra will be enough to illustrate this).

As examples consider the contractions of a $(1,2)$ tensor T, say, with itself and with a vector V. With indices one would write $T^\mu{}_{\nu\lambda}$ and V^μ and the possible contractions would be written as

$$\begin{aligned} T^\mu{}_{\nu\lambda} \quad &\to \quad T^\mu{}_{\mu\lambda} \quad , \quad T^\mu{}_{\nu\mu} \\ (T^\mu{}_{\nu\lambda}, V^\mu) \quad &\to \quad T^\mu{}_{\nu\lambda} V^\nu \quad , \quad T^\mu{}_{\nu\lambda} V^\lambda \quad , \end{aligned} \tag{4.168}$$

the first line indicating the two distinct covectors one obtains as contractions of T itself, and the second the two distinct possibilities of contracting T and V to obtain a $(1,1)$-tensor. In an index-free notation one would have to invent some operation like C^m_n to indicate a contraction over the m'th upper and n'th lower index.[11] In this notation, the four objects above would then be written as

$$\begin{aligned} T \quad &\to \quad C^1_1(T) \quad , \quad C^1_2(T) \\ (T,V) \quad &\to \quad C^2_1(T\otimes V) \quad , \quad C^2_2(T\otimes V) \ . \end{aligned} \tag{4.169}$$

Is this superior? It does not even allow one to read off the tensor type of the resulting objects unless one remembers what the tensor types of T and V were to begin with, whereas this is completely manifest in (4.168).

Moreover, imagine how untransparent this would become were one to perform even the simplest sequence of such elementary operations: compare

$$A_{\alpha\beta}V^\alpha W^\beta \quad \longleftrightarrow \quad C^1_1C^2_2(A\otimes V\otimes W) \ . \tag{4.170}$$

If you prefer the right-hand side, or some variant of it, feel free to use it. However, you should be aware of the fact that the left-hand side contains an equivalent amount of information, simply packaged in a more digestible way that is both more informative ("it's a scalar!") and easier to manipulate. For most intents and purposes the index notation is really extremely convenient and it is for this reason that we will continue to make use of it in these notes.

One other reason for concern may be that by exclusively working with local coordinates and coordinate bases one may be missing some global aspects of a space or space-time. This is certainly true to a certain extent but is not primarily a notational issue. Rather, it means that in addition one needs to make use of more advanced notions from topology, global analysis etc. This is not something I will attempt here (cf. the book by Hawking and Ellis in the previous footnote for a description of the groundbreaking early applications of global analysis to general relativity). One related, but more elementary, issue is the introduction and use of the term *manifold* when referring to spaces or space-times of the kind we are dealing with in these notes. This is something I will very briefly come back to in section 5.11 below.

Let me, to conclude this ~~rant~~ section, come back to the issue of the notational ambiguity when one writes something like V^μ, which can occasionally be a source of confusion. Even though, as mentioned above, usually it is clear from the context what one means, one might imagine wanting to write down a couple of equations with indices which are only valid in spherical coordinates, say, and are therefore not to be understood as tensorial equations. Then it might be helpful to have a notation which reveals that information as well.

[11] I am not making this up - see e.g. section 2.2 of *The large scale structure of space-time* by S. Hawking and G. Ellis, in all other respects a wonderful book.

This can for instance be accomplished by inventing a new notation like $\stackrel{*}{=}$ (or whatever) to indicate an equality only in a special or specified coordinate system, but while this may add clarity it does not address the fundamental issue that just writing V^α does not unambiguously specify what one has in mind.

Alternatively, and more elegantly and attractively, this can e.g. be accomplished with very little effort with the help of what is known as the *Penrose abstract index notation*. The idea is to still indicate the tensor type of an object by a certain kind of indices, but with these indices only serving that purpose and not simultaneously referring to any particular kind of basis. Thus for example, one would indicate a vector by an object V^a, where the fact that one has a single upper index a just means that this is a $(1,0)$-tensor, and nothing else (exactly as in interpretation 2 above). For the components of this vector with respect to some basis (coordinates x^μ) one could then continue to use the traditional V^μ.

The advantage of this "abstract index" notation is that for tensorial operations one never needs to specify a basis anyway, so they can all be performed at the level of the abstract indices and tensorial equations look identical when written with these abstract indices or when written with concrete component indices. Thus $V^a W_a$ is used to indicate the scalar one obtains by contraction of a vector V^a with a covector W_a. Likewise, instead of $T^\mu_{\ \mu\lambda}$ (which may look basis dependent) one would write $T^a_{\ ab}$, and this is completely equivalent to writing something like $C^1_1(T)$,

$$T^a_{\ ab} \quad \longleftrightarrow \quad C^1_1(T) \tag{4.171}$$

but much more informative and user-friendly, and all the usual rules of tensor algebra apply to these abstract indices.

Whenever one wants or needs to specify a basis or coordinate system, this can be accomplished by using other kinds of indices. Thus g_{ab} could e.g. be used to refer to the metric tensor in general, while $g_{\mu\nu}$ could then be used to refer to its components in the basis x^μ. From this we see that

> "[...] the distinction between the index notation and the component notation is much more one of spirit (i.e., how one thinks of the quantities appearing) than of substance (i.e., the physical form the equations take)."[12]

While I will not make use of the abstract index notation in these notes (with the hope that this will not cause any confusion), the use of abstract indices appears to be an ideal ("eat the cake and have it too") compromise combining the best of both worlds

[12] R. Wald, *General Relativity*. See section 2.4 of this book for a more detailed explanation of the abstract index notation, which is systematically used throughout the book. For a detailed treatment of the abstract index notation and a discussion of some minor subtleties with this notation see R. Penrose, W. Rindler, *Spinors and Space-Time*, Vol. 1: *Two-Spinor Calculus and Relativistic Fields*.

and should actually keep both camps happy. It does not yet appear to have found widespread acceptance among mathematicians, however.

An alternative compromise solution is the already mentioned use of differential forms (in an orthonormal basis, say), which is manifestly covariant and minimises clutter, displaying only the (essential and informative) Lorentz Lie algebra indices while suppressing the component indices of forms (anti-symmetric tensors).

5 Tensor Analysis (Generally Covariant Differentiation)

Tensors transform in a nice and simple way under general coordinate transformations. Thus these appear to be the right objects to construct equations from that satisfy the Principle of General Covariance.

However, the laws of physics are differential equations, so we need to know how to differentiate tensors. This is not an issue of particular concern in Special Relativity, because (cf. (1.49)) the partial derivative of a Lorentz tensor

$$T^{a_1\ldots a_p}_{c_1\ldots c_q}(\xi) \quad \to \quad \partial_a T^{a_1\ldots a_p}_{c_1\ldots c_q}(\xi) \tag{5.1}$$

is again a Lorentz tensor. This relies on 2 facts, namely first that the partial derivative transforms as a covector under Lorentz transformations and secondly that the associated Jacobi matrix of Lorentz transformations is constant.

The former generalises to arbitrary coordinate transformations and implies, in particular, that the partial derivative of a scalar field is a covector field (4.18). However, because in general the Jacobi matrix is not constant, the ordinary partial derivative does not map tensors to tensors.

This is easy to see: take for example a vector V^μ. Under a coordinate transformation $x^\mu \to y^\alpha$, its partial derivative transforms as

$$\begin{aligned} \partial_\beta V^\alpha &= \frac{\partial x^\nu}{\partial y^\beta}\frac{\partial}{\partial x^\nu}\frac{\partial y^\alpha}{\partial x^\mu}V^\mu \\ &= \frac{\partial x^\nu}{\partial y^\beta}\frac{\partial y^\alpha}{\partial x^\mu}\partial_\nu V^\mu + \frac{\partial x^\nu}{\partial y^\beta}\frac{\partial^2 y^\alpha}{\partial x^\mu \partial x^\nu}V^\mu \quad . \end{aligned} \tag{5.2}$$

The appearance of the second term shows that the partial derivative of a vector is not a tensor.

As the second term is zero for linear transformations, you see that partial derivatives transform in a tensorial way e.g. under Lorentz transformations, so that partial derivatives are all one usually needs in special relativity.

We also see that the lack of covariance of the partial derivative is very similar to the lack of covariance of the equation $\ddot{x}^\mu = 0$, and this suggests that the problem can be cured in the same way - by introducing Christoffel symbols. This is indeed the case.

5.1 Covariant Derivative for Vector Fields

Let us define the *covariant derivative* $\nabla_\nu V^\mu$ of a vector field V^μ by

$$\nabla_\nu V^\mu = \partial_\nu V^\mu + \Gamma^\mu_{\nu\lambda} V^\lambda \quad . \tag{5.3}$$

It follows from the non-tensorial behaviour (2.78), (2.79) of the Christoffel symbols under coordinate transformations $x^\mu \to y^\alpha$ that $\nabla_\nu V^\mu$, as defined above, is indeed a $(1,1)$ tensor.

In order to establish this, we transform

$$\nabla_\beta V^\alpha = \partial_\beta V^\alpha + \Gamma^\alpha_{\beta\gamma} V^\gamma \tag{5.4}$$

to x-coordinates, using the tensorial transformation behaviour of ∂_α and V^α and the non-tensorial transformation behaviour (2.79) of the Christoffel symbols, to arrive at

$$\nabla_\beta V^\alpha = J^\mu_\beta \partial_\mu (J^\alpha_\rho V^\rho) + (J^\alpha_\mu J^\nu_\beta J^\lambda_\gamma \Gamma^\mu_{\ \nu\lambda} + J^\alpha_\mu \partial_\beta J^\mu_\gamma) J^\gamma_\rho V^\rho \ . \tag{5.5}$$

The obstructions to tensoriality are the 2 terms involving the derivatives of the Jacobi matrix, but these cooperatively combine to give

$$\begin{aligned} J^\mu_\beta \partial_\mu J^\alpha_\rho + J^\alpha_\mu (\partial_\beta J^\mu_\gamma) J^\gamma_\rho &= J^\mu_\beta \partial_\rho J^\alpha_\mu + J^\alpha_\mu (\partial_\gamma J^\mu_\beta) J^\gamma_\rho \\ &= J^\mu_\beta \partial_\rho J^\alpha_\mu - (\partial_\gamma J^\alpha_\mu) J^\mu_\beta J^\gamma_\rho \\ &= J^\mu_\beta \partial_\rho J^\alpha_\mu - (\partial_\rho J^\alpha_\mu) J^\mu_\beta = 0 \ . \end{aligned} \tag{5.6}$$

Here we have used the symmetry

$$\partial_\mu J^\alpha_\rho = \frac{\partial^2 y^\alpha}{\partial x^\mu \partial x^\rho} = \partial_\rho J^\alpha_\mu \tag{5.7}$$

and

$$J^\alpha_\mu J^\mu_\beta = \delta^\alpha_\beta \quad \Rightarrow \quad J^\alpha_\mu (\partial_\gamma J^\mu_\beta) = -(\partial_\gamma J^\alpha_\mu) J^\mu_\beta \ . \tag{5.8}$$

The remaining terms in (5.5) then just give rise to the tensorial transformation of a (1,1)-tensor. Thus we have shown that (5.3) indeed defines a tensor. Moreover, in a locally inertial coordinate system this reduces to the ordinary partial derivative, and we have thus, as desired, arrived at an appropriate tensorial generalisation of the partial derivative operator.

REMARKS:

1. Analysing the above argument for the tensoriality of the covariant derivative, we see that it relies exclusively on the specific non-tensorial form of the transformation behaviour of the Christoffel symbols, not on the explicit form of the Christoffel symbols themselves.

   Thus any other object $\tilde{\Gamma}^\mu_{\ \nu\lambda}$ could also be used to define a covariant derivative (generalising the partial derivative and mapping tensors to tensors) provided that it transforms in the same way as the Christoffel symbols, i.e. provided that one has

   $$\tilde{\Gamma}^\alpha_{\ \beta\gamma} = J^\alpha_\mu J^\nu_\beta J^\lambda_\gamma \tilde{\Gamma}^\mu_{\ \nu\lambda} + J^\alpha_\mu \partial_\beta J^\mu_\gamma \ . \tag{5.9}$$

This implies (and is equivalent to the fact) that the difference

$$C^{\mu}_{\ \nu\lambda} = \tilde{\Gamma}^{\mu}_{\ \nu\lambda} - \Gamma^{\mu}_{\ \nu\lambda} \tag{5.10}$$

transforms as a tensor. Thus, any such $\tilde{\Gamma}$ is of the form

$$\tilde{\Gamma}^{\mu}_{\ \nu\lambda} = \Gamma^{\mu}_{\ \nu\lambda} + C^{\mu}_{\ \nu\lambda} \tag{5.11}$$

where $C^{\mu}_{\ \nu\lambda}$ is a (1,2)-tensor, and could be used to define a corresponding covariant derivative $\tilde{\nabla}_{\mu}$.

Therefore the question arises if the covariant derivative defined in terms of the Christoffel symbols is somehow singled out or preferred. Indeed it is, and we will return to this question on various occasions below, in particular in section 5.4.

2. We could have arrived at the above definition of the covariant derivative (using the Christoffel symbols) in a somewhat more systematic way by appealing to the equivalence principle and/or general covariance. Namely, let $\{\xi^a\}$ be an inertial coordinate system. In an inertial coordinate system we can just use the ordinary partial derivative $\partial_b V^a$. We now define the new (improved, covariant) derivative $\nabla_\nu V^\mu$ in any other coordinate system $\{x^\mu\}$ by demanding that it transforms as a (1,1)-tensor, i.e. we define

$$\nabla_\nu V^\mu := \frac{\partial x^\mu}{\partial \xi^a}\frac{\partial \xi^b}{\partial x^\nu}\partial_b V^a \ . \tag{5.12}$$

By a straightforward calculation one finds that

$$\nabla_\nu V^\mu = \partial_\nu V^\mu + \gamma^{\mu}_{\ \nu\lambda} V^\lambda \ , \tag{5.13}$$

where $\gamma^{\mu}_{\ \nu\lambda}$ is our old friend (1.98)

$$\gamma^{\mu}_{\ \nu\lambda} = \frac{\partial x^\mu}{\partial \xi^a}\frac{\partial^2 \xi^a}{\partial x^\nu \partial x^\lambda} \ . \tag{5.14}$$

One would then also be led to adopt (5.13) with $\gamma \to \Gamma$ as a definition of the *covariant derivative* in a general metric space or space-time (with the Christoffel symbols calculated from the metric in the usual way).

That $\nabla_\mu V^\nu$, defined in this way, is indeed a $(1,1)$ tensor, now follows directly from the way we arrived at the definition of the covariant derivative. Indeed, imagine transforming from inertial coordinates to another coordinate system $\{y^\alpha\}$. Then (5.12) is replaced by

$$\nabla_\beta V^\alpha := \frac{\partial y^\alpha}{\partial \xi^a}\frac{\partial \xi^b}{\partial y^\beta}\partial_b V^a \ . \tag{5.15}$$

Comparing this with (5.12), we see that the two are related by

$$\nabla_\beta V^\alpha := \frac{\partial y^\alpha}{\partial x^\mu}\frac{\partial x^\nu}{\partial y^\beta}\nabla_\nu V^\mu \ , \tag{5.16}$$

as required.

3. Notation: frequently, the covariant derivative $\nabla_\nu V^\mu$ is also denoted by a semi-colon,

$$\nabla_\nu V^\mu = V^\mu{}_{;\nu} \quad . \tag{5.17}$$

Since covariant derivatives do not necessarily commute (as we will discuss in detail in section 8), when using this notation one has to pay attention to the order (and reversal of the order) of indices,

$$\nabla_\lambda \nabla_\nu V^\mu = V^\mu{}_{;\nu\,;\lambda} \quad . \tag{5.18}$$

One can also define the covariant directional derivative of a vector field V along another vector field X^μ by

$$\nabla_X V^\mu \equiv X^\nu \nabla_\nu V^\mu \quad . \tag{5.19}$$

4. The appearance of the Christoffel-term in the definition of the covariant derivative may at first sight appear a bit unusual (even though it also appears when one just transforms Cartesian partial derivatives to polar coordinates etc.). There is a more invariant way of explaining the appearance of this term, related to the more coordinate-independent way of looking at tensors explained in section 4.6. Namely, since the $V^\mu(x)$ are really just the coefficients of the vector field $V(x) = V^\mu(x)\partial_\mu$ when expanded in the basis ∂_μ, a meanigful definition of the derivative of a vector field must take into account not only the change in the coefficients but must also include a prescription how bases at (infinitesimally) neighbouring points are related (or *connected*). Such a prescription is provided by the *Levi-Civita connection* $\Gamma^\mu_{\nu\lambda}$ (or a general *connection* $\tilde{\Gamma}^\mu_{\nu\lambda}$).

Indeed, writing

$$\begin{aligned} \nabla_\nu V &= \nabla_\nu (V^\mu \partial_\mu) \\ &= (\partial_\nu V^\mu)\partial_\mu + V^\lambda (\nabla_\nu \partial_\lambda) \quad , \end{aligned} \tag{5.20}$$

we see that the covariant derivative of the coordinate basis vector ∂_λ (i.e. $V^\lambda = 1$, $V^\mu = 0$ otherwise), is the linear transformation (a prescription for a change of basis)

$$\nabla_\nu \partial_\lambda = \Gamma^\mu_{\nu\lambda} \partial_\mu \quad . \tag{5.21}$$

5.2 Extension of the Covariant Derivative to Other Tensor Fields

So far we have defined the covariant derivative for vector fields, and we now want to extend the definition of the covariant derivative to other tensor fields. In order to achieve this, we now adopt a more systematic and axiomatic approach.

Our basic postulates for the covariant derivative are the following:

1. Linearity and Tensoriality

   ∇_μ is a linear operator that maps (p,q)-tensors to $(p,q+1)$-tensors

2. Generalisation of the Partial Derivative

   On scalars ϕ, the covariant derivative ∇_μ reduces to the ordinary partial derivative (since $\partial_\mu\phi$ is already a covector),

$$\nabla_\mu\phi = \partial_\mu\phi \ . \tag{5.22}$$

3. Leibniz Rule (or Product Rule)

   Acting on the direct product of tensors, ∇_μ satisfies a generalised Leibniz rule,

$$\nabla_\mu(A^{\mu_1\ldots\mu_p}_{\nu_1\ldots\nu_q}B^{\rho_1\ldots\rho_r}_{\lambda_1\ldots\lambda_s}) = \nabla_\mu(A^{\mu_1\ldots\mu_p}_{\nu_1\ldots\nu_q})B^{\rho_1\ldots\rho_r}_{\lambda_1\ldots\lambda_s} + A^{\mu_1\ldots\mu_p}_{\nu_1\ldots\nu_q}\nabla_\mu B^{\rho_1\ldots\rho_r}_{\lambda_1\ldots\lambda_s} \tag{5.23}$$

We will now see that, demanding the above properties, in particular the Leibniz rule, there is a unique extension of the covariant derivative on vector fields to a differential operator on general tensor fields, mapping (p,q)- to $(p,q+1)$-tensors.

To define e.g. the covariant derivative for covectors U_μ, we note that $U_\mu V^\mu$ is a scalar for any vector V^μ so that

$$\nabla_\mu(U_\nu V^\nu) = \partial_\mu(U_\nu V^\nu) = (\partial_\mu U_\nu)V^\nu + U_\nu(\partial_\mu V^\nu) \tag{5.24}$$

(since the partial derivative satisfies the Leibniz rule), and we demand

$$\nabla_\mu(U_\nu V^\nu) = (\nabla_\mu U_\nu)V^\nu + U_\nu\nabla_\mu V^\nu \ . \tag{5.25}$$

As we know $\nabla_\mu V^\nu$, these two equations determine $\nabla_\mu U_\nu$ uniquely to be

$$\nabla_\mu U_\nu = \partial_\mu U_\nu - \Gamma^\lambda_{\mu\nu}U_\lambda \ . \tag{5.26}$$

That this is indeed a $(0,2)$-tensor can either be checked directly or, alternatively, is a consequence of the quotient theorem.

The extension to other (p,q)-tensors is now immediate. Here are two ways to proceed:

1. If the (p,q)-tensor is the direct product of p vectors and q covectors, then we already know its covariant derivative (using the Leibniz rule again). We simply adopt the same resulting formula for an arbitrary (p,q)-tensor.

2. Alternatively, contract the (p,q)-tensor with p covectors and q vectors to turn it into a scalar, and proceed as above for a covector.

Either way, the result is that the covariant derivative of a general (p,q)-tensor is the sum of the partial derivative, a Christoffel symbol with a positive sign for each of the p upper indices, and a Christoffel with a negative sign for each of the q lower indices. In equations

$$\begin{aligned}\nabla_\mu T^{\nu_1\cdots\nu_p}{}_{\rho_1\cdots\rho_q} &= \partial_\mu T^{\nu_1\cdots\nu_p}{}_{\rho_1\cdots\rho_q} \\ &+ \underbrace{\Gamma^{\nu_1}_{\mu\lambda} T^{\lambda\nu_2\cdots\nu_p}{}_{\rho_1\cdots\rho_q} + \ldots + \Gamma^{\nu_p}_{\mu\lambda} T^{\nu_1\cdots\nu_{p-1}\lambda}{}_{\rho_1\cdots\rho_q}}_{p \text{ terms}} \\ &- \underbrace{\Gamma^{\lambda}_{\mu\rho_1} T^{\nu_1\cdots\nu_p}{}_{\lambda\rho_2\cdots\rho_q} - \ldots - \Gamma^{\lambda}_{\mu\rho_q} T^{\nu_1\cdots\nu_p}{}_{\rho_1\cdots\rho_{q-1}\lambda}}_{q \text{ terms}}\end{aligned} \tag{5.27}$$

Having defined the covariant derivative for arbitrary tensors, we are also ready to define it for tensor densities. For this we recall that if T is a $(p,q;w)$ tensor density, then $g^{-w/2}T$ is a (p,q)-tensor. Thus $\nabla_\mu(g^{-w/2}T)$ is a $(p,q+1)$-tensor. To map this back to a tensor density of weight w, we multiply this by $g^{w/2}$, arriving at the definition

$$\nabla_\mu T := g^{+w/2}\nabla_\mu(g^{-w/2}T) \ . \tag{5.28}$$

Working this out explictly, one finds

$$\nabla_\mu T = -\frac{w}{2g}(\partial_\mu g)T + \nabla^{\text{tensor}}_\mu T \ , \tag{5.29}$$

where $\nabla^{\text{tensor}}_\mu$ just means the usual covariant derivative for (p,q)-tensors defined above. For example, for a scalar density ϕ one has

$$\nabla_\mu \phi = \partial_\mu \phi - \frac{w}{2g}(\partial_\mu g)\phi \ . \tag{5.30}$$

In particular, since the determinant g is a scalar density of weight $+2$, it follows that

$$\nabla_\mu g = 0 \ , \tag{5.31}$$

which obviously simplifies integrations by parts in integrals defined with the measure $\sqrt{g}d^4x$. However, it should be kept in mind that the crucial property that makes an integral like $\int \sqrt{g}\nabla_\alpha(\ldots)^\alpha$ a total derivative is *not* this fact but the fact that in this expression the $\sqrt{g}$ cancels and the integrand becomes an ordinary total derivative (cf. the discussion of the Gauss Theorem (5.61) below).

5.3 Main Properties of the Covariant Derivative

The main properties of the covariant derivative, in addition to those that were part of our postulates (like linearity and the Leibniz rule) are the following:

1. ∇_μ Commutes with Contraction

   This means that if A is a (p,q)-tensor and B is the $(p-1,q-1)$-tensor obtained by contraction over two particular indices, then the covariant derivative of B is the same as the covariant derivative of A followed by contraction over these two indices. This comes about because of a cancellation between the corresponding two Christoffel symbols with opposite signs. Consider e.g. a (1,1)-tensor $A^\nu{}_\rho$ and its contraction $A^\nu{}_\nu$. The latter is a scalar and hence its covariant derivative is just the partial derivative. This can also be obtained by taking first the covariant derivative of A,

$$\nabla_\mu A^\nu{}_\rho = \partial_\mu A^\nu{}_\rho + \Gamma^\nu{}_{\mu\lambda} A^\lambda{}_\rho - \Gamma^\lambda{}_{\mu\rho} A^\nu{}_\lambda \ , \tag{5.32}$$

   and then contracting:

$$\nabla_\mu A^\nu{}_\nu = \partial_\mu A^\nu{}_\nu + \Gamma^\nu{}_{\mu\lambda} A^\lambda{}_\nu - \Gamma^\lambda{}_{\mu\nu} A^\nu{}_\lambda = \partial_\mu A^\nu{}_\nu \ . \tag{5.33}$$

   The most transparent way of stating this property is that the Kronecker delta is covariantly constant, i.e. that

$$\nabla_\mu \delta^\nu{}_\lambda = 0 \ . \tag{5.34}$$

   To see this, we use the Leibniz rule to calculate

$$\begin{aligned} \nabla_\mu A^{\nu\ldots}{}_{\nu\ldots} &= \nabla_\mu (A^{\nu\ldots}{}_{\rho\ldots} \delta^\rho{}_\nu) \\ &= (\nabla_\mu A^{\nu\ldots}{}_{\rho\ldots}) \delta^\rho{}_\nu + A^{\nu\ldots}{}_{\rho\ldots} \nabla_\mu \delta^\rho{}_\nu \\ &= (\nabla_\mu A^{\nu\ldots}{}_{\rho\ldots}) \delta^\rho{}_\nu \end{aligned} \tag{5.35}$$

   which is precisely the statement that covariant differentiation and contraction commute. To establish that the Kronecker delta is covariantly constant, we follow the rules to find

$$\begin{aligned} \nabla_\mu \delta^\nu{}_\lambda &= \partial_\mu \delta^\nu{}_\lambda + \Gamma^\nu{}_{\mu\rho} \delta^\rho{}_\lambda - \Gamma^\rho{}_{\mu\lambda} \delta^\nu{}_\rho \\ &= \Gamma^\nu{}_{\mu\lambda} - \Gamma^\nu{}_{\mu\lambda} = 0 \ . \end{aligned} \tag{5.36}$$

   This property does not rely on the specific form of the $\Gamma^\mu{}_{\nu\lambda}$, and is thus true for any covariant derivative defined by some choice of connection $\tilde{\Gamma}^\mu{}_{\nu\lambda}$,

2. The Metric is Covariantly Constant: $\nabla_\mu g_{\nu\lambda} = 0$

   This is one of the key properties of the covariant derivative ∇_μ we have defined. I will give two arguments to establish this:

   (a) Since $\nabla_\mu g_{\nu\lambda}$ is a tensor, we can choose any coordinate system we like to establish if this tensor is zero or not at a given point x. Choose an inertial coordinate system at x. Then the partial derivatives of the metric and the Christoffel symbols are zero there. Therefore the covariant derivative of the metric is zero. Since $\nabla_\mu g_{\nu\lambda}$ is a tensor, this is then true in every coordinate system.

(b) The other argument is by direct calculation. Recalling the identity

$$\partial_\mu g_{\nu\lambda} = \Gamma_{\nu\lambda\mu} + \Gamma_{\lambda\nu\mu} \ , \tag{5.37}$$

we calculate

$$\begin{aligned} \nabla_\mu g_{\nu\lambda} &= \partial_\mu g_{\nu\lambda} - \Gamma^\rho{}_{\mu\nu} g_{\rho\lambda} - \Gamma^\rho{}_{\mu\lambda} g_{\nu\rho} \\ &= \Gamma_{\nu\lambda\mu} + \Gamma_{\lambda\nu\mu} - \Gamma_{\lambda\mu\nu} - \Gamma_{\nu\mu\lambda} \\ &= 0 \ . \end{aligned} \tag{5.38}$$

3. ∇_μ Commutes with Raising and Lowering of Indices

   This is really a direct consequence of the covariant constancy of the metric. For example, if V_μ is the covector obtained by lowering an index of the vector V^μ, $V_\mu = g_{\mu\nu} V^\nu$, then

   $$\nabla_\lambda V_\mu = \nabla_\lambda (g_{\mu\nu} V^\nu) = g_{\mu\nu} \nabla_\lambda V^\nu \ . \tag{5.39}$$

4. Covariant Derivatives Commute on Scalars

   This is of course a familiar property of the ordinary partial derivative, but it is also true for the second covariant derivatives of a scalar and is a consequence of the symmetry of the Christoffel symbols in the second and third indices and is also knowns as the *no torsion* property of the covariant derivative. Namely, we have

   $$\begin{aligned} \nabla_\mu \nabla_\nu \phi - \nabla_\nu \nabla_\mu \phi &= \nabla_\mu \partial_\nu \phi - \nabla_\nu \partial_\mu \phi \\ &= \partial_\mu \partial_\nu \phi - \Gamma^\lambda{}_{\mu\nu} \partial_\lambda \phi - \partial_\nu \partial_\mu \phi + \Gamma^\lambda{}_{\nu\mu} \partial_\lambda \phi = 0 \ . \end{aligned} \tag{5.40}$$

   Note that *the second covariant derivatives on higher rank tensors do not commute* - we will come back to this in our discussion of the curvature tensor later on.

5.4 Uniqueness of the Levi-Civita Connection (Christoffel symbols)

We noted before that the postulates for a covariant derivative (a linear tensorial operator reducing to the partial derivative on scalars and satisfying the Leibniz rule) do not determine it uniquely but only up to the addition of a tensor to the connection,

$$\Gamma^\mu{}_{\nu\lambda} \to \tilde{\Gamma}^\mu{}_{\nu\lambda} = \Gamma^\mu{}_{\nu\lambda} + C^\mu{}_{\nu\lambda} \ , \tag{5.41}$$

where $C^\mu{}_{\nu\lambda}$ is a (1,2)-tensor. Is there anything special or preferred about the Levi-Civita connection using the Christoffel symbols?

In some sense, the answer is an immediate yes because it is this particular covariant derivative (or connection) that enters in determining the paths of freely falling particles (the geodesics which extremise proper time).

Not unrelated to this is the fact that it is the unique connection that can be built from only the metric and its 1st derivatives (and which thus vanishes in an inertial coordinate system in Minkowski space or at the origin of an inertial coordinate system in an arbitrary gravitational field).

Moreover, as we have seen, this covariant derivative has two important properties, namely that

1. the metric is covariantly constant, $\nabla_\mu g_{\nu\lambda} = 0$, and
2. the torsion is zero, i.e. the second covariant derivatives of a scalar commute.

In fact, it turns out that these two conditions uniquely determine the $\tilde{\Gamma}$ to be the Christoffel symbols. The second condition implies that the $\tilde{\Gamma}^{\mu}_{\ \nu\lambda}$ are symmetric in the two lower indices,

$$[\tilde{\nabla}_\mu, \tilde{\nabla}_\nu]\phi = 0 \quad \Leftrightarrow \quad \tilde{\Gamma}^{\lambda}_{\ \mu\nu} = \tilde{\Gamma}^{\lambda}_{\ \nu\mu} \ . \tag{5.42}$$

The first condition now allows one to express the $\tilde{\Gamma}^{\lambda}_{\ \mu\nu}$ in terms of the derivatives of the metric, leading uniquely to the familiar expression for the Christoffel symbols $\Gamma^{\mu}_{\ \nu\lambda}$: First of all, by definition / construction one has (e.g. from demanding the Leibniz rule for $\tilde{\nabla}_\mu$)

$$\tilde{\nabla}_\mu g_{\nu\lambda} = \partial_\mu g_{\nu\lambda} - \tilde{\Gamma}^{\rho}_{\ \nu\mu} g_{\rho\lambda} - \tilde{\Gamma}^{\rho}_{\ \lambda\mu} g_{\nu\rho} \equiv \partial_\mu g_{\nu\lambda} - \tilde{\Gamma}_{\lambda\nu\mu} - \tilde{\Gamma}_{\nu\lambda\mu} \ . \tag{5.43}$$

Requiring that this be zero implies in particular that

$$\begin{aligned} 0 &= \tilde{\nabla}_\mu g_{\nu\lambda} + \tilde{\nabla}_\nu g_{\mu\lambda} - \tilde{\nabla}_\lambda g_{\mu\nu} \\ &= \partial_\mu g_{\nu\lambda} + \partial_\nu g_{\mu\lambda} - \partial_\lambda g_{\mu\nu} - \tilde{\Gamma}_{\lambda\nu\mu} - \tilde{\Gamma}_{\nu\lambda\mu} - \tilde{\Gamma}_{\lambda\mu\nu} - \tilde{\Gamma}_{\mu\lambda\nu} + \tilde{\Gamma}_{\nu\mu\lambda} + \tilde{\Gamma}_{\mu\nu\lambda} \\ &= 2(\Gamma_{\lambda\mu\nu} - \tilde{\Gamma}_{\lambda\mu\nu}) \end{aligned} \tag{5.44}$$

(where the cancellations are entirely due to the assumed symmetry of the coefficients in the last two indices). Thus $\tilde{\Gamma} = \Gamma$. This unique metric-compatible and torsion-free connection is also known as the *Levi-Civita connection*. It is the connection canonically associated to a space-time (manifold) equipped with a metric tensor, and it is the connection used in general relativity.

It is possible to relax either of the conditions (1) or (2), or both of them and this will be discussed in section 11.5, and subsequently also in section 20.7.

5.5 Tensor Analysis: Some Special Cases

In this section we will look at some common and useful special cases of the Levi-Civita covariant derivative (simply "*the* covariant derivative" in the following), such as the covariant curl and divergence etc.

1. The Covariant Curl of a Covector

   One has

   $$\nabla_\mu U_\nu - \nabla_\nu U_\mu = \partial_\mu U_\nu - \partial_\nu U_\mu \ , \tag{5.45}$$

   because the symmetric Christoffel symbols drop out in this anti-symmetric linear combination. Thus in particular the Maxwell field strength

   $$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu \tag{5.46}$$

   is a tensor under general coordinate transformations, no metric or covariant derivative is needed to make it a tensor in a general space-time. The fact that the (ordinary) curl of a covector is, i.e. transforms like, a tensor under general coordinate transformations can of course also be checked directly (and then "explains" the above identity). We will come back to this below.

2. The Covariant Curl of an Antisymmetric Tensor

   Let $A_{\nu\lambda\ldots}$ be completely anti-symmetric. Then, as for the curl of covectors, the metric and Christoffel symbols drop out of the expression for the curl, i.e. one has

   $$\nabla_{[\mu} A_{\nu\lambda\cdots]} = \partial_{[\mu} A_{\nu\lambda\cdots]} \ . \tag{5.47}$$

   Here the square brackets on the indices denote complete anti-symmetrisation. In particular, the Bianchi identity for the Maxwell field strength tensor is independent of the metric also in a general metric space-time.

3. The Covariant Divergence of a Vector

   By the covariant divergence of a vector field one means the scalar

   $$\nabla_\mu V^\mu = \partial_\mu V^\mu + \Gamma^\mu_{\ \mu\lambda} V^\lambda \ . \tag{5.48}$$

   Now a useful identity for the contracted Christoffel symbol is

   $$\Gamma^\mu_{\ \mu\lambda} = \frac{1}{\sqrt{g}} \partial_\lambda \sqrt{g} \ . \tag{5.49}$$

   I will give a proof of this identity in an appendix to this section (subsection 5.6).

   Thus the covariant divergence can be written compactly as

   $$\nabla_\mu V^\mu = \frac{1}{\sqrt{g}} \partial_\mu (\sqrt{g} V^\mu) \ , \tag{5.50}$$

   and one only needs to calculate g and its derivative, not the Christoffel symbols themselves, to calculate the covariant divergence of a vector field.

   This formula is also useful (and provides the quickest way of arriving at the result) if one just wants to write the ordinary flat space divergence of vector calculus on $\mathbb{R}^3$ in, say, polar or cylindrical coordinates.

In Cartesian coordinates (x^1, x^2, x^3), the divergence of a 3-vector $\vec{V}$ is of course given by the familiar expression

$$\mathsf{div}\vec{V} = \partial_1 V^1 + \partial_2 V^2 + \partial_3 V^3 \quad . \tag{5.51}$$

However, as you also know, e.g. in spherical coordinates (r, θ, ϕ) the divergence is not simply of this form,

$$\mathsf{div}\vec{V} \neq \partial_r V^r + \partial_\theta V^\theta + \partial_\phi V^\phi \quad . \tag{5.52}$$

Rather, going through the coordinate transformation and Jacobians etc., one finds that calculating the divergence in spherical coordinates one picks up additional terms, the result taking the somewhat unintuitive form

$$\mathsf{div}\vec{V} = \partial_r V^r + \partial_\theta V^\theta + \partial_\phi V^\phi + \frac{2}{r} V^r + \cot\theta V^\theta \quad . \tag{5.53}$$

The easy and quick way to obtain this, which provides a rationale for and explanation of the origin of these additional terms, is from the result (5.50). Using $\sqrt{g} = r^2 \sin\theta$, one has

$$\begin{aligned} \mathsf{div}\vec{V} &= \frac{1}{r^2 \sin\theta} \left[\partial_r (r^2 \sin\theta V^r) + \partial_\theta (r^2 \sin\theta V^\theta) + \partial_\phi (r^2 \sin\theta V^\phi) \right] \\ &= \partial_r V^r + \partial_\theta V^\theta + \partial_\phi V^\phi + \frac{2}{r} V^r + \cot\theta V^\theta \quad . \end{aligned} \tag{5.54}$$

This thus produces the correct result on the nose and with very little effort.

4. The Covariant Laplacian of a Scalar

   How should the Laplacian be defined? Well, the obvious guess (something that is covariant and reduces to the ordinary Laplacian for the Minkowski metric) is $\Box = g^{\mu\nu}\nabla_\mu \nabla_\nu$, which can alternatively be written as

$$\Box = g^{\mu\nu}\nabla_\mu \nabla_\nu = \nabla^\mu \nabla_\mu = \nabla_\mu \nabla^\mu = \nabla_\mu g^{\mu\nu} \nabla_\nu \tag{5.55}$$

   etc. Note that, even though the covariant derivative on scalars reduces to the ordinary partial derivative, so that one can write

$$\Box\phi = \nabla_\mu g^{\mu\nu} \partial_\nu \phi \quad , \tag{5.56}$$

   it makes no sense to write this as $\nabla_\mu \partial^\mu \phi$: since ∂_μ does not commute with the metric in general, the notation ∂^μ is at best ambiguous as it is not clear whether this should represent $g^{\mu\nu}\partial_\nu$ or $\partial_\nu g^{\mu\nu}$ or something altogether different. This ambiguity does not arise for the Minkowski metric, but of course it is present in general.

A compact yet explicit expression for the Laplacian follows from the expression for the covariant divergence of a vector:

$$\begin{aligned} \Box\phi &:= g^{\mu\nu}\nabla_\mu\nabla_\nu\phi \\ &= \nabla_\mu(g^{\mu\nu}\partial_\nu\phi) \\ &= g^{-1/2}\partial_\mu(g^{1/2}g^{\mu\nu}\partial_\nu\phi) \ . \end{aligned} \tag{5.57}$$

Again, this formula is also useful (and provides the quickest way of arriving at the result) if one just wants to write the ordinary flat space Laplacian on $\mathbb{R}^3$ in, say, polar or cylindrical coordinates.

To illustrate this, let us calculate the Laplacian for the standard metric on $\mathbb{R}^{n+1}$ in polar coordinates. The standard procedure would be to first determine the coordinate transformation $x^i = x^i(r, \text{angles})$, then calculate $\partial/\partial x^i$, and finally assemble all the bits and pieces to calculate $\Delta = \sum_i (\partial/\partial x^i)^2$. This is a pain.

To calculate the Laplacian, we do not need to know the coordinate transformation, all we need is the metric. In polar coordinates, this metric takes the form

$$ds^2(\mathbb{R}^{n+1}) = dr^2 + r^2 d\Omega_n^2 \ , \tag{5.58}$$

where $d\Omega_n^2$ is the standard line-element on the unit n-sphere S^n. The determinant of this metric is $g \sim r^{2n}$ (times a function of the coordinates (angles) on the sphere). Thus, for $n = 1$ one has $ds^2 = dr^2 + r^2 d\phi^2$ and therefore

$$\Delta = r^{-1}\partial_\mu(r g^{\mu\nu}\partial_\nu) = r^{-1}\partial_r(r\partial_r) + r^{-2}\partial_\phi^2 = \partial_r^2 + r^{-1}\partial_r + r^{-2}\partial_\phi^2 \ . \tag{5.59}$$

In general, denoting the angular part of the Laplacian, i.e. the Laplacian of S^n, by Δ_{S^n}, one finds analogously

$$\Delta = \partial_r^2 + n r^{-1}\partial_r + r^{-2}\Delta_{S^n} \ . \tag{5.60}$$

I hope you agree that this method is superior to the standard procedure.

5. The Covariant Form of the Gauss Theorem

Let V^μ be a vector field, $\nabla_\mu V^\mu$ its divergence and recall that integrals in curved space are defined with respect to the integration measure $\sqrt{g}d^4x$. Thus one has

$$\int \sqrt{g}d^4x \nabla_\mu V^\mu = \int d^4x \partial_\mu(\sqrt{g}V^\mu) \ . \tag{5.61}$$

Now the integrand on the right-hand side is an ordinary total derivative and thus, by the usual fundamental theorem of calculus, the integral of this over some region R can be written as an integral over the boundary ∂R of that region,

$$\int_R \sqrt{g}d^4x \nabla_\mu V^\mu = \int_{\partial R} d^3x(\ldots) \ . \tag{5.62}$$

A somewhat more precise statement of this theorem, including the precise boundary contribution to the integral (involving the component of V^μ normal to the boundary), will be given in section 16.3.

In particular, if

- either the integral is over all of space-time and V^μ vanishes sufficiently rapidly at infinity,
- or the integral is over some region R with (finite) boundary ∂R, and the integrand vanishes on ∂R,

one has

$$\int \sqrt{g} d^4x \nabla_\mu V^\mu = 0 \ . \tag{5.63}$$

In these circumstances, one also has the integration by parts formula

$$\int \sqrt{g} d^4x \nabla_\mu (f V^\mu) = 0 \quad \Rightarrow \quad \int \sqrt{g} d^4x f \nabla_\mu V^\mu = -\int \sqrt{g} d^4x (\nabla_\mu f) V^\mu \tag{5.64}$$

for a scalar f and a vector field V^μ.

6. The Covariant Divergence of an Antisymmetric Tensor

For a $(p,0)$-tensor $T^{\mu\nu\cdots}$ one has

$$\begin{aligned} \nabla_\mu T^{\mu\nu\cdots} &= \partial_\mu T^{\mu\nu\cdots} + \Gamma^\mu_{\mu\lambda} T^{\lambda\nu\cdots} + \Gamma^\nu_{\mu\lambda} T^{\mu\lambda\cdots} + \dots \\ &= g^{-1/2} \partial_\mu (g^{1/2} T^{\mu\nu\cdots}) + \Gamma^\nu_{\mu\lambda} T^{\mu\lambda\cdots} + \dots \ . \end{aligned} \tag{5.65}$$

In particular, if $A^{\mu\nu\cdots}$ is completely anti-symmetric, the Christoffel terms disappear and one is left with

$$\nabla_\mu A^{\mu\nu\cdots} = g^{-1/2} \partial_\mu (g^{1/2} A^{\mu\nu\cdots}) \ . \tag{5.66}$$

7. The Lie derivative of the Metric

In section 3.2 we had encountered the expression (3.34) for the variation of the metric under an infinitesimal coordinate transformation $\delta x^\alpha = \epsilon V^\alpha$,

$$\delta_V g_{\alpha\beta} = V^\gamma \partial_\gamma g_{\alpha\beta} + (\partial_\alpha V^\gamma) g_{\gamma\beta} + (\partial_\beta V^\gamma) g_{\alpha\gamma} \ . \tag{5.67}$$

While we saw that this expression could be understood and deduced from the requirement that the variation of the metric is itself a tensorial object that transforms like the metric, the tensorial nature of the above expression is far from manifest. However, it has a very nice and simple expression in terms of covariant derivatives of V, namely

$$\delta_V g_{\alpha\beta} = \nabla_\alpha V_\beta + \nabla_\beta V_\alpha \tag{5.68}$$

(as is easily verified).

Thus a vector field K^α generates a symmetry of the metric (such vectors are called *Killing vectors*) if it satisfies the *Killing equation*

$$K \quad \text{Killing Vector} \quad \Leftrightarrow \quad \nabla_\alpha K_\beta + \nabla_\beta K_\alpha = 0 \ . \tag{5.69}$$

We can also obtain this condition as the covariantisation of the statement that in a particular coordinate system the coefficients of the metric do not depend on one of these coordinates, say y,

$$\partial_y g_{\alpha\beta} = 0 \ , \tag{5.70}$$

so that the metric is then manifestly invariant under translations in y. In such a coordinate system adapted to the symmetry at hand, these translations are generated by $K = \partial_y$, and for a vector of this form (in particular, thus, with constant coefficients) one has

$$\begin{aligned} K = \partial_y \quad &\Rightarrow \quad \nabla_\alpha K^\beta = \Gamma^\beta_{\alpha y} \\ &\Rightarrow \quad \nabla_\alpha K_\beta = \Gamma_{\beta\alpha y} \\ &\Rightarrow \quad \nabla_\alpha K_\beta + \nabla_\beta K_\alpha = \partial_y g_{\alpha\beta} \end{aligned} \tag{5.71}$$

(where in the last step the basic relation (2.68) was used). Thus we find that the fact that the metric is y-translation invariant can be characterised covariantly as the statement that $K = \partial_y$ satisfies

$$\partial_y g_{\alpha\beta} = 0 \quad \Leftrightarrow \quad \nabla_\alpha K_\beta + \nabla_\beta K_\alpha = 0 \ . \tag{5.72}$$

This is again the Killing equation (5.69). As this equation is now tensorial it is valid in any coordinate system, in particular independently of whether or not the coordinate system is adapted to K in the way described above.

The expressions (5.68) and (5.72) will be rederived (and placed into the general context of Lie derivatives and Killing vectors) in section 9 - see in particular section 9.5.

You will have noticed that many equations simplify considerably for completely anti-symmetric tensors. In particular, their curl can be defined in a tensorial way without reference to any metric. This observation is at the heart of the coordinate independent calculus of *differential forms*. In this context, the curl is known as the *exterior derivative*.

Indeed, it is also straightforward to show directly, i.e. without going through the illogical loop of introducing the covariant derivative in order to obtain something manifestly tensorial only to find it disappear again from the final expression, that $\partial_{[\mu} A_{\mu_1 \ldots \mu_p]}$ is a tensor, i.e. transforms as a tensor under coordinate transformations: what happens is that the possible obstructions to the tensorial behaviour, namely derivatives of Jacobians, drop out after anti-symmetrisations because they are are really 2nd partial

derivatives of the coordinates, which are symmetric and thus do not survive the anti-symmetrisation.

To see this completely explicitly, consider a covector $A_\mu(x)$ and a coordinate transformation $x^\mu = x^\mu(y^\alpha)$, with Jacobi matrix

$$J^\mu_\alpha = \frac{\partial x^\mu}{\partial y^\alpha} \quad . \tag{5.73}$$

As a covector, A_μ transforms as $A_\alpha = J^\mu_\alpha A_\mu$, and therefore its derivative transforms as (using $\partial_\beta = J^\nu_\beta \partial_\nu$)

$$A_\alpha = J^\mu_\alpha A_\mu \quad \Rightarrow \quad \partial_\beta A_\alpha = J^\mu_\alpha J^\nu_\beta \partial_\nu A_\mu + (\partial_\beta J^\mu_\alpha) A_\mu \quad . \tag{5.74}$$

Because of

$$\partial_\beta J^\mu_\alpha = \frac{\partial^2 x^\mu}{\partial y^\alpha \partial y^\beta} = \partial_\alpha J^\mu_\beta \quad , \tag{5.75}$$

for the anti-symmetrised derivative one finds the tensorial transformation behaviour

$$\partial_\beta A_\alpha - \partial_\alpha A_\beta = J^\mu_\alpha J^\nu_\beta (\partial_\nu A_\mu - \partial_\mu A_\nu) \quad . \tag{5.76}$$

Likewise, Lie derivatives of tensors in general (section 9) are, as the special case of the Lie derivative of the metric mentioned above - see (5.68), automatically tensorial objects (and one can, but need not, make their tensorial nature manifest by writing these derivatives in terms of covariant derivatives).

5.6 Appendix: A Formula for the Variation of the Determinant

Here is an elementary proof of the identity (5.49), and a useful more general formula for the variation of the determinant of the metric, namely

$$\delta g = g g^{\mu\nu} \delta g_{\mu\nu} \quad \text{or} \quad g^{-1} \delta g = g^{\mu\nu} \delta g_{\mu\nu} \quad . \tag{5.77}$$

This proof is based on the standard cofactor or minor expansion of the determinant of a matrix (an alternative standard proof can, as also outlined below, be based on the identity $\det G = \exp \operatorname{tr} \log G$ and its derivative or variation). The cofactor expansion formula for the determinant is

$$g = \sum_\nu (-1)^{\mu+\nu} g_{\mu\nu} |m_{\mu\nu}| \quad , \tag{5.78}$$

for a fixed (but arbitrary) value of the index μ. Here $|m_{\mu\nu}|$ is the determinant of the minor of $g_{\mu\nu}$, i.e. of the matrix one obtains by removing the μ'th row and ν'th column from $g_{\mu\nu}$.

As a consequence of (5.78) one also has

$$\sum_\nu (-1)^{\mu+\nu} g_{\lambda\nu} |m_{\mu\nu}| = 0 \qquad \lambda \neq \mu \quad , \tag{5.79}$$

since this is, in particular, the determinant of a matrix with $g_{\mu\nu} = g_{\lambda\nu}$, i.e. of a matrix with two equal rows. Together, these two results can be written as

$$\sum_\nu (-1)^{\mu+\nu} g_{\lambda\nu} |m_{\mu\nu}| = \delta_{\lambda\mu} g \quad . \tag{5.80}$$

This shows that the coefficients of the inverse metric $g^{\mu\nu}$ are given by

$$g^{\nu\mu} = (-1)^{\mu+\nu} \frac{|m_{\mu\nu}|}{g} \quad , \tag{5.81}$$

a formula that should also be familiar from linear algebra.

In order to now determine the variation (or derivative) of the determinant with respect to the matrix elements $g_{\mu\nu}$, for each value of μ we can use (5.78), and we then have to also perform the summation over μ. Noting that, by construction, $m_{\mu\nu}$ does not depend on $g_{\mu\nu}$, one then finds

$$\delta g = \sum_\mu \sum_\nu (-1)^{\mu+\nu} \delta g_{\mu\nu} |m_{\mu\nu}| = g g^{\nu\mu} \delta g_{\mu\nu} \quad . \tag{5.82}$$

For a symmetric matrix, in particular for the metric, this reduces to the formula (5.77) we set out to establish. Here are some variations and applications of this formula:

1. When the determinant g is viewed as a (smooth) function of the coefficients $g_{\mu\nu}$, this shows that
$$\frac{\partial g}{\partial g_{\mu\nu}} = g g^{\nu\mu} \quad . \tag{5.83}$$

2. It also implies
$$\delta\sqrt{g} = \tfrac{1}{2}\sqrt{g} g^{\nu\mu} \delta g_{\mu\nu} \quad , \tag{5.84}$$
a particularly useful result that we will repeatedly make use of.

3. An equally useful variant of this equation is an expression for the variation of $\sqrt{g}$ expressed in terms of the variations $\delta g^{\mu\nu}$ of the components of the inverse metric. As a consequence of
$$g^{\mu\nu} g_{\nu\lambda} = \delta^\mu_\lambda \quad \Rightarrow \quad \delta g^{\mu\nu} = -g^{\mu\lambda} \delta g_{\lambda\rho} g^{\rho\nu} \tag{5.85}$$
or
$$g^{\mu\nu} g_{\mu\nu} = 4 \quad \Rightarrow \quad (\delta g^{\mu\nu}) g_{\mu\nu} = -g^{\mu\nu} \delta g_{\mu\nu} \tag{5.86}$$
one can equivalently write (5.84) as
$$\delta\sqrt{g} = \tfrac{1}{2}\sqrt{g} g^{\nu\mu} \delta g_{\mu\nu} = -\tfrac{1}{2}\sqrt{g} g_{\nu\mu} \delta g^{\mu\nu} \quad . \tag{5.87}$$

4. It follows from (5.83) that if the variation is the partial derivative ("how does the determinant $g = g(x)$ of the metric vary with x?") one has

$$\partial_\lambda g = \frac{\partial g}{\partial g_{\mu\nu}} \partial_\lambda g_{\mu\nu} = g g^{\mu\nu} \partial_\lambda g_{\mu\nu} \ . \tag{5.88}$$

or

$$g^{-1} \partial_\lambda g = g^{\mu\nu} \partial_\lambda g_{\mu\nu} \ , \tag{5.89}$$

and therefore also

$$\partial_\lambda \sqrt{g} = \tfrac{1}{2} \sqrt{g} g^{\mu\nu} \partial_\lambda g_{\mu\nu} \ . \tag{5.90}$$

5. On the other hand, the contracted Christoffel symbol is

$$\Gamma^\mu_{\ \mu\lambda} = \frac{1}{2} g^{\mu\nu} \partial_\lambda g_{\mu\nu} \ . \tag{5.91}$$

Therefore

$$\Gamma^\mu_{\ \mu\lambda} = \frac{1}{\sqrt{g}} \partial_\lambda \sqrt{g} \tag{5.92}$$

which establishes the identity (5.49).

The result (5.82) can also be written in matrix form, with G denoting the matrix with components $(G)_{\mu\nu} = g_{\mu\nu}$, as

$$\delta \log \det G = \operatorname{tr} G^{-1} \delta G \ . \tag{5.93}$$

In this form, the result can also be derived from variation of the remarkably useful identity

$$\det G = \mathrm{e}^{\operatorname{tr} \log G} \tag{5.94}$$

This identity, in turn, can be derived in an elementary way for diagonalisable G by noting that it holds trivially for diagonal matrices, and therefore, by the conjugation invariance of det and tr, also for diagonalisable matrices (like the metric). [And if desired, this can in turn be extended to all matrices by topological arguments involving extensions of continuous functionals from the dense set of diagonalisable matrices to the space of all matrices ...]

5.7 Covariant Differentiation Along a Curve

So far, we have defined covariant differentiation for tensors defined everywhere in space-time. Frequently, however, one encounters tensors that are only defined on curves - like the momentum of a particle which is only defined along its world line. In this section we will see how to define covariant differentiation along a curve. Thus consider a curve $x^\mu(\tau)$ (where τ could be, but need not be, proper time) and the tangent vector field $X^\mu(x(\tau)) = \dot{x}^\mu(\tau)$. Now define the covariant derivative D_τ along the curve, covariantising $d/d\tau$, by

$$\frac{d}{d\tau} = \dot{x}^\mu \partial_\mu \quad \to \quad D_\tau = X^\mu \nabla_\mu = \dot{x}^\mu \nabla_\mu \ . \tag{5.95}$$

Frequently one also uses the (suggestive, but ugly) notation

$$D_\tau = D/D\tau \quad \text{or} \quad D/d\tau \ . \tag{5.96}$$

For example, for a vector one has

$$\begin{aligned} D_\tau V^\mu &= \dot{x}^\nu \partial_\nu V^\mu + \dot{x}^\nu \Gamma^\mu_{\ \nu\lambda} V^\lambda \\ &= \frac{d}{d\tau} V^\mu(x(\tau)) + \Gamma^\mu_{\ \nu\lambda}(x(\tau)) \dot{x}^\nu(\tau) V^\lambda(x(\tau)) \ . \end{aligned} \tag{5.97}$$

For this to make sense, V^μ needs to be defined only along the curve and not necessarily everywhere in space-time.

This notion of covariant derivative along a curve permits us, in particular, to define the (covariant) acceleration a^μ of a curve $x^\mu(\tau)$ as the covariant derivative of the velocity $u^\mu = \dot{x}^\mu$ along the curve,

$$a^\mu = D_\tau \dot{x}^\mu = \ddot{x}^\mu + \Gamma^\mu_{\ \nu\lambda} \dot{x}^\nu \dot{x}^\lambda = u^\nu \nabla_\nu u^\mu \ . \tag{5.98}$$

Thus we can characterise (affinely parametrised) geodesics as those curves whose covariant acceleration is zero,

$$\text{Geodesics:} \quad a^\mu = u^\nu \nabla_\nu u^\mu = 0 \ , \tag{5.99}$$

a reasonable and natural statement regarding the movement of freely falling particles. If they are not affinely parametrised, as in (2.136), then instead of $u^\nu \nabla_\nu u^\mu = 0$ one has

$$u^\nu \nabla_\nu u^\mu = \kappa u^\mu \ . \tag{5.100}$$

5.8 Parallel Transport and Geodesics

We now come to the important notion of *parallel transport* of a tensor along a curve. Note that, in a general (curved) metric space-time, it does not make sense to ask if two vectors defined at points x and y are parallel to each other or not. However, given a metric and a curve connecting these two points, one can compare the two by dragging one along the curve to the other using the covariant derivative.

We say that a tensor $T^{\cdots}_{\ \ \cdots}$ is *parallel transported along the curve* $x^\mu(\tau)$ if

$$D_\tau T^{\cdots}_{\ \ \cdots} = 0 \ . \tag{5.101}$$

Here are some immediate consequences of this definition:

1. In a locally inertial coordinate system along the curve, this condition reduces to $dT/d\tau = 0$, i.e. to the statement that the tensor does not change along the curve. Thus the above is indeed an appropriate tensorial generalisation of the intuitive notion of parallel transport to a general metric space-time.

2. The parallel transport condition is a first order differential equation along the curve and thus defines $T^{\cdots}_{\cdots}(\tau)$ given an initial value $T^{\cdots}_{\cdots}(\tau_0)$.

3. Taking T to be the tangent vector $u^\mu = \dot{x}^\mu$ to the curve itself, the condition for parallel transport becomes

$$D_\tau u^\mu = 0 \quad \Leftrightarrow \quad \ddot{x}^\mu + \Gamma^\mu_{\nu\lambda}\dot{x}^\nu\dot{x}^\lambda = 0 \ , \tag{5.102}$$

   i.e. precisely the geodesic equation. We have already seen that geodesics are precisely the curves with zero acceleration. We can now equivalently characterise them by the property that their tangent vectors are parallel transported (do not change) along the curve. For this reason geodesics are also known as *autoparallels*.

4. Since the metric is covariantly constant, it is parallel along any curve. Thus, in particular, if V^μ is parallel transported, also its length remains constant along the curve,

$$D_\tau V^\mu = 0 \quad \Rightarrow \quad \frac{d}{d\tau}(g_{\mu\nu}V^\mu V^\nu) = D_\tau(g_{\mu\nu}V^\mu V^\nu) = 0 \ . \tag{5.103}$$

   In particular, we rediscover the fact claimed in (2.100) that the quantity $g_{\mu\nu}\dot{x}^\mu\dot{x}^\nu$ is constant along a geodesic,

$$D_\tau \dot{x}^\mu = 0 \quad \Rightarrow \quad \frac{d}{d\tau}(g_{\mu\nu}\dot{x}^\mu\dot{x}^\nu) = 0 \ . \tag{5.104}$$

5. Now let $x^\mu(\tau)$ be a geodesic and V^μ parallel along this geodesic. Then, as one might intuitively expect, also the angle between V^μ and the tangent vector to the curve u^μ remains constant. This is a consequence of the fact that both the norm of V and the norm of u are constant along the curve and that

$$\frac{d}{d\tau}(g_{\mu\nu}u^\mu V^\nu) = D_\tau(g_{\mu\nu}u^\mu V^\nu) = g_{\mu\nu}(D_\tau u^\mu)V^\nu + g_{\mu\nu}u^\mu D_\tau V^\nu = 0 \tag{5.105}$$

5.9 Example: Parallel Transport on the 2-Sphere

As usual, the simplest non-trivial example is provided by the 2-sphere with its standard line element

$$ds^2 = d\Omega^2 = d\theta^2 + \sin^2\theta \, d\phi^2 \ , \tag{5.106}$$

with the non-zero Christoffel symbols (determined e.g. from the geodesic equation, as in (3.16) - (3.21))

$$\Gamma^\theta_{\phi\phi} = -\sin\theta \, \cos\theta \quad , \quad \Gamma^\phi_{\theta\phi} = \Gamma^\phi_{\phi\theta} = \cot\theta \ . \tag{5.107}$$

Let us consider parallel transport of some vector $V = V^\alpha\partial_\alpha$ along a circle with $\theta = \theta_0$ constant, choosing the angle ϕ to parametrise the curve, i.e. we consider the family of paths

$$x^\alpha(\tau) = (\theta(\tau), \phi(\tau)) = (\theta_0, \tau) \tag{5.108}$$

with tangent vector

$$\dot{x}^\alpha = (0,1) \ . \tag{5.109}$$

Note that this is not normalised in the standard way,

$$g_{\alpha\beta}\dot{x}^\alpha\dot{x}^\beta = \sin^2\theta_0 \ , \tag{5.110}$$

so proper distance would be measured not by ϕ but, as is also pictorially evident, by $s = \phi\sin\theta_0$ (which agrees with ϕ on the equator $\theta_0 = \pi/2$).

A vector $V^\alpha\partial_\alpha$ with coordinate components (V^θ, V^ϕ) parallel transported along such a curve thus satisfies the equations

$$\begin{aligned} 0 &= \partial_\phi V^\alpha + \Gamma^\alpha_{\beta\gamma}\dot{x}^\beta V^\gamma = \partial_\phi V^\alpha + \Gamma^\alpha_{\phi\gamma}V^\gamma \\ &= \partial_\phi V^\alpha + \Gamma^\alpha_{\phi\phi}V^\phi + \Gamma^\alpha_{\phi\theta}V^\theta \ . \end{aligned} \tag{5.111}$$

Using the explicit form of the Christoffel symbols, the parallel transport equations are thus

$$\begin{aligned} 0 &= \partial_\phi V^\theta - \sin\theta\ \cos\theta\ V^\phi \\ 0 &= \partial_\phi V^\phi + \cot\theta\ V^\theta \ . \end{aligned} \tag{5.112}$$

Differentiating once more, these equations can be decoupled and take the form of harmonic oscillator equations with frequency $\cos\theta_0$,

$$(\partial_\phi^2 + \cos^2\theta_0)V^\alpha = 0 \ . \tag{5.113}$$

The general solution of this 2nd order differential equation is of course

$$V^\alpha = A^\alpha\sin(\phi\cos\theta_0) + B^\alpha\cos(\phi\cos\theta_0) \ . \tag{5.114}$$

Plugging this into the 1st order equations to reduce the spurious 4 to 2 integration constants, and relating them to the intial values at $\phi = 0$, say,

$$V^\alpha(\theta_0, \phi = 0) = v^\alpha \ , \tag{5.115}$$

one finally finds the result

$$\begin{aligned} V^\theta(\theta_0,\phi) &= v^\phi\sin\theta_0\ \sin(\phi\cos\theta_0) + v^\theta\ \cos(\phi\cos\theta_0) \\ V^\phi(\theta_0,\phi) &= -(v^\theta/\sin\theta_0)\ \sin(\phi\cos\theta_0) + v^\phi\ \cos(\phi\cos\theta_0) \ . \end{aligned} \tag{5.116}$$

Remarks:

1. In the special case of parallel transport along the equator $\theta_0 = \pi/2$, one has $\cos\theta_0 = 0$, and therefore

$$\theta_0 = \pi/2 \quad \Rightarrow \quad V^\alpha(\pi/2,\phi) = v^\alpha \ . \tag{5.117}$$

In other words, the components are constant under parallel transport along the equator. This is inuitively obvious on the basis of spherical symmetry. Since among the family of constant $\theta = \theta_0$ curves only the equator is a geodesic (great circle), this is also in agreement with the general results obtained above, which imply that upon parallel transport along the equator the angle between the vector and the equator remains constant. In 2 dimensions, this condition, together with the fact that the lenght of a vector remains invariant under parallel transport in general, is sufficient to imply that the parallel transported components are constant along the path.

2. While the above is not unexpected, perhaps the most interesting consequence of the above result (5.116) is that, in general, not only are the components not constant but that actually, after having completed the 2π-circuit along the path to return to the starting point, the parallel transported vector will not agree with the initial vector. Indeed, the components at $\phi = 2\pi$ are related to the components v^α at $\phi = 0$ by

$$\begin{aligned} V^\theta(\theta_0, \phi = 2\pi) &= v^\phi \sin\theta_0 \ \sin(2\pi \ \cos\theta_0) + v^\theta \ \cos(2\pi \ \cos\theta_0) \\ V^\phi(\theta_0, \phi = 2\pi) &= -(v^\theta / \sin\theta_0) \ \sin(2\pi \ \cos\theta_0) + v^\phi \ \cos(2\pi \ \cos\theta_0) \ . \end{aligned} \tag{5.118}$$

3. As we will see in section 11.1, this fact that parallel transport along closed paths is non-trivial (equivalently that parallel transport from one point to another depends on the path) can be directly attributed to (and is the smoking gun of) the presence of *curvature*.

4. If desired, the result can be written in terms of proper distance s along the circle, rather than the angle ϕ, by the substitution

$$\phi \ \cos\theta_0 = s \ \cot\theta_0 \ . \tag{5.119}$$

5. The result (5.116) takes on a more transparent form when written in terms of the components of V and v with respect to an orthonormal basis (section 4.8) E_α rather than the coordinate basis ∂_α. Such an orthonormal basis is provided by

$$E_\theta = \partial_\theta \quad , \quad E_\phi = (\sin\theta)^{-1}\partial_\phi \ , \tag{5.120}$$

since one evidently has

$$g_{\alpha\beta} E^\alpha_\theta E^\beta_\theta = g_{\alpha\beta} E^\alpha_\phi E^\beta_\phi = 1 \quad , \quad g_{\alpha\beta} E^\alpha_\theta E^\beta_\phi = 0 \ . \tag{5.121}$$

The components with respect to this orthonormal basis are related to the coordinate components by

$$V = V^\alpha \partial_\alpha = \hat{V}^\alpha E_\alpha \quad \Rightarrow \quad \hat{V}^\theta = V^\theta \quad , \quad \hat{V}^\phi = \sin\theta \, V^\phi \tag{5.122}$$

(and likewise for $v = v^\alpha \partial_\alpha = \hat{v}^\alpha E_\alpha$). Then (5.116) can be written in matrix form as a rotation (orthogonal transformation)

$$\begin{pmatrix} \hat{V}^\theta(\theta_0, \phi) \\ \hat{V}^\phi(\theta_0, \phi) \end{pmatrix} = \begin{pmatrix} \cos(\phi \cos\theta_0) & \sin(\phi \cos\theta_0) \\ -\sin(\phi \cos\theta_0) & \cos(\phi \cos\theta_0) \end{pmatrix} \begin{pmatrix} \hat{v}^\theta \\ \hat{v}^\phi \end{pmatrix} \tag{5.123}$$

by the angle

$$\alpha(\phi) = \phi \cos\theta_0 \quad . \tag{5.124}$$

Thus parallel transport amounts to a continuous rotation of the orthonormal components along the path.

6. In particular, the angle that one picks up after a 2π-rotation,

$$\alpha(2\pi) = 2\pi \cos\theta_0 \quad , \tag{5.125}$$

is known as the *deficit angle* or *holonomy* of the parallel transport along the given loop. With this terminology we can say that the holonomy along the equator is trivial.

7. At the other extreme, we see that there is a non-trivial holonomy as $\theta_0 \to 0$, i.e. for parallel transport along an infinitesimal loop around the north pole, along which the parallel transported vector performs a complete 2π-rotation, $\alpha(2\pi) = 2\pi$. As shown in section 11.1, parallel transport along infinitesimal loops at or around a point provides a precise measure of the curvature at that point.

8. Curiously, as shown by Rothman, Ellis and Murugan, the holonomy along circular equatorial orbits in the Schwarzschild geometry (such orbits are geodesics at the critical points of the effective potential for geodesic motion, to be discussed in section 25.6), is non-trivial, even though again intuitive reasoning based on spherical symmetry might have led one to expect a trivial result (and would thus have led one astray).[13]

5.10 Fermi-Walker Parallel Transport

The properties of parallel transport established in section 5.8 show that this is a natural prescription for transporting tensorial objects along a geodesic. However, it is important to keep in mind that this is just one possible description, obtained by imposing the differential equation (5.101), e.g. for a vector

$$D_\tau V^\alpha = 0 \quad . \tag{5.126}$$

[13] T. Rothman, G. Ellis, J. Murugan, *Holonomy in the Schwarzschild-Droste Geometry*, `arXiv:gr-qc/0008070`.

If the curve is not a geodesic,

$$a^\alpha = D_\tau u^\alpha = \dot{x}^\beta \nabla_\beta \dot{x}^\alpha \neq 0 \ , \tag{5.127}$$

however, this prescription has some shortcomings. For example, parallel transport of a tangent vector to the curve at a point to another point at the curve will not give rise to the tangent vector at the second point, simply because $D_\tau V^\alpha = 0$ with initial condition $V^\alpha(\tau_0) = u^\alpha(\tau_0)$, say (parallel transport) is evidently not the same as $D_\tau u^\alpha = a^\alpha$ (the equation satisfied by the tangent vector). Likewise, the scalar product between the tangent vector to the (non-geodesic) curve and some parallel-transported vector along it will not remain constant in general,

$$D_\tau u^\alpha = a^\alpha \quad , \quad D_\tau V^\alpha = 0 \quad \Rightarrow \quad \frac{d}{d\tau}(g_{\alpha\beta} u^\alpha V^\beta) = a_\alpha V^\alpha \ . \tag{5.128}$$

A vivid illustration of this is provided by the example of the previous section:

- As we have seen, parallel transporting a pair of orthonormal vectors along a circle $\theta = \theta_0 \neq \pi/2$ results in a continuous rotation of these two basis vectors.
- On the other hand, it is clearly possible to transport an orthonormal basis along the circle in such a way that, for example, one basis vector always points forwards along the latitude (a tangent vector to the curve), and the other always points northwards along the longitude.

The latter procedure appears to be much more natural in this case than rotating one's basis as one goes around the sphere. Analogously, for an observer along a timelike curve it would be desirable to be able to set up once and for all a local reference system on the worldline, consisting of the (unit) tangent vector $E_0 = u^\mu \partial_\mu$ in the time-direction, and three orthogonal and mutually orthogonal vectors E_k in the spatial directions (the laboratory system of the observer), regardless of whether the observer is in free fall or not (indeed, for perfectly good reasons most laboratories are not ...).

This procedure can be formalised by replacing the parallel transport condition (5.126) along a timelike curve by the *Fermi-Walker Transport* prescription

$$F_\tau V^\alpha \equiv D_\tau V^\alpha + \mathcal{F}^\alpha_{\ \beta} V^\beta = 0 \ , \tag{5.129}$$

with

$$\mathcal{F}^\alpha_{\ \beta} = a^\alpha u_\beta - u^\alpha a_\beta \ . \tag{5.130}$$

Indeed, parallel transport according to this prescription has the following desirable features:

1. Fermi-Walker transport evidently reduces to parallel transport if the curve is a geodesic, i.e. for $a^\alpha = 0$. It obligingly does this even when the geodesic is not affinely parametrised, i.e. if one has $a^\alpha \sim u^\alpha$,

$$a^\alpha = u^\beta \nabla_\beta u^\alpha \sim u^\alpha \quad \Rightarrow \quad \mathcal{F}^\alpha_{\ \beta} = 0 \ . \tag{5.131}$$

2. The tangent vector to a curve is Fermi-Walker transported for any curve,

$$F_\tau u^\alpha = 0 \ . \tag{5.132}$$

Proof:

$$\begin{aligned} F_\tau u^\alpha &= D_\tau u^\alpha + \mathcal{F}^\alpha_{\ \beta} u^\beta \\ &= a^\alpha + (a^\alpha u_\beta - u^\alpha a_\beta) u^\beta = a^\alpha - a^\alpha = 0 \end{aligned} \tag{5.133}$$

because $u_\beta u^\beta = -1$ and $a_\beta u^\beta = 0$. Thus the solution to the Fermi-Walker transport prescription for $V^\alpha(\tau_0) = u^\alpha(\tau_0)$ is just the tangent vector u^α,

$$F_\tau V^\alpha = 0 \quad , \quad V^\alpha(\tau_0) = u^\alpha(\tau_0) \quad \Rightarrow \quad V^\alpha(\tau) = u^\alpha(\tau) \ . \tag{5.134}$$

3. If V^α is Fermi-Walker transported along the curve, then instead of (5.128) one obtains

$$D_\tau u^\alpha = a^\alpha \quad , \quad D_\tau V^\alpha = -\mathcal{F}^\alpha_{\ \beta} V^\beta \quad \Rightarrow \quad \frac{d}{d\tau}(g_{\alpha\beta} u^\alpha V^\beta) = 0 \ . \tag{5.135}$$

Proof:

$$\begin{aligned} \frac{d}{d\tau}(g_{\alpha\beta} u^\alpha V^\beta) = D_\tau(g_{\alpha\beta} u^\alpha V^\beta) &= g_{\alpha\beta}(a^\alpha V^\beta - u^\alpha \mathcal{F}^\beta_{\ \gamma} V^\gamma) \\ &= a_\alpha V^\alpha - a_\alpha u^\alpha u_\gamma V^\gamma + u_\alpha u^\alpha a_\gamma V^\gamma = a_\alpha V^\alpha - a_\gamma V^\gamma = 0 \ , \end{aligned} \tag{5.136}$$

beause $a_\alpha u^\alpha = 0$ and $u_\alpha u^\alpha = -1$.

4. Similarly, if V and W are Fermi-Wallker transported, one has

$$F_\tau V^\alpha = F_\tau W^\alpha = 0 \quad \Rightarrow \quad \frac{d}{d\tau}(g_{\alpha\beta} V^\alpha W^\beta) = 0 \ . \tag{5.137}$$

Proof:

$$\begin{aligned} \frac{d}{d\tau}(g_{\alpha\beta} V^\alpha W^\beta) &= g_{\alpha\beta}(-\mathcal{F}^\alpha_{\ \gamma} V^\gamma W^\beta - \mathcal{F}^\beta\ _\gamma V^\alpha W^\gamma) \\ &= -(\mathcal{F}_{\alpha\gamma} + \mathcal{F}_{\gamma\alpha}) V^\alpha W^\gamma = 0 \end{aligned} \tag{5.138}$$

because $\mathcal{F}_{\alpha\gamma}$ is anti-symmetric.

<u>Remarks:</u>

1. The signs chosen here are appropriate for timelike curves with $u^\alpha u_\alpha = -1$. As the proofs of the above statements show, in the spacelike case one needs to replace $\mathcal{F}^\alpha_{\ \beta} \to -\mathcal{F}^\alpha_{\ \beta}$.

2. The above manipulations can be formalised (and then subsequently trivialised) by

   - extending the action of F_τ to arbitrary rank tensors in the same way as the covariant derivative, i.e. by requiring that on scalars it reduces to the ordinary derivative,

$$F_\tau f = \frac{d}{d\tau} f \ , \tag{5.139}$$

- extending it to arbitrary tensors by requiring the Leibniz rule, so that e.g. on covectors one has

$$F_\tau A_\beta = D_\tau A_\beta - \mathcal{F}^\alpha_{\ \beta} A_\alpha \quad , \tag{5.140}$$

- and then showing that as a consequence

$$\begin{aligned} F_\tau g_{\alpha\beta} &= D_\tau g_{\alpha\beta} - \mathcal{F}^\gamma_{\ \alpha} g_{\gamma\beta} - \mathcal{F}^\gamma_{\ \beta} g_{\alpha\gamma} \\ &= -(\mathcal{F}_{\alpha\beta} + \mathcal{F}_{\beta\alpha}) = 0 \quad . \end{aligned} \tag{5.141}$$

Then assertions like (5.135),

$$F_\tau u^\alpha = F_\tau V^\alpha = 0 \quad \Rightarrow \quad \frac{d}{d\tau}(u_\alpha V^\alpha) = 0 \quad . \tag{5.142}$$

or (5.137) become a triviality.

3. Note that the properties 2-4 in the above list rely on the 3 properties

$$\mathcal{F}^\alpha_{\ \beta} u^\beta = -a^\alpha \quad , \quad u_\alpha \mathcal{F}^\alpha_{\ \beta} = a_\beta \quad , \quad \mathcal{F}_{\alpha\beta} + \mathcal{F}_{\beta\alpha} = 0 \tag{5.143}$$

of $\mathcal{F}^\alpha_{\ \beta}$ respectively. These conditions determine $\mathcal{F}^\alpha_{\ \beta}$ up to rotations in the plane orthogonal to u^α, i.e. up to the ambiguity

$$\mathcal{F}^\alpha_{\ \beta} \to \mathcal{F}^\alpha_{\ \beta} + \omega^\alpha_{\ \beta} \tag{5.144}$$

with

$$u^\alpha \omega_{\alpha\beta} = \omega_{\alpha\beta} u^\beta = 0 \quad , \quad \omega_{\alpha\beta} + \omega_{\beta\alpha} = 0 \quad . \tag{5.145}$$

Since there is no such rotation term in the prescription for Fermi-Walker transport, and no natural candidate for it either with only u^α and a^α at one's disposal, it is natural to think of Fermi-Walker transport as a prescription for transporting objects in a non-rotating way.

4. In particular, if one uses the Fermi-Walker prescription to construct an orthonormal basis (E_0, E_k) along the worldline, the spatial vectors can be interpreted as providing a *non-rotating* choice of axes.[14]

5.11 Epilogue: Manifolds? Think Globally, Act Locally!

In section 4.9 I had already briefly discussed some issues regarding the use of indices (and thus in some sense of local coordinates), and had advocated them as a useful bookkeeping device that also provides a transparent way of performing algebraic operations (tensor algebra). In the meantime we have seen that this extends to tensor analysis, and I can only reiterate that for most purposes and in most cases it is much more convenient to

[14] And according to S. Hawking, G. Ellis, *The large-scale structure of space-time*, section 4.1, these "could be realised physically by small gyroscopes pointing in the direction of each vector".

perform calculations in this notation than in some supposedly more elegant index-free notation.

There is one issue, however, that is worth commenting upon, and that in the end actually provides further justification for being allowed to adopt this procedure. Namely, in using local (Cartesian, say) coordinates x^μ to describe a space or space-time (I will use "space" in the following) one is implicitly assuming the following 3 things:

1. first of all, that one can always locally introduce Cartesian coordinates on that space (so as to then be able to perform tensor algebra, tensor analysis etc.);
2. secondly, that different choices of local coordinates will give compatible descriptions of that space;
3. and finally, that in principle one can obtain complete information about the space by covering it with such local coordinate systems.

When these assumptions are satisfied, then one is justified in using local coordinates to describe such a space. The point of this brief section is just to point out that (modulo some topological fine-points) these conditions amount precisely to the definition of a (differentiable or smooth) *manifold* in mathematics.

Thus while I could have started off these notes with an introduction to and definition of smooth manifolds (and numerous textbooks do), for all local intents and purposes this is then really equivalent to (consistently) working in local coordinates, as we have done and will continue to do. It is true that the notion of manifolds, of vector bundles on them etc. becomes indispensable for certain more advanced questions dealing with the global structure of a space-time, or theorems about the existence and uniqueness of solutions to differential equations on some manifold, say, but these are not topics that will be addressed in these notes.

The idea of a manifold is that an n-dimensional manifold is a sufficiently nice topological space that locally looks like (i.e. can be modelled on) the simple and nice topological space $\mathbb{R}^n$, and that this allows one to do calculus on this space by importing the relevant concepts from $\mathbb{R}^n$.

The usual textbook definition of a manifold consists essentially of the following steps:[15]

1. Topological Spaces

   A topological space is a set S together with a collection of subsets U of S (called open sets) which includes S and the empty set, and which is closed under union

[15] This presentation is adapted from the concise and clear description in S. Mukhi, N. Mukunda, *Lectures on Advanced Mathematical Methods for Physicists*, but equivalent descriptions can be found in many other places.

and finite intersection. This set of open sets defines the topology of the space and a corresponding notion of continuous maps (the inverse image of any open set is open) and homeomorphisms (bijective maps ϕ such that both ϕ and ϕ^{-1} are continuous) between topological spaces. In particular there is a notion of continuity for (real-valued, say) functions

$$f: \quad S \to \mathbb{R} \tag{5.146}$$

(with $\mathbb{R}$ equipped with its standard topology).

2. Charts

   However, in this context there is no notion of differentiability or differentiation. In order to have such things at one's disposal one needs topological spaces that locally "look like" $\mathbb{R}^n$. The essential building blocks of such a topological space are "charts":

   A chart C on a topological space S is the pair $C = (U, \phi)$ where $U \subset S$ is an open set of S and ϕ is a homeomorphism

   $$U \subset S \to \phi(U) \subset \mathbb{R}^n \ . \tag{5.147}$$

   The homeomorphism condition implies in particular that $\phi(U)$ is open in $\mathbb{R}^n$. The integer n is then known as the dimension of U (it does not depend on ϕ).

3. Topologial Manifolds

   A topological manifold is a topological space M that is locally homeomorphic to $\mathbb{R}^n$ in the sense that for each point p there is a chart $C = (U, \phi)$ with $p \in U$ (and that satisfies some further topological regularity conditions we are not interested in, such as Hausdorff and usually either second countable or paracompact). Equivalently, a topological space has the structure of a topological manifold when it possesses a covering by open sets U_a with charts $C_a = (U_a, \phi_a)$.

4. Local Coordinates and Local Coordinate Transformations

   The notion of a chart allows (and is equivalent to and the formalisation of) the introduction of local coordinates on the open set $U \subset M$. The coordinates of a point $p \in U$ in this chart are by definition simply the Cartesian coordinates $\vec{x}_p$ of the point $\phi(p) \in \mathbb{R}^n$.

   If one has two charts on M, $C_1 = (U_1, \phi_1)$ and $C_2 = (U_2, \phi_2)$, and $U_1 \cap U_2 \neq \emptyset$, then the "transition functions"

   $$\begin{aligned} \phi_1 \circ \phi_2^{-1}: &\quad \phi_2(U_1 \cap U_2) \to \phi_1(U_1 \cap U_2) \\ \phi_2 \circ \phi_1^{-1}: &\quad \phi_1(U_1 \cap U_2) \to \phi_2(U_1 \cap U_2) \end{aligned} \tag{5.148}$$

   are automatically continous maps between open subsets of $\mathbb{R}^n$. These can be interpreted as local coordinate transformations,

   $$(\phi_2 \circ \phi_1^{-1})(\vec{x}_p^1) = \phi_2(p) = \vec{x}_p^2 \ . \tag{5.149}$$

5. Local Functions and Differentiation

   In particular, with the help of charts we can express functions on M in terms of "local oordinates" on $\mathbb{R}^n$. More precisely, given a (continuous) function

$$f: \quad M \to \mathbb{R} \tag{5.150}$$

   and a chart $C = (U, \phi)$, we can associate to the restriction of f to U the function

$$f_U = f \circ \phi^{-1}: \quad \phi(U) \subset \mathbb{R}^n \to \mathbb{R} \ . \tag{5.151}$$

   i.e.

$$p \in U \quad \Rightarrow \quad f(p) = f_U(\vec{x}_p) \ . \tag{5.152}$$

   For such functions on $\mathbb{R}^n$ we now not only have a notion of continuity at our disposal, but also the notions of differentiability, smoothness, differentiation etc.

   On the intersection of 2 charts we can represent the function f in 2 different ways in terms of local coordinates, namely by the functions $f_{U_a} \equiv f_a$ for $a = 1, 2$,

$$\begin{aligned} \text{on } \ U_1 \cap U_2: \quad f = f_1 \circ \phi_1 = f_2 \circ \phi_2 \quad \Rightarrow \quad & f_2 = f_1 \circ (\phi_1 \circ \phi_2^{-1}) \\ & f_1 = f_2 \circ (\phi_2 \circ \phi_1^{-1}) \end{aligned} \tag{5.153}$$

   This is just the change of variables formula for a function (scalar), namely

$$f_2(\vec{x}_p^2) = f_1(\vec{x}_p^1) \ . \tag{5.154}$$

6. Compatibility of Charts

   In order to be able to extend the notion of smoothness (C^∞-differentiability), say, of a function from a local chart consistently to all of M, we need to impose compatibility conditions on intersecting charts.

   It is evident from (5.153) that the notion of smoothness of a function around a point p will only be independent of the chart if the transition functions $\phi_1 \circ \phi_2^{-1}$ and $\phi_2 \circ \phi_1^{-1}$ (i.e. the coordinate transformations) are also smooth. Thus we define 2 charts to be smoothly compatible if either $U_1 \cap U_2$ is empty or, otherwise, if these maps are smooth.

   Note that for topological manifolds and the condition of continuity any 2 charts are automatically compatible since the transition functions are continuous.

7. Smooth Atlas and Compatibility and Equivalence of Atlases

   A smooth atlas $\mathcal{A}(M)$ of M is now naturally a family of charts $C_a = (U_a, \phi_a)$ which cover M and such that all charts are mutually smoothly compatible.

   2 smooth atlases $\mathcal{A}_1(M)$ and $\mathcal{A}_2(M)$ for the same topological manifold M are said to be compatible with each other if all the charts of $\mathcal{A}_1$ are compatible with all the charts of $\mathcal{A}_2$. This defines an equivalence relation on atlases.

8. Smooth Structure and Smooth Manifold

   A smooth structure on a topological manifold M is an equivalence class

   $$\mathcal{S}(M) = [\mathcal{A}(M)] \tag{5.155}$$

   of smooth atlases on M. A smooth manifold is a topological manifold M equipped with a smooth structure $\mathcal{S}$.

9. Smooth Functions and Smooth Maps

   A function

   $$f: \quad M \to \mathbb{R} \tag{5.156}$$

   on a smooth manifold M is then said to be smooth if all its local coordinate representatives

   $$f_a = f \circ \phi_a^{-1}: \quad \phi(U_a) \to \mathbb{R} \tag{5.157}$$

   are smooth, and a map

   $$\mu: \quad M \to N \tag{5.158}$$

   from a smooth manifold M (with charts (U_a, ϕ_a)) of dimension m to a smooth manifold N (with charts $(V_{b'}, \psi_{b'})$) of dimension n is said to be smooth if all of its local coordinate representatives

   $$\mu_{ab'} = \psi_{b'} \circ \mu \circ \phi_a^{-1}: \quad \phi_a(U_a) \subset \mathbb{R}^m \to \psi_{b'}(V_{b'}) \subset \mathbb{R}^n \tag{5.159}$$

   are smooth. Such smooth functions can be differentiated by differentiating their local coordinate representatives and mapping the result back to M using the charts (and likewise for maps).

Topological fine-points aside we see that a smooth manifold is by definition a space on which one can consistently do calculus in local coordinates. Hence in these notes we were, are and will be dealing with (smooth) manifolds, regardless of whether or not we state this explicitly.

Analogously one can define C^k-differentiable manifolds (transition functions are required to be of degree C^k), real analytic manifolds (transition functions are required to be real analytic), complex manifolds (modelled on open subsets of $\mathbb{C}^n$, with holomorphic transition functions), etc., as well as submanifolds (modelled on subspaces of $\mathbb{R}^n$), manifolds with boundary (modelled on the half-space $\mathbb{R}^n_+$) etc.

6 Physics in a Gravitational Field and Minimal Coupling

6.1 Principle (or Algorithm) of Minimal Coupling

Recall that the Principle of General Covariance (section 4.1) says that, by virtue of the Einstein Equivalence Principle, a generally covariant equation will be valid in an arbitrary gravitational field provided that it is valid in Minkowski space in inertial coordinates (i.e. in the absence of gravity and/or acceleration).

We now have all the tools at our disposal to construct such equations. In particular, the fact that the covariant derivative ∇ maps tensors to tensors and reduces to the ordinary partial derivative in a locally inertial coordinate system suggests the following procedure or algorithm for obtaining equations that satisfy the Principle of General Covariance:

1. Write down the Lorentz invariant equations or expressions of Special Relativity you are interested in (e.g. those of relativistic mechanics, Maxwell theory, relativistic hydrodynamics, ...) in terms of inertial coordinates ξ^a, the Minkowski metric η_{ab} and other Lorentz tensors $T^{a\cdots}_{b\cdots}$.

2. Replace the coordinates ξ^a by arbitrary coordinates x^μ,
$$\xi^a \mapsto x^\mu \ . \tag{6.1}$$

3. Wherever the Minkowski metric η_{ab} appears, replace it by the metric $g_{\mu\nu}$ describing the gravitational field,
$$\eta_{ab} \mapsto g_{\mu\nu}(x) \ . \tag{6.2}$$

4. Promote the Lorentz tensors $T^{a\cdots}_{b\cdots}$ to tensors $T^{\mu\cdots}_{\nu\cdots}$ under general coordinate transformations,
$$T^{a\cdots}_{b\cdots}(\xi) \mapsto T^{\mu\cdots}_{\nu\cdots}(x) \ . \tag{6.3}$$

5. Wherever a partial derivative $\partial_a = \partial_{\xi^a}$ appears, replace it by the covariant derivative ∇_μ,
$$\partial_a \mapsto \nabla_\mu \ . \tag{6.4}$$

6. In particular, for the proper-time derivative along a curve this entails replacing $d/d\tau$ by D_τ,
$$\frac{d}{d\tau} \mapsto D_\tau = \dot{x}^\mu \nabla_\mu \ . \tag{6.5}$$

7. Wherever an integral $\int d^4\xi$ appears, replace it by $\int \sqrt{g} d^4x$,
$$\int d^4\xi \mapsto \int \sqrt{g} d^4x \ . \tag{6.6}$$

By construction, the resulting equations or expressions are tensorial (generally covariant) and true in the absence of gravity and hence satisfy the conditions for the Principle of General Covariance to apply. As a consequence they will be true in the presence of gravitational fields, at least on scales small compared to those of the gravitational fields. This procedure can thus be regarded as providing us with a prescription how to couple matter (particles, fields) to the gravitational field.

Remarks:

1. This procedure is analogous to the perhaps more familiar "minimal coupling" algorithm for the coupling of matter to gauge fields ("replace partial by gauge covariant derivatives"), and hence also in the current context this procedure is referred to as minimal coupling.

2. The reasons for the "at least on small scales" *caveat* in the paragraph above is that if one considers higher derivatives of the metric tensor then there are other equations that one can write down, involving e.g. the curvature tensor, that are tensorial but reduce to the same equations in the absence of gravity.

3. Thus "minimal coupling", as formulated here, is not a unique and unambiguous description, but it is nevertheless a pragmatic and effective procedure. We will see an example of the ambiguity in the minimal coupling prescription in the discussion of Maxwell theory in a gravitational field in section 6.6, and we will briefly return to the issue in section 8.10.

6.2 Particle Mechanics in a Gravitational Field Revisited

We can see the power of the formalism we have developed so far by rederiving the laws of particle mechanics in a general gravitational field. In Special Relativity (SR), the motion of a free particle with mass m is governed by the equation

$$\text{SR:} \quad a^a = \frac{du^a}{d\tau} = 0 \ , \tag{6.7}$$

where $u^a = d\xi^a/d\tau$ is the 4-velocity and a^a the 4-acceleration. Thus, using the principle of minimal coupling, the equation of motion of a free particle in a general gravitational field is

$$\text{GR:} \quad a^\mu = D_\tau u^\mu = 0 \quad \Leftrightarrow \quad \ddot{x}^\mu + \Gamma^\mu_{\nu\lambda}\dot{x}^\nu\dot{x}^\lambda = 0 \ , \tag{6.8}$$

where $u^\mu = dx^\mu/d\tau$. Thus we rediscover the familiar geodesic equation, but we see that it follows much faster from demanding general covariance (as made precise by the principle of minimal coupling) than from our previous somewhat more convoluted and roundabout considerations based e.g. on the equivalence principle.

We could also have arrived at this equation for a free particle in a gravitational field by applying the minimal coupling description not at the level of the equations of motion but rather (and perhaps conceptually more satisfactorily) at the level of the action, i.e. by replacing

$$S = -m\int d\tau = -m\int \sqrt{-\eta_{ab}d\xi^a d\xi^b} \quad \to \quad -m\int d\tau = -m\int \sqrt{-g_{\mu\nu}dx^\mu dx^\nu} \ , \tag{6.9}$$

and this is exactly what we already did back in section 2.3 where we showed that this also leads to the geodesic equation (6.8).

6.3 Klein-Gordon Scalar Field in a Gravitational Field

Here is where the formalism we have developed really pays off. We will see once again that, using the minimal coupling rule, we can immediately rewrite the equations for a scalar field (here) and the Maxwell equations (in section 6.6 below) in a form in which they are valid in an arbitrary gravitational field.

1. The action for a (real) free massive scalar field ϕ in Special Relativity is

$$\text{SR:} \qquad S[\phi] = \int d^4\xi \left[-\tfrac{1}{2}\eta^{ab}\partial_a\phi\partial_b\phi - \tfrac{1}{2}m^2\phi^2\right] \ . \tag{6.10}$$

   To covariantise this, we replace $d^4\xi \to \sqrt{g}d^4x$, $\eta^{ab} \to g^{\alpha\beta}$, and we can replace ∂_a by ∇_α or ∂_α (since this makes no difference on scalars). Therefore, the covariant action in a general gravitational field is

$$\text{GR:} \qquad S[\phi, g_{\alpha\beta}] = \int \sqrt{g}d^4x \left[-\tfrac{1}{2}g^{\alpha\beta}\partial_\alpha\phi\partial_\beta\phi - \tfrac{1}{2}m^2\phi^2\right] \ . \tag{6.11}$$

   Here I have also indicated the dependence of the action on the metric $g_{\alpha\beta}$. This is not (yet) a dynamical field, though, just the gravitational background field.

2. The equations of motion for ϕ one derives from this are

$$\frac{\delta}{\delta\phi}S[\phi, g_{\alpha\beta}] = 0 \quad \Rightarrow \quad \left(\Box_g - m^2\right)\phi = 0 \tag{6.12}$$

   where $\Box_g = g^{\alpha\beta}\nabla_\alpha\nabla_\beta$, the Laplacian associated to the metric $g_{\alpha\beta}$. This is precisely what one would have obtained by applying the minimal coupling description to the Minkowski Klein-Gordon equation $(\Box_\eta - m^2)\phi = 0$.

   Remarks:

   (a) A comment on how to derive this: if one thinks of the ∂_α in the action as covariant derivatives, $\partial_\alpha \to \nabla_\alpha$, then one can use the covariant intgration by

parts formula (5.64) to conclude (dropping the boundary term, as usual in variational calculus)

$$-\int \sqrt{g}d^4x\ (\nabla_\alpha\delta\phi)g^{\alpha\beta}\nabla_\beta\phi = \int \sqrt{g}d^4x\ \delta\phi\ g^{\alpha\beta}\nabla_\alpha\nabla_\beta\phi = \int \sqrt{g}d^4x\ \delta\phi\ \Box_g\phi\ . \tag{6.13}$$

Thus essentially the calculation is identical to that in Minkowski space. If one sticks with the ordinary partial derivatives, then upon the usual integration by parts one picks up a term $\sim \partial_\alpha(\sqrt{g}g^{\alpha\beta}\partial_\beta\phi)$ which then evidently leads to the Laplacian in the form (5.57).

(b) If the relative sign of $\Box_\eta$ (or $\Box_g$) and m^2 in the Klein-Gordon equation looks unfamiliar to you, then this is probably due to the fact that in a course where you first encountered the Klein-Gordon equation the opposite (particle physicists') sign convention for the Minkowski metric was used, with its negative definite spatial metric.

(c) All of this generalises in a straightforward way to (self-)interacting scalar fields, described by a potential $V(\phi)$. In particular, the action is

$$S[\phi, g_{\alpha\beta}] = \int \sqrt{g}d^4x\ \left[-\tfrac{1}{2}g^{\alpha\beta}\partial_\alpha\phi\partial_\beta\phi - V(\phi)\right]\ . \tag{6.14}$$

Logically the next thing to discuss would be the energy-momentum tensor, e.g. the minimally coupled counterpart of the special relativistic (Noether) energy-momentum tensor

$$\text{SR:}\quad T_{ab} = \partial_a\phi\partial_b\phi + \eta_{ab}L \tag{6.15}$$

and its properties. However, it turns out that there is more to say about this than meets the eye, and we will therefore return to this issue in more detail in section 7.

6.4 Interlude: General Covariance in Minkowski Space?

Before turning to our next example, I want to briefly comment on the issue of general covariance in Minkowski space, as this tends to generate quite a bit of confusion and unnecessary debates. I will discuss this issue in the context of the above example of a scalar field, but the discussion is valid more generally.

On the one hand, the action (6.10) is generally considered to be invariant (only) under Lorentz or Poincaré transformations, while by construction the action (6.11) is invariant under arbitrary coordinate transformations. Does this really mean that the theory of a scalar field in a non-trivial gravitational background has more invariances than that in a Minkowski background?

On the other hand, certainly nothing prevents one from using e.g. spherical (and thus in particular non-inertial) coordinates in Minkowski space to write down the Klein-Gordon

equation or action. But does this mean that the action (6.10) is actually (secretly) invariant also under such non-Lorentz transformations?

Well, that depends ... While this sounds like (and generally is correctly considered to be) a somewhat unsatisfactory answer, I can be more specific:

- it depends on what one means by "invariance" (or "covariance")
- and it depends on how one treats or regards the Minkowski metric.

From the current point of view, the natural answer is that the action (6.11) is generally covariant in *any* gravitational field, in particular therefore also in the absence of a true gravitational field, i.e. in a purely fictitious gravitational field or, equivalently, in Minkowski space. If we specialise the action (6.11) to such a gravitational field, i.e. to the Minkowski metric written in some perhaps non-inertial coordinates, we get

$$S[\phi, \eta_{\alpha\beta}] = \int \sqrt{\eta} d^4x \left[-\tfrac{1}{2}\eta^{\alpha\beta}\partial_\alpha\phi\partial_\beta\phi - \tfrac{1}{2}m^2\phi^2 \right] \quad . \tag{6.16}$$

Here it is important to keep in mind that $\eta_{\alpha\beta}$ refers to the components of the Minkowski metric in the not-necessarily inertial coordinates x^α, as in

$$\eta_{\alpha\beta} = \frac{\partial\xi^a}{\partial x^\alpha}\frac{\partial\xi^b}{\partial x^\beta}\eta_{ab} \quad . \tag{6.17}$$

As a consequence, also $\sqrt{\eta}$ is not necessarily equal to 1. This action is invariant under arbitrary coordinate transformations, provided that one transforms the fields *and the metric* appropriately.

If one now chooses to write this action in inertial coordinates, $x^\alpha \to \xi^a$, with $\eta_{\alpha\beta} \to \eta_{ab}$ and thus $\sqrt{\eta} \to 1$, then the action (6.16) appears to reduce to the special relativistic action (6.10). So is this action, which is simply a generally covariant action written in some particular coordinates, invariant under Lorentz (or Poincaré) transformations only or under all coordinate transformations?

1. If one looks for the transformations of the coordinates ξ^a and the fields ϕ that leave the action invariant (with fixed metric components η_{ab}) then none too surprisingly one finds that the action is invariant under Poincaré transformations of the coordinates provided that the scalar fields transform as scalars, but not under more general transformations.

2. If one looks for the transformations of the coordinates ξ^a and the fields ϕ *and the metric* η_{ab} that leave the action invariant, then one finds that the action is invariant under arbitrary coordinate transformations

   - provided that one also transforms $\eta_{ab} \to \eta_{\alpha\beta}$ like a (0,2)-tensor

- and provided that one either thinks of $d^4\xi$ as the invariant volume element $\sqrt{\eta}d^4\xi$, or equivalently one treats the Lagrangian L as a scalar density $\sqrt{\eta}L$.

Sometimes option (1) is taken to define the *invariance group* (Poincaré transformations) while option (2) refers to the *covariance group*. In this sense, special relativity is *invariant* under Poincaré transformations but is at the same time *generally covariant*. In philosophy of science or epistemological terms whether one has option (1) or option (2) is related to the question whether or not the Minkowski metric is regarded as an *absolute element* of the theory. With η_{ab} promoted to an absolute element, general covariance is reduced to Poincaré invariance (those transformations that, from the generally covariant "$\eta_{\alpha\beta}$ transforms" point of view, leave η_{ab} invariant).[16] Unfruitful discussions ensue when tacitly conflicting assumptions are made about what are considered to be the absolute elements of a theory.

6.5 Lorentz-Covariant Formulation of Maxwell Theory (Review)

In order to discuss the formulation of Maxwell theory in a gravitational field, we will need to quickly recall the Lorentz-covariant formulation of Maxwell theory in Minkowski space. This will also fix our conventions for Maxwell theory.

In the traditional non-covariant formulation one has

1. the homogeneous equations

$$\vec{\nabla}\cdot\vec{B}=0 \quad , \quad \vec{\nabla}\times\vec{E}+\partial_t\vec{B}=0 \tag{6.18}$$

2. the inhomogeneous equations

$$\vec{\nabla}\cdot\vec{E}=\rho/\epsilon_0 \quad , \quad \vec{\nabla}\times\vec{B}-\frac{1}{c^2}\partial_t\vec{E}=\mu_0\vec{J} \tag{6.19}$$

3. the ensuing continuity equation

$$\partial_t\rho+\vec{\nabla}.\vec{J}=0 \tag{6.20}$$

4. the vector and scalar potentials $\vec{A}$ and ϕ,

$$\vec{B}=\vec{\nabla}\times\vec{A} \quad , \quad \vec{E}=-\vec{\nabla}\phi-\partial_t\vec{A} \tag{6.21}$$

5. and the corresponding gauge transformations leaving $\vec{E}$ and $\vec{B}$ invariant,

$$\vec{A}\to\vec{A}+\vec{\nabla}\Psi \quad , \quad \phi\to\phi-\partial_t\Psi \quad \Rightarrow \quad \vec{E}\to\vec{E} \quad , \quad \vec{B}\to\vec{B} \ . \tag{6.22}$$

[16] For more erudite discussions of these and related issues, see e.g. section 2.5 ("On Covariance and Invariance") of N. Straumann, *General Relativity* or D. Giulini, *Some remarks on the notions of general covariance and background independence*, `arXiv:gr-qc/0603087`.

The charge density and current can be packaged into a Lorentz vector

$$J^a = (c\rho, \vec{J}) \tag{6.23}$$

(note that in signature (-+++) one has to choose whether to identify J^0 or $J_0 = -J^0$ with the charge density, here we choose the former), and the continuity equation can be written in the manifestly Lorentz-invariant form

$$\partial_a J^a = 0 \ . \tag{6.24}$$

Likewise, the scalar and vector potential can be packaged into a Lorentz covector

$$A_a = (-\phi/c, \vec{A}) \ , \tag{6.25}$$

and the gauge transformations can be compactly written as

$$A_a \to A_a + \partial_a \Psi \ . \tag{6.26}$$

The gauge invariant Maxwell field strength tensor F_{ab} is defined by

$$F_{ab} = \partial_a A_b - \partial_b A_a \tag{6.27}$$

and has the components

$$F_{0k} = -F_{k0} = -E_k/c \quad , \quad F_{ik} = \epsilon_{ik\ell} \, B_\ell \tag{6.28}$$

or, in matrix form,

$$(F_{ab}) = \begin{pmatrix} 0 & -E_1/c & -E_2/c & -E_3/c \\ +E_1/c & 0 & +B_3 & -B_2 \\ +E_2/c & -B_3 & 0 & +B_1 \\ +E_3/c & +B_2 & -B_1 & 0 \end{pmatrix} \tag{6.29}$$

and

$$(F^{ab}) = \begin{pmatrix} 0 & +E_1/c & +E_2/c & +E_3/c \\ -E_1/c & 0 & +B_3 & -B_2 \\ -E_2/c & -B_3 & 0 & +B_1 \\ -E_3/c & +B_2 & -B_1 & 0 \end{pmatrix} \tag{6.30}$$

In terms of these Lorentz tensors, the homogeneous Maxwell equations can be written as

$$\partial_{[a} F_{bc]} = 0 \quad \Leftrightarrow \quad \partial_a F_{bc} + \partial_c F_{ab} + \partial_b F_{ca} = 0 \ , \tag{6.31}$$

and these equations are identically satisfied if F_{ab} derives from a potential,

$$F_{ab} = \partial_a A_b - \partial_b A_a \quad \Rightarrow \quad \partial_{[a} F_{bc]} = 0 \ . \tag{6.32}$$

The inhomogeneous Maxwell equations can (suppressing μ_0, i.e. setting $\mu_0 = 1$) be written as

$$\partial_a F^{ab} = -J^b \quad \Leftrightarrow \quad \Box A_a - \partial_a (\partial_b A^b) = -J_a \ . \tag{6.33}$$

These equations can be derived from the Lorentz-invariant action

$$S[A,J] = S[A] + S_I[A,J] = -\tfrac{1}{4}\int d^4\xi\ F_{ab}F^{ab} + \int d^4\xi\ A_a J^a \ , \tag{6.34}$$

with the Maxwell Lagrangian

$$-\tfrac{1}{4}F_{ab}F^{ab} = -\tfrac{1}{2}F_{0k}F^{0k} - \tfrac{1}{4}F_{ik}F^{ik} = \tfrac{1}{2}(\vec{E}^2/c^2 - \vec{B}^2) \ . \tag{6.35}$$

This is essentially all we will need (some facts regarding the Noether versus covariant energy-momentum tensor of Maxwell theory will be recalled below).

6.6 Maxwell Theory in a Gravitational Field

Mutatis mutandis we can now proceed in the same way as for a scalar field.

1. The basic dynamical field is the vector potential A_a. Given the vector potential A_μ, the Maxwell field strength tensor in Special Relativity is

$$\text{SR:} \quad F_{ab} = \partial_a A_b - \partial_b A_a \ . \tag{6.36}$$

   Therefore in a general metric space-time (gravitational field) one is led to (or tempted to) define the field strength tensor as

$$\text{GR:} \quad F_{\mu\nu} = \nabla_\mu A_\nu - \nabla_\nu A_\mu = \partial_\mu A_\nu - \partial_\nu A_\mu \ . \tag{6.37}$$

   - A cautionary remark: Actually, this is a bit misleading. The field strength tensor (two-form) in any, Abelian or non-Abelian, gauge theory is always given in terms of the gauge-covariant exterior derivative of the vector potential (i.e. it is the curvature of the connection), and as such has nothing whatsoever to do with a metric on space-time. So you should not really regard the first equality in the above equation as the definition of $F_{\mu\nu}$, but you should regard the second equality as a proof that $F_{\mu\nu}$, *always* defined by $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$, is a tensor.

     The mistake of adopting $\nabla_\mu A_\nu - \nabla_\nu A_\mu$ as the definition of $F_{\mu\nu}$ in a curved space-time has led some poor souls to believe, and even claim in published papers, that in a space-time with torsion, for which the second equality does not hold, the Maxwell field strength tensor is invariably modified by the torsion. This is nonsense.

     If there is torsion, one is of course free to consider non-minimal couplings of the torsion tensor to other tensor fields (like the Maxwell field strength tensor), but this is not required by either gauge invariance or general covariance.

2. In Special Relativity, the Maxwell equations read

$$\begin{aligned} \text{SR:} \qquad & \partial_a F^{ab} = -J^b \\ & \partial_{[a} F_{bc]} = 0 \ . \end{aligned} \tag{6.38}$$

Thus in a general gravitational field (curved space-time) these equations become

$$\begin{aligned} \text{GR:} \quad & \nabla_\mu F^{\mu\nu} = -J^\nu \\ & \nabla_{[\mu} F_{\nu\lambda]} = 0 \ , \end{aligned} \tag{6.39}$$

where now of course all indices are raised and lowered with the metric $g_{\mu\nu}$,

$$F^{\mu\nu} = g^{\mu\lambda} g^{\nu\rho} F_{\lambda\rho} \ . \tag{6.40}$$

Remarks:

(a) Regarding the use of the covariant derivative in the second equation, the same *caveat* as above applies.

(b) In particular, using the results derived in section 5.5, we can rewrite these two equations as

$$\begin{aligned} \text{GR:} \qquad & \partial_\mu (\sqrt{g} F^{\mu\nu}) = -\sqrt{g} J^\nu \\ & \partial_{[\mu} F_{\nu\lambda]} = 0 \ . \end{aligned} \tag{6.41}$$

(c) It is clear from the first of these equations that the Maxwell equations imply that the current is covariantly conserved: since

$$\partial_\nu \partial_\mu (\sqrt{g} F^{\mu\nu}) = 0 \tag{6.42}$$

by anti-symmetry of $F^{\mu\nu}$, it follows that

$$\partial_\nu (\sqrt{g} J^\nu) = 0 \quad \Leftrightarrow \quad \nabla_\nu J^\nu = 0 \ . \tag{6.43}$$

From the covariant version $\nabla_\mu F^{\mu\nu} = -J^\nu$ this follows in the seemingly more roundabout way from the identity (8.47) for the commutator of covariant derivatives that we will establish later, in the context of our discussion of the Riemann curvature tensor in section 8.

(d) In Special Relativity, the inhomogeneous Maxwell equations can be decoupled by imposing the Loren(t)z (see footnote 65 in section 23.5) gauge condition $\partial_a A^a = 0$,

$$\partial_a A^a = 0 \quad \Rightarrow \quad \partial_a F^{ab} = -J^b \ \rightarrow \ \Box A_b = -J_b \ . \tag{6.44}$$

This gauge condition has the virtue of preserving Lorentz invariance. Similarly, its covariantised version

$$\nabla_\mu A^\mu = 0 \tag{6.45}$$

has the virtue of preserving general covariance, because $\nabla_\mu A^\mu$ is a scalar. However, the inhomogeneous Maxwell equations in this gauge do not take the form $\Box A_\nu = -J_\nu$ one might perhaps have anticipated on the basis of minimal coupling. Rather, using the covariant Lorenz gauge condition (6.45), the covariant divergence of the Maxwell field strength tensor can be written as

$$\nabla_\mu A^\mu = 0 \quad \Rightarrow \quad \nabla_\mu F^{\mu\nu} = \nabla_\mu(\nabla^\mu A^\nu - \nabla^\nu A^\mu) = \Box A^\nu - [\nabla_\mu, \nabla_\nu] A^\mu \ , \tag{6.46}$$

where $\Box A_\mu = \nabla^\nu \nabla_\nu A_\mu$ is the "naive" Laplacian on scalars. The second term would of course be zero in Minkowski space, but here it is not. Indeed, as we will see in section 8, the quintessence of a non-trivial geometry is that covariant derivatives do not commute on tensors other than scalars. In particular, here one finds that as a consequence of (8.40) the Maxwell equations in the covariant Lorenz gauge can be written as

$$\Box A^\nu - R^\nu_{\ \mu} A^\mu = -J^\nu \ , \tag{6.47}$$

where $R_{\mu\nu}$ is the *Ricci tensor*, a particular contraction of the Riemann curvature tensor, constructed from the second derivatives of the metric.

(e) Thus, these equations appear to display a *non-minimal coupling to the gravitational field*, even though we started off with the minimally coupled equations which we can also derive, see below, from the minimally coupled action. We will return to a discussion of this issue in section 8.10.

3. The electromagnetic force acting on a particle of charge e is given in Special Relativity by the Lorentz force

$$\text{SR:} \quad f^a = e F^a_{\ b} \dot{\xi}^b \ . \tag{6.48}$$

Thus in General Relativity it becomes

$$\text{GR:} \quad f^\mu = e g^{\mu\lambda} F_{\lambda\nu} \dot{x}^\nu \ . \tag{6.49}$$

4. The Lorentz-invariant action of (vacuum) Maxwell theory is

$$\text{SR:} \quad S[A_a] = -\tfrac{1}{4} \int d^4\xi \ F_{ab} F^{ab} \tag{6.50}$$

in Special Relativity, and thus becomes

$$\text{GR:} \quad S[A_\alpha, g_{\alpha\beta}] = -\tfrac{1}{4} \int \sqrt{g} d^4x \ F_{\mu\nu} F^{\mu\nu} \equiv -\tfrac{1}{4} \int \sqrt{g} d^4x \ g^{\mu\lambda} g^{\nu\rho} F_{\mu\nu} F_{\lambda\rho} \tag{6.51}$$

in General Relativity.

As for the scalar field, depending on whether one writes the field strength tensor as $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$ or as $F_{\mu\nu} = \nabla_\mu A_\nu - \nabla_\nu A_\mu$, by varying this action with respect to the A_μ one obtains the vacuum Maxwell equations $\nabla_\mu F^{\mu\nu} = 0$ in either of the 2 forms

$$\frac{\delta}{\delta A_\nu} S[A_\alpha, g_{\alpha\beta}] = 0 \quad \Rightarrow \quad \left\{ \begin{array}{c} \partial_\mu(\sqrt{g} F^{\mu\nu}) = 0 \\ \nabla_\mu F^{\mu\nu} = 0 \end{array} \right. \tag{6.52}$$

REMARKS:

(a) Writing out explicitly the Lagrangian in terms of its components (with respect to some coordinate system $x^\alpha = (t, x^k)$) one finds

$$\begin{aligned} -\tfrac{1}{4} F_{\alpha\beta} F^{\alpha\beta} = & -\tfrac{1}{2} F_{0i} F_{0k} (g^{00} g^{ik} - g^{0i} g^{0k}) \\ & -\tfrac{1}{4} F_{ij} F_{kl} g^{ik} g^{jl} \\ & -\tfrac{1}{2} F_{0i} F_{kl} (g^{0k} g^{il} - g^{0l} g^{ik}) \end{aligned} \tag{6.53}$$

While the 1st and 2nd lines look just like "gravitationally dressed" standard terms $\sim \vec{E}^2$ and $\sim \vec{B}^2$, the last line appears to suggest a gravitationally induced coupling between the electric and magnetic fields. This, however, is misleading and simply not a meaningful way of expressing things. After all, even in Minkowski space the decomposition of the electro-magnetic field into electric and magnetic fields depends on the choice of inertial reference system.

(b) In order to add sources, one can add $\int \sqrt{g} d^4x \; A_\mu J^\mu$ to the Maxwell action, thus coupling the matter current to the Maxwell gauge field. Instead of just adding such a (phenomenological) source-term by hand, a more coherent microscopic approach (which also provides the sources with their own dynamics) is to consider a matter action (minimally) coupled to the Maxwell field,

$$S_M[\phi] \to S_M[\phi, A_\alpha] \quad . \tag{6.54}$$

The combined Maxwell + matter action will then give rise to the Maxwell equations with a source provided that one defines the current J^α as the variation of the matter action with respect to the gauge field,

$$J^\alpha \sim \frac{\delta S_M[\phi, A_\alpha]}{\delta A_\alpha} \quad . \tag{6.55}$$

As in the case of scalar fields, we will postpone a discussion of the energy-momentum tensor and how to properly define it (something that is already an issue in Minkowski space because for Maxwell theory the Noether energy-momentum tensor turns out to be neither symmetric nor gauge-invariant!) to section 7.

In anticipation of this I just want to point out that, by the same rationale as that leading to (6.55), perhaps we should define the source term for the gravitational field by the variation of the gravitationally minimally coupled matter action with respect to the metric. If we now call this source term the energy-momentum tensor, then we have a candidate definition of the energy-momentum tensor which is natural and appropriate from the gravitational point of view. We will pursue this point of view in section 7.6.

6.7 Minimal Coupling and (quasi-)Topological Couplings

In all the cases considered so far, the minimal coupling prescription resulted in a minimally coupled matter action that depends explicitly on the metric - this is as it should be and is not a surprise. What would be more of a surprise would be to find minimally coupled and hence generally covariant contributions to an action that do not depend on the metric, but such examples do indeed exist (and play an important role in many branches of physics and even mathematics, ranging from the strong-CP problem in QCD to high-T_c superconductors to topology). Such terms in the action are usually referred to as "topological terms" in the physics literature but as they need not be (and usually are not) purely topological in the mathematics sense, for lack of a better name I refer to them as "quasi-topological".

Here are 2 prototypical examples illustrating this phenomenon:

1. Axionic Coupling in (3+1) Dimensions

   The first toy-model we will consider consists of Maxwell-theory coupled to a neutral scalar field through what is known as an axionic coupling only (with analogous considerations for the more interesting case of a non-Abelian Yang-Mills field),

   $$S[\phi, A] = S_s[\phi] + S_m[A] + S_a[\phi, A] \quad , \tag{6.56}$$

   with

   $$S_s[\phi] = \int d^4x \; L_s(\phi, \partial_\alpha \phi) \tag{6.57}$$

   some arbitrary standard scalar field action (of the type already discussed), $S_m[A]$ the usual Maxwell action,

   $$S_m[A] = \int d^4x \; L_m(\partial_\alpha A_\beta) = -\tfrac{1}{4} \int d^4x \; F^{\alpha\beta} F_{\alpha\beta} \tag{6.58}$$

   and the axionic coupling term is

   $$S_a[\phi, A] = -\tfrac{1}{4} \int d^4x f(\phi) \tilde{F}^{\alpha\beta} F_{\alpha\beta} \tag{6.59}$$

   where

   $$\tilde{F}^{\alpha\beta} = \tfrac{1}{2} \in^{\alpha\beta\gamma\delta} F_{\gamma\delta} \tag{6.60}$$

with $\in^{\alpha\beta\gamma\delta} = 0, \pm 1$ the Levi-Civita symbol, a tensor density (cf. remark 2 in section 4.5) and $f(\phi)$ some function of the scalar field ϕ. Note that for $f(\phi) = 1$ (or in the absence of a scalar field) the axionic term would be (locally) a total derivative and would hence not contribute to the equations of motion. For non-trivial $f(\phi)$, on the other hand, the axionic term is itself non-trivial.

Minimal coupling for the first two (standard) terms proceeds as already discussed above. For the third, axionic term, we make the usual replacement $d^4x \to \sqrt{g}d^4x$ and recall from (4.83) that

$$\epsilon^{\alpha\beta\gamma\delta} \equiv \frac{1}{\sqrt{g}} \in^{\alpha\beta\gamma\delta} \tag{6.61}$$

is a (4,0) tensor, so that the generally covariant generalisation of the axionic action is

$$\begin{aligned} S_a[\phi, A, g_{\alpha\beta}] &= -\tfrac{1}{8} \int \sqrt{g} d^4x f(\phi) \epsilon^{\alpha\beta\gamma\delta} F_{\gamma\delta} F_{\alpha\beta} \\ &= -\tfrac{1}{8} \int d^4x f(\phi) \in^{\alpha\beta\gamma\delta} F_{\gamma\delta} F_{\alpha\beta} = S_a[\phi, A] \end{aligned} \tag{6.62}$$

We see that, as announced, the metric dependence drops out of the minimally coupled generally covariant action. The reason for this is that the axionic Lagrangian is already all by itself a scalar density of weight $w = +1$, and that therefore its integral (4.76) is well-defined and generally covariant without having to take recourse to a metric to construct an auxiliary weight-one object like $\sqrt{g}$.

2. Maxwell - Chern-Simons Theory in (2+1) Dimensions

The second prominent example involves the addition of what is known as an Abelian Chern-Simons term to the Maxwell action in (2+1) dimensions (with analogous considerations for the more interesting case of a non-Abelian Yang-Mills field). The Minkowski space Lagrangian of this model is

$$L = L_m + k\, L_{cs} = -\tfrac{1}{4} F^{\alpha\beta} F_{\alpha\beta} + \tfrac{1}{2} k \;\in^{\alpha\beta\gamma} A_\alpha F_{\beta\gamma} \;\; . \tag{6.63}$$

Minimal coupling for the first term is standard and for the 2nd term one finds, as above, that the generally covariant minimally coupled Chern-Simons action is actually metric independent (since the Chern-Simons Lagrangian is a density of weight $w = 1$),

$$\begin{aligned} S_{cs}[A, g_{\alpha\beta}] &= \tfrac{1}{2} k \int \sqrt{g} d^3x \, \epsilon^{\alpha\beta\gamma} A_\alpha F_{\beta\gamma} \\ &= \tfrac{1}{2} k \int d^3x \;\in^{\alpha\beta\gamma} A_\alpha F_{\beta\gamma} = S_{cs}[A] \end{aligned} \tag{6.64}$$

As an aside note that the above theory is also known as topologically massive Maxwell theory, since the CS term provides a gauge-invariant mass term for the photon. One quick way to see this is to note that the equations of motion are

$$\partial_\alpha F^{\alpha\beta} + k \;\in^{\beta\gamma\delta} F_{\gamma\delta} = 0 \;\; . \tag{6.65}$$

and that in terms of the dual field strength

$$G^\beta = \tfrac{1}{2} \in^{\beta\gamma\delta} F_{\gamma\delta} \tag{6.66}$$

the equations of motion and the Bianchi identity take the form

$$\partial_\alpha G_\beta - \partial_\beta G_\alpha = 2k \ \in_{\alpha\beta\gamma} G^\gamma \quad , \quad \partial_\beta G^\beta = 0 \tag{6.67}$$

respectively. Acting with ∂^α on the equation of motion and using the Bianchi identity and again the equation of motion one finds

$$\begin{aligned} \Box G_\beta &= 2k \ \in_{\alpha\beta\gamma} \partial^\alpha G^\gamma = k \ \in_{\alpha\beta\gamma} (\partial^\alpha G^\gamma - \partial^\gamma G^\alpha) \\ &= 2k^2 \ \in_{\alpha\beta\gamma} \in^{\alpha\gamma\delta} \ G_\delta = 4k^2 G_\beta \end{aligned} \tag{6.68}$$

so that the theory describes excitations of mass $m^2 = 4k^2$.

These quasi-topological terms modify the equations of motion. Moreover, since they depend on the derivatives of the fields, they will contribute to the canonical Noether energy-momentum tensor. On the other hand, since they do not depend on the metric, they do not contribute to the covariant energy-momentum tensor, defined in section 7 in terms of the variation of the matter action with respect to the metric (and as such playing the role of the source term for the Einstein gravitational field equations).

How it nevertheless conspires that this tensor is conserved on-shell (meaning: for a solution to the matter equations of motion) even though the equations of motion have been modified and how the improved canonical energy-momentum tensor nevertheless ends up agreeing with the covariant energy-momentum tensor on-shell will be explored and explained in section 22.5.

6.8 Conserved Charges from Covariantly Conserved Currents

In Special Relativity a conserved current J^a is characterised by the vanishing of its divergence, i.e. by $\partial_a J^a = 0$. It leads to a conserved charge Q by integrating J^a over a spacelike hypersurface, say the one described by $t = t_0$,

$$Q = \int_{t=t_0} d^3x \ J^0 \ . \tag{6.69}$$

That Q is conserved, i.e. independent of t_0, is a consequence of the fact that by virtue of the Gauss theorem

$$Q(t_1) - Q(t_0) = \int_V d^4\xi \ \partial_a J^a = 0 \ , \tag{6.70}$$

where V is the four-volume $\mathbb{R}^3 \times [t_0, t_1]$. This holds provided that J vanishes at spatial infinity.

Now in General Relativity, the conservation law will be replaced by the covariant conservation law $\nabla_\mu J^\mu = 0$, and one may wonder if this also leads to some conserved charges in the ordinary sense. The answer is yes because, recalling the formula for the covariant divergence of a vector,

$$\nabla_\mu J^\mu = g^{-1/2}\partial_\mu(g^{1/2}J^\mu) \ , \tag{6.71}$$

we see that

$$\nabla_\mu J^\mu = 0 \Leftrightarrow \partial_\mu(g^{1/2}J^\mu) = 0 \ , \tag{6.72}$$

so that $g^{1/2}J^\mu$ is a conserved current in the ordinary sense. We then obtain conserved quantities in the ordinary sense by integrating J^μ over a spacelike hypersurface Σ. We will develop a more precise formula for this, an appropriate version of the Gauss theorem for hypersurfaces in curved space-times, in section 16.3.

The factor $g^{1/2}$ apearing in the current conservation law can be understood physically. To see what it means, split J^μ into its space-time direction u^μ, with $u^\mu u_\mu = -1$, and its magnitude ρ as

$$J^\mu = \rho u^\mu \ . \tag{6.73}$$

This defines the average four-velocity of the conserved quantity represented by J^μ and its density ρ measured by an observer moving at that average velocity (rest mass density, charge density, number density, ...). Since u^μ is a vector, in order for J^μ to be a vector, ρ has to be a scalar. Therefore this density is defined as per unit proper volume. The factor of $g^{1/2}$ transforms this into density per coordinate volume and this quantity is conserved (in a comoving coordinate system where $J^0 = \rho, J^i = 0$).

We will come back to this in the context of cosmology later on in this course, but for now just think of the following picture (Figure 44 in section 34): take a balloon, draw lots of dots on it at random, representing particles or galaxies. Next choose some coordinate system on the balloon and draw the coordinate grid on it. Now inflate or deflate the balloon. This represents a time dependent metric, roughly of the form $ds^2 = r^2(t)(d\theta^2 + \sin^2\theta d\phi^2)$. You see that the number of dots per coordinate volume element (area element in this case) does not change, whereas the number of dots per unit proper volume (area) will.

7 Energy-Momentum Tensor I: Basics

7.1 Introduction

Newton's gravitational field equation for the gravitational potential ϕ is the Poisson equation $\Delta\phi = 4\pi G_N \mu$, with μ the mass density. Thus in Newton's theory, mass is the source of gravity. We can also more usefully, and thinking relativistically, write this in terms of the energy density $\rho = \mu c^2$ as

$$\Delta\phi = \frac{4\pi G_N}{c^2}\rho \overset{(c=1)}{=} 4\pi G_N \ \rho \ . \tag{7.1}$$

Now we already noted in section 1.1 that in Special Relativity ρ is not a scalar but rather just one component of a tensor, the energy-momentum tensor

$$T_{ab}: \ \ T_{00} = \rho \ , \tag{7.2}$$

with the components T_{ab} transforming into each other under Lorentz transformations according to the transformation rules for Lorentz tensors.

It is therefore entirely plausible that in a relativistic theory of gravity the source of gravity should be the entire energy-momentum tensor. In particular, also the other components of T_{ab}, T_{0k} ($\sim$ energy flux), T_{k0} ($\sim$ momentum density) and T_{ik} ($\sim$ stresses or pressure) are a source of gravity. Clearly, therefore, the notion of energy-momentum tensor will play a crucial role in the following. This then immediately raises the question how to find or define an energy-momentum tensor.

Within the framework of special relativity and relativistic field theories there are (at least) 2 common approaches to constructing or defining an energy-momentum tensor, namely

1. a Macroscopic Phenomenological Description
2. a Microscopic Lagrangian Prescription

and we will now briefly discuss these in turn.

7.2 Perfect Fluid Energy-Momentum Tensor in Special Relativity

A macroscopic phenomenological description is useful when one does not know (or does not care about) the microscopic description of the matter one is dealing with but rather tries to characterise its properties in terms of the specification of some macroscopic (thermodynamic, hydrodynamic) parameters such as energy (density), pressure, viscosity etc. For many purposes this is the appropriate language for describing e.g. gases or fluids.

In this case, one constructs the energy-momentum tensor in such a way that it encodes the physics one is trying to describe (primarily conservation laws and dynamics). As a simple example of this (not by coincidence the one which is of most relevance for gravitational physics and thus also later on in these notes), we consider a *perfect fluid*.

By definition, a perfect fluid is one in which a comoving observer (i.e. an observer in a local rest-frame of the fluid) sees the fluid around him as isotropic (rotation-invariant). This means that in this reference system the components of the energy-momentum tensor have the form (any non-zero T_{0k} would break rotation invariance, and δ_{ik} is the unique rotation-invariant symmetric $(0,2)$-tensor)

$$T_{00} = \rho \quad , \quad T_{0k} = 0 \quad , \quad T_{ik} = p\delta_{ik} \ . \tag{7.3}$$

Here ρ and p are any functions of the coordinates, interpreted as the energy density and the pressure of the fluid .

To specify the kind of fluid one is working with, one should supplement this by an *equation of state* which provides a relation between ρ and p. Typically this amounts to specifying p as a function of ρ,

$$\text{Equation of State:} \quad p = p(\rho) \tag{7.4}$$

(and possibly other parameters).

In terms of the 4-velocity u^a of the fluid, which in the local rest frame has the components

$$u^a = (1,0,0,0) \ , \tag{7.5}$$

one can combine the components of the energy-momentum tensor into the expression,

$$T_{ab} = (\rho + p)u_a u_b + p\eta_{ab} \ , \tag{7.6}$$

(note that energy density and pressure = force per unit area have the same dimensions). As this is now a tensorial equation it is now valid in any inertial system. It defines the energy-momentum tensor of a perfect fluid. The conditions

$$\partial^a T_{ab} = 0 \tag{7.7}$$

imply a continuity equation and (as we will see below) a relativistic generalisation of the Euler equations for a perfect fluid. These are usually supplemented by a further continuity equation for the fluid density current

$$j^a = nu^a \tag{7.8}$$

with n e.g. the number density or particle density, say, namely

$$\partial_a j^a = 0 \ . \tag{7.9}$$

Now let us look at the consequences of these equations. Since

$$u^a u_a = -1 \quad \Rightarrow \quad (\partial_a u_b) u^b = \partial_a (u_b u^b)/2 = 0 \ , \tag{7.10}$$

the u-component of (7.7) can be written as

$$(\partial^a T_{ab}) u^b = 0 \quad \Leftrightarrow \quad u^a \partial_a \rho + (\rho + p) \partial_a u^a = 0 \ . \tag{7.11}$$

With the help of the current conservation equation, this equation can be recast into the form

$$\begin{aligned} 0 &= u^a \partial_a \rho + (\rho + p) \partial_a (j^a / n) \\ &= u^a \partial_a \rho + (\rho + p) j^a \partial_a (1/n) \\ &= u^a [\partial_a \rho + (\rho + p) n \partial_a (1/n)] \\ &= n u^a [p \partial_a (1/n) + \partial_a (\rho/n)] \ . \end{aligned} \tag{7.12}$$

The point of rewriting the equation in this way is that (assuming a situation of thermodynamic equilibrium) the 2nd law of thermodynamics says that pressure p, energy density ρ and the volume per particle $(1/n)$ are related by

$$T ds = p d(1/n) + d(\rho/n) \ . \tag{7.13}$$

where T is the temperature and s the specific entropy, i.e. the entropy per particle.[17] Thus the above equation says that the specific entropy s is constant along the flow,

$$u^a \partial_a s = 0 \ . \tag{7.14}$$

The significance of the spatial (transverse to u) components of (7.7) is easier to decipher if one writes the equations non-covariantly by setting

$$u^0 = \gamma(v) \quad , \quad u^i = \gamma(v) v^i = v^i u^0 \ , \tag{7.15}$$

so that

$$u^a \partial_a = \gamma(v) (\partial_t + \vec{v}.\vec{\nabla}) \tag{7.16}$$

is ($\gamma(v)$ times) the usual *convective derivative* or comoving time-derivative, and the above equation for the conservation of the specific entropy can be written as

$$(\partial_t + \vec{v}.\vec{\nabla}) s = 0 \ . \tag{7.17}$$

Moreover, the continuity equation for the current j^a with components

$$j^0 = \gamma(v) n \quad , \quad j^i = \gamma(v) n v^i \tag{7.18}$$

becomes

$$\partial_t (\gamma(v) n) + \vec{\nabla}.(\gamma(v) n \vec{v}) = 0 \ , \tag{7.19}$$

[17] See e.g. J. van Holten, *Relativistic Fluid Dynamics*, `http://www.nikhef.nl/~t32/relhyd.pdf` for a derivation of this and further discussion.

and the time-component of (7.7) can be written as

$$\partial_t(p-\gamma(v)^2(\rho+p))-\vec{\nabla}.[\gamma(v)^2(\rho+p)\vec{v}]=0 \ . \tag{7.20}$$

Using this equation the spacelike components of (7.7) can then be written as

$$\gamma(v)^2(\rho+p)(\partial_t\vec{v}+\vec{v}.\vec{\nabla}\vec{v})+\vec{v}\partial_t p+\vec{\nabla}p=0 \ . \tag{7.21}$$

In a suitable non-relativistic limit ($v \ll 1, p \ll \rho$), this latter equation reduces to

$$\rho(\partial_t\vec{v}+\vec{v}.\vec{\nabla}\vec{v})+\vec{\nabla}p=0 \ , \tag{7.22}$$

which is the non-relativistic Euler equation for a perfect fluid.

As we will see, it is straightforward to promote such a perfect fluid energy-momentum tensor by minimal coupling to the energy-momentum tensor describing a perfect fluid in a gravitational field, and we will come back to this below.

For a covariant rendition and elementary covariant derivation of the ensuing equations of motion in a general gravitational field from the conservation of the energy-momentum tensor, see e.g. the derivation of (35.81) and (35.82) in sections 35.4 and 35.5.

7.3 Noether Energy-Momentum Tensor in Special Relativity (Review)

A microscopic Lagrangian description is the method of choice when one has a Poincaré-invariant Lagrangian field theory description of the matter one is trying to describe. In particular, this applies to the scalar and Maxwell field theories we have already discussed and, more generally, to the modern microscopic and action-based description of the fundamental interactions of particle physics.

In this case, there is a canonical procedure for constructing an energy-momentum tensor, namely from Noether's theorem applied to translations, resulting in what is then appropriately known as the *Noether energy-momentum tensor* or the *canonical energy-momentum tensor* Θ_{ab}.

For a Lagrangian $L = L(\phi, \partial_a\phi)$ depending on some fields ϕ and their 1st derivatives (these could be scalar, vector, ... fields), this tensor is defined by

$$\Theta^a_{\ b}=-\frac{\partial L}{\partial(\partial_a\phi)}\partial_b\phi+\delta^a_{\ b}L \tag{7.23}$$

(sign conventions are such that Θ_{00} rather than $\Theta^0_{\ 0}$ is the energy density). It is built from the 4 Noether currents

$$\Theta^a_{\ b}\equiv J^a_{(b)} \tag{7.24}$$

associated to translation invariance in the x^b-direction, $\delta_{(b)}\phi=\partial_b\phi$. By calculating its divergence, one finds

$$\partial_a\Theta^a_{\ b}=\frac{\delta L}{\delta\phi}\partial_b\phi \ , \tag{7.25}$$

where $\delta L/\delta\phi$ is the Euler-Lagrange variational derivative,

$$\frac{\delta L}{\delta\phi} = \frac{\partial L}{\partial\phi} - \partial_a \frac{\partial L}{\partial(\partial_a\phi)} \ . \tag{7.26}$$

Thus Θ_{ab} is on-shell (meaning: for a solution to the matter equations of motion) conserved,

$$\partial_a \Theta^a_{\ b} = 0 \quad \text{on-shell} \ , \tag{7.27}$$

and leads to the conserved energy-momentum 4-vector

$$P_b = \int d^3x \ J^0_{(b)} = \int d^3x \ \Theta^0_{\ b} \ . \tag{7.28}$$

This procedure and prescription is perfectly adequate and sufficient for scalar (spin 0) fields, but it turns out to be far from satisfactory and far from the end of the story for other fields (e.g. for Maxwell theory, for which Θ_{ab} turns out to be neither symmetric nor gauge invariant). In this more general situation one is then required to "improve" this prescription in order to obtain an energy-momentum tensor T_{ab} with the desired properties.

As a first example where everything works out nicely, consider the energy-momentum tensor of a Klein-Gordon scalar field in Minkowski space. In this case,

$$\Theta_{ab} = \partial_a\phi\partial_b\phi + \eta_{ab}L = \partial_a\phi\partial_b\phi - \tfrac{1}{2}\eta_{ab}\left(\eta^{cd}\partial_c\phi\partial_d\phi + m^2\phi^2\right) \tag{7.29}$$

with

$$\Theta_{00} = \tfrac{1}{2}(\dot\phi^2 + (\vec\nabla\phi)^2 + m^2\phi^2) \ . \tag{7.30}$$

This energy-momentum tensor is conserved for ϕ a solution to the equations of motion,

$$\left(\Box_\eta - m^2\right)\phi = 0 \quad \Rightarrow \quad \partial^a\Theta_{ab} = 0 \ . \tag{7.31}$$

This energy-momentum tensor is also manifestly symmetric (off-shell, i.e. without using the equations of motion),

$$\Theta_{ba} = \Theta_{ab} \ . \tag{7.32}$$

In particular, this implies that the angular momentum current associated to an infinitesimal Lorentz transformation (1.28) with parameters $\omega_{bc} = -\omega_{cb}$, namely

$$L^a = \tfrac{1}{2}\omega_{bc}L^{abc} \tag{7.33}$$

with

$$L^{abc} = x^b\Theta^{ac} - x^c\Theta^{ab} \ , \tag{7.34}$$

is on-shell conserved,

$$\partial_a L^{abc} = \Theta^{bc} - \Theta^{cb} = 0 \ . \tag{7.35}$$

Since Θ_{ab} is symmetric (and gauge invariance is not an issue), in this example there is no need to "improve" the Noether energy-momentum tensor, and we thus denote it by T_{ab},

$$T_{ab} = \Theta_{ab} = \partial_a\phi\partial_b\phi + \eta_{ab}L \tag{7.36}$$

As we will see below, it is also straightforward to promote this tensor by minimal coupling to a (covariantly conserved) energy-momentum tensor of a scalar field in a gravitational field,

Now let us take a look at Maxwell theory in Minkowski space. In this case the canonical Noether energy-momentum tensor is

$$\Theta_{ab} = -\frac{\partial L}{\partial(\partial^a A_c)}\partial_b A_c + \eta_{ab}L = F_a{}^c\partial_b A_c - \tfrac{1}{4}\eta_{ab}F_{cd}F^{cd} \ . \tag{7.37}$$

It is of course on-shell conserved by construction,

$$\partial^a\Theta_{ab} = 0 \quad \text{on-shell} \tag{7.38}$$

(note that both sets of Maxwell equations are required to derive this), but it is neither symmetric nor gauge-invariant. In particular, therefore, the angular momentum current (7.34) is not conserved (even though Maxwell theory is Lorentz invariant), and the expression for the energy-density is not gauge-invariant and does not agree with the standard expression

$$\Theta_{00} \neq \tfrac{1}{2}(\vec{E}^2 + \vec{B}^2) \ . \tag{7.39}$$

This can be rectified by manipulating Θ_{ab} as

$$\begin{aligned}\Theta_{ab} &= F_a{}^c(\partial_b A_c - \partial_c A_b) - \tfrac{1}{4}\eta_{ab}F_{cd}F^{cd} + F_{ac}\partial^c A_b \\ &= F_a{}^c F_{bc} - \tfrac{1}{4}\eta_{ab}F_{cd}F^{cd} + F_{ac}\partial^c A_b \ ,\end{aligned} \tag{7.40}$$

and noting that the last term can be written as a sum of two terms,

$$F_{ac}\partial^c A_b = \partial^c(F_{ac}A_b) - (\partial^c F_{ac})A_b \tag{7.41}$$

the first of which is identically conserved because of $F_{ac} = -F_{ca}$,

$$\partial^a\partial^c(F_{ac}A_b) = 0 \ , \tag{7.42}$$

and the second of which vanishes on-shell,

$$(\partial^c F_{ac})A_b = 0 \quad \text{on-shell.} \tag{7.43}$$

Therefore one can redefine the energy-momentum tensor in a first step to

$$\hat{\Theta}_{ab} = \Theta_{ab} - \partial^c(F_{ac}A_b) \tag{7.44}$$

and notes that this energy-momentum tensor is still conserved on-shell,

$$\partial^a\hat{\Theta}_{ab} = 0 \quad \text{on-shell} \ , \tag{7.45}$$

as well as on-shell gauge invariant,

$$\begin{aligned}\hat{\Theta}_{ab} &= F_{ac}F_b{}^c - \tfrac{1}{4}\eta_{ab}F_{cd}F^{cd} - (\partial^c F_{ac})A_b \\ &= F_{ac}F_b{}^c - \tfrac{1}{4}\eta_{ab}F_{cd}F^{cd} \quad \text{on-shell} \ .\end{aligned} \tag{7.46}$$

Therefore one can define the "improved" energy-momentum tensor

$$T_{ab} = F_{ac}F_b{}^c - \tfrac{1}{4}\eta_{ab}F_{cd}F^{cd} \ , \tag{7.47}$$

in such a way that

- T_{ab} is still on-shell conserved,

$$\partial^a T_{ab} = 0 \quad \text{on-shell} \tag{7.48}$$

  (again both sets of Maxwell equations are required to establish this; with an external source,

$$\partial_{[a}F_{bc]} = 0 \quad , \quad \partial_a F^{ab} = -J^b \tag{7.49}$$

  one has the non-conservation law

$$\partial^a T_{ab} = J^a F_{ab} \tag{7.50}$$

  instead, which becomes a conservation law when one adds to T_{ab} the energy-momentum tensor of the source fields + interaction terms);

- T_{ab} is off-shell symmetric,

$$T_{ab} = T_{ba} \tag{7.51}$$

- T_{ab} is gauge-invariant and correctly gives the gauge-invariant and positive-definite energy-density as

$$T_{00} = \tfrac{1}{2}(\vec{E}^2 + \vec{B}^2) \ . \tag{7.52}$$

Moreover, the components of T_{0k} are the components of the Poynting vector and the spatial components T_{ik} are the components of the Maxwell stress tensor. Thus T_{ab} is *the* correct energy-momentum tensor of Maxwell theory.

This procedure to obtain T_{ab} from Θ_{ab} can be understood in a more general and systematic way, via the so-called Belinfante improvement (or symmetrisation) procedure. A brief synopsis of this construction will be provided in section 7.4 below.

One of the many useful properties of a symmetric, conserved energy-momentum tensor, and one that is occasionally used in general relativity, e.g. in the discussion of the energy and energy flux of gravitational waves, is the *Laue Theorem* (or *tensor virial theorem*). It states that for such an energy-momentum tensor and a localised source (so that one can integrate by parts with impunity) one has the relation

$$\left.\begin{array}{c}\partial^a T_{ab} = 0 \quad , \quad T_{ab} = T_{ba} \\ \text{localised source}\end{array}\right\} \quad \Rightarrow \quad \int d^3x \, T^{ik} = \tfrac{1}{2}(\partial_0)^2 \int d^3x \, T_{00} x^i x^k \tag{7.53}$$

between the integrated spatial components T_{ik} and the "quadrupole moments"

$$Q^{ik}(t) = \int d^3x\; T_{00} x^i x^k \tag{7.54}$$

of the energy density T_{00},

$$\int d^3x\; T^{ik} = \tfrac{1}{2}\ddot{Q}^{ik} \quad . \tag{7.55}$$

The proof of this identity is a straightforward repeated application of the conservation law and integration by parts. Indeed, using the symmetry, the time and space components of the conservation law

$$\partial^a T_{a0} = 0 \quad \Leftrightarrow \quad \partial_0 T_{00} = \partial^i T_{i0} \quad , \quad \partial^a T_{ak} = 0 \quad \Leftrightarrow \quad \partial_0 T_{0i} = \partial^k T_{ki} \tag{7.56}$$

and discarding boundary terms, one calculates

$$\begin{aligned}
\tfrac{1}{2}(\partial_0)^2 \int d^3x\; T_{00} x^i x^k &= +\tfrac{1}{2}\partial_0 \int d^3x\; \partial_0 T_{00} x^i x^k \\
&= +\tfrac{1}{2}\partial_0 \int d^3x\; (\partial_j T^j_{\;0}) x^i x^k \\
&= -\tfrac{1}{2}\partial_0 \int d^3x\; (T^i_{\;0} x^k + T^k_{\;0} x^i) \\
&= -\tfrac{1}{2} \int d^3x\; (\partial_0 T^i_{\;0} x^k + \partial_0 T^k_{\;0} x^i) \\
&= -\tfrac{1}{2} \int d^3x\; ((\partial_j T^{ij}) x^k + (\partial_j T^{kj}) x^i) \\
&= + \int d^3x\; T^{ik} \quad .
\end{aligned} \tag{7.57}$$

7.4 Synopsis of the Belinfante Improvement Procedure (Review)

The procedure to obtain a symmetric and conserved T_{ab} from the canonical Noether energy-momentum tensor Θ_{ab} of a Poincaré-invariant field theory, illustrated above in the case of Maxwell theory, can be understood in a more general and systematic, but also somewhat round-about way by appealing to the Lorentz-invariance of the action and taking into account the non-trivial transformation behaviour of the fields with spin $\neq 0$ under Lorentz transformations. This recipe is known as the *Belinfante improvement procedure*.[18] Here is, just for reference purposes, a brief description of the general features of this construction:

[18] This is explained in many places, with varying degree of comprehensibility or comprehension. For a detailed explanation, geared also towards applications to general relativity, see section 2 of T. Ortin, *Gravity and Strings*; for a succinct description, and an extension of the usual procedure to Lagrangians depending also on second derivatives of the fields, see section II of D. Bak, D. Cangemi, R. Jackiw, *Energy-Momentum Conservation in General Relativity*, `arXiv:hep-th/9310025`.

- In general (with the exception of spin zero scalar fields), $\Theta_{ab} = \eta_{ac}\Theta^c{}_b$ is not symmetric,
$$\Theta_{ab} \neq \Theta_{ba} \quad . \tag{7.58}$$

- As a consequence, the would-be angular momentum current (7.34) is now not on-shell conserved,
$$\partial_a L^{abc} = \Theta^{bc} - \Theta^{cb} \neq 0 \quad . \tag{7.59}$$

- By Lorentz invariance of the action and Noether's theorem, the total (orbital + spin) angular momentum should be conserved, and the above (purely orbital) angular momentum current fails to be conserved because it does not take into account the spin, i.e. the fact that the ϕ are possibly non-trivial Lorentz tensors (an irrelevant fact as far as the translational symmetries and hence the Noether energy-momentum tensor are concerned).

- This can be rectified by constructing the conserved total angular momentum current J^{abc} directly from Noether's theorem applied to Lorentz transformations $\delta\phi = \delta_L\phi$ of the fields and coordinates. This gives rise to an additional (spin) contribution to the current, schematically of the form
$$J^a = J^a_{\text{orbit}} + \frac{\partial L}{\partial(\partial_a\phi)}\delta_L\phi \quad , \quad J^{a[bc]}_{\text{orbit}} = L^{abc} \quad . \tag{7.60}$$
From the conservation of this current one can then via some gymnastics deduce and extract a candidate energy-momentum tensor $\hat{\Theta}_{ab}$ which is such that the total angular momentum current J^a takes the form
$$J^{abc} = x^b\hat{\Theta}^{ac} - x^c\hat{\Theta}^{ab} \quad . \tag{7.61}$$
Note that the spin-contribution to the total angular momentum has in this way been transformed into an orbital contribution with respect to the new energy-momentum tensor $\hat{\Theta}_{ab}$.

- This tensor $\hat{\Theta}_{ab}$ turns out to differ from the canonical energy-momentum tensor Θ_{ab} by an identically conserved term,
$$\hat{\Theta}_{ab} = \Theta_{ab} + \partial_c\Psi^{cab} \quad , \tag{7.62}$$
with
$$\Psi^{cab} = -\Psi^{acb} \quad \Rightarrow \quad \partial_a\partial_c\Psi^{cab} \equiv 0 \tag{7.63}$$
so that
$$\partial_a\Theta^a{}_b = 0 \quad \text{on-shell} \quad \Rightarrow \quad \partial_a\hat{\Theta}^a{}_b = 0 \quad \text{on-shell} \quad . \tag{7.64}$$

- Addition of such a term to the energy-momentum tensor is always possible as it does not violate the conservation law. While this changes the definition of the

local energy and momentum densities, with suitable fall-off conditions on the Ψ^{abc} this has no effect on the total energy-momentum P_b (7.28),

$$\begin{aligned} P^b \to P^b + \int d^3x\, \partial_c \Psi^{c0b} &= P^b + \int d^3x\, \partial_k \Psi^{k0b} \quad . \\ &= P^b + \oint dS_k \Psi^{k0b} \quad . \end{aligned} \tag{7.65}$$

- Angular momentum conservation together with (7.64) now implies

$$\partial_a J^{abc} = 0 \quad \text{on-shell} \quad \Rightarrow \quad \hat{\Theta}_{ab} = \hat{\Theta}_{ba} \quad \text{on-shell} \quad . \tag{7.66}$$

- Thus on-shell $\hat{\Theta}_{ab}$ agrees with a tensor T_{ab}, which can be chosen to be symmetric (off-shell) and on-shell conserved,

$$\begin{aligned} \hat{\Theta}_{ab} \to T_{ab} : \quad & T_{ab} = T_{ba} \quad \text{off-shell} \\ & \partial_a T^a_{\ b} = 0 \quad \text{on-shell} \quad . \end{aligned} \tag{7.67}$$

- This tensor T_{ab} (or occasionally just $\hat{\Theta}_{ab}$) is known as the *Belinfante improvement* of the energy-momentum tensor, and $\partial_c \Psi^{cab}$ as the (identically conserved) *improvement term*,

$$T_{ab} = \{\text{Improvement of } \Theta\}_{ab} \quad . \tag{7.68}$$

T_{ab} is then generally considered to be the "correct" choice of energy-momentum tensor for the Lagrangian field thory at hand, but it should be clear from the above discussion that this somewhat round-about procedure for finding and obtaining it leaves something to be desired (to put it mildly), already in the framework of Special Relativity.

7.5 Energy-Momentum Tensor from Minimal Coupling?

Given the success of the minimal coupling prescription, it is natural to try to define the matter energy-momentum tensor in a gravitational field in the same way. While this is certainly possible to a certain extent (as the examples will show), this procedure also leaves something to be desired (as the examples will also show).

Let us start by considering the "phenomenological" perfect fluid energy-momentum tensor (7.6),

$$T_{ab} = (\rho + p) u_a u_b + p \eta_{ab} \quad . \tag{7.69}$$

Following the minimal coupling rules, we promote this to the energy-momentum tensor

$$T_{\alpha\beta} = (\rho + p) u_\alpha u_\beta + p g_{\alpha\beta} \quad , \tag{7.70}$$

where u^α denotes the proper-time normalised velocity field of the fluid, $g_{\alpha\beta} u^\alpha u^\beta = -1$.

The covariantisation of the conservation law (7.7) evidently reads

$$\partial^a T_{ab} = 0 \quad \rightarrow \quad \nabla^\alpha T_{\alpha\beta} = 0 \ . \tag{7.71}$$

This generalises the continuity equation and the relativistic Euler equations to a fluid moving in a gravitational field and reduces to the special relativistic laws at the origin of a freely falling coordinate system, as it should.

There are neither conceptual nor technical complications in this example, and we will adopt this perfect fluid energy-momentum tensor, supplemented by an appropriate equation of state, to model the interior of a star (section 24.7) and the matter content of the universe (in our discussion of cosmology). In both of these examples, such a phenomenological description is quite appropriate and sufficient (although for more detailed investigations one may need to go beyond the perfect fluid approximation). For a detailed analysis of the conservation equations in the context of cosmology, see sections 35.4 and 35.6.

Let us now turn to energy-momentum tensors for Lagrangian field theories, starting with the example of the Klein-Gordon scalar field. As we saw above, in Minkowski space its (Noether = improved) energy-momentum tensor is given by

$$T_{ab} = \partial_a\phi\partial_b\phi + \eta_{ab}L = \partial_a\phi\partial_b\phi - \tfrac{1}{2}\eta_{ab}\left(\eta^{cd}\partial_c\phi\partial_d\phi + m^2\phi^2\right) \ . \tag{7.72}$$

The corresponding minimally coupled energy-momentum tensor in a gravitational field is then evidently

$$T_{\alpha\beta} = \partial_\alpha\phi\partial_\beta\phi + g_{\alpha\beta}L = \partial_\alpha\phi\partial_\beta\phi - \tfrac{1}{2}g_{\alpha\beta}\left(g^{\mu\nu}\partial_\mu\phi\partial_\nu\phi + m^2\phi^2\right) \ , \tag{7.73}$$

and it is easy to check that it is covariantly conserved for ϕ a solution to the equations of motion in a gravitational background,

$$\left(\Box_g - m^2\right)\phi = 0 \quad \Rightarrow \quad \nabla^\alpha T_{\alpha\beta} = 0 \ . \tag{7.74}$$

For the action (6.14) with a potential $V(\phi)$, the energy-momentum tensor of course also has the form (7.73) with $m^2\phi^2/2$ unsurprisingly replaced by $V(\phi)$,

$$T_{\alpha\beta} = \partial_\alpha\phi\partial_\beta\phi - \tfrac{1}{2}g_{\alpha\beta}\, g^{\mu\nu}\partial_\mu\phi\partial_\nu\phi - g_{\alpha\beta}V(\phi) \ , \tag{7.75}$$

with

$$\Box_g\phi = V'(\phi) \quad \Rightarrow \quad \nabla^\alpha T_{\alpha\beta} = 0 \ . \tag{7.76}$$

So far so good. However, the significance of this energy-momentum tensor outside the realm of special relativity is not clear. In special relativity, it encodes the conserved quantities associated to translation invariance, but in a general gravitational field there is no translation invariance (or other symmetry). In particular, in a general gravitational field

- one cannot even derive the energy-momentum tensor (7.73) from Noether's theorem applied to translations
- and, related to this is the fact that one does not obtain an ordinary conservation law but the covariant conservation law $\nabla_\alpha T^{\alpha\beta} = 0$.

Regarding the first point, it is fair to wonder if $T_{\alpha\beta}$ possesses an intrinsic gravitational significance beyond being merely the non-conserved minimally coupled counterpart of something that happens to have a significance in the absence of gravity. We will see below that, yes indeed, it is precisely the *source of gravity* arising from scalar fields.

Regarding the second point, we will see in sections 7.9 and 10.1 below that to any continous symmetry of a gravitational field (metric) and the covariantly conserved energy-momentum tensor one *can* associate a covariantly conserved current and thus also (as discussed in section 6.8) a conserved charge.

Now let us turn to Maxwell theory. Here the situation is a priori a bit murkier, because in principle we have both the canonical Noether energy-momentum tensor Θ_{ab} (7.37),

$$\Theta_{ab} = F_a{}^c \partial_b A_c - \tfrac{1}{4}\eta_{ab} F_{cd} F^{cd} \quad , \tag{7.77}$$

and its Belinfante-improved symmetric gauge-invariant sibling T_{ab} (7.47),

$$T_{ab} = F_{ac} F_b{}^c - \tfrac{1}{4}\eta_{ab} F_{cd} F^{cd} \quad , \tag{7.78}$$

at our disposal. Let us start with the latter, not only because it is the nicer object but also because it turns out to give the "correct" result. Applying the rules of minimal coupling, one finds the tensor

$$T_{\alpha\beta} = F_{\alpha\gamma} F_\beta{}^\gamma - \tfrac{1}{4} g_{\alpha\beta} F_{\gamma\delta} F^{\gamma\delta} \quad , \tag{7.79}$$

where indices of the (metric independent) field strength tensor $F_{\alpha\beta}$ are of course raised with the aid of the inverse metric $g^{\alpha\beta}$. This object turns out to have all the right properties to qualify as a candidate energy-momentum tensor of Maxwell theory in a gravitational field. In particular, it is off-shell symmetric and moreover on-shell covariantly conserved,

$$\nabla_\alpha T^{\alpha\beta} = 0 \quad \text{on-shell} \quad , \tag{7.80}$$

where "on-shell" of course refers to the equations of motion (6.39) in a gravitational field. Again both sets of vacuum Maxwell equations are required to verify this. In the presence of an external current, this is modified to

$$\nabla_\alpha T^{\alpha\beta} = J_\alpha F^{\alpha\beta} \quad \text{on-shell} \quad . \tag{7.81}$$

While one may have anticipated these last two equations on the basis of the minimal coupling recipe, it is important (and a useful exercise) to verify by direct calculation that

they indeed hold. The point of this verification is to make sure that no commutators of covariant derivatives, i.e. "curvature terms", arise in and mess up this equation, as they will in the calculation below involving the Noether energy-momentum tensor.

So let us take a brief look at the covariantised or minimally coupled Noether energy-momentum tensor, namely

$$\Theta_{\alpha\beta} = F_{\alpha}^{\ \gamma}\nabla_{\beta}A_{\gamma} - \tfrac{1}{4}g_{\alpha\beta}F_{\gamma\delta}F^{\gamma\delta} \quad . \tag{7.82}$$

While the canonical energy-momentum tensor in Minkowski space had some undesirable properties, its one redeeming feature was that it was on-shell conserved. In contrast to this, $\Theta_{\alpha\beta}$ is neither on-shell conserved nor on-shell covariantly conserved. In order to establish $\partial^a\Theta_{ab} = 0$ in Minkowski space, one uses the fact that partial derivatives commute. Thus, analogously, in calculating $\nabla^{\alpha}\Theta_{\alpha\beta}$ one encounters the *commutator of covariant derivatives*. Explicitly on-shell one finds

$$\nabla_{\alpha}\Theta^{\alpha}_{\ \beta} = \tfrac{1}{2}F^{\alpha\gamma}[\nabla_{\alpha},\nabla_{\gamma}]A_{\beta} \quad , \tag{7.83}$$

However, as we will discuss at length in section 8, the characteristic and defining feature of a non-trivial curved space-time is that these covariant derivatives do not commute when acting on tensors other than scalars (their commutator defining the curvature tensor of the space-time).

Likewise the covariant version of the improvement term in

$$\hat{\Theta}_{ab} = \Theta_{ab} - \partial^{c}(F_{ac}A_{b}) \quad , \tag{7.84}$$

namely $\nabla^{\gamma}(F_{\alpha\gamma}A_{\beta})$ is not identically conserved anymore, rather one has

$$\nabla^{\alpha}\nabla^{\gamma}(F_{\alpha\gamma}A_{\beta}) = \tfrac{1}{2}F^{\alpha\gamma}[\nabla_{\alpha},\nabla_{\gamma}]A_{\beta} \quad , \tag{7.85}$$

so that it would not qualify as an "improvement term" in the standard sense. Nevertheless, subtracting this term from the (non-conserved) Noether energy-momentum tensor, one finds that this indeed cancels the commutator term arising form (7.83), thus giving rise to an on-shell covariantly conserved $\hat{\Theta}_{\alpha\beta}$ or $T_{\alpha\beta}$. From the present perspective, however, this must be considered to be somewhat of a miracle or fluke. For some more comments on this, see section 22.2.

Thus we adopt (7.79),

$$T_{\alpha\beta} = F_{\alpha\gamma}F_{\beta}^{\ \gamma} - \tfrac{1}{4}g_{\alpha\beta}F_{\gamma\delta}F^{\gamma\delta} \quad , \tag{7.86}$$

as our (preliminary) definition of the energy-momentum tensor of Maxwell theory in a gravitational field, but we now face the same issue as in the case of scalar fields, namely the question what, if any, is the intrinsic gravitational significance of this energy-momentum tensor.

7.6 Covariant Energy-Momentum Tensor: the Source of Gravity

As we have seen, there are some irritating conceptual and technical issues associated with the "Noether + minimal coupling" procedure in general. These irritants turn out to be a good thing, though, because they motivate us to rethink this issue from scratch, and this will now lead us to a much more compelling and both conceptually and technically perfectly satisfactory general definition of the energy-momentum tensor of any Lagrangian field theory in a gravitational field.

Thus let us think about this issue from a Lagrangian, action-based, perspective. So far we have discussed what is the appropriate form of the action for matter fields in a gravitational field, namely a generally covariant action

$$S_{matter} = S_M[\phi; g_{\alpha\beta}] = \int \sqrt{g} d^4x \; L_M(\phi, \partial_\alpha \phi, \ldots, g_{\alpha\beta}, \ldots) \tag{7.87}$$

for the matter fields ϕ in a gravitational background $g_{\alpha\beta}$, obtained e.g. by the minimal coupling description and thus describing the dynamics of the fields in a gravitational background and encoding the coupling of the matter fields to gravity. Ultimately, this action should then be one part of the total gravitational + matter action describing the dynamics of the matter fields *and* of the gravitational field,

$$S_{total} = S_{gravity} + S_{matter} \quad . \tag{7.88}$$

Since the gravitational field is described by the (now dynamical) variables $g_{\alpha\beta}(x)$, we can write this marginally more explicitly as

$$S[g_{\alpha\beta}, \phi] = S_g[g_{\alpha\beta}] + S_M[\phi; g_{\alpha\beta}] \quad . \tag{7.89}$$

The precise form of the gravitational action S_g will not be relevant here - this is something that we will discuss at length in section 20. All we need to keep in mind is that this action is to provide us with the gravitational part of the gravitational field equations, i.e. with the appropriate tensorial generalisation of the left-hand side $\Delta\phi$ of the Newtonian field equation $\Delta\phi = 4\pi G_N \rho$.

Variation of this total action with respect to the matter fields ϕ is equivalent to the variation of the matter action S_M alone with respect to the matter fields,

$$\frac{\delta S[g_{\alpha\beta}, \phi]}{\delta \phi} = 0 \quad \Leftrightarrow \quad \frac{\delta S_M[\phi; g_{\alpha\beta}]}{\delta \phi} = 0 \quad , \tag{7.90}$$

and will thus simply give rise to the equations of motion of the matter fields in a gravitational field, as required.

Now let us consider the variation of the total action with respect to the gravitational dynamical variables $g_{\alpha\beta}$,

$$\frac{\delta S[g_{\alpha\beta}, \phi]}{\delta g_{\alpha\beta}} = \frac{\delta S_g[g_{\alpha\beta}]}{\delta g_{\alpha\beta}} + \frac{\delta S_M[\phi; g_{\alpha\beta}]}{\delta g_{\alpha\beta}} \stackrel{!}{=} 0 \tag{7.91}$$

Variation of the gravitational action with respect to the gravitational field $g_{\alpha\beta}$ will give us the gravitational part of the field equations. Thus variation of the matter action with respect to the gravitational field will give us the source term for the gravitational field equations provided by the matter fields,

$$\frac{\delta S_M[\phi; g_{\alpha\beta}]}{\delta g_{\alpha\beta}} = \text{Source of Gravity} \quad . \tag{7.92}$$

On the other hand, as recalled in the introduction to this section (section 7.1), we expect the energy-momentum tensor to act as the source of gravity. Therefore we should simply define the energy-momentum tensor by this relation,

$$\begin{aligned} & T^{\alpha\beta} := \text{Source of Gravity} \\ \Rightarrow \quad & T^{\alpha\beta} \sim \frac{\delta S_M[\phi; g_{\alpha\beta}]}{\delta g_{\alpha\beta}} \quad . \end{aligned} \tag{7.93}$$

We will fix the proportionality factor momentarily.

Note that this is precisely analogous to the way a source term for the Maxwell equations, a current J^α, arises from the variation of the coupled matter-Maxwell action with respect to the gauge field A_α (6.55),

$$J^\alpha \sim \frac{\delta S_M[\phi, A_\alpha]}{\delta A_\alpha} \quad . \tag{7.94}$$

In order to test this suggestion, let us take a look at our two standard examples, a scalar field and Maxwell theory. For a scalar field, the minimally coupled action is (6.14)

$$S[\phi, g_{\alpha\beta}] = \int \sqrt{g} d^4x \left(-\tfrac{1}{2} g^{\alpha\beta} \partial_\alpha \phi \partial_\beta \phi - V(\phi) \right) \quad . \tag{7.95}$$

Since the action depends explicitly on the inverse metric, it is more convenient to determine the variation of the action under variations

$$g^{\alpha\beta} \to g^{\alpha\beta} + \delta g^{\alpha\beta} \tag{7.96}$$

of the inverse metric. Under such a variation, the volume factor $\sqrt{g}$ varies as (5.87)

$$\delta\sqrt{g} = -\tfrac{1}{2}\sqrt{g} g_{\alpha\beta} \delta g^{\alpha\beta} \quad . \tag{7.97}$$

Thus the metric-variation of the scalar field action is

$$\delta S_M[\phi, g_{\alpha\beta}] = -\tfrac{1}{2} \int \sqrt{g} d^4x \left(\partial_\alpha \phi \partial_\beta \phi + g_{\alpha\beta} (-\tfrac{1}{2} g^{\mu\nu} \partial_\mu \phi \partial_\nu \phi - V(\phi)) \right) \delta g^{\alpha\beta} \quad , \tag{7.98}$$

Comparison with the minimally coupled energy-momentum tensor (7.75) of a scalar field,

$$T_{\alpha\beta} = \partial_\alpha \phi \partial_\beta \phi - \tfrac{1}{2} g_{\alpha\beta}\, g^{\mu\nu} \partial_\mu \phi \partial_\nu \phi - g_{\alpha\beta} V(\phi) \quad , \tag{7.99}$$

shows that this is precisely what we have obtained from the metric-variation of the matter action,

$$\delta S_M[\phi, g_{\alpha\beta}] = -\tfrac{1}{2} \int \sqrt{g} d^4x\, T_{\alpha\beta} \delta g^{\alpha\beta} \quad . \tag{7.100}$$

Now let us look at Maxwell theory, our litmus test. In this case, the action is (6.51)

$$S[A_\alpha, g_{\alpha\beta}] = -\tfrac{1}{4}\int \sqrt{g} d^4x \; g^{\mu\lambda} g^{\nu\rho} F_{\mu\nu} F_{\lambda\rho} \quad . \tag{7.101}$$

The variation of $\sqrt{g}$ is as before, and as regards the variation of the inverse metric, there is now an additional relative factor of two compared with the calculation for the scalar fields because the action depends quadratically on the inverse metric. Thus one has

$$\delta S[A_\alpha, g_{\alpha\beta}] = -\tfrac{1}{2}\int \sqrt{g} d^4x \; \left(g^{\nu\rho} F_{\alpha\nu} F_{\beta\rho} - \tfrac{1}{4} g_{\alpha\beta} F_{\mu\nu} F^{\mu\nu} \right) \delta g^{\alpha\beta} \quad . \tag{7.102}$$

Comparing with (7.79),

$$T_{\alpha\beta} = F_{\alpha\gamma} F_\beta{}^\gamma - \tfrac{1}{4} g_{\alpha\beta} F_{\gamma\delta} F^{\gamma\delta} \quad , \tag{7.103}$$

we see that we once again have

$$\delta S_M[\phi, g_{\alpha\beta}] = -\tfrac{1}{2}\int \sqrt{g} d^4x \; T_{\alpha\beta} \delta g^{\alpha\beta} \quad . \tag{7.104}$$

Thus the metric variation of the matter action has given us on the nose the symmetric, gauge invariant, on-shell conserved energy-momentum tensor of Maxwell theory, without any need to appeal to any improvement procedures!

Thus, when it comes to defining the energy-momentum tensor for Maxwell theory, the above approach based on the variation of the matter action with respect to the metric wins hands down over the painful canonical definition based on Noether's theorem for translations and the Belinfante improvement procedure combined with minimal coupling.

Encouraged by this, we now define the energy-momentum tensor $T_{\alpha\beta}$ in general by

$$\delta_{\text{metric}} S_M[\phi, g_{\alpha\beta}] = -\tfrac{1}{2}\int \sqrt{g} d^4x \; T_{\alpha\beta} \delta g^{\alpha\beta} \quad , \tag{7.105}$$

or, equivalently,

$$T_{\alpha\beta} := -\frac{2}{\sqrt{g}} \frac{\delta}{\delta g^{\alpha\beta}} S_M[\phi, g_{\alpha\beta}] \quad . \tag{7.106}$$

Even though, as we have seen, there are other definitions of the energy-momentum tensor, this is the modern, and by far the most useful, definition of the energy-momentum tensor, namely as the response of the matter action to a variation of the metric (equivalently, as the source of gravity).

Moreover, crucially for the present context, whatever the virtues of other definitions may be, from the variational principle for general relativity it is this energy-momentum tensor that plays the role of the source term for the Einstein equations.

REMARKS:

1. The energy-momentum tensor as defined by (7.105) or (7.106) is frequently called the *metric* energy-momentum (or stress-energy) tensor, or also the *Hilbert* or

Rosenfeld energy-momentum tensor. It is sometimes also referred to as the gravitational energy-momentum tensor, but that is confusing as it does not describe the energy-momentum of the gravitational field itself, a more mysterious and elusive quantity we will briefly look at and for in section 22.6.

I prefer the attribute *covariant*, to distinguish it from what is usually called the *canonical* Noether energy-momentum tensor. Thus, even though this terminology is not standard, I will henceforth refer to $T_{\alpha\beta}$ as defined by (7.105) or (7.106), as the *Covariant Energy-Momentum Tensor*.

2. One of the many advantages of this definition is that it automatically and in general gives a symmetric and gauge invariant tensor (no improvement terms or similar gymnastics required). This is obvious from the definition.

3. This energy-momentum tensor turns out to also automatically be covariantly conserved (on-shell, i.e. for matter fields satisfying their Euler-Lagrange equations of motion). We will establish this latter fact in section 20.6 below where we will see that this is simply a consequence of the general covariance of the matter action S_M,

$$\text{general covariance of } S_M \quad \Rightarrow \quad \nabla^\alpha T_{\alpha\beta} = 0 \quad \text{on-shell} \ . \tag{7.107}$$

4. When the minimally coupled matter Lagrangian depends only on the metric and not on the first derivatives of the metric (i.e. not on the Christoffel symbols),

$$L_M(x) = L_M(\phi(x), \partial_\mu\phi(x), g_{\mu\nu}(x)) \ , \tag{7.108}$$

as in the case of scalar or Maxwell gauge fields, then more explicitly the covariant energy-momentum tensor can be written as (and calculated from)

$$T_{\mu\nu}(x) = -\frac{2}{\sqrt{g}}\frac{\partial(\sqrt{g}L_M(x))}{\partial g^{\mu\nu}(x)} = -2\frac{\partial L_M(x)}{\partial g^{\mu\nu}(x)} + g_{\mu\nu}(x)L_M(x) \tag{7.109}$$

or

$$T^{\mu\nu}(x) = 2\frac{\partial L_M(x)}{\partial g_{\mu\nu}(x)} + g^{\mu\nu}(x)L_M(x) \ . \tag{7.110}$$

Here the sign change is due to the fact that $\delta g^{\mu\nu}$ denotes the variation of the inverse metric, not the contravariant components of $\delta g_{\mu\nu}$. Thus it is not the same as $g^{\mu\lambda}g^{\nu\rho}\delta g_{\lambda\rho}$, but rather minus this expression,

$$\delta g^{\mu\nu} = -g^{\mu\lambda}g^{\nu\rho}\delta g_{\lambda\rho} \ , \tag{7.111}$$

as can be seen by varying $g^{\mu\nu}g_{\nu\lambda} = \delta^\mu_\lambda$,

$$0 = \delta(g^{\mu\nu}g_{\nu\lambda}) = (\delta g^{\mu\nu})g_{\nu\lambda} + g^{\mu\rho}\delta g_{\rho\lambda} \quad \Leftrightarrow \quad \delta g^{\mu\nu} = -g^{\mu\lambda}g^{\nu\rho}\delta g_{\lambda\rho} \ . \tag{7.112}$$

5. The definition (7.105) or the explicit expression (7.109) also provides an efficient strategy to determine the energy-momentum tensor even if one is just interested in Poincaré-invariant field theories in Minkowski space:

   In order to determine a symmetric, gauge invariant, and on-shell conserved energy-momentum tensor T_{ab} for such a theory, one

   - temporarily minimally couples the theory to a metric $g_{\alpha\beta}(x)$,
   - uses (7.105) or (7.106) or (7.109) to determine $T_{\alpha\beta}$,
   - and then replaces $g_{\alpha\beta} \to \eta_{ab}$ etc. again at the end.

   In equations, one defines T_{ab} by

   $$T_{ab} := (T_{\alpha\beta})|_{x^\alpha \to \xi^a, g_{\alpha\beta} \to \eta_{ab}} \quad . \tag{7.113}$$

   It can be shown that for fields of any spin this energy-momentum tensor agrees on-shell with what one could have also obtained by invoking the Belinfante improvement procedure of the Noether energy-momentum tensor,

   $$\hat{\Theta}_{ab} = T_{ab} \quad \text{on-shell} \tag{7.114}$$

   (see the discussion and references in section 7.4).

6. When the minimally coupled matter action depends also on the first derivatives of the metric, through the covariant derivative $\nabla_\mu \psi$ of some (non-scalar) field ψ, say, by the usual rules of variational calculus there will be additional contributions to the energy-momentum tensor, arising from an integration by parts of

   $$\int \sqrt{g} d^4x \, \left(-2 \frac{\partial L_M(x)}{\partial \nabla_\mu \psi(x)} (\delta \nabla_\mu \psi)(x) \right)$$

   where $\delta(\nabla_\mu \psi) = (\delta \nabla_\mu)\psi$ denotes the variation of the covariant derivative induced by the metric-variation (e.g. via the corresponding variation (20.14) of the Christoffel symbols). The precise form of the resulting contribution to the energy-momentum tensor depends on the tensorial type of ψ, is rarely needed, and it is unedifying to attempt to write down a general formula for this.

7.7 On the Energy-Momentum Tensor for Weyl-invariant Actions

Another general feature of the energy-momentum tensor that is readily understood by adopting the definition

$$T_{\alpha\beta} = -\frac{2}{\sqrt{g}} \frac{\delta S_{\text{matter}}}{\delta g^{\alpha\beta}} \quad . \tag{7.115}$$

is the relation between Weyl invariance and the trace of the energy-momentum tensor.

We consider the situation where the minimally coupled matter action happens to be invariant under *Weyl rescalings*, i.e. under rescalings of the metric

$$g_{\alpha\beta}(x) \to \mathrm{e}^{2\omega(x)} g_{\alpha\beta}(x) \tag{7.116}$$

by a positive definite function, or infinitesimally

$$\delta_\omega g_{\alpha\beta}(x) = 2\omega(x) g_{\alpha\beta}(x) \ . \tag{7.117}$$

In particular, thus, we consider the (admittedly very special) situation where one has such a symmetry without any accompanying transformation of the matter fields. The discussion can be extended to the case where also a transformation of the matter fields is required, but for present purposes this special case is good enough (see the end of this section for a comment on the general case).

Examples of such actions are e.g. the action of a massless scalar field (6.11) in $D = 2$ (space-time) dimensions

$$S[\phi, g_{\alpha\beta}] = -\tfrac{1}{2} \int d^2x \ \sqrt{g} g^{\alpha\beta} \partial_\alpha \phi \partial_\beta \phi \tag{7.118}$$

and that of Maxwell theory (6.51) in $D = 4$ dimensions,

$$S[A_\alpha, g_{\alpha\beta}] = -\tfrac{1}{4} \int d^4x \ \sqrt{g} g^{\alpha\beta} g^{\gamma\delta} F_{\alpha\gamma} F_{\beta\delta} \ . \tag{7.119}$$

Indeed, in that case the metric dependence of the action is precisely such that the combination of of the determinant $\sqrt{g}$ and the inverse metric that appears is invariant under Weyl rescalings,

$$g_{\alpha\beta} \to \mathrm{e}^{2\omega} g_{\alpha\beta} \quad \Rightarrow \quad \begin{cases} D = 2 & \sqrt{g} g^{\alpha\beta} \to \sqrt{g} g^{\alpha\beta} \\ D = 4 & \sqrt{g} g^{\alpha\beta} g^{\gamma\delta} \to \sqrt{g} g^{\alpha\beta} g^{\gamma\delta} \end{cases} \tag{7.120}$$

This is reflected in the fact that the corresponding energy-momentum tensor is traceless precisely in these dimensions: from (7.73) and (7.79) one finds

$$\begin{aligned} T_{\alpha\beta} = \partial_\alpha \phi \partial_\beta \phi - \tfrac{1}{2} g_{\alpha\beta} \left(g^{\mu\nu} \partial_\mu \phi \partial_\nu \phi \right) \quad &\Rightarrow \quad T^\alpha_{\ \alpha} = -\tfrac{1}{2}(D-2) g^{\mu\nu} \partial_\mu \phi \partial_\nu \phi \\ T_{\alpha\beta} = F_{\alpha\lambda} F_\beta^{\ \lambda} - \tfrac{1}{4} g_{\alpha\beta} F_{\lambda\sigma} F^{\lambda\sigma} \quad &\Rightarrow \quad T^\alpha_{\ \alpha} = -\tfrac{1}{4}(D-4) F_{\mu\nu} F^{\mu\nu} \ . \end{aligned} \tag{7.121}$$

The relation between these two observations / assertions is provided by noting that if the matter action is invariant under Weyl rescalings one has

$$\begin{aligned} 0 = \delta_\omega S_{\text{matter}} &= -\tfrac{1}{2} \int \sqrt{g} d^D x \ T_{\alpha\beta}(x) \delta_\omega g^{\alpha\beta}(x) \\ &= \int \sqrt{g} d^D x \ T_{\alpha\beta}(x) g^{\alpha\beta}(x) \omega(x) = \int \sqrt{g} d^D x \ T^\alpha_{\ \alpha}(x) \omega(x) \ . \end{aligned} \tag{7.122}$$

Since this is to be zero for all functions $\omega(x)$, this proves

$$\text{invariance under Weyl rescalings of the metric} \quad \Rightarrow \quad T^\alpha_{\ \alpha} = 0 \ . \tag{7.123}$$

In the special case that we have considered here (invariance under scalings of the metric alone, without transforming the matter fields), this is true off-shell, i.e. without using the equations of motion for the matter fields. In the more general case of an invariance under joint Weyl rescalings of the metric and accompanying scalings of the matter fields, in the above chain of arguments one would need to also vary the matter action with respect to the matter fields to establish the invariance of the action. The term arising from the variation of the matter fields is evidently proportional to the Euler-Lagrange equations of the matter fields, and therefore in that case one could only conclude that $T^\alpha_\alpha = 0$ on-shell,

$$\left.\begin{array}{c}\text{invariance under joint Weyl rescalings}\\ \text{of the metric and the matter fields}\end{array}\right\} \quad \Rightarrow \quad T^\alpha_\alpha = 0 \ \text{ on-shell.} \tag{7.124}$$

An example of this is provided by the so-called *conformally coupled scalar field*. This conformal coupling involves a space-time dependent mass term that represents a non-minimal coupling of the scalar field to the scalar curvature (a contraction of the Riemann curvature tensor to be introduced in section 8), and understanding the Weyl invariance of this model requires a formula for the variation of the scalar curvature with respect to the metric which we will derive in section 20.2. Therefore we will need to postpone a discussion of this model to section 22.3.

7.8 Klein-Gordon Scalar Field in (1+1) Minkowski and Rindler Space

As an aside, but as a concrete, and the simplest non-trivial, example, and an illustration of the above remarks regarding Weyl invariance, let us consider a massless scalar field in (1+1)-dimensions, in either the usual Minkowski coordinates, or in the Rindler coordinates discussed in sections 1.3 and 3.4 (we will in particular make use of the results in section 3.4).

In inertial coordinates in (1+1)-dimensional Minkowski space-time, $ds^2 = -dt^2 + dx^2$, the action and equation of motion of a massless scalar field are

$$S_M[\phi] = -\tfrac{1}{2}\int dtdx\ \eta^{\alpha\beta}\partial_\alpha\phi\partial_\beta\phi \quad \Rightarrow \quad \Box\phi = (-\partial_t^2 + \partial_x^2)\phi = 0\ . \tag{7.125}$$

Thus a natural basis of solutions to this equation is provided by the plane waves $f_k \sim \exp(-i\omega t + ikx)$, with $k^2 = \omega^2$, i.e. $k = \pm\omega, \omega > 0$, and their complex conjugates. For a given ω there are thus two linearly-independent positive frequency solutions,

$$\begin{aligned} f_\omega(t,x) &= \frac{1}{(4\pi\omega)^{1/2}}\mathrm{e}^{-i\omega(t-x)} \\ g_\omega(t,x) &= \frac{1}{(4\pi\omega)^{1/2}}\mathrm{e}^{-i\omega(t+x)} \end{aligned} \tag{7.126}$$

(the normalisation factors are inserted for QFT-pedantry reasons only and are irrelevant for the following). Thus the basis of solutions splits into *right-movers* or *right-moving*

modes f_ω and *left-movers* g_ω. It is thus convenient to introduce the corresponding null coordinates $u_M = t - x, v_M = t + x$ as in (3.94), in terms of which the solutions can be written as

$$\begin{aligned} f_\omega = f_\omega(u_M) &= \frac{1}{(4\pi\omega)^{1/2}} \mathrm{e}^{-i\omega u_M} \\ g_\omega = g_\omega(v_M) &= \frac{1}{(4\pi\omega)^{1/2}} \mathrm{e}^{-i\omega v_M} \quad . \end{aligned} \tag{7.127}$$

That the solutions split in this way could have also been deduced from the form of the wave operator in these lightcone (null) coordinates, namely $\Box = -4\partial_{u_M}\partial_{v_M}$, and the ensuing solutions to the equation of motion,

$$\Box\phi = 0 \quad \Rightarrow \quad \phi = f(u_M) + g(v_M) \quad . \tag{7.128}$$

Here f and g can now be arbitrary wave packets constructed from the solutions f_ω and g_ω respectively.

The energy-density $\rho_M = T_{tt}$ of the scalar field with respect to Minkowski time is

$$\rho_M = \tfrac{1}{2}((\partial_t\phi)^2 + (\partial_x\phi)^2) \tag{7.129}$$

and in terms of lightcone coordinates this splits into a sum of left-moving and right-moving contributions,

$$\rho_M = (\partial_{u_M}\phi)^2 + (\partial_{v_M}\phi)^2 \quad , \tag{7.130}$$

with $f(u_M)$ evidently only contributing to the former and $g(v_M)$ to the latter.

Now let us consider the same issue in Rindler coordinates. In terms of the coordinates (η, ξ) (3.89), the metric takes the form (3.87)

$$ds^2 = \mathrm{e}^{2a\xi}(-d\eta^2 + d\xi^2) \quad . \tag{7.131}$$

Note that, as mentioned in section 3.4, the metric in these coordinates is conformally flat. Thus, by the reasoning above, in section 7.7, in particular the discussion around equation (7.120), we know that the action and equation of motion for a scalar field in Rindler coordinates will look just like those in Minkowski coordinates, with the replacement $(t, x) \to (\eta, \xi)$,

$$S_R[\phi] = -\tfrac{1}{2}\int \sqrt{g}d\eta d\xi \; g^{\alpha\beta}\partial_\alpha\phi\partial_\beta\phi = -\tfrac{1}{2}\int d\eta d\xi \; \eta^{\alpha\beta}\partial_\alpha\phi\partial_\beta\phi \quad , \tag{7.132}$$

and

$$\Box_g\phi = 0 \quad \Leftrightarrow \quad (-\partial_\eta^2 + \partial_\xi^2)\phi = 0 \quad . \tag{7.133}$$

Thus by the same reasoning as above, the solutions can be split into left- and right-movers and are conveniently written in terms of the Rindler lightcone coordinates (3.95)

$$(u_R, v_R) = \eta \mp \xi \quad , \tag{7.134}$$

i.e. one has

$$\Box_g\phi = 0 \quad \Leftrightarrow \quad \partial_{u_R}\partial_{v_R}\phi = 0 \quad \Leftrightarrow \quad \phi = f(u_R) + g(v_R) \ . \tag{7.135}$$

The energy-density $\rho_R = T_{\eta\eta}$ of the scalar field with respect to Rindler time is

$$\rho_R = \tfrac{1}{2}((\partial_\eta\phi)^2 + (\partial_\xi\phi)^2) \tag{7.136}$$

and in terms of lightcone coordinates this splits into a sum of left-moving and right-moving contributions,

$$\rho_R = (\partial_{u_R}\phi)^2 + (\partial_{v_R}\phi)^2 \ , \tag{7.137}$$

with $f(u_R)$ evidently only contributing to the former and $g(v_R)$ to the latter.

The interest in these (fairly trivial) considerations lies in the fact that the exponential relation (3.97) between the Minkowski and Rindler null coordinates

$$u_M = -a^{-1}\mathrm{e}^{-au_R} \quad , \quad v_M = +a^{-1}\mathrm{e}^{+av_R} \ , \tag{7.138}$$

reflecting the exponential redshift of a Rindler relative to an inertial observer (and vice-versa) has a number of non-trivial and remarkable implications. I will just mention 2 of them here:

1. The exponential redshift expressed by (7.138) implies that the right-moving energy densities in Minkowski and Rindler coordinates are related by

$$\frac{\partial u_M}{\partial u_R} = -au_M \quad \Rightarrow \quad (\partial_{u_M}\phi)^2 = \frac{1}{a^2u_M^2}(\partial_{u_R}\phi)^2 \tag{7.139}$$

   (and likewise for the left-movers). Thus essentially any classical solution that is regarded as regular by the Rindler observer (finite and non-zero ρ_R) corresponds to a divergent Minkowski energy-density as $u_M \to 0$, i.e. on the future boundary (horizon) $t = x$ of the Rindler wedge.

2. The exponential redshift expressed by (7.138) also implies that the notions of *positive frequency* with respect to Minkowski and Rindler time are inequivalent, e.g. in the sense that $f_\omega(u_M)$, restricted to the right Rindler-wedge $u_M < 0$, say, cannot be written as a superposition of Rindler right-moving positive frequency waves alone,

$$f_\omega(u_M) \neq \int_0^\infty d\omega'\alpha(\omega,\omega')f_{\omega'}(u_R) \ . \tag{7.140}$$

   Of course, the $f_\omega(u_R)$ and their complex conjugates $f^*_\omega(u_R)$ provide a basis of solutions for the right-moving modes (in the right Rindler-wedge), so that one can certainly expand the Minkowski plane waves as

$$f_\omega(u_M) = \int_0^\infty d\omega' \ \left(\alpha(\omega,\omega')f_{\omega'}(u_R) + \beta(\omega,\omega')f^*_{\omega'}(u_R)\right) \ , \tag{7.141}$$

but necessarily with some of the $\beta(\omega, \omega') \neq 0$.

If you know a little bit of quantum field theory, you will be able to anticipate that this means that the notions of creation and annihilation operators are inequivalent, and that therefore what is the vacuum, say, for an inertial observer, will not be seen as the vacuum by the accelerating observer (and vice-versa).

Combining the two facts, one also arrives at the conclusion that the "Rindler vacuum" is singular both at the future horizon (from right-movers) and at the past horizon (from left-movers).

In the spirit of the equivalence principle ("before studying gravity, let us study accelerations in flat space"), this *Unruh Effect* is a fascinating and rewarding first step towards understanding (or appreciating the difficulties encountered by) quantum field theory in curved space-times, i.e. in non-trivial gravitational fields. For more on this see the references given in section 27.7.

As further examples of scalar fields in particular gravitational backgrounds, in section 26.8 we will consider scalar fields in the Schwarzschild space-time, and in section 34.10 we will look at the equations of motion of scalar fields in a cosmological gravitational background.

7.9 Conserved Currents from the Energy-Momentum Tensor?

In section 6.8 we had discussed how to obtain conserved charges from covariantly conserved currents. Now in special relativity one can construct conserved currents (corresponding to the generators of Poincaré transformations) from the conserved energy-momentum tensor, and hence from there the corresponding conserved charges like energy, momentum and angular momentum. In this section we will take a first look at the question if or to which extent we can also obtain such conserved currents from the covariantly conserved energy-momentum tensor in a gravitational field.

To set the stage, recall that in Special Relativity, if T^{ab} is the energy-momentum tensor of a physical system, it generally satisfies an equation of the form

$$\partial_a T^{ab} = G^b \quad , \tag{7.142}$$

where G^b represents the density of the external forces acting on the system. In particular, if there are no external forces, the divergence of the energy-momentum tensor is zero. For example, in the case of Maxwell theory and a current corresponding to a charged particle we have

$$G^b = J_a F^{ab} = -F^a_{\ b} J^b \sim -F^a_{\ b} \dot{\xi}^b \quad , \tag{7.143}$$

which is indeed the relevant external (Lorentz) force density (in writing this I have suppressed the δ-function that localises the current to the worldline $\xi^a = \xi^a(\tau)$ of the particle).

When there are no external forces, i.e. when one has taken into account the complete matter action, the total energy-momentum tensor is conserved. In that case, $T^{ab} = J^{(b)a}$ defines four conserved currents, more or less (modulo Belinfante improvement terms, see e.g. the discussion in sections 7.4 and 22.2 and the references given there) the currents associated to translation invariance of the action via Noether's theorem. One is thus in the setting of conserved currents of section 6.8, and one can define conserved quantities like total energy and momentum, P^a, and angular momentum J^{ab}, by integrals of T^{0a} or $\xi^a T^{0b} - \xi^b T^{0a}$ (the latter being conserved if T_{ab} is symmetric) over spacelike hypersurfaces.

The situation in general relativity is somewhat different (exactly how different it is perceived to be is partly a matter of personal preconceptions or desires). In particular, in general relativity, and assuming that $T_{\mu\nu}$ is the complete matter energy-momentum tensor (otherwise we certainly cannot expect to derive any conservation law), we will have a "conservation law" of the form

$$\nabla_\mu T^{\mu\nu} = g^{-1/2}\partial_\mu(g^{1/2}T^{\mu\nu}) + \Gamma^\nu{}_{\mu\lambda}T^{\mu\lambda} = 0 \ . \tag{7.144}$$

We see that, due to the second term, this does not define four conserved currents in the ordinary or covariant sense (and we will return to the interpretation of this equation, and the related issue of energy and energy density of the gravitational field, in section 22.6).

Nevertheless, in analogy with special relativity, one might like to attempt to define conserved quantities like total energy and momentum, P^μ, and angular momentum $J^{\mu\nu}$, by integrals of $T^{0\mu}$ or $x^\mu T^{0\nu} - x^\nu T^{0\mu}$ over spacelike hypersurfaces. However, these quantities are rather obviously not covariant, and nor are they conserved.

This should perhaps not be too surprising because, after all, for a Poincaré-invariant field theory in Minkowski space these quantities are preserved as a consequence of Poincaré invariance, i.e. because of the symmetries (isometries) of the Minkowski metric (as well as of the action).

A generic metric has no isometries whatsoever (the explicit examples of metrics in these notes not withstanding, all of which exhibit at least some symmetries). As it has no symmetries, we have no reason to expect to find associated conserved quantities in general.

However, if there are symmetries then one should indeed be able to define conserved quantities (think of Noether's theorem again), one for each symmetry generator. In order to implement this we need to understand how to define and detect isometries of

the metric. For this we need the concepts of Lie derivatives and Killing vectors. These already made occasional brief appearances in previous sections and will be discussed more systematically in section 9, the corresponding conserved charges then being the subject of section 10.

Alternatively, one might try to just go ahead optimistically and attempt to construct a covariant current-like object (with a corresponding conservation law and the ensuing possibility to define conserved charges) by contracting the energy-momentum tensor not with the coordinates but with a vector field V^λ, along the lines of

$$J^\mu_V = T^\mu_\lambda V^\lambda \ . \tag{7.145}$$

At least this now has the merit of clearly being a vector field, but is it conserved? Calculating its covariant divergence, and using the fact that $T^{\mu\nu}$ is symmetric and conserved, one finds

$$\nabla_\mu J^\mu_V = \tfrac{1}{2} T^{\mu\nu}(\nabla_\mu V_\nu + \nabla_\nu V_\mu) \ . \tag{7.146}$$

Thus we would have a conserved current (and associated conserved charge by the previous section) for any conserved energy-momentum tensor if the vector field V^λ were such that it satisfies

$$\nabla_\mu V_\nu + \nabla_\nu V_\mu = 0 \quad \Rightarrow \quad \nabla_\mu (T^\mu_\lambda V^\lambda) = 0 \ . \tag{7.147}$$

The link between this observation and the one in the preceding paragraph regarding symmetries is that this is precisely the condition characterising (infinitesimal) symmetries of metric:

- First of all, this is the condition we already found and encountered in (3.35), as reformulated in (5.68), for the infinitesimal coordinate transformation $\delta x^\mu = \epsilon V^\mu$ to generate a symmetry of the metric, thus leading to a conserved charge for geodesics.
- More generally, as we will discuss in detail in section 9 below, vector fields satisfying the equation $\nabla_\mu V_\nu + \nabla_\nu V_\mu = 0$ are indeed in one-to-one correspondence with infinitesimal generators of continuous symmetries of a metric (isometries).

Thus this gives a satisfactory and coherent overall picture of symmetries and conservation laws in a gravitational field.

8 Curvature I: The Riemann Curvature Tensor

8.1 Curvature: Preliminary Remarks

We now come to one of the most important concepts of General Relativity and Riemannian Geometry, that of curvature and how to describe it in tensorial terms. Among other things, this will finally allow us to decide unambiguously if a given metric is just the (flat) Minkowski metric in disguise or the metric of a genuinely curved space (but a proof of this statement is postponed to section 11). More importantly (for present purposes) it will allow us to construct tensors that depend on the 2nd derivatives of the metric and will thus allow us to construct tensorial (generally covariant) differential equations for the metric. In particular, this will then lead us fairly directly to the Einstein equations (section 19), i.e. to the field equations for the gravitational field.

Recall that the equations that describe the behaviour of particles and fields in a gravitational field involve the metric and the Christoffel symbols determined by the metric. Thus the equations for the gravitational field should be generally covariant (tensorial) differential equations for the metric.

At first, here we seem to face a dilemma. How can we write down covariant differential equations for the metric when the covariant derivative of the metric is identically zero? Having come to this point, Einstein himself reached an impasse and required the help of his mathematician friend Marcel Grossmann ("Grossmann, you have to help me, or else I'll go crazy!") whom he had asked to investigate if there were any tensors that could be built from the second derivatives of the metric.

Grossmann soon found that this problem had indeed been addressed and solved in the mathematics literature, in particular by Riemann (generalising work of Gauss on curved surfaces), Ricci-Curbastro and Levi-Civita. It was shown by them that there are indeed non-trivial tensors that can be constructed from (ordinary) derivatives of the metric. These can then be used to write down covariant differential equations for the metric.[19]

The most important among these are the Riemann curvature tensor and its various contractions. In fact, it is known that these are the only tensors that can be constructed from the metric and its first and second derivatives, and they will therefore play a central role in all that follows.

Technically the most straightforward way of introducing the Riemann curvature tensor is via the commutator of covariant derivatives. In this section we will adopt this pragmatic (and relatively streamlined) approach, as it is sufficient to

[19] Of course, the story is not as simple and straightforward as that. For an account of Marcel Grossmann's (often overlooked) contributions to tensor calculus and the development of general relativity, see T. Sauer, *Marcel Grossmann and his contribution to the general theory of relativity*, `arXiv:1312.4068 [physics.hist-ph]`.

- determine the most important algebraic and differential properties of the curvature tensor (symmetries and Bianchi identities)
- assess its physical significance (gravitational tidal forces) via the influence of the curvature tensor on the motion of (families of) freely falling particles
- and to thus provide us with all the information and ingredients we need to then discuss the Einstein equations (section 19) and their formulation in terms of an action principle (section 20).

However, this is not geometrically the most intuitive way to introduce the concept of curvature, and it downplays the extent to which the curvature tensor reflects and encodes the geometric properties of space-time and, more generally, does not do justice to the fundamental differential geometric notion and significance of curvature. Some of these aspects are discussed in Part B of these notes, in particular in sections 11, 12, 13 and 14.

8.2 Riemann Tensor from the Commutator of Covariant Derivatives

As mentioned before, second covariant derivatives do not commute on (p,q)-tensors unless $p = q = 0$. However, the fact that they *do* commute on scalars has the pleasant consequence that e.g. the commutator of covariant derivatives acting on a vector field V^μ does not involve any derivatives of V^μ. In fact, I will first show, without actually calculating the commutator, that

$$[\nabla_\mu, \nabla_\nu](\phi V^\lambda) = \phi[\nabla_\mu, \nabla_\nu]V^\lambda \tag{8.1}$$

for any scalar field ϕ. This implies that $[\nabla_\mu, \nabla_\nu]V^\lambda$ cannot depend on derivatives of V because if it did it would also have to depend on derivatives of ϕ.

Hence, the commutator can be expressed purely algebraically in terms of V. As the dependence on V is clearly linear, the commutator of covariant derivatives must then act like a linear transformation. There must therefore be an object $R^\lambda{}_{\sigma\mu\nu}$ such that

$$[\nabla_\mu, \nabla_\nu]V^\lambda = R^\lambda{}_{\sigma\mu\nu}V^\sigma \quad . \tag{8.2}$$

This can of course also be verified by a direct calculation, and we will come back to this below. For now let us just note that, since the left hand side of this equation is clearly a tensor for any V, the quotient theorem implies that the quantities $R^\lambda_{\sigma\mu\nu}$ are the components of a tensor.

Let us first verify (8.1). We have

$$\nabla_\mu\nabla_\nu\phi V^\lambda = (\nabla_\mu\nabla_\nu\phi)V^\lambda + (\nabla_\nu\phi)(\nabla_\mu V^\lambda) + (\nabla_\mu\phi)(\nabla_\nu V^\lambda) + \phi\nabla_\mu\nabla_\nu V^\lambda \quad . \tag{8.3}$$

Thus, upon taking the commutator the 2nd and 3rd terms drop out (because the 3rd is the symmetrisation of the 2nd), and we are left with

$$\begin{aligned}[\nabla_\mu, \nabla_\nu]\phi V^\lambda &= ([\nabla_\mu, \nabla_\nu]\phi)V^\lambda + \phi[\nabla_\mu, \nabla_\nu]V^\lambda \\ &= \phi[\nabla_\mu, \nabla_\nu]V^\lambda \ , \end{aligned} \tag{8.4}$$

where the last line follows from the fact that 2nd covariant derivatives do commute on scalars. Thus we have established (8.1).

By explicitly calculating the commutator, one can confirm the structure displayed in (8.2). This explicit calculation shows that the *Riemann-Christoffel Curvature Tensor* (or Riemann tensor for short) is given by

$$\boxed{R^\lambda{}_{\sigma\mu\nu} = \partial_\mu \Gamma^\lambda{}_{\sigma\nu} - \partial_\nu \Gamma^\lambda{}_{\sigma\mu} + \Gamma^\lambda{}_{\mu\rho}\Gamma^\rho{}_{\nu\sigma} - \Gamma^\lambda{}_{\nu\rho}\Gamma^\rho{}_{\mu\sigma}} \tag{8.5}$$

REMARKS:

1. Note how useful the quotient theorem is in this case. It would be quite unpleasant to have to verify the tensorial nature of this expression by explicitly checking its behaviour under coordinate transformations.

2. Note also that this tensor is clearly zero for the Minkowski metric written in Cartesian coordinates. Hence it is also zero for the Minkowski metric written in any other coordinate system. We will prove the converse, that vanishing of the Riemann curvature tensor implies that the metric is (locally) equivalent to the Minkowski metric, in section 11.2.

3. In the above we have defined the Riemann tensor by the relation (8.2) and then deduced the explicit expression (8.5). While this is, pragmatically speaking, a useful way of proceeding, it may be more logical to initially define the Riemann tensor in a different way, e.g. directly by (8.5) (for instance because by painful calculations one has discovered that this particular combination of non-tensorial objects miraculously happens to transform as a tensor). In that case, (8.2) is a result rather than a definition, known as the *Ricci identity*.

It is straightforward to extend the above to an action of the commutator $[\nabla_\mu, \nabla_\nu]$ on arbitrary tensors. For covectors we have, since we can raise and lower the indices with the metric with impunity,

$$\begin{aligned}[\nabla_\mu, \nabla_\nu]V_\rho &= g_{\rho\lambda}[\nabla_\mu, \nabla_\nu]V^\lambda \\ &= g_{\rho\lambda}R^\lambda{}_{\sigma\mu\nu}V^\sigma \\ &= R_{\rho\sigma\mu\nu}V^\sigma \\ &= R_\rho{}^\sigma{}_{\mu\nu}V_\sigma \ . \end{aligned} \tag{8.6}$$

We will see later that the Riemann tensor is anti-symmetric in its first two indices. Hence we can also write

$$[\nabla_\mu, \nabla_\nu] V_\rho = -R^\sigma{}_{\rho\mu\nu} V_\sigma \quad . \tag{8.7}$$

The extension to arbitrary (p, q)-tensors now follows the usual pattern, with one Riemann curvature tensor, contracted as for vectors, appearing for each of the p upper indices, and one Riemann curvature tensor, contracted as for covectors, for each of the q lower indices. Thus, e.g. for a $(2, 0)$-tensor $T^{\alpha\beta}$ one has

$$[\nabla_\mu, \nabla_\nu] T^{\alpha\beta} = R^\alpha{}_{\gamma\mu\nu} T^{\gamma\beta} + R^\beta{}_{\gamma\mu\nu} T^{\alpha\gamma} \tag{8.8}$$

and for a (1,1)-tensor A^λ_ρ one has

$$[\nabla_\mu, \nabla_\nu] A^\lambda_\rho = R^\lambda{}_{\sigma\mu\nu} A^\sigma_\rho - R^\sigma{}_{\rho\mu\nu} A^\lambda_\sigma \quad . \tag{8.9}$$

I will give two other versions of the fundamental formula (8.2) which are occasionally useful and used.

1. Instead of looking at the commutator $[\nabla_\mu, \nabla_\nu]$ of two derivatives in the coordinate directions x^μ and x^ν, we can look at the commutator $[\nabla_X, \nabla_Y]$ of two directional covariant derivatives. Evidently, in calculating this commutator one will pick up new terms involving $\nabla_X Y^\mu - \nabla_Y X^\mu$. Denoting this vector field by $[X, Y]^\mu$ (the rationale for this notation will be explained in section 9.3), we can also write the formula for the curvature tensor as

$$([\nabla_X, \nabla_Y] - \nabla_{[X,Y]}) V^\lambda = R^\lambda{}_{\sigma\mu\nu} X^\mu Y^\nu V^\sigma \quad . \tag{8.10}$$

2. Secondly, one can consider a net of curves $x^\mu(s_1, s_2)$ parametrising, say, a two-dimensional surface, and look at the commutators of the covariant derivatives along the s_1- and s_2-curves. The formula one obtains in this case (it can be obtained from (8.10) by noting that X and Y commute in this case) is

$$(D_{s_1} D_{s_2} - D_{s_2} D_{s_1})\, V^\lambda = R^\lambda{}_{\sigma\mu\nu} \frac{dx^\mu}{ds_1} \frac{dx^\nu}{ds_2} V^\sigma \quad , \tag{8.11}$$

   where D_{s_k} denotes the covariant derivative along the curve parametrised by s_k, i.e. (section 5.7)

$$D_{s_k} = \frac{\partial x^\mu(s_1, s_2)}{\partial s_k} \nabla_\mu \quad . \tag{8.12}$$

8.3 Symmetries and Algebraic Properties of the Riemann Tensor

A priori, the Riemann tensor has $256 = 4^4$ components in 4 dimensions. However, because of a large number of symmetries, the actual number of independent components is much smaller.

In general, to read off all the symmetries from the formula (8.5) is difficult. One way to simplify things is to look at the Riemann curvature tensor at the origin x_0 of a Riemann normal coordinate system (or some other inertial coordinate system). In that case, all the first derivatives of the metric disappear and only the first two terms of (8.5) contribute. One finds

$$\begin{aligned} R_{\alpha\beta\gamma\delta}(x_0) &= g_{\alpha\lambda}(\partial_\gamma \Gamma^\lambda_{\beta\delta} - \partial_\delta \Gamma^\lambda_{\beta\gamma})(x_0) \\ &= (\partial_\gamma \Gamma_{\alpha\beta\delta} - \partial_\delta \Gamma_{\alpha\beta\gamma})(x_0) \\ &= \tfrac{1}{2}(g_{\alpha\delta,\beta\gamma} + g_{\beta\gamma,\alpha\delta} - g_{\alpha\gamma,\beta\delta} - g_{\beta\delta,\alpha\gamma})(x_0) \ . \end{aligned} \tag{8.13}$$

In principle, this expression is sufficiently simple to allow one to read off all the symmetries of the Riemann tensor. However, it is more insightful to derive these symmetries in a different way, one which will also make clear *why* the Riemann tensor has these symmetries.

1. Anti-symmetry in the second pair of indices:

$$R_{\alpha\beta\gamma\delta} = -R_{\alpha\beta\delta\gamma} \tag{8.14}$$

   This is obviously true from the definition or by construction.

2. Anti-symmetry in the first pair of indices:

$$R_{\alpha\beta\gamma\delta} = -R_{\beta\alpha\gamma\delta} \tag{8.15}$$

   This is a consequence of the fact that the metric is covariantly constant. In fact, we can calculate

$$\begin{aligned} 0 &= [\nabla_\gamma, \nabla_\delta] g_{\alpha\beta} \\ &= R_{\alpha\ \gamma\delta}^{\ \lambda} g_{\lambda\beta} + R_{\beta\ \gamma\delta}^{\ \lambda} g_{\alpha\lambda} \\ &= (R_{\alpha\beta\gamma\delta} + R_{\beta\alpha\gamma\delta}) \ . \end{aligned} \tag{8.16}$$

   As mentioned before, this implies that we can write the commutator of covariant derivatives on a covector as

$$[\nabla_\mu, \nabla_\nu] V_\rho = R_{\rho\ \mu\nu}^{\ \sigma} V_\sigma = -R^\sigma_{\ \rho\mu\nu} V_\sigma \ . \tag{8.17}$$

3. Cyclic permutation symmetry (or first Bianchi identity)

$$R_{\alpha[\beta\gamma\delta]} = 0 \quad \Leftrightarrow \quad R_{\alpha\beta\gamma\delta} + R_{\alpha\delta\beta\gamma} + R_{\alpha\gamma\delta\beta} = 0 \tag{8.18}$$

   This Bianchi identity is a consequence of the fact that there is no torsion. In fact, applying $[\nabla_\gamma, \nabla_\delta]$ to the covector $\nabla_\beta \phi$, ϕ a scalar, one has

$$\nabla_{[\gamma} \nabla_\delta \nabla_{\beta]} \phi = 0 \quad \Rightarrow \quad R^\lambda_{\ [\beta\gamma\delta]} \nabla_\lambda \phi = 0 \ . \tag{8.19}$$

As this has to be true for all scalars ϕ, this implies $R_{\alpha[\beta\gamma\delta]} = 0$ (to see this you could e.g. choose the (locally defined) coordinate functions $\phi^{(\mu)}(x) = x^\mu$ with $\nabla_\lambda \phi^{(\mu)} = \delta^\mu_\lambda$).

In turn this identity now implies that for any covector (not necessarily a gradient) one has the identity (also called Bianchi identity)

$$\nabla_{[\gamma}\nabla_\delta V_{\beta]} = -R^\alpha_{\ [\beta\gamma\delta]} V_\alpha = 0 \ . \tag{8.20}$$

4. Symmetry under exchange of the two pairs of indices

$$R_{\alpha\beta\gamma\delta} = R_{\gamma\delta\alpha\beta} \tag{8.21}$$

This identity, stating that the Riemann tensor is symmetric in its two pairs of indices, is not an independent symmetry but can be deduced from the three other symmetries by some not particularly interesting algebraic manipulations. One (quite possibly not optimal or minimal) possibility is

$$\begin{aligned} R_{\gamma\delta\alpha\beta} &\overset{(3)}{=} -R_{\gamma\alpha\beta\delta} - R_{\gamma\beta\delta\alpha} \\ &\overset{(2)}{=} R_{\alpha\gamma\beta\delta} + R_{\beta\gamma\delta\alpha} \\ &\overset{(3)}{=} -R_{\alpha\delta\gamma\beta} - R_{\alpha\beta\delta\gamma} - R_{\beta\alpha\gamma\delta} - R_{\beta\delta\alpha\gamma} \\ &\overset{(1,2)}{=} 2R_{\alpha\beta\gamma\delta} + R_{\delta\alpha\gamma\beta} + R_{\delta\beta\alpha\gamma} \\ &\overset{(3)}{=} 2R_{\alpha\beta\gamma\delta} - R_{\delta\gamma\beta\alpha} \\ &\overset{(1,2)}{=} 2R_{\alpha\beta\gamma\delta} - R_{\gamma\delta\alpha\beta} \ , \end{aligned} \tag{8.22}$$

from which the claim follows.

Slightly more elegant (but equally obtuse) is the following argument.[20] Consider the matrix

$$\mathcal{R} = \begin{pmatrix} R_{\alpha\beta\gamma\delta} & R_{\alpha\delta\beta\gamma} & R_{\alpha\gamma\delta\beta} \\ R_{\delta\alpha\beta\gamma} & R_{\delta\gamma\alpha\beta} & R_{\delta\beta\gamma\alpha} \\ \vdots & \dots & \dots \end{pmatrix} \tag{8.23}$$

where the first column consists of the 4 cyclic permutations of all 4 indices of the Riemann tensor, while each row consists of the 3 cyclic permutations of the last 3 indices. Thus the sum σ_k of the entries of the k'th row is zero for all k (by symmetry (3)),

$$\sigma_k = \sum_{l=1}^{3} \mathcal{R}_{kl} \overset{(3)}{=} 0 \ , \tag{8.24}$$

while

$$\sigma_1 + \sigma_2 - \sigma_3 - \sigma_4 \overset{(1,2)}{=} 2R_{\alpha\beta\gamma\delta} - 2R_{\gamma\delta\alpha\beta} \ . \tag{8.25}$$

[20] See e.g. D. Bleecker, *Gauge Theory and Variational Principles*.

We can now count how many independent components the Riemann tensor really has. (1) implies that the second pair of indices can only take $N = (4\times 3)/2 = 6$ independent values. (2) implies the same for the first pair of indices. (4) thus says that the Riemann curvature tensor behaves like a symmetric (6×6) matrix and therefore has $(6\times 7)/2 = 21$ components. We now come to the remaining condition (3): if two of the indices in (3) are equal, (3) is equivalent to (4) and (4) we have already taken into account. With all indices unequal, (3) then provides one and only one more additional constraint. We conclude that the total number of independent components is 20.

REMARKS:

1. Note that this agrees precisely with our previous counting in section 3.6 of how many of the second derivatives of the metric cannot be set to zero by a coordinate transformation: the second derivative of the metric has 100 independent components, to be compared with the $4 \times (4 \times 5 \times 6)/(2 \times 3) = 80$ components of the third derivatives of the coordinates. This also leaves 20 components. We thus see very explicitly that the Riemann curvature tensor contains all the coordinate independent information about the geometry up to second derivatives of the metric. In fact, it can be shown that in a Riemann normal coordinate system one has

$$g_{\mu\nu}(x) = \eta_{\mu\nu} + 0 + \tfrac{1}{3} R_{\mu\lambda\sigma\nu}(x_0)(x-x_0)^\lambda (x-x_0)^\sigma + \mathcal{O}((x-x_0)^3) \quad . \tag{8.26}$$

2. Just for the record, I note here that in general dimension $D = d+1$ the Riemann tensor has $D^2(D^2-1)/12$ independent components. This number arises as

$$\begin{aligned} \frac{D^2(D^2-1)}{12} &= \frac{N(N+1)}{2} - \binom{D}{4} \\ N &= \frac{D(D-1)}{2} \end{aligned} \tag{8.27}$$

and describes (as above) the number of independent components of a symmetric $(N\times N)$-matrix, now subject to $\binom{D}{4}$ conditions which arise from all the possibilities of choosing 4 out of D possible distinct values for the indices in (3). Just as for $D = 4$, this number of components of the Riemann tensor coincides with the number of second derivatives of the metric minus the number of independent components of the third derivatives of the coordinates determined in (3.179),

$$\frac{D(D+1)}{2} \times \frac{D(D+1)}{2} - D \times \frac{D(D+1)(D+2)}{2\times 3} = \frac{D^2(D^2-1)}{12} \quad . \tag{8.28}$$

For $D = 2$ this formula predicts one independent component, and this is as it should be. Rather obviously the only independent non-vanishing component of the Riemann tensor in this case is R_{1212}. We will discuss curvature in 2 dimensions in more detail in sections 8.6 and 11.3 below.

Finally, a word of warning: there are a large number of sign conventions involved in the definition of the Riemann tensor (and its contractions we will discuss below), so whenever reading a book or article, in particular when you want to use results or equations presented there, make sure what conventions are being used and either adopt those or translate the results into some other convention. As a check: the conventions used here are such that $R_{\phi\theta\phi\theta}$ as well as the curvature scalar (to be introduced below) are *positive* for the standard metric on the two-sphere.

8.4 Tidal Forces: Influence of Curvature on Particle Trajectories

In a certain sense the main effect of curvature (or gravity) is that initially parallel trajectories of freely falling non-interacting particles (dust, pebbles,...) do not remain parallel, i.e. that gravity is an attractive force that has the tendency to focus matter. This statement find its mathematically precise formulation in equations describing the influence of space-time curvature on the behaviour of (families of) geodesics.

Let us, as we will need this later anyway, recall first the situation in the Newtonian theory. One particle moving under the influence of a gravitational field is governed by the equation

$$\tfrac{d^2}{dt^2}x^i = -\partial^i\phi(x) \ , \tag{8.29}$$

where ϕ is the potential. Now consider a family of particles, or just two nearby particles, one at $x^i(t)$ and the other at $x^i(t) + \delta x^i(t)$. The other particle will of course obey the equation

$$\tfrac{d^2}{dt^2}(x^i + \delta x^i) = -\partial^i\phi(x + \delta x) \ . \tag{8.30}$$

From these two equations one can deduce an equation for δx itself, namely

$$\tfrac{d^2}{dt^2}\delta x^i = -\partial^i\partial_j\phi(x)\delta x^j \ . \tag{8.31}$$

It describes the effect of *gravitational tidal forces* (the gradient of the gravitational force) on a family of particles moving in a gravitational field.

In particular, when there is no gravitational force, and the trajectories are straight lines, one has

$$\tfrac{d^2}{dt^2}\delta x^i = 0 \quad \Rightarrow \quad \delta x^i = (\delta x^i)_0 + (\delta v^i)t \ . \tag{8.32}$$

Thus one recovers Euclid's parallel axiom, that two straight lines intersect at most once (for suitable choices of $\delta v^i \neq 0$) and that they never intersect when they are initially parallel ($\delta v^i = 0$). Any departure from this equation or its Minkowskian counterpart

$$\tfrac{d^2}{d\tau^2}\delta\xi^a = 0 \ , \tag{8.33}$$

will therefore indicate a departure from Euclidean geometry!

It is the counterpart of (8.31) that we will be seeking in the context of General Relativity. One derivation of this can be modelled on the Newtonian derivation above. It is elementary but looks non-covariant (and therefore somewhat messy) at intermediate stages of the calculation (see section 12.1 for a manifestly covariant derivation).

The starting point is of course the geodesic equation for x^μ and for its nearby partner $x^\mu + \delta x^\mu$,

$$\frac{d^2}{d\tau^2}x^\mu + \Gamma^\mu_{\ \nu\lambda}(x)\frac{d}{d\tau}x^\nu \frac{d}{d\tau}x^\lambda = 0 \quad , \tag{8.34}$$

and

$$\frac{d^2}{d\tau^2}(x^\mu + \delta x^\mu) + \Gamma^\mu_{\ \nu\lambda}(x+\delta x)\frac{d}{d\tau}(x^\nu + \delta x^\nu)\frac{d}{d\tau}(x^\lambda + \delta x^\lambda) = 0 \quad . \tag{8.35}$$

As above, from these one can deduce an equation for δx, namely

$$\frac{d^2}{d\tau^2}\delta x^\mu + 2\Gamma^\mu_{\ \nu\lambda}(x)\frac{d}{d\tau}x^\nu \frac{d}{d\tau}\delta x^\lambda + \partial_\rho \Gamma^\mu_{\ \nu\lambda}(x)\delta x^\rho \frac{d}{d\tau}x^\nu \frac{d}{d\tau}x^\lambda = 0 \quad . \tag{8.36}$$

Now this does not look particularly covariant. Thus instead of in terms of $d/d\tau$ we would like to rewrite this in terms of the covariant operator D_τ, with

$$D_\tau \delta x^\mu = \frac{d}{d\tau}\delta x^\mu + \Gamma^\mu_{\ \nu\lambda}\frac{dx^\nu}{d\tau}\delta x^\lambda \quad . \tag{8.37}$$

Calculating $(D_\tau)^2\delta x^\mu$, replacing $\ddot{x}^\mu$ appearing in that expression by $-\Gamma^\mu_{\ \nu\lambda}\dot{x}^\nu\dot{x}^\lambda$ (because x^μ satisfies the geodesic equation) and using (8.36), one eventually finds the nice covariant *geodesic deviation equation*

$$(D_\tau)^2\delta x^\mu = R^\mu_{\ \nu\lambda\rho}\dot{x}^\nu\dot{x}^\lambda\delta x^\rho \tag{8.38}$$

REMARKS:

1. This shows very clearly that curvature, as captured by the Riemann curvature tensor, leads to non-Euclidean geometry in which e.g. the parallel axiom is not necessarily satisifed.

2. In general, solutions to the geodesic deviation equation are called *Jacobi fields*. They describe the difference between the given geodesic and a (hypothetical) infinitely close neighbouring geodesic.

3. Clearly the present derivation of this result leaves something to be desired. It is also possible to give a manifestly covariant, and thus perhaps slightly more satisfactory, derivation of the above geodesic deviation equation, and we will return to this in section 12.1.

8.5 Contractions of the Riemann Tensor: Ricci Tensor and Ricci Scalar

The Riemann tensor, as we have seen, is a four-index tensor. For many purposes this is not the most useful object, but we can create new tensors by contractions of the Riemann tensor. Due to the symmetries of the Riemann tensor, there is essentially only one possibility, namely the *Ricci tensor*

$$R_{\mu\nu} := R^{\lambda}{}_{\mu\lambda\nu} = g^{\lambda\sigma} R_{\sigma\mu\lambda\nu} \quad . \tag{8.39}$$

It arises naturally from the definition (8.2) of the Riemann tensor in terms of commutators of covariant derivatives, when one considers a contracted commutator,

$$[\nabla_\mu, \nabla_\nu] V^\lambda = R^{\lambda}{}_{\sigma\mu\nu} V^\sigma \quad \Rightarrow \quad [\nabla_\mu, \nabla_\nu] V^\mu = R^{\mu}{}_{\sigma\mu\nu} V^\sigma \equiv R_{\sigma\nu} V^\sigma \quad . \tag{8.40}$$

In particular, this identity explains why the Maxwell equations in the covariant Lorenz gauge (6.46) take the non-minimally coupled form (6.47).

It follows from the symmetries of the Riemann tensor that $R_{\mu\nu}$ is symmetric. Indeed

$$R_{\nu\mu} = g^{\lambda\sigma} R_{\sigma\nu\lambda\mu} = g^{\lambda\sigma} R_{\lambda\mu\sigma\nu} = R^{\sigma}{}_{\mu\sigma\nu} = R_{\mu\nu} \quad . \tag{8.41}$$

Thus, for $D = 4$, the Ricci tensor has 10 independent components, for $D = 3$ it has 6, while for $D = 2$ there is only 1 because there is only one independent component of the Riemann curvature tensor to start off with.

There is one more contraction of the Riemann tensor we can perform, namely on the Ricci tensor itself, to obtain what is called the *Ricci scalar* or *curvature scalar*

$$R := g^{\mu\nu} R_{\mu\nu} \quad . \tag{8.42}$$

Remarks:

1. One might have thought that at least in four dimensions there is another way of constructing a (pseudo-)scalar, by contracting the Riemann tensor with the Levi-Civita tensor, but
$$\epsilon^{\mu\nu\rho\sigma} R_{\mu\nu\rho\sigma} = 0 \tag{8.43}$$
because of the Bianchi identity (cyclic symmetry of the Riemann tensor).

2. Note that for $D = 2$ the Riemann curvature tensor has as many independent components as the Ricci scalar, namely one, and that for $D = 3$ the Ricci tensor has as many components as the Riemann tensor, namely 6. Thus in $D = 2$ one can express the entire Riemann tensor in terms of the Ricci scalar (and the metric) alone, and one has
$$D = 2 : \quad R_{\alpha\beta\gamma\delta} = \tfrac{1}{2}(g_{\alpha\gamma} g_{\beta\delta} - g_{\alpha\delta} g_{\beta\gamma}) R \tag{8.44}$$

(we will establish this relation in section 11.3, see (11.29)), while in $D = 3$ one has

$$D = 3: \quad R_{\alpha\beta\gamma\delta} = (g_{\alpha\gamma}R_{\beta\delta} + R_{\alpha\gamma}g_{\beta\delta} - g_{\alpha\delta}R_{\beta\gamma} - R_{\alpha\delta}g_{\beta\gamma}) + \tfrac{1}{2}(g_{\alpha\delta}g_{\beta\gamma} - g_{\alpha\gamma}g_{\beta\delta})R \tag{8.45}$$

(and we will prove this in section 11.4).

3. It is thus only in four (and more) dimensions that there are strictly less components of the Ricci tensor than of the Riemann tensor. This has profound implications for the dynamics of gravity in these dimensions. In fact, we will see that it is only in dimensions $D > 3$ that gravity becomes truly dynamical, where empty space can be curved, where gravitational waves can exist etc.

4. Contracting (8.8), one consequence of the symmetry of the Ricci tensor is the useful general result

$$[\nabla_\mu, \nabla_\nu]T^{\mu\nu} = R_{\mu\nu}(T^{\mu\nu} - T^{\nu\mu}) = 0 \tag{8.46}$$

for any tensor $T^{\mu\nu}$. If $T^{\mu\nu} = F^{\mu\nu}$ is anti-symmetric, $F^{\mu\nu} = -F^{\nu\mu}$, it is not necessary to take the commutator, so one also has

$$F^{\mu\nu} = -F^{\nu\mu} \quad \Rightarrow \quad \nabla_\mu(\nabla_\nu F^{\mu\nu}) = \tfrac{1}{2}[\nabla_\mu, \nabla_\nu]F^{\mu\nu} = 0 \ . \tag{8.47}$$

Note that this can also be deduced (without knowing anything about curvature in general or the Ricci tensor in particular) from the general expression (5.66) for the divergence of an anti-symmetric tensor,

$$\nabla_\mu F^{\mu\nu} = g^{-1/2}\partial_\mu(g^{1/2}F^{\mu\nu}) \quad \Rightarrow \quad \nabla_\nu\nabla_\mu F^{\mu\nu} = g^{-1/2}\partial_\nu\partial_\mu(g^{1/2}F^{\mu\nu}) = 0 \ . \tag{8.48}$$

This is how we had shown in section 6.6 that the Maxwell equations imply covariant current conservation,

$$\nabla_\mu F^{\mu\nu} = -J^\nu \quad \Rightarrow \quad \nabla_\nu J^\nu = 0 \ . \tag{8.49}$$

Now we see that we can alternatively directly use the identity (8.47) to arrive at this result.

5. There are other scalars that can be built from the curvature tensor, but these are necessarily of higher order in the curvature tensor, such as (trivially) R^2 or (somewhat less trivially) $R_{\mu\nu}R^{\mu\nu}$ or the square of the Riemann tensor, the so-called *Kretschmann scalar*

$$K = R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma} \ . \tag{8.50}$$

Analogously, scalars can be built from higher powers of the Riemann tensor and or from powers of covariant derivatives of the Riemann tensor ($\Box R$ being the simplest example).

6. Such scalars are useful in analysing a given metric because, since they are scalars they are invariant under coordinate transformations. Thus they directly provide coordinate-invariant information about a metric. For instance if K is singular at some point in some coordinate system then it will be singular at that point in all coordinate systems, and thus such a singularity is not an artefact of a bad choice of coordinate system but a property of the space(-time) itself described by that metric. A prominent example is the singularity at the origin $r = 0$ of the Schwarzschild metric, unambiguously unveiled by the singularity of its Kretschmann scalar (27.163).

7. Contracting (8.40) with V^ν, one finds

$$V^\nu \nabla_\mu \nabla_\nu V^\mu - V^\nu \nabla_\nu \nabla_\mu V^\mu = R_{\mu\nu} V^\mu V^\nu \ . \tag{8.51}$$

Rewriting the first term as

$$V^\nu \nabla_\mu \nabla_\nu V^\mu = \nabla_\mu (V^\nu \nabla_\nu V^\mu) - (\nabla_\mu V^\nu)(\nabla_\nu V^\mu) \tag{8.52}$$

this identity can be written as

$$V^\nu \nabla_\nu (\nabla_\mu V^\mu) + (\nabla_\mu V_\nu)(\nabla^\nu V^\mu) - \nabla_\mu (V^\nu \nabla_\nu V^\mu) + R_{\mu\nu} V^\mu V^\nu = 0 \ . \tag{8.53}$$

This is a very useful and versatile "master equation" which provides valuable information about the relation between vector fields and curvature when specialised e.g. to geodesic vector fields, $V^\nu \nabla_\nu V^\mu = 0$, or Killing vector fields, $\nabla_\mu V_\nu = -\nabla_\nu V_\mu$ and $\nabla_\mu V^\mu = 0$. Various specialisations of this equation will therefore appear later on in these notes, and even though we will then usually rederive them from scratch in the case at hand, it is good to keep in mind that e.g. (12.22) (our starting point for the discussion of the Raychaudhuri equation in section 12.2) and (13.12) (a useful identity relating Killing vectors and curvature) are special cases of (8.53).

8. As an a(far)side, and as an illustration of what one can do with (8.53), assume that V is such that its curl $\nabla_\mu V_\nu - \nabla_\nu V_\mu = 0$ and its divergence $\nabla_\mu V^\mu = 0$ are zero. Locally, the first condition has the solution $V_\mu = \partial_\mu f$, and then the second condition says that $\Box f = 0$, i.e. that f is *harmonic*. Therefore let us call a V^μ *harmonic* if it satisfies the above two conditions (to the mathematically more sophisticated: yes, I know that this is backwards, but we will specialise to the compact Riemannian case below ...).

For V harmonic in this sense, (8.53) reduces to

$$(\nabla_\mu V_\nu)(\nabla^\mu V^\nu) + R_{\mu\nu} V^\mu V^\nu = \nabla_\mu (V^\nu \nabla_\nu V^\mu) \ . \tag{8.54}$$

The simplest (albeit perhaps not of most direct relevance for physics) situation where one can deduce something of substance from this equation is when one

has a Riemannian (i.e. positive-definite) metric and the space one is dealing with is compact, without boundary. Then (a) the first term is non-negative, and (b) upon integration over the space the total derivative term on the right-hand side gives zero upon use of the Gauss theorem (5.63) (discussed in some more detail in section 16.3).

This implies that for a harmonic V to exist on such a space, the integral of $R_{\mu\nu}V^\mu V^\nu$ must be non-positive. In particular,

- if the Ricci tensor is positive (as a quadratic form), there are no harmonic vector fields at all,
- and if $R_{\mu\nu}V^\mu V^\nu$ =0, then a harmonic vector field is necessarily covariantly constant, $\nabla_\mu V_\nu = 0$.

In more mathematical terms this means that the first Betti number of a compact manifold admitting a metric with positive Ricci curvature is equal to zero. A variant of this kind of argument for Killing vectors will be given in section 13.3.[21]

8.6 Example: Curvature Tensor of the 2-Sphere

To see how calculations of the curvature tensor can be done in practice, let us work out the example of the two-sphere of unit radius, i.e. with line element

$$ds^2 = d\theta^2 + \sin^2\theta d\phi^2 \equiv g_{ab}dx^a dx^b \quad . \tag{8.55}$$

We already know that the non-zero Christoffel symbols necessarily have two ϕ-indices and one θ-index (from $g_{\phi\phi} = \sin^2\theta$), and are given by

$$\Gamma^\phi_{\ \phi\theta} = \cot\theta \quad , \qquad \Gamma^\theta_{\ \phi\phi} = -\sin\theta\cos\theta \quad . \tag{8.56}$$

We also know that the Riemann curvature tensor has only one independent component. Let us therefore work out $R^\theta_{\ \phi\theta\phi}$. From the definition we find

$$R^\theta_{\ \phi\theta\phi} = \partial_\theta\Gamma^\theta_{\ \phi\phi} - \partial_\phi\Gamma^\theta_{\ \theta\phi} + \Gamma^\theta_{\ \theta c}\Gamma^c_{\ \phi\phi} - \Gamma^\theta_{\ \phi c}\Gamma^c_{\ \theta\phi} \quad . \tag{8.57}$$

The second and third terms are manifestly zero, and we are left with

$$R^\theta_{\ \phi\theta\phi} = \partial_\theta(-\sin\theta\cos\theta) + \sin\theta\cos\theta\cot\theta = \sin^2\theta \quad . \tag{8.58}$$

Thus we have

$$\begin{aligned} R^\theta_{\ \phi\theta\phi} &= R_{\theta\phi\theta\phi} = \sin^2\theta \\ R^\phi_{\ \theta\phi\theta} &= 1 \quad . \end{aligned} \tag{8.59}$$

[21] Theorems of this kind, and proved with the help of these kinds of techniques, were pioneered by S. Bochner and K. Yano in the 1930s - 1950s. See e.g. S. Bochner, K. Yano, *Curvature and Betti Numbers* or W. Poor, *Differential Geometric Structures* for more details, rigour and applications.

Therefore the Ricci tensor R_{ab} has the components

$$\begin{aligned} R_{\theta\theta} &= 1 \\ R_{\theta\phi} &= 0 \\ R_{\phi\phi} &= \sin^2\theta \ . \end{aligned} \tag{8.60}$$

These equations can succinctly be written as

$$R_{ab} = g_{ab} \ , \tag{8.61}$$

showing that the standard metric on the two-sphere is what we will later call an *Einstein metric*. The Ricci scalar R is

$$R = g^{\theta\theta}R_{\theta\theta} + g^{\phi\phi}R_{\phi\phi} = 1 + \frac{1}{\sin^2\theta}\sin^2\theta = 2 \ . \tag{8.62}$$

In particular, we have here our first concrete example of a space with non-trivial, in fact positive, curvature.

The result for the Riemann tensor can be written succinctly as

$$R^a{}_{bcd} = \delta^a{}_c g_{bd} - \delta^a{}_d g_{bc} \ , \tag{8.63}$$

which also immediately implies (8.61),

$$R_{bd} = R^a{}_{bad} = g_{bd} \ . \tag{8.64}$$

We will see later on, in section 14, that this form of the curvature tensor, or its equivalent,

$$R_{abcd} = g_{ac}g_{bd} - g_{ad}g_{bc} \ , \tag{8.65}$$

is characteristic of the curvature tensor of the sphere in any dimension.

8.7 More Examples: Curvature Tensor and Polar/Spherical Coordinates

We now turn to some variations of the above theme (and some other generalisations are discussed in section 11.3 below).

1. First of all, let us address the question what is the curvature (scalar) of a sphere of radius L, i.e. of the space with line element

$$ds^2 = L^2(d\theta^2 + \sin^2\theta d\phi^2) \ . \tag{8.66}$$

   There are at least 3 ways to answer this question:

   - The first is to simply and blindly redo the above calculations in this case and to see what one gets.

- Alternatively, and somewhat more insightfully, rather than redoing the calculation in that case one can argue as follows. Let us observe first of all that the Christoffel symbols are invariant under constant rescalings of the metric because they are schematically of the form $g^{-1}\partial g$. Therefore the Riemann curvature tensor, which only involves derivatives and products of Christoffel symbols, is also invariant. Hence the Ricci tensor, which is just a contraction of the Riemann tensor, is also invariant:

$$g_{ab} \to L^2 g_{ab} \quad \Rightarrow \quad \Gamma^a_{\ bc} \to \Gamma^a_{\ bc} \quad \Rightarrow \quad R^a_{\ bcd} \to R^a_{\ bcd} \quad \Rightarrow \quad R_{bd} \to R_{bd} \ . \tag{8.67}$$

  However, to construct the Ricci scalar, one needs the inverse metric. This introduces an explicit L-dependence and the result is that the curvature scalar of a sphere of radius L is $R = 2/L^2$,

$$R(L^2 g_{ab}) = L^{-2} R(g_{ab}) = 2/L^2 \ . \tag{8.68}$$

  In particular, the curvature scalar of a large sphere is smaller than that of a small sphere, something which makes intuitve sense, a very large sphere locally "looking flatter" than a small sphere. However, one should use this intuition with care since, as we have seen, e.g. the Ricci tensor is independent of the size of the sphere.

- Finally, this result could also have been obtained on purely dimensional grounds. The curvature scalar is constructed from second derivatives of the metric. Hence it has length-dimension (-2). Therefore for a sphere of radius L, R has to be proportional to $1/L^2$. Comparing with the known result for $L = 1$ determines $R = 2/L^2$, as before.

2. Now let us consider, instead of the unit 2-sphere, the unit hyperboloid H^2 with metric (2.31)

$$ds^2(H^2) = d\sigma^2 + \sinh^2\sigma \, d\phi^2 \ . \tag{8.69}$$

   It is clear that, apart from a few sign changes here and there, the calculation of the Riemann curvature tensor is identical to that for S^2. These sign changes ultimately lead to the conclusion that the curvature scalar of H^2 is (-2). While the sphere is the prototypical example of a space with positive curvature, the hyperboloid is the prototypical example of a space with negative curvature.

   Instead of just doing the calculation for this specific example, it is slightly more instructive to do it for the class of metrics

$$ds^2 = dx^2 + f(x)^2 d\phi^2 \ , \tag{8.70}$$

   for some (for the time being unspecified) function $f = f(x)$. Denoting the derivative with respect to x by a prime, $f'(x) = df/dx$, one finds (this is a simple but

constructive exercise) that the Ricci tensor and Ricci scalar are

$$R_{ab} = -(f''/f)g_{ab} \quad , \quad R(x) = -2f''(x)/f(x) \ . \tag{8.71}$$

In particular, for the Euclidean metric and the standard metrics on the sphere and the hyperboloid one finds

$$f(x) = \begin{cases} x & (R^2) \\ \sin x & (S^2) \\ \sinh x & (H^2) \end{cases} \quad \Rightarrow \quad R = \begin{cases} 0 \\ +2 \\ -2 \end{cases} \tag{8.72}$$

In 2 dimensions, R is related to the *Gauss Curvature* K of a surface by $K = R/2$ so that $K = 0, \pm 1$ in these examples. See section 11.3 for some more information.

3. Now let us promote the constant radius L of S^2 to a new radial coordinate r and ask the question what is the curvature tensor of the 3-dimensional space with coordinates $(r, x^a) = (r, \theta, \phi)$ and line element

$$ds^2 = dr^2 + r^2(d\theta^2 + \sin^2\theta d\phi^2) \ . \tag{8.73}$$

On the one hand, because one seems to have just added a trivial r-direction to the 2-sphere, one might be tempted to suspect that also this 3-dimensional space has non-trivial curvature. On the other hand, we recognise the above metric as the Euclidean metric on $\mathbb{R}^3$, written in spherical coordinates, and as such we expect its curvature (in fact, all components of the Riemann tensor) to be zero.

The latter expectation is of course borne out, but it is instructive to see explicitly how this cancellation occurs. In fact, it will be even more instructive to consider an apparently harmless and innocuous modification of the above metric which consists in replacing dr^2 by some constant multiple of dr^2,

$$ds^2 = p\, dr^2 + r^2(d\theta^2 + \sin^2\theta d\phi^2) \ . \tag{8.74}$$

Equivalently, up to a truly harmless overall constant factor, we can think of this as the Euclidean metric, but with the metric on the unit-sphere replaced by that of a sphere of radius $1/\sqrt{p} \neq 1$),

$$ds^2 = p\ \left(dr^2 + (r^2/p)(d\theta^2 + \sin^2\theta d\phi^2)\right) \ . \tag{8.75}$$

Proceeding in a pedestrian way, we thus have a metric $g_{\alpha\beta}$ with components

$$g_{rr} = p \quad , \quad g_{ar} = 0 \quad , \quad g_{ab} = r^2\gamma_{ab} \ , \tag{8.76}$$

with γ_{ab} in this example denoting the components of the metric on the unit sphere (and with $\gamma^a_{\ bc}$ and $r^a_{\ bcd}$ its associated Christoffel symbols and components of the Riemann curvature tensor determined in the previous section). From these we can deduce that for $r > 0$ the non-trivial Christoffel symbols are

$$\Gamma^r_{\ ab} = -p^{-1}r\gamma_{ab} \quad , \quad \Gamma^a_{\ br} = r^{-1}\delta^a_{\ b} \quad , \quad \Gamma^a_{\ bc} = \gamma^a_{\ bc} \ . \tag{8.77}$$

From this, in turn, one finds that all the components of the Riemann tensor involving at least one r-index are zero, whereas for the purely angular components one finds

$$R^a{}_{bcd} = r^a{}_{bcd} + \Gamma^a{}_{cr}\Gamma^r{}_{bd} - \Gamma^a{}_{dr}\Gamma^r{}_{bc} \ . \tag{8.78}$$

Using (8.63) and (8.77), one sees that

$$R^a{}_{bcd} = (1-p^{-1})r^a{}_{bcd} = (1-p^{-1})(\delta^a{}_c\gamma_{bd} - \delta^a{}_d\gamma_{bc}) \ . \tag{8.79}$$

Therefore precisely for $p = 1$ the two contributions to the curvature tensor indeed cancel and the curvature tensor is identically zero, as expected.

Equally interesting is the fact that for $p \neq 1$ the curvature is non-zero even away from $r = 0$ (in addition, there is a conical deficit angle singularity at $r = 0$, as in the next example below, but this shall not be our concern here). In particular it follows from the above result that the only non-vanishing components of the Ricci tensor of this 3-dimensional space are

$$R^a{}_{bcd} = (1-p^{-1})r^a{}_{bcd} \quad \Rightarrow \quad R_{bd} = (1-p^{-1})r_{bd} = (1-p^{-1})\gamma_{bd} \ . \tag{8.80}$$

Therefore also its Ricci scalar is non-zero,

$$R = g^{\alpha\beta}R_{\alpha\beta} = g^{ab}R_{ab} = 2(1-p^{-1})r^{-2} \ . \tag{8.81}$$

We also see from this that this space actually has a curvature singularity as $r \to 0$. Since the Ricci scalar is a scalar (under coordinate transformations), this divergence cannot be an artefact of a bad choice of coordinates, and indicates that there is a genuine geometric singularity for $r \to 0$.

Extended to a four-dimensional space-time metric via

$$ds^2 = -dt^2 + p\, dr^2 + r^2(d\theta^2 + \sin^2\theta d\phi^2) \ , \tag{8.82}$$

this describes the gravitational field outside a "monopole".[22]

4. As a final variation of this theme, we consider the above example in one dimension less, i.e. we look at the metric one obtains if one replaces the Euclidean metric on $\mathbb{R}^2$ written in polar coordinates by

$$dr^2 + r^2 d\phi^2 \to p\, dr^2 + r^2 d\phi^2 \ , \tag{8.83}$$

where the angle ϕ has period 2π.

In this case there is an interesting twist (pun intended) and the situation is somewhat different. Pulling out the factor of p, one sees that (up to this irrelevant overall constant factor) the metric can be written as

$$ds^2 = dr^2 + r^2 d(\phi/\sqrt{p})^2 \equiv dr^2 + C^2 r^2 d\phi^2 \ . \tag{8.84}$$

[22] M. Barriola, A. Vilenkin, *Gravitational Field of a Global Monopole*, Phys. Rev. Lett. 63 (1989) 341-343.

This would be the standard Euclidean metric on $\mathbb{R}^2$ either for $p = 1$ or if the angle ϕ had periodicity $2\pi\sqrt{p}$, but since ϕ has period 2π, this results in a misidentification of the points in a plane, like when one rolls up a flat piece of paper into a cone. Away from $r = 0$, this space is intrinsically flat (all the components of the Riemann curvature tensor are zero, as one can easily calculate - see section 11.1 for an explanation of this use of the word "intrinsic"). There is, however, a conical singularity at the tip of the cone $r = 0$, which can be thought of as a δ-function contribution to the curvature localised at $r = 0$. Extended to a four-dimensional space-time metric,

$$ds^2 = -dt^2 + dz^2 + dr^2 + C^2 r^2 d\phi^2 \quad , \tag{8.85}$$

it can be interpreted as the space-time metric of an idealised cosmic string extended in the z-direction.[23]

8.8 Bianchi Identities and the Einstein Tensor

So far, we have discussed algebraic properties of the Riemann tensor. The Riemann tensor also satisfies some differential identities which, in particular in their contracted form, will be of fundamental importance in the following.

The first identity is easy to derive. As a (differential) operator the covariant derivative clearly satisfies the Jacobi identity

$$[\nabla_{[\mu}, [\nabla_\nu, \nabla_{\lambda]}]] = 0 \tag{8.86}$$

(total anti-symmetrisation over all 3 indices). Since the commutator $[\nabla_\nu, \nabla_\lambda]$ is already anti-symmetric in the indices ν, λ, this anti-symmetrisation is equivalent to cyclic permutation of the 3 indices,

$$[\nabla_{[\mu}, [\nabla_\nu, \nabla_{\lambda]}]] = 0 \quad \Leftrightarrow \quad [\nabla_\mu, [\nabla_\nu, \nabla_\lambda]] + \circlearrowleft (\mu, \nu, \lambda) = 0 \ . \tag{8.87}$$

If you do not believe this identity (valid for any 3 associative linear operators), you can just write out the twelve relevant terms explicitly to see that there is indeed a complete

[23] It is far from straightforward, however, to find a formalism which allows one to calculate and derive the distributional Riemann tensor of this space-time - see R. Geroch, J. Traschen, *Strings and other distributional sources in general relativity*, Phys. Rev. D36 (1987) 1017-1031 for a general analysis of the problem and issues arising in this and related contexts, C. Clarke, J. Vickers, J. Wilson, *Generalized functions and distributional curvature of cosmic strings*, Class. Quantum Grav. 13 (1996) 2485-2498 for one approach (based on the Colombeau algebra of distributions), and D. Garfinkle, *Metrics with distributional curvature*, `arXiv:gr-qc/9906053` for a different approach. We will (mostly) stay away from distributional curvatures in these notes.

cancellation:

$$\begin{aligned}[\nabla_{[\mu},[\nabla_\nu,\nabla_{\lambda]}]] &\sim \nabla_\mu\nabla_\nu\nabla_\lambda - \nabla_\mu\nabla_\lambda\nabla_\nu - \nabla_\nu\nabla_\lambda\nabla_\mu + \nabla_\lambda\nabla_\nu\nabla_\mu \\ &+ \nabla_\lambda\nabla_\mu\nabla_\nu - \nabla_\lambda\nabla_\nu\nabla_\mu + \nabla_\nu\nabla_\mu\nabla_\lambda - \nabla_\mu\nabla_\nu\nabla_\lambda \\ &+ \nabla_\nu\nabla_\lambda\nabla_\mu - \nabla_\nu\nabla_\mu\nabla_\lambda - \nabla_\lambda\nabla_\mu\nabla_\nu + \nabla_\mu\nabla_\lambda\nabla_\nu \\ &= 0 \ . \end{aligned} \tag{8.88}$$

To determine the implications of this identity for the Riemann tensor, we apply it to a vector field V, say. The first term in (8.87) is

$$\begin{aligned}[\nabla_\mu,[\nabla_\nu,\nabla_\lambda]]V^\rho &= \nabla_\mu(R^\rho{}_{\sigma\nu\lambda}V^\sigma) - [\nabla_\nu,\nabla_\lambda](\nabla_\mu V^\rho) \\ &= (\nabla_\mu R^\rho{}_{\sigma\nu\lambda})V^\sigma + R^\rho{}_{\sigma\nu\lambda}\nabla_\mu V^\sigma - R^\rho{}_{\sigma\nu\lambda}\nabla_\mu V^\sigma + R^\sigma{}_{\mu\nu\lambda}\nabla_\sigma V^\rho \\ &= (\nabla_\mu R^\rho{}_{\sigma\nu\lambda})V^\sigma + R^\sigma{}_{\mu\nu\lambda}\nabla_\sigma V^\rho \ . \end{aligned} \tag{8.89}$$

Upon taking the cyclic permutations, the sum of the 2nd terms vanishes by the cyclic symmetry of the Riemann tensor, and therefore one finds

$$(\nabla_\mu R^\rho{}_{\sigma\nu\lambda})V^\sigma + \circlearrowright(\mu,\nu,\lambda) = 0 \ . \tag{8.90}$$

Since this holds for any V, one deduces the *Bianchi identity*

$$\nabla_\mu R^\rho{}_{\sigma\nu\lambda} + \circlearrowright(\mu,\nu,\lambda) = 0 \quad \Leftrightarrow \quad \nabla_{[\mu}R^\rho{}_{|\sigma|\nu\lambda]} = 0 \tag{8.91}$$

(where $|\sigma|$ indicates that this index is to be excluded from the anti-symmetrisation). Using the symmetry (IV) of the Riemann tensor, this can equivalently be written as

$$\nabla_{[\mu}R_{\nu\lambda]\rho\sigma} = 0 \ . \tag{8.92}$$

We will mainly be interested in a (double) contraction of this identity. To that end we write out (8.91) explicitly as

$$\nabla_\lambda R_{\alpha\beta\mu\nu} + \nabla_\nu R_{\alpha\beta\lambda\mu} + \nabla_\mu R_{\alpha\beta\nu\lambda} = 0 \ . \tag{8.93}$$

By contracting this with $g^{\alpha\mu}$ we obtain

$$\nabla_\lambda R_{\beta\nu} - \nabla_\nu R_{\beta\lambda} + \nabla_\mu R^\mu{}_{\beta\nu\lambda} = 0 \ . \tag{8.94}$$

This is not yet particularly useful. To also turn the last term into a Ricci tensor we contract once more, with $g^{\beta\lambda}$ to obtain the contracted Bianchi identity

$$\nabla_\lambda R^\lambda{}_\nu - \nabla_\nu R + \nabla_\mu R^\mu{}_\nu = 0 \ , \tag{8.95}$$

or

$$\nabla^\mu(R_{\mu\nu} - \tfrac{1}{2}g_{\mu\nu}R) = 0 \ . \tag{8.96}$$

The tensor appearing in this equation is the so-called *Einstein tensor* $G_{\mu\nu}$,

$$G_{\mu\nu} = R_{\mu\nu} - \tfrac{1}{2}g_{\mu\nu}R \ . \tag{8.97}$$

It is the unique divergence-free tensor that can be built from the metric and its first and second derivatives (apart from $g_{\mu\nu}$ itself, of course),

$$\nabla^\mu G_{\mu\nu} = 0 \quad , \tag{8.98}$$

and this is why it will play *the* central role in the Einstein equations for the gravitational field.

A minor *caveat* regarding the above statement about the uniqueness of the Einstein tensor is that, as it stands, it is only true in $D = 4$ space-time dimensions. In $D > 4$, there are other tensors with this property, but they are non-linear in 2nd derivatives of the metric. The uniqueness statement continues to be true for $D > 4$ if one adds the requirement that the tensor is linear in 2nd derivatives of the metric. I will briefly come back to this in the discussion of the action principle for general relativity in section 20.1.

8.9 Riemann Normal Coordinates Revisited

In section 3.6 we had introduced Riemann normal coordinates as a special class of inertial coordinate systems, based on geodesics. The main idea was to introduce new coordinates $x^\alpha \to \xi^a$ in such a way that the coordinate lines of the new coordinates ξ^a are geodesics passing through the point p at which one wants to erect this coordinate system.

In particular,

1. we considered the Taylor expansion (3.170)

   $$x^\alpha(\tau) = x_0^\alpha + \tau u_0^\alpha - \tfrac{1}{2}\tau^2 \Gamma^\alpha_{\beta\gamma}(x_0) u_0^\beta u_0^\gamma + \dots \quad , \tag{8.99}$$

   of a solution to the geodesic equation;

2. this led us to consider the coordinate transformation (3.172)

   $$x^\alpha(\xi) = x_0^\alpha + (\xi - \xi_0)^a e_a^\alpha - \tfrac{1}{2}(\xi - \xi_0)^b (\xi - \xi_0)^c \Gamma^\alpha_{\beta\gamma}(x_0) e_b^\beta e_c^\gamma + \dots \tag{8.100}$$

   which has the property that the lines

   $$\xi^a(\tau) = \xi_0^a + \tau \lambda^a \tag{8.101}$$

   are geodesics for any constant λ^a;

3. for these geodesics the geodesic equation reduces to

   $$\ddot{\xi}^a + \Gamma^a_{bc} \dot{\xi}^b \dot{\xi}^c = 0 \quad \Rightarrow \quad \Gamma^a_{bc}(\xi_0^a + \tau\lambda^a)\lambda^b\lambda^c = 0 \quad , \tag{8.102}$$

   implying at $\tau = 0$

   $$\Gamma^a_{bc}(\xi_0)\lambda^b\lambda^c = 0 \quad \forall \lambda^a \quad \Rightarrow \Gamma^a_{bc}(\xi_0) = 0 \tag{8.103}$$

   (and we will look at the implications of the next term in the Taylor expansion of (8.102) below).

Therefore the Taylor expansion of the metric around $\xi = \xi_0$ has the form

$$g_{ab}(\xi) = \eta_{ab} + \tfrac{1}{2}(\xi - \xi_0)^c(\xi - \xi_0)^d g_{ab,\,cd}(\xi_0) + \dots \quad , \tag{8.104}$$

and we will now determine the quadratic term in this expansion (and be able to express it in terms of the components of the Riemann tensor $R_{abcd}(\xi_0)$ at the point p in these coordinates). To that end we look at the next term in the Taylor expansion of (8.102). Thus we differentiate (8.102) along the geodesic, i.e. with respect to τ, and evaluate the results at $\tau = 0$ to deduce

$$\partial_d \Gamma^a{}_{bc}(\xi_0)\lambda^d\lambda^b\lambda^c = 0 \quad \forall \ \ \lambda^b \tag{8.105}$$

or, equivalently

$$\partial_{(d}\Gamma^a{}_{bc)}(\xi_0) = 0 \quad \Leftrightarrow \quad \partial_d \Gamma^a{}_{bc}(\xi_0) + \partial_b \Gamma^a{}_{cd}(\xi_0) + \partial_c \Gamma^a{}_{db}(\xi_0) = 0 \ . \tag{8.106}$$

This condition, and the analogous conditions

$$\partial_{(d\dots e}\Gamma^a{}_{bc)}(\xi_0) = 0 \tag{8.107}$$

arising from the higher-order terms in the Taylor expansion of (8.102) impose constraints on the Christoffel symbols and their derivatives that are satisfied in Riemann normal coordinates (but not in general inertial coordinate systems).

A useful way of reexpressing the condition (8.106) is the following (a certain amount of hindsight or trial-and-error is required for this): because $\Gamma^a{}_{bc}(\xi_0) = 0$, from the definition of the Riemann tensor we have

$$\begin{aligned} R^a{}_{bcd}(\xi_0) + R^a{}_{cbd}(\xi_0) &= \partial_c \Gamma^a{}_{bd}(\xi_0) - \partial_d \Gamma^a{}_{bc}(\xi_0) + \partial_b \Gamma^a{}_{cd}(\xi_0) - \partial_d \Gamma^a{}_{cb}(\xi_0) \\ &= \partial_c \Gamma^a{}_{bd}(\xi_0) + \partial_b \Gamma^a{}_{cd}(\xi_0) - 2\partial_d \Gamma^a{}_{bc}(\xi_0) \ , \end{aligned} \tag{8.108}$$

and using (8.106) this can be written as

$$\partial_d \Gamma^a{}_{bc}(\xi_0) = -\tfrac{1}{3}(R^a{}_{bcd}(\xi_0) + R^a{}_{cbd}(\xi_0)) \ . \tag{8.109}$$

On the other hand, from

$$g_{ab,\,c}(\xi) = \Gamma_{abc}(\xi) + \Gamma_{bac}(\xi) \tag{8.110}$$

we have

$$g_{ab,\,cd}(\xi) = \partial_d \Gamma_{abc}(\xi) + \partial_d \Gamma_{bac}(\xi) \tag{8.111}$$

and at ξ_0 we can use (8.109) and the symmetries of the Riemann tensor to deduce

$$\begin{aligned} g_{ab,\,cd}(\xi_0) &= -\tfrac{1}{3}(R_{abcd} + R_{acbd} + R_{bacd} + R_{bcad})(\xi_0) \\ &= -\tfrac{1}{3}(R_{acbd} + R_{bcad})(\xi_0) = -\tfrac{1}{3}(R_{acbd} + R_{adbc})(\xi_0) \ . \end{aligned} \tag{8.112}$$

We have thus found that, to quadratic order in a Taylor expansion of the metric around the origin of a Riemann normal coordinate system, the metric can be written as

$$\begin{aligned} g_{ab}(\xi) &= \eta_{ab} - \tfrac{1}{6}(R_{acbd}(\xi_0) + R_{adbc}(\xi_0))(\xi - \xi_0)^c(\xi - \xi_0)^d + \mathcal{O}(\xi^3) \\ &= \eta_{ab} - \tfrac{1}{3}R_{acbd}(\xi_0)(\xi - \xi_0)^c(\xi - \xi_0)^d + \mathcal{O}(\xi^3) \ . \end{aligned} \tag{8.113}$$

If required, higher order terms can be determined analogously with the help of the higher order terms in the Taylor expansion of (8.102), and (with a steady hand) can be expressed in terms of the covariant derivatives of the Riemann tensor at ξ_0.

8.10 Principle of Minimal Coupling Revisited

In sections 4.1 and 6.1 on the principles of general covariance and minimal coupling respectively, I mentioned that these do not necessarily fix the equations uniquely. In other words, there could be more than one generally covariant equation which reduces to a given equation in Minkowski space. Having the curvature tensor at our disposal now, we can construct examples of this kind.

Given some tensorial equation, obtained by the minimal coupling prescription, say, one can always contemplate the possibility to add additional terms to it involving the curvature tensor. Since such terms take the form of higher derivative corrections to the original equation, multiplied by appropriate dimensionful constants, one can usually get away with ignoring such terms when dealing with weak fields and other low-energy phenomena, and under such conditions the minimal coupling rule can usually be trusted. However, such terms are not negligible under extreme conditions involving e.g. very strong or strongly fluctuating gravitational fields.

An example which shows very clearly that the minimal coupling prescription, at least the way we have formulated it, is itself ambiguous is, as already briefly pointed out in section 6.6, provided by Maxwell theory. In that case, we saw that in the covariant Lorenz gauge one has (6.46)

$$\nabla_\mu A^\mu = 0 \quad \Rightarrow \quad \nabla_\mu F^{\mu\nu} = \nabla_\mu(\nabla^\mu A^\nu - \nabla^\nu A^\mu) = \Box A^\nu - [\nabla_\mu, \nabla^\nu] A^\mu \ , \tag{8.114}$$

where $\Box A^\nu = \nabla^\mu \nabla_\mu A^\nu$. It thus follows from (8.40) that the Maxwell equations in the covariant Lorenz gauge can be written as (6.47)

$$\nabla_\mu A^\mu = 0 \quad \Rightarrow \quad \nabla_\mu F^{\mu\nu} = -J^\nu \quad \rightarrow \quad \Box A^\nu - R^\nu{}_\mu A^\mu = -J^\nu \ . \tag{8.115}$$

What this shows is that "minimal coupling" all by itself is not a unique prescription, as we would have obtained (8.115) without the curvature terms by applying the minimal coupling prescription to the special relativity Maxwell equation in the Lorenz gauge, namely just $\Box A_\nu = -J_\nu$.

In the present situation, (6.47) is superior to the equation without the curvature term because

- it follows from a variational principle (involving the minimally coupled counterpart of the Maxwell action)

- and (related to this) because (8.115) implies that the current is covariantly conserved (as we had verified in section 6.6 in an arbitrary gauge), while for the equation without the curvature term covariant current conservation would then be violated by a curvature term (as can easily be verified).

Thus occasionally some such additional criteria can be used to eliminate (or reduce) the ambiguity in the minimal coupling prescription, but this need not always be the case.

As another example, consider the wave equation for a (massless, say) scalar field Φ. In Minkowski space, this is the Klein-Gordon equation which has the obvious curved space analogue (5.56)

$$\Box\Phi = 0 \tag{8.116}$$

obtained by the minimal coupling description. However, one could equally well postulate the equation

$$(\Box + \xi R)\Phi = 0 \ , \tag{8.117}$$

where ξ is a (dimensionless) constant and R is the scalar curvature. This equation is generally covariant, and reduces to the ordinary Klein-Gordon equation in Minkowski space, so this is an acceptable curved-space extension of the wave equation for a scalar field. This equation of motion arises (in D sapce-time dimensions) from the action

$$S_\xi[\phi, g_{\alpha\beta}] = -\tfrac{1}{2}\int \sqrt{g} d^D x \ \left(g^{\mu\nu}\partial_\mu\phi\partial_\nu\phi + \xi R\phi^2\right) \ , \tag{8.118}$$

exhibiting the non-minimal (yet generally covariant) coupling of the scalar field to gravity via the term $\xi R\phi^2$ (which acts as an x-dependent mass term for the scalar field).

Moreover, ξ is dimensionless, so one cannot argue that on dimensional grounds this ambiguity is irrelevant for weak fields. Indeed, one frequently postulates a specific non-zero value for ξ which makes the wave equation *conformally invariant* (invariant under position-dependent Weyl rescalings of the metric) for massless fields, and this criterion can be imposed to select a particular non-zero value for ξ (e.g. for a 4-dimensional space-time this turns out to be the value $\xi = 1/6$). This will be discussed and explained in section 22.3.

Thus in general such ambiguities are present and are something one has to live with.

19 The Einstein Equations

19.1 Heuristics

We expect the gravitational field equations to be non-linear second order partial differential equations for the metric. If we knew more about the weak field equations of gravity (which should thus be valid near the origin of an inertial coordinate system) we could use the Einstein equivalence principle (or the principle of general covariance) to deduce the equations for strong fields.

However, we do not know a lot about gravity beyond the Newtonian limit of weak time-independent fields and low velocities, simply because gravity is so 'weak'. Hence, we cannot find the gravitational field equations in a completely systematic way and some guesswork will be required.

Nevertheless we will see that with some very natural assumptions (and the benefit of hindsight) we will arrive at an essentially unique set of equations. Further theoretical (and aesthetical) confirmation for these equations will then come from the fact that they turn out to be the Euler-Lagrange equations of the absolutely simplest action principle for the metric imaginable.

Recall that, way back, in section 1.1, we had briefly discussed the possibility of a scalar relativistic theory of gravity described by an equation of the form (1.3)

$$\Delta\phi = 4\pi G_N \rho \quad \longrightarrow \quad \Box\phi = 4\pi G_N \rho \ . \tag{19.1}$$

We had noted there that one way to render this equation (tensorially) consistent is to think of both the left and the right hand side as (00)-components of some tensor, which we expressed in (1.6) as

$$\{\text{Some tensor generalising } \Delta\phi\}_{\alpha\beta} \sim 4\pi G_N T_{\alpha\beta} \ . \tag{19.2}$$

While this appeared to be an exotic proposal back in section 1.1, we now understand that this is exactly what is required, and we have a fairly precise idea of what this tensor on the left-hand side should be.

Indeed, recall from our discussion of the Newtonian limit of the geodesic equation that the weak static field produced by a non-relativistic mass density ρ is

$$g_{00} = -(1 + 2\phi) \ , \tag{19.3}$$

With the identification

$$T_{00} = \rho \ , \tag{19.4}$$

the Newtonian field equation $\Delta\phi = 4\pi G_N \rho$ can now also be written as

$$\Delta g_{00} = -8\pi G_N T_{00} \ . \tag{19.5}$$

This suggests that the weak-field equations for a general energy-momentum tensor take the form

$$E_{\alpha\beta} = \quad \{\text{Some tensor generalising } (-\Delta g_{00})\}_{\alpha\beta} \quad = 8\pi G_N T_{\alpha\beta} \ , \tag{19.6}$$

where $E_{\alpha\beta}$ is constructed from the metric and its first and second derivatives.

By the Einstein equivalence principle, if this equation is valid for weak fields (i.e. near the origin of an inertial coordinate system) then also the equations which govern gravitational fields of arbitrary strength must be of this form, with $E_{\mu\nu}$ a tensor constructed from the metric and its first and second derivatives.

Another way of anticipating what form the field equations for gravity may take is via an analogy, a comparison of the geodesic deviation equations in Newton's theory and in General Relativity. Recall that in Newton's theory we have

$$\begin{aligned} \frac{d^2}{dt^2}\delta x^i &= -K^i{}_j \delta x^j \\ K^i{}_j &= \partial^i \partial_j \phi \ , \end{aligned} \tag{19.7}$$

whereas in General Relativity we have

$$\begin{aligned} (D_\tau)^2 \delta x^\mu &= -K^\mu_{\ \nu} \delta x^\nu \\ K^\mu_{\ \nu} &= R^\mu_{\ \lambda\nu\rho} \dot{x}^\lambda \dot{x}^\rho \ . \end{aligned} \tag{19.8}$$

Now Newton's field equation is

$$\mathrm{Tr}\, K \equiv \Delta\phi = 4\pi G_N \rho \ , \tag{19.9}$$

while in General Relativity we have

$$\mathrm{Tr}\, K = R_{\mu\nu} \dot{x}^\mu \dot{x}^\nu \ . \tag{19.10}$$

This suggests that somehow in the gravitational field equations of General Relativity, $\Delta\phi$ should be replaced by the Ricci tensor $R_{\mu\nu}$. Note that, at least roughly, the tensorial structure of this identification is compatible with the relation between ϕ and g_{00} in the Newtonian limit, the relation between ρ and the 0-0 component T_{00} of the energy momentum tensor, and the fact that for small velocities $R_{\mu\nu}\dot{x}^\mu\dot{x}^\nu \sim R_{00}$.

We will now turn to a somewhat more precise argument along these lines which will enable us to determine $E_{\mu\nu}$.

19.2 More Systematic Approach

Let us take stock of what we know about $E_{\mu\nu}$.

1. $E_{\mu\nu}$ is a tensor

2. $E_{\mu\nu}$ has the dimensions of a second derivative. If we assume that no new dimensionful constants enter in $E_{\mu\nu}$ then it has to be a linear combination of terms which are either linear in second derivatives of the metric or quadratic in the first derivatives of the metric. (Later on, we will see that there is the possibility of a zero derivative term, but this requires a new dimensionful constant, the *cosmological constant* Λ. Higher derivative terms or higher non-linearities could in principle appear but would only be relevant at very high energies.)

3. $E_{\mu\nu}$ is symmetric since $T_{\mu\nu}$ is symmetric.

4. Since $T_{\mu\nu}$ is covariantly conserved, the same has to be true for $E_{\mu\nu}$,

$$\nabla_\mu T^{\mu\nu} = 0 \Rightarrow \nabla_\mu E^{\mu\nu} = 0 \ . \tag{19.11}$$

5. Finally, for a weak static gravitational field and non-relativistic matter we should find

$$E_{00} = -\Delta g_{00} \ . \tag{19.12}$$

Now it turns out that these conditions (1)-(5) determine $E_{\mu\nu}$ *uniquely*! First of all, (1) and (2) tell us that $E_{\mu\nu}$ has to be a linear combination

$$E_{\mu\nu} = aR_{\mu\nu} + bg_{\mu\nu}R \ , \tag{19.13}$$

where $R_{\mu\nu}$ is the Ricci tensor and R the Ricci scalar. Then condition (3) is automatically satisfied.

To implement (4), we rewrite the above as a linear combination of the Einstein tensor (8.97) and $g_{\mu\nu}R$,

$$E_{\mu\nu} = aG_{\mu\nu} + cg_{\mu\nu}R \equiv a(R_{\mu\nu} - \tfrac{1}{2}g_{\mu\nu}R) + cg_{\mu\nu}R \tag{19.14}$$

and recall the contracted Bianchi identity (8.95,8.96),

$$\nabla^\mu G_{\mu\nu} = 0 \ . \tag{19.15}$$

It follows that (4) is satisfied iff $c\nabla_\nu R = c\partial_\nu R = 0$. We therefore have to require either $\nabla_\nu R = 0$ or $c = 0$. That the first possibility is ruled out (inconsistent) can be seen by taking the trace of (19.6),

$$E^\mu_{\ \mu} = (4c - a)R = 8\pi G_N T^\mu_{\ \mu} \ . \tag{19.16}$$

Thus, R is proportional to $T^\mu_{\ \mu}$ and since this quantity need certainly not be constant for a general matter configuration, we are led to the conclusion that $c = 0$. Thus we find

$$E_{\mu\nu} = aG_{\mu\nu} \ . \tag{19.17}$$

We can now use condition (5) to determine the constant a.

By the above considerations we have determined the field equations to be of the form

$$aG_{\mu\nu} = 8\pi G_N T_{\mu\nu} \ , \tag{19.18}$$

with a some, as yet undetermined, constant. We will now consider the Newtonian weak-field limit of this equation. We need to find that G_{00} is proportional to Δg_{00} and we can then use the condition (5) to fix the value of a. The following manipulations are somewhat analogous to those we performed in section 3.3 when considering the Newtonian limit of the geodesic equation. The main difference is that now we are dealing with second derivatives of the metric rather than with just its first derivatives entering in the geodesic equation.

As in section 3.3, let us begin by stating the assumptions that we make when considering the Newtonian limit:

1. Weak Fields

   We take this to mean that there exists a coordinate system $x^\alpha = (x^0, x^i)$ in which the metric takes the form

   $$g_{\alpha\beta} = \eta_{\alpha\beta} + h_{\alpha\beta} \tag{19.19}$$

   with $\eta_{\alpha\beta}$ the standard form of the Minkowski metric and $h_{\alpha\beta}$ and its derivatives small. In practice this means that in the following we will neglect terms that are quadratic or of higher order in $h_{\alpha\beta}$.

2. Time-independent Fields

   We assume that in these coordinates the gravitational field is time-independent, i.e. that one has

   $$\partial_0 g_{\alpha\beta} = \partial_0 h_{\alpha\beta} = 0 \ . \tag{19.20}$$

3. Non-relativistic Matter Source

   This replaces the condition that particles move non-relativistically (with coordinate speeds $v \ll c$), and we take this to mean that the only non-negligible contribution to the energy-momentum tensor $T_{\alpha\beta}$ comes from the energy density $T_{00} = \rho$,

   $$T_{00} = \rho \neq 0 \quad , \quad T_{\alpha\beta} = 0 \quad \text{otherwise} \ . \tag{19.21}$$

So we need to determine

$$G_{00} = R_{00} - \tfrac{1}{2} g_{00} R \ . \tag{19.22}$$

Since the scalar curvature is at least linear in $h_{\alpha\beta}$, to leading order in $h_{\alpha\beta}$ we can replace $g_{00} \to \eta_{00} = -1$ to obtain

$$G_{00} = R_{00} + \tfrac{1}{2} R \ , \tag{19.23}$$

where here and in the remainder of this calculation equality signs signify equalities to leading order in $h_{\alpha\beta}$.

To bootstrap the calculation of G_{00}, we start from

$$T_{ij} = 0 \quad \Rightarrow \quad G_{ij} = 0 \quad \Leftrightarrow \quad R_{ij} = \tfrac{1}{2} g_{ij} R \ , \tag{19.24}$$

and by the same reasoning as above we can write this as

$$R_{ij} = \tfrac{1}{2} \delta_{ij} R \ . \tag{19.25}$$

Therefore, for the scalar curvature we find

$$R = g^{\alpha\beta} R_{\alpha\beta} = \eta^{\alpha\beta} R_{\alpha\beta} = -R_{00} + \delta^{ij} R_{ij} = -R_{00} + \tfrac{3}{2} R \ , \tag{19.26}$$

or

$$R = 2R_{00} \ . \tag{19.27}$$

Thus

$$G_{00} = R_{00} + \tfrac{1}{2} R = 2R_{00} \tag{19.28}$$

and it just remains to calculate this one component of the Ricci tensor. In the weak field limit, R_{00} is given by

$$R_{00} = R^{k}_{\ 0k0} = \delta^{ik} R_{i0k0} \ . \tag{19.29}$$

Moreover, in this limit only the linear (second derivative) part of $R_{\mu\nu\lambda\sigma}$ will contribute, not the terms quadratic in first derivatives. Thus we can use the expression (8.13) for the curvature tensor. Additionally, in the static case we can ignore all time derivatives. Then only one term (the third) of (8.13) contributes and we find

$$R_{i0k0} = -\tfrac{1}{2} g_{00,ik} \ , \tag{19.30}$$

and therefore

$$R_{00} = -\tfrac{1}{2} \Delta g_{00} \ . \tag{19.31}$$

Putting everything together, we get

$$G_{00} = 2R_{00} = -\Delta g_{00} \ . \tag{19.32}$$

Thus we obtain the correct functional form of E_{00} and comparison with condition (5) determines $a = +1$ and therefore $E_{\alpha\beta} = G_{\alpha\beta}$. See also section 23.3 for a somewhat more streamlined and covariant derivation of this result from the linearised Einstein equations.

19.4 Einstein Equations

We have finally arrived at the Einstein equations for the gravitational field (metric) of a matter-energy configuration described by the energy-momentum tensor $T_{\mu\nu}$. It is

$$\boxed{R_{\mu\nu} - \tfrac{1}{2} g_{\mu\nu} R = 8\pi G_N \, T_{\mu\nu}} \tag{19.33}$$

These are the equations that replace the Newtonian (Poisson) equation for the gravitational potential.

Another common way of writing the Einstein equations is obtained by taking the trace of (19.33), which yields

$$R - 2R = 8\pi G_N \, T^{\mu}_{\mu} \equiv 8\pi G_N \, T \quad , \tag{19.34}$$

and substituting this back into (19.33) to obtain

$$R_{\mu\nu} = 8\pi G_N (T_{\mu\nu} - \tfrac{1}{2} g_{\mu\nu} T) \quad . \tag{19.35}$$

In particular, for the vacuum, $T_{\mu\nu} = 0$, the Einstein equations are

$$T_{\mu\nu} = 0 \quad \Rightarrow \quad R_{\mu\nu} = 0 \quad , \tag{19.36}$$

and this condition is equivalent to the vanishing of the Einstein tensor,

$$G_{\mu\nu} = 0 \quad \Leftrightarrow \quad R_{\mu\nu} = 0 \quad . \tag{19.37}$$

A space-time metric satisfying this equation is, for obvious reasons, said to be Ricci flat.

A priori, the Einstein equations constitute 10 coupled non-linear (actually quasi-linear, since they are linear in second derivatives) second order partial differential equations for the metric $g_{\mu\nu}(x)$, which appears both in the Einstein tensor on the left-hand side of these equations as well as usually also on the right-hand side in the matter energy-momentum tensor (we will see below, in section 19.7, that these 10 equations are linked by 4 differential identities, the contracted Bianchi identities).

This is a tremendously complicated set of equations, and trying to learn and say something about general properties of solutions to these equations is very challenging.[34]

[34] The current state of knowledge and understanding of the mathematical structure of the Einstein equations, in particular regarding the properties of the Cauchy (initial value) problem for the Einstein equations, is described in detail in the awe-inspiring œuvre *General Relativity and the Einstein Equations* by Y. Choquet-Bruhat (warning: not for the faint of heart). A readable historical introduction to the Cauchy problem for the Einstein equations is given by her in Y. Choquet-Bruhat, *Beginnings of the Cauchy problem*, `arXiv:1410.3490 [gr-qc]`.

Even the vacuum Einstein equations still constitute a complicated set of non-linear coupled partial differential equations whose general solution is not, and probably will never be, known. Usually one makes some assumptions, in particular regarding the symmetries of the metric, that reduce the number of independent variables from 10 functions $g_{\alpha\beta}(x)$ of 4 variables to a smaller number of functions depending on a smaller number of variables, and which then simplify the equations to the extent that they can be analysed explicitly, either analytically, or at least qualitatively or numerically. How to do this in practice (in the simplest non-trivial situations), will be explained in detail later on in these notes.

REMARKS:

1. With c not set equal to one, and with the convention that T_{00} is normalised such that it gives the energy-density rather than the mass-density, one finds that the factor $8\pi G_N$ on the right hand side should be replaced by

$$8\pi G_N \quad \to \quad \frac{8\pi G_N}{c^4} \ . \tag{19.38}$$

   A note on dimensions: Newton's constant has dimensions (M mass, L length, T time) $[G_N] = \mathsf{M}^{-1}\mathsf{L}^3\mathsf{T}^{-2}$ so that

$$[G_N] = \mathsf{M}^{-1}\mathsf{L}^3\mathsf{T}^{-2} \quad \Rightarrow \quad [G_N/c^4] = \mathsf{L}^{-1}\mathsf{M}^{-1}\mathsf{T}^2 \ . \tag{19.39}$$

   Moreover, an energy density $\rho = \mu c^2$, μ a mass density, has dimensions

$$[\rho] = [\mu c^2] = \mathsf{M}\mathsf{L}^{-3}\mathsf{L}^2\mathsf{T}^{-2} = \mathsf{M}\mathsf{L}^{-1}\mathsf{T}^{-2} \ . \tag{19.40}$$

   Thus

$$[\rho G_N/c^4] = \mathsf{L}^{-2} = [R_{\mu\nu}] \ , \tag{19.41}$$

   as it should be. Frequently, an alternative (and equally reasonable) convention is used in which T_{00} is a mass density, so that then $T_{tt} = c^2 T_{00}$ is the energy density. In that case, the factor on the right-hand side of the Einstein equations is $8\pi G_N/c^2$.

2. The streamlined "derivation" of the Einstein equations given here may give the misleading impression that also for Einstein this was a straighforward affair. Nothing could be further from the truth. Not only do we have the benefit of hindsight. We also have a much more systematic and advanced understanding of Riemannian geometry and tensor calculus than was available to Einstein at the time. This concerns in particular things like the contracted Bianchi identities and their importance for energy-momentum (non-)conservation and for general covariance (to be briefly discussed in section 19.7 below).[35]

[35] For an illuminating brief account of the torturous and convoluted route and crucial final stages that led Einstein (and Grossmann) to the correct field equations, see N. Straumann, *Einstein's 'Zürich Notebook' and his Journey to General Relativity*, `arXiv:1106.0900v1 [physics.hist-ph]`.

3. As an aside, note that the trace (19.34) of the Einstein equations

$$R = -8\pi G_N \, T \quad , \tag{19.42}$$

is a scalar generally covariant differential equation for the metric (but it is of course far from sufficient to determine 10 independent components of the metric up to coordinate transformations). If one assumes, however, that the space-time metric can be parametrised by a single scalar ψ, say (somewhat like in the Newtonian limit), e.g. by stipulating that it only differs from the Minkowski metric by a conformal factor, as in

$$g_{\mu\nu} = \psi^2 \eta_{\mu\nu} \quad , \tag{19.43}$$

then a scalar equation like (19.42) (the numerical constant needs to be adjusted appropriately in order to obtain the correct Newtonian limit) provides a differential equation for ψ and thus a generally covariant scalar theory of gravity. A theory of this kind, a geometrisation and covariantisation of previous scalar theories of gravity, was proposed by Einstein and Fokker in 1913/14, some two years before Einstein arrived at the final (tensorial) form of the field equations. In this theory, there is no coupling of gravity and Maxwell theory (which has a traceless energy-momentum tensor), and null lines are identical to null lines in Minkowski space (because of conformal flatness), so for either of these reasons there is no bending of lightrays by the gravitational field in such a theory.

4. As we saw before, in two and three dimensions, vanishing of the Ricci tensor implies the vanishing of the Riemann tensor. Thus in these cases, space-times are necessarily flat away from where there is matter, i.e. at points at which $T_{\mu\nu}(x) = 0$. Thus there are no true gravitational fields and no gravitational waves.

   In four dimensions, however, the situation is completely different. As we saw, the Ricci tensor has 10 independent components whereas the Riemann tensor has 20. Thus there are 10 components of the Riemann tensor which can curve the vacuum, as e.g. in the field around the sun, and a lot of interesting physics is already contained in the vacuum Einstein equations.

5. If for whatever reason one is interested in studying solutions to the matter + Einstein equations in dimensions other than $D = 3 + 1$, this is straightforward and there are just a a few small points to pay attention to:

   - The Einstein tensor, i.e. the (unique) rank-2 tensor that can be constructed from the Riemann curvature tensor which has vanishing covariant divergence, has the same form in any dimension, $G_{\mu\nu} = R_{\mu\nu} - (1/2)g_{\mu\nu}R$.
   - Likewise, what will appear on the right-hand side of the equations is the appropriate generally covariant energy-momentum tensor.

- However, in the higher-dimensional analogue of the Einstein equations the constant of proportionality between the Einstein and energy-momentum tensors should not be called $8\pi G_N$. After all, this factor was determined from the Newtonian limit of the $(3+1)$-dimensional Einstein equations where e.g. a factor 4π has, via the Poisson equation for a point mass, its origin in the fact that the area of a unit 2-sphere is 4π. Thus we will just call it κ (which is then related in a dimension-dependent way to however one wants to normalise the D-dimensional gravitational coupling constant). Thus precisely as in 4 dimensions one can write the Einstein equations as

$$G_{\mu\nu} \equiv R_{\mu\nu} - \frac{1}{2}\, g_{\mu\nu} R = \kappa\, T_{\mu\nu} \quad . \tag{19.44}$$

- If one wants to use the analogue of (19.35), one should pay attention to the fact that it is less symmetric with respect to (19.44) than its 4-dimensional counterpart since in $D = n + 1$ dimensions it takes the form

$$R_{\mu\nu} = \kappa(T_{\mu\nu} - \frac{1}{D-2}\, g_{\mu\nu} T^{\lambda}_{\ \lambda}) \quad . \tag{19.45}$$

19.5 Cosmological Constant

As mentioned before, there is one more term that can be added to the Einstein equations provided that one relaxes the condition (2) that only terms quadratic in derivatives should appear. This term takes the form $\Lambda g_{\mu\nu}$. This is compatible with the condition (4) (the conservation law) provided that Λ is a constant, the *cosmological constant*. It is a dimensionful parameter with dimension $[\Lambda] = \mathsf{L}^{-2}$ one over length squared.

The Einstein equations with a cosmological constant now read

$$R_{\mu\nu} - \tfrac{1}{2} g_{\mu\nu} R + \Lambda g_{\mu\nu} = 8\pi G_N T_{\mu\nu} \quad . \tag{19.46}$$

To be compatible with condition (5) ((1), (3) and (4) are obviously satisfied), Λ has to be quite small (and observationally it is very small indeed).

Remarks:

1. The vacuum Einstein equations with a cosmological constant read

$$R_{\mu\nu} - \tfrac{1}{2} g_{\mu\nu} R = -\Lambda g_{\mu\nu} \quad . \tag{19.47}$$

   Taking traces, this implies (and is equivalent to)

$$R_{\mu\nu} = \Lambda g_{\mu\nu} \quad , \tag{19.48}$$

   which is the counterpart of the Ricci-flatness condition for vacuum solutions of the Einstein equations without a cosmological constant. In general, solutions to the equation $R_{\mu\nu} = c g_{\mu\nu}$ for some constant c (and either Riemannian or Lorentzian signature) are known as *Einstein manifolds* in the mathematics literature.

2. Λ gives a contribution to the energy-momentum tensor that, in Minkowski space, would be proportional to the Minkowski metric and Lorentz-invariant, thus compatible with the symmetries of the vacuum, and Λ is often said to play the role of a vacuum energy density (more precisely vacuum energy should perhaps be considered as one possible contribution to the cosmological constant - see section 38.4 for further discussion of this issue).

3. Comparing with the energy-momentum tensor of, say, a perfect fluid (see (7.70) in section 7.5 or section 35.2),

$$T_{\mu\nu} = (\rho + p)u_\mu u_\nu + p g_{\mu\nu} \quad , \tag{19.49}$$

we see that Λ corresponds to the energy density and pressure values

$$\rho_\Lambda = -p_\Lambda = \frac{\Lambda}{8\pi G_N} \quad , \tag{19.50}$$

and to an energy-momentum tensor

$$T^\Lambda_{\mu\nu} = -\rho_\Lambda g_{\mu\nu} \quad . \tag{19.51}$$

Thus, depending on the sign of Λ either the energy density or the pressure is negative,

$$\Lambda < 0 \quad \Rightarrow \quad \rho_\Lambda < 0 \quad , \quad \Lambda > 0 \quad \Rightarrow \quad p_\Lambda < 0 \quad . \tag{19.52}$$

4. The cosmological constant was originally introduced by Einstein because he was unable to find static cosmological solutions without it. We will review this *Einstein Static Universe* in section 37.2. After Hubble's discovery of the expansion of the universe, a static universe fell out of fashion and the cosmological constant was no longer required.

5. However, things are not as simple as that. Just because it is not required does not mean that it is not there. In fact, one of the biggest puzzles in theoretical physics today is why the cosmological constant is so small. According to standard quantum field theory lore, the vacuum energy density should be many many orders of magnitude larger than astrophysical observations allow. Now usually in quantum field theory one does not worry too much about the vacuum energy as one can normal-order it away. However, as we know, gravity is unlike any other theory in that not only energy-differences but absolute energies matter (and cannot just be dropped).

   The question why the observed cosmological constant is so small (and recent astrophysical observations appear to favour a tiny non-zero value) is one aspect of what is known as the *Cosmological Constant Problem*. See section 38.4 for a brief discussion of this profound issue and some references.

6. We will consider the possibility that $\Lambda \neq 0$ only in the sections on cosmology (in all other applications, Λ can indeed be neglected).

The Einstein equations

$$G_{\mu\nu} = 8\pi G_N \, T_{\mu\nu} \tag{19.53}$$

can, taken at face value, be regarded as ten algebraic equations for certain traces of the Riemann tensor $R_{\mu\nu\rho\sigma}$. $R_{\mu\nu\rho\sigma}$ has, as we know, twenty independent components, so how are the other ten determined? The obvious answer, already given above, is of course that we solve the Einstein equations for the metric $g_{\mu\nu}$ and then calculate the Riemann curvature tensor of that metric.

However, this answer leaves something to be desired because it does not really provide an explanation of how the information about these other components is encoded in the Einstein equations. It is interesting to understand this because it is precisely these components of the Riemann tensor wich represent the effects of gravity in vacuum, i.e. where $T_{\mu\nu} = 0$, like tidal forces and gravitational waves.

The more insightful answer is that the information is encoded in the Bianchi identities which serve as propagation equations for the trace-free parts of the Riemann tensor away from the regions where $T_{\mu\nu} \neq 0$.

Let us see how this works. Recall from section 11.4 the decomposition of the Riemann tensor into the traceless Weyl tensor and the trace parts, the Ricci tensor and Ricci scalar,

$$\begin{aligned} C_{\mu\nu\rho\sigma} = {} & R_{\mu\nu\rho\sigma} \\ & - \frac{1}{D-2}(g_{\mu\rho}R_{\nu\sigma} + R_{\mu\rho}g_{\nu\sigma} - g_{\nu\rho}R_{\mu\sigma} - R_{\nu\rho}g_{\mu\sigma}) \\ & + \frac{1}{(D-1)(D-2)}R(g_{\mu\rho}g_{\nu\sigma} - g_{\nu\rho}g_{\mu\sigma}) \ . \end{aligned}$$

In the vacuum, $R_{\mu\nu} = 0$, and therefore at points where the energy-momentum tensor vanishes one has

$$T_{\mu\nu}(x) = 0 \quad \Rightarrow \quad R_{\mu\nu\rho\sigma}(x) = C_{\mu\nu\rho\sigma}(x) \ . \tag{19.54}$$

As anticipated, the Weyl tensor thus encodes the information about the gravitational field in vacuum.

The question thus is how $C_{\mu\nu\rho\sigma}$ is determined everywhere in space-time by an energy-momentum tensor which may be localised in some finite region of space-time. To address that question we make use of the relation (11.88) derived in section 11.4,

$$\nabla^\mu C_{\mu\nu\rho\sigma} = (D-3)\,(\nabla_\rho P_{\nu\sigma} - \nabla_\sigma P_{\nu\rho}) \ . \tag{19.55}$$

Here $P_{\mu\nu}$ is the Schouten tensor (11.81),

$$P_{\mu\nu} = \frac{1}{D-2}\left(R_{\mu\nu} - \frac{1}{2(D-1)}g_{\mu\nu}R\right) \ . \tag{19.56}$$

Using the D-dimensional Einstein equations (19.44), (19.45) to replace the Ricci tensor and Ricci scalar by the energy-momentum tensor, one now obtains a propagation equation for the Weyl tensor of the form

$$\nabla^\mu C_{\mu\nu\rho\sigma} = J_{\nu\rho\sigma} \quad , \tag{19.57}$$

where $J_{\nu\rho\sigma}$ depends only on the energy-momentum tensor and its derivatives,

$$J_{\nu\rho\sigma} = \kappa \, \frac{D-3}{D-2} \left[\nabla_\rho T_{\nu\sigma} - \nabla_\sigma T_{\nu\rho} - \frac{1}{D-1} \left[\nabla_\rho T^\lambda{}_\lambda g_{\nu\sigma} - \nabla_\sigma T^\lambda{}_\lambda g_{\nu\rho} \right] \right] \quad . \tag{19.58}$$

This is the equation which determines the Weyl tensor components in terms of the sources. It is reminiscent of the Maxwell equation

$$\nabla^\mu F_{\mu\nu} = -J_\nu \quad , \tag{19.59}$$

and provides an intuitive (as well as, if required, detailed analytical) understanding of the propagation properties of the gravitational field.

19.7 General Covariance and Significance of the Bianchi Identities

Let us try to understand in a bit more detail, but necessarily at a very superficial and unsophisticated level, the structure of the Einstein equations.

As a first step, let us do something that we should have perhaps done rightaway, namely count the number of dynamical variables and the number of equations we have:

- the dynamical variables are the components $g_{\alpha\beta}(x)$ of the metric, i.e. 10 functions of 4 variables.
- the Ricci or Einstein tensor is symmetric; therefore the Einstein equations consitute a set of ten algebraically independent second order differential equations for the metric $g_{\alpha\beta}$.

At first, this "ten 2nd order equations for ten unknowns" looks exactly right: specifying the values of the metric and its first time-derivative as initial values on some (constant "time") hypersurface, say, this should then uniquely determine the ten components of the metric in some region to the future of that hypersurface.

At second sight, however, this cannot possibly be right and the end of the story and, if true, would actually be a major disaster. After all, the Einstein equations are generally covariant. Thus, given one metric that is a solution to the Einstein equations, one should be able to perform an arbitrary coordinate transformation and still have a (physically equivalent) solution to the Einstein equations. That means that the (ten?) Einstein equations should not determine the ten components of the metric uniquely but only

up to arbitrary coordinate transformations, i.e. up to four arbitrary functions of four variables.

Phrased in terms of initial values, one should be able to perform arbitrary time-dependent coordinate transformations on a solution, but if these coordinate transformations happen to be the identity transformation on the initial hypersurface, then these solutions related by (future) coordinate transformations should arise from the same initial data.

Either way we should expect only six independent generally covariant equations for the metric, determining the 10 components of the metric up to 4 arbitary functions. How does that happen? Here we should recall the contracted Bianchi identities. They tell us that

$$\nabla^\alpha G_{\alpha\beta} = 0 \ . \tag{19.60}$$

We see that, even though the ten Einstein equations are algebraically independent, there are actually four differential relations among them, so this is just right.

It is no coincidence, by the way, that the Bianchi identities come to the rescue of general covariance. We will see in section 20.6 that the Bianchi identities can in fact be understood as a consequence of the general covariance of the Einstein equations (and of the corresponding action principle).

The general covariance of the Einstein equations is reflected in the fact that only six of the ten equations are truly dynamical 2nd-order differential equations while four of them constrain the initial values of the fields on some spacelike hypersurface. Indeed, in terms of some choice $(x^\alpha) = (t, x^k)$ of time and space coordinates, the Bianchi identities

$$\nabla_\alpha G^{\alpha\beta} = \partial_\alpha G^{\alpha\beta} + \Gamma^\alpha_{\ \alpha\gamma} G^{\beta\gamma} + \Gamma^\beta_{\ \alpha\gamma} G^{\alpha\gamma} = 0 \tag{19.61}$$

can be written as

$$\partial_t G^{t\beta} = -\partial_k G^{k\beta} - \Gamma^\alpha_{\ \alpha\gamma} G^{\beta\gamma} - \Gamma^\beta_{\ \alpha\gamma} G^{\alpha\gamma} \ . \tag{19.62}$$

Since the 3 terms on the right-hand side contain at most 2nd time derivatives of the metric, the 4 components $G_{t\beta}$ of the Einstein tensor can contain at most 1st time derivatives of the metric. Thinking of initial data as being given by the metric and its 1st time-derivative on some initial hypersurface, this means that the components $G_{t\beta} = 0$ of the Einstein equations (or their counterpart in the presence of matter) impose constraints on these initial data and do not provide evolution equations for these initial data.

The perhaps more familiar counterpart of these constraints in the case of Maxwell theory is the Gauss Law constraint $\vec{\nabla}.\vec{E} = 0$, which arises as the 0-component of the Maxwell equations $\partial_a F^{ab} = 0$,

$$\partial_a F^{a0} = \partial_k F^{k0} = -\vec{\nabla}.\vec{E} = 0 \ , \tag{19.63}$$

and which also involves at most 1st time-derivatives of the dynamical field (the gauge field), and thus constitutes a constraint on the initial conditions rather than a true evolution equation.

In this case, the obvious ("contracted Bianchi") identity

$$\partial_a \partial_b F^{ab} = 0 \quad \Leftrightarrow \quad \partial_0(\partial_i F^{i0}) = -\partial_k(\partial_a F^{ak}) \tag{19.64}$$

implies

1. that the 4 Maxwell equations are not independent (as required by gauge invariance as they should only determine the 4 components A_a of the gauge field up to gauge transformations) and

2. that the Gauss Law contraint equation is "propagated", i.e. that by virtue of the true equations of motion it will hold at all times if it holds initially:

$$(\partial_a F^{ak})|_{t=0} = 0 \quad \Rightarrow \quad (\partial_k(\partial_a F^{ak}))_{t=0} = 0 \quad \Rightarrow \quad (\partial_0(\partial_i F^{i0}))_{t=0} = 0 \ , \tag{19.65}$$

   and likewise for the higer t-derivatives,

$$(\partial_0 \partial_a F^{ak})|_{t=0} = 0 \quad \Rightarrow \quad (\partial_k \partial_0(\partial_a F^{ak}))_{t=0} = 0 \quad \Rightarrow \quad (\partial_0^2(\partial_i F^{i0}))_{t=0} = 0 \ , \tag{19.66}$$

   etc. Thus if the true dynamical equations are satisfied at all times, the constraints will be satisfied at all times provided that they are satisfied initially.

Analogously, for the Einstein equations the contracted Bianchi identity in the form (19.62) implies not only 4 relations among the 10 field equations (as required by general covariance) but also that the constraints of general relativity are again "propagated" in this sense. One simple way to see this (or that this is plausible, at least - in order to prove a theorem one would need te be more precise about the initial value formulation and make sure that it leads to a well-defined time evolution etc.) is to note that by (19.62) $G^{\alpha\beta} = 0$ at $t = t_0$ (thus also $\partial_k G^{\alpha\beta} = 0$ at $t = t_0$) implies

$$G^{\alpha\beta}|_{t=t_0} = 0 \quad \Rightarrow \quad (\partial_t G^{t\beta})|_{t=t_0} = 0 \tag{19.67}$$

(and likewise for higher t-derivatives).

We will discuss this and related issues in some more detail from a slightly different (Hamiltonian) perspective in section 21.

20 Einstein Equations from an Action Principle

20.1 Einstein-Hilbert Action

To increase our confidence that the Einstein equations we have derived above are in fact reasonable and almost certainly correct, we can adopt a more modern point of view. We can ask if the Einstein equations follow from an action principle or, alternatively, what would be a natural action principle for the metric.

After all, for example in the construction of the Standard Model, one also does not start with the equations of motion but one writes down the simplest possible Lagrangian with the desired field content and symmetries.

We will start with the gravitational part, i.e. the Einstein tensor $G_{\alpha\beta}$ of the Einstein equations, and deal with the matter part, the energy-momentum tensor $T_{\alpha\beta}$, later.

By general covariance, an action for the metric $g_{\alpha\beta}$ will have to take the form

$$S = \int \sqrt{g} d^4x \; \Phi(g_{\alpha\beta}) \;\; , \tag{20.1}$$

where Φ is a scalar constructed from the metric. So what is Φ going to be? Clearly, the simplest choice is the Ricci scalar R, and this is also the unique choice if one is looking for a scalar constructed from not higher than second derivatives of the metric. Therefore we postulate the beautifully simple and elegant action

$$\boxed{S_{EH}[g_{\alpha\beta}] = \int \sqrt{g} d^4x \; R} \tag{20.2}$$

known as the Einstein-Hilbert action. It was presented by Hilbert practically on the same day that Einstein presented his final form (19.33) of the gravitational field equations. Discussions regarding who did what first and who deserves credit for what have been a favourite occupation of historians of science ever since. However, Hilbert's work would certainly not have been possible without Einstein's realisation that gravity should be regarded not as a force but as a property of space-time and his physical insight that Riemannian geometry and tensor analysis provide the correct framework for embodying the equivalence principle. Regarding the action principle for general relativity, in his superb scientific biography of Einstein, A. Pais says

> Hilbert was not the first to apply this principle to gravitation. Lorentz had done it before him. So had Einstein, a few weeks earlier. Hilbert was the first, however, to state this principle correctly.[36]

[36] A. Pais, *Subtle is the Lord* (chapter 14.d, which also contains a detailed account of the interaction between Einstein and Hilbert in the crucial November 1915 period).

We will now prove that the Euler-Lagrange equations following from the Einstein-Hilbert Lagrangian indeed give rise to the Einstein tensor and the vacuum Einstein equations. It is truly remarkable, that such a simple Lagrangian is capable of explaining practically all known gravitational, astrophysical and cosmological phenomena.

Before turning to a proof of this statement, I need to make one preliminary remark:

In this discussion we will at first ignore total derivative (or boundary) terms that one picks up from integration by parts of the variations and concentrate on the bulk Euler-Lagrange equations of motion. In standard variational problems one usually justifies this by appealing to the fact that one can e.g. choose the variations of the fields to vanish on the boundary and that therefore such boundary terms are zero. In the case at hand, things are a bit more complicated since the boundary terms that one picks up in the process of performing the variations turns out to depend both on the variation of the field (i.e. the metric) on the boundary and on its normal derivative, and it is not consistent to require both to be zero (i.e. to impose both Dirichlet and Neumann boundary conditions). This whole issue is interesting in its own right and warrants a separate discussion, and therefore we will deal with it afterwards, in sections 20.4 and 20.5.

Returning to the Einstein-Hilbert action, we now need to determine its behaviour under a variation of the metric. Since the Ricci scalar is $R = g^{\alpha\beta}R_{\alpha\beta}$, it turns out to be more convenient to consider variations $\delta g^{\alpha\beta}$ of the inverse metric instead of $\delta g_{\alpha\beta}$. This is of course equivalent, the two variations being related by

$$\delta(g^{\alpha\beta}g_{\beta\gamma}) = \delta(\delta^{\alpha}_{\ \gamma}) = 0 \quad \Rightarrow \quad \delta g^{\alpha\beta} = -g^{\alpha\gamma}(\delta g_{\gamma\delta})g^{\delta\beta} \ . \tag{20.3}$$

Thus, as a first step we write

$$\begin{aligned} \delta S_{EH} &= \delta \int \sqrt{g} d^4x \; g^{\alpha\beta}R_{\alpha\beta} \\ &= \int d^4x \; \Big((\delta\sqrt{g}) g^{\alpha\beta}R_{\alpha\beta} + \sqrt{g}(\delta g^{\alpha\beta})R_{\alpha\beta} + \sqrt{g}g^{\alpha\beta}\delta R_{\alpha\beta} \Big) \ . \end{aligned} \tag{20.4}$$

Now we make use of the identity (5.77)

$$\delta\sqrt{g} = \tfrac{1}{2}\sqrt{g}g^{\alpha\beta}\delta g_{\alpha\beta} = -\tfrac{1}{2}\sqrt{g}g_{\alpha\beta}\delta g^{\alpha\beta} \ . \tag{20.5}$$

to deduce

$$\begin{aligned} \delta S_{EH} &= \int \sqrt{g} d^4x \; [(-\tfrac{1}{2}g_{\alpha\beta}R + R_{\alpha\beta})\delta g^{\alpha\beta} + g^{\alpha\beta}\delta R_{\alpha\beta}] \\ &= \int \sqrt{g} d^4x \; (R_{\alpha\beta} - \tfrac{1}{2}g_{\alpha\beta}R)\delta g^{\alpha\beta} + \int \sqrt{g} d^4x \; g^{\alpha\beta}\delta R_{\alpha\beta} \ . \end{aligned} \tag{20.6}$$

The first term all by itself would already give the Einstein tensor. Thus we need to show that the second term is a total derivative. I do not know of any particularly elegant

argument to establish this (in a coordinate basis - written in terms of differential forms this would be completely obvious), so this will require a little bit of work, but it is not difficult.

Postponing the proof of this statement to the next section 20.2, we have established that (ignoring boundary terms) the variation of the Einstein-Hilbert action gives the gravitational part (left hand side) of the Einstein equations,

$$\delta S_{EH}[g_{\alpha\beta}] = \delta \int \sqrt{g} d^4x \; R = \int \sqrt{g} d^4x \; G_{\alpha\beta} \delta g^{\alpha\beta} + \oint \dots \quad . \tag{20.7}$$

Remarks:

1. If one wants to include the cosmological constant Λ, then the action gets modified to

$$S_{EH,\Lambda} = \int \sqrt{g} d^4x \; (R - 2\Lambda) \quad . \tag{20.8}$$

Indeed, the only effect of including Λ is to replace $R \to R - 2\Lambda$ in the Einstein equations, so that

$$G_{\alpha\beta} = R_{\alpha\beta} - \tfrac{1}{2} g_{\alpha\beta} R \to R_{\alpha\beta} - \tfrac{1}{2} g_{\alpha\beta}(R - 2\Lambda) = G_{\alpha\beta} + \Lambda g_{\alpha\beta} \quad , \tag{20.9}$$

which gives rise to the modified Einstein equation (19.46).

2. Of course, once one is working at the level of the action, it is easy to come up with covariant generalisations of the Einstein-Hilbert action, such as

$$S = \int \sqrt{g} d^4x \; (R + c_1 R^2 + c_2 R_{\alpha\beta} R^{\alpha\beta} + c_3 R_{\alpha\beta\gamma\delta} R^{\alpha\beta\gamma\delta} + c_4 R \Box R + \dots) \quad , \tag{20.10}$$

with dimensionful coefficients c_k, but these invariably involve higher-derivative terms and/or higher non-linearities and are therefore irrelevant for low-energy physics and thus the world we live in. Such terms could be relevant for the early universe, however, and are also typically predicted by quantum theories of gravity like string theory.

3. A particular class of such higher-order actions has attracted some attention. As already briefly mentioned at the end of section 8.8, in $D > 4$ space-time dimensions there are other candidate tensors that could replace the Einstein tensor, provided that one is willing to give up linearity of the 2nd derivative terms of the metric. These tensors can be obtained from a variational principle involving very special linear combinations of higher order terms in the action, e.g. the *Gauss-Bonnet* term

$$L_{GB} = R^{\alpha\beta\gamma\delta} R_{\alpha\beta\gamma\delta} - 4 R^{\alpha\beta} R_{\alpha\beta} + R^2 \quad . \tag{20.11}$$

In $D = 4$ this term is (locally) a total derivative, and thus does not contribute to the equations of motion. It is non-trivial in $D > 4$, however, but nevertheless (a

priori totally non-obviously) leads to equations of motion that are no higher than 2nd order in derivatives.[37]

20.2 Appendix: A Formula for the Variation of the Ricci Tensor

The purpose of this technical appendix to the previous section is to derive a formula for the metric variation of the Ricci tensor which shows that indeed $g^{\alpha\beta}\delta R_{\alpha\beta}$ is a total derivative.

First of all, we need the explicit expression for the Ricci tensor in terms of the Christoffel symbols, which can be obtained by contraction of (8.5),

$$R_{\mu\nu} = \partial_\lambda \Gamma^\lambda_{\ \mu\nu} - \partial_\nu \Gamma^\lambda_{\ \mu\lambda} + \Gamma^\lambda_{\ \lambda\rho}\Gamma^\rho_{\ \nu\mu} - \Gamma^\lambda_{\ \nu\rho}\Gamma^\rho_{\ \lambda\mu} \ . \tag{20.12}$$

Now we need to calculate the variation of $R_{\mu\nu}$. We will not require the explicit expression in terms of the variations of the metric, but only in terms of the variations $\delta\Gamma^\mu_{\ \nu\lambda}$ induced by the variations of the metric. This simplifies things considerably.

Obviously, $\delta R_{\mu\nu}$ will then be a sum of six terms,

$$\delta R_{\mu\nu} = \partial_\lambda \delta\Gamma^\lambda_{\ \mu\nu} - \partial_\nu \delta\Gamma^\lambda_{\ \mu\lambda} + \delta\Gamma^\lambda_{\ \lambda\rho}\Gamma^\rho_{\ \nu\mu} + \Gamma^\lambda_{\ \lambda\rho}\delta\Gamma^\rho_{\ \nu\mu} - \delta\Gamma^\lambda_{\ \nu\rho}\Gamma^\rho_{\ \lambda\mu} - \Gamma^\lambda_{\ \nu\rho}\delta\Gamma^\rho_{\ \lambda\mu} \ . \tag{20.13}$$

Now the crucial observation is that $\delta\Gamma^\mu_{\ \nu\lambda}$ is a tensor. This follows from the arguments given in section 5.4, but I will repeat it here in the present context. Of course, we know that the Christoffel symbols themselves are not tensors, because of the inhomogeneous (second derivative) term appearing in the transformation rule under coordinate transformations, but this term is independent of the metric. Thus the metric variation of the Christoffel symbols indeed transforms as a tensor.

This can also be confirmed by explicit calculation. Just for the record, I will give an expression for $\delta\Gamma^\mu_{\ \nu\lambda}$ which is easy to remember as it takes exactly the same form as the definition of the Christoffel symbol, only with the metric replaced by the metric variation and the partial derivatives by covariant derivatives, i.e.

$$\delta\Gamma^\mu_{\ \nu\lambda} = \tfrac{1}{2} g^{\mu\rho}(\nabla_\nu \delta g_{\rho\lambda} + \nabla_\lambda \delta g_{\rho\nu} - \nabla_\rho \delta g_{\nu\lambda}) \ . \tag{20.14}$$

It turns out, none too surprisingly, that $\delta R_{\mu\nu}$ can be written rather compactly in terms of covariant derivatives of $\delta\Gamma^\mu_{\ \nu\lambda}$, namely as

$$\delta R_{\mu\nu} = \nabla_\lambda \delta\Gamma^\lambda_{\ \mu\nu} - \nabla_\nu \delta\Gamma^\lambda_{\ \mu\lambda} \ . \tag{20.15}$$

Thus one simply needs to replace the partial derivatives in (20.13) by covariant derivatives and drop the other terms that involve the undifferentiated Christoffel symbols.

[37] For a review of these so-called *Lanczos-Lovelock* models, see e.g. T. Padmanabhan, D. Kothawala, *Lanczos-Lovelock models of gravity*, `arXiv:1302.2151 [gr-qc]`.

In fact, this could not have been otherwise as (20.13) depends on the partial derivatives of the $\delta\Gamma$ but must at the same time be tensorial. The expression (20.15) is the unique possibility that fulfills these requirements. If you don't trust this argument (which essentially amounts to working at the origin of an inertial coordinate system where partial = covariant derivatives), you can also check this in detail (and thus perhaps in this way learn to trust and appreciate the quick argument):

As a first check on (20.15), note that the first term on the right hand side is manifestly symmetric and that the second term is also symmetric because of (5.49) and (6.71). To establish (20.15), one simply has to use the definition of the covariant derivative. The first term is

$$\nabla_\lambda \delta\Gamma^\lambda{}_{\mu\nu} = \partial_\lambda \delta\Gamma^\lambda{}_{\mu\nu} + \Gamma^\lambda{}_{\lambda\rho}\delta\Gamma^\rho{}_{\mu\nu} - \Gamma^\rho{}_{\mu\lambda}\delta\Gamma^\lambda{}_{\rho\nu} - \Gamma^\rho{}_{\nu\lambda}\delta\Gamma^\lambda{}_{\rho\mu} \quad , \tag{20.16}$$

which takes care of the first, fourth, fifth and sixth terms of (20.13). The remaining terms are

$$-\partial_\nu \delta\Gamma^\lambda{}_{\mu\lambda} + \delta\Gamma^\lambda{}_{\lambda\rho}\Gamma^\rho{}_{\nu\mu} = -\nabla_\nu \delta\Gamma^\lambda{}_{\mu\lambda} \quad , \tag{20.17}$$

which establishes (20.15).

What we really need is the contraction $g^{\mu\nu}\delta R_{\mu\nu}$, which we can now write as

$$\begin{aligned} g^{\mu\nu}\delta R_{\mu\nu} &= \nabla_\lambda(g^{\mu\nu}\delta\Gamma^\lambda{}_{\mu\nu}) - \nabla_\nu(g^{\mu\nu}\delta\Gamma^\lambda{}_{\lambda\mu}) \\ &= \nabla_\lambda\left(g^{\mu\nu}\delta\Gamma^\lambda{}_{\mu\nu} - g^{\mu\lambda}\delta\Gamma^\nu{}_{\nu\mu}\right) \quad . \end{aligned} \tag{20.18}$$

This establishes the claim that this term is a total derivative and hence gives rise to a boundary term in the variation of the Einstein-Hilbert action, a boundary term that does, however, require further discussion - see sections 20.4 and 20.5 below.

Using the explicit expression for $\delta\Gamma^\mu{}_{\nu\lambda}$ given above, we see that we can also write (20.18) rather neatly and explicitly as

$$\begin{aligned} g^{\mu\nu}\delta R_{\mu\nu} &= (\nabla^\mu\nabla^\nu - g^{\mu\nu}\Box)\delta g_{\mu\nu} \\ &= (g^{\mu\alpha}g^{\nu\beta} - g^{\mu\nu}g^{\alpha\beta})\nabla_\mu\nabla_\nu\delta g_{\alpha\beta} \\ &= \nabla_\lambda\left((g^{\lambda\alpha}g^{\nu\beta} - g^{\lambda\nu}g^{\alpha\beta})\nabla_\nu\delta g_{\alpha\beta}\right) \quad . \end{aligned} \tag{20.19}$$

This result will turn out to be useful on a couple of occasions later on in these notes, e.g. for the discussion of Noether currents associated to general covariance in section 20.6 and for the derivation of the energy-momentum tensor of a non-minimally coupled scalar field in section 22.3.

One can also use the identity (20.19) to rather painlessly determine the metric variation of some more complicated Lagrangians of the type (20.10). Consider e.g. the class of Lagrangians known as "$F(R)$ Lagrangians" where the Lagrangian is (none too surprisingly) some function $F(R)$ of the scalar curvature R,

$$S = \int \sqrt{g}d^4x\ F(R) \tag{20.20}$$

(for no particularly compelling reason, at least as far as I can see ("it can be done" is *not* a compelling reason ...), a lot of work has been dedicated to such Lagrangians in the last ten years, as a quick look at the `arXiv` will reveal).

The metric variation of this action is evidently

$$\delta S = \int \sqrt{g} d^4x \, \sqrt{g} \left(-\frac{1}{2} g_{\mu\nu} F(R) \delta g^{\mu\nu} + F'(R) \delta R \right) \quad . \tag{20.21}$$

Using (20.19) in the form

$$\delta R = R_{\mu\nu} \delta g^{\mu\nu} - (\nabla_\mu \nabla_\nu - g_{\mu\nu} \Box) \delta g^{\mu\nu} \quad , \tag{20.22}$$

and assuming there are no (or ignoring) boundary terms, so that we can integrate by parts the differential operator acting on $\delta g^{\mu\nu}$ and let it act on $F'(R)$ instead, one finds

$$\delta S = \int \sqrt{g} \left(-\frac{1}{2} g_{\mu\nu} F(R) + F'(R) R_{\mu\nu} - (\nabla_\mu \nabla_\nu - g_{\mu\nu} \Box) F'(R) \right) \delta g^{\mu\nu} \quad . \tag{20.23}$$

From this one can immediately read off the vacuum field equations.

One evident consequence of this is that for non-pathological choices of $F(R)$, a solution of the vacuum Einstein equations ($R_{\mu\nu} = 0, R = 0$, such as the Schwarzschild solution) will continue to be a solution of this $F(R)$-gravity theory, so that such proposed modifications of the Einstein-Hilbert action do not immediately run afoul of precision solar system tests of general relativity.

20.3 Matter Action and the Covariant Energy-Momentum Tensor

In order to obtain the non-vacuum Einstein equations, we need to decide what the matter Lagrangian should be. Now there is an obvious choice for this. If we have matter, then in addition to the Einstein equations we also want the equations of motion for the matter fields. Therefore we should add to the Einstein-Hilbert action the standard minimally coupled matter action

$$S_M[\phi, g_{\alpha\beta}] = \int \sqrt{g} d^4x \; L_M(\phi(x), \partial_\lambda \phi(x), \ldots; g_{\mu\nu}(x), \partial_\lambda g_{\mu\nu}(x), \ldots) \quad , \tag{20.24}$$

ϕ representing any kind of (scalar, vector, tensor, ...) matter field, obtained by suitable covariantisation of the corresponding matter action in Minkowski space via the principle of minimal coupling (section 5). Thus e.g. the matter action for a Klein-Gordon field would be (6.11),

$$S_M[\phi, g_{\alpha\beta}] = \int \sqrt{g} d^4x \left[-\tfrac{1}{2} g^{\alpha\beta} \partial_\alpha \phi \partial_\beta \phi - \tfrac{1}{2} m^2 \phi^2 \right] \quad , \tag{20.25}$$

and that for Maxwell theory would be (6.51),

$$S[A_\alpha, g_{\alpha\beta}] = -\tfrac{1}{4} \int \sqrt{g} d^4x \; g^{\mu\lambda} g^{\nu\rho} F_{\mu\nu} F_{\lambda\rho} \quad . \tag{20.26}$$

Of course, the variation of the matter action with respect to the matter fields will give rise to the covariant equations of motion of the matter fields. If we now want to derive the coupled gravity-matter equations from a variational principle, then the matter contribution to the gravitational field equations (i.e. the source terms for the gravitational field) will necessarily be given by the metric variation of the matter action. As already discussed in detail in section 7.6, we may as well simply define the *covariant energy-momentum tensor* to be the source of the gravitational field equations (7.105),

$$\delta_{\text{metric}} S_M[\phi, g_{\alpha\beta}] = -\tfrac{1}{2} \int \sqrt{g} d^4x \, T_{\mu\nu} \delta g^{\mu\nu} \quad \Leftrightarrow \quad T_{\mu\nu} := -\frac{2}{\sqrt{g}} \frac{\delta}{\delta g^{\mu\nu}} S_M[\phi, g_{\alpha\beta}] \ , \tag{20.27}$$

In particular, we had already seen in section 7.6 that this definition reproduces the known results in the case of a scalar field or Maxwell theory, and that in general it automatically gives a symmetric and gauge invariant tensor without the need for some improvement procedure. It is also automatically covariantly conserved on-shell as a consequence of general covariance of the matter action (cf. section 20.6).

Therefore, the complete gravity-matter action for General Relativity is

$$S[g_{\alpha\beta}, \phi] = \frac{1}{16\pi G_N} S_{EH}[g_{\alpha\beta}] + S_M[\phi, g_{\alpha\beta}] \tag{20.28}$$

with

$$\frac{\delta S[g_{\alpha\beta}, \phi]}{\delta g^{\mu\nu}} = 0 \quad \Leftrightarrow \quad G_{\mu\nu} = 8\pi G_N T_{\mu\nu} \ . \tag{20.29}$$

REMARKS:

1. If one were to try to deduce the gravitational field equations by starting from a variational principle, i.e. by constructing the simplest generally covariant action for the metric and the matter fields (and this would be the modern approach to the problem, had Einstein not already solved it for us a 100 years ago), then one would also invariably be led to the above action.

   The relative numerical factor $16\pi G_N$ between the two terms would of course then not be fixed a priori, because this approach will not (and cannot possibly be expected to) determine Newton's constant. The prefactor could once again be determined by looking at the Newtonian limit of the resulting equations of motion.

2. As we saw above, a cosmological constant term can be included by adding a constant term to the Einstein-Hilbert Lagrangian. One can equally well add a constant term to the matter Lagrangian instead (and this clearly reveals its interpretation as a constant shift of the energy, e.g. by a vacuum energy contribution, of the matter fields).

As we have seen above, the variation of the Einstein-Hilbert action leads to a boundary term that depends not just on the metric but also on the derivatives of the metric. This is related to the fact that the action itself depends also on the second derivatives of the metric. Indeed, it follows from the explicit expression for the scalar curvature in terms of the Christoffel symbols and the metric, obtained by contracting (8.5),

$$R = g^{\mu\nu}(\partial_\lambda \Gamma^\lambda{}_{\mu\nu} - \partial_\nu \Gamma^\lambda{}_{\mu\lambda} + \Gamma^\lambda{}_{\lambda\rho}\Gamma^\rho{}_{\nu\mu} - \Gamma^\lambda{}_{\nu\rho}\Gamma^\rho{}_{\lambda\mu}) \ , \tag{20.30}$$

that the Einstein-Hilbert action contains terms that are quadratic in first derivatives of the metric, as well as terms that depend linearly on the second derivatives.

In ordinary Lagrangian field theory, such linear second derivative terms can usually be introduced or eliminated (depending on what one wants to achieve) by the addition of suitable boundary terms to the action. As an example, consider the action of a free scalar field (in Minkowski space, say):

- The standard action is
$$S_0[\phi] = -\tfrac{1}{2}\int \partial^\alpha \phi \partial_\alpha \phi \ . \tag{20.31}$$
When one uses this action, the boundary term arising from the variation of the action will depend on $\delta\phi$ but not on its derivatives,
$$\delta S[\phi] = \int (\Box\phi)\delta\phi - \int \partial_\alpha(\delta\phi\partial^\alpha\phi) = \int (\Box\phi)\delta\phi - \oint_\Sigma d\sigma_\alpha(\delta\phi\partial^\alpha\phi) \ , \tag{20.32}$$
where the boundary of the integration region is denoted by Σ. This is thus the appropriate action for
$$\text{Dirichlet boundary conditions:} \quad (\delta\phi)|_\Sigma = 0 \ . \tag{20.33}$$

- One can also consider the action
$$S_1[\phi] = \tfrac{1}{2}\int \phi\Box\phi \ . \tag{20.34}$$
This action differs from S_0 by a total derivative (or boundary) term,
$$S_1[\phi] = S_0[\phi] + \tfrac{1}{2}\int \partial_\alpha(\phi\partial^\alpha\phi) = S_0[\phi] + \tfrac{1}{2}\oint d\sigma_\alpha(\phi\partial^\alpha\phi) \ . \tag{20.35}$$
It will therefore give rise to the same Euler-Lagrange bulk equations of motion. In this case, however, the boundary term arising from the variation of the action will depend both on $\delta\phi$ and its derivatives,
$$\delta S_1[\phi] = \int (\Box\phi)\delta\phi + \tfrac{1}{2}\oint d\sigma_\alpha(\phi\partial^\alpha\delta\phi - \delta\phi\partial^\alpha\phi) \ . \tag{20.36}$$
There are no obvious (or at least no obviously useful) boundary conditions compatible with this form of the action.

- Another option is the action

$$\begin{aligned} S_2[\phi] &= S_0[\phi] + \int \partial_\alpha(\phi\partial^\alpha\phi) = \tfrac{1}{2}\int \partial^\alpha\phi\partial_\alpha\phi + \int \phi\Box\phi \\ &= 2S_1[\phi] - S_0[\phi] \ . \end{aligned} \tag{20.37}$$

  Its variation is

$$\delta S_2[\phi] = \int (\Box\phi)\delta\phi + \oint d\sigma_\alpha(\phi\partial^\alpha\delta\phi) \ . \tag{20.38}$$

  In this case, the boundary term only depends on the normal derivative of the variation,

$$d\sigma_\alpha(\phi\partial^\alpha\delta\phi) \sim \phi N^\alpha\partial_\alpha\delta\phi \ , \tag{20.39}$$

  and therefore $S_2[\phi]$ is the appropriate choice of action for

$$\text{Neumann boundary conditions:} \quad (N^\alpha\partial_\alpha\delta\phi)|_\Sigma = 0 \ . \tag{20.40}$$

Let us now return to the case at hand, the action for gravity. In this case second derivative terms are required by general covariance of the action (since there is no scalar that can be constructed solely from the metric and its first derivatives). However, the fact that that the second derivatives appear linearly and that $g^{\alpha\beta}\delta R_{\alpha\beta}$ is a total derivative reflect the fact that these second derivatives are spurious in the sense that they can be eliminated by an integration by parts or by adding a suitable boundary or total derivative term, albeit at the expense of general covariance.

In fact, by straightforward manipulation of (20.30), using identities such as

$$\partial_\lambda g^{\mu\nu} = -g^{\mu\alpha}\ \partial_\lambda g_{\alpha\beta}\ g^{\beta\nu} = -(\Gamma^\mu_{\beta\lambda}g^{\beta\nu} + \Gamma^\nu{}_{\beta\lambda}g^{\beta\mu}) \tag{20.41}$$

and $\partial_\lambda(\sqrt{g}) = \sqrt{g}\ \Gamma^\mu{}_{\mu\lambda}$, one finds that the Lagrangian density $\sqrt{g}R$ can be written as

$$\sqrt{g}R = 2\sqrt{g}\ g^{\alpha\beta}(\Gamma^\mu_{\nu\alpha}\Gamma^\nu{}_{\mu\beta} - \Gamma^\mu_{\alpha\beta}\Gamma^\nu{}_{\nu\mu}) + \partial_\lambda(\sqrt{g}\ B^\lambda) \tag{20.42}$$

where

$$B^\lambda = g^{\mu\nu}\Gamma^\lambda_{\mu\nu} - g^{\mu\lambda}\Gamma^\nu{}_{\nu\mu} \ . \tag{20.43}$$

With due care, one can also write the total derivative term as

$$\partial_\lambda(\sqrt{g}\ B^\lambda) = \sqrt{g}\ \nabla_\lambda B^\lambda \ , \tag{20.44}$$

as long as one remembers that B^λ is not a tensor. Either way, we see that instead of the generally covariant Einstein-Hilbert action one can use the non-covariant but quadratic action

$$S_{EH}[g_{\alpha\beta}] \to S_E[g_{\alpha\beta}] = \int \sqrt{g}d^4x\ 2g^{\alpha\beta}(\Gamma^\mu_{\nu\alpha}\Gamma^\nu{}_{\mu\beta} - \Gamma^\mu_{\alpha\beta}\Gamma^\nu{}_{\nu\mu}) \ . \tag{20.45}$$

This action was originally considered by Einstein himself and is therefore also known as the *Einstein action*.

It will be useful for the following to write this in the form

$$S_E[g_{\alpha\beta}] = S_{EH}[g_{\alpha\beta}] - \int \sqrt{g} d^4x \; \nabla_\lambda B^\lambda = S_{EH}[g_{\alpha\beta}] - \oint d\sigma_\lambda B^\lambda \quad , \tag{20.46}$$

in which the non-covariant terms are now manifestly confined to the boundary of the region of integration. This is now a reasonably respectable action, but there is a more attractive variant of this construction which we will discuss in the next section.

20.5 Gibbons-Hawking-York Boundary Term

In section 20.2 we have seen that the metric variation of the Einstein-Hilbert action has the form

$$\delta S_{EH}[g_{\alpha\beta}] = \int \sqrt{g} d^4x \; G_{\alpha\beta} \delta g^{\alpha\beta} + \int \sqrt{g} d^4x \; \nabla_\lambda (\Delta B)^\lambda \quad , \tag{20.47}$$

where

$$\begin{aligned} (\Delta B)^\lambda &= g^{\mu\nu} \delta \Gamma^\lambda_{\mu\nu} - g^{\mu\lambda} \delta \Gamma^\nu_{\nu\mu} \\ &= (g^{\lambda\alpha} g^{\nu\beta} - g^{\lambda\nu} g^{\alpha\beta}) \nabla_\nu \delta g_{\alpha\beta} \quad . \end{aligned} \tag{20.48}$$

The reason for this notation, and the relation between this object $(\Delta B)^\lambda$ and the quantity B^λ introduced in (20.43) will be explained below. The first (bulk) term gives us as the Euler-Lagrange equations the vacuum Einstein equations $G_{\alpha\beta} = 0$, while the second term is a total derivative.

Thus, when one performs the integral over a space-time region $\mathcal{V}$ bounded by the hypersurface $\partial\mathcal{V} = \Sigma$ (which we shall assume to be spacelike or timelike), upon use of the Gauss integral formula (16.47) one finds that the second (total derivative) term can be written as

$$\begin{aligned} \int_{\mathcal{V}} \sqrt{g} d^4x \; \nabla_\lambda (\Delta B)^\lambda &= \oint_\Sigma d\sigma_\lambda (\Delta B)^\lambda \\ &= \epsilon \oint_\Sigma d^n y \; \sqrt{h} \; N_\lambda (\Delta B)^\lambda \quad , \end{aligned} \tag{20.49}$$

where N_λ is the normal vector to the boundary Σ in $\mathcal{V}$ and h_{ab} is the induced metric on the boundary. Explicitly the boundary integrand is

$$N_\lambda (\Delta B)^\lambda = (N^\rho g^{\mu\nu} - N^\mu g^{\rho\nu}) \nabla_\mu \delta g_{\rho\nu} \quad . \tag{20.50}$$

Using the decomposition

$$g^{\mu\nu} = h^{\mu\nu} + \epsilon N^\mu N^\nu \tag{20.51}$$

of the metric on Σ (with $N^\mu N_\mu = \epsilon$), one sees that the terms with 3 N's cancel, and one is left with

$$N_\lambda (\Delta B)^\lambda = N^\rho h^{\mu\nu} \nabla_\mu \delta g_{\rho\nu} - N^\mu h^{\rho\nu} \nabla_\mu \delta g_{\rho\nu} \quad . \tag{20.52}$$

The first term only depends on the variations $\delta g_{\rho\nu}$ on the boundary, and its tangential derivatives $h^{\mu\nu} \nabla_\mu \delta g_{\rho\nu}$. Therefore that term is zero if one imposes standard Dirichlet boundary conditions

$$\delta g_{\alpha\beta}|_\Sigma = 0 \tag{20.53}$$

on the metric at Σ. The second term, on the other hand, depends on $\delta g_{\rho\nu}$ and its normal derivative $N^\mu \nabla_\mu \delta g_{\rho\nu}$, and is therefore non-zero,

$$\delta g_{\alpha\beta}|_\Sigma = 0 \quad \Rightarrow \quad N_\lambda (\Delta B)^\lambda|_\Sigma = -N^\mu h^{\rho\nu} \nabla_\mu \delta g_{\rho\nu}|_\Sigma = -h^{\rho\nu} N^\mu \partial_\mu \delta g_{\rho\nu}|_\Sigma \ . \tag{20.54}$$

Therefore with Dirichlet boundary conditions the variation of the Einstein-Hilbert action gives rise to a non-zero boundary term,

$$\delta S_{EH}[g_{\alpha\beta}] = \int \sqrt{g} d^4x \, G_{\alpha\beta} \delta g^{\alpha\beta} + \epsilon \oint_\Sigma \sqrt{h} d^3y \, (-h^{\rho\nu} N^\mu \partial_\mu \delta g_{\rho\nu}) \ . \tag{20.55}$$

It is therefore not true that the variation of the Einstein-Hilbert action is the Einstein tensor,

$$\frac{\delta}{\delta g^{\alpha\beta}} S_{EH} \neq \sqrt{g} G_{\alpha\beta} \ . \tag{20.56}$$

In fact, the presence of this boundary term means that the functional $S_{EH}[g_{\alpha\beta}]$ is not even differentiable (in the sense of variational calculus).

The way to resolve this issue is, as for the scalar field discussed above, to add a suitable boundary term to the action itself. This will not change the bulk variation, and it turns out that e.g. for Dirichlet boundary conditions the boundary term can be chosen in such a way that its variation cancels the boundary term above.

Actually, we already have one candidate for the boundary term, namely the one relating the Einstein and Einstein-Hilbert actions in (20.46). The variation of the Einstein action is

$$\begin{aligned} \delta S_E[g_{\alpha\beta}] &= \delta S_{EH}[g_{\alpha\beta}] - \delta \oint d\sigma_\lambda B^\lambda \\ &= \int \sqrt{g} d^4x \, G_{\alpha\beta} \delta g^{\alpha\beta} + \oint_\Sigma d^n y \, \sqrt{h} \, \epsilon N_\lambda (\Delta B)^\lambda - \delta \Big(\oint_\Sigma d^n y \, \sqrt{h} \, \epsilon N_\lambda B^\lambda \Big) \ . \end{aligned} \tag{20.57}$$

Calculating the variation in the second term for Dirichlet boundary conditions on Σ, one finds

$$\delta \Big(\oint_\Sigma d^n y \, \sqrt{h} \, \epsilon N_\lambda B^\lambda \Big) = \oint_\Sigma d^n y \, \sqrt{h} \, \epsilon N_\lambda \delta B^\lambda \ , \tag{20.58}$$

with

$$\delta B^\lambda = \delta (g^{\mu\nu} \Gamma^\lambda_{\mu\nu} - g^{\mu\lambda} \Gamma^\nu{}_{\nu\mu}) \quad = g^{\mu\nu} \delta \Gamma^\lambda_{\mu\nu} - g^{\mu\lambda} \delta \Gamma^\nu{}_{\nu\mu} = (\Delta B)^\lambda \ . \tag{20.59}$$

Thus for Dirichlet boundary conditions, the variation of B^λ on the boundary equals the quantity $(\Delta B)^\lambda$ (20.48) arising from the variation of the Einstein-Hilbert action. As a consequence, there is no boundary term in the variation of the Einstein action for Dirichlet boundary conditions,

$$\delta S_E[g_{\alpha\beta}] = \int \sqrt{g} d^4x \, G_{\alpha\beta} \delta g^{\alpha\beta} \tag{20.60}$$

and the Einstein action leads to a well-defined variational principle (with a differentiable action).

Although this is progress, the boundary term that one adds to the Einstein-Hilbert action to obtain the Einstein action is not particularly attractive, in particular as it is non-covariant not only with respect to bulk coordinate transformations but also with respect to boundary coordinate transformations (i.e. the integrand is not a Σ-scalar in the terminology of section 15.1).

This is something one would have to live with if one could not do better. However, such a boundary term achieving this is not unique as it is evidently only defined up to terms whose variations vanish for Dirichlet boundary conditions, in particular up to terms that only depend on the intrinsic geometry of Σ. Among these candidates there is a preferred, geometrically transparent, boundary term, the *Gibbons-Hawking-York* boundary term,

$$S_{GHY}[g_{\alpha\beta}] = 2\epsilon \oint_\Sigma \sqrt{h} d^3y \; K \quad . \tag{20.61}$$

Here

$$\begin{aligned} K &= h^{ab}K_{ab} = h^{\alpha\beta}\nabla_\alpha N_\beta \\ &= (g^{\alpha\beta} + N^\alpha N^\beta)\nabla_\alpha N_\beta \\ &= g^{\alpha\beta}\nabla_\alpha N_\beta = \nabla_\alpha N^\alpha \end{aligned} \tag{20.62}$$

is the trace of the extrinsic curvature K_{ab} of Σ (discussed in section 18).

One way to prove that this is a good boundary term is to determine its variation and to show that it cancels against the variation arising from the Einstein-Hilbert action for Dirichlet boundary conditions. Alternatively one should show that the above boundary term differs from the boundary term for the Einstein action only by expressions that depend on the metric $g_{\alpha\beta}$ and its tangential derivatives.

Since usually one does the former, let us do the latter. The difference between the two boundary integrands is

$$2K + N_\lambda B^\lambda = 2\nabla_\lambda N^\lambda + N_\lambda B^\lambda \quad . \tag{20.63}$$

A calculation identical to the one leading to $N_\lambda(\Delta B)^\lambda$ in (20.52) shows that

$$N_\lambda B^\lambda = N^\rho h^{\mu\nu}\partial_\mu g_{\rho\nu} - N^\mu h^{\rho\nu}\partial_\mu g_{\rho\nu} \quad . \tag{20.64}$$

Here the first term only depends on the metric and its tangential derivatives, while the second term involves normal derivatives of the metric. On the other hand, for K we have

$$2K = h^{\nu\rho}L_N g_{\nu\rho} = h^{\nu\rho}N^\mu\partial_\mu g_{\nu\rho} + 2h^\mu_{\ \nu}\partial_\mu N^\nu \quad , \tag{20.65}$$

where we have used the non-covariant way (9.39) of writing the Lie derivative,

$$L_N g_{\mu\nu} = N^\lambda\partial_\lambda g_{\mu\nu} + \partial_\mu N^\lambda g_{\lambda\nu} + \partial_\nu N^\lambda g_{\mu\lambda} \quad . \tag{20.66}$$

Thus the first term cancels the normal derivative term in (20.64) and the remaining terms in (20.64) and (20.65) only involve fields $g_{\alpha\beta}$, i.e. $h_{\alpha\beta}$ and N_α, and their tangential derivatives that are fixed on the boundary.

This establishes that the Gibbons-Hawking-York boundary term is an acceptable choice of boundary term. This is also the standard choice, and gives rise to the standard gravitational action

$$\begin{aligned} S_g[g_{\alpha\beta}] &= S_{EH}[g_{\alpha\beta}] + S_{GHY}[g_{\alpha\beta}] \\ &= \int \sqrt{g} d^4x \; R + 2\epsilon \oint_\Sigma \sqrt{h} d^3y \; K \quad , \end{aligned} \tag{20.67}$$

expressed purely in terms of the intrinsic and extrinsic scalar curvatures R and K, with

$$\delta S_g[g_{\alpha\beta}] = \int \sqrt{g} d^4x \; G_{\alpha\beta} \delta g^{\alpha\beta} \tag{20.68}$$

for Dirichlet boundary conditions.

Remarks:

1. In addition, one can add terms to the action that do not depend on the dynamical fields (as this will certainly change neither the variation with respect to the dynamical fields nor the equations of motion). A common choice is a kind of "background subtraction", designed to associate the numerical value $S = 0$ to a particular background metric $g^0_{\alpha\beta}$.[38] Thus one could define the "physical" action to be

$$S[g_{\alpha\beta}] = S_g[g_{\alpha\beta}] - S_g[g^0_{\alpha\beta}] \quad . \tag{20.69}$$

   In particular, if one is interested in asymptotically flat space-times, say, then the appropriate reference background metric is just the flat Minkowski metric, and then (20.69) takes the simple form

$$S[g_{\alpha\beta}] = \int \sqrt{g} d^4x \; R + 2\epsilon \oint_\Sigma \sqrt{h} d^3y \; (K - K^0) \quad , \tag{20.70}$$

   where K^0 is the trace of the extrinsic curvature of the boundary (isometrically) embedded into Minkowski space. In section 21.12 we will be led to a similar subtraction prescription at the level of the Hamiltonian.

2. Another way to motivate (or arrive at) the Gibbons-Hawking-York boundary term is to start from the decomposition (18.73)

$$R = \bar{R} + \epsilon(K^2 - K^{\alpha\beta}K_{\alpha\beta}) + 2\epsilon\nabla_\alpha(N^\beta \nabla_\beta N^\alpha - N^\alpha \nabla_\beta N^\beta) \quad . \tag{20.71}$$

   of the Ricci scalar provided by the Gauss-Codazzi equations. This turns out to be a convenient starting point for the canonical (Hamiltonian) formulation of general relativity, and we will therefore discuss this in that context in section 21.2.

[38] See e.g. S. Hawking, G. Horowitz, *The Gravitational Hamiltonian, Action, Entropy, and Surface Terms*, `arXiv:gr-qc/9501014`, for a brief discussion.

20.6 General Covariance and Noether Identities

The variational (i.e. action or Lagrangian based) formulation of general relativity has a number of significant conceptual and technical advantages, and we will explore some of them in this section.

- I mentioned before, in section 19.7, that it is no accident that the Bianchi identities come to the rescue of the general covariance of the Einstein equations in the sense that they reduce the number of independent equations from ten to six. We will now see that indeed the Bianchi identities are a consequence of the general covariance of the Einstein-Hilbert action.
- Virtually the same calculation will show that the covariant (metric, Hilbert) energy-momentum tensor, as defined above, is automatically conserved (on shell) by virtue of the general covariance of the matter action.
- This argument can also be turned around to show that (generically, at least) the Einstein equations imply the matter equations of motion, a very characteristic feature of generally covariant gravitational field equations.
- Simple variants of these arguments will also provide us with the Noether currents associated with the general covariance of the Einstein-Hilbert action.
- Analogous considerations, but now applied to the minimally coupled generally covariant matter action, will provide us with some insight into the relation between the (Belinfante improved) canonical and covariant energy-momentum tensors introduced in section 7.

To set the stage, we need to discuss how to express general covariance of an action, either of the Einstein-Hilbert gravitational action $S_{EH}[g_{\alpha\beta}]$ or of some (minimally) gravitationally coupled general covariant matter action $S_M[\phi, g_{\alpha\beta}]$, in a form that allows us to explore its consequences in a Lagrangian formalism.

At first, the statement that an action is generally covariant means that it is invariant under transformations $x^\alpha \to x'^\alpha$ of the coordinates (for present purposes it will for once be more convenient to use the same indices on the old and new coordinates and to distinguish transformed objects by primes), and the accompanying tensorial transformation of the fields, given e.g. by

$$\begin{aligned} x \to x' \quad \Rightarrow \quad & \phi(x) \to \phi'(x') = \phi(x) \\ & g_{\alpha\beta}(x) \to g'_{\alpha\beta}(x') = \frac{\partial x^\gamma}{\partial x'^\alpha} \frac{\partial x^\delta}{\partial x'^\beta}\, g_{\gamma\delta}(x) \quad . \end{aligned} \tag{20.72}$$

So far, this is utterly familiar, but since in the action one is integrating over the coordinates x^α, say, we would like to express this invariance not as a statement between

transformed fields at x' and the old fields at x but in terms of transformations of the fields at a point x (which we can then plug into the Lagrangian or action). This means that we do not want to consider the transformation $\phi(x) \to \phi'(x')$ (and its counterpart for the metric) but rather the transformations

$$\phi(x) \to \phi'(x) \quad , \quad g_{\alpha\beta}(x) \to g'_{\alpha\beta}(x) \ . \tag{20.73}$$

In particular, for infinitesimal coordinate transformations of the form

$$x'^{\alpha} = x^{\alpha} + \epsilon\xi^{\alpha}(x) \tag{20.74}$$

or (suppressing the ϵ)

$$\delta_{\xi} x^{\alpha} = \xi^{\alpha}(x) \ , \tag{20.75}$$

the infinitesimal variations of the fields are then precisely the Lie derivatives of the fields discussed in section 9,

$$\delta_{\xi}\phi(x) = L_{\xi}\phi(x) = \xi^{\alpha}\partial_{\alpha}\phi(x) \tag{20.76}$$

for a scalar field,

$$\delta_{\xi} g_{\alpha\beta}(x) = L_{\xi} g_{\alpha\beta}(x) = \nabla_{\alpha}\xi_{\beta}(x) + \nabla_{\beta}\xi_{\alpha}(x) \ , \tag{20.77}$$

for the metric, etc.

As a reminder, a quick way to derive this transformation of the metric is to start with the tensorial transformation behaviour in the form

$$(g_{\alpha\beta}(x') - g'_{\alpha\beta}(x'))dx'^{\alpha}dx'^{\beta} = g_{\alpha\beta}(x')dx'^{\alpha}dx'^{\beta} - g_{\alpha\beta}(x)dx^{\alpha}dx^{\beta} \ , \tag{20.78}$$

and to then apply this to the infinitesimal transformation (20.74). Expanding the differentials

$$dx'^{\alpha} = dx^{\alpha} + \epsilon(\partial_{\gamma}\xi^{\alpha})dx^{\gamma} \tag{20.79}$$

and the components $g_{\alpha\beta}(x')$ of the metric to first order in ϵ,

$$g_{\alpha\beta}(x') = g_{\alpha\beta}(x) + \epsilon\xi^{\gamma}\partial_{\gamma}g_{\alpha\beta}(x) \ , \tag{20.80}$$

one finds (20.77) (in its not manifestly covariant form (9.39)).

To see that this indeed leads to a symmetry for any generally covariant action, i.e. any action of the form

$$S = \int \sqrt{g}d^4x \ L(x) \tag{20.81}$$

where L is a scalar, note that for any density $\sqrt{g}F$, F a scalar, one has, by the by now familiar identity for the variation of $\sqrt{g}$,

$$\begin{aligned} \delta_{\xi}(\sqrt{g}F) &= (\delta_{\xi}\sqrt{g})F + \sqrt{g}\delta_{\xi}F = \tfrac{1}{2}\sqrt{g}g^{\alpha\beta}(L_{\xi}g_{\alpha\beta})F + \sqrt{g}L_{\xi}F \\ &= \sqrt{g}(\nabla_{\alpha}\xi^{\alpha})F + \sqrt{g}\xi^{\alpha}\partial_{\alpha}F = \partial_{\alpha}(\sqrt{g}\ \xi^{\alpha}F) \ , \end{aligned} \tag{20.82}$$

i.e.

$$\delta_\xi(\sqrt{g}F) = \partial_\alpha(\sqrt{g}\,\xi^\alpha F) = \sqrt{g}\,\nabla_\alpha(\xi^\alpha F) \ , \tag{20.83}$$

a result previously obtained in (9.69). Thus the variation of a generally covariant action,

$$\delta_\xi S = \int \sqrt{g}d^4x\,\nabla_\alpha(\xi^\alpha L) \ , \tag{20.84}$$

is a total derivative and $\delta_\xi = L_\xi$ certainly generates a symmetry in the usual Lagrangian / Noether sense. On the other hand, $\delta_\xi(\sqrt{g}L)$ can also be expressed in terms of the δ_ξ-variations of the fields, and combining this with the invariance of the action (up to boundary terms), one can now deduce the consequences of general covariance for a given theory, either for ξ that vanish at the boundary of the integration region or, more generally, for ξ that are non-trivial there. We will now discuss these possibilities in turn for the Einstein-Hilbert and matter actions.

1. Contracted Bianchi Identities

   As our first application we consider the Einstein-Hilbert action, and its associated invariance under all infinitesimal coordinate transformations generated by ξ that vanish on the boundary. In that case, from the considerations above we know that the Einstein-Hilbert action is strictly invariant,

   $$\delta_\xi S_{EH} = 0 \ . \tag{20.85}$$

   On the other hand we also know that, modulo boundary terms,

   $$\delta S_{EH} = \int \sqrt{g}d^4x\,G_{\alpha\beta}\delta g^{\alpha\beta} = -\int \sqrt{g}d^4x\,G^{\alpha\beta}\delta g_{\alpha\beta} \tag{20.86}$$

   for any metric variation. Combining these two facts we arrive at the conclusion that

   $$\begin{aligned} 0 = \delta_\xi S_{EH} &= -\int \sqrt{g}d^4x\,G^{\alpha\beta}\delta_\xi g_{\alpha\beta} \\ &= -2\int \sqrt{g}d^4x\,G^{\alpha\beta}\nabla_\alpha\xi_\beta \\ &= +2\int \sqrt{g}d^4x\,(\nabla_\alpha G^{\alpha\beta})\xi_\beta \ . \end{aligned} \tag{20.87}$$

   Since this has to hold for all ξ (vanishing on the boundary), we deduce

   $$\delta_\xi S_{EH} = 0 \quad \forall\,\xi \quad \Rightarrow \quad \nabla^\alpha G_{\alpha\beta} = 0 \ . \tag{20.88}$$

   As promised and anticipated, the contracted Bianchi identities are a consequence of the general covariance of the Einstein-Hilbert action.

   To further strengthen the analogy with the gauge invariance of Maxwell theory emphasised in the discussion in section 19.7, note that also the relevant "contracted Bianchi" identity $\partial_a\partial_b F^{ab} = 0$ can be derived from gauge invariance of the Maxwell

action (although this is of course not the most economical way of arriving at this identity in the case at hand).

Indeed, the general variation of the Maxwell action (for variations vanishing on the boundary) is

$$\delta S[A] = \int d^4x (\partial_a F^{ab}) \delta A_b \ . \tag{20.89}$$

The action is invariant under gauge transformations

$$\delta_\Psi A_a = \partial_a \Psi \ , \tag{20.90}$$

i.e. one has

$$0 = \delta_\Psi S[A] = \int d^4x (\partial_a F^{ab}) \partial_b \Psi = \int d^4x (\partial_a \partial_b F^{ab}) \Psi \tag{20.91}$$

(for any Ψ vanishing at the boundary), and thus $\partial_a \partial_b F^{ab} = 0$.

2. Identically Conserved Noether Currents

The fact that one obtains kinematical identities rather than non-trivially conserved currents (non-trivial in the sense that their conservation requires the validity of some dynamical equations of motion) is a characteristic feature of Noether's theorem applied to local (gauge) symmetries. We can also see this when we consider general vector fields ξ (not constrained by the requirement that they give rise to vanishing boundary terms for the given integration domain). In that case, $\delta_\xi S_{EH}$ will not be identically zero but will be a total derivative, and also the corresponding Noether currents will turn out to be identically conserved.

Thus we now consider again the Einstein-Hilbert action, but now with ξs that are allowed to be non-zero on the boundary,

$$\delta_\xi S_{EH} = \int \sqrt{g} d^4x \ \nabla_\alpha (\xi^\alpha R) \ . \tag{20.92}$$

By explicitly performing this variation, as above, the bulk (Einstein tensor) term is identically zero by the contracted Bianchi identity, but we obtain one total derivative term from (20.19),

$$\begin{aligned} g^{\alpha\beta} \delta_\xi R_{\alpha\beta} &= \nabla_\mu [(g^{\mu\alpha} g^{\nu\beta} - g^{\mu\nu} g^{\alpha\beta}) \nabla_\nu L_\xi g_{\alpha\beta}] \\ &= \nabla_\mu [(g^{\mu\alpha} g^{\nu\beta} - g^{\mu\nu} g^{\alpha\beta}) \nabla_\nu (\nabla_\alpha \xi_\beta + \nabla_\beta \xi_\alpha)] \ , \end{aligned} \tag{20.93}$$

and a second total derivative term

$$\nabla_\mu (-2 G^\mu_{\ \nu} \xi^\nu) = \nabla_\mu (-2 R^\mu_{\ \nu} \xi^\nu + \xi^\mu R) \tag{20.94}$$

from the integration by parts performed in the course of the calculation in (20.87). The term involving the scalar curvature is identical to, and cancels against, the

scalar curvature term arising from (20.92). One is thus left with the statement that for any vector field ξ^μ and any integration domain one has

$$\int \sqrt{g} d^4x \nabla_\mu [-2R^\mu_\nu \xi^\nu + (g^{\mu\alpha} g^{\nu\beta} - g^{\mu\nu} g^{\alpha\beta}) \nabla_\nu (\nabla_\alpha \xi_\beta + \nabla_\beta \xi_\alpha)] = 0 \ . \qquad (20.95)$$

Thus for any ξ^μ one has the conserved Noether current

$$J^\mu(\xi) = R^\mu_\nu \xi^\nu + \tfrac{1}{2}(g^{\mu\nu} g^{\alpha\beta} - g^{\mu\alpha} g^{\nu\beta}) \nabla_\nu (\nabla_\alpha \xi_\beta + \nabla_\beta \xi_\alpha) \ . \qquad (20.96)$$

This is precisely the identically conserved current (13.51), (13.53)

$$J^\mu(\xi) = \nabla_\nu (\nabla^{[\mu} \xi^{\nu]}) \quad \Rightarrow \quad \nabla_\mu J^\mu(\xi) = 0 \qquad (20.97)$$

already mentioned in section 13.7. We thus learn that the generalised Komar currents of that section are indeed, as anticipated there, precisely the identically conserved Noether currents associated to the general covariance of the Einstein-Hilbert action.

Note that, as mentioned above, it is a general feature of Noether currents associated to local (gauge) symmetries that they are in fact identically conserved: the current $J^\mu(\xi)$ (or its conterpart for some local symmetry of another theory) cannot possibly be conserved for all possible $\xi^\mu(x)$ unless it is actually identically conserved.[39] In particular, this implies that the conserved charges associated with these currents can always be expressed as surface integrals.

3. On-Shell Covariant Conservation of the Energy-Momentum Tensor

Now let us play the same game with the matter action S_M (20.24). Once again, the variation $\delta_\xi S_M$, expressed in terms of the Lie derivatives $L_\xi g_{\mu\nu}$ and $\delta_\xi \phi = L_\xi \phi$ of the matter fields should be identically zero, by general covariance of the matter action (for the time being we again at first only consider ξ which are such that any boundary terms vanish). Proceeding as before, and using the definition (7.105) of the energy-momentum tensor, we find

$$\begin{aligned} 0 &= \delta_\xi S_M \\ &= \int \sqrt{g} d^4x \, (-\tfrac{1}{2} T_{\mu\nu} \delta_\xi g^{\mu\nu} + \frac{\delta L_M}{\delta \phi} \delta_\xi \phi) \\ &= -\int \sqrt{g} d^4x \, (\nabla^\mu T_{\mu\nu}) \xi^\nu + \int \sqrt{g} d^4x \, \frac{\delta L_M}{\delta \phi} \delta_\xi \phi \ . \end{aligned} \qquad (20.98)$$

[39] See e.g. B. Julia, S. Silva, *Currents and Superpotentials in classical gauge invariant theories I: Local results with applications to Perfect Fluids and General Relativity*, `arXiv:gr-qc/9804029` for a rather explicit elementary argument, and R. Wald, *On identically closed forms locally constructed from a field*, J. Math. Phys. 31 (1990) 2378, R. Wald, *Black Hole Entropy is Noether Charge*, `arXiv:gr-qc/9307038`, V. Iyer and R. Wald, *Some properties of Noether charge and a proposal for dynamical black hole entropy*, `arXiv:gr-qc/9403028` and references therein for related considerations and further developments.

Now once again this has to hold for all ξ, and as the second term is identically zero 'on-shell', i.e. for ϕ satisfying the matter Euler-Lagrange equations of motion $\delta L_M/\delta\phi = 0$, we deduce that

$$\delta_\xi S_M = 0 \quad \forall\, \xi \quad \Rightarrow \quad \nabla^\mu T_{\mu\nu} = 0 \qquad \text{on-shell} \ . \tag{20.99}$$

This should be contrasted with the contracted Bianchi identities which are valid 'off-shell'. The more general situation with the ξ not restricted to vanish on the boundary will be analysed in detail in section 22.2.

4. Einstein Equations Imply (generically) the Matter Equations of Motion

   Note that, to a certain extent, this argument can also be turned around to show that *the equations of motion of the gravitational field generated by some matter fields imply the equations of motion of these matter fields*!

   Indeed, we already know that the Einstein equations imply covariant conservation of the energy-momentum tensor,

$$G_{\alpha\beta} = 8\pi G_N\, T_{\alpha\beta} \quad \Rightarrow \quad \nabla_\alpha T^{\alpha\beta} = 0 \ . \tag{20.100}$$

   Hence, by the above reasoning, the Einstein equations imply

$$G_{\alpha\beta} = 8\pi G_N\, T_{\alpha\beta} \quad \Rightarrow \quad \int \sqrt{g} d^4x\, \frac{\delta L_M}{\delta\phi}\delta_\xi\phi = 0 \tag{20.101}$$

   for all compactly supported ξ. Excluding certain non-generic cases (like for example a constant scalar field for which $\delta_\xi\phi = 0$), one sees that

$$G_{\alpha\beta} = 8\pi G_N\, T_{\alpha\beta} \quad \overset{\text{generically}}{\Longrightarrow} \quad \frac{\delta L_M}{\delta\phi} = 0 \ . \tag{20.102}$$

   This should be contrasted with the Maxwell equations in the presence of (charged) matter fields, say, which only imply current conservation,

$$\partial_\mu F^{\mu\nu} \sim j^\nu \quad \Rightarrow \quad \partial_\nu j^\nu = 0 \ , \tag{20.103}$$

   but not the complete equations of motion of the matter fields.

   To explain what is special about the generally covariant gravitational field equations in this respect, I will conclude this section with a quote by Misner, Thorne and Wheeler, since I could not possibly state this more eloquently:

> The Maxwell equations are so constructed that they automatically fulfill and demand the conservation of charge; but not everything has charge. The Einstein field equation is so constructed that it automatically fulfills and demands the conservation of momentum-energy; and everything does have energy. The Maxwell field equations are indifferent to the interposition of an "external" force, because that force in no way threatens

the principle of conservation of charge. The Einstein field equation cares about every force, because every force is a medium for the exchange of energy.

Electromagnetism has the motto, "I count all the electric charge that's here." All that bears no charge escapes its gaze.

"I weigh all that's here" is the motto of spacetime curvature. No physical entity escapes this surveillance.[40]

20.7 First Order Form of the Action, Torsion and the Palatini Principle

For certain purposes (e.g. as a precursor to a Hamiltonian formulation) it can be useful to put an action leading to 2nd order differential equations into "1st order form" by the introduction of some auxiliarly variables. The prototypical example is Maxwell theory, whose original action (cf. section 6.5)

$$S[A] = -\tfrac{1}{4}\int d^4\xi \; F_{ab}F^{ab} \tag{20.104}$$

depends quadratically on the 1st derivatives of the gauge field A_a, and leads to the 2nd order equations of motion

$$\partial_a F^{ab} = 0 \ . \tag{20.105}$$

To put this into 1st order form, one treats A_a and $F_{ab} \to \mathcal{F}_{ab}$ as a priori *independent* fields and considers the action

$$\begin{aligned} S[A,\mathcal{F}] &= \int d^4\xi \; \left(-(\partial_a A_b)\mathcal{F}^{ab} + \tfrac{1}{4}\mathcal{F}_{ab}\mathcal{F}^{ab}\right) \\ &= \int d^4\xi \; \left(-\tfrac{1}{2}(\partial_a A_b - \partial_b A_a)\mathcal{F}^{ab} + \tfrac{1}{4}\mathcal{F}_{ab}\mathcal{F}^{ab}\right) \ , \end{aligned} \tag{20.106}$$

which depends purely algebraically on $\mathcal{F}_{ab}$ and only linearly on the 1st derivatives of A_a. The equations of motion arising from the variation of A_a are the 1st order equations

$$\delta A \quad \Rightarrow \quad \partial_a \mathcal{F}^{ab} = 0 \ . \tag{20.107}$$

However these are not (yet) the vacuum Maxwell equations because $\mathcal{F}_{ab}$ is not (yet) to be identified with the Maxwell field stength tensor of A_a. This identification now results from the (purely algebraic) equation of motion associated with variations of $\mathcal{F}_{ab}$,

$$\delta\mathcal{F} \quad \Rightarrow \quad \mathcal{F}_{ab} = \partial_a A_b - \partial_b A_a = F_{ab} \ . \tag{20.108}$$

Plugging this result into the previous equations then gives rise to the standard Maxwell equations (and plugging it into the 1st order action $S[A,\mathcal{F}]$ reduces it to the standard Maxwell action $S[A]$).

[40] C. Misner, K. Thorne, J. Wheeler, *Gravitation*, section 20.6.

Something similar (but more interesting and somewhat more subtle) can be done in the case of general relativity.

Recall from sections 5.4 and 11.5 that a priori a metric $g_{\mu\nu}$ and a connection $\tilde{\Gamma}^{\lambda}_{\ \mu\nu}$ are independent concepts, and that the notion of curvature (curvature and Ricci tensors) can be defined for an arbitrary connection,

$$R^{\lambda}_{\ \sigma\mu\nu}(\tilde{\Gamma}) = \partial_\mu \tilde{\Gamma}^{\lambda}_{\ \sigma\nu} - \partial_\nu \tilde{\Gamma}^{\lambda}_{\ \sigma\mu} + \tilde{\Gamma}^{\lambda}_{\ \mu\rho}\tilde{\Gamma}^{\rho}_{\ \nu\sigma} - \tilde{\Gamma}^{\lambda}_{\ \nu\rho}\tilde{\Gamma}^{\rho}_{\ \mu\sigma} \quad , \quad R_{\mu\nu}(\tilde{\Gamma}) = R^{\lambda}_{\ \mu\lambda\nu}(\tilde{\Gamma}) \ . \tag{20.109}$$

General relativity employs and is formulated in terms of the canonical Levi-Civita connection described by the Christoffel symbols $\tilde{\Gamma}^{\lambda}_{\ \mu\nu} = \Gamma^{\lambda}_{\ \mu\nu}$, characterised by the fact that the connection is compatible with the metric and has no torsion. It is thus easy to come up with various generalisations of general relativity in which these requirements are relaxed. We will not get into these matters here.[41]

However, it is a curious, and occasionally calculationally or conceptually useful, fact that it is possible to relax somewhat the a priori identification of the connection with the Levi-Civita connection and nevertheless reproduce general relativity by treating the connection and metric as independent variables.

Specifically, we will consider an action of the generalised Einstein-Hilbert-like form

$$S[g_{\mu\nu}, \tilde{\Gamma}^{\lambda}_{\ \mu\nu}] = \int \sqrt{g} d^4x \ R(\tilde{\Gamma}) = \int \sqrt{g} d^4x \ g^{\mu\nu} R_{\mu\nu}(\tilde{\Gamma}) \tag{20.110}$$

for a (yet to be specified) class of connections $\tilde{\Gamma}^{\lambda}_{\ \mu\nu}$, with $g_{\mu\nu}$ and $\tilde{\Gamma}^{\lambda}_{\ \mu\nu}$ to be treated as independent variables. Since $R_{\mu\nu}(\tilde{\Gamma})$ does then not depend on the metric, the action depends purely algebraically on the metric, and on at most 1st derivatives of the connection (linearly!).

One key simplification of this kind of action is that the variation with respect to the metric is elementary (and identical to the variation of the $\sqrt{g}g^{\mu\nu}$ terms of the Einstein-Hilbert Lagrangian density $\sqrt{g}g^{\mu\nu}R_{\mu\nu}$), namely

$$\delta_g S[g_{\mu\nu}, \tilde{\Gamma}^{\lambda}_{\ \mu\nu}] = \int \sqrt{g} d^4x \ \left(R_{\mu\nu}(\tilde{\Gamma}) - \tfrac{1}{2} g_{\mu\nu} R(\tilde{\Gamma}) \right) \delta g^{\mu\nu} \tag{20.111}$$

(no integration by parts or identification of total derivative terms required). Thus, in the absence of the coupling of the metric (or of gravity) to other fields we find the equations of motion

$$G_{\mu\nu}(\tilde{\Gamma}) \equiv R_{\mu\nu}(\tilde{\Gamma}) - \tfrac{1}{2} g_{\mu\nu} R(\tilde{\Gamma}) = 0 \ . \tag{20.112}$$

These are, however, not yet the vacuum Einstein equations because the independent connection $\tilde{\Gamma}$ is not the Levi-Civita connection.

[41] Many of these generalisations (including theories with non-symmetric metrics) were originally explored by Einstein and his collaborators in their futile and (at least by the 1930s) ill-motivated attempts to find a unified field theory of gravity and Maxwell theory. For details, see e.g. the review H. Goenner, *On the History of Unified Field Theories*, Living Rev. Relativity 7 (2004) 2, `http://www.livingreviews.org/lrr-2004-2`.

It remains to look at the equations of motion imposed by stationarity of the action with respect to variations of $\tilde{\Gamma}$. It turns out (the *Palatini principle*) that

1. if one chooses the connections to be torsion-free and imposes the $\tilde{\Gamma}$-equations of motion, then the connections are forced to also be compatible with the metric and thus $\tilde{\Gamma}$ is uniquely determined to be the Levi-Civita connection
2. if one chooses the connections to be compatible with the metric and imposes the $\tilde{\Gamma}$-equations of motion, then the connections are forced to also be torsion-free and thus $\tilde{\Gamma}$ is uniquely determined to be the Levi-Civita connection

In terms of the notation introduced in section 11.5, this amounts to the assertions

$$\begin{array}{ccccccc} T_{\lambda\mu\nu}(\tilde{\Gamma})=0 & \text{and} & \delta_{\tilde{\Gamma}}S[g,\tilde{\Gamma}]=0 & \Rightarrow & Q_{\lambda\mu\nu}(\tilde{\Gamma})=0 & \Rightarrow & \tilde{\Gamma}^{\lambda}_{\ \mu\nu}=\Gamma^{\lambda}_{\ \mu\nu} \\ Q_{\lambda\mu\nu}(\tilde{\Gamma})=0 & \text{and} & \delta_{\tilde{\Gamma}}S[g,\tilde{\Gamma}]=0 & \Rightarrow & T_{\lambda\mu\nu}(\tilde{\Gamma})=0 & \Rightarrow & \tilde{\Gamma}^{\lambda}_{\ \mu\nu}=\Gamma^{\lambda}_{\ \mu\nu} \end{array} \tag{20.113}$$

In either case, the metric equations of motion (20.112) then reduce to the vacuum Einstein equations.

In order to establish the assertions (20.113), we need two preparatory results. The first is that the generalisation of the formula (20.15),

$$\delta R_{\mu\nu}(\Gamma)=\nabla_\lambda\delta\Gamma^{\lambda}_{\ \mu\nu}-\nabla_\nu\delta\Gamma^{\lambda}_{\ \mu\lambda} \tag{20.114}$$

for the variation of the Ricci tensor in terms of the variation of the connection is

$$\delta R_{\mu\nu}(\tilde{\Gamma})=\tilde{\nabla}_\lambda\delta\tilde{\Gamma}^{\lambda}_{\ \mu\nu}-\tilde{\nabla}_\nu\delta\tilde{\Gamma}^{\lambda}_{\ \mu\lambda}+\left(\tilde{\Gamma}^{\rho}_{\ \nu\lambda}-\tilde{\Gamma}^{\rho}_{\ \lambda\nu}\right)\delta\tilde{\Gamma}^{\lambda}_{\ \mu\rho}\ . \tag{20.115}$$

The second is that when the connection is not the Levi-Civita connection, an expression like $\tilde{\nabla}_\lambda J^\lambda$ is not a total derivative in the integral, this being only true for the Levi-Civita connection thanks to the identity

$$\int\sqrt{g}d^4x\ \nabla_\lambda J^\lambda=\int d^4x\ \partial_\lambda(\sqrt{g}J^\lambda)\ . \tag{20.116}$$

Writing

$$\tilde{\Gamma}^{\lambda}_{\ \mu\nu}=\Gamma^{\lambda}_{\ \mu\nu}+C^{\lambda}_{\ \mu\nu}\ , \tag{20.117}$$

we have

$$\tilde{\nabla}_\lambda V^\mu=\nabla_\lambda V^\mu+C^{\mu}_{\ \nu\lambda}V^\nu \tag{20.118}$$

and, in particular,

$$\tilde{\nabla}_\lambda J^\lambda=\nabla_\lambda J^\lambda+C^{\lambda}_{\ \nu\lambda}J^\nu\ , \tag{20.119}$$

only the first term on the right-hand side giving rise to a total derivative.

What we are interested in is $g^{\mu\nu}\delta R_{\mu\nu}(\tilde{\Gamma})$, and with the above results and notation we can write this as

$$\begin{aligned} g^{\mu\nu}\delta R_{\mu\nu}(\tilde{\Gamma})=&g^{\mu\nu}\left(C^{\lambda}_{\ \rho\lambda}\delta C^{\rho}_{\ \mu\nu}-C^{\rho}_{\ \mu\lambda}\delta C^{\lambda}_{\ \rho\nu}-C^{\rho}_{\ \lambda\nu}\delta C^{\lambda}_{\ \mu\rho}+C^{\rho}_{\ \mu\nu}\delta C^{\lambda}_{\ \rho\lambda}\right) \\ &+\text{total derivative terms} \end{aligned} \tag{20.120}$$

Thus the variation of the action with respect to the $\tilde{\Gamma}$ is

$$\begin{aligned}\delta_{\tilde{\Gamma}} S[g,\tilde{\Gamma}] &= \int \sqrt{g} d^4x \; g^{\mu\nu} \left(C^{\lambda}_{\rho\lambda} \delta C^{\rho}_{\mu\nu} - C^{\rho}_{\mu\lambda} \delta C^{\lambda}_{\rho\nu} - C^{\rho}_{\lambda\nu} \delta C^{\lambda}_{\mu\rho} + C^{\rho}_{\mu\nu} \delta C^{\lambda}_{\rho\lambda} \right) \\ &= \int \sqrt{g} d^4x \; \left(g^{\alpha\beta} C^{\lambda\gamma}{}_{\lambda} + g^{\gamma\beta} C^{\alpha\lambda}{}_{\lambda} - C^{\beta\gamma\alpha} - C^{\alpha\beta\gamma} \right) \delta C_{\gamma\alpha\beta}\end{aligned} \tag{20.121}$$

If we were to consider arbitrary $\tilde{\Gamma}$ and hence unconstrained variations $\delta C_{\gamma\alpha\beta}$, the condition for the action to be stationary with respect to variations of $\tilde{\Gamma}$ would be

$$\delta_{\tilde{\Gamma}} S[g,\tilde{\Gamma}] = 0 \quad \Leftrightarrow \quad g^{\alpha\beta} C^{\lambda\gamma}{}_{\lambda} + g^{\gamma\beta} C^{\alpha\lambda}{}_{\lambda} - C^{\beta\gamma\alpha} - C^{\alpha\beta\gamma} = 0 \ . \tag{20.122}$$

However, these equations do not determine the $C^{\alpha}_{\beta\gamma}$ uniquely (we will explicitly parametrise this non-uniqueness below), and hence in this case the Einstein-Hilbert-like action (20.110) alone does not give rise to acceptable equations of motion for the fields.

The situation changes if one imposes some a priori constraints on the allowed $\tilde{\Gamma}$, and hence on their variations $\delta C_{\gamma\alpha\beta}$. We now consider separately the two cases mentioned above:

1. $\tilde{\Gamma}$ are restricted to be symmetric (torsion-free)

   In terms of the coefficients $C^{\gamma}_{\alpha\beta}$, this amounts to the condition

   $$C^{\gamma}_{\alpha\beta} = C^{\gamma}_{\beta\alpha} \tag{20.123}$$

   and the same condition should be imposed on their variations,

   $$\delta C^{\gamma}_{\alpha\beta} = \delta C^{\gamma}_{\beta\alpha} \tag{20.124}$$

   Thus, symmetrising appropriately, from (20.121) one obtains the constraints

   $$2 g^{\alpha\beta} C^{\lambda\gamma}{}_{\lambda} + g^{\gamma\beta} C^{\alpha\lambda}{}_{\lambda} + g^{\gamma\alpha} C^{\beta\lambda}{}_{\lambda} - 2 C^{\beta\alpha\gamma} - 2 C^{\alpha\beta\gamma} = 0 \ . \tag{20.125}$$

   Taking traces, once by contraction with $g_{\alpha\beta}$ and once by contraction with $g_{\alpha\gamma}$ (or, equivalently, with $g_{\beta\gamma}$), one obtains two linearly independent conditions on the traces $C^{\lambda\gamma}{}_{\lambda}$ and $C^{\alpha\lambda}{}_{\lambda}$ requiring both to vanish,

   $$\text{traces} \quad \Rightarrow \quad C^{\lambda\gamma}{}_{\lambda} = 0 \quad , \quad C^{\alpha\lambda}{}_{\lambda} = 0 \ . \tag{20.126}$$

   Then, upon symmetrisation, (20.125) reduces to the condition

   $$C^{\beta\alpha\gamma} + C^{\alpha\beta\gamma} = 0 \quad \Leftrightarrow \quad C_{\beta\alpha\gamma} + C_{\alpha\beta\gamma} = 0 \tag{20.127}$$

   As we have seen in (11.103) of section 11.5, this is precisely the condition that the non-metricity tensor is zero,

   $$Q_{\alpha\beta\gamma} = C_{\beta\alpha\gamma} + C_{\alpha\beta\gamma} = 0 \ , \tag{20.128}$$

i.e. that the connection is compatible with the metric.

Since we started off with a torsion-free (symmetric) connection, this means that the equations of motion fix the connection $\tilde{\Gamma}$ to be the Levi-Civita connection. Alternatively, (20.123) and (20.127) imply that $C_{\gamma\alpha\beta} = 0$ This concludes the proof of the first assertion in (20.113).

2. $\tilde{\Gamma}$ are restricted to be compatible with the metric

   This is largely analogous. In terms of the coefficients $C^{\gamma}{}_{\alpha\beta}$, metric compatibility amounts (as just recalled) to the condition

$$C_{\gamma\alpha\beta} = -C_{\alpha\gamma\beta} \tag{20.129}$$

   and the same condition should be imposed on their variations,

$$\delta C_{\gamma\alpha\beta} = -\delta C_{\alpha\gamma\beta} \tag{20.130}$$

   Thus

$$C_{\gamma\alpha\beta} = C_{\gamma(\alpha\beta)} + C_{\gamma[\alpha\beta]} = T_{(\alpha\beta)\gamma} + \tfrac{1}{2} T_{\gamma\alpha\beta} \tag{20.131}$$

   is the contorsion tensor. In this case, anti-symmetrisation of (20.121) leads to

$$g^{\alpha\beta} C^{\lambda\gamma}{}_{\lambda} - g^{\gamma\beta} C^{\lambda\alpha}{}_{\lambda} + C^{\beta\alpha\gamma} - C^{\beta\gamma\alpha} = 0 \ . \tag{20.132}$$

   Taking traces, one finds $C_{\lambda\gamma}{}^{\lambda} = 0$ (the other trace $C^{\lambda}_{\lambda\gamma}$ is identically zero because of anti-symmetry), and thus (20.132) reduces to

$$C^{\beta\alpha\gamma} - C^{\beta\gamma\alpha} = 0 \quad \Leftrightarrow \quad C_{\beta\alpha\gamma} = C_{\beta\gamma\alpha} \ , \tag{20.133}$$

   which is precisely the symmetry (no torsion) condition

$$T_{\beta\alpha\gamma} = C_{\beta\alpha\gamma} - C_{\beta\gamma\alpha} = 0 \ . \tag{20.134}$$

   Since we started off with metric-compatible connection, this means that the equations of motion fix the connection $\tilde{\Gamma}$ to be the Levi-Civita connection. Alternatively, (20.129) and (20.133) imply that $C_{\gamma\alpha\beta} = 0$. This concludes the proof of the second assertion in (20.113).

Alternatively, and perhaps somewhat more insightfully, one can first determine the general solution to the (under-determined) equation (20.122),

$$g^{\alpha\beta} C^{\lambda\gamma}{}_{\lambda} + g^{\gamma\beta} C^{\alpha\lambda}{}_{\lambda} - C^{\beta\gamma\alpha} - C^{\alpha\beta\gamma} = 0 \ , \tag{20.135}$$

and then analyse the properties of the solution and the consequences of imposing some conditions on $C_{\alpha\beta\gamma}$.[42] To disentangle this equation, we proceed as in the proof of the

[42] See e.g. A. Bernal et al., *On the (non-)uniqueness of the Levi-Civita solution in the Einstein-Hilbert-Palatini formalism*, `arXiv:1606.08756 [gr-qc]` and references therein.

uniqueness of the Levi-Civita connection in section 5.4 and take sums and differences of cyclic permutations of the above equation. Then one ends up with the equation

$$2C^{\alpha\beta\gamma} = g^{\alpha\beta}(A^\gamma + B^\gamma) + g^{\beta\gamma}(A^\alpha - B^\alpha) + g^{\gamma\alpha}(B^\beta - A^\beta) \ , \tag{20.136}$$

where

$$A^\alpha = C^{\alpha\beta}{}_\beta \quad , \quad B^\alpha = C^{\beta\alpha}{}_\beta \tag{20.137}$$

are two of the (a priori independent) traces of $C^{\alpha\beta\gamma}$. Performing either of these contractions in (20.136), one finds the condition

$$A^\alpha = B^\alpha \ , \tag{20.138}$$

and therefore

$$C^{\alpha\beta\gamma} = g^{\alpha\beta}A^\gamma \quad \Leftrightarrow \quad C^\alpha{}_{\beta\gamma} = \delta^\alpha_\beta A_\gamma \ . \tag{20.139}$$

Thus the general solution to the equation of motion (20.122) is

$$\tilde{\Gamma}^\alpha{}_{\beta\gamma} = \Gamma^\alpha{}_{\beta\gamma} + \delta^\alpha_\beta A_\gamma \ , \tag{20.140}$$

with A_γ an arbitrary covector.

This family of connections has the properties

$$\tilde{\nabla}_\gamma g_{\alpha\beta} = -2A_\gamma g_{\alpha\beta} \tag{20.141}$$

and

$$\tilde{\Gamma}^\alpha{}_{\beta\gamma} - \tilde{\Gamma}^\alpha{}_{\gamma\beta} = \delta^\alpha_\beta A_\gamma - \delta^\alpha_\gamma A_\beta \ . \tag{20.142}$$

It is now obvious that requiring either metric compatibility or the symmetry of the connection enforces $A_\gamma = 0$ and thus $\tilde{\Gamma} = \Gamma$.

Remarks:

1. When one couples either of these theories to matter, one will find the standard Einstein equations with source the usual matter energy-momentum tensor, provided that the minimally coupled matter action depends only on the metric and not on the connection. As we have seen, this is satisfied in the case of scalar or Maxwell gauge fields (for which the minimally coupled action in the usual setting could be written in such a way that it depends only on the metric but not on the derivatives of the metric). However, typically the connection appears explicitly in the action for spinors, and in this case variation of the matter action will produce a non-zero contribution to the torsion, say. Thus in that case the Einstein-Hilbert approach (no torsion as a kinematical constraint) and the Palatini approach (torsion determined dynamically) are inequivalent.

2. Within the present framework, it is not possible to relax both the no-torsion and the metricity constraints simultaneously and to simultaneously regain them dynamically, but one can attempt to achieve this either with the aid of additional auxiliary (Lagrange multiplier) fields, or "spontaneously" by adding potentials that force the connection in the ground state to either zero torsion or zero non-metricity.[43]

3. Once one contemplates and permits the presence of non-metricity and/or torsion, there are many more terms that one can in principle use to build an action (by using scalars built from the torsion and non-metricity tensors $T_{\mu\nu\lambda}$ and $Q_{\mu\nu\lambda}$ and their covariant derivatives). Thus, unless one imposes additional symmetry requirements, say, there is no good reason to focus attention exclusively on the Einstein-Hilbert-like action (20.110), and many generalisations of general relativity suggested and discussed in the literature can and should be either rejected or amended simply on these grounds.[44]

[43] See e.g. R. Percacci, *Geometry of Nonlinear Field Theories* for an exploration of some of these ideas.

[44] In his review of gauge theories of gravity, F. Hehl, *Gauge Theories of Gravity and Spacetime*, `arXiv:1204.3672 [gr-qc]`, also emphasises this: "Numerous pages of printed pages could be saved if our colleagues would [...] just motivate their choice of the unknown constants." In that review it is also pointed out that what is generally referred to (and I also called and will continue to call) the Palatini formalism should properly also be attributed to Einstein (1925).

23 Linearised Gravity and Gravitational Waves

23.1 Preliminary Remarks

While it is evidently of interest to apply general relativity to situations where the gravitational field is so strong that the Newtonian approximation fails (and we will consider such situations later on), in most ordinary situations, the gravitational field is weak, very weak, and then it is legitimate to work with a linearisation of the Einstein equations.

When we first derived the Einstein equations we checked that we were doing the right thing by deriving the Newtonian theory in the limit where

1. the gravitational field is weak
2. the gravitational field is static
3. matter is non-relativistic

The fact that the 2nd condition had to be imposed in order to recover the Newtonian theory is interesting in its own right, as it suggests that there are novel features in general relativity, even for weak fields, when the 2nd condition is not imposed. Indeed, as we will see, in this more general setting one discovers that general relativity also predicts the existence of *gravitational waves*, i.e. linearised perturbations of the gravitational field propagating like ordinary waves on the (Minkowski) background geometry. Our principal aim in this section will be to derive these linearised equations, show that they are wave equations, and study some of their more elementary consequences.

An important next step would be to study and understand how or under which circumstances gravitational waves are created and how they can be detected. These, unfortunately, are rather complicated questions in general and I will not enter into this. The things I will cover in the following are much more elementary, both technically and conceptually, than other applications of general relativity discussed elsewhere in these notes.

23.2 Linearised Einstein Equations

We express the weakness of the gravitational field by the condition that the metric be 'close' to that of Minkowski space, i.e. that

$$g_{\mu\nu} = g^{(1)}_{\mu\nu} \equiv \eta_{\mu\nu} + h_{\mu\nu} \tag{23.1}$$

with $|h_{\mu\nu}| \ll 1$. This means that we will drop terms which are quadratic or of higher power in $h_{\mu\nu}$. Here and in the following the superscript $^{(1)}$ indicates that we keep only up to linear (first order) terms in $h_{\mu\nu}$. In particular, the inverse metric is

$$g^{(1)\mu\nu} = \eta^{\mu\nu} - h^{\mu\nu} \tag{23.2}$$

where indices are raised with $\eta^{\mu\nu}$.

As one has thus essentially chosen a background metric, the Minkowski metric, one can think of the linearised version of the Einstein equations (which are field equations for $h_{\mu\nu}$) as a Lorentz-invariant theory of a symmetric tensor field propagating in Minkowski space-time. I won't dwell on this but it is good to keep this in mind. It gives rise to the field theorist's picture of gravity as the theory of an interacting spin-2 field (which is useful for many purposes but which I do not subscribe to unconditionally because it is an inherently perturbative and background dependent picture).

It is straightforward to work out the Christoffel symbols and curvature tensors in this approximation. The terms quadratic in the Christoffel symbols do not contribute to the Riemann curvature tensor and one finds

$$\begin{aligned} \Gamma^{(1)\mu}_{\nu\lambda} &= \eta^{\mu\rho}\tfrac{1}{2}(\partial_\lambda h_{\rho\nu} + \partial_\nu h_{\rho\lambda} - \partial_\rho h_{\nu\lambda}) \\ R^{(1)}_{\mu\nu\rho\sigma} &= \tfrac{1}{2}(\partial_\rho\partial_\nu h_{\mu\sigma} + \partial_\mu\partial_\sigma h_{\rho\nu} - \partial_\rho\partial_\mu h_{\nu\sigma} - \partial_\nu\partial_\sigma h_{\rho\mu}) \end{aligned} \tag{23.3}$$

(this result for the Riemann tensor can also be inferred directly from the expression (8.13) of the Riemann tensor at the origin of an inertial coordinate system). Hence

- the linearised Ricci tensor is

$$R^{(1)}_{\mu\nu} = \tfrac{1}{2}(\partial_\sigma\partial_\nu h^\sigma_{\ \mu} + \partial_\sigma\partial_\mu h^\sigma_{\ \nu} - \partial_\mu\partial_\nu h - \Box h_{\mu\nu}) \ , \tag{23.4}$$

  where

$$h \equiv h^\mu_{\ \mu} = \eta^{\mu\nu}h_{\mu\nu} = -h_{00} + \delta^{ik}h_{ik} \tag{23.5}$$

  is the trace of $h_{\mu\nu}$ and

$$\Box = \partial^\mu\partial_\mu = -(\partial_0)^2 + \Delta \tag{23.6}$$

  is the Minkowski wave operator;

- the linearised Ricci scalar is

$$R^{(1)} = \eta^{\mu\nu}R^{(1)}_{\mu\nu} = \partial_\mu\partial_\nu h^{\mu\nu} - \Box h \ ; \tag{23.7}$$

- the linearised Einstein tensor is

$$\begin{aligned} G^{(1)}_{\mu\nu} &= R^{(1)}_{\mu\nu} - \tfrac{1}{2}\eta_{\mu\nu}R^{(1)} \\ &= \tfrac{1}{2}(\partial_\sigma\partial_\nu h^\sigma_{\ \mu} + \partial_\sigma\partial_\mu h^\sigma_{\ \nu} - \partial_\mu\partial_\nu h - \Box h_{\mu\nu} - \eta_{\mu\nu}\partial_\rho\partial_\sigma h^{\rho\sigma} + \eta_{\mu\nu}\Box h) \ ; \end{aligned} \tag{23.8}$$

- and the linearised vacuum Einstein equations

$$R^{(1)}_{\mu\nu} = 0 \quad \Leftrightarrow \quad G^{(1)}_{\mu\nu} = 0 \tag{23.9}$$

  are thus

$$R^{(1)}_{\mu\nu} = 0 \quad \Leftrightarrow \quad \partial_\sigma\partial_\nu h^\sigma_{\ \mu} + \partial_\sigma\partial_\mu h^\sigma_{\ \nu} - \partial_\mu\partial_\nu h - \Box h_{\mu\nu} = 0 \ , \tag{23.10}$$

  or, equivalently (in the form $G^{(1)}_{\mu\nu} = 0$)

$$\partial_\sigma\partial_\nu h^\sigma_{\ \mu} + \partial_\sigma\partial_\mu h^\sigma_{\ \nu} - \partial_\mu\partial_\nu h - \Box h_{\mu\nu} - \eta_{\mu\nu}\partial_\rho\partial_\sigma h^{\rho\sigma} + \eta_{\mu\nu}\Box h = 0 \ . \tag{23.11}$$

The latter can be derived from the quadratic Minkowski space (Poincaré invariant) field theory action (the *Fierz-Pauli action* (1939))

$$\begin{aligned} S[h_{\mu\nu}] &= \int d^4x\, \mathcal{L}(h_{\mu\nu}, \partial_\sigma h_{\mu\nu}) \\ \mathcal{L}(h_{\mu\nu}, \partial_\sigma h_{\mu\nu}) &= -\tfrac{1}{4} h_{\mu\nu,\sigma} h^{\mu\nu,\sigma} + \tfrac{1}{2} h_{\mu\nu,\sigma} h^{\sigma\mu,\nu} + \tfrac{1}{4} h^{,\sigma} h_{,\sigma} - \tfrac{1}{2} h_{,\sigma} h^{\mu\sigma}{}_{,\mu} \end{aligned} \tag{23.12}$$

for a free massless spin-2 field described by the Lorentz tensor $h_{\mu\nu}$. Indeed, variation of the action with respect to the $h_{\mu\nu}$ (and the usual integration by parts) leads to

$$h_{\mu\nu} \to h_{\mu\nu} + \delta h_{\mu\nu} \quad \Rightarrow \quad \delta S[h_{\mu\nu}] = -\int d^4x\, G^{(1)}_{\mu\nu} \delta h^{\mu\nu} \ . \tag{23.13}$$

Remarks:

1. Signs 1: note that from (23.2) one has

   $$g^{(1)\mu\nu} = \eta^{\mu\nu} - h^{\mu\nu} \quad \Rightarrow \quad \delta g^{(1)\mu\nu} = -\delta h^{\mu\nu} \ , \tag{23.14}$$

   so that the sign in (23.13) is the standard one if expressed in terms of the variation of $g^{\mu\nu}$,

   $$\delta S[h_{\mu\nu}] = +\int d^4x\, G^{(1)}_{\mu\nu} \delta g^{(1)\mu\nu} \ . \tag{23.15}$$

2. Signs 2: the sign and overall normalisation of the Lagrangian are also such that one obtains the canonically normalised kinetic term

   $$\mathcal{L} = +\tfrac{1}{4} (\partial_t h_{ik})^2 + \dots \tag{23.16}$$

   for a 2-index field.

3. Note that the Lagrangian is *not* the linearised Einstein-Hilbert Lagrangian, i.e. the linearised Ricci scalar (23.7) (as the latter is, by construction, linear in $h_{\mu\nu}$). Rather,

   $$\mathcal{L} = \tfrac{1}{2} h^{\mu\nu} G^{(1)}_{\mu\nu} \ + \ \text{total derivative} \ , \tag{23.17}$$

   where the total derivative contributions serve to turn the terms of the form $h\partial^2 h$ arising from $h^{\mu\nu} G^{(1)}_{\mu\nu}$ into the standard $(\partial h)^2$-form, so that $\mathcal{L}$ is of a standard (quadratic) form in the derivatives of $h_{\mu\nu}$, as it should be.

4. Note that we can indeed take the integration measure to be the flat Minkowski measure d^4x, as $\mathcal{L}$ is already quadratic in $h_{\mu\nu}$ and its derivatives, so that any contribution of $h_{\mu\nu}$ to the measure would give a subleading contribution.

5. This theory, just like its spin-1 Maxwell counterpart, has a local gauge invariance which we will return to and make use of below.

6. In the presence of matter the linearised Einstein equations are

$$G^{(1)}_{\mu\nu} = 8\pi G_N T^{(0)}_{\mu\nu} \quad . \tag{23.18}$$

Note that only the zero'th order term in the h-expansion appears on the right hand side of this equation. This is due to the fact that $T_{\mu\nu}$ must itself already be small in order for the linearised approximation to be valid, i.e. $T^{(0)}_{\mu\nu}$ should be of order $h_{\mu\nu}$. Therefore, any terms in $T_{\mu\nu}$ depending on $h_{\mu\nu}$ would already be of order $(h_{\mu\nu})^2$ and can be dropped.

7. The conservation law for the energy-momentum tensor is therefore just

$$\partial_\mu T^{(0)\mu\nu} = 0 \quad , \tag{23.19}$$

and this is indeed compatible with the linearised Bianchi identity

$$\partial_\mu G^{(1)\mu\nu} = 0 \quad , \tag{23.20}$$

which can easily be verified, and which reflects the invariance of the theory under the linearised coordinate transformations to be discussed below.

23.3 Newtonian Limit Revisited

In section 19.3 we verified that the Newtonian (weak field, static, non-relativistic matter) limit of the Einstein equations reduces to the Newtonian (Poisson) equation $\Delta\phi = 4\pi G_N \rho$. This is a special case of the above general weak-field limit equations (23.18), with $G^{(1)}_{\mu\nu}$ given in (23.8), and it will be instructive to redo the calculation of section 19.3 from this more general perspective.

We thus assume an energy-momentum tensor whose only non-negligible component is the energy density $T_{00} = \rho$, with ρ static and $G_N\rho \ll 1$. Then we can assume that the deviations of the space-time geometry from the Minkowski metric are small and time-independent,

$$g_{\mu\nu} = g^{(1)}_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu} \quad , \quad \partial_0 h_{\mu\nu} = 0 \quad . \tag{23.21}$$

Then the (00)-component of the Einstein tensor reduces (after some cancellations) to

$$G^{(1)}_{00} = -\tfrac{1}{2}\Delta(\delta^{ik}h_{ik}) + \tfrac{1}{2}\partial_i\partial_k h^{ik} \quad . \tag{23.22}$$

In particular, h_{00} and its derivatives have dropped out of this expression, which appears to be at odds with the desired $G_{00} = -\Delta g_{00}$ for Newtonian fields (section 19.3). However, we have not yet used at all the condition that $T_{ik} = 0 \to G_{ik} = 0$. In particular, for static perturbations we find from (23.8) that the trace of the spatial components of the Einstein tensor is

$$\delta^{ik}G^{(1)}_{ik} = \tfrac{1}{2}\Delta(\delta^{ik}h_{ik}) - \tfrac{1}{2}\partial_i\partial_k h^{ik} - \Delta h_{00} \tag{23.23}$$

so that

$$G^{(1)}_{ik} = 0 \quad \Rightarrow \quad G^{(1)}_{00} = -\Delta h_{00} \ . \tag{23.24}$$

This is precisely the relation required (and verified) in the analysis of section 19, in order to have the correct Newtonian limit.

23.4 ADM and Komar Energies of an Isolated System

As we are in the realm of a Poincaré-invariant classical field theory, we can define the total energy of the system as the integral of the energy-density $T_{00} = \rho$ over a spatial constant time $x^0 = \text{const.}$ slice Σ,

$$E = \int_\Sigma d^3x \; T_{00} \tag{23.25}$$

(another way to see that we can take the integration measure to be the flat Euclidean measure d^3x is to recall that T_{00} is small by assumption). If the linearised Einstein equations are satisfied, we can express the total energy in terms of the Einstein tensor,

$$E = \frac{1}{8\pi G_N} \int_\Sigma d^3x \; G^{(1)}_{00} \ . \tag{23.26}$$

Now we have two distinct expressions for $G^{(1)}_{00}$ in terms of the derivatives of the metric and, interestingly, both of them are spatial total derivatives, namely

$$G^{(1)}_{00} = -\tfrac{1}{2}\Delta(\delta^{ik}h_{ik}) + \tfrac{1}{2}\partial_i\partial_k h^{ik} = \partial_i(-\tfrac{1}{2}\partial^i h^k_{\ k} + \tfrac{1}{2}\partial_k h^{ik}) \tag{23.27}$$

and

$$G^{(1)}_{00} = -\Delta h_{00} = -\partial_i(\partial^i h_{00}) \ . \tag{23.28}$$

This allows us to write the total energy E of the system as a boundary integral over the boundary

$$\partial\Sigma = S^2_\infty \tag{23.29}$$

of the spatial slice "at infinity" in two different ways, namely as

$$E = \frac{1}{16\pi G_N} \oint_{S^2_\infty} dS_i \; (\partial_k h^{ik} - \partial^i h^k_{\ k}) \tag{23.30}$$

or as

$$E = -\frac{1}{8\pi G_N} \oint_{S^2_\infty} dS_i \; \partial^i h_{00} \ . \tag{23.31}$$

Note that these final expressions only depend on the asymptotics of the gravitational field at spatial infinity (the assumption that the gravitational field is weak there translating into the statement that the metric is asymptotically flat). It is thus tempting to adopt E as the definition of the total energy of an isolated (i.e. asymptotically flat) system in general, even when the field in the interior is not weak:

1. Thus in general the *ADM Energy* of such a system is defined by

$$E_{ADM} = \frac{1}{16\pi G_N} \oint_{S^2_\infty} dS_i \, (\partial_k h_{ik} - \partial_i h_{kk}) \ , \tag{23.32}$$

this expression relying on a Cartesian coordinate system at infinity (which is why we can permit ourselves to be careless with the positioning of the indices - repeated indices are to be summed, as usual). Somewhat more covariantly one can write this as

$$E_{ADM} = \frac{1}{16\pi G_N} \oint_{S^2_\infty} dS_i \, (\bar{\nabla}^k h_{ik} - \bar{\nabla}_i h) \ , \tag{23.33}$$

where $\bar{\nabla}$ is the covariant derivative with respect to a background metric $\bar{g}_{ik}$,

$$h_{ik} = g_{ik} - \bar{g}_{ik} \ , \tag{23.34}$$

and

$$h = \bar{g}^{ik} h_{ik} \tag{23.35}$$

is the trace of h_{ik} with respect to the background metric. If the background metric is the Minkowski metric, *and* one works in Cartesian coordinates, then (23.33) reduces to (23.32).

The justification for the identification of this quantity with the energy is substantially strengthened by the canonical (Hamiltonian) ADM analysis of general relativity discussed in section 21, in particular in section 21.12. Indeed, in section 21.12 we deduced another candidate expression for the energy from the boundary term in the ADM Hamiltonian, namely (21.191)

$$E_{ADM} = -\frac{1}{8\pi G_N} \oint_{S_\infty} d^2x \, \sqrt{s}(k_S - k^0_S) \tag{23.36}$$

At first sight, beyond the fact that both are expressed as integrals over a 2-sphere at infinity and that both depend in some way on a choice of reference (background) metric, these two expresssions appear to have little in common. However, it is possible to show that (23.33) is in fact equal to (23.36) provided that the metrics on S_∞ induced by the metrics $\bar{g}_{ik}$ (appearing in (23.33)) and g^0_{ik} (implicitly appearing in (23.36)) are the same. The proof of this statement I am aware of relies on a convenient choice of coordinate system and is therefore not particularly insightful *per se* and will not be given here.[64]

2. The second expression is actually a special case of the Komar charges briefly mentioned in section 13.7, here applied to the (asymptotic) timelike Killing vector ∂_t (the generator of time-translations, and thus naturally associated with the

[64] For this proof, see e.g. S. Hawking, G. Horowitz, *The Gravitational Hamiltonian, Action, Entropy, and Surface Terms*, `arXiv:gr-qc/9501014`, or Exercise 4.5.7 of E. Poisson, *A Relativist's Toolkit* (modelled on the cited article).

energy or Hamiltonian). Indeed, recall that for any Killing vector K^μ we had the conserved current $J^\mu = R^\mu_{\ \nu} K^\nu$ (13.49), which was itself a divergence of an anti-symmetric tensor, $J^\mu = \nabla_\nu A^{\mu\nu}$, with $A^{\mu\nu} = -A^{\nu\mu} = \nabla^\mu K^\nu$. Thus the corresponding conserved charge $Q_K(V)$ contained in a volume V can be written as a surface integral of $\nabla^\mu K^\nu$ over the boundary ∂V of the volume V,

$$Q_K(V) \sim \oint_{\partial V} dS_{\mu\nu} \nabla^\mu K^\nu \ . \tag{23.37}$$

Applying this to $V = \Sigma$, $\partial V = S^2_\infty$, and $K = \partial_t$, we find (asymptotically)

$$\begin{aligned} \nabla^i K^0 &= \nabla_i K^0 = \Gamma^0_{i\alpha} K^\alpha = \Gamma^0_{i0} = -\Gamma_{0i0} = -\tfrac{1}{2}\partial_i h_{00} \\ \nabla^0 K^i &= -\nabla_0 K^i = -\Gamma^i_{0\alpha} K^\alpha = -\Gamma^i_{00} = -\Gamma_{i00} = +\tfrac{1}{2}\partial_i h_{00} \ . \end{aligned} \tag{23.38}$$

Thus the second expression (23.31) can be written as

$$\begin{aligned} E &= -\frac{1}{8\pi G_N} \oint_{S^2_\infty} dS_i \ \partial_i h_{00} = -\frac{1}{4\pi G_N} \oint_{S^2_\infty} dS_{0i} \ \nabla^0 K^i \\ &= -\frac{1}{8\pi G_N} \oint_{S^2_\infty} dS_{\mu\nu} \ \nabla^\mu K^\nu \ . \end{aligned} \tag{23.39}$$

This fixes the normalisation of the Komar charge $Q_K(V)$ (23.37) in this case,

$$E_{\text{Komar}}(V) \equiv Q_{K=\partial_t}(V) = -\frac{1}{8\pi G_N} \oint_{\partial V} dS_{\mu\nu} \ \nabla^\mu K^\nu \tag{23.40}$$

and we can identify (23.31) with the total Komar energy ($V = \Sigma$) associated to the Killing vector $K = \partial_t$,

$$E_{\text{Komar}}(\Sigma) = -\frac{1}{8\pi G_N} \oint_{S^2_\infty} dS_i \ \partial_i h_{00} = -\frac{1}{8\pi G_N} \oint_{S^2_\infty} dS_{\mu\nu} \ \nabla^\mu K^\nu \ . \tag{23.41}$$

As our primary litmus test, in section 24.8 we will apply these expressions to the Schwarzschild metric, the unique spherically symmetric asymptotically flat solution of the vacuum Einstein equations. In particular, it therefore describes the gravitational field in the exterior of a star, and it depends on a single parameter m, related to the mass of the star by $m = G_N M/c^2$, and it is then of interest to see what the ADM and Komar energies have to say about this. Suffice it to say here that reassuringly one indeed finds $E_{ADM} = E_{\text{Komar}} = M$.

In the same way one can also introduce various notions of momentum or angular momentum of an isolated system (the latter for instance being non-zero for the Kerr metric (30.3) describing a rotating star or black hole), but we will not pursue this here.

23.5 Wave Equations and Gauge Conditions in Maxwell Theory

We will now abandon the assumption of static non-relativistic fields and return to the general weak-field linearised Einstein equations (23.18). In order to understand how to proceed from there, in this section we will briefly recall in a condensed way the analogous steps in the case of Maxwell theory.

1. The Maxwell equations (in a Minkowski background) are already linear (no need to linearise) and read

$$\partial^\alpha F_{\alpha\beta} = -J_\beta \ . \tag{23.42}$$

In terms of the potentials A_α these equations are

$$\Box A_\beta - \partial_\beta(\partial_\alpha A^\alpha) = -J_\beta \ . \tag{23.43}$$

These are to be thought of as the analogue of the linearised Einstein equations (23.18), with the linearised Einstein tensor given in (23.8). These are wave equations (with a source) for the components A_α, but due to the second term on the left-hand side the individual components of the potential are not yet completely decoupled in this equation.

2. Maxwell theory has the gauge invariance

$$A_\alpha \to A_\alpha + \partial_\alpha V \tag{23.44}$$

(in particular, the field strenghts $F_{\alpha\beta}$ are invariant) which allows one to choose for instance the Loren(t)z gauge condition[65]

$$\partial_\alpha A^\alpha = 0 \ . \tag{23.45}$$

Indeed, if V is chosen to satisfy the differential equation

$$\Box V = -\partial_\alpha A^\alpha \ , \tag{23.46}$$

then

$$\partial^\alpha(A_\alpha + \partial_\alpha V) = 0 \ . \tag{23.47}$$

With this choice of gauge the Maxwell equations become decoupled wave equations,

$$\Box A_\beta = -J_\beta \ , \tag{23.48}$$

and can now be straightforwardly solved in terms of Green functions etc. One particular solution to the inhomogeneous equation is the retarded potential

$$A_\alpha(t,\vec{x}) = (4\pi)^{-1} \int d^3x' \, \frac{J_\alpha(t-|\vec{x}-\vec{x}'|,\vec{x}')}{|\vec{x}-\vec{x}'|} \ . \tag{23.49}$$

Note that, as a consequence of $\partial_\alpha J^\alpha = 0$, this solution is automatically in the Lorenz gauge. As usual, the general solution is then a sum of this particular solution of the inhomogeneous equation and the general solution of the homogeneous equation.

[65] Credit for this should really go the Danish physicist Ludvig V. Lorenz (who used this condition in its Lorent(!)z non-invariant form already in 1867, just 3 years after the publicaton of Maxwell's treatise) rather than the more famous Dutch physicist Hendrik A. Lorentz. See J. Jackson's *Electrodynamics* (note at the end of chapter 6).

3. The homogeneous equation, i.e. the vacuum Maxwell equations, can now be solved in terms of plane waves,

$$A_\alpha = \epsilon_\alpha \mathrm{e}^{ik_\beta x^\beta} \quad \text{with} \quad k_\beta k^\beta = 0 \tag{23.50}$$

(or wave packets constructed from them), and the Lorenz gauge constrains the *polarisation vector* ϵ_α by

$$k^\alpha \epsilon_\alpha = 0 \ . \tag{23.51}$$

For a wave travelling in the x^3-direction, this implies for the polarisation vector

$$k^\alpha = (\omega, 0, 0, k^3 = \omega) \quad \Rightarrow \quad \epsilon_3 = -\epsilon_0 \ . \tag{23.52}$$

4. One still has the residual gauge invariance

$$A'_\alpha = A_\alpha + \partial_\alpha \xi \quad , \quad \Box \xi = 0 \tag{23.53}$$

leaving the Lorenz gauge condition invariant. Choosing a particular solution of this homegenous wave equation, say

$$\xi = \xi_0 \mathrm{e}^{ik_\beta x^\beta} \tag{23.54}$$

now completely fixes this residual gauge invariance. Under this residual gauge transformation the polarisation vector transforms as

$$\epsilon'_\alpha = \epsilon_\alpha + i\xi_0 k_\alpha \ , \tag{23.55}$$

in particular

$$\epsilon'_0 = \epsilon_0 - i\xi_0 \omega \ . \tag{23.56}$$

Thus choosing $\xi_0 = \epsilon_0 / i\omega$, one has $\epsilon'_0 = 0$, implying $\epsilon'_3 = 0$, and one is left with the polarisation vector

$$\epsilon^\alpha = (0, \epsilon^1, \epsilon^2, 0) \tag{23.57}$$

displaying the two physical transverse polarisations of an electromagnetic wave.

23.6 Linearised Gravity: Gauge Invariance and Coordinate Choices

We now proceed analogously in the case of linearised gravity. First of all we need to understand the gauge invariance (or the counterpart of gauge invariance) in the present case.

The original non-linear Einstein equations have a local invariance consisting of general coordinate transformations. What remains of general coordinate invariance in the linearised approximation are, naturally, linearised general coordinate transformations. Indeed, $h_{\mu\nu}$ and

$$h'_{\mu\nu} = h_{\mu\nu} + L_V \eta_{\mu\nu} \tag{23.58}$$

represent the same physical perturbation because $\eta_{\mu\nu} + L_V \eta_{\mu\nu}$ is just an infinitesimal coordinate transform of the Minkowski metric $\eta_{\mu\nu}$. Therefore linearised gravity has the gauge freedom

$$h_{\mu\nu} \to h_{\mu\nu} + \partial_\mu V_\nu + \partial_\nu V_\mu \ . \tag{23.59}$$

For example, the linearised Riemann tensor $R^{(1)}_{\mu\nu\rho\sigma}$ is, rather obviously, invariant under this transformation. As a consequence, in particular also the Einstein tensor, and hence the vacuum linearised Einstein equations, have this local gauge invariance. This can be understood without any calculation by noting that under the variation

$$\delta_V h_{\mu\nu} = L_V \eta_{\mu\nu} \tag{23.60}$$

the linearised Riemann tensor transforms into the Riemann tensor for the Minkowski metric,

$$\delta_V R^{(1)}_{\mu\nu\rho\sigma} = L_V R_{\mu\nu\rho\sigma}(\eta_{\alpha\beta}) = 0 \ . \tag{23.61}$$

However, in contrast to the Lagrangian of Maxwell theory, say, the Lagrangian (23.12) for linearised gravity is *not* strictly gauge invariant, but only invariant up to a total derivative.

Given this gauge invariance of the linearised theory, for explicit calculations it is useful to make a particular gauge choice and, as in Maxwell theory, a good choice of gauge can simplify things considerably (and a bad choice of gauge can have the opposite effect).

It turns out that, in general, a very useful gauge condition is

$$g^{\mu\nu}\Gamma^\rho_{\mu\nu} = 0 \ . \tag{23.62}$$

It is called the *harmonic gauge condition*, or Fock, or de Donder gauge condition (even though harmonic coordinates were used extensively by Einstein himself until just before the discovery of the final, truly generally covariant, formulation of general relativity). The name harmonic derives from the fact that in this gauge the coordinate functions x^μ are harmonic:

$$\Box x^\mu \equiv g^{\nu\rho}\nabla_\rho \partial_\nu x^\mu = -g^{\nu\rho}\Gamma^\mu_{\nu\rho} \ , \tag{23.63}$$

and thus

$$\Box x^\mu = 0 \ \Leftrightarrow \ g^{\nu\rho}\Gamma^\mu_{\nu\rho} = 0 \ . \tag{23.64}$$

It is the analogue of the Lorenz gauge $\partial_\mu A^\mu = 0$ in Maxwell theory. Moreover, in flat space Cartesian coordinates are obviously harmonic, and in general harmonic coordinates are (like geodesic coordinates) a nice and useful curved space counterpart of Cartesian coordinates.

In the linearised theory, this gauge condition becomes

$$\partial_\mu h^\mu_\lambda - \tfrac{1}{2}\partial_\lambda h = 0 \ . \tag{23.65}$$

The gauge parameter V_μ which will achieve this is the solution to the equation

$$\Box V_\lambda = -(\partial_\mu h^\mu_\lambda - \tfrac{1}{2}\partial_\lambda h) \ . \tag{23.66}$$

Indeed, with this choice one has

$$\partial_\mu(h^\mu_\lambda + \partial^\mu V_\lambda + \partial_\lambda V^\mu) - \tfrac{1}{2}\partial_\lambda(h + 2\partial^\mu V_\mu) = 0 \ . \tag{23.67}$$

Note for later that, as in Maxwell theory, this gauge choice does not necessarily fix the gauge completely. Any transformation $x^\mu \to x^\mu + \xi^\mu$ with $\Box\xi^\mu = 0$ will leave the harmonic gauge condition invariant.

23.7 Wave Equation

Now let us use this gauge condition in the linearised Einstein equations. In this gauge they simplify somewhat to

$$\Box h_{\mu\nu} - \tfrac{1}{2}\eta_{\mu\nu}\Box h = -16\pi G_N T^{(0)}_{\mu\nu} \ . \tag{23.68}$$

In particular, the vacuum equations (or the equations in a source-free region of space-time) are just

$$T^{(0)}_{\mu\nu} = 0 \ \ \Rightarrow \ \ \Box h_{\mu\nu} = 0 \ , \tag{23.69}$$

which is the standarad relativistic wave equation. Together, the equations

$$\begin{aligned} \Box h_{\mu\nu} &= 0 \\ \partial_\mu h^\mu_\lambda - \tfrac{1}{2}\partial_\lambda h &= 0 \end{aligned} \tag{23.70}$$

determine the evolution of a disturbance in a gravitational field in vacuum in the harmonic gauge.

It is often convenient to consider the linear combination

$$\bar{h}_{\mu\nu} = h_{\mu\nu} - \tfrac{1}{2}\eta_{\mu\nu}h \ . \tag{23.71}$$

This is also commonly known as the *trace reversed perturbation*, because in 4 space-time dimensions (but only there) one has

$$\bar{h}^\mu_\mu = -h^\mu_\mu \ . \tag{23.72}$$

Note, as an aside, that with this notation and terminology the Einstein tensor (again in 4 space-time dimensions only) is the trace reversed Ricci tensor,

$$\bar{R}_{\mu\nu} = R_{\mu\nu} - \tfrac{1}{2}g_{\mu\nu}R = G_{\mu\nu} \ . \tag{23.73}$$

In terms of $\bar{h}_{\mu\nu}$, the linearised Einstein equations and the harmonic gauge condition (in any dimension) are just

$$\begin{aligned} \Box\bar{h}_{\mu\nu} &= -16\pi G_N T^{(0)}_{\mu\nu} \\ \partial_\mu \bar{h}^\mu_{\ \nu} &= 0 \ . \end{aligned} \tag{23.74}$$

This way of writing the linearised Einstein equations sharpens the analogy with the Maxwell equations in the Lorenz gauge:

- the Lorenz gauge $\partial_\mu A^\mu = 0$ decouples the Maxwell equations for A_μ which in this gauge read $\Box A_\mu = -j_\mu$
- the gauge condition $\partial_\mu \bar{h}^\mu_{\ \nu} = 0$ decouples the linearised Einstein equations for the variables $\bar{h}_{\mu\nu}$ which in this gauge read $\Box\bar{h}_{\mu\nu} = -16\pi G_N T^{(0)}_{\mu\nu}$.

One solution is, of course, the retarded solution

$$\bar{h}_{\mu\nu}(t,\vec{x}) = 4G_N \int d^3x' \ \frac{T^{(0)}_{\mu\nu}(t-|\vec{x}-\vec{x}'|,\vec{x}')}{|\vec{x}-\vec{x}'|} \ . \tag{23.75}$$

Note that, as a consequence of $\partial_\mu T^{(0)\mu\nu} = 0$, this solution is automatically in the harmonic gauge. As usual, the general solution is then a sum of this particular solution of the inhomogeneous equation and the general solution of the homogeneous equation.

23.8 Polarisation Tensor and the Metric of a Gravitational Wave

The homogeneous equation is the linearised vacuum Einstein equation in the harmonic gauge,

$$\Box\bar{h}_{\mu\nu} = 0 \ . \tag{23.76}$$

This is clearly solved by

$$\bar{h}_{\mu\nu} = \epsilon_{\mu\nu} \mathrm{e}^{ik_\alpha x^\alpha} \ , \tag{23.77}$$

where $\epsilon_{\mu\nu}$ is a constant, symmetric *polarisation tensor* and k^α is a constant *wave vector*, provided that k^α is null, $k^\alpha k_\alpha = 0$. In order to obtain real metrics from this one should of course use real linear combinations of such solutions in the end.

Thus plane waves are solutions to the linearised equations of motion and the Einstein equations predict the existence of gravitational waves travelling along null geodesics (at the speed of light). The timelike component of the wave vector is often referred to as the *frequency* ω of the wave, and we can write $k^\mu = (\omega, k^i)$. Plane waves are of course not the most general solutions to the wave equations but any solution can be written as a superposition of plane wave solutions (wave packets).

So far, we have ten parameters $\epsilon_{\mu\nu}$ and four parameters k^μ to specify the wave, but many of these are spurious, i.e. can be eliminated by using the freedom to perform linearised coordinate transformations and Lorentz rotations.

First of all, the harmonic gauge condition implies that

$$\partial_\mu \bar{h}^\mu_{\ \nu} = 0 \quad \Rightarrow \quad k^\mu \epsilon_{\mu\nu} = 0 \ . \tag{23.78}$$

Now we can make use of the residual gauge freedom $x^\mu \to x^\mu + \xi^\mu$ with $\Box \xi^\mu = 0$ to impose further conditions on the polarisation tensor. Since this is a wave equation for ξ^μ, once we have specified a solution for ξ^μ we will have fixed the gauge completely. Alternatively, note that under $h_{\mu\nu} \to h_{\mu\nu} + \partial_\mu V_\nu + \partial_\nu V_\mu$ one has

$$\begin{aligned} & \bar{h}_{\mu\nu} \to \bar{h}_{\mu\nu} + \partial_\mu V_\nu + \partial_\nu V_\mu - \eta_{\mu\nu} \partial_\lambda V^\lambda \\ \Rightarrow \quad & \partial^\mu \bar{h}_{\mu\nu} \to \partial^\mu \bar{h}_{\mu\nu} + \Box V_\nu \end{aligned} \tag{23.79}$$

so that the gauge condition is invariant precisely under linearised coordinate transformations with $\Box V_\mu = 0$. Taking the solution of this equation to be of the form

$$V_\mu = v_\mu \mathrm{e}^{ik_\alpha x^\alpha} \ , \tag{23.80}$$

the polarisation tensor transforms according to

$$\epsilon_{\alpha\beta} \to \epsilon_{\alpha\beta} + i(k_\alpha v_\beta + k_\beta v_\alpha) - i\eta_{\alpha\beta} k^\gamma v_\gamma \ . \tag{23.81}$$

One can now choose the v_μ in such a way (see the example below) that the new polarisation tensor satisfies $k^\mu \epsilon_{\mu\nu} = 0$ (as before) as well as

$$\epsilon_{\mu 0} = \epsilon^\mu_{\ \mu} = 0 \ . \tag{23.82}$$

All in all, we appear to have nine conditions on the polarisation tensor $\epsilon_{\mu\nu}$ but as both (23.78) and the first of (23.82) imply $k^\mu \epsilon_{\mu 0} = 0$, only eight of these are independent. Therefore, there are two independent polarisations for a gravitational wave.

Together with $k^\mu \epsilon_{\mu\nu} = 0$, these gauge conditions thus impose the conditions

$$k^\mu \bar{h}_{\mu\nu} = 0 \quad , \quad \bar{h}_{\mu 0} = 0 \quad , \quad \bar{h}^\mu_{\ \mu} = 0 \ . \tag{23.83}$$

This is known as the *transverse traceless gauge*, and a field satisfying this gauge is frequently denoted by $\bar{h}^{TT}_{\mu\nu}$.

For example, let us consider a wave travelling in the x^3-direction,

$$k^\mu = (\omega, 0, 0, k^3) = (\omega, 0, 0, \omega) \ . \tag{23.84}$$

Then

- the condition $k^\mu \epsilon_{\mu\nu} = 0$ becomes $\epsilon_{3\nu} = -\epsilon_{0\nu}$

- the condition $\epsilon_{00} = 0$ determines $v_0 - v_3$, and also implies $\epsilon_{30} = 0$
- the other linear combination $v_0 + v_3 \sim k^\alpha v_\alpha$ can be used to achieve $\epsilon^\mu_{\ \mu} = 0$
- the conditions $\epsilon_{10} = \epsilon_{20} = 0$ determine v_1 and v_2

Therefore, the only independent components are ϵ_{ab} with $a, b = 1, 2$. As ϵ_{ab} is symmetric and traceless, this wave is completely characterised by $\epsilon_{11} = -\epsilon_{22}$, $\epsilon_{12} = \epsilon_{21}$ and the frequency ω.

Now we should not forget that, when talking about the polarisation tensor of a gravitational wave, we are actually talking about the space-time metric itself. Namely, since for a traceless perturbation we have

$$\epsilon^\mu_{\ \mu} = 0 \quad \Rightarrow \quad \bar{h}_{\alpha\beta} = h_{\alpha\beta} \ , \tag{23.85}$$

we have deduced that the metric describing a gravitational wave travelling in the x^3-direction can always be put into the form

$$ds^2 = -dt^2 + (\delta_{ab} + h_{ab}) dx^a dx^b + (dx^3)^2 \ , \tag{23.86}$$

with $h_{ab} = h_{ab}(t \mp x^3)$ symmetric and traceless. This neatly encodes and describes the distortion of the space-time geometry in the directions transverse to the gravitational wave.

Remarks:

1. Introducing null coordinates

$$u = t - x^3 \quad , \quad v = t + x^3 \ , \tag{23.87}$$

the gravitational wave metric can be written as

$$ds^2 = -du\, dv + (\delta_{ab} + h_{ab}(u)) dx^a dx^b \ . \tag{23.88}$$

Explicitly, with $x^a = (x, y)$, the linearised perturbation can be written as

$$h_{ab}(u) dx^a dx^b = h_+(u)(dx^2 - dy^2) + 2h_\times(u) dx dy \ , \tag{23.89}$$

with two arbitrary functions $h_{11}(u) \equiv h_+(u)$ and $h_{12}(u) \equiv h_\times(u)$ (see the next section 23.9 for an explanation of this notation).

2. This analysis was rather evidently independent of the dimension. In D dimensions the polarisation states of a graviton are described by a symmetric, transverse, and traceless tensor h_{ab} where $a, b = 1, \ldots, D-2$. Thus the number of physical polarisation states of a graviton in D dimensions are

$$\#[h_{ab}] = \frac{(D-2)(D-1)}{2} - 1 = \frac{D(D-3)}{2} \ . \tag{23.90}$$

Note that this gives zero for $D = 3$, in agreement with the fact, noted before, that in 3 space-time dimensions there is no gravitational vacuum dynamics since vanishing of the Ricci tensor is equivalent to vanishing of the Riemann tensor.

3. By abandoning the assumption that $h_{ab}(u)$ be small, one can look for (and easily find) solutions of the full non-linear Einstein equations of the form

$$ds^2 = -du\, dv + g_{ab}(u)dx^a dx^b \quad . \tag{23.91}$$

These are known as exact gravitational plane waves, and are discussed in some detail in section 43.

23.9 Physical Effects of Gravitational Waves

To determine the physical effect of a gravitational wave racing by, we cannot just look at the gravitational field (23.86) at a point (by the equivalence principle), i.e. we cannot detect the presence of such a wave (ultra-)locally. However, we can consider its influence on the relative motion of nearby particles. In other words, we look at the geodesic deviation equation (8.38).

Consider a family of nearby particles described by the *velocity field* $u^\mu(x)$ and separation (deviation) vector $S^\mu(x)$. Then the change of the deviation vector along the flow lines of the velocity field is determined by

$$(D_\tau)^2 S^\mu = R^\mu{}_{\nu\rho\sigma} u^\nu u^\rho S^\sigma \quad . \tag{23.92}$$

We consider the situation where the test particles are initially, in the absence of the gravitational wave, at rest, $u^\mu = (1, 0, 0, 0)$. Then the gravitational wave will lead, to lowest order in the perturbation $h_{\mu\nu}$, to a 4-velocity

$$u^\mu = (1, 0, 0, 0) + \mathcal{O}(h) \quad . \tag{23.93}$$

However, because the Riemann tensor is already of order h, to lowest order the right hand side of the geodesic deviation equation reduces to

$$R^{(1)}_{\mu 00\sigma} = \tfrac{1}{2}\partial_0\partial_0 h_{\mu\sigma} \tag{23.94}$$

(because $h_{0\mu} = 0$). On the other hand, to lowest order the left hand side is just the ordinary time derivative. Thus the geodesic deviation equation becomes (an overdot denoting a t-derivative)

$$\ddot{S}^\mu = \tfrac{1}{2}\ddot{h}^\mu_\sigma\, S^\sigma \quad . \tag{23.95}$$

In particular, we see immediately that the gravitational wave is *transversally polarised*, i.e. the component S^3 of S^μ in the longitudinal direction of the wave is unaffected and

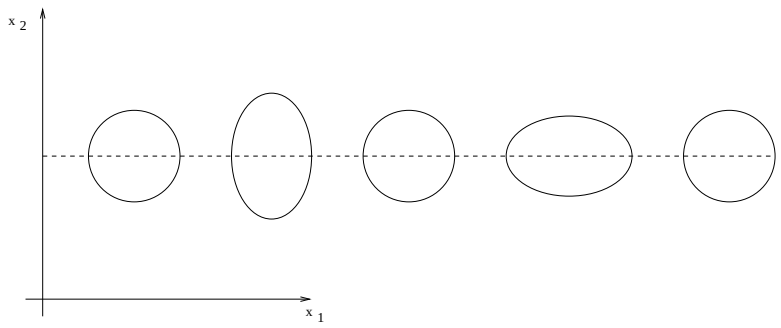


Figure 9: Effect of a gravitational wave with polarisation ϵ_{11} moving in the x^3-direction, on a ring of test particles in the $x^1 - x^2$-plane.

the particles are only disturbed in directions perpendicular to the wave. The movement of the particles in the 1-2 plane is then governed by

$$\ddot{S}^a = \tfrac{1}{2}\ddot{h}^a_{\ b}\, S^b \equiv -(\Omega^2)^a_{\ b}\, S^b \ , \tag{23.96}$$

which is the equation of a 2-dimensional time-dependent harmonic oscillator, giving rise to characteristic oscillating movements of the test particles in the 1-2 plane. With

$$h^a_{\ b} = \mathrm{e}^{\,-i\omega(t\,-\,x^3)}\epsilon^a_{\ b} \ , \tag{23.97}$$

as above, one has

$$(\Omega^2)^a_{\ b} = \tfrac{1}{2}\omega^2 h^a_{\ b} \ , \tag{23.98}$$

and we can consider separately the two cases (1) $\epsilon_{12} = 0$ and (2) $\epsilon_{11} = -\epsilon_{22} = 0$.

1. For $\epsilon_{12} = 0$ one has

$$\begin{aligned} \ddot{S}^1(t) &= -\tfrac{1}{2}\epsilon_{11}\omega^2 \mathrm{e}^{\,-i\omega t} S^1(t)\, \mathrm{e}^{\,i\omega x^3} \\ \ddot{S}^2(t) &= +\tfrac{1}{2}\epsilon_{11}\omega^2 \mathrm{e}^{\,-i\omega t} S^2(t)\, \mathrm{e}^{\,i\omega x^3} \end{aligned} \tag{23.99}$$

   Recalling that ϵ_{ab} is small, the solution to lowest order is simply

$$\begin{aligned} S^1(t) &= (1 + \tfrac{1}{2}\epsilon_{11}\mathrm{e}^{\,-i\omega(t\,-\,x^3)})S^1(0) \\ S^2(t) &= (1 - \tfrac{1}{2}\epsilon_{11}\mathrm{e}^{\,-i\omega(t\,-\,x^3)})S^2(0) \end{aligned} \tag{23.100}$$

   Given the interpretation of S^μ as a separation vector, this means that particles originally separated in the x^1-direction will oscillate back and forth in the x^1-direction and likewise for x^2. A nice (and classical) way to visualise this (see Figure 9) is to start off with a ring of particles in the $1-2$ plane. As the wave passes by the particles will start bouncing in such a way that the ring bounces in the shape of a cross $+$. For this reason, ϵ_{11} is also frequently written as ϵ_+.

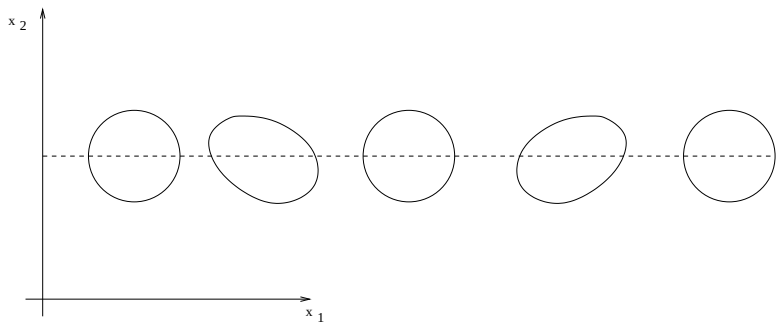


Figure 10: Effect of a gravitational wave with polarisation ϵ_{12} moving in the x^3-direction, on a ring of test particles in the $x^1 - x^2$-plane.

2. If, on the other hand, $\epsilon_{11} = 0$ but $\epsilon_{12} = \epsilon_{21} \neq 0$, then the lowest order solution is

$$\begin{aligned} S^1(t) &= S^1(0) + \tfrac{1}{2}\epsilon_{12} \mathrm{e}^{-i\omega(t-x^3)} S^2(0) \\ S^2(t) &= S^2(0) + \tfrac{1}{2}\epsilon_{12} \mathrm{e}^{-i\omega(t-x^3)} S^1(0) \quad . \end{aligned} \tag{23.101}$$

   This time the deplacement in the x^1-direction is governed by the original deplacement in the x^2-direction and vice-versa, and the ring of particles will bounce in the shape of a $\times$ ($\epsilon_{12} = \epsilon_\times$) - see Figure 10.

3. Of course, one can also construct circularly polarised waves by using

$$\epsilon_{R,L} = \frac{1}{\sqrt{2}}(\epsilon_{11} \pm i\epsilon_{12}) \quad . \tag{23.102}$$

These solutions display the characteristic behaviour of *quadrupole radiation*, and this is something that we might have anticipated on general grounds. First of all, we know from Birkhoff's theorem (see section 24.6) that a spherically symmetric vacuum solution of the Einstein field equations is necessarily static. Thus there can be no radial oscillations, and thus no monopole (s-wave) radiation. Moreover, dipole radiation is due to oscillations of the center of charge. While this is certainly possible for electric charges, an oscillation of the center of mass would violate momentum conservation and is therefore ruled out. Thus the lowest possible mode of gravitational radiation is quadrupole radiation, just as we have found.

23.10 Brief Comments on Production and Energy of Gravitational Waves

Now that we have found the solutions to the vacuum equations, we should include sources and study the production of gravitational waves, characterise the type of radiation that is emitted, estimate the radiated energy etc. However, this is quite a delicate

and both technically and conceptually quite challenging subject, and I will just develop this to the extent that the quadrupole property of the radiation becomes plausible.[66]

In order to study the production of gravitational waves, we need to include sources, i.e. we need to go back to the retarded solution (23.75)

$$\bar{h}_{\mu\nu}(t,\vec{x}) = 4G_N \int d^3y \, \frac{T^{(0)}_{\mu\nu}(t-|\vec{x}-\vec{y}|,\vec{y})}{|\vec{x}-\vec{y}|} \quad . \tag{23.103}$$

At large distances, and if the source does not oscillate too rapidly (the wavelength should be much larger than the size of the source), one can approximate this by

$$\bar{h}_{\mu\nu}(t,\vec{x}) \approx \frac{4G_N}{r} \int d^3y \, T^{ret}_{\mu\nu}(t,\vec{y}) \quad , \tag{23.104}$$

where $r = |\vec{x}|$ and the retarded source is

$$T^{ret}_{\mu\nu}(t,\vec{y}) = T^{(0)}_{\mu\nu}(t-r,\vec{y}) \quad . \tag{23.105}$$

This is the gravitational analogue of the dipole approximation to the multipole expansion in electrodynamics (and, as we will see, here this turns out to be a quadrupole approximation).

Next, since $T^{(0)}_{\mu\nu}$ is conserved, also

$$\bar{h}_{\mu 0} \sim \int d^3y \, T^{(ret)}_{\mu 0} \tag{23.106}$$

is conserved, i.e. time-independent. Therefore, in this approximation the leading $(1/r)$-part of $\bar{h}_{\mu 0}$ will not lead to gravitational waves. We can thus concentrate on the spatial components

$$\bar{h}_{ik}(t,\vec{x}) \approx \frac{4G_N}{r} \int d^3y \, T^{ret}_{ik}(t,\vec{y}) \quad , \tag{23.107}$$

Using the Laue theorem (7.53),

$$\int d^3x \, T^{ik} = \tfrac{1}{2}(\partial_0)^2 \int d^3x \, T_{00} x^i x^k \equiv \tfrac{1}{2}\ddot{Q}^{ik} \quad , \tag{23.108}$$

we thus have

$$\bar{h}_{ik}(t,\vec{x}) \approx \frac{2G_N}{r} \ddot{Q}^{ret}_{ik} \quad , \tag{23.109}$$

where (7.54)

$$Q^{ret}_{ik}(t) = \int d^3x \, \rho^{ret} x_i x_k \tag{23.110}$$

is the quadrupole moment tensor of the retarded energy density $T^{ret}_{00} = \rho^{ret}$. Thus, if the source has a time-dependence

$$\rho(t) \sim \mathrm{e}^{-i\Omega t} \quad , \tag{23.111}$$

[66] See e.g. S. Weinberg, *Gravitation and Cosmology*, B. Schutz, *A first course in general relativity*, or S. Carroll, *Spacetime and Geometry*, and of course C. Misner, K. Thorne, J. Wheeler, *Gravitation* for detailed discussionss.

say (of course, one should in the end take real superpositions of such modes), then

$$\bar{h}_{ik}(t,r) \approx -2G_N\Omega^2 Q^{ret}_{ik}\ \frac{\mathrm{e}^{-i\Omega(t-r)}}{r} \tag{23.112}$$

clearly describes an outgoing spherical wave.

As noted before, the retarded solution is automatically in the harmonic gauge, but it is not yet in the transverse traceless gauge. Transforming the above solution to the transverse traceless gauge, one finds that the (transverse, traceless) components

$$\bar{h}^{TT}_{ab} = h^{TT}_{ab} \tag{23.113}$$

are naturally expressed not in terms of the quadrupole moments Q_{ik} but in terms of the so-called "reduced" (traceless) quadrupole moments

$$\begin{aligned} \mathcal{Q}^{ret}_{ik} &= \int d^3x\ \rho^{ret}(x_i x_k - \tfrac{1}{3}\delta_{ik} r^2) \\ &= Q^{ret}_{ik} - \tfrac{1}{3}\delta_{ik}(Q^{ret})^j{}_j \ . \end{aligned} \tag{23.114}$$

These formulae can now in principle be applied to various specific situations of interest by specifying the source term appropriately.

Finally, one quantity of particular interest is of course the energy radiated away by the source. However, as discussed in section 22.6, the notion of "gravitational energy" or "energy of the gravitational field" is not in general well defined and raises numerous conceptual and technical issues. One might perhaps have hoped that these issues can be completely side-stepped in the linearised theory we are dealing with here, which is after all much more like a standard field theory in Minkowski space. And indeed, several strategies are available, and they all lead to expressions for the energy-density which are of the standard form "quadratic in the derivatives of the fields". For example one can

- proceed by analogy with quadrupole radiation in Maxwell theory,
- use the (Belinfante-improved) Noether energy-momentum tensor of the quadratic Fierz-Pauli action (23.12),
- expand the Einstein equations not only to linear but to quadratic order in the fluctuations $h_{\mu\nu}$ and interpret the quadratic terms as the gravitational contribution to the energy-momentum tensor,
- ...

E.g. in the transverse traceless gauge the Fierz-Pauli action reduces to a "standard" quadratic action

$$\mathcal{L} = -\tfrac{1}{2}(\eta^{\alpha\beta}\partial_\alpha h_{11}\partial_\beta h_{11} + \eta^{\alpha\beta}\partial_\alpha h_{12}\partial_\beta h_{12}) \ . \tag{23.115}$$

For $h_{ab} = h_{ab}(t - x^3)$, say, this gives rise to an energy density and energy flux

$$\Theta_{00} = -\Theta_{03} = (\dot{h}_{11})^2 + (\dot{h}_{12})^2 \sim r^{-2}(\dddot{\mathcal{Q}}^{ret})^2 \quad . \tag{23.116}$$

On the basis of such considerations one might expect or anticipate the total radiated energy to be proportional to something like

$$dE/dt \sim -(\dddot{Q}^{ret})^2 \quad . \tag{23.117}$$

However, dealing with quadratic terms in a linearised theory is somewhat dodgy and not strictly speaking internally consistent. As a consequence, equally plausible strategies may not necessarily lead to equivalent results. Nevertheless, there appears to be some consensus that a formula like this is indeed correct, and more specifically that (with certain approximations and averaging) one has the remarkable formula

$$\frac{dE}{dt} = -\frac{G_N}{5}(\dddot{\mathcal{Q}}^{ret})_{ik}(\dddot{\mathcal{Q}}^{ret})^{ik} \quad . \tag{23.118}$$

While this formula (with its 3rd derivative squared) may look unfamiliar, it is precisely analogous to the corresponding formula for the radiated power of an electric quadrupole in Maxwell theory (also proportional to $\dddot{Q}^2$). The main difference between gravitational and electro-magentic radiation lies in the fact that in the Maxwell case the leading (lowest multipole) contribution arises from dipole radiation, while in the gravitational case the leading contribution is quadrupole radiation.

23.11 Even Briefer Comments on Detection of Gravitational Waves

I will conclude this section with some very general comments on the detection of gravitational waves.

In principle, this ought to be straightforward. In practice, however, because of the extreme weakness of gravitational fields, this is about as far from straightforward as one can possibly imagine. For example, on the basis of the calculations done in section 23.9, one might like to simply try to track the separation of two freely suspended masses (and this is indeed part of the principle of the interferometers I will briefly return to below).

Alternatively, the particles need not be free but could be connected by a solid piece of material. Then gravitational tidal forces will stress the material. If the resonant frequency of this "antenna" equals the frequency of the gravitational wave, this should lead to a detectable oscillation. This is the principle of the so-called *Weber detectors* or *Weber bars* (1966-...). While fine in principle, in practice gravitational waves are extremely weak. To the best of my knowledge, such detectors have not produced conclusive results so far, and other detection techniques are favoured in modern generations of detectors.

More sensitive modern experiments are not fine-tuned to a particular resonant frequency but can in principle detect a continuous range of frequencies. These use detectors based on huge laser Michelson interferometers (arms several kilometers long), e.g. LIGO (Laser Interferometer Gravitational Wave Observatory) and VIRGO. These or their upgrades, or the space-based LISA (Laser Interferometer Space Antenna), are widely expected to have reached sufficient sensitivity to finally directly detect gravitational waves in the next couple of years.[67] **See the 2016 Update at the end of this section!**

However, in spite of the absence of direct evidence for gravitational waves, reassuringly there is indirect (and very compelling) evidence for gravitational waves. A binary system of stars rotating around its common center of mass should radiate gravitational waves (much like electro-magnetic synchroton radiation). For two stars of equal mass M at distance $2r$ from each other, the prediction of General Relativity is that the power radiated by the binary system according to the general formula (23.118) is

$$P = dE/dt = -\frac{2}{5}\frac{G_N^4 M^5}{r^5} \quad . \tag{23.119}$$

This energy loss has actually been observed. In 1974, Hulse and Taylor discovered a binary system, affectionately known as PSR1913+16, in which both stars are small neutron stars, both roughly of solar mass, one of them being a *pulsar*, a rapidly spinning neutron star. The period of the orbit is only eight hours, and the fact that one of the stars is a pulsar provides a highly accurate clock with respect to which a change in the period as the binary loses energy can be measured. The observed value is in good agreement with the theoretical prediction for loss of energy by gravitational radiation and Hulse and Taylor were rewarded for these discoveries with the 1993 Nobel Prize. These observations have been confirmed and refined by the discovery and precise measurements and observations of other (even more extreme) binary systems.

Other situations in which gravitational waves might be either detected directly or inferred indirectly are extreme situations like gravitational collapse (supernovae) or matter orbiting black holes.

2016 Update[68]

On February 11, 2016, the LIGO Scientific Collaboration and Virgo Collaboration teams announced that they had made the first observation of gravitational waves, originating from a pair of merging black holes using the Advanced LIGO detectors. On June 15, 2016, a second detection of gravitational waves from coalescing black holes was announced.

[67] If you want (your PC) to contribute to the search for (continous) gravitational wave sources such as pulsars, take a look at the Einstein@Home project at `http://einstein.phys.uwm.edu/`.

[68] Quoted from `https://en.wikipedia.org/wiki/Gravitational_wave`. For more details, see e.g. `https://en.wikipedia.org/wiki/First_observation_of_gravitational_waves`.

25 Particle and Photon Orbits in the Schwarzschild Geometry

We have now accumulated all the ingredients to address an issue of fundamental interest in general relativity, namely the study of planetary orbits and lightrays in the gravitational field of the sun, i.e. the properties of timelike and null geodesics of the Schwarzschild geometry.

We shall see that, by making good use of the symmetries of the problem, we can reduce the (initially somewhat complicated looking) set of coupled 2nd order geodesic equations to a single 1st order differential equation in one variable, analogous to that for a one-dimensional particle moving in a particular ("effective") potential. Solutions to this equation can then readily be discussed qualitatively and also quantitatively (analytically).

In order to be able to read this section as a concrete ("real-life", so to speak) application and illustration of the general formalism for geodesics introduced in sections 2 and 3 (without knowledge of all the intervening material on tensor analysis and the Einstein equations, say), this section starts with a brief summary of the key features and main properties of the Schwarzschild metric that we will make use of in the following and that were established in detail in the previous section 24.

25.1 Schwarzschild Metric: Summary of its Key Properties

In standard Schwarzschild coordinates (t, r, θ, ϕ), the Schwarzschild metric is

$$ds^2 = -f(r)dt^2 + f(r)^{-1}dr^2 + r^2 d\Omega^2 \quad , \tag{25.1}$$

where $d\Omega^2 = d\theta^2 + \sin^2\theta d\phi^2$ is the line element of the standard (rotationally invariant) metric on the 2-sphere S^2, and $f(r)$ is the function

$$f(r) = 1 - \frac{2m}{r} \quad , \tag{25.2}$$

with m an integration constant (with the dimension of length).

This metric has the following origin / interpretation / properties:

1. As shown in section 24.2, via a suitable choice of coordinates, any metric that is static and spherically symmetric (and thus describes a time-independent spherically symmetric gravitational field) can be put into the form

   $$ds^2 = -A(r)dt^2 + B(r)dr^2 + r^2 d\Omega^2 \quad , \tag{25.3}$$

   for some (positive) functions $A(r)$ and $B(r)$.

   Provided that $A(r) \to 1$ and $B(r) \to 1$ (sufficiently rapidly) as $r \to \infty$, this metric is asymptotically Minkowskian (usually referred to as *asymptotically flat*).

An anymptotically flat metric can be considered to describe the gravitational field of an isolated/localised object in space.

In particular, therefore, the Schwarzschild metric is static, spherically symmetric and asymptotically flat.

2. What singles out the Schwarzschild metric among all such metrics is the crucial fact (established in section 24.3) that

   The Schwarzschild metric is the unique static, spherically symmetric and asymptotically flat solution of the vacuum Einstein field equations!

   [This statement can be sharpened somewhat: the assumption that the metric is static turns out to be unnecessary - see the discussion of Birkhoff's theorem in section 24.6.]

   Here "vacuum Einstein field equations" refers to the field equations for the metric in the absence of matter or in matter-free regions of space-time. Thus the physical interpretation of the Schwarzschild metric is that it describes the gravitational field in the exterior region $r > r_0$ of a static spherically symmetric star of radius r_0.

3. Comparison of the Schwarzschild

$$g_{00} = -\left(1 - \frac{2m}{r}\right) \tag{25.4}$$

   with the Newtonian limit, where one has (cf. the discussion in section 3.3)

$$g_{00} = -\left(1 + \frac{2\phi}{c^2}\right) \tag{25.5}$$

   shows that this physical interpretation is perfectly compatible with the Newtonian limit, where the appropriate Newtonian potential ϕ for a star with mass M has the form

$$\phi(r) = -\frac{G_N M}{r} \quad . \tag{25.6}$$

   In particular, the physical interpretation of the integration constant m appearing in the Schwarzschild solution is that it is related to the mass M by

$$m = \frac{G_N M}{c^2} \quad , \tag{25.7}$$

   The Schwarzschild metric is thus the extension of the Newtonian $1/r$-potential, the unique exact spherically symmetric solution of the Newtonian "vacuum" field equation

$$\Delta\phi(r) = 0 \quad \text{for} \quad r > r_0 \tag{25.8}$$

   vanishing as $r \to \infty$, to an exact solution of the general relativistic vacuum field equations for the metric.

4. Here is what we can say about the coordinates:

   - The Schwarzschild coordinates are adapted to the symmetries of the metric in the sense that invariance under time translations $t \to t + b$ and rotations of the angular coordinates (θ, ϕ) is manifest, or as manifest as possible.
   - The coordinates (θ, ϕ) have the standard interpretation and range as coordinates on the 2-sphere.
   - Even though r is not a measure of proper radial distance, it does have a clean geometric interpretation, namely that r is such that spheres of radius r are those that have area $4\pi r^2$.
   - A priori the range of r is limited by the requirement $r > r_0$ (because the Schwarzschild metric only describes the gravitational field to the exterior of the star). One might also be concerned about the behaviour of the metric as $r \to 2m$ but, as recalled below, for standard astrophysical objects like planets or stars one has $r_0 \gg 2m$, so that the condition $r > r_0$ automatically excludes this potentially dangerous region.
   - Since the metric is independent of t, the range of the coordinate t can be taken to be $(-\infty, +\infty)$. Up to affine transformations $t \to at + b$, t is also characterised by the fact that the time-translation symmetry is realised by translations $t \to t + b$ of the coordinate t (and not some more complicated transformation of the coordinates).
   - The physical interpretation of t is that it is the proper time of a Minkowskian observer "at infinity" (where $f(r) \to 1$) and that, up to a constant conversion factor $\sqrt{f(r)} < 1$ it is the proper time of observers hovering at fixed values of the spatial coordinates (r, θ, ϕ).

   Since symmetries are manifest, and the coordinates have a simple interpretation, these coordinates are ideally suited for studying the geodesics, i.e. the motion of particles and light, in the gravitational field of an object like the sun, with $r_0 \gg 2m$, and we will embark on this below.

   However, for other purposes other coordinates may be more useful or even necessary and can provide additional insight. We will look at this in quite some detail in subsequent sections.

5. As shown in (24.40), the apparently problematic behaviour of the Schwarzschild metric as $r \to 2m$ is irrelevant for applications to the solar system (or the description of the gravitational field of other standard benign astrophysical objects), since for such objects m is much less than the radius of the object, $m \ll r_0$ (while the Schwarzschild metric only applies to the exterior region $r > r_0$).

   E.g. for the earth m is of the order of a centimeter, while for an object with a mass approximately that of the sun it is of the order of a couple of kilometers.

However, we will take a closer look at what happens at (and beyond) $r = 2m$ in section 26 and subsequent sections, when exploring the physics of black holes.

25.2 Symmetries and Effective Potential for the Kepler Problem

To set the stage for the subsequent discussion of the motion of particles (and light) in the Schwarzschild geometry, let us very briefly recall how one goes about this in the corresponding (Kepler) problem in Newtonian mechanics, namely the motion of particles in the potential (25.6).

The crucial point here is to make judicious use of the symmetries of the problem to reduce the problem of the 3-dimensional motion of a particle in the spherically symmetric potential $V(r)$ to that of a 1-dimensional radial motion in a suitable *effective potential*. Then the problem has been reduced to that of solving a single ordinary differential equation (and this can be done in closed form for the Kepler problem), and even without knowledge of the exact solutions the qualitative behaviour of the orbits in the Schwarzschild geometry can essentially be determined "by inspection" from drawing the effective potential.

The steps are the following:

1. Because of time-translation invariance, the Newtonian energy E_N of a particle moving in an arbitary time-independent potential $V(\vec{x})$ is conserved. This energy is the sum of the kinetic energy and the potential energy. Passing straightaway to spherical coordinates (r, θ, ϕ), and just looking at a particle of unit mass (since we already know that in the case of the Kepler problem the mass of the particle will drop out of all equations anyway) this energy is

$$E_N = \frac{1}{2}\left(\dot{r}^2 + r^2\dot{\theta}^2 + r^2\sin^2\theta\dot{\phi}^2\right) + V(r,\theta,\phi) \quad , \tag{25.9}$$

   where (in this Newtonian context) an overdot refers to an ordinary time-derivative, $\dot{r} = dr/dt$ etc.

2. When the potential is purely radial (spherically symmetric), $V = V(r)$, the problem is invariant under spatial rotations, and correspondingly angular momentum $\vec{L}$ is conserved. Thus motion takes place in the plane orthogonal to $\vec{L}$. For any choice of $\vec{L}$ one can choose the coordinate system such that $\vec{L}$ points into the x^3-direction or, equivalently, such that the motion is restricted to the equatorial plane $\theta = \pi/2$,

$$\theta = \pi/2 \quad \Rightarrow \quad \vec{L} = (0,0,L) \quad , \quad L = r^2\dot{\phi} \ . \tag{25.10}$$

3. Plugging this back into the energy, one finds

$$E_N = \frac{1}{2}\dot{r}^2 + \frac{L^2}{2r^2} + V(r) \equiv \frac{1}{2}\dot{r}^2 + V_{eff}(r) \quad , \tag{25.11}$$

where $V_{eff}(r)$ is known as the *effective potential*. It governs the radial motion of the particle in the potential $V(r)$ and differs from $V(r)$ by the usual centrifugal / angular momentum barrier term (which acts as a potential for the corresponding centrifugal "pseudo-force").

Thus the symmetries have allowed us to reduce the original set of 3 2nd-order differential equations for the coordinates $\vec{x} = \vec{x}(t)$ to a single ordinary 1st-order differential equation for $r = r(t)$,

$$\frac{dr}{dt} = \pm\sqrt{2E_N - 2V_{eff}(r)} \ . \tag{25.12}$$

4. To obtain an equation for the *orbit*, i.e. for $r = r(\phi)$, one can use

$$\frac{dr}{d\phi} = \frac{\dot{r}}{\dot{\phi}} = \pm\frac{r^2}{L}\sqrt{2E_N - 2V_{eff}(r)} \ . \tag{25.13}$$

As is well known, for the Kepler problem, with $V(r) \sim 1/r$, this equation can be solved in closed form in terms of conical sections (this is recalled at the beginning of section 25.8).

25.3 Symmetries and Effective Potential for Schwarzschild Geodesics

Our aim is now to try to develop a similarly efficient strategy to deal with the corresponding general relativistic problem, i.e. for describing the solutions of the geodesic equation in the Schwarzschild geometry.

A convenient starting point in general for discussing geodesics is, as I stressed repeatedly, the Lagrangian $\mathcal{L} = g_{\mu\nu}\dot{x}^\mu\dot{x}^\nu$. For the Schwarzschild metric this is

$$\mathcal{L} = -(1-2m/r)\dot{t}^2 + (1-2m/r)^{-1}\dot{r}^2 + r^2(\dot{\theta}^2 + \sin^2\theta\dot{\phi}^2) \ , \tag{25.14}$$

where $2m = 2MG_N/c^2$.

Rather than writing down and solving the (second order) geodesic equations, we will make use of the conserved quantities associated with the continuous symmetries of the Lagrangian (or the metric). After all, conserved quantities correspond to first integrals of the equations of motion and if there are a sufficient number of them (there are) we can directly reduce the second order differential equations to first order equations.

1. Spherical Symmetry and Conserved Angular Momentum

   Since the gravitational field is spherically symmetric, or isotropic, there is conservation of angular momentum. Thus, exactly as in the Newtonian problem, the orbits of the particles or planets are planar. Without loss of generality, we can choose our coordinates in such a way that this plane is the equatorial plane $\theta = \pi/2$, so in particular $\dot{\theta} = 0$ at all times.

In case this is not obvious (even though it should be), here are two explicit ways to establish this:

- One can certainly choose one's coordinates in such a way that at some initial time $\tau = \tau_0$ one has $\theta(\tau_0) = \pi/2$ and that the angular velocity $\dot\theta(\tau_0) = 0$. It then follows from the Euler-Lagrange equations of motion for $\theta(\tau)$ that they are solved at all times by $\theta(\tau) = \pi/2$, $\dot\theta(\tau) = 0$.
- If you already know about Killing vectors (which made their first appearance in this context in sections 3.2), you may find it somewhat more insightful to use the conserved angular momenta

$$L_{(a)} = g_{\alpha\beta}\dot x^\alpha V^\beta_{(a)} \tag{25.15}$$

associated to the Killing vectors $V_{(a)}$ (9.55),

$$\begin{aligned} L_{(1)} &= r^2\left(-\cos\phi\,\dot\theta + \cot\theta\sin\phi(\sin^2\theta\,\dot\phi)\right) \\ L_{(2)} &= r^2\left(+\sin\phi\,\dot\theta + \cot\theta\cos\phi(\sin^2\theta\,\dot\phi)\right) \\ L_{(3)} &= r^2\sin^2\theta\,\dot\phi \ , \end{aligned} \tag{25.16}$$

and to use spherical symmetry to rotate the $L_{(a)}$ into the form

$$(L_{(1)}, L_{(2)}, L_{(3)}) = (0, 0, L) \ . \tag{25.17}$$

Using the explicit expressions (25.16), it is straightforward to see that $L_{(1)} = L_{(2)} = 0$ implies $\theta = \pi/2$, $\dot\theta = 0$.

Either way, we have fixed the direction of $\vec L$, the motion is now restricted to the equatorial plane, with

$$\theta = \pi/2 \quad \Rightarrow \quad \vec L = (0, 0, L) \ , \tag{25.18}$$

and the residual Lagrangian to deal with is

$$\mathcal{L} = -(1 - 2m/r)\dot t^2 + (1 - 2m/r)^{-1}\dot r^2 + r^2\dot\phi^2 \ . \tag{25.19}$$

The magnitude L (or z-component of $\vec L$) of the angular momentum (per unit rest mass) is now the conserved quantity

$$L = r^2\dot\phi \tag{25.20}$$

associated to the cyclic variable ϕ of the residual Lagrangian,

$$\frac{\partial\mathcal{L}}{\partial\phi} = 0 \Rightarrow \frac{d}{d\tau}\frac{\partial\mathcal{L}}{\partial\dot\phi} = 0 \ . \tag{25.21}$$

2. Time Translation Invariance and Conserved Energy

   The Lagrangian (both the original Lagrangian and the reduced Lagrangian) is invariant under translations of t, which is also a cyclic variable,

$$\frac{\partial \mathcal{L}}{\partial t} = 0 \Rightarrow \frac{d}{d\tau}\frac{\partial \mathcal{L}}{\partial \dot{t}} = 0 \ . \tag{25.22}$$

   This gives rise to the conserved energy (with a conventional choice of normalisation)

$$E = (1 - 2m/r)\dot{t} \ . \tag{25.23}$$

   Calling L the angular momentum (per unit rest mass) required no further justification, but let me pause to explain in what sense E is an energy (per unit rest mass).

   - On the one hand, it is the conserved quantity (3.7) associated to time-translation invariance. As such, it certainly deserves to be called the energy.
   - It is moreover true that for a particle at infinity ($r \to \infty$) E is just the special relativistic energy $E = \gamma(v_\infty)c^2$, with $\gamma(v) = (1 - v^2/c^2)^{-1/2}$ the usual relativistic γ-factor, and v_∞ the coordinate velocity dr/dt at infinity. This can be seen in two ways. First of all, for a particle that reaches $r = \infty$, the constant E can be determined by evaluating it at $r = \infty$. It thus follows from the definition of E that

$$E = \dot{t}_\infty \ . \tag{25.24}$$

     In Special Relativity, the relation between proper and coordinate time is given by (setting $c = 1$ again)

$$d\tau = \sqrt{1 - v^2}dt \quad \Rightarrow \quad \dot{t} = \gamma(v) \ , \tag{25.25}$$

     suggesting the identification

$$E = \gamma(v_\infty) \qquad (E = \gamma(v_\infty)c^2 \quad \text{if} \quad c \neq 1) \tag{25.26}$$

     Another argument for this identification will be given below, once we have introduced the effective potential.

3. τ-Translation Invariance and the Conserved Lagrangian $\mathcal{L}$

   As we have seen in section 2.5 (and again in section 5.8), there is also always one more integral of the geodesic equation, namely $\mathcal{L}$ itself,

$$\frac{d}{d\tau}\mathcal{L} = 2g_{\mu\nu}\dot{x}^\mu D_\tau \dot{x}^\nu = 0 \tag{25.27}$$

   (this can be interpreted as the conserved "Hamiltonian" associated with the invariance of $\mathcal{L}$ under translations of the affine parameter). Thus we set

$$\mathcal{L} = \epsilon \ , \tag{25.28}$$

where $\epsilon = -1$ for timelike geodesics and $\epsilon = 0$ for null geodesics. We thus have

$$-(1-2m/r)\dot{t}^2 + (1-2m/r)^{-1}\dot{r}^2 + r^2\dot{\phi}^2 = \epsilon \ , \tag{25.29}$$

Putting everything together, we can now express $\dot{t}$ and $\dot{\phi}$ in terms of the conserved quantities E and L to obtain a first order differential equation for r alone, namely

$$-(1-2m/r)^{-1}E^2 + (1-2m/r)^{-1}\dot{r}^2 + \frac{L^2}{r^2} = \epsilon \ . \tag{25.30}$$

Multiplying by $(1-2m/r)/2$ and rearranging the terms, one obtains

$$\frac{E^2+\epsilon}{2} = \frac{\dot{r}^2}{2} + \epsilon\frac{m}{r} + \frac{L^2}{2r^2} - \frac{mL^2}{r^3} \ . \tag{25.31}$$

Now this equation is of the familiar Newtonian form

$$E_{eff} = \frac{\dot{r}^2}{2} + V_{eff}(r) \ , \tag{25.32}$$

with

$$\begin{aligned} E_{eff} &= \frac{E^2+\epsilon}{2} \\ V_{eff}(r) &= \epsilon\frac{m}{r} + \frac{L^2}{2r^2} - \frac{mL^2}{r^3} \ , \end{aligned} \tag{25.33}$$

describing the energy conservation for the 1-dimensional motion in an effective potential. Except for $t \to \tau$, this is exactly the same as the Newtonian equation of motion in a potential

$$V(r) = \epsilon\frac{m}{r} - \frac{mL^2}{r^3} \ , \tag{25.34}$$

the effective angular momentum term $L^2/r^2 = r^2\dot{\phi}^2$ arising, as usual, from the change to polar coordinates. As in the corresponding Newtonian one-dimensional (radial) problem, the qualitative behaviour of the orbits in the Schwarzschild geometry can thus essentially be determined "by inspection".

Given that in principle we started off with the four coupled non-linear geodesic differential equations, this is an enormous and enormously useful simplification, and the main result of this section.

REMARKS:

1. In particular, for $\epsilon = -1$, the general relativistic motion (as a function of τ) is exactly the same as the Newtonian motion (as a function of t) in the potential

$$\epsilon = -1 \ \Rightarrow \ V(r) = -\frac{m}{r} - \frac{mL^2}{r^3} \ . \tag{25.35}$$

The first term is just the ordinary Newtonian potential, so the second term is apparently a general relativistic correction. We will later on treat this as a perturbation but note that the above is an exact result, not an approximation (so, for example, there are no higher order corrections proportional to higher powers of m/r). We expect observable consequences of this general relativistic correction because many properties of the Newtonian orbits (Kepler's laws) depend sensitively on the fact that the Newtonian potential is precisely $\sim 1/r$.

2. In order to see that the new 3rd term really is a (very) small correction to the Newtonian effective potential, note first of all that, restoring a factor of c^2 in the first (Newtonian) potential term, we can write the effective potential as

$$V_{eff}(r) = -\frac{G_N M}{r} + \frac{L^2}{2r^2} - \frac{m}{r}\frac{L^2}{r^2} \quad . \tag{25.36}$$

For the planetary orbits, there is a balance between the attractive 1st and repulsive 2nd term, so these two terms are of the same order of magnitude. The 3rd term is then smaller than these by a factor (m/r). Since r (the radial coordinate of the planet) is much larger than r_0 (the radius of the sun) and, as we had already seen before, for standard astrophysical objects like the sun the radius r_0 is much larger than the Schwarzschild radius $2m$, we have

$$r \gg r_0 \gg m \quad \Rightarrow \quad \frac{m}{r} \ll 1 \quad . \tag{25.37}$$

Thus, the general relativistic corrction term $\sim mL^2/r^3$ indeed provided a very small (but important) correction to the Newtonian potential.

3. Looking at the equation for $\epsilon = -1$,

$$\tfrac{1}{2}\dot{r}^2 + V_{eff}(r) = \tfrac{1}{2}(E^2 - 1) \tag{25.38}$$

and noting that $V_{eff}(r) \to 0$ for $r \to \infty$, we can read off that for a particle that reaches $r = \infty$ we have the relation

$$\dot{r}_\infty^2 = E^2 - 1 \quad . \tag{25.39}$$

This implies, in particular, that for such (scattering) trajectories one necessarily has $E \geq 1$, with $E = 1$ corresponding to a particle initially or finally at rest at infinity. For $E > 1$ the coordinate velocity at infinity can be computed from

$$v_\infty^2 = \frac{\dot{r}_\infty^2}{\dot{t}_\infty^2} \quad . \tag{25.40}$$

Using (25.24) and (25.39), one finds

$$v_\infty^2 = \frac{E^2 - 1}{E^2} \quad \Leftrightarrow \quad E = (1 - v_\infty^2)^{-1/2} \quad , \tag{25.41}$$

thus confirming the result claimed in (25.26).

4. From the effective potential we can deduce the radial geodesic equation of motion, namely

$$\epsilon = -1 \quad \Rightarrow \quad \ddot{r} = -V'_{eff}(r) = -\frac{m}{r^2} + (r - 3m)\frac{L^2}{r^4} \ . \tag{25.42}$$

5. For null geodesics, $\epsilon = 0$, on the other hand, the Newtonian part of the potential is zero, as one might expect for massless particles, but in General Relativity a photon with $L \neq 0$ feels a non-trivial potential

$$\begin{aligned} \epsilon = 0 \quad \Rightarrow \quad V(r) &= -\frac{mL^2}{r^3} \\ V_{eff}(r) &= f(r)\frac{L^2}{2r^2} \ . \end{aligned} \tag{25.43}$$

6. It will also turn out to be useful to have the full radial part of the null geodesic equation at our disposal. As in the timelike case above, this can simply be read off from the effective potential and one finds

$$\epsilon = 0 \quad \Rightarrow \quad \ddot{r} = -V'_{eff}(r) = \frac{L^2}{r^4}(r - 3m) \ . \tag{25.44}$$

Obviously something potentially interesting is happening at $r = 3m$, provided the star is sufficiently small so that its radius $r_0 < 3m$ of course - see section 25.10). Also note that this differs from its timelike counterpart (25.42) precisely by the absence of the Newtonian term $-m/r^2$.

7. Finally, we note that the success of the above analysis relied only on the symmetries of the metric, not on the fact that the particular metric we were looking at satisfies the vacuum Einstein equations. It is indeed reasonably straightforward to generalise the preceding analysis to arbitrary static spherically symmetric metrics. As mentioned before, among these general static spherically symmeric metrics the class of metrics (24.75)

$$ds^2 = -f(r)dt^2 + f(r)^{-1}dr^2 + r^2 d\Omega^2 \tag{25.45}$$

is of particular importance and interest. For these metrics one finds that the geodesic equation can still be reduced to the effective potential form (25.32), with E_{eff} precisely as in (25.33), and V_{eff} now given by

$$V_{eff}(r) = -\epsilon\phi(r) + \frac{L^2}{2r^2} + \phi(r)\frac{L^2}{r^2} \ , \tag{25.46}$$

where we have written $f(r)$ in terms of the corresponding "Newtonian" potential $\phi(r)$ as

$$f(r) = 1 + 2\phi(r) \ . \tag{25.47}$$

For the Schwarzschild metric one has $\phi(r) = -m/r$ and (25.46) reduces to (25.33).

25.4 Caveat: Effective Potential versus Routhian

Because this tends to cause some confusion, not just among students but also occasionally in the literature, before turning to a more detailed analysis of the consequences of the equations we have just derived, I want to insert a cautionary remark, a *caveat*, regarding the above derivation of the effective potential equation:

Namely, starting from the Lagrangian $\mathcal{L}$ (25.14), considerations about spherical symmetry led us to choose $\theta = \pi/2$, and imposing this condition in the Lagrangian we were led to the reduced Lagrangian in (25.19). This is legitimate, since such *holonomic* constraints can always be inserted into the Lagrangian itself (this is one of the main virtues and advantages of the Lagrangian formalism).

Subsequently we used the conserved quantities E and L associated to the cyclic variables t and ϕ in order to eliminate from the Lagrangian the quantities $\dot{t}$ and $\dot{\phi}$. Here one needs to be more careful. In general such non-holonomic constraints like $r^2\dot{\phi} = L$ cannot be inserted into the Lagrangian to obtain a (reduced) Lagrangian from which the equations of motion for the remaining variables (here r) can be obtained as the Euler-Lagrange equations. One correct way to do this is to pass to what is known as the *Routhian*, a partial Legendre transform of the Lagrangian on the cyclic variables.

As a simple example, consider the Lagrangian of a (unit mass) free particle in 2 dimensions, expressed in polar coordinates. Its Lagrangian is

$$\mathcal{L} = \tfrac{1}{2}(\dot{r}^2 + r^2\dot{\phi}^2) \ . \tag{25.48}$$

The variable ϕ is cyclic, leading to the conserved angular momentum

$$p_\phi = r^2\dot{\phi} = L \ , \tag{25.49}$$

and the equation of motion for r is the Euler-Lagrange equation

$$\ddot{r} = r\dot{\phi}^2 \ . \tag{25.50}$$

It is legitimate to eliminate $\dot{\phi}$ in this equation, leading to the centrifugal force equation

$$\ddot{r} = L^2/r^3 \ . \tag{25.51}$$

Had one eliminated $\dot{\phi}$ in the Lagrangian instead, one would have found the reduced Lagrangian and resulting Euler-Lagrange equations

$$\mathcal{L} \to \tfrac{1}{2}(\dot{r}^2 + L^2/r^2) \quad \Rightarrow \quad \ddot{r} = -L^2/r^3 \qquad \text{(wrong!)} \ . \tag{25.52}$$

However, even though you might have gained that impression, this is of course not what we were doing in our derivation of the radial effective potential equation. There we were using this elimination in conjunction with the condition that $\mathcal{L}$ is constant on

solutions, to obtain a first-order equation for r (the effective potential equation), and this *is* legitimate. This is a 1st integral of the 2nd order radial equation, the 2nd order equations for r follow from this simply by differentiating with respect to proper time τ, and *not* by treating $\dot{r}^2/2 + V_{eff}(r)$ as a Lagrangian and looking at its Euler-Lagrange equations.

And finally, just to illustrate the claim about the Routhian in the above example, the partial Legendre transform with respect to the cyclic variable ϕ of $\mathcal{L}$ is (expressing $\dot{\phi}$ in terms of p_ϕ)

$$\mathcal{R} = \mathcal{L} - p_\phi \dot{\phi} = \tfrac{1}{2}(\dot{r}^2 - p_\phi^2/r^2) \to \tfrac{1}{2}(\dot{r}^2 - L^2/r^2) \tag{25.53}$$

(note the sign flip compared with the "wrong" Lagrangian above). This is the correct reduced Lagrangian, whose Euler-Lagrange equations give rise to the correct radial equation of motion.

25.5 Equation for the Shape of the Orbit

Typically, one is primarily interested in the shape of an orbit, that is in the radius r as a function of ϕ, $r = r(\phi)$, rather than in the dependence of, say, r on some extraterrestrial's proper time τ. In this case, the above mentioned difference between t (in the Newtonian theory) and τ (here) is irrelevant: In the Newtonian theory one uses $L = r^2 d\phi/dt$ to express t as a function of ϕ, $t = t(\phi)$ to obtain $r(\phi)$ from $r(t)$. In General Relativity, one uses the analogous equation $L = r^2 d\phi/d\tau$ to express τ as a function of ϕ, $\tau = \tau(\phi)$. Hence the shapes of the General Relativity orbits are precisely the shapes of the Newtonian orbits in the potential (25.34). Thus we can use the standard methods of Classical Mechanics to discuss these general relativistic orbits and of course this simplifies matters considerably.

To obtain r as a function of ϕ we proceed as indicated above. Thus we use

$$\left(\frac{dr}{d\phi}\right)^2 = \frac{\dot{r}^2}{\dot{\phi}^2} \tag{25.54}$$

to combine (25.32),

$$\dot{r}^2 = 2E_{eff} - 2V_{eff}(r) \ , \tag{25.55}$$

and (25.20),

$$\dot{\phi}^2 = \frac{L^2}{r^4} \tag{25.56}$$

into

$$\frac{r'^2}{r^4} L^2 = 2E_{eff} - 2V_{eff}(r) \tag{25.57}$$

where a prime denotes a ϕ-derivative.

In the examples to be discussed below, we will be interested in the angle $\Delta\phi$ swept out by the object in question (a planet or a photon) as it travels along its trajectory between

the farthest distance r_2 from the star (sun) ($r_2 = \infty$ for scattering trajectories) and the position of closest approach to the star r_1 (the perihelion or, more generally, if we are not talking about our own solar system, periastron), and back again,

$$\Delta\phi = 2\int_{r_1}^{r_2} \frac{d\phi}{dr} dr \quad . \tag{25.58}$$

In the Newtonian case, these integrals can be evaluated in closed form. With the general relativistic correction term, however, these are elliptic integrals which cannot be expressed in closed form. A perturbative evaluation of these integrals (treating the exact general relativistic correction as a small perturbation) also turns out to be somewhat delicate since e.g. the limits of integration depend on the perturbation.

It is somewhat simpler to deal with this correction term not at the level of the solution (integral) but at the level of the corresponding differential equation. As in the Kepler problem, it is convenient to make the change of variables

$$u = \frac{1}{r} \qquad u' = -\frac{r'}{r^2} \quad . \tag{25.59}$$

Then (25.57) becomes

$$u'^2 = L^{-2}(2E_{eff} - 2V_{eff}(r)) \quad . \tag{25.60}$$

Upon inserting the explicit expression for the effective potential, this becomes

$$u'^2 + u^2 = \frac{E^2 + \epsilon}{L^2} - \frac{2\epsilon m}{L^2}u + 2mu^3 \quad . \tag{25.61}$$

This can be used to obtain an equation for $d\phi(u)/du = u'^{-1}$, leading to

$$\Delta\phi = 2\int_{u_2}^{u_1} \frac{d\phi}{du} du \quad . \tag{25.62}$$

Differentiating (25.61) once more, one finds

$$u'(u'' + u) = u'(-\frac{\epsilon m}{L^2} + 3mu^2) \quad . \tag{25.63}$$

Thus either $u' = 0$, which corresponds to a circular orbit of constant radius (i.e. the solution is $u(\phi) = u_0$ or $r(\phi) = r_0$)), and this is not only a trivial but also an irrelevant solution since neither the planets nor the photons of interest to us travel on circular orbits, or

$$u'' + u = -\frac{\epsilon m}{L^2} + 3mu^2 \quad . \tag{25.64}$$

The unperturbed ("Newtonian") equation is the linear equation obtained by dropping the last term, and its solutions $u_0(\phi)$,

$$u_0'' + u_0 = -\frac{\epsilon m}{L^2} \quad , \tag{25.65}$$

give the familiar conical sections for $\epsilon = -1$ (cf. section 25.8) and straight lines for $\epsilon = 0$ (section 25.11). Treating the last term as a small perturbation, one can then expand the solution u as

$$u = u_0 + u_1 \tag{25.66}$$

with u_1 small, leading to the linear equation

$$u_1'' + u_1 = 3mu_0^2 \tag{25.67}$$

for the perturbation u_1, with u_0 providing the source term (or external force on the harmonic oscillator).

Equations (25.65) and (25.67) are the equations that we will study below to determine the perihelion shift and the bending of light by a star. In the latter case, which is a bit simpler, I will also sketch two other derivations of the result, based on different perturbative evaluations of the elliptic integral.

25.6 Timelike Geodesics

We will first try to gain a qualitative understanding of the behaviour of geodesics in the effective potential

$$V_{eff}(r) = -\frac{m}{r} + \frac{L^2}{2r^2} - \frac{mL^2}{r^3} \quad . \tag{25.68}$$

The standard way to do this is to plot this potential as a function of r for various values of the parameters L and m. The basic properties of $V_{eff}(r)$ are the following:

1. Asymptotically, i.e. for $r \to \infty$, the potential tends to the Newtonian potential,

$$V_{eff}(r) \stackrel{r\to\infty}{\longrightarrow} -\frac{m}{r} \quad . \tag{25.69}$$

2. At the Schwarzschild radius $r_s = 2m$, nothing special happens and the potential is completely regular there,

$$V_{eff}(r = 2m) = -\frac{1}{2} \quad . \tag{25.70}$$

   For the discussion of planetary orbits in the solar system we can safely assume that the radius of the sun is much larger than its Schwarzschild radius, $r_0 \gg r_s$, but the above shows that even for these highly compact objects with $r_0 < r_s$ geodesics are perfectly regular as one approaches r_s. Of course the particular numerical value of $V_{eff}(r = 2m)$ has no special significance because $V(r)$ can always be shifted by a constant.

3. The extrema of the potential, i.e. the points at which $dV_{eff}/dr = 0$, are at

$$mr^2 - L^2 r + 3mL^2 = 0 \quad \Rightarrow \quad r_\pm = (L^2/2m)[1 \pm \sqrt{1 - 12(m/L)^2}] \quad , \tag{25.71}$$

   and the potential has a maximum at r_- and a local minimum at r_+. Thus there are qualitative differences in the shapes of the orbits between $L/m < \sqrt{12}$ and $L/m > \sqrt{12}$.

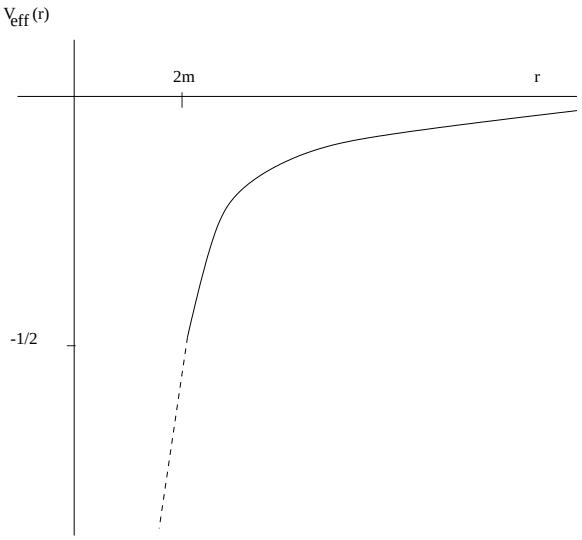


Figure 12: Effective potential for a massive particle with $L/m < \sqrt{12}$. The extrapolation to values of $r < 2m$ has been indicated by a dashed line.

Let us discuss these two cases in turn. When $L/m < \sqrt{12}$, then there are no critical points and the potential looks approximately like that in Figure 12. Note that for the time being we should be careful with extrapolating to values of r with $r < 2m$ because we know that the Schwarzschild metric has a coordinate singularity there. However, the picture (in fact the entire effective potential) turns out to be correct also for $r < 2m$.

From this picture we can read off that there are no bounded orbits for these values of the parameters. Any inward bound particle with $L < \sqrt{12}m$ will continue to fall inwards (provided that it moves on a geodesic). This should be contrasted with the Newtonian situation in which for any $L \neq 0$ there is always the centrifugal barrier reflecting incoming particles since the repulsive term $L^2/2r^2$ will dominate over the attractive $-m/r$ for small values of r. In General Relativity, on the other hand, it is the attractive term $-mL^2/r^3$ that dominates for small r.

Fortunately for the stability of the solar system, the situation is qualitatively quite different for sufficiently large values of the angular momentum, namely $L > \sqrt{12}m$ (see Figure 13).

In that case, there is a minimum and a maximum of the potential. The critical radii correspond to exactly circular orbits, unstable at r_- (on top of the potential) and stable at $r_+ > r_-$ (the minimum of the potential). We will briefly discuss these separately in section 25.7 below.

For given L, for sufficiently large values of E_{eff} a particle will fall all the way down the potential. For $E_{eff} < 0$, there are bound orbits which are not circular and which range between the radii r_1 and r_2, the turning points at which $\dot{r} = 0$ and therefore

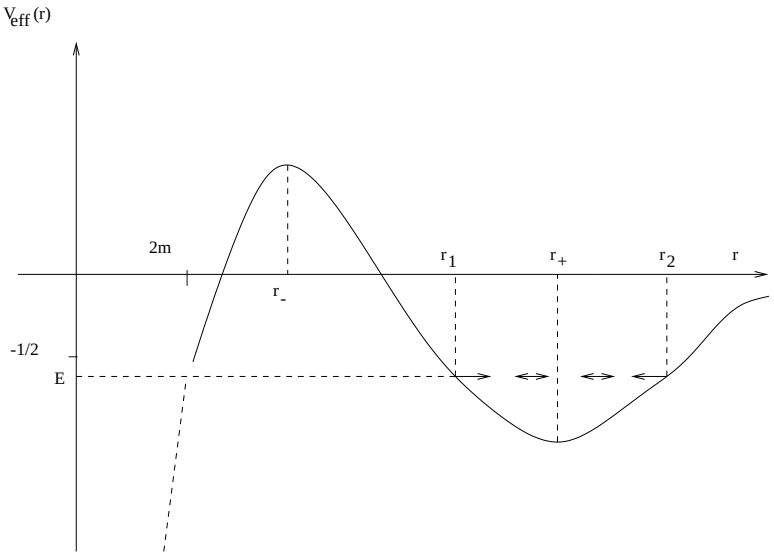


Figure 13: Effective potential for a massive particle with $L/m > \sqrt{12}$. Shown are the maximum of the potential at r_- (an unstable circular orbit), the minimum at r_+ (a stable circular orbit), and the orbit of a particle with $E_{eff} < 0$ with turning points r_1 and r_2.

$E_{eff} = V_{eff}(r_{1,2})$. We will take a closer look at these bound (but not closed) orbits, with their characteristic precession of the perihelion, in section 25.8.

25.7 Some Comments on Circular Timelike Orbits

Recall from above that, provided that $(L/m) > \sqrt{12}$, for a given L there are 2 circular orbits, at the critical points $r_\pm$ of the potential, given by (25.71)

$$r_\pm = (L^2/2m)[1 \pm \sqrt{1 - 12(m/L)^2}] \ , \tag{25.72}$$

For $L \to \sqrt{12}m$ these two circular orbits approach each other, the critical radius tending to $r_\pm \to 6m$. Thus the innermost stable circular orbit (known affectionately as the ISCO in astrophysics) is located at

$$r_{ISCO} = 3r_s = 6m \ . \tag{25.73}$$

This is of course only a relevant quantity when r_{ISCO} lies outside the star, i.e. for $r_0 < 3r_s$ (and thus mainly for black holes or possibly other quite extreme objects like extremely dense neutron stars).

On the other hand, for very large values of L the critical radii are (expand the square root to first order) to be found at

$$(r_+, r_-) \overset{L\to\infty}{\longrightarrow} (L^2/m, 3m) \ . \tag{25.74}$$

Thus letting L range over $[\sqrt{12}m, \infty)$, one sees that circular orbits can occur at any radius $r > 3m$ (for a suitable finetuned choice of angular momentum L). However, such an orbit is stable only for $r > 6m$.

REMARKS:

1. Another way to understand the appearance and significance of the minimal circular radius at $r = 3m$ is to look directly at the radial geodesic equation (25.42)

$$\ddot{r} = -\frac{m}{r^2} + (r - 3m)\frac{L^2}{r^4} \quad . \tag{25.75}$$

   We see that at $r = 3m$ the "centrifugal" term changes sign, and that we can only compensate the first (Newtonian) term on the right-hand side by this term for $r > 3m$.

   We will see in section 25.10 that $r = 3m$ also has a special (and more prominent) significance for *null* geodesics, namely as the location of (unstable) circular null geodesics (the *photon sphere*).

2. From (25.20) one finds that the angular coordinate velocity on a circular orbit $r_\pm$ is

$$\omega_\pm = d\phi/dt = \dot{\phi}/\dot{t} = (L/E)(1 - 2m/r_\pm)/r_\pm^2 \quad . \tag{25.76}$$

   Since this frequency is constant, this immediately gives $\phi = \phi(t)$,

$$\phi(t) = \omega_\pm t + \phi_0 \quad . \tag{25.77}$$

   However, since the quantities $m, E, L, r_\pm$ are not independent of each other, this is not a particularly useful way of writing $\omega_\pm$. First of all, from the effective potential equation and $\dot{r}_\pm = 0$ one finds that the energy E can be written in terms of the other parameters as

$$E^2 = 1 + 2V_{eff}(r_\pm) = (1 - 2m/r_\pm)(1 + L^2/r_\pm^2) \quad . \tag{25.78}$$

   Similarly, from (25.71) L^2 can be written in terms of m and $r_\pm$ as

$$L^2 = mr_\pm^2/(r_\pm - 3m) \quad . \tag{25.79}$$

   Thus one finds that

$$\frac{L^2}{E^2} = \frac{mr_\pm^3}{(r_\pm - 2m)^2} \quad , \tag{25.80}$$

   and the angular velocity (as a function of m and $r_\pm$) is

$$(\omega_\pm)^2 = \frac{m}{r_\pm^3} \quad \Leftrightarrow \quad \omega_\pm^2 r_\pm^3 = m \quad . \tag{25.81}$$

   In particular, for the ISCO one has the characteristic orbital frequency

$$m\omega_{ISCO} = 6^{-3/2} \quad . \tag{25.82}$$

Since ω is inversely proportional to the period of the motion, (25.81) looks very much like Kepler's law which says that the period-squared is proportional to the radius-cubed of an orbit. However, this is a bit of a fluke because this relation is not coordinate-independent but relies on having expressed time and radius in terms of the Schwarzschild coordinates (rather than in terms of proper time along the orbit, say).

25.8 Anomalous Precession of the Perihelia of the Planetary Orbits

Because of the general relativistic correction $\sim 1/r^3$, the bound orbits will not be closed (elliptical). In particular, the position of the perihelion, the point of closest approach of the planet to the sun where the planet has distance r_1, will not remain constant. However, because r_1 is constant, and the planetary orbit is planar, this point will move on a circle of radius r_1 around the sun.

As described in section 25.5, in order to calculate this perihelion shift one needs to calculate the total angle $\Delta\phi$ swept out by the planet during one revolution by integrating this from r_1 to r_2 and back again to r_1, or

$$\Delta\phi = 2\int_{r_1}^{r_2} \frac{d\phi}{dr} dr \quad . \tag{25.83}$$

Rather than trying to evaluate the above integral via some sorcery, we will determine $\Delta\phi$ by analysing the orbit equation (25.64) for $\epsilon = -1$,

$$u'' + u = \frac{m}{L^2} + 3mu^2 \quad . \tag{25.84}$$

In the Newtonian approximation, this equation reduces to that of a displaced harmonic oscillator,

$$u_0'' + u_0 = \frac{m}{L^2} \quad \Leftrightarrow \quad (u_0 - m/L^2)'' + (u_0 - m/L^2) = 0 \quad , \tag{25.85}$$

and the solution is a Kepler ellipse described parametrically by

$$u_0(\phi) = \frac{m}{L^2}(1 + e\cos\phi) \tag{25.86}$$

where e is the eccentricity ($e = 0$ means constant radius and hence a circular orbit). Plugging this back into the Newtonian non-linear 1st-order equation (cf. (25.61))

$$u_0'^2 + u_0^2 = \frac{E^2 - 1}{L^2} + \frac{2m}{L^2}u_0 \quad , \tag{25.87}$$

one finds that the integration constant e is related to the energy by

$$e^2 = 1 + \frac{L^2}{m^2}(E^2 - 1) = 1 + \frac{2L^2}{m^2}E_{eff} \quad . \tag{25.88}$$

In particular, $e^2 < 1$ for bound states (bounded orbits) with $E_{eff} < 0$, and we will concentrate on these orbits. The perihelion (aphelion) is then at $\phi = 0$ ($\phi = \pi$), with

$$r_{1,2} = \frac{L^2}{m}\frac{1}{1 \pm e} \quad . \tag{25.89}$$

Thus the semi-major axis a of the ellipse,

$$2a = r_1 + r_2 \ , \tag{25.90}$$

is

$$a = \frac{L^2}{m}\frac{1}{1-e^2} \ . \tag{25.91}$$

In particular, in the Newtonian theory, one has

$$(\Delta\phi)_0 = 2\int_0^\pi d\phi = 2\pi \ . \tag{25.92}$$

The anomalous perihelion shift due to the effects of General Relativity is thus

$$\delta\phi = \Delta\phi - 2\pi \ . \tag{25.93}$$

In order to determine $\delta\phi$, we now seek a solution to (25.84) of the form

$$u = u_0 + u_1 \tag{25.94}$$

where u_1 is a small deviation. This leads to the equation

$$u_1'' + u_1 = 3mu_0^2 \ . \tag{25.95}$$

The general solution of this inhomogenous differential equation is the general solution of the homogeneous equation (we are not interested in) plus a special solution of the inhomogeneous equation. Noting that this is a forced harmonic oscillator equation, with the frequency of the force commensurate with the frequency of the oscillator, we expect to encounter a resonance phenomenon leading to a non-periodic (and linearly growing) contribution to the solution (just as a special solution of $\ddot{x}(t) + x(t) = \sin t$ is $x(t) = (t/2)\sin t$).

This expectation is indeed borne out. Writing

$$(1 + e\cos\phi)^2 = (1 + \tfrac{1}{2}e^2) + 2e\cos\phi + \tfrac{1}{2}e^2\cos 2\phi \tag{25.96}$$

and noting that

$$\begin{aligned} (\phi\sin\phi)'' + \phi\sin\phi &= 2\cos\phi \\ (\cos 2\phi)'' + \cos 2\phi &= -3\cos 2\phi \end{aligned} \tag{25.97}$$

one sees that a special solution is

$$u_1(\phi) = \frac{3m^3}{L^4}((1 + \tfrac{1}{2}e^2) - \tfrac{1}{6}e^2\cos 2\phi + e\phi\sin\phi) \ . \tag{25.98}$$

The term of interest to us is the third (non-periodic, resonance) term which provides a cumulative non-periodic effect over successive orbits. Focussing on this term, we can write the approximate solution to the orbit equation as

$$u(\phi) \approx \frac{m}{L^2}(1 + e\cos\phi + \frac{3m^2e}{L^2}\phi\sin\phi) \ . \tag{25.99}$$

If the first perihelion is at $\phi = 0$, the next one will be at a point $\Delta\phi = 2\pi + \delta\phi$ close to 2π which is such that $u'(\Delta\phi) = 0$ or

$$\sin\delta\phi = \frac{3m^2}{L^2}(\sin\delta\phi + (2\pi + \delta\phi)\cos\delta\phi) \quad . \tag{25.100}$$

Using that $\delta\phi$ is small, and keeping only the lowest order terms in this equation, one finds the result

$$\delta\phi = \frac{6\pi m^2}{L^2} = 6\pi(\frac{G_N M}{cL})^2 \quad . \tag{25.101}$$

An alternative way to obtain this result is to observe that (25.99) can be approximately written as

$$u(\phi) \approx \frac{m}{L^2}\left(1 + e\cos[(1 - \frac{3m^2}{L^2})\phi]\right) \tag{25.102}$$

From this equation it is manifest that during each orbit the perihelion advances by

$$\delta\phi = 2\pi\frac{3m^2}{L^2} \tag{25.103}$$

$(2\pi(1 - 3m^2/L^2)(1 + 3m^2/L^2) \approx 2\pi)$ in agreement with the above result. In terms of the eccentricity e and the semi-major axis a (25.91) of the elliptical orbit, this can be written as

$$\delta\phi = \frac{6\pi G_N}{c^2}\frac{M}{a(1 - e^2)} \quad . \tag{25.104}$$

Thus general relativity predicts a deviation from the Kepler orbits of the planets, manifesting itself in a precession of the perihelia. As the parameters entering (25.104) are known for the planetary orbits, $\delta\phi$ can be evaluated. Can this also be observed and tested?[78]

At first sight, this seems difficult. Even for Mercury, where this effect is largest (because it has the largest eccentricity) one only finds a $\delta\phi$ of the order of $0,1''$ per revolution. This is of course a tiny effect (1 second, $1''$, is one degree divided by 3600) and not *per se* detectable. However,

1. this effect is cumulative, i.e. after N revolutions one has an anomalous perihelion shift $N\delta\phi$;
2. Mercury has a very short solar year, with about 415 revolutions per century;
3. and accurate observations of the orbit of Mercury go back over more than 200 years.

Thus the cumulative effect is approximately 10^3 $\delta\phi$ and this *is* sufficiently large to be observable in principle. And indeed such an effect is observed (and had for a long time presented a puzzle, an anomaly, for astronomers).

[78] Since I have no independent way of checking them, the following facts and numbers are taken from S. Weinberg, *Gravitation and Cosmology*, section 8.6.

In actual fact, the perihelion of Mercury's orbit, as observed from the Earth, shows a rather significant precession rate of $5601''$ per century, which seems to flatly contradict the Newtonian result that there should be no such precession at all.

However,

- about $5025''$ are due to fact that one is using a non-inertial geocentric coordinate system (precession of the equinoxes)
- $532''$ are due to perturbations of Mercury's orbit caused by the (Newtonian) gravitational attraction of the other planets of the solar system (chiefly Venus, earth and Jupiter).

This much was known prior to General Relativity and left an unexplained anomalous perihelion shift of

$$\delta\phi_{\text{anomalous}} = 43,11'' \pm 0,45''/\text{century} \quad . \tag{25.105}$$

Various ad hoc explanations for this tiny discrepancy between theory and observation (like postulating an additional planet, called *Vulcan*, on an orbit closer to the sun than that of Mercury) were proposed and dismissed.

From (25.104) the prediction of General Relativity for the precession of the perihelion due the the general relativistic correction to the Newtonian theory can be calculated to be

$$\delta\phi_{\text{GR}} = 43,03''/\text{century} \quad . \tag{25.106}$$

Thus general relativity, with its general relativistic correction to the (unperturbed) Kepler orbit of Mercury, appears to account precisely for the observed "anomaly". This is quite a striking, remarkable and impressive confirmation of a prediction of general relativity.

Other observations, involving e.g. the mini-planet Icarus, discovered in 1949, with a huge eccentricity $e \sim 0,827$, binary pulsar systems, and more recently obervations of highly eccentric stars close to the galactic (black hole) center have provided further confirmation of the agreement between General Relativity and observations.

25.9 Null Geodesics

To study the behaviour of massless particles (photons) in the Schwarzschild geometry, we need to study the effective potential

$$V_{eff}(r) = \frac{L^2}{2r^2} - \frac{mL^2}{r^3} = \frac{L^2}{2r^2}(1 - \frac{2m}{r}) \quad . \tag{25.107}$$

The following properties are immediate:

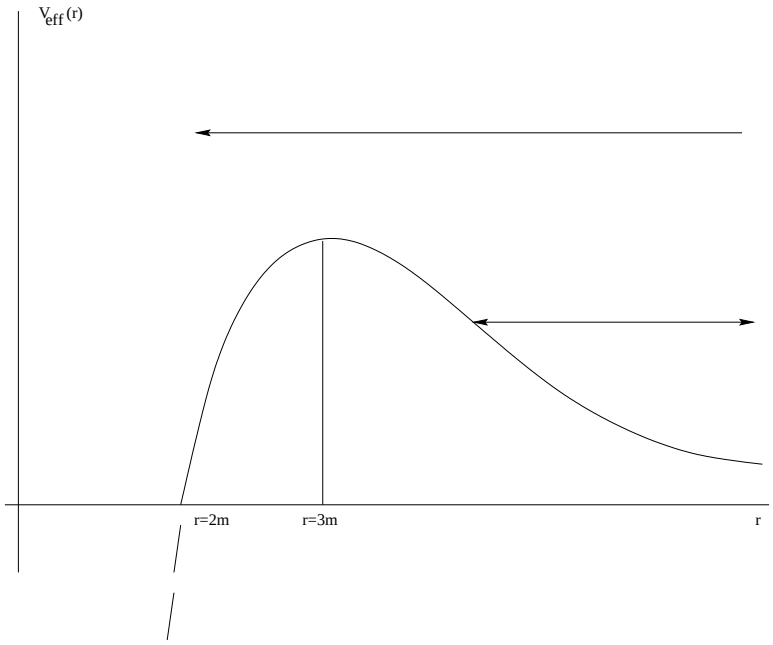


Figure 14: Effective potential for a massless particle. Displayed is the location of the unstable circular orbit at $r = 3m$. A photon with an energy $E^2 < L^2/27m^2$ will be deflected (lower arrow), photons with $E^2 > L^2/27m^2$ will be captured by the star.

1. For $r > 2m$, the potential is positive, $V(r) > 0$.
2. For $r \to \infty$, one has $V_{eff}(r) \to 0$.
3. $V_{eff}(r = 2m) = 0$.
4. When $L = 0$, the photons feel no potential at all.
5. There is one critical point of the potential, at $r = 3m$, with $V_{eff}(r = 3m) = L^2/54m^2$.

Thus the potential has the form sketched in Figure 14, with the following consequences:

1. For energies $E^2 > L^2/27m^2$, photons are captured by the star and will spiral into it. For energies $E^2 < L^2/27m^2$, on the other hand, there will be a turning point, and lightrays will be deflected by the star.

   As this may sound a bit counterintuitive (shouldn't a photon with higher energy be more likely to zoom by the star without being forced to spiral into it?), think about this in the following way. $L = 0$ corresponds to a photon falling radially towards the star, L small corresponds to a slight deviation from radial motion, while L large (thus $\dot{\phi}$ large) means that the photon is travelling along a trajectory that will not bring it very close to the star at all (see the next subsection for the

precise relation between the angular momentum L and the impact parameter b of the photon). It is then not surprising that photons with small L are more likely to be captured by the star (this happens for $L^2 < 27m^2E^2$) than photons with large L which will only be deflected in their path. We will study this in more detail below.

2. Now let us also consider the opposite situation, that of light from or near the star (and we are of course assuming that $r_0 > r_s$). Then for $r_0 < 3m$ and $E^2 < L^2/27m^2$, the light cannot escape to infinity but falls back to the star, whereas for $E^2 > L^2/27m^2$ light will escape. Thus for a path sufficiently close to radial (L small, because $\dot{\phi}$ is then small) light can always escape as long as $r_0 > 2m$.

3. Finally, there is one critical point of the potential, at $r = 3m$. We will (briefly) discuss this case separately below.

25.10 Some Comments on Circular Null Orbits (The Photon Sphere)

As noted above, the effective potential has one critical point, a maximum, at $r = 3m$. This can also be seen from the equation of motion (25.44)

$$\ddot{r} = \frac{L^2}{r^4}(r - 3m) \quad . \tag{25.108}$$

Thus there exists one unstable circular orbit for photons at $r = 3m$ (more precisely one in each "equatorial plane").

While these orbits have some properties in common with the circular timelike orbits discussed in section 25.7, there are also some interesting differences. The first and crucial difference is that this circular orbit can only arise at one particular value of r, namely at $r = 3m$, while (as we have seen) circular timelike orbits can exist for any $r > 3m$. Here are some more brief comments and observations.

Remarks:

1. While not relevant for the applications to the solar system in this section, where we are dealing with objects with a size much larger than $3m$, the existence of this sphere of unstable photon orbits, the so-called *photon sphere*, turns out to be of some interest in black hole astrophysics (as a possibly observable signature of black holes or other very compact objects).

2. Since

$$r = 3m \quad \Rightarrow \quad \dot{t} = E/f(r = 3m) = 3E \quad , \quad \dot{\phi} = L/(3m)^2 = L/(9m^2) \quad , \tag{25.109}$$

these circular null geodesics are characterised by the null wave vector

$$(k^\alpha) = (\dot{t}, \dot{r}, \dot{\phi}, \dot{\theta}) = (3E, 0, L/9m^2, 0) \ , \tag{25.110}$$

with E and L related either by the null condition or, of course equivalently, by the effective potential equation,

$$\dot{r}|_{r=3m} = 0 \quad \Rightarrow \quad E^2 = 2V_{eff}(r = 3m) = L^2/(27m^2) \ . \tag{25.111}$$

Thus the wave vector can e.g. be written as

$$(k^\alpha) = (L/\sqrt{3}m, 0, L/9m^2, 0) \ . \tag{25.112}$$

Of course, the overall normalisation is irrelevant (as it depends on the choice of affine parameter).

3. In this case, the angular velocity (frequency) is given by

$$\omega = d\phi/dt = \dot{\phi}/\dot{t} = (L/E)(1 - 2m/r)/r^2|_{r=3m} = L/(27Em^2) \ . \tag{25.113}$$

Using the relation (25.111), this can be written as

$$\omega = E/L = 1/(3\sqrt{3}m) \quad \Leftrightarrow \quad m\omega = 3^{-3/2} \tag{25.114}$$

This agrees with the $r_- \to 3m$ ($L \to \infty$) limit of the frequency ω_- of the unstable circular timelike orbit (25.81) at $r = r_-$,

$$\lim_{r_- \to 3m} (m\omega_-) = 3^{-3/2} \tag{25.115}$$

(and is structurally similar to, but different from, the ISCO frequency $m\omega_{ISCO} = 6^{-3/2}$ (25.82)).

25.11 Bending of Light by a Star: 3 Derivations

To study the bending of light by a star, we consider an incoming photon (or lightray) with impact parameter b (see Figure 15) and we need to calculate $\phi(r)$ for a trajectory with turning point at $r = r_1$. At that point we have $\dot{r} = 0$. Here the dot can, as usual, be taken to be the derivative with repect to some affine parameter σ. However, noting that the condition $g_{\alpha\beta}\dot{x}^\alpha\dot{x}^\beta = 0$ is reparametrisation-invariant (unlike its massive cousin $g_{\alpha\beta}\dot{x}^\alpha\dot{x}^\beta = -1$), we can equally well choose to parametrise the lightrays by the coordinate time t even though this is not an affine parameter (this matters at the level of the 2nd order geodesic differential equations but not at the level of the 1st integrals and the effective potential).

Either way r_1 is determined by

$$E_{eff} = V_{eff}(r_1) \quad \Leftrightarrow \quad r_1^2 = \frac{L^2}{E^2}(1 - \frac{2m}{r_1}) \ . \tag{25.116}$$

The first thing we need to establish is the relation between b and the other parameters E and L. Consider the ratio

$$\frac{L}{E} = \frac{r^2\dot{\phi}}{(1-2m/r)\dot{t}} \quad . \tag{25.117}$$

For large values of r, $r \gg 2m$, this reduces to

$$\frac{L}{E} = r^2 \frac{d\phi}{dt} \quad . \tag{25.118}$$

On the other hand, for large r we can approximate $b/r = \sin\phi$ by ϕ. Since we also have $dr/dt = -1$ (for an incoming lightray), we deduce

$$\frac{L}{E} = r^2 \frac{d}{dt}\frac{b}{r} = b \quad . \tag{25.119}$$

In terms of the variable $u = 1/r$ the equation for the shape of the orbit (25.64) is

$$u'' + u = 3mu^2 \tag{25.120}$$

and the elliptic integral (25.62) for $\Delta\phi$ is

$$\Delta\phi = 2\int_{r_1}^{\infty} \frac{d\phi}{dr} dr = 2\int_0^{u_1} du\, [b^{-2} - u^2 + 2mu^3]^{-1/2} \quad . \tag{25.121}$$

Moreover, in terms of u we can write the equation (25.116) for $u_1 = 1/r_1$ as

$$b^{-2} = u_1^2 - 2mu_1^3 \quad . \tag{25.122}$$

In the absence of the general relativistic correction (calling this 'Newtonian' is perhaps not really appropriate since we are dealing with photons/lightrays) one has $b^{-1} = u_1$ or $b = r_1$ (no deflection). The orbit equation

$$u_0'' + u_0 = 0 \tag{25.123}$$

has the solution

$$u_0(\phi) = \frac{1}{b}\sin\phi \quad , \tag{25.124}$$

describing the straight line

$$r_0(\phi)\sin\phi = b \quad . \tag{25.125}$$

Obligingly the integral gives

$$(\Delta\phi)_0 = 2\int_0^1 dx(1-x^2)^{-1/2} = 2\arcsin 1 = \pi \quad . \tag{25.126}$$

Thus the deflection angle is related to $\Delta\phi$ by

$$\delta\phi = \Delta\phi - \pi \quad . \tag{25.127}$$

We will now determine $\delta\phi$ in three different ways,

- by perturbatively solving the orbit equation (25.120);
- by perturbatively evaluating the elliptic integral (25.121);
- by performing a perturbative expansion (linearisation) of the Schwarzschild metric.

Derivation I: Perturbative Solution of the Orbit Equation

In order to solve the orbit equation (25.120), we proceed as in section 25.8. Thus the equation for the (small) deviation $u_1(\phi)$ is

$$u_1'' + u_1 = 3mu_0^2 = \frac{3m}{b^2}(1-\cos^2\phi) = \frac{3m}{2b^2}(1-\cos 2\phi) \tag{25.128}$$

which has the particular solution (cf. (25.97))

$$u_1(\phi) = \frac{3m}{2b^2}(1+\tfrac{1}{3}\cos 2\phi) \quad . \tag{25.129}$$

Therefore

$$u(\phi) = \frac{1}{b}\sin\phi + \frac{3m}{2b^2}(1+\tfrac{1}{3}\cos 2\phi) \quad . \tag{25.130}$$

By considering the behaviour of this equation as $r \to \infty$ or $u \to 0$, one finds an equation for (minus) half the deflection angle, namely

$$\frac{1}{b}(-\delta\phi/2) + \frac{3m}{2b^2}\frac{4}{3} = 0 \quad , \tag{25.131}$$

leading to the result

$$\delta\phi = \frac{4m}{b} = \frac{4MG_N}{bc^2} \quad . \tag{25.132}$$

Derivation II: Perturbative Evaluation of the Elliptic Integral

The perturbative evaluation of (25.121) is rather tricky when it is regarded as a function of the independent variables m and b, with r_1 determined by (25.116) (try this!). The trick to evaluate (25.121) is (see R. Wald, *General Relativity*) to regard the integral as a function of the independent variables r_1 and m, with b eliminated via (25.122). Thus (25.121) becomes

$$\Delta\phi = 2\int_0^{u_1} du\, [u_1^2 - u^2 - 2m(u_1^3 - u^3)]^{-1/2} \quad . \tag{25.133}$$

The first order correction

$$\Delta\phi = (\Delta\phi)_0 + m(\Delta\phi)_1 + \mathcal{O}(m^2) \tag{25.134}$$

is therefore

$$(\Delta\phi)_1 = \left(\frac{\partial}{\partial m}\Delta\phi\right)_{m=0} = 2\int_0^{b^{-1}} du\, \frac{b^{-3}-u^3}{(b^{-2}-u^2)^{3/2}} \quad . \tag{25.135}$$

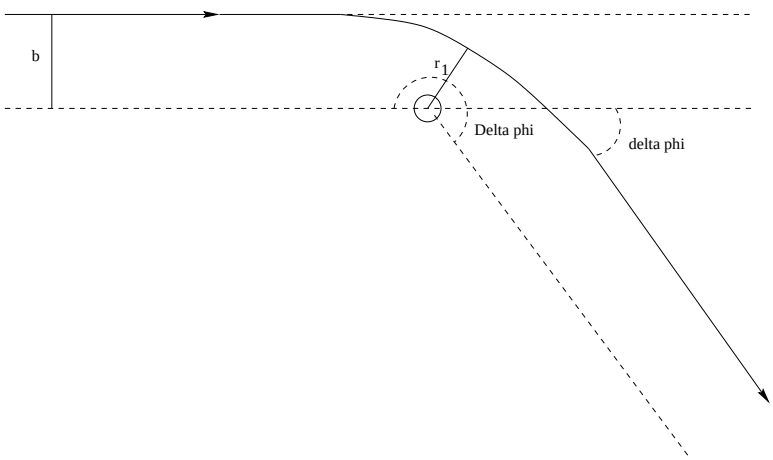


Figure 15: Bending of light by a star. Indicated are the definitions of the impact parameter b, the perihelion r_1, and of the angles $\Delta\phi$ and $\delta\phi$.

This integral is elementary,

$$\int dx\, \frac{1-x^3}{(1-x^2)^{3/2}} = -(x+2)\left(\frac{1-x}{1+x}\right)^{1/2} \quad , \tag{25.136}$$

and thus

$$(\Delta\phi)_1 = 4b^{-1} \quad , \tag{25.137}$$

leading to

$$\delta\phi = \frac{4m}{b} = \frac{4MG_N}{bc^2} \quad , \tag{25.138}$$

in agreement with the result (25.132) obtained above.

Derivation III: Linearising the Schwarzschild Metric

It is instructive to look at the second derivation from another point of view. As we will see, in some sense this derivation 'works' because the bending of light is accurately described by the *linearised solution*, i.e. by the metric that one obtains from the Schwarzschild metric by the approximation

$$\begin{aligned} A(r) = 1 - \frac{2m}{r} \quad &\to \quad 1 - \frac{2m}{r} \\ B(r) = (1 - \frac{2m}{r})^{-1} \quad &\to \quad 1 + \frac{2m}{r} \quad . \end{aligned} \tag{25.139}$$

I will only sketch the main steps in this calculation, so you should think of this subsection as an annotated exercise.

First of all, redoing the analysis of sections 25.3 and 25.5 for a general spherically symmetric static metric (24.6),

$$ds^2 = -A(r)dt^2 + B(r)dr^2 + r^2(d\theta^2 + \sin^2\theta d\phi^2) \quad , \tag{25.140}$$

it is easy to see that the orbit equation can be written as

$$B(r)\frac{r'^2}{r^4} + \frac{1}{r^2} = \frac{\epsilon}{L^2} + \frac{E^2}{L^2 A(r)} \tag{25.141}$$

or, in terms of $u = 1/r$, as (abusing notation by writing $A(r = 1/u)$ as $A(u)$ etc., apologies if this causes allergic reactions)

$$B(u)\ u'^2 + u^2 = \frac{\epsilon}{L^2} + \frac{E^2}{L^2 A(u)} \quad . \tag{25.142}$$

We will concentrate on the lightlike case $\epsilon = 0$,

$$B(u)\ u'^2 + u^2 = \frac{E^2}{L^2 A(u)} \quad , \tag{25.143}$$

and express the impact parameter $b = L/E$ in terms of the turning point $r_1 = 1/u_1$ of the trajectory. At this turning point, $u' = 0$, and thus

$$\frac{E^2}{L^2} = A(u_1)u_1^2 \quad , \tag{25.144}$$

leading to

$$B(u)\ u'^2 = \frac{A(u_1)}{A(u)}u_1^2 - u^2 \quad . \tag{25.145}$$

We thus find

$$\frac{d\phi}{du} = \pm B(u)^{1/2}\left[\frac{A(u_1)}{A(u)}u_1^2 - u^2\right]^{-1/2} \quad . \tag{25.146}$$

For the (linearised) Schwarzschild metric the term in square brackets is

$$\begin{aligned} \frac{A(u_1)}{A(u)}u_1^2 - u^2 &= u_1^2(1 + 2m(u - u_1)) - u^2 \\ &= (u_1^2 - u^2)(1 - 2m\frac{u_1^2}{u_1 + u}) \quad . \end{aligned} \tag{25.147}$$

Using this and the approximate (linearised) value for $B(u)$,

$$B(u) \approx 1 + 2mu \tag{25.148}$$

one finds that $d\phi/du$ is given by

$$\begin{aligned} \frac{d\phi}{du} &= \pm B(u)^{1/2}\left[(u_1^2 - u^2)(1 - 2m\frac{u_1^2}{u_1 + u})\right]^{-1/2} \\ &\approx (u_1^2 - u^2)^{-1/2}\left(1 + m(\frac{u_1^2}{u_1 + u} + u)\right) \\ &= (u_1^2 - u^2)^{-1/2} + m\frac{u_1^3 - u^3}{(u_1^2 - u^2)^{3/2}} \quad . \end{aligned} \tag{25.149}$$

The first term now gives us the Newtonian result and, comparing with Derivation II, we see that the second term agrees precisely with the integrand of (25.135) with $b \to r_1$

(which, in a term that is already of order m, makes no difference). We thus conclude that the deflection angle is, as before,

$$\delta\phi = 2\int_0^{u_1} du\; m\frac{u_1^3 - u^3}{(u_1^2 - u^2)^{3/2}} = 4mu_1 \approx \frac{4m}{b} \quad . \tag{25.150}$$

REMARKS:

1. This effect is physically measurable and was one of the first true tests of Einstein's new theory of gravity. For light just passing the sun the predicted value is

$$\delta\phi \sim 1,75'' \quad . \tag{25.151}$$

   Quick sanity check on this estimate: $1'' = \pi/(180 \times 3600) \approx 0.5 \times 10^{-5}$ radians, while using the (rough) values for $r_s = 2m$ and $r_0 \approx b$ for the sun given in (24.40), one has

$$\frac{4m}{b} \approx (6/7) \times 10^{-5} \approx (12/7)'' \approx 1.7'' \quad . \tag{25.152}$$

   Experimentally this is a bit tricky to observe because one needs to look at light from distant stars passing close to the sun. Under ordinary circumstances this would not be observable, but in 1919 a test of this was performed during a total solar eclipse, by observing the effect of the sun on the apparent position of stars in the direction of the sun. The observed value was rather imprecise, yielding $1,5'' < \delta\phi < 2,2''$ which is, if not a confirmation of, at least consistent with General Relativity.

2. More recently, it has also been possible to measure the deflection of radio waves by the gravitational field of the sun. These measurements rely on the fact that a particular Quasar, known as 3C275, is obscured annually by the sun on October 8th, and the observed result (after correcting for diffraction effects by the corona of the sun) in this case is $\delta = 1,76'' \pm 0,02''$.

3. The value predicted by General Relativity is, interestingly enough, exactly twice the value that would have been predicted by the Newtonian approximation of the geodesic equation alone (but the Newtonian approximation is not valid anyway because it applies to slowly moving objects, and light certainly fails to satisfy this condition). A calculation leading to this wrong value had first been performed by Soldner in 1801 (!) (by cancelling the mass m out of the Newtonian equations of motion before setting $m = 0$) and also Einstein predicted this wrong result in 1908 (his equivalence principle days, long before he came close to discovering the field equations of General Relativity now carrying his name).

   This result can be obtained from the above calculation by setting $B(u) = 1$ instead of (25.148), as in the Newtonian approximation only g_{00} is non-trivial.

4. More generally, one can calculate the deflection angle for a metric with the approximate behaviour

$$B(u) \approx 1 + 2\gamma m u \ , \tag{25.153}$$

for γ a real parameter, with the result

$$\delta\phi \approx \frac{1+\gamma}{2}\,\frac{4m}{b} \ . \tag{25.154}$$

This reproduces the previous result for $\gamma = 1$, half its value for $\gamma = 0$, and checking to which extent measured deflection angles agree with the theoretical prediction of general relativity ($\gamma = 1$) constitutes an experimental test of general relativity. In this context γ is known as one of the PPN parameters (PPN for *parametrised post-Newtonian approximation*).

25.12 A Unified Description in terms of the Runge-Lenz Vector

The perhaps slickest way to obtain the orbits of the Kepler problem is to make use of the so-called *Runge-Lenz vector* (even though it was discovered neither by Runge nor by Lenz).[79]

Recall that, due to conservation of angular momentum $\vec{L}$, the orbits in any spherically symmetric potential are planar. The bound orbits of the Kepler problem, however, have the additional property that they are closed, i.e. that the perihelion is constant. This suggests that there is a further hidden symmetry in the Kepler problem, with the position of the perihelion the corresponding conserved charge. This is indeed the case.

Consider, for a spherically symmetric potential $W(r)$, the vector

$$\vec{A} = \dot{\vec{x}} \times \vec{L} + W(r)\vec{x} \tag{25.155}$$

or, in components,

$$A_i = \epsilon_{ijk}\dot{x}_j L_k + W(r)x_i \ . \tag{25.156}$$

A straightforward calculation, using the Newtonian equations of motion in the potential $W(r)$, shows that

$$\frac{d}{dt}A_i = (r\partial_r W(r) + W(r))\dot{x}_i \ . \tag{25.157}$$

Thus $\vec{A}$ is conserved if and only if $W(r)$ is homogeneous of degree (-1),

$$\frac{d}{dt}\vec{A} = 0 \ \Leftrightarrow \ W(r) = \frac{c}{r} \ . \tag{25.158}$$

In our notation, $c = \epsilon m$, and we will henceforth refer to the vector

$$\vec{A} = \dot{\vec{x}} \times \vec{L} + \frac{\epsilon m}{r}\vec{x} \tag{25.159}$$

[79] For an account of the history, with further references, see e.g. the Wikipedia entry `https://en.wikipedia.org/wiki/Laplace-Runge-Lenz_vector`.

as the Runge-Lenz vector.

In addition to being preserved for a $(1/r)$-potential, the Runge-Lenz vector has the following properties:

1. $\vec{A}$ is orthogonal to $\vec{L}$ and hence lies in the plane of the orbit,

$$\vec{x}.\vec{L} = 0 \quad \Rightarrow \quad \vec{A}.\vec{L} = 0 \ . \tag{25.160}$$

2. The norm A of $\vec{A}$ can be expressed in terms of the other conserved quantities and parameters (energy E, angular momentum L, mass m) of the problem. In the notation of section 25.3 one has

$$A^2 = E^2 L^2 + \epsilon(L^2 + \epsilon m^2) \ . \tag{25.161}$$

3. Thus, even though a priori $\vec{A}$ has 3 components, the only new information is contained in the constant direction of $\vec{A}$, which (since it lies in the orbital plane) is just one angle, a single real number. Thus all in all, in the Kepler problem one has 5 independent constants of motion, $E, \vec{L}$ and the direction of $\vec{A}$.

4. It is well known, and can be shown e.g. by determining the Poisson brackets among the L_i and A_j (the calculation of $\{A_i, A_j\}$ is a bit messy), and a suitable rescaling of the A_i by a function of the (conserved) energy, that $\vec{A}$ extends the manifest symmetry group of rotations $SO(3)$ of the Kepler problem to the (hidden, phase space) symmetry group $SO(4)$ for bound orbits and $SO(3,1)$ for scattering orbits. We will not need to make use of this fact here, though.

Given all this information, it is now straightforward to determine the Keplerian orbits from $\vec{A}$. Let us choose the constant direction of $\vec{A}$ to be in the direction $\phi = 0$. Then $\vec{A}.\vec{x} = Ar\cos\phi$ and from (25.159) one finds

$$Ar\cos\phi = L^2 + \epsilon m r \ . \tag{25.162}$$

Now we consider the two cases $\epsilon = -1$ and $\epsilon = 0$.

For $\epsilon = -1$, (25.162) can be written as

$$\frac{1}{r(\phi)} = \frac{m}{L^2}(1 + \frac{A}{m}\cos\phi) \ . \tag{25.163}$$

Comparing with (25.86), we recognise this as the equation for an ellipse with eccentricity e and semi-major axis a (25.91) given by

$$e = \frac{A}{m} \qquad \frac{m}{L^2} = \frac{1}{a(1-e^2)} \ . \tag{25.164}$$

Moreover, we see that the perihelion is at $\phi = 0$ which establishes that the Runge-Lenz vector points from the center of attraction to the (constant) position of the perihelion. During one revolution the angle ϕ changes from 0 to 2π.

For $\epsilon = 0$ (i.e. no potential), on the other hand, (25.162) reduces to

$$\frac{1}{r(\phi)} = \frac{A}{L^2}\cos\phi \tag{25.165}$$

This describes a straight line (25.124) with impact parameter

$$b = \frac{L^2}{A} = \frac{L}{E} \ . \tag{25.166}$$

In this case, ϕ runs from $-\pi/2$ to $\pi/2$ and the point of closest approach is again at $\phi = 0$ (distance b).

We see that the Runge-Lenz vector captures precisely the information that in the Newtonian theory bound orbits are closed and lightrays are not deflected. The Runge-Lenz vector will no longer be conserved in the presence of the general relativistic correction to the Newtonian motion, and this non-constancy is a precise measure of the deviation of the general relativistic orbits from their Newtonian counterparts. As shown e.g. in an article by Brill and Goel[80] this provides a very elegant and quick way of (re-)deriving the results about perihelion precession and deflection of light in the solar system.

Calculating the time-derivative of the Newtonian Runge-Lenz $\vec{A}$ (25.159), but now for a particle moving in the general relativistic potential (25.34)

$$V(r) = \epsilon\frac{m}{r} - \frac{mL^2}{r^3} \ , \tag{25.167}$$

one finds one additional term arising from substituting the equation of motion into $\ddot{x}_j$, leading to (of course we now switch from t to τ)

$$\frac{d}{d\tau}\vec{A} = \frac{3mL^2}{r^2}\frac{d}{d\tau}\vec{n} \ , \tag{25.168}$$

where $\vec{n} = \vec{x}/r = (\cos\phi, \sin\phi, 0)$ is the unit vector in the equatorial plane $\theta = \pi/2$ of the orbit. Thus $\vec{A}$ rotates in the ϕ-direction in the equatorial plane. If $\vec{A}$ is originally pointing in the x^1-direction $\phi = 0$, then its initial angular velocity in the x^2-direction is

$$\omega = \frac{3mL^2\cos\phi}{Ar^2}\dot{\phi} \ . \tag{25.169}$$

In principle, here A refers to the norm of the Newtonian Runge-Lenz vector (25.159) calculated for a trajectory $\vec{x}(\tau)$ in the general relativistic potential (25.167). This norm is now no longer constant,

$$A^2 = E^2L^2 + \epsilon(L^2 + \epsilon m^2) + \frac{2mL^4}{r^3} \ . \tag{25.170}$$

[80] D. Brill, D. Goel, *Light bending and perihelion precession: A unified approach*, Am. J. Phys. 67 (1999) 316, `arXiv:gr-qc/9712082`

However, assuming that the change in $\vec{A}$ is small, we obtain an approximate expression for ω by substituting the unperturbed orbit $r_0(\phi)$ from (25.162),

$$r_0(\phi) = \frac{L^2}{A\cos\phi - \epsilon m} \quad , \tag{25.171}$$

as well as the unperturbed norm (25.161) in (25.169) to find

$$\omega \approx \frac{3m}{AL^2}(A\cos\phi - \epsilon m)^2 \cos\phi \, \dot{\phi} \quad . \tag{25.172}$$

Now the total change in the direction of $\vec{A}$ when the object moves from ϕ_1 to ϕ_2 can be calculated from

$$\begin{aligned} \delta\phi &= \int_{\phi_1}^{\phi_2} \omega d\tau \\ &= \frac{3m}{AL^2} \int_{\phi_1}^{\phi_2} d\phi \, (A\cos\phi - \epsilon m)^2 \cos\phi \quad . \end{aligned} \tag{25.173}$$

For $\epsilon = -1$, and $(\phi_1, \phi_2) = (0, 2\pi)$, this results in (only the $\cos^2\phi$-term gives a non-zero contribution)

$$\delta\phi = 2\pi \frac{3m^2}{L^2} = \frac{6\pi m^2}{L^2} \quad , \tag{25.174}$$

in precise agreement with (25.101,25.103).

For $\epsilon = 0$, on the other hand, one has

$$\delta\phi = \frac{3mA}{L^2} \int_{-\pi/2}^{\pi/2} d\phi \, \cos^3\phi \quad . \tag{25.175}$$

Using

$$\int \cos^3\phi = \sin\phi - \tfrac{1}{3}\sin^3\phi \quad , \tag{25.176}$$

one finds

$$\delta\phi = \frac{4mA}{L^2} = \frac{4m}{b} \quad , \tag{25.177}$$

which agrees precisely with the results of section 25.11.

26 Black Holes I: Approaching the Schwarzschild Radius r_s

Recall that the Schwarzschild metric, given in standard coordinates by

$$ds^2 = -f(r)dt^2 + f(r)^{-1}dr^2 + r^2 d\Omega^2 \quad , \quad f(r) = 1 - \frac{2m}{r} \quad , \tag{26.1}$$

is the unique spherically symmetric vacuum solution of the Einstein equations. Associated with this solution there is a characteristic length scale, the *Schwarzschild radius* $r = r_s \equiv 2m$. More generally, this is the characteristic lenght scale associated to an object of mass M via the formula

$$r_s = 2m = \frac{2G_N M}{c^2} \quad . \tag{26.2}$$

In our previous discussions of this solution and its properties, in sections 24 and 25, we had considered standard astrophysical objects ("stars") of a size larger (in practice much larger) than their Schwarzschild radius, $r_0 > r_s = 2m$. As a consequence we did not have to address the question what happens to the metric or the geometry as $r \to r_s$ (which was in the deep interior of the star, not desribed by the Schwarzshild metric). We will now contemplate the existence of objects with radius $r_0 < r_s$.

26.1 Preliminary Considerations

Thus we now need to understand the behaviour of the Schwarzschild geometry / gravitational field as one approaches or crosses $r = r_s$. At that radius something special appears to happen to the metric, even though it is not clear what precisely happens there. On the one hand, we noted that the effective potential $V_{eff}(r)$ is perfectly well behaved at r_s. On the other hand, some components of the metric evidently become singular (zero or infinite) there.

Whether this indicates a true singularity of the geometry or the gravitational field (such as infinite tidal forces) or is perhaps simply due to an unfortunate choice of coordinates is usually not something that can just be decided by superficial inspection but requires a more detailed investigation.

Just to illustrate this point, it is a familiar fact that the (apparent) degeneracy of the Euclidean metric on $\mathbb{R}^2$ in polar coordinates,

$$ds^2 = dx^2 + dy^2 = dr^2 + r^2 d\phi^2 \quad , \tag{26.3}$$

at $r = 0$ is simply a coordinate singularity. Likewise, the fact that $g_{\phi\phi} \to \infty$ as $r \to \infty$ is not an indication of a singular behaviour of the metric there. By changing variables, $r = 1/u$, one finds that the line element takes the form

$$ds^2 = u^{-4}du^2 + u^{-2}d\phi^2 \quad . \tag{26.4}$$

In this case, both the non-zero components of the metric diverge as $u \to 0$, but again we know that this is just due to an unfortunate choice of coordinates. However, if one had just been given the above metric in the coordinates (u, ϕ) without an explanation how it was obtained, one might not have realised immediately that this is just the Euclidean metric in disguise. In fact, one can compare and contrast this e.g. with the metric

$$ds^2 = du^2 + u^{-2} d\phi^2 \quad , \tag{26.5}$$

which may look somewhat less singular than the previous metric but is actually genuinely singular as $u \to 0$. This can be verified by calculating its curvature following e.g. the examples given in section 11.3: then one finds that its Gauss curvature or Ricci scalar is proportional to u^{-2} (while the Gauss curvature of the Euclidean metric on $\mathbb{R}^2$ is of course identically zero, regardless of which coordinates one uses).

Thus in order to decide what happens or does not happen as $r \to r_s$, we need to take a closer look at the metric. We will do this in several steps, which you can think of as the various stages of an expedition, from the familiar region at $r \gg r_s$, with a map provided by the Schwarzschild coordinates (t, r, θ, ϕ), into potentially dangerous new and uncharted territory:

- First we will consider observers that don't quite dare to cross r_s and who try to remain static at a fixed value of r close to r_s.
- Then we will consider observers that fall freely (and radially) in this geometry, and describe their voyage both from their point of view and from that of a distant static observer (which will turn out to be quite different).
- Then we will study the geometry of the Schwarzschild metric near $r = r_s$, and show that the geometry is completely non-singular (and is in fact closely related to the geometry of the Rindler metric for Minkowski space-time discussed in section 1.3).
- These considerations will indicate that (and explain why) the usual Schwarzschild coordinates (specifically the time-coordinate t) are inadequate for describing the physics across the radius r_s.
- Encouraged by this, in section 27 we then begin to explore the region near and beyond $r = r_s$. To that end we first introduce coordinates that are adapted to infalling observers. This is an illuminating exercise which is interesting in its own right and which will already tell us something about the hidden geometry behind r_s.
- In order to learn more about this region, we next study the behaviour of lightcones and lightrays in this geometry. These considerations will then lead us to the introduction of corresponding adapted coordinates in which the Schwarzschild

metric is non-singular for all $0 < r < \infty$, and then we will be in a position to understand what actually happens (and what characterises) $r = r_s$.

26.2 Static Observers

Some insight into the Schwarzschild geometry, and the difference between Newtonian gravity and general relativistic gravity, is provided by looking at static observers, i.e. observers hovering at fixed values of (r, θ, ϕ). Thus their 4-velocity $u^\alpha = \dot{x}^\alpha$ has the form $u^\alpha = (u^0, 0, 0, 0)$ with $u^0 > 0$. The normalisation $u^\alpha u_\alpha = -1$ then implies

$$u^\alpha = \Big(f(r)^{-1/2}, 0, 0, 0\Big) = \left(\frac{1}{\sqrt{1-2m/r}}, 0, 0, 0\right) \quad . \tag{26.6}$$

Note that, as in (3.125), we could have written this more invariantly as

$$u^\alpha = V^\alpha / \mathbb{V} \tag{26.7}$$

where $V = \partial_t$ is the timelike Killing vector and $\mathbb{V}$ its norm (3.124), but for present (pragmatic and calculational) purposes the explicit coordinate expression is more useful.

The worldline of a static observer is clearly not a geodesic (that would be the worldline of an observer freely falling in the gravitational field), and we can calculate its covariant acceleration (5.98)

$$a^\alpha = D_\tau u^\alpha = u^\beta \nabla_\beta u^\alpha = (d/d\tau) u^\alpha + \Gamma^\alpha_{\beta\gamma} u^\beta u^\gamma \quad . \tag{26.8}$$

Noting that u^α is time-independent, because the observer is at fixed r by hypothesis, one finds

$$a^\alpha = \Gamma^\alpha_{00} (u^0)^2 = \Gamma^\alpha_{00} (1 - 2m/r)^{-1} \quad . \tag{26.9}$$

Thus only a^r is non-zero, and one finds

$$a^\alpha = (0, m/r^2, 0, 0) \quad . \tag{26.10}$$

This looks nicely Newtonian, with a force in the radial direction designed to precisely cancel the gravitational attraction. However, this is a bit misleading since this is still a coordinate dependent statement. A coordinate-invariant quantity (scalar) is the norm of the acceleration,

$$\mathsf{a}(r) \equiv \Big(g_{\alpha\beta} a^\alpha a^\beta\Big)^{1/2} = \frac{m}{r^2} \left(1 - \frac{2m}{r}\right)^{-1/2} \quad . \tag{26.11}$$

While this approaches the Newtonian value as $r \to \infty$, it diverges as $r \to 2m$, indicating that static observers will find it harder and harder to remain static close to $r = 2m$.

REMARKS:

1. At the same time, the discrepancy (gravitational time delay) between the static observer's proper time and the coordinate time (proper time of a static observer at $r \to \infty$) becomes more and more pronounced,

$$\Delta\tau = (1 - \frac{2m}{r})^{1/2}\Delta t \ , \tag{26.12}$$

and diverges as $r \to 2m$. This leads to an infinite gravitational redshift (see section 26.5 below). It is also the first (but all by itself not conclusive) indication, that the time coordinate t may not be appropriate for understanding the physics at and beyond $r = 2m$.

2. While $\mathsf{a}(r)$ diverges as $r \to 2m$, we will see later (section 27.10) that the finite quantity $\lim_{r\to 2m} f(r)^{1/2}\mathsf{a}(r) = 1/4m$ can be regarded as a measure of the strength of the gravitational field at $r = 2m$ (*surface gravity*).

3. One can also think of $\mathsf{a}(r) = \mathsf{a}^r(r)$ as the radial component of the acceleration with respect to an orthonormal frame (see section 4.8) at that point: a radial unit vector is

$$e_r = (1 - 2m/r)^{1/2}\partial_r : \quad g_{\alpha\beta}e_r^\alpha e_r^\beta = 1 \ , \tag{26.13}$$

and the acceleration vector can be written as

$$a = a^r\partial_r = a^r(1 - 2m/r)^{-1/2}((1 - 2m/r)^{1/2}\partial_r) \equiv \mathsf{a}^r e_r \ . \tag{26.14}$$

4. The covariant components of the acceleration can be written as the gradient

$$a_\alpha = \partial_\alpha(\tfrac{1}{2}\ln f(r)) \equiv \partial_\alpha\Phi(r) \ , \tag{26.15}$$

so that here (and in the static spherically symmetric case in general) $\Phi(r)$ can be regarded as the general relativistic analogue of the Newtonian potential $\phi(r)$, to which it reduces in the Newtonian limit,

$$\Phi(r) = \tfrac{1}{2}\ln f(r) \approx \tfrac{1}{2}\ln(1 + 2\phi(r)) = \phi(r) + \dots \tag{26.16}$$

5. Using the covariant characterisation (26.7) of the 4-velocity of a static observer, this result can be derived and generalised as follows. First of all the acceleration of u^α is

$$a_\beta = (V^\alpha/\mathbb{V})\nabla_\alpha(V_\beta/\mathbb{V}) = (V^\alpha/\mathbb{V}^2)\nabla_\alpha V_\beta \ , \tag{26.17}$$

because $\mathbb{V}$ is constant along V (9.61). Using the Killing condition this can be written as

$$a_\beta = -(V^\alpha/\mathbb{V}^2)\nabla_\beta V_\alpha = \tfrac{1}{2}\nabla_\beta\log(-V_\alpha V^\alpha) \ , \tag{26.18}$$

which provides the appropriate generalisation of (26.15), suggesting that a suitable general relativistic analogue of the scalar Newtonian potential in space-times admitting a timelike Killing vector is provided by the logarithm of the redshift factor $\mathbb{V}$,

$$\Phi = \tfrac{1}{2}\log(-V_\alpha V^\alpha) = \log\mathbb{V} \ . \tag{26.19}$$

26.3 Vertical Free Fall

We will now consider an object with $r_0 < r_s$ and an observer who is freely falling vertically (radially) towards such an object. "Vertical" means that $\dot{\phi} = 0$, and therefore there is no angular momentum, $L = 0$. Hence the effective potential equation (25.31) becomes

$$E^2 - 1 = \dot{r}^2 - \frac{2m}{r} \quad , \tag{26.20}$$

where the conserved energy E (per unit mass) is given by

$$E = (1 - \frac{2m}{r})\dot{t} \quad . \tag{26.21}$$

In particular, if r_i is the point at which the particle (observer) A was initially at rest,

$$\frac{dr}{d\tau}|_{r=r_i} = 0 \quad , \tag{26.22}$$

we have the relation

$$E^2 = 1 - \frac{2m}{r_i} = f(r_i) \tag{26.23}$$

between the constant of motion E and the initial condition r_i. In particular, $E = 1$ for an observer following a trajectory of an object that would have initially been at rest at infinity. In that case, (26.20) is readily integrated to give

$$r(\tau) \sim (\tau_0 - \tau)^{2/3} \tag{26.24}$$

so that the observer reaches $r = 0$ at $\tau = \tau_0$.

For $r_i < \infty$, we obtain

$$\dot{r}^2 = \frac{2m}{r} - \frac{2m}{r_i} = f(r_i) - f(r) \tag{26.25}$$

and, upon differentiation (for any E)

$$\ddot{r} + \frac{m}{r^2} = 0 \quad . \tag{26.26}$$

This is just like the Newtonian equation (which should not come as a surprise as V_{eff} coincides with the Newtonian potential for zero angular momentum $L = 0$), apart from the fact that r is *not* radial distance and the familiar $\tau \neq t$. Nevertheless, calculation of the time τ along the path proceeds exactly as in the Newtonian theory. For the proper time required to reach the point with coordinate value $r = r_f$ we obtain

$$\tau = -(2m)^{-1/2} \int_{r_i}^{r_f} dr \, \left(\frac{r_i r}{r_i - r} \right)^{1/2} \quad . \tag{26.27}$$

Since this is just the Newtonian integral, we know, even without calculating it, that it is finite as $r_f \to r_s$ (and even as $r_f \to 0$). This integral can also be calculated in closed form, e.g. via the change of variables

$$r(\eta) = \tfrac{1}{2} r_i \, (1 + \cos \eta) \quad \text{with} \quad \eta_i = 0 \leq \eta \leq \eta_f \leq \pi \quad , \tag{26.28}$$

leading (after some convenient cancellations) to

$$\tau = \left(\frac{r_i}{2m}\right)^{1/2} \int_0^{\eta_f} d\eta\, r(\eta) = \left(\frac{r_i^3}{8m}\right)^{1/2} (\eta_f + \sin\eta_f) \ . \tag{26.29}$$

In particular, this is finite as $r_f \to 2m$ and our freely falling observer can reach and cross the Schwarzschild radius r_s in finite proper time.

Coordinate time, on the other hand, becomes infinite at $r_f = 2m$. This can be anticipated from the relation (26.21) between t and τ for a freely falling observer, which diverges as $r \to 2m$. We will address this in a more quantitative way in section 26.4 below.

26.4 Vertical Free Fall as seen by a Distant Observer

We will now investigate how the above situation presents itself to a distant observer hovering at a fixed radial distance r_∞. He will observe the trajectory of the freely falling observer as a function of his proper time τ_∞. Up to a constant factor $(1 - 2m/r_\infty)^{1/2}$, this is the same as coordinate time t, and we will lose nothing by expressing r as a function of t rather than as a function of τ_∞.

From (26.20),

$$\dot{r}^2 + (1 - \frac{2m}{r}) = E^2 \ , \tag{26.30}$$

which expresses r as a function of the freely falling observer's proper time τ, and the definition of E,

$$E = \dot{t}(1 - \frac{2m}{r}) \ , \tag{26.31}$$

which relates τ to the coordinate time t, one finds an equation for r as a function of t,

$$\frac{dr}{dt} = -E^{-1}(1 - \frac{2m}{r})(E^2 - (1 - \frac{2m}{r}))^{1/2} \tag{26.32}$$

(the minus sign has been chosen because r decreases as t increases). At large distances, the difference between t and τ, and hence between $r = r(t)$ and $r = r(\tau)$ is small and not particularly interesting in the present context. We want to analyse the behaviour of the solution of this equation as the freely falling observer approaches the Schwarzschild radius, $r \to 2m$. In that region, we can approximate

$$1 - \frac{2m}{r} = \frac{r - 2m}{r} \approx \frac{r - 2m}{2m} \ , \tag{26.33}$$

and we get

$$\begin{aligned} \frac{dr}{dt} &= -E^{-1}(\frac{r - 2m}{r})(E^2 - \frac{r - 2m}{r})^{1/2} \\ &\approx -E^{-1}(\frac{r - 2m}{2m})(E^2)^{1/2} = -(\frac{r - 2m}{2m}) \ . \end{aligned} \tag{26.34}$$

We can write this equation as

$$\frac{d}{dt}(r-2m) = -\frac{1}{2m}(r-2m) \quad , \tag{26.35}$$

which obviously has the solution

$$(r-2m)(t) \propto \mathrm{e}^{-t/2m} \quad . \tag{26.36}$$

This shows that, from the point of view of the observer at infinity, the freely falling observer reaches $r = 2m$ only as $t \to \infty$. In particular, the distant observer will never actually see the infalling observer cross the Schwarzschild radius.

This is clearly an indication that there is something wrong with the time coordinate t which runs too fast as one approaches the Schwarzschild radius. We can also see this by looking at the coordinate velocity $v = dr/dt$ as a function of r. Let us choose $r_i = \infty$ for simplicity - other choices will not change our conclusions as we are interested in the behaviour of $v(r)$ as $r \to r_s$. Then $E^2 = 1$ and from (26.32) we find (now dropping the minus sign)

$$v(r) = (2m)^{1/2}\frac{r-2m}{r^{3/2}} \tag{26.37}$$

As a function of r, $v(r)$ reaches a maximum at the critical radius $r_c = 6m = 3r_s$,

$$\frac{d}{dr}v(r) = 0 \quad \Rightarrow \quad r = r_c = 6m \quad , \tag{26.38}$$

where the velocity is (restoring the speed of light c)

$$v(r_c) = \frac{2c}{3\sqrt{3}} \quad . \tag{26.39}$$

The fact that this radius agrees with the ISCO (innermost stable circular orbit) (25.73) mentioned in section 25.6 should (presumably) be regarded as a coincidence.

Beyond that point, i.e. for $r < r_c$, $v(r)$ decreases again and clearly goes to zero as $r \to 2m$. The fact that the coordinate velocity goes to zero is simply another manifestation of the fact that coordinate time goes to infinity, and that the Schwarzschild coordinates are simply not suitable for describing the physics at or beyond the Schwarzschild radius because the time coordinate one has chosen is running too fast. This is the crucial insight that will later on allow us to construct "better" coordinates, which are also valid for $r < r_s$.

Remarks:

1. Sometimes (actually all too frequently) the fact that the coordinate velocity is decreasing as one approaches $r \to 2m$ is claimed to provide evidence for some "repulsive" nature of the gravitational field as one approaches $r = 2m$. While it is understandable that such things caused some confusion in the early days of general relativity, this is of course utter nonsense for which there can be no excuse today.

2. As an aside, if one repeats the above "calculation" for arbitrary E, from (26.32) one finds that the critical radius is

$$r_c = \frac{2m}{1 - 2E^2/3} \quad , \tag{26.40}$$

with the E-dependent maximal velocity

$$v(r_c) = \frac{2c}{3\sqrt{3}} E^2 \quad . \tag{26.41}$$

This reproduces the above result for $E = 1$, shows that $r_c \to 2m$ for $E \to 0$, and moreover curiously shows that there is a maximal E, $E_{max} = \sqrt{3/2}$, for which such a critical point of the coordinate velocity will occur, at $r_c \to \infty$, with maximal velocity

$$E \to E_{max} = \sqrt{3/2} \quad \Rightarrow \quad r_c \to \infty \quad \text{and} \quad v(r_c) \to v_{max} = c/\sqrt{3} \ . \tag{26.42}$$

Thus for larger values of E, the coordinate velocity will monotonically decrease from its initial value

$$v(r \to \infty) = E^{-1}(E^2 - 1)^{1/2} \tag{26.43}$$

to $v(r) = 0$ as $r \to 2m$.

[And if anybody has an intuitive explanation for these facts, please let me know.]

26.5 Gravitational Redshift in the Schwarzschild Geometry

One dramatic aspect of what is happening at (or, better, near) the Schwarzschild radius for very (very!) compact objects with $r_s > r_0$ is the following. Recall the formula (3.108) for the gravitational redshift, which gave us the ratio between the frequency of light ν_e emitted at the radius r_e and the frequency ν_∞ received at the radius $r_\infty > r_e$ for static observers in a static spherically symmetric gravitational field.

The result, which is in particular also valid for the Schwarzschild metric, was

$$\frac{\nu_\infty}{\nu_e} = \frac{(-g_{00}(r_e))^{1/2}}{(-g_{00}(r_\infty))^{1/2}} \quad . \tag{26.44}$$

In the case of the Schwarzschild metric, this is

$$\frac{\nu_\infty}{\nu_e} = \frac{(1 - 2m/r_e)^{1/2}}{(1 - 2m/r_\infty)^{1/2}} \quad , \tag{26.45}$$

in accordance with the relation (26.12) between the coordinate time and the proper time of a static observer. As $r_e \to r_s$, one clearly finds

$$\frac{\nu_\infty}{\nu_e} \to 0 \quad . \tag{26.46}$$

Expressed in terms of the gravitational redshift factor z,

$$1 + z = \frac{\nu_e}{\nu_\infty} \tag{26.47}$$

this means that there is an *infinite gravitational redshift* as $r_e \to r_s$,

$$r_e \to r_s \quad \Rightarrow \quad z \to \infty \ . \tag{26.48}$$

This is for static observers. For a freely falling observer whose position is described by $r_e = r(\tau)$ or $r(t)$ and the receiver the static observer at $r_\infty \gg r_s$ we cannot directly apply the above formula (because the emitter is not static but freely falling), and the actual redshift of the emitted lightray will be larger because of an additional Doppler effect due to the observer's proper motion away from the receiver.

If (for the time being) we ignore this additional Doppler contribution, and just use the above formula (this will thus underestimate the actual redshift) in conjunction with the late-time behaviour for $r = r(t)$ derived in (26.36), we can obtain the late-time behaviour of the redshift factor $z = z(t)$ as a function of the distant observer's proper time t, namely

$$1 + z \propto (r - 2m)(t)^{-1/2} \propto \mathrm{e}^{t/4m} \ . \tag{26.49}$$

Taking into account the proper motion of the freely falling observer one finds (we will derive this below) that, essentially due to the difference between $u^0 = \dot{t} = f^{-1/2}$ for a static observer and $u^0 = \dot{t} = E/f$ for a geodesic observer, as $r \to 2m$ the redshift is enhanced by a further factor $f^{-1/2}$, so that all in all one has $(1 + z) \propto (r - 2m)(t)^{-1}$ instead of (26.49). See (26.67) below.

Either way for the distant observer at late times there is an exponentially growing redshift and the distant observer will never actually see the unfortunate emitter crossing the Schwarzschild radius: he/she will see the freely falling observer's signals becoming dimmer and dimmer and arriving at greater and greater intervals, and the freely falling observer will completely disappear from the distant observer's sight as $r_e \to r_s$. Note that the time-scale t_z for this exponential redshift at late times is set by $t_z \sim m/c$, which is of the order of

$$t_z \sim 10^{-5}\mathsf{s} \left(\frac{M}{M_{\mathsf{sun}}} \right) \ , \tag{26.50}$$

so that this is pretty much instantaneous for an object the mass of an ordinary star. We will come back to this estimate later on when (briefly) talking about gravitational collapse in section 29.3.

In order to take into account the proper freely falling motion of the emitter, one can use the method described in the second derivation of the gravitational redshift given in section 3.5, based on the equation (3.117),

$$\omega = -u^\alpha k_\alpha \ , \tag{26.51}$$

for the frequency ω of a lightray with wave vector k^α as seen (or emitted) by an arbitrary (not necessarily static) observer with 4-velocity u^α.

We describe lightrays by null curves $x^\alpha = x^\alpha(\lambda)$ with wave vector

$$k^\alpha = \frac{d}{d\lambda} x^\alpha(\lambda) \equiv x'^\alpha \tag{26.52}$$

(for some affine parameter λ) and the motion of timelike observers (as usual) by $x^\alpha = x^\alpha(\tau)$ with 4-velocity

$$u^\alpha = \frac{d}{d\tau} x^\alpha(\tau) \equiv \dot{x}^\alpha \quad . \tag{26.53}$$

Let us now apply this formula to the case at hand (Schwarzschild geometry and radial motion). The relevant wave and velocity vectors are the following:

- Radial Lightrays

  For lightrays, there is a conserved energy

  $$e = f(r) t' \tag{26.54}$$

  and for radial lightrays the effective potential is zero, so that the radial equation of motion for r is simply

  $$(r')^2 = e^2 \quad \Rightarrow \quad r' = \pm e \tag{26.55}$$

  (for out/in-going lightrays respectively). Thus out/in-going radial lightrays have wave vector

  $$(k^\alpha)_{out/in} = (t', r', 0, 0) = (e/f, \pm e, 0, 0) : \quad g_{\alpha\beta} k^\alpha k^\beta = 0 \quad . \tag{26.56}$$

- Static Observers

  For static observers one has (26.6)

  $$(u^\alpha)_s = (f^{-1/2}, 0, 0, 0) : \quad g_{\alpha\beta} u^\alpha u^\beta = -1 \quad . \tag{26.57}$$

- Radial Freely Falling Observers

  For geodesic (freely falling) obervers, there is a conserved energy

  $$E = f(r) \dot{t} \tag{26.58}$$

  and for radial free fall the effective potential is just the Newtonian potential, so that the radial equation of motion is simply (26.20)

  $$\dot{r}^2 = E^2 - f \quad \Rightarrow \quad \dot{r} = -(E^2 - f)^{1/2} \tag{26.59}$$

  (for infalling observers). Thus their 4-velocity is

  $$(u^\alpha)_{ff} = (\dot{t}, \dot{r}, 0, 0) = (E/f, -(E^2 - f)^{1/2}, 0, 0) : \quad g_{\alpha\beta} u^\alpha u^\beta = -1 \quad . \tag{26.60}$$

From these and (26.51) we can now read off the following results:

1. Static Observers and In/Outgoing Lightrays

   The frequency of an in/out-going lightray as seen by a static observer at the radius r is

   $$\omega_{s,in/out}(r) = -g_{\alpha\beta}(u^\alpha)_s(k^\beta)_{in/out} = f(r)f(r)^{-1/2}(e/f(r)) = ef(r)^{-1/2} \ . \tag{26.61}$$

   Note that this only depends on the time component of k^α and is thus the same for in- and outgoing lightrays. In particular,

   - this identifies the energy e as the frequency as seen by an observer "at infinity",
   $$\omega_\infty \equiv \lim_{r\to\infty} \omega_{s,in/out} = e \ , \tag{26.62}$$
   - and reproduces the standard result that the redshift of light emitted by a static observer at radius r relative to a distant static observer scales like $f(r)^{-1/2}$,
   $$\frac{\omega_\infty}{\omega_{s,in/out}} = f(r)^{1/2} \ . \tag{26.63}$$

2. Freely Falling Observers and In- versus Outgoing Lightrays

   For a freely falling observer instead, one has

   $$\omega_{ff,in/out} = -g_{\alpha\beta}(u^\alpha)_{ff}k^\beta_{in/out} = f(E/f)(e/f) \mp f^{-1}(E^2 - f)^{1/2}e \tag{26.64}$$

   and thus (restoring the r-dependence)

   $$\omega_{ff,in/out}(r) = \frac{e}{f(r)}(E \mp (E^2 - f(r))^{1/2}) \ . \tag{26.65}$$

   Note that, roughly speaking, and as anticipated above, compared to the result for a static emitter or observer, the redshift/blueshift is enhanced by another factor $f(r)^{\pm 1/2}$ due to a Doppler-like effect caused by the motion of the observer or emitter.

   This effect is of course particularly pronounced if either emitter or observer approaches $r = 2m$. More specifically, we can consider the following two situations:

   (a) Outgoing Lightrays

   Taking the limit $f(r) \to 0$, one finds that

   $$f(r) \to 0 \quad \Rightarrow \quad \omega_{ff,out}(r) \approx \frac{2eE}{f(r)} \quad \Rightarrow \quad \frac{\omega_\infty}{\omega_{ff,out}(r)} \approx \frac{f(r)}{2E} \to 0 \ . \tag{26.66}$$

   Thus from the perspective of a static observer an outgoing lightray sent out by an infalling observer will undergo an infinite redshift proportional to

$(r-2m)^{-1}$ as the emitter approaches $r \to 2m$. In particular, here we are now allowed to insert the time dependence (26.36) and instead of (26.49) we will obtain

$$1+z \propto (r-2m)(t)^{-1} \propto \mathrm{e}^{\,t/2m} \quad . \tag{26.67}$$

(b) Ingoing Lightrays

We can also consider the opposite situation where an ingoing lightray reaches an infalling observer (from above/behind) as $r \to 2m$. In that case, the lightray is detected by the infalling observer with a finite redshift,

$$f(r) \to 0 \quad \Rightarrow \quad \omega_{ff,in}(r) \approx \frac{e}{2E} \quad \Rightarrow \quad \frac{\omega_\infty}{\omega_{ff,in}} = 2E \quad . \tag{26.68}$$

In particular, for a freely falling observer with $E = 1$, i.e. an oberver who falls as if he/she had started off at rest at infinity, the frequency he/she measures is precisely 1/2 the frequency ω_∞.

This is perhaps one "practical" way for this infalling observer to determine when precisely he/she reaches $r = 2m$: just after the agreed upon almost-UV violet light with emitter frequency 790 THz (at the upper end of the visible spectrum) arrives as the almost-IR red light with frequency 405 THz (at the lower end of the visible spectrum).

26.6 Geometry near r_s and Minkowski Space in Rindler Coordinates

We have now seen in two different ways why the Schwarzschild coordinates are not suitable for exploring the physics in the region $r \leq 2m$: in these coordinates the metric becomes singular at $r = 2m$ and the coordinate time becomes infinite. On the other hand, we have seen no indication that the local physics, expressed in terms of covariant quantities like proper time or the geodesic equation, becomes singular as well. So we have good reasons to suspect that the singular behaviour we have found is really just an artefact of a bad choice of coordinates.

In fact, the situation regarding the Schwarzschild coordinates is quite similar to that of the Rindler coordinates for Minkowski space we discussed (way back) in section 1.3.

Recall from section 1.3 that

- in the coordinates

$$\xi^0(\eta,\rho) = \rho \sinh \eta \qquad \xi^1(\eta,\rho) = \rho \cosh \eta \quad . \tag{26.69}$$

the $(1+1)$-dimensional Minkowski metric takes the form (1.74)

$$ds^2 = -\rho^2 d\eta^2 + d\rho^2 \quad ; \tag{26.70}$$

- that the lines of constant ρ are hyperbolas and that these are the worldlines of observers with constant acceleration $1/\rho$;
- that these coordinates are adapted to observers with constant acceleration in the same way as inertial coordinates are adapted to static observers: they stay at fixed values of their spatial coordinate, and the coordinate time is a direct measure of their proper time;
- the null lines $\xi^0 = \pm\xi^1$ correspond to $\rho = 0$ (infinite acceleration) or $\eta = \pm\infty$, so that the Rindler coordinates cover only the first quadrant $\xi^1 > |\xi^0|$ of Minkowski space;
- inertial (geodesic, freely falling) observers could exit this region in finite proper time.

All this is of course quite reminiscent of the things that we have discovered so far about the Schwarzschild geometry, and this is in fact more than just a loose analogy: as we will see now, remarkably the Rindler metric (26.70) gives an accurate description of the geometry of the Schwarzschild metric close to the Schwarzschild radius.

To confirm this, let us temporarily introduce the variable $\tilde{r} = r - 2m$ measuring the coordinate distance from the critical radius $r = 2m$. In term of $\tilde{r}$ the (t, r)-part of the Schwarzschild metric reads

$$ds^2 = -\left(\frac{\tilde{r}}{\tilde{r}+2m}\right)dt^2 + \left(\frac{\tilde{r}+2m}{\tilde{r}}\right)d\tilde{r}^2 \quad . \tag{26.71}$$

Close to $r_s = 2m$, i.e. for small $\tilde{r} << 2m$, we can approximate

$$\frac{\tilde{r}}{\tilde{r}+2m} \approx \frac{\tilde{r}}{2m} \qquad\qquad \frac{\tilde{r}+2m}{\tilde{r}} \approx \frac{2m}{\tilde{r}} \quad , \tag{26.72}$$

so that the metric becomes

$$ds^2 = -\frac{\tilde{r}}{2m}dt^2 + \frac{2m}{\tilde{r}}d\tilde{r}^2 \quad . \tag{26.73}$$

Introducing the new radial variable ρ (proper radial distance from $r = r_s$) via

$$d\rho^2 = \frac{2m}{\tilde{r}}d\tilde{r}^2 \quad \Rightarrow \quad \rho = \sqrt{8m\tilde{r}} \quad , \tag{26.74}$$

one finds

$$ds^2 = -\frac{1}{16m^2}\rho^2 dt^2 + d\rho^2 \quad . \tag{26.75}$$

Finally a simple rescaling of t, $\eta = t/4m$, leads to

$$ds^2 = -\rho^2 d\eta^2 + d\rho^2 \quad , \tag{26.76}$$

which, remarkably, is identical to the Rindler metric (1.74). Keeping track of the transverse 2-sphere and using $r^2 \approx (2m)^2$ in the near-r_s approximation, the complete metric in this limit and in these coordinates reads

$$ds^2 = -\rho^2 d\eta^2 + d\rho^2 + (2m)^2 d\Omega^2 \tag{26.77}$$

If we further restrict to just a small angular region on the sphere, we can approximate

$$d\theta^2 + \sin^2\theta d\phi^2 \approx d\theta^2 + \theta^2 d\phi^2 = (dx^2)^2 + (dx^3)^2 \tag{26.78}$$

which gives us the complete 4-dimensional Rindler metric.

In any case, we see that, up to the harmless redefinition $(r,t) \to (\rho,\eta)$ that we just performed, which is just a reparametrisation

$$\begin{aligned} \rho &= \rho(r) = \sqrt{8m(r-2m)} \\ \eta &= \eta(t) = t/4m \end{aligned} \tag{26.79}$$

Schwarzschild coordinates for the Schwarzschild geometry near $r = 2m$ are just like Rindler coordinates for Minkowski space. This leads to a much improved understanding of the Schwarzschild geometry in general and the observations that we made regarding static and freely falling observers in particular:

1. The first, and most crucial, thing we learn from this is that the singularity of the Schwarzschild metric at $r = 2m$ in the Schwarzschild coordinates (t,r) is, as anticipated, a mere coordinate singularity. Indeed, $r = 2m$ corresponds to $\tilde{r} = 0 \Leftrightarrow \rho = 0$, and we already know that what appears to be a possibly singular point of the Rindler metric at $\rho = 0$ is just a coordinate singularity which can be eliminated e.g. by passing to standard inertial Minkowski coordinates via (1.73).

   Specifically, from (26.79) one could introduce Minkowski-like coordinates (t_M, x_M) (say), via

   $$t_M = \sqrt{8m(r-2m)}\sinh t/4m \quad , \quad x_M = \sqrt{8m(r-2m)}\cosh t/4m \ , \tag{26.80}$$

   in terms of which the metric in the region $r \gtrsim 2m$ takes the standard Minkowskian form $-dt_M^2 + dx_M^2$. The reason for not pursuing this here is that we will actually be able to do much better than this in various ways in section 27, where we will construct a variety of coordinate systems which exhibit the non-singular nature of the geometry at $r = 2m$ but which are not restricted to only a tiny (strictly speaking infinitesimal) neighbourhood of $r = 2m$. In particular, the above coordinate transformation (26.80) can be understood as the $r \gtrsim 2m$ approximation of the coordinate transformation from Schwarzschild coordinates to *Kruskal coordinates* (to be introduced in section 27.7).

2. Moreover, we can now understand physically why the Schwarzschild coordinates break down at $r = 2m$: they are adapted to accelerating observers, in the present context the static observers in the Schwarzschild geometry at constant r, with proper time proportional to the Schwarzschild coordinate time t. Referring back to Figure 7 in section 1.3, these are the observers with constant ρ hyperbolic world lines.

   The problem is evidently that the required acceleration of these observers becomes infinite as $\rho \to 0 \Leftrightarrow r \to 2m$ (as we have calculated in section 26.2). That these observers appear to see a singular metric is then not the geometry's fault but can be attributed to a bad choice of observers whose perception of the geometry is distorted by their acceleration and their desperate attempt to stay at constant values of r even when they are very close to $r = 2m$.

3. The situation is quite different for the freely falling observers. Their worldlines look like the vertical line labelled "worldline of a static observer" in Figure 7, they cross the "horizon" in finite proper time, experiencing no strong acceleration or gravitational fields. As already noted in section 1.3, they evidently become invisible to observers in the Rindler quadrant (now "Schwarzschild patch") of the geometry, static outside observers noting an infinite gravitational redshift affecting the signals sent out by the freely falling observer.

4. Introducing coordinates adapted to freely falling observers (so that e.g. time is their proper time) would be tantamount to passing from Rindler coordinates to ordinary Minkowski coordinates, and we will consider that option in section 27.2 below (Painlevé-Gullstrand coordinates). These will provide us, as we will see, with a coordinate system that extends in a non-singular way across the Schwarzschild radius.

5. In order to elucidate the significance of the Schwarzschild radius, it will then turn out to be useful to base the construction of new coordinates not on the behaviour of timelike freely falling observers but on ingoing lightrays instead (which evidently also do not suffer from the problems of the static observers). These are the Eddington-Finkelstein coordinates to be discussed in section 27.4.

6. Finally, we can anticipate that upon introduction of suitable analogues of the Minkowski coordinates for the Rindler space-time, we may perhaps uncover not just one new region (quadrant) of space-time (the one lying to the "future" of $r = 2m$), but also counterparts of the other two quadrants of Minkowski space. This expectation will indeed be borne out.

In section 26.3 we had seen that observers following timelike radial ingoing geodesics reach $r = 2m$ at finite values of the affine parameter (proper time), and in section 26.4 that they do so at infinite values of the coordinate time t. The same is *mutatis mutandis* true for ingoing lightrays travelling along null geodesics, and the argument in this case is even simpler.

Indeed, since for radial lightrays the effective potential (25.43) is zero, radial lightrays are governed by the equation

$$\tfrac{1}{2}\dot{r}^2 = E_{eff} = \tfrac{1}{2}E^2 \quad \Rightarrow \quad \dot{r} = \pm E \ , \tag{26.81}$$

the lower sign corresponding to ingoing lightrays. The solution for ingoing lightrays is thus evidently

$$r(\lambda) = r(0) - E\lambda \tag{26.82}$$

so that from any finite initial position $r(0) > 2m$, the Schwarzschild radius $r = 2m$ is reached for the finite value

$$\lambda = \frac{r(0) - 2m}{E} \tag{26.83}$$

of the affine parameter.

Note, by the way, that (26.82) or $\ddot{r} = 0$ shows that r and λ are related by an affine transformation so that one can equally well use r as the affine parameter along ingoing null geodesics, making it even more manifest that $r = 2m$ arises at a finite value of that affine parameter. To reduce clutter, let us choose $r(0) = 2m$ so that $r = 2m$ is reached at $\lambda = 0$. It then follows from $f(r)\dot{t} = E$, i.e.

$$\dot{t} = E/f(r) = Er/(r - 2m) = E + 2mE/(r - 2m) \ , \tag{26.84}$$

that

$$\dot{t}(\lambda) = E + 2mE/(-E\lambda) = E - 2m/\lambda \quad \Rightarrow \quad t(\lambda) = E\lambda - 2m\log(-\lambda) \ . \tag{26.85}$$

This makes it manifest that

$$r \to 2m \quad \Rightarrow \quad \lambda \to 0_- \quad \Rightarrow \quad t \to +\infty \tag{26.86}$$

also for null geodesics. Thus t is evidently not a good coordinate to describe physics at (or beyond) $r = 2m$.

To improve our understanding of the Schwarzschild geometry, it is important to study its *causal structure*, i.e. the lightcones. Radial null curves satisfy

$$(1 - 2m/r)dt^2 = (1 - 2m/r)^{-1}dr^2 \ . \tag{26.87}$$

Thus

$$\frac{dt}{dr} = \pm(1 - 2m/r)^{-1} \ . \tag{26.88}$$

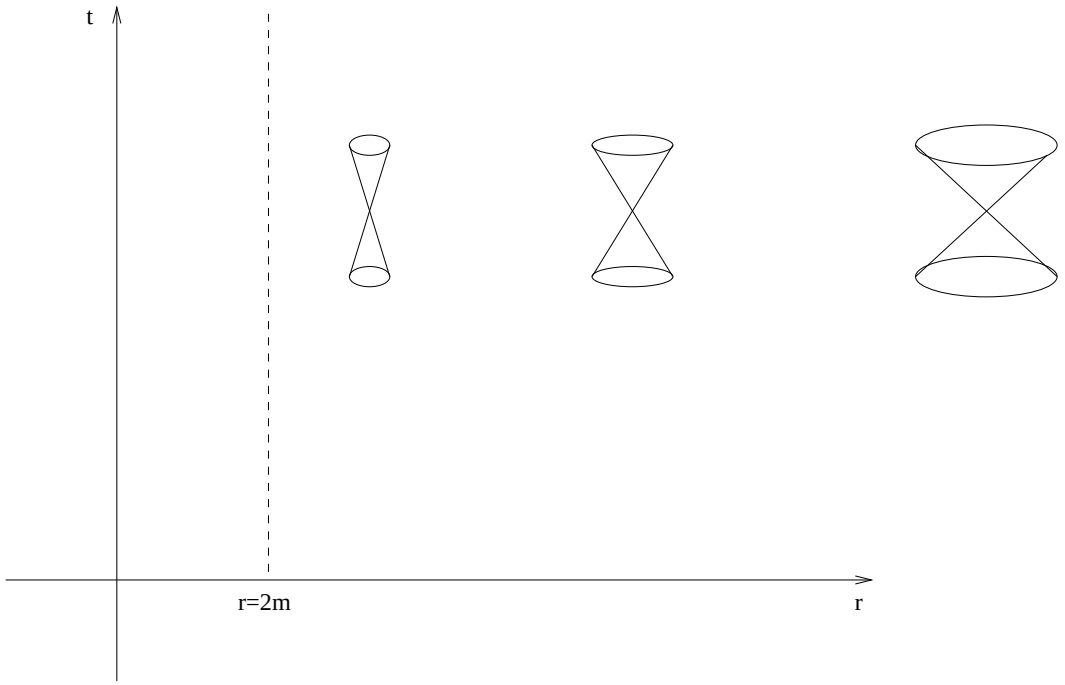


Figure 16: Causal structure of the Schwarzschild geometry in the Schwarzschild coordinates (r, t). As one approaches $r = 2m$, the lightcones become narrower and narrower and eventually fold up completely.

Recall that, as noted above, this equation for $t = t(r)$ can be equivalently regarded as an equation for $t = t(\lambda)$ as a function of the affine parameter.

In the (t, r)-diagram of Figure 16, dt/dr represents the slope of the lightcones at a given value of r. Now, as $r \to 2m$, one has

$$\frac{dt}{dr} \stackrel{r\to 2m}{\longrightarrow} \pm\infty \ , \tag{26.89}$$

so the light cones 'close up' as one approaches the Schwarzschild radius. This is the same statement as before regarding the fact that the coordinate velocity goes to zero at $r = 2m$, but this time for null rather than timelike geodesics.

As our first step towards introducing coordinates that are more suitable for describing the region around r_s, let us write the Schwarzschild metric in the form

$$ds^2 = (1 - 2m/r)(-dt^2 + (1 - 2m/r)^{-2} dr^2) + r^2 d\Omega^2 \ . \tag{26.90}$$

We see that it is convenient to introduce a new radial coordinate r^* via

$$dr^* = (1 - 2m/r)^{-1} dr = \frac{r}{r - 2m} dr = (1 + \frac{2m}{r - 2m}) dr \ . \tag{26.91}$$

The solution to this equation is (up to an arbitrary finite constant)

$$r^* = r + 2m \log(r/2m - 1) \ . \tag{26.92}$$

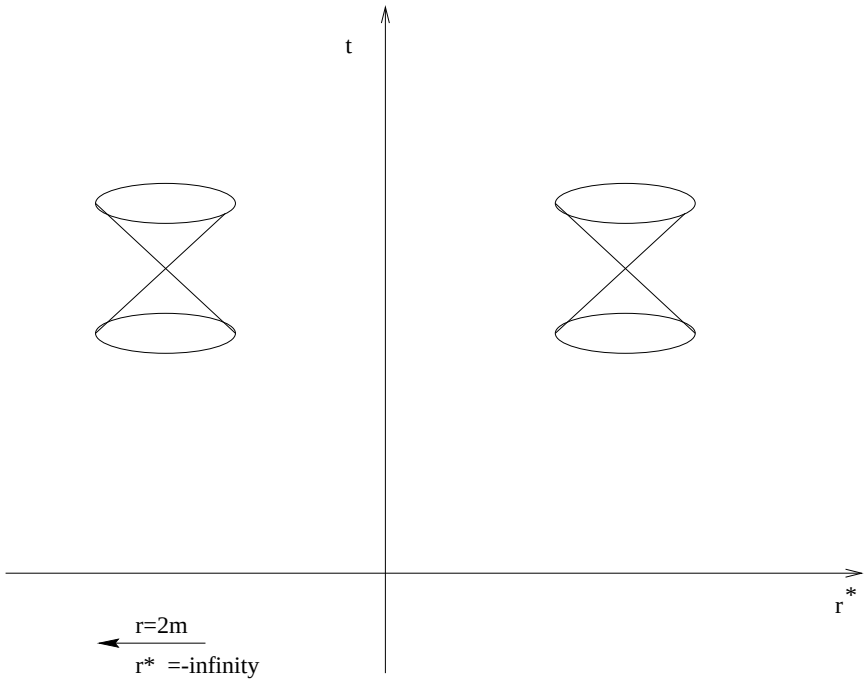


Figure 17: Causal structure of the Schwarzschild geometry in the tortoise coordinates (r^*, t). The lightcones look like the lightcones in Minkowski space and no longer fold up as $r \to 2m$ (which now sits at $r^* = -\infty$).

This new radial coordinate r^*, known as the *Regge - Wheeler radial coordinate* or *tortoise coordinate*, also provides us with the solution

$$t = \pm r^* + C_\pm \tag{26.93}$$

(with $C_\pm$ constants of integration) to the equation (26.88) describing the radial lightcones:

- lines of constant $C_+ = t - r^*$ describe *outgoing* lightrays;
- lines of constant $C_- = t + r^*$ describe *ingoing* lightrays.

In terms of r^* the metric simply reads

$$ds^2 = (1 - 2m/r)(-dt^2 + dr^{*2}) + r^2 d\Omega^2 \ , \tag{26.94}$$

where r is to be thought of as a function of r^*.

Now the lightcones, defined by

$$dt^2 = dr^{*2} \ , \tag{26.95}$$

do not seem to fold up as the lightcones have the constant slope $dt/dr^* = \pm 1$ (see Figure 17), and there is no singularity at $r = 2m$. However, r^* is still only defined for

$r > 2m$ and the surface $r = 2m$ has been pushed infinitely far away ($r = 2m$ is now at $r^* = -\infty$). Moreover, even though non-singular, the metric components g_{tt} and $g_{r^*r^*}$ (as well as $\sqrt{g}$) vanish at $r = 2m$.

Thus the tortoise coordinate has so far not really allowed us to make dramatic progress in our exploration of the region near or behind r_s, but we will substantially improve this situation in section 27.

26.8 Massless Klein-Gordon Scalar Field in the Schwarzschild Geometry

As an aside, and to conclude this section, I just want to point out that the tortoise coordinate r^* and the corresponding retarded and advanced time-coordinates $u, v = t \mp r^*$ (which we will reencounter in section 27.4 as part of the Eddington Finkelstein coordinates) are not only useful for clarifying the causal structure of the Schwarzschild geometry, but also for the analysis of the propagation of scalar (and other) fields. This is principally due to the fact that in these coordinates the (t, r)-part of the metric is conformally flat - see (26.94) or (27.141) below. Combined with the observation of section 7.7 that the massless Klein-Gordon action is conformally invariant in $(1 + 1)$-dimensions, this leads to a canonical form for the (3+1) wave operator in the (t, r)-sector (and the (θ, ϕ)-sector is standard anyway).

Specifically, we start with the action for a massless scalar field ϕ in the metric (26.94),

$$\begin{aligned} S[\phi] &\sim \int \sqrt{g} d^4x \; g^{\alpha\beta} \partial_\alpha \phi \partial_\beta \phi \\ &= \int dt \; dr^* \; d\Omega \; r^2 f(r) \left[f(r)^{-1} (-(\partial_t \phi)^2 + (\partial_{r^*} \phi)^2) - r^{-2} \phi \; \Delta_{S^2} \phi \right] \\ &= \int dt \; dr^* \; d\Omega \; \left[-(r\partial_t \phi)^2 + (r\partial_{r^*} \phi)^2) - f(r) \phi \; \Delta_{S^2} \phi \right] \quad , \end{aligned} \tag{26.96}$$

where $f(r) = 1 - 2m/r$, $d\Omega = \sin\theta d\theta \; d\phi$ denotes the solid angle on the 2-sphere, and Δ_{S^2} is the Laplace operator on the 2-sphere. Separating variables according to

$$\phi(x) = r^{-1} \sum_{\ell,m} \psi_{\ell m}(t, r^*) Y_{\ell m}(\theta, \phi) \quad , \tag{26.97}$$

using

$$\Delta_{S^2} Y_{\ell m} = -\ell(\ell + 1) Y_{\ell m} \quad , \tag{26.98}$$

and using

$$\partial_{r^*} r = f(r) \quad \Rightarrow \quad r\partial_{r^*}(\psi_{\ell m}/r) = \partial_{r^*} \psi_{\ell m} - r^{-1} f(r) \psi_{\ell m} \quad , \tag{26.99}$$

one finds for $\psi = \psi_{\ell m}$ the equation of motion

$$(\partial_t^2 - \partial_{r^*}^2)\psi + V_\ell(r^*)\psi = 0 \quad , \tag{26.100}$$

where

$$V_\ell(r^*) = f(r)\left(\frac{\ell(\ell+1)}{r^2} + \frac{2m}{r^3}\right) \quad . \tag{26.101}$$

REMARKS:

1. Note that this potential is quite similar to the effective potential (25.43)

$$V_{eff}(r) = f(r)\frac{L^2}{2r^2} \tag{26.102}$$

for massless particles in the Schwarzschild geometry derived in section 25.3. Indeed, in the large ℓ limit the last term in (26.101) can be neglected and then V_ℓ reduces to V_{eff} with the identification $\ell(\ell+1) \to L^2$. This is as it should be and as one expects (and can show) on general grounds: in a suitable geometric optics (large ℓ, high frequency) limit the massless Klein-Gordon equation reduces to the Hamilton-Jacobi equation for null geodesics.

2. This potential is non-negative for all $2m < r < \infty$ and goes to zero quite rapidly as $r \to \infty$ or $r \to 2m$, i.e. as $r^* \to \pm\infty$,

$$\begin{aligned} r^* \to +\infty \Leftrightarrow r \to \infty: &\quad V_\ell(r) \sim (r^*)^{-2} \\ r^* \to -\infty \Leftrightarrow r \to 2m: &\quad V_\ell(r) \sim \mathrm{e}^{r^*/2m} \quad . \end{aligned} \tag{26.103}$$

This means that at infinity (in r) and near the horizon, the solutions of this equation can be chosen to have the standard right-moving (outgoing) / left-moving (ingoing) form

$$\psi(t, r^*) \sim \mathrm{e}^{\pm i\omega(t-r^*)} = \mathrm{e}^{\pm i\omega u} \quad \text{or} \quad \psi(t, r^*) \sim \mathrm{e}^{\pm i\omega(t+r^*)} = \mathrm{e}^{\pm i\omega v} \quad . \tag{26.104}$$

3. However, this does not mean that a mode having the above form near infinity, say, evolved from a mode that had also had such a form near the horizon. Rather, the almost infinite exponential gravitational redshift between the near-horizon and asymptotic regions discussed in section 26.5 leads to an exponential relation between the parameter u, say, labelling an outgoing wave at infinity, and the corresponding parameter near the horizon. This exponential relation is analogous to that encountered for a scalar field in Rindler versus inertial Minkowski coordinates (section 7.8). For the Schwarzschild metric it is encoded in the precisely analogous relation (27.156) between the coordinates u, v and the Kruskal coordinates u_K, v_K to be introduced below, the latter being the analogues of Minkowski inertial coordinates, and the former the analogues of Rindler coordinates. These observations are at the heart of the so-called *Hawking Effect*, i.e. the quantum radiation of black holes. See the references given in section 27.7 for introductions to these topics.

4. By separating out the time-dependence,

$$\psi(t, r^*) = \mathrm{e}^{-i\omega t}\psi(r^*) \ , \tag{26.105}$$

the exact equation to be solved takes the form of a standard time-independent Schrödinger equation,

$$-\partial_{r^*}^2\psi + V_\ell(r^*)\psi = \omega^2\psi \ . \tag{26.106}$$

It plays an important role in numerous aspects of Black Hole physics, e.g. in the analysis of the stability of the Schwarzschild solution. In this context, the above equation and its counterparts for vectors and symmetric tensors are known as the Regge-Wheeler(-Zerilli) equations.

27 Black Holes II: the Schwarzschild Black Hole

27.1 Preliminary Remarks

The primary purpose of this section is to understand the significance and physics of the Schwarzschild radius and the region $r < 2m$. We will acccomplish this principally via a construction of appropriate, physically motivated coordinate systems and, indeed, a secondary purpose of this section is to illustrate how to go about constructing such coordinate systems in a systematic way (instead of just introducing them without further explanations).

Even though the details will differ, the general principles of how to construct coordinates to explore and understand a given space-time (by constructing and using coordinates adapted to preferred classes of observers or geodesics) can be applied to other metrics, e.g. some of those listed in section 30.

Let me also make clear from the outset what the issue is *not*. Namely, the issue is not one of constructing appropriate coordinates solely in the region $0 < r < 2m$. Indeed, we have known such coordinates all along, simply the original Schwarzschild coordinates (t, r). The Schwarzschild metric is a vacuum solution of the Einstein equations also in that region, and the coordinates (t, r) give a valid non-singular description of the metric there. Since $f(r) = 1 - 2m/r < 0$, their interpretation differs, i.e. r is a timelike coordinate, and t plays the role of a radial coordinate, but this notational issue can easily be rectified by renaming $r = T, t = R$, and writing the Schwarzschild metric in the region $0 < r = T < 2m$ as

$$\begin{aligned} & f(r) = 1 - \frac{2m}{r} \equiv -(\frac{2m}{T} - 1) \\ \Rightarrow \quad & ds^2 = -(\frac{2m}{T} - 1)^{-1} dT^2 + (\frac{2m}{T} - 1) dR^2 + T^2 d\Omega_2^2 \ . \end{aligned} \tag{27.1}$$

While this provides some minimal insight (e.g. that for $r < 2m$ surfaces of constant r are spacelike and that the metric looks time-dependent), what we are looking for is a way of describing the physics of the Schwarzschild solution that encompasses both the region $r > 2m$ and the region $r < 2m$ (and that therefore e.g. provides a valid continuous map for the freely falling observer as he crosses $r = 2m$).

The way we will go about this is to use either the worldlines of ingoing timelike geodesic observers (along with their proper time) or those of ingoing lightrays (along with an affine parameter along them), to provide us with coordinates in the region $r < 2m$ (recall that in both cases $r = 2m$ lies at a finite value of that affine parameter so that this affine parameter is also a good coordinate beyond $r = 2m$).

We will explore the 1st option in sections 27.2 and 27.3 and the 2nd option in section 27.4 and subsequent sections. The latter, based on null geodesics, is technically somewhat simpler than the former (and is therefore also the approach commonly adopted in

the literature), and will lead us rather painlessly and quickly to the maximal analytic extension of the Schwarzschild geometry in section 27.8.

I have also included the former option here, based on timelike goedesics, precisely because the resulting coordinate systems, which are quite interesting in their own right and are useful for certain more advanced applications, are usually not dealt with in any detail in the standard textbooks (and I therefore had to work out many of these details myself at some point). However, it is possible to skip sections 27.2 and 27.3 and go directly to section 27.4 and continue from there (and this is what I usually do in a first course on General Relativity).

27.2 Crossing r_s with Painlevé-Gullstrand Coordinates

We had seen that the Schwarzschild time coordinate t is adapted to static observers (whose proper time is proportional to t), and therefore not useful for describing the region $r < 2m$. We had also seen that freely falling observers cross $r = 2m$ in finite proper time τ. This suggests to choose some family of freely falling observers (geodesics) and to use their proper time $\tau = T(t, r, \theta, \phi)$ as the new time coordinate, i.e. to perform the coordinate transformation

$$(t, r, \theta, \phi) \to (T, r, \theta, \phi) \tag{27.2}$$

(retaining, for the time being, the coordinates (r, θ, ϕ)).

A natural (and the simplest) choice is to consider the family of freely falling observers which fall radially (angular momentum $L = 0$) and start off at rest from infinity (energy $E = 1$). In this case, the geodesic equations (26.20) and (26.21) take the form

$$\dot{r} = -(2m/r)^{1/2} \quad , \quad \dot{t} = (1 - 2m/r)^{-1} \quad , \tag{27.3}$$

which also imply that

$$dt/dr = -(r/2m)^{1/2}(1 - 2m/r)^{-1} \quad . \tag{27.4}$$

The solutions for $r = r(\tau)$ and $t = t(r)$ are

$$\begin{aligned} r(\tau) &= [3\sqrt{2m}(\tau_0 - \tau)/2]^{2/3} \quad \Leftrightarrow \quad \tau - \tau_0 = -(2/3)(2m)^{-1/2} r^{3/2} \\ t(r) &= t_0 - (2/3)(2m)^{-1/2} r^{3/2} - \Theta(r) \quad , \end{aligned} \tag{27.5}$$

where

$$\Theta(r) = 2m \left(2(r/2m)^{1/2} + \ln \left| \frac{(r/2m)^{1/2} - 1}{(r/2m)^{1/2} + 1} \right| \right) \tag{27.6}$$

is the solution of

$$d\Theta(r)/dr = (2m/r)^{1/2}(1 - 2m/r)^{-1} \quad . \tag{27.7}$$

In particular, we see that, up to the irrelevant constant $\tau_0 - t_0$, $\tau(r)$ and $t(r)$ are related by

$$\tau(r) = t(r) + \Theta(r) \quad . \tag{27.8}$$

We are thus led to introduce the new time coordinate $T(t, r)$ by

$$T(t,r) = t + \Theta(r) = t + 2(2mr)^{1/2} + 2m \ln \left| \frac{(r/2m)^{1/2} - 1}{(r/2m)^{1/2} + 1} \right| \quad . \tag{27.9}$$

Since the metric does not depend explicitly on t, to determine the metric in the coordinates (T, r, θ, ϕ) we only need to substitute dt by

$$dt = dT - d\Theta = dT - (r/2m)^{-1/2}(1 - 2m/r)^{-1} dr \quad , \tag{27.10}$$

or

$$dt = dT - \frac{\sqrt{r/2m}}{(r/2m) - 1} dr \quad . \tag{27.11}$$

Then one immediately finds the simple result

$$\begin{aligned} ds^2 &= -f(r)dt^2 + f(r)^{-1}dr^2 + r^2 d\Omega^2 \\ &= -f(r)dT^2 + 2\sqrt{2m/r}dTdr + (dr^2 + r^2 d\Omega^2) \\ &= -dT^2 + \left(dr + \sqrt{2m/r}dT\right)^2 + r^2 d\Omega^2 \quad . \end{aligned} \tag{27.12}$$

This is the Schwarzschild metric in *Painlevé-Gullstrand Coordinates* (Painlevé (1921), Gullstrand (1922)), abbreviated to PG coordinates in the following, and this form of the metric already reveals a number of important properties of the Schwarzschild geometry.

Most importantly, we see that due to the non-singular off-diagonal term the metric in these coordinates is well-defined and non-degenerate for all $0 < r < \infty$, in particular at $r = 2m$. This is the definitive proof that the singularity at $r = 2m$ in Schwarzschild coordinates is really just a coordinate singularity.

REMARKS:

1. The metric has the characteristic property that the metric induced on the slices of constant T is just the flat Euclidean metric for any T,

$$ds^2|_{T=T_1} = dr^2 + r^2 d\Omega^2 = d\vec{x}^2 \quad . \tag{27.13}$$

   In particular, even though the radial coordinate distance of the coordinate r is most certainly not the proper radial distance at constant Schwarzschild time t, $ds = f(r)^{-1/2} dr \neq dr$, the above form of the metric shows that the coordinate r has the property that it does measure the proper radial distance on surfaces of constant T. In this sense, r appears to be (perhaps somewhat surprisingly) adapted to freely falling observers with $E = 1$.

In any case, this makes this form of the metric (and choice of time coordinate) particularly convenient e.g. for the canoncial quantisation of fields in the Schwarzschild space-time, and it is also for this reason that this coordinate system has become increasingly popular in recent years.[81]

2. It is easy to verify / confirm directly in the PG coordinates that the new time-coordinate T really has the interpretation as measuring the proper time τ of radially freely falling observers starting off at rest at infinity. To that end note that in PG coordinates a radial timelike geodesic satisfies

$$-\dot{T}^2 + \left(\dot{r} + \sqrt{2m/r}\dot{T}\right)^2 = -1 \quad , \tag{27.14}$$

while the conserved energy associated with the T-translation invariance of the metric has the form

$$E = f\dot{T} - \sqrt{2m/r}\dot{r} \quad . \tag{27.15}$$

It immediately follows that

$$T = \tau \quad \Rightarrow \quad \dot{T} = +1 \quad \Rightarrow \quad \dot{r} = -\sqrt{2m/r} \quad \Rightarrow \quad E = 1 \quad . \tag{27.16}$$

These are precisely the geodesic paths with which we began the construction.

The coordinate transformation (27.9) is of the general form $T(t,r) = t + \psi(r)$ (24.2) discussed previously, and preserves the t-independence and manifest spherical symmetry of the metric. It leads to the metric in the form anticipated in (24.8).

Turning this around, once one has found the Schwarzschild metric in Schwarzschild coordinates, say, one can of course "discover" an infinite number of new coordinate systems for the Schwarzschild metric by performing such a (or a more general) coordinate transformation. Only in special cases, however, will one find a coordinate system that is actually useful. Let us see how to recover the PG coordinate system in this way, and how to "precover" another coordinate system that we will discuss in more detail in section 27.4.

Starting with the Schwarzschild metric in Schwarzschild coordinates, and performing the coordinate transformation $t \to T(t,r) = t + \psi(r)$, one finds the metric

$$\begin{aligned} ds^2 &= -f(r)dt^2 + f(r)^{-1}dr^2 + r^2 d\Omega^2 \\ &= -f(r)(dT - \psi'(r)dr)^2 + f(r)^{-1}dr^2 + r^2 d\Omega^2 \\ &= -f(r)dT^2 + 2f(r)\psi'(r)dTdr + f(r)^{-1}(1 - f(r)^2(\psi'(r))^2)dr^2 + r^2 d\Omega^2 \\ &\equiv -f(r)dT^2 + 2C(r)dTdr + f(r)^{-1}(1 - C(r)^2)dr^2 + r^2 d\Omega^2 \quad , \end{aligned} \tag{27.17}$$

[81] See e.g. P. Kraus, F. Wilczek, *A Simple Stationary Line Element for the Schwarzschild Geometry, and Some Applications*, `arXiv:gr-qc/9406042` and citations thereto; A. Nielsen, M. Visser, *Production and decay of evolving horizons*, `arXiv:gr-qc/0510083` for PG-like metrics for general time-dependent spherically symmetric metrics; C. Barcelo, S. Liberati, M. Visser, *Analogue Gravity*, `arXiv:gr-qc/0505065` for a detailed discussion of PG-like metrics in the context of so-called analogue models of gravity.

where

$$C(r) = f(r)\psi'(r) \tag{27.18}$$

is an essentially arbitrary function of r. This metric represents the Schwarzschild metric for any choice of $C(r)$. In particular, the vacuum Einstein equations do not impose any constraints on $C(r)$ - this can be checked explicitly, but it would be silly to do so since we have just seen explicitly that $C(r)$ is not determined and just corresponds to the freedom of performing a particular class of coordinate transformations. One can therefore now choose $C(r)$ at will.

1. One natural choice is

$$g_{Tr} = 0 \quad \Leftrightarrow \quad C(r) = 0 \ . \tag{27.19}$$

This makes the metric diagonal, corresponds to $\psi(r)$ constant, and evidently returns one to Schwarzschild coordinates. Any other choice of $C(r)$ will lead to a non-diagonal metric in the coordinates (T, r).

2. Another attractive choice is

$$\begin{aligned} g_{rr} = 1 \quad \Leftrightarrow \quad ds^2|_{T=T_0} = dr^2 + r^2 d\Omega^2 \quad &\Leftrightarrow \quad 1 - C(r)^2 = f(r) \\ &\Leftrightarrow \quad C(r) = \pm\sqrt{2m/r} \end{aligned} \tag{27.20}$$

We see that with the upper sign this is precisely the choice giving rise to Painlevé-Gullstrand coordinates introduced at the beginning of this section, confirmed by the fact that for $\psi(r)$ this implies the differential equation

$$C(r) = +\sqrt{2m/r} \quad \Rightarrow \quad \psi'(r) = \sqrt{2m/r} f(r)^{-1} \quad \Rightarrow \quad \psi(r) = \Theta(r) \ . \tag{27.21}$$

The same argument shows that the lower sign gives a corresponding set of PG coordinates based on outgoing rather than on ingoing geodesics.

3. Yet another appealing choice is

$$g_{rr} = 0 \quad \Leftrightarrow \quad C(r) = \pm 1 \quad \Leftrightarrow \quad \psi'(r) = \pm f(r)^{-1} \ , \tag{27.22}$$

so that the metric has the particularly simple (and non-singular at $r = 2m$) form

$$ds^2 = -f(r)dT^2 \pm 2dTdr + r^2 d\Omega^2 \ . \tag{27.23}$$

Referring back to the discussion of section 26.7, in particular (26.91), we see that the solution to this equation is

$$\psi(r) = \pm r^* \quad \Rightarrow \quad T(t, r) = t \pm r^* \ . \tag{27.24}$$

These are just the advanced and retarded time coordinates (u, v) of section 26.8, and we will encounter them in section 27.4 as Eddington-Finkelstein coordinates.

4. Finally, the Kerr-Schild coordinates to be discussed in section 27.6 can be obtained by the simple choice

$$C(r) = 1 - f(r) \quad \Rightarrow \quad T(t,r) = t + r^* - r = t + 2m\log(r/2m - 1) \ . \tag{27.25}$$

This procedure can in principle be applied to any static, spherically symmetric metric, and we will also make use of it later, e.g. in section 39.2 when constructing coordinates for de Sitter space.

A geometrically more satisfactory construction of such adapated coordinates can be based on the attempt to find a function T such that the 4-velocity of a family (congruence) of observers u^α is orthogonal to the slices of constant T, in the sense that $u_\alpha = -\partial_\alpha T$. The two conditions $u_\alpha u^\alpha = -1$ (for a timelike congruence, say) and $u_\alpha = -\partial_\alpha T$ imply that u^α is geodesic,

$$u^\alpha \nabla_\alpha u_\beta = u^\alpha \nabla_\beta u_\alpha = \tfrac{1}{2}\nabla_\beta(u^\alpha u_\alpha) = 0 \ , \tag{27.26}$$

(this is a special case of the general result established in (9.65) that a gradient vector field is geodesic iff it is of constant length), and that the metric component g^{TT} with respect to this new coordinate function T is

$$g^{TT} = g^{\alpha\beta}\partial_\alpha T \partial_\beta T = g^{\alpha\beta} u_\alpha u_\beta = -1 \ , \tag{27.27}$$

and the construction can proceed from there.[82]

Conversely, this kind of reasoning may allow one to discover the geometric interpretation of a coordinate system that one has selected through other criteria, e.g. via a convenient choice of the function $C(r)$ in (27.17). In particular, applied to PG coordinates, one can argue as follows:

1. Given the metric in PG coordinates, one seeks the interpretation of the gradient covector

$$u_\alpha = -\partial_\alpha T \ , \tag{27.28}$$

which is correctly normalised for a 4-velocity field, $u_\alpha u^\alpha = -1$, because $g^{TT} = -1$ in PG coordinates;

2. in Schwarzschild coordinates, u_α has the components

$$(-\partial_t T - \partial_r T, -\partial_\theta T, -\partial_\phi T) = (-1, -(r/2m)^{-1/2} f(r)^{-1}, 0, 0) \ ; \tag{27.29}$$

[82] See K. Martel, E. Poisson, *Regular coordinate systems for Schwarzschild and other spherical spacetimes*, `arXiv:gr-qc/0001069`, which also introduces generalised PG coordinates adapted to geodesic observers with $E > 1$, i.e. with non-zero velocity at infinity. Corresponding coordinates for $E < 1$, i.e. adapted to freely falling observers with maximal radius $r_i < \infty$, have been constructed by Gautreau and Hoffmann (see the references in this article).

3. thus its contravariant components give the 4-velocity field

$$u^\alpha = (f(r)^{-1}, -(r/2m)^{-1/2}, 0, 0) \ , \tag{27.30}$$

and comparison with (27.3) shows that this is precisely the family of tangent vectors

$$u^\alpha = \dot{x}^\alpha = (\dot{t}, \dot{r}, \dot{\theta}, \dot{\phi}) \tag{27.31}$$

characterising the radial ingoing geodesics with $E = 1$.

Even though we now have a coordinate system that extends in a non-singular way across $r = 2m$, so that $r = 2m$ is not a true singularity, this does not mean that nothing interesting at all happens at that locus. Indeed, the (legitimate) static observers at large radii $r \gg 2m$ are still waiting for an explanation for their observations, described in sections 26.4 and 26.5. Simply telling them that their coordinates are no good near $r = 2m$ will hardly be considered by them to be a satisfactory explanation of what they observe.

In order to address this issue, we now look at the behaviour of (radial) lightrays in PG coordinates, characterised by

$$-f(r)dT^2 + 2\sqrt{2m/r}dTdr + dr^2 = 0 \ . \tag{27.32}$$

Parametrising the lightrays by $r = r(T)$, with a prime denoting a T-derivative, $r' = dr/dT$, we can write this equation as

$$(r')^2 + 2\sqrt{2m/r}r' = f \quad \Leftrightarrow \quad (r' + \sqrt{2m/r})^2 = 1 \ . \tag{27.33}$$

Thus this has the two solutions

$$r' + \sqrt{2m/r} = \pm 1 \quad \Leftrightarrow \quad r'_\pm = \pm 1 - \sqrt{2m/r} \ , \tag{27.34}$$

and we can now understand the role that $r = 2m$ plays.

For $r > 2m$ one has

$$r > 2m: \quad r'_+ > 0 \quad \text{and} \quad r'_- < 0 \ . \tag{27.35}$$

This is the usual situation. There is one outgoing direction along which $r = r(T)$ grows, and one ingoing direction along which $r(T)$ decreases. In particular, while remaining within the future lightcone one can choose to go to either larger or smaller values of r.

For $r < 2m$, on the other hand, one has

$$r < 2m: \quad r'_+ < 0 \quad \text{and} \quad r'_- < 0 \ . \tag{27.36}$$

Thus along both directions lightrays (and therefore massive particles as well) must move to smaller values of r. In particular, *no observer, no lightray and no information can escape from the region* $r < 2m$. No wonder that the asymptotic static observers never

"see" the infalling observer cross $r = 2m$. This new region is a future extension of the Schwarzschild "patch" $r > 2m$ of the space-time, uncovered by ingoing radial geodesics.

With the opposite choice of sign in (27.20), corresponding to outgoing rather than ingoing radial geodesics, one would instead have discovered a region $r < 2m$ in which $r'_{\pm} > 0$. This can therefore not possibly be the "same" region $r < 2m$, and indeed is a past extension of the Schwarzschild patch, uncovered by the back-tracking of outgoing radial geodesics.

While it is possible to further explore the consequences of all this in terms of the present PG coordinates, since we have been led to consider the structure of the lightcones, i.e. null geodesics, it turns out to be more convenient, also for the following, to discuss this in terms of coordinates that are adapted to lightrays rather than to the timelike geodesics. These are the Eddington-Finkelstein coordinates to be discussed in section 27.4 below.

27.3 Lemaître and Novikov Coordinates

Before turning to Eddington-Finkelstein coordinates, I want to briefly mention two more coordinate systems that are instructive and occasionally useful, and that combine one of the attractive feature of PG coordinates (namely that the time coordinate is physical proper time for suitable geodesic observers) with one of the useful features of Schwarzschild coordinates (namely that the metric is diagonal), albeit at the expense of a time-dependent coordinate transformation rendering also the metric time-dependent.

This will be accomplished by constructing *comoving coordinates* for the geodesic observers underlying the (generalised) PG coordinate system, where comoving means that these observers remain at fixed values of all the spatial coordinates and only evolve in time = proper time. Since radial geodesics already remain at fixed values of the angular coordinates (θ, ϕ), what this amounts to is to trade the coordinate r for another coordinate that simply labels the individual geodesics (and that is therefore, tautologically, guaranteed to also remain constant along such a geodesic). Such a label is provided by an integration constant appearing in the solution to the radial geodesic equation since (tauto-)logically such an integration constant is constant along the geodesic.

Let us first consider the case $E = 1$. In this case, the effective potential for radial motion gives us the radial equation

$$\dot{r}^2 = \frac{2m}{r} \quad , \tag{27.37}$$

with the ingoing solution (27.5),

$$\dot{r} = -\sqrt{2m/r} \quad \Rightarrow \quad r(\tau) = [3\sqrt{2m}(\tau_0 - \tau)/2]^{2/3} \quad . \tag{27.38}$$

We can thus immediately identify τ_0 as a candidate for a comoving radial coordinate. This label τ_0 can be interpreted as the proper time at which the geodesic ends up at

$r = 0$, $r(\tau_0) = 0$. Alternatively and preferably, if one is (understandably) reluctant to label geodesics by their behaviour at $r = 0$, τ_0 can be thought of as being related to the value r_0 of $r(\tau)$ at $\tau = 0$,

$$r_0 = r(\tau = 0) = [3\sqrt{2m}\tau_0/2]^{2/3} \quad . \tag{27.39}$$

We therefore introduce a new radial variable $\rho = \rho(r, \tau)$ by

$$r(\tau, \rho) = [3\sqrt{2m}(\rho - \tau)/2]^{2/3} \quad . \tag{27.40}$$

This satisfies

$$dr = \dot{r}d\tau + (\partial r/\partial\rho)d\rho = \dot{r}(d\tau - d\rho) = -\sqrt{2m/r}(d\tau - d\rho) \quad , \tag{27.41}$$

so that (denoting the PG proper time coordinate T now by τ) the Schwarzschild line element in PG form (27.12) turns into

$$\begin{aligned} ds^2 &= -d\tau^2 + \left(dr + \sqrt{2m/r}d\tau\right)^2 + r^2 d\Omega^2 \\ &= -d\tau^2 + \left(\sqrt{2m/r}d\rho\right)^2 + r^2 d\Omega^2 \\ &= -d\tau^2 + \frac{2m}{r(\tau,\rho)}d\rho^2 + r(\tau,\rho)^2 d\Omega^2 \quad , \end{aligned} \tag{27.42}$$

with $r(\tau, \rho)$ given explicitly by (27.40). This is the Schwarzschild metric in *Lemâitre Coordinates* (Lemaître (1938)).

REMARKS:

1. These coordinates have a clear physical interpretation and also manifestly extend in a non-singular way across $r = 2m$, the Schwarzschild radius being located at the innocuous value
$$r = 2m \quad \Rightarrow \quad \rho - \tau = 4m/3 \tag{27.43}$$
of the Lemaître coordinates, to all $r > 0$.

2. This is the first (but will not be the last) time that we see that it can be useful to work with a radial coordinate, here $\rho = \rho(r, \tau)$, that is not equal to the standard (aereal radius) radial coordinate r.

3. For comparison purposes with the metric in Novikov coordinates to be discussed below, note that from the explicit change of variables (27.40) one has
$$\frac{2m}{r(\tau,\rho)} = (\partial_\tau r)^2 = (\partial_\rho r)^2 \quad , \tag{27.44}$$
so that the metric (27.42) can also be written as
$$\begin{aligned} ds^2 &= -d\tau^2 + \frac{2m}{r(\tau,\rho)}d\rho^2 + r(\tau,\rho)^2 d\Omega^2 \\ &= -d\tau^2 + (\partial_\rho r)^2 d\rho^2 + r(\tau,\rho)^2 d\Omega^2 \quad . \end{aligned} \tag{27.45}$$

This also shows that the form of the metric does not depend on the precise choice of integration constant used to label the geodesics, since it is manifestly invariant under transformations

$$\rho \to \sigma = F(\rho) \quad \Rightarrow \quad (\partial_\rho r)^2 d\rho^2 = (\partial_\sigma r)^2 d\sigma^2 \ . \tag{27.46}$$

4. Since

$$dr = 0 \quad \Rightarrow \quad d\tau = d\rho \ , \tag{27.47}$$

the metric induced on surfaces of constant $r = r_1$ is given by

$$ds^2|_{r=r_1} = -\left(1 - \frac{2m}{r_1(\tau,\rho)}\right) d\tau^2 + r_1(\tau,\rho)^2 d\Omega^2 \ . \tag{27.48}$$

Thus, as could have (partially) been anticipated from the Schwarzschild form of the metric, a hypersurface of constant $r = r_1$ is timelike for $r_1 > 2m$, null for $r_1 = 2m$, and spacelike for $r_1 < 2m$.

5. The surfaces of constant time $\tau = \tau_1$, on the other hand, are manifestly spacelike everywhere,

$$ds^2|_{\tau=\tau_1} = \frac{2m}{r(\tau_1,\rho)} d\rho^2 + r(\tau_1,\rho)^2 d\Omega^2 \ . \tag{27.49}$$

They run into the spacelike singularity at $r = 0$ at $\rho = \tau_1$.

6. Since from (27.41) one sees that at constant $\tau = \tau_1$ one has

$$\sqrt{\frac{2m}{r}} d\rho|_{\tau=\tau_1} = dr \ , \tag{27.50}$$

this spatial metric induced on surfaces of constant τ is again, as in the case of PG coordinates (27.13), just the Euclidean metric on $\mathbb{R}^3$,

$$ds^2|_{\tau=\tau_1} = dr^2 + r^2 d\Omega^2 \ . \tag{27.51}$$

This should not come as a surprise since the PG coordinate T is equal to the Lemaître coordinate τ.

7. In these coordinates, the volume element $\sqrt{g}$ has the simple form

$$\sqrt{g} = \sqrt{2m} r^{3/2} \sin\theta = 3m(\rho - \tau)\sin\theta \ . \tag{27.52}$$

8. Even though the metric is explicitly time-dependent in Lemaître coordinates, the time-translation invariance of the Schwarzschild metric is still manifest in these coordinates. Indeed, since $r(\tau,\rho)$ only depends on the difference $\rho - \tau$, the metric is invariant under simultanous constant translations

$$(\tau,\rho) \to (\tau + c, \rho + c) \tag{27.53}$$

of τ and ρ. Equivalently, in Lemaître coordinates the Killing vector ∂_t takes the form

$$\partial_t = \partial_\tau + \partial_\rho \ . \tag{27.54}$$

The norm of this Killing vector is

$$g_{\alpha\beta}(\partial_t)^\alpha(\partial_t)^\beta = -(1 - \frac{2m}{r}) \ , \tag{27.55}$$

and thus ∂_t is timelike for $r(\tau,\rho) > 2m$, null for $r = 2m$, and spacelike for $0 < r < 2m$.

9. Note that $g_{\tau\tau} = -1$ and $g_{\tau\rho} = 0$ are automatic consequences of this construction: by definition, in comoving coordinates the velocity field has the form $u^\alpha = (1,0,0,0)$, and $u_\alpha = -\partial_\alpha\tau$ implies

$$\begin{aligned} u^\alpha = -g^{\alpha\beta}\partial_\beta\tau = -g^{\alpha\tau} \stackrel{!}{=} (1,0,0,0) \quad &\Rightarrow \quad g^{\tau\tau} = -1 \quad , \quad g^{\tau\rho} = 0 \\ &\Rightarrow \quad g_{\tau\tau} = -1 \quad , \quad g_{\tau\rho} = 0 \ . \end{aligned} \tag{27.56}$$

We will discuss comoving coordinates, and metrics employing them, in much more detail in the context of cosmology in sections 33-38, cf. in particular section 34.2 for geodesics of comoving observers in comoving coordinates,

10. Somewhat surprisingly, Lemaître coordinates appear not to have attracted a lot of attention or found widespread use. However, the fact that in Lemaître coordinates the metric extends all the way to $r = 0$, is diagonal, and is explicit (unlike the Novikov coordinates, briefly discussed below, or the Kruskal-Szekeres coordinates to be described in detail later on) again makes them an attractive coordinate system to use e.g. when studying quantum field theory in a Schwarzschild background.[83]

Closely related to Lemaître coordinates are the so-called *Novikov coordinates*, comoving coordinates based on radial geodesics with a finite maximal radius, $r_i < \infty$ (i.e. $E < 1$) in the notation of section 26.3. In this case, the equation to solve is

$$\dot{r}^2 = \frac{2m}{r} + 2E_{eff} = \frac{2m}{r} + E^2 - 1 \ , \tag{27.57}$$

where E_{eff} or E depends on the choice of geodesic via a choice of integration constant (one could even choose E itself, say, as a comoving coordinate). Using r_i instead, related to the energy by

$$E(r_i)^2 = 1 - \frac{2m}{r_i} = f(r_i) = 1 + 2E_{eff}(r_i) \ , \tag{27.58}$$

[83] See e.g. K. Melnikov, M. Weinstein, *On Unitary Evolution of a Massless Scalar Field In A Schwarzschild Background: Hawking Radiation and the Information Paradox*, `arXiv:hep-th/0205223`.

the trajectories are implicitly defined by (26.28) and (26.29), i.e.

$$\begin{aligned} r(\eta) &= \tfrac{1}{2} r_i \, (1+\cos\eta) \\ \tau(\eta) &= \left(\frac{r_i^3}{8m}\right)^{1/2} (\eta + \sin\eta) \quad . \end{aligned} \tag{27.59}$$

We now think of these relations, together with the usual relation $E = f\dot{t}$ (26.21) as defining a change of variables

$$r = r(\tau, r_i) \quad , \quad t = t(\tau, r_i) \quad . \tag{27.60}$$

Note that r_i is clearly a comoving coordinate, as it can be used to label the geodesic. Note also that, if required/desired, from (27.59) one can solve for $\tau = \tau(r, r_i)$,

$$\begin{aligned} & \eta = \arccos(2r/r_i - 1) \\ \Rightarrow \quad & \tau(r, r_i) = \left(\frac{r_i^3}{8m}\right)^{1/2} \left(\arccos(2r/r_i - 1) + 2\sqrt{(r/r_i) - (r/r_i)^2}\right) \quad . \end{aligned} \tag{27.61}$$

However, even without having to solve or invert these implicit equation, it is easy to see that the resulting metric will have the form

$$ds^2 = -d\tau^2 + f(r_i)^{-1} r'(\tau, r_i)^2 dr_i^2 + r(\tau, r_i)^2 d\Omega^2 \quad , \tag{27.62}$$

where $r'(\tau, r_i) = \partial r(\tau, r_i)/\partial r_i$. This is the metric in *Novikov Coordinates* (Novikov, 1964).

In order to establish (27.62), we proceed as follows:

- Writing

  $$dt = \dot{t} d\tau + t' dr_i \quad , \quad dr = \dot{r} d\tau + r' dr_i \quad , \tag{27.63}$$

  and plugging this into the Schwarzschild metric, one finds

  $$ds^2 = -(f\dot{t}^2 - f^{-1}\dot{r}^2) d\tau^2 + 2(f^{-1}\dot{r}r' - f\dot{t}t') d\tau dr_i + (f^{-1}(r')^2 - f(t')^2) dr_i^2 \quad . \tag{27.64}$$

- The first term in brackets is equal to 1 because $(t(\tau), r(\tau))$ parametrise a timelike radial geodesic,

  $$\text{timelike radial geodesic} \quad \Rightarrow \quad -f(r)\dot{t}^2 + f(r)^{-1}\dot{r}^2 = -1 \quad . \tag{27.65}$$

  Thus $g_{\tau\tau} = -1$, as expected for comoving coordinates.

- Moreover, for comoving coordinates the second term in brackets, i.e. the off-diagonal term, is zero, $g_{\tau r_i} = 0$ (27.56). Thus we can deduce

  $$g_{\tau r_i} = 0 \quad \Rightarrow \quad f^{-1}\dot{r}r' - f\dot{t}t' = 0 \quad . \tag{27.66}$$

Using

$$\dot{r} = -(f_i - f)^{1/2} \quad , \quad f\dot{t} = E = f_i^{1/2} \qquad (f_i \equiv f(r_i)) \ , \tag{27.67}$$

this allows us to eliminate t' in favour of r',

$$(t')^2 = f^{-2}(1 - f/f_i)(r')^2 \ . \tag{27.68}$$

- This implies for the third term in brackets that

$$g_{r_i r_i} = f^{-1}(r')^2 - f(t')^2 = f_i^{-1}(r')^2 \ , \tag{27.69}$$

leading to the result given in (27.62).

REMARKS:

1. Note that, just as the Lemaître metric in (27.45), the form of the Novikov metric does not depend on the precise choice of integration constants used to label the geodesics, as the term $(\partial_{r_i} r)^2 (dr_i)^2$ is invariant under transformations

$$r_i \to \rho = F(r_i) \ . \tag{27.70}$$

Both metrics can then uniformly be presented as

$$ds^2 = -d\tau^2 + \frac{(\partial_\rho r)^2}{1 + 2E_{eff}(\rho)} d\rho^2 + r(\tau, \rho)^2 d\Omega^2 \ , \tag{27.71}$$

where $r(\tau, \rho)$ is the solution to the radial equation of motion

$$\left(\frac{\partial r}{\partial \tau}\right)^2 = \frac{2m}{r} + 2E_{eff}(\rho) \ . \tag{27.72}$$

For $E_{eff}(\rho) = 0$ this reduces to the metric in Lemaître coordinates, for $E_{eff} < 0$ and $\rho = r_i$ this reduces to the above metric in Novikov coordinates, and one can likewise consider Novikov coordinates based on radial geodesics with $E_{eff} > 0$, i.e. with a non-zero velocity at infinity.

2. Usually, the metric is expressed not in terms of the variable r_i but in terms of

$$R = (r_i/2m - 1)^{1/2} \quad \Leftrightarrow \quad r_i = 2m(R^2 + 1) \quad \Leftrightarrow \quad E^2 = f_i = \frac{R^2}{R^2 + 1} \tag{27.73}$$

so that the metric takes the form

$$ds^2 = -d\tau^2 + \frac{R^2 + 1}{R^2} \left(\frac{\partial r}{\partial R}\right)^2 dR^2 + r(\tau, R)^2 d\Omega^2 \ . \tag{27.74}$$

In terms of R, the relation (27.61) can also, using

$$\arccos(2r/r_i - 1) = 2 \arccos(r/2m(R^2 + 1))^{1/2} \ , \tag{27.75}$$

which in turn can be established by using the trigonometric identity

$$\cos 2\varphi = 2\cos^2\varphi - 1 \quad , \tag{27.76}$$

be written as

$$\frac{\tau(r,R)}{2m} = (R^2+1)\left[\frac{r}{2m} - \frac{(r/2m)^2}{R^2+1}\right]^{1/2} + (R^2+1)^{3/2}\arccos\left[\left(\frac{r/2m}{R^2+1}\right)^{1/2}\right] \quad . \tag{27.77}$$

This expression is occasionally found and used in the literature.

3. The fact that $r(\tau, r_i)$ or $r(\tau, R)$ is only determined implicitly makes Novikov coordinates somewhat more awakward to use in practice than Lemaître coordinates.[84] Nevertheless, this is compensated by their clear physical interpretation. In particular, they are useful e.g. in numerical simulations and other investigations of gravitational collapse, and the Novikov metric will indeed naturally arise in this context in our brief discussion of gravitational collapse in section 29, in particular first in section 29.3.

4. Since the timelike geodesics that Novikov coordinates are based on oscillate in finite proper time from $r = 0$ in the past through the maximal radius r_i to $r = 0$ in the future, Novikov coordinates cover both the past and future extensions of the metric in the Schwarzschild patch discovered in terms of PG coordinates in section 27.2, and to be discussed again below in section 27.5. The reflection symmetry $R \to -R$ of the metric also exchanges the Schwarzschild patch and its "mirror region" (first encountered in the context of isotropic coordinates at the end of section 24.5, and to be discussed in detail in section 27.8), and thus Novikov coordinates actually turn out to provide a complete covering of the fully extended Kruskal-Schwarzschild space-time.[85]

27.4 Eddington-Finkelstein Coordinates I: Regularity of the Metric

In previous sections we have used the worldlines of freely falling observers and in particular the proper time along such worldlines, to construct (in various ways) coordinates in which the Schwarzschild metric extends in a non-singular way across $r = 2m$.

Instead of singling out some class of timelike observers in order to construct coordinates, it is at least equally (if not more) natural to introduce coordinates that are adapted to null geodesics.

[84] See e.g. the appendix of J. Makela, A. Peltola, *Thermodynamical Properties of Horizons*, `arXiv:gr-qc/0205128` for somewhat more explicit expressions, including a Taylor expansion of r'.

[85] See e.g. Figure 31.2 of C. Misner, K. Thorne, J. Wheeler, *Gravitation*.

We can easily accomplish this by promoting the integration constants $C_\pm$ in (26.93) labelling the lightray to new coordinates, namely the retarded or advanced time coordinate

$$C_+ \to u = t - r^* \quad , \quad C_- \to v = t + r^* \ , \tag{27.78}$$

where

$$r^* = r + 2m \log |r/2m - 1| \tag{27.79}$$

is the solution of

$$\frac{dr^*}{dr} = f(r)^{-1} \ . \tag{27.80}$$

Then ingoing radial null geodesics ($dr^*/dt = -1$) are characterised by $v = const.$ and outgoing radial null geodesics by $u = const.$ (and u and v can be thought of as "comoving" coordinates for outgoing resp. ingoing lightrays).

Then we can label space-time points (in addition to by their angular coordinates) e.g. in terms of ingoing lightrays by specifying the lightray (i.e. v) and the affine parameter λ indicating how far one has to travel along that lightray to reach the point. As we had seen explicitly e.g. in (26.82), this affine parameter can conveniently be chosen to be the radial coordinate r itself, and evidently extends across $r = 2m$.

Thus we now pass to the ingoing or outgoing *Eddington-Finkelstein* coordinates (v, r, θ, ϕ) or (u, r, θ, ϕ) (i.e. we keep r but eliminate t). Note that this coordinate transformation $u(t, r) = t - r^*$ or $v(t, r) = t + r^*$ is also of the general form $T(t, r) = t + \psi(r)$ whose consequences were already explored in section 27.2. Since the metric does not depend explicitly on t, we only need to substitute dt using

$$dt = dv - dr^* = dv - dr/f(r) \tag{27.81}$$

(and likewise for u). It follows that in terms of these coordinates the Schwarzschild metric reads

$$ds^2 = -(1 - 2m/r)dv^2 + 2dv\ dr + r^2 d\Omega^2 \tag{27.82}$$

in ingoing (advanced) Eddington-Finkelstein coordinates, and

$$ds^2 = -(1 - 2m/r)du^2 - 2du\ dr + r^2 d\Omega^2 \tag{27.83}$$

in outgoing (retarded) Eddington-Finkelstein coordinates, as already anticipated in (27.23). This way of writing the Schwarzschild metric turns out to be extremely informative and useful. We will focus on the metric and its regularity in this section, and discuss the causal structure (lightcones) and its implications in section 27.5 below.

REMARKS:

1. Even though the metric coefficent g_{uu} or g_{vv} vanishes at $r = 2m$, as for the PG coordinates of section 27.2 there is no real degeneracy, the two-dimensional

metric in the (v,r)- or (u,r)-directions having the completely non-singular and non-degenerate form

$$(g_{(v,r),(v,r)}) = \begin{pmatrix} -f(r) & \pm 1 \\ \pm 1 & 0 \end{pmatrix} \tag{27.84}$$

Indeed, the determinant and inverse of this (2×2)-block of the metric are

$$|\det(g_{(v,r),(v,r)})| = 1 \tag{27.85}$$

and

$$(g^{(v,r),(v,r)}) = \begin{pmatrix} 0 & \pm 1 \\ \pm 1 & f(r) \end{pmatrix} \ , \tag{27.86}$$

both of which are completely regular for all $r > 0$, in particular at $r = 2m$.

Therefore we can now extend the range of r to the region $0 < r < 2m$ with impunity. Thus this provides another explicit proof that the singularity of the Schwarzschild metric in the Schwarzschild coordinates at $r = 2m$ is a removable (pure coordinate) singularity.

2. Given the Eddington-Finkelstein coordinates, in which the metric is regular at and across $r = 2m$, it is now instructive to revisit some of the preliminary calculations that we performed in section 26 while exploring the Schwarzschild metric in the region $r \gtrsim 2m$.

   In particular, let us reconsider the calculation of the acceleration of a static observer (section 26.2). In Schwarzschild (SS) coordinates (which for the purposes of this remark we will denote by y^μ, reserving x^α for the Eddington-Finkelstein (EF) coordinates), the acceleration vector of a static oberver with 4-velocity (26.6)

$$(u^\mu)_{SS} = (u^t, u^r, u^\theta, u^\phi) = (f(r)^{-1/2}, 0, 0, 0) \tag{27.87}$$

   was found to be (26.10)

$$(a^\mu)_{SS} = (a^t, a^r, a^\theta, a^\phi) = (0, m/r^2, 0, 0) \ , \tag{27.88}$$

   which looks nicely regular and unspectacularly "Newtonian". This was misleading, however, as the norm of the acceleration vector involves the component $g_{rr} = f(r)^{-1}$ of the metric, leading to

$$g_{\mu\nu}(a^\mu)_{SS}(a^\nu)_{SS} = f(r)^{-1}(m/r^2)^2 \ , \tag{27.89}$$

   which diverges as $r \to 2m$.

   Let us now look at how this calculation presents itself in EF coordinates in which, as we have seen, the metric is regular at $r = 2m$. In EF coordinates, a static observer has 4-velocity

$$(u^\alpha)_{EF} = (u^v, u^r, u^\theta, u^\phi) = (u^v, 0, 0, 0) \ , \tag{27.90}$$

with

$$g_{\alpha\beta}(u^\alpha)_{EF}(u^\beta)_{EF} = -f(r)(u^v)^2 = -1 \quad \Rightarrow \quad u^v = f(r)^{-1/2} \ . \tag{27.91}$$

Thus

$$(u^\alpha)_{EF} = (f(r)^{-1/2}, 0, 0, 0) \ . \tag{27.92}$$

This agrees with the result in SS coordinates, as could have also been deduced from the vectorial transformation behaviour

$$(u^v)_{EF} = \frac{\partial v}{\partial y^\mu}(u^\mu)_{SS} = \frac{\partial v}{\partial t}(u^t)_{SS} = (u^t)_{SS} \ . \tag{27.93}$$

under the coordinate transformation

$$dv = dt + dr_* = dt + dr/f(r) \quad \Rightarrow \quad \frac{\partial v}{\partial t} = 1 \quad , \quad \frac{\partial v}{\partial r} = f(r)^{-1} \ . \tag{27.94}$$

The acceleration vector can now be determined

- either by calculating

$$a^\alpha = \Gamma^\alpha_{\beta\gamma} u^\beta u^\gamma = \Gamma^\alpha_{vv}(u^v)^2 = f(r)^{-1}\Gamma^\alpha_{vv} \ . \tag{27.95}$$

  in EF coordinates (Exercise!),

- or by transforming $(a^\mu)_{SS}$ to EF coordinates,

$$(a^\alpha)_{EF} = \frac{\partial x^\alpha}{\partial y^\mu}(a^\mu)_{SS} = \frac{\partial x^\alpha}{\partial r}(a^r)_{SS} \quad \Rightarrow \quad \begin{cases} (a^v)_{EF} = f(r)^{-1} m/r^2 \\ (a^r)_{EF} = m/r^2 \end{cases} \tag{27.96}$$

Either way, the result is

$$(a^\alpha)_{EF} = (a^v = f(r)^{-1} m/r^2, a^r = m/r^2, 0, 0) \ . \tag{27.97}$$

Thus the (non-singular, "Newtonian") r-component agrees with that of the acceleration in SS coordinates, but in addition in EF coordinates there is now a v-component which is singular as $r \to 2m$.

Thus the norm of the acceleration in EF coordinates is

$$g_{\alpha\beta}(a^\alpha)_{EF}(a^\beta)_{EF} = -f(r)(a^v)^2 + 2a^v a^r = f(r)^{-1}(m/r^2)^2 \ , \tag{27.98}$$

in complete agreement with the result in SS coordinates (as it should, since this is a scalar).

We see that, in these coordinates in which the metric is regular, the divergence of the acceleration shows up in the components of that quantity itself (whereas in the "singular" SS coordinates that divergence was misleadingly hidden in the metric).

3. One can of course check directly that the Eddington-Finkelstein metric (27.82) is a solution of the vacuum Einstein equations. One can also obtain the metric directly from integrating the Einstein vacuum equations if one starts not with the standard form of a static isotropic metric (24.6) but makes an ansatz of the form

$$ds^2 = -f(v,x)dv^2 + 2dv\,dx + r(v,x)^2 d\Omega^2 \tag{27.99}$$

in terms of two unknown functions $f(v,x)$ and $r(v,x)$ of two variables. The same arguments as in the discussion of ingoing null geodesics in section 27.5 below show that this general form of the metric is characterised by the fact that that the integral curves of the null vector ∂_x (i.e. the curves with constant (v,θ,ϕ)) are null geodesics, with affine parameter x,

$$g_{xx} = 0 \quad , \quad \nabla_{\partial_x}\partial_x = 0 \ . \tag{27.100}$$

As one can always choose to build a coordinate system in such a way, this is a valid a priori ansatz for the metric, and from this perspective one never encounters the question if the singularity at $r = 2m$ is real or not, since it is not even a coordinate singularity in these coordinates.

Alternatively, one can start with the metric in the Bondi gauge (40.30),

$$ds^2 = -\mathrm{e}^{2h(v,r)} f(v,r)dv^2 + 2\mathrm{e}^{h(v,r)} dv\,dr + r^2 d\Omega^2 \ . \tag{27.101}$$

This is tantamount to introducing the aereal radius $r(v,x)$ as a new coordinate instead of x, and this ansatz is the Eddington-Finkelstein-like counterpart of the Schwarzschild-Birkhoff ansatz (24.68). It is shown in section 40.3 that solving the vacuum Einstein equations in this gauge, one also obtains directly the Schwarzschild solution in Eddington-Finkelstein coordinates.

4. In the Eddington-Finkelstein coordinates, the Killing vector $\xi = \partial_t$ generating the time translation invariance of the metric in Schwarzschild coordinates can be written as

$$\xi = \partial_t = (\partial_t r)\partial_r + (\partial_t v)\partial_v = \partial_v \tag{27.102}$$

or

$$\xi = \partial_t = (\partial_t r)\partial_r + (\partial_t u)\partial_u = \partial_u \ . \tag{27.103}$$

That this is indeed a Killing vector is obvious from the fact that in Eddington-Finkelstein coordinates the components of the metric do not depend on v or u.

In particular, ξ now extends smoothly across $r = 2m$, with norm

$$g_{\alpha\beta}\xi^\alpha\xi^\beta = -(1 - 2m/r) \ . \tag{27.104}$$

Thus ξ is timelike for $r > 2m$, null on $r = 2m$ and spacelike for $0 < r < 2m$. This is a crucial and characteristic feature we will come back to on various occasions in subsequent sections.

27.5 Eddington-Finkelstein Coordinates II: Event Horizons and Black Holes

We now take a closer look at the causal structure of the extended Schwarzschild geometry in Eddington-Finkelstein coordinates. To determine the lightcones in ingoing Eddington-Finkelstein coordinates we again look at radial null geodesics which this time are solutions to

$$(1-2m/r)dv^2 = 2dv\,dr \quad . \tag{27.105}$$

There are thus two possibilities:

1. Ingoing null geodesics are described by $dv/dr = 0$ or $v = const.$. These null geodesics can also be understood in a different way. Note that from the Eddington-Finkelstein form (27.82) of the metric it is evident that ∂_r is a null vector, since $g_{rr} = 0$. Moreover, its integral curves, i.e. the curves

$$x^\alpha(\lambda) = (r = r_0 - \lambda, v = v_0, \theta = \theta_0, \phi = \phi_0) \tag{27.106}$$

are geodesics. The sign $-\lambda$ has been chosen so that the tangent vector $\dot{x}^\alpha = dx^\alpha/d\lambda$ is future-oriented, i.e. such that its scalar product with $\partial_t = \partial_v$ is negative,

$$g_{\alpha\beta}\dot{x}^\alpha(\partial_t)^\beta = g_{\alpha v}\dot{x}^\alpha = \dot{r} \overset{!}{<} 0 \quad . \tag{27.107}$$

Thus the radius indeed decreases along future-oriented ingoing null geodesics (as the name was meant to suggest) and the radial coordinate is an affine parameter along these geodesics (as we also already saw in (26.82)).

That these curves are geodesics can be seen explicitly by calculating the acceleration of $u^\alpha = (0, -1, 0, 0)$ (in the coordinates (v, r, θ, ϕ)),

$$u^\alpha\nabla_\alpha u^\beta = -\nabla_r u^\beta = -\Gamma^\beta_{r\gamma}u^\gamma = \Gamma^\beta_{rr} \quad , \tag{27.108}$$

or more concisely (see (5.21)) by

$$\nabla_{\partial_r}\partial_r = \Gamma^\beta_{rr}\partial_\beta \quad . \tag{27.109}$$

Either way this shows that ∂_r is geodesic, as $\Gamma^\beta_{rr} = 0$ (since $g_{rr} = 0$, the only possible contribution to $\Gamma_{\beta rr}$ could have arisen from $\partial_r g_{vr}$, but $g_{vr} = 1$ is constant).

2. Outgoing null geodesics are described by the solutions to the equation

$$\frac{dv}{dr} = 2(1-2m/r)^{-1} \quad . \tag{27.110}$$

The solution to this equation is evidently

$$v(r) = 2r^* + C \quad \Leftrightarrow \quad u = t - r^* = C \quad . \tag{27.111}$$

We thus reassuringly recover the fact that outgoing lightrays are described by lines of constant u.

Thus the metric and the lightcones remain well-behaved (do not fold up) at $r = 2m$, the surface $r = 2m$ is at a finite coordinate distance, namely (to reiterate the obvious) at $r = 2m$, and there is no problem with following geodesics beyond $r = 2m$.

In particular, this means that we now encounter no difficulties when entering the region $r < 2m$, e.g. along lines of constant v and this region should be included as part of the physical space-time. Note that because $v = t + r^*$ and $r^* \to -\infty$ for $r \to 2m$, we see that decreasing r along lines of constant v amounts to $t \to \infty$. Thus the new region at $r \leq 2m$ we have discovered is in some sense a future extension of the original Schwarzschild space-time.

To understand the nature of this new region, we now take a more detailed look at the behaviour of the lightcones. Even though the lightcones do not fold up at $r = 2m$, something interesting is certainly happening there. Whereas, in a (v, r)-diagram (see Figure 18), one side of the lightcone always remains horizontal (at $v = const.$), the other side becomes vertical at $r = 2m$ ($dv/dr = \infty$) and then tilts over to the other side. In particular, beyond $r = 2m$ all future-directed paths, those within the forward lightcone, now have to move in the direction of decreasing r: clearly the ingoing null geodesics move towards smaller values of r, but so do those that for $r > 2m$ were outgoing,

$$(27.110) \quad \Rightarrow \quad dr/dv < 0 \quad \text{for} \quad r < 2m \ . \tag{27.112}$$

There is thus no way to turn back to larger values of r, not on a geodesic but also not on any other path (i.e. not even with a powerful rocket) once one has gone past $r = 2m$.

Thus, even though locally the physics at $r = 2m$ is well behaved, globally the surface $r = 2m$ is very significant as it is a point of no return:

- Since $r = 2m$ is a null surface, once one has reached the event horizon one has to travel at the speed of light to stay there and not be forced further towards smaller values of r.
- Once one has passed the Schwarzschild radius, there is no turning back to larger values of r.

Therefore nothing, absolutely nothing, no information, no lightray, no particle, can escape from the region $r < 2m$. Thus we have a *Black Hole*, an object that is (classically) completely invisible from the outside. In particular, no information about any events occurring in the black hole region $r < 2m$ can reach the asymptotic ($r >> 2m$) region. The boundary of this region is the surface $r = 2m$, and a (necessarily null) surface with this property is known as an *Event Horizon*.

Even though the Eddington-Finkelstein coordinates (more precisely the closely related Eddington time coordinate - cf. section 27.6 below) were already introduced by Eddington back in 1924, this full significance of the Schwarzschild radius and its interpretation as a "one-way membrane" were only understood much later (Finkelstein, 1958).

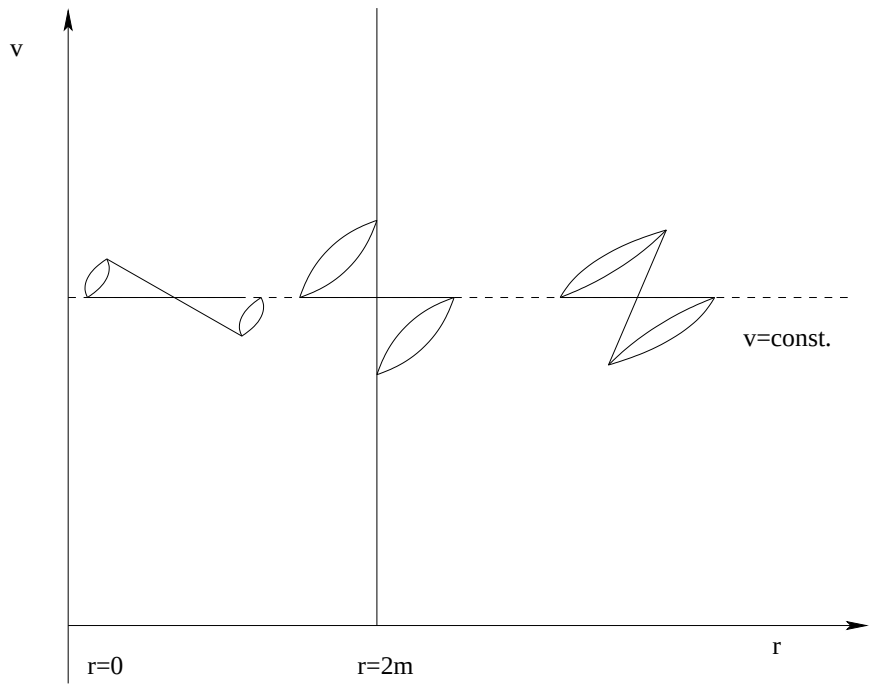


Figure 18: Behaviour of lightcones in ingoing Eddington-Finkelstein coordinates. Lightcones do not fold up at $r = 2m$ but tilt over so that for $r < 2m$ only movement in the direction of decreasing r towards the singularity at $r = 0$ is allowed.

REMARKS:

1. In the above (v, r) coordinate system we can cross the event horizon only on future directed paths, not on past directed ones, and only in the direction of decreasing r. However, clearly this cannot be the whole story: the Schwarzschild metric in the Schwarzschild coordinates is invariant under time-reflections $t \to -t$. Hence there must also exist a time-reversed version of the future extension and its event horizon. Noting that $t \to -t$ implies

$$t \to -t \quad \Rightarrow \quad v \to -u \quad , \quad dr/dt \to -dr/dt \ , \tag{27.113}$$

   it is clear that one will have access to this new region when working with the outgoing Eddington-Finkelstein coordinates (u, r) instead, and backtracking outgoing lightrays beyond $t = -\infty$.

   Indeed, when one uses the coordinates (u, r) instead of (v, r), the lightcones in Figure 18 are flipped (either up-down or left-right), and one can now pass through the horizon along future directed paths only in the outgoing direction of increasing r.

   The new region of space-time covered by the coordinates (u, r) is thus definitely different from the new region we uncovered using (v, r) even though both of them lie 'behind' $r = 2m$. In fact, this one is a past extension (beyond $t = -\infty$) of

the original Schwarzschild 'patch' of space-time. In this patch, the region behind $r = 2m$ acts like the opposite (time-reversal) of a black hole (a white hole) which cannot be entered on any future-directed path.

As we will discuss later, in section 29.3, this white hole region is unphysical, i.e. an artefact of the idealisation of an eternal black hole metric. The future black hole region, however, is definitely of relevance as it can be created by gravitational collapse.

2. The above statement about the behaviour of lightrays for $r = r_s = 2m$ and $r < r_s$ can be phrased somewhat more invariantly and geometrically in terms of the *expansion* (12.101), (12.102)

$$\theta_\ell = \tfrac{1}{2} s^{\alpha\beta} L_\ell s_{\alpha\beta} = \frac{1}{\sqrt{s}} L_\ell \sqrt{s} \quad . \tag{27.114}$$

of a null geodesic congruence introduced in connection with the Raychaudhuri equation in section 12.4, and measuring the change in the cross-sectional area element $\sqrt{s}$ of the congruence along the congruence.

To that end let us introduce the null vector fields

$$n = -\partial_r \quad , \quad \ell = \partial_v + \tfrac{1}{2} f(r) \partial_r \quad . \tag{27.115}$$

It is easy to check that these are 2 linearly independent future-pointing null vector-fields, n corresponding to ingoing lightrays and ℓ to (would-be) outgoing lightrays, cross-normalised to $n_\alpha \ell^\alpha = -1$,

$$n_\alpha n^\alpha = \ell_\alpha \ell^\alpha = 0 \quad , \quad n_\alpha \ell^\alpha = -1 \quad . \tag{27.116}$$

With $\sqrt{s} = r^2 \sin\theta$, one finds from (27.114) that

$$\theta_n = -\frac{2}{r} \quad , \quad \theta_\ell = \frac{r f(r)}{r^2} = \frac{r - 2m}{r^2} \quad . \tag{27.117}$$

In particular, in Minkowski space one has the standard behaviour that the ingoing radial null congruence always has negative expansion (it is contracting) while the outgoing radial null congruence has positive expansion (it is expanding),

$$f(r) = 1 \quad \Rightarrow \quad \theta_n = -\frac{2}{r} < 0 \quad , \quad \theta_\ell = +\frac{1}{r} > 0 \quad . \tag{27.118}$$

While one still has $\theta_n < 0$ in the Schwarzschild geometry (ingoing lightrays are contracting),

$$f(r) = 1 - \frac{2m}{r} \quad \Rightarrow \quad \theta_n < 0 \quad \forall\, r \quad , \tag{27.119}$$

for the congruence ℓ one has

$$f(r) = 1 - \frac{2m}{r} \quad \Rightarrow \quad \begin{cases} \theta_\ell > 0 & r > r_s \\ \theta_\ell = 0 & r = r_s \\ \theta_\ell < 0 & r < r_s \end{cases} \tag{27.120}$$

Thus $r = r_s$ is characterised by the fact that the "outgoing" lightrays have zero expansion. We will come back to this, and the related notion of *trapped surfaces*, briefly in the context of the discussion of horizons of black holes in section 32.

Since we now have at our disposal

- the defining feature of a black hole, namely the existence of an event horizon which causally seals off the interior (black hole) region from the outside (we will discuss this in more general terms in section 32.4),
- and the prime example of a black hole gravitational field, namely that described by the future extension of the Schwarzschild metric,

let us briefly, to conclude this section, discuss (and equally briefly dispose of) several popular misconceptions about black holes that are unfortunately quite common in the pop-sci and sci-fi literature:

1. black holes as cosmic vacuum cleaners

   Often a black hole is pictured as an object travelling through space and in the process violently sucking in everything around it. This is quite misleading: at a coordinate distance r in the Schwarzschild metric, the gravitational field (i.e. the metric) is the same regardless of whether in the interior there is a star of mass $M = m/G_N$ or a black hole of the same mass (there is no extra "black hole force" that enters the scene, it is all just gravity).

   In particular, if our sun, say, suddenly collapsed to form a black hole (as we will discuss in section 29, this is the way black holes are believed to form in astrophysical processes, even though this will not be the fate of our sun), then quite a number of things will change for humanity but what will not change appreciably (if the mass is appropriately conserved in this process) is the outside gravitational field and thus e.g. the orbit of the earth.

2. "locking in" of lightrays by black holes as a consequence of extremely strong gravitational fields

   It is natural to think that the properties of black holes are due to extremely strong gravitational effects at or near the horizon which prevent even lightrays from escaping the gravitational field. This point of view is of course not completely incorrect, and is occasionally reinforced by a classical "derivation" of the Schwarzschild radius as the radius at which the escape velocity from an object of mass M with that radius equals the speed of light ($v^2/2 = G_N M/r$ and $v = c$ imply $r = 2G_N M/c^2 = 2m$; I hope I do not have to tell you that you should not take this "calculation" seriously). More to the point, perhaps, we had seen that

static observers close to $r = 2m$ require very large accelerations to balance what these observers perceive to be the gravitational attraction.

Nevertheless, this is somewhat misleading since this gravitational attraction is observer-dependent (e.g. there is none for a freely falling observer) and thus does not provide us with an objective measure of the strength of the gravitational field. Objective information about the gravitational field is contained in the tidal forces, as encoded in the Riemann curvature tensor via the geodesic deviation equation discussed e.g. in section 8.4. For instance, the Kretschmann scalar (8.50) of the Schwarzschild metric is (27.163)

$$R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma} \sim \frac{m^2}{r^6} \ . \tag{27.121}$$

This shows that the strength of the gravitational field in the Schwarzschild geometry, as measured by tidal forces, is $\sim m/r^3$ (as in the Newtonian theory). Near the horizon at $r = 2m$ this is

$$m/r^3 \sim 1/m^2 \tag{27.122}$$

and can thus be arbitrarily weak for sufficiently massive (and large) black holes. Related to this, we will estimate in section 29.3 the average density of an object with the size of its Schwarzschild radius and will see that for sufficiently massive objects (e.g. galaxies) this density can be as small as one likes.

The crucial point, as we will discuss in more detail in section 32, is that by definition a black hole is characterised in terms of global properties of space-time (in the sense of a region of space-time that is invisible to an asymptotic observer), and these are not necessarily unambiguously detectable by local observers. In particular, as we will see, event horizons can exist in (and emerge from) flat Minkowskian regions of space-time, in which there is definitely no trace of a strong gravitational field (at that time!).

27.6 Eddington Time Coordinate and Kerr-Schild Form of the Metric

Ordinary space-time diagrams are more familiar (and therefore more intuitive) than space-null diagrams such as the above (r, v)-diagram (in which, for example, in the asymptotically flat regime $r \to \infty$ the lightcone has slopes 0 and 2 rather than the usual $\pm 45°$ slopes ± 1). In the present case this can easily be rectified by introducing a new time-coordinate $\tilde{t}$ instead of v by the relation

$$v = t + r^* = \tilde{t} + r \ , \tag{27.123}$$

i.e.

$$\tilde{t} = t + 2m \log(r/2m - 1) \ . \tag{27.124}$$

This time coordinate is also known as the *Eddington time coordinate* (it was discovered and used first by Eddington, and then rediscovered by Finkelstein, but neither of them wrote down explicitly the null form of the metric that is now known as the Eddington-Finkelstein form of the metric and which we discussed above).

Initially the metric looks perhaps somewhat less illuminating in the $(\tilde{t}, r)$-coordinates,

$$ds^2 = -(1-2m/r)d\tilde{t}^2 + (4m/r)d\tilde{t}\,dr + (1+2m/r)dr^2 + r^2 d\Omega^2 \quad , \tag{27.125}$$

but the lightcones and the horizon now have the following simple and easy to visualise description, which follows from factorising the line element as

$$ds^2 = (d\tilde{t} + dr)(-(1-2m/r)d\tilde{t} + (1+2m/r)dr) + r^2 d\Omega^2 \quad , \tag{27.126}$$

namely:

- the ingoing side of the lightcone is described by

$$d\tilde{t} + dr = 0 \quad \Leftrightarrow \quad d\tilde{t}/dr = -1 \quad \Leftrightarrow \quad v = \text{const.} \tag{27.127}$$

  (as we already knew) and is thus at $-45°$ everywhere in a $(\tilde{t}, r)$-diagram.

- the "outgoing" side of the lightcone is described by

$$ds^2 = -(1-2m/r)d\tilde{t} + (1+2m/r)dr = 0 \quad \Leftrightarrow \quad d\tilde{t}/dr = (1+2m/r)/(1-2m/r) \quad . \tag{27.128}$$

  Therefore $d\tilde{t}/dr = +1$ for $r \to \infty$ (slope $+1$), so lightcones have the standard Minkowskian form there, $d\tilde{t}/dr \to \infty$ for $r \to 2m$, and $d\tilde{t}/dr \to -1$ for $r \to 0$.

- in particular, the horizon is (again) vertical in such a diagram, with the (would-be) outgoing side of the lightcone vertical and tangent to the horizon, while the lightcone degenerates as $r \to 0$.

Nice diagrams that you can find in many places depicting the collapse of a spherically symmetric star to a black hole and the formation of the horizon typically (either implicitly or explicitly) use these $(\tilde{t}, r)$-coordinates.

Remarks:

1. This coordinate transformation is once again of the general form $T(t, r) = t + \psi(r)$, whose effect on the Schwarzschild metric was studied in some detail in section 27.2. In particlar, the time coordinate $\tilde{t}$ (27.124) already appeared in (27.25).

2. Note that in these coordinates the metric can be split as

$$\begin{aligned} ds^2 &= -(1-2m/r)d\tilde{t}^2 + (4m/r)d\tilde{t}\,dr + (1+2m/r)dr^2 + r^2 d\Omega^2 \\ &= \left(-d\tilde{t}^2 + dr^2 + r^2 d\Omega^2\right) + \frac{2m}{r} dv(\tilde{t}, r)^2 \quad . \end{aligned} \tag{27.129}$$

Thus it has the curious property that in components it has the form

$$g_{\alpha\beta} = \eta_{\alpha\beta} + \frac{2m}{r}\partial_\alpha v \partial_\beta v \tag{27.130}$$

where $\eta_{\alpha\beta}$ is the Minkowski metric (here written in spatial spherical coordinates) and $\partial_\alpha v$ is null with respect to the inverse Minkowski metric $\eta^{\alpha\beta}$,

$$\eta^{\alpha\beta}\partial_\alpha v \partial_\beta v = \eta^{vv} = 0 \tag{27.131}$$

because $v = \tilde{t} + r$ is a null coordinate for the Minkowski metric $-d\tilde{t}^2 + dr^2 + (\ldots)$. This is the Schwarzschild metric in what is known as *Kerr-Schild form*. I will therefore also occasionally refer to $\tilde{t}$ as the Kerr-Schild time coordinate (but, as mentioned above, $\tilde{t}$ is commonly also known as the Eddington time coordinate).

3. By introducing standard inertial (Cartesian) coordinates $\tilde{x}^\alpha = (\tilde{t}, \vec{x})$ for Minkowski space, the metric can now also be written in the form

$$ds^2 = -d\tilde{t}^2 + d\vec{x}^2 + \frac{2m}{r}(\ell_\alpha d\tilde{x}^\alpha)^2 \tag{27.132}$$

where $r^2 = \vec{x}^2$, and ℓ_α has the components

$$\ell_\alpha = \partial_\alpha(\tilde{t} + r) = (1, x^k/r) \ . \tag{27.133}$$

4. There is evidently also a corresponding Kerr-Schild form of the metric in outgoing Eddington-Finkelstein coordinates, now with

$$u = t - r^* = \tilde{t} - r \quad \Rightarrow \quad \tilde{t} = t - 2m\log(r/2m - 1) \ , \tag{27.134}$$

so this is not the same $\tilde{t}$ as before, namely

$$ds^2 = \left(-d\tilde{t}^2 + dr^2 + r^2 d\Omega^2\right) + \frac{2m}{r}du(\tilde{t}, r)^2 \ . \tag{27.135}$$

More generally, metrics of the form

$$g_{\alpha\beta} = \eta_{\alpha\beta} + f(x)N_\alpha N_\beta \quad \text{with} \quad \eta^{\alpha\beta}N_\alpha N_\beta = \eta_{\alpha\beta}N^\alpha N^\beta = 0 \tag{27.136}$$

are known as *Kerr-Schild metrics*. This ansatz for the metric played an important role in the search for other exact solutions of the Einstein equations.

The form (27.136) implies that

1. $N^\alpha \equiv \eta^{\alpha\beta}N_\beta$ is also null with respect to $g_{\alpha\beta}$:

$$g_{\alpha\beta}N^\alpha N^\beta = \eta_{\alpha\beta}N^\alpha N^\beta + f(x)(N_\alpha N^\alpha)(N_\beta N^\beta) = 0 \tag{27.137}$$

2. the inverse metric is given by

$$g^{\alpha\beta} = \eta^{\alpha\beta} - f(x)N^\alpha N^\beta \tag{27.138}$$

Indeed, for $g^{\alpha\beta}$ of this form one has

$$\begin{aligned} g^{\alpha\beta}g_{\beta\gamma} &= (\eta^{\alpha\beta} - f(x)N^\alpha N^\beta)(\eta_{\beta\gamma} + f(x)N_\beta N_\gamma) \\ &= \delta^\alpha_{\ \gamma} - f(x)N^\alpha N_\gamma + f(x)N^\alpha N_\gamma = \delta^\alpha_{\ \gamma} \ . \end{aligned} \tag{27.139}$$

3. the contravariant components of N_α with respect to $g_{\alpha\beta}$ are the same as those with respect to $\eta_{\alpha\beta}$:

$$g^{\alpha\beta}N_\beta = \eta^{\alpha\beta}N_\beta - f(x)N^\alpha N^\beta N_\beta = \eta^{\alpha\beta}N_\beta \ , \tag{27.140}$$

so that the notation N^α is unambiguous.

27.7 Kruskal-Szekeres Coordinates

Using Eddington-Finkelstein coordinates, we have discovered two "new" regions of the space-time. Are there still other regions of space-time to be discovered? The answer is yes. Not only is this suggested by the analogy with Rindler and Minkowski space (but this is after all just an analogy at the moment). We had actually already seen a "doubling" of the Schwarzschild patch in the discussion of the Schwarzschild metric in isotropic coordinates (see the comment at the end of section 24.5). One way to rediscover this region from the present perspective would be to study spacelike rather than null geodesics. Alternatively, let us try to short-cut this somewhat and let us guess how one might be able to describe the maximal extension of space-time.

The first guess might be to use the coordinates u and v simultaneously, instead of r and t. In these coordinates, the metric takes the form

$$ds^2 = -(1 - 2m/r)du\ dv + r^2 d\Omega^2 \ , \tag{27.141}$$

with $r = r(u, v)$. While this is a good idea, the problem is that in these coordinates the horizon is once again infinitely far away, at $u = +\infty$ or $v = -\infty$ (i.e. at $2r^* = v - u = -\infty$). We can rectify this by introducing coordinates U and V with

$$U = -\mathrm{e}^{-u/4m} \quad , \quad V = \mathrm{e}^{v/4m} \tag{27.142}$$

say, so that the horizon is now at either $U = 0$ or $V = 0$. To better understand why we choose these coordinates and why exactly this factor in the exponent (and not any other positive number, which so far would have had the same effect of moving the horizon to $U = 0$ or $V = 0$), note that

$$\frac{v - u}{4m} = \frac{r}{2m} + \log\left(\frac{r}{2m} - 1\right) \ , \tag{27.143}$$

so that the prefactor $f(r) = 1 - 2m/r$ can be written as

$$1 - \frac{2m}{r} = \frac{2m}{r}\left(\frac{r}{2m} - 1\right) = \frac{2m}{r}\mathrm{e}^{-r/2m}\mathrm{e}^{(v-u)/4m} \quad . \tag{27.144}$$

Thus the metric can then be written as

$$ds^2 = \frac{2m}{r}\mathrm{e}^{-r/2m}\left(\mathrm{e}^{v/4m}dv\right)\left(-\mathrm{e}^{-u/4m}du\right) + r(u,v)^2 d\Omega^2 \quad . \tag{27.145}$$

It is now clear why we made the choice (27.142). The metric now takes the simple form

$$ds^2 = -\frac{32m^3}{r}\mathrm{e}^{-r/2m}dUdV + r(U,V)^2 d\Omega^2 \quad , \tag{27.146}$$

with $r = r(U,V)$ given implicitly by

$$UV = -\mathrm{e}^{(v-u)/4m} = -\mathrm{e}^{r^*/2m} = -(r/2m-1)\mathrm{e}^{r/2m} \tag{27.147}$$

and $t = t(U,V)$ explicitly by

$$U/V = -\mathrm{e}^{-(u+v)/4m} = -\mathrm{e}^{-t/2m} \quad . \tag{27.148}$$

Finally, we pass from the null coordinates (U,V) (meaning that ∂_U and ∂_V are null vectors) to more familiar timelike and spacelike coordinates (T,X) defined, in analogy with $(u,v) = t \mp r^*$, by

$$U = T - X \quad , \quad V = T + X \quad , \tag{27.149}$$

in terms of which the metric is

$$ds^2 = \frac{32m^3}{r}\mathrm{e}^{-r/2m}(-dT^2 + dX^2) + r^2 d\Omega^2 \quad . \tag{27.150}$$

Here $r = r(T,X)$ is now implicitly given by

$$X^2 - T^2 = (r/2m-1)\mathrm{e}^{r/2m} \quad . \tag{27.151}$$

Chasing through the above sequence of coordinate transformations

$$(t,r) \to (t,r^*) \to (u,v) \to (U,V) \to (T,X) \quad , \tag{27.152}$$

one finds that the coordinate transformation $(t,r) \to (T,X)$ is explicitly, and in its full glory, given by

$$\begin{aligned} X(t,r) &= \tfrac{1}{2}(V-U) = (r/2m-1)^{1/2}\mathrm{e}^{r/4m}\cosh t/4m \\ T(t,r) &= \tfrac{1}{2}(U+V) = (r/2m-1)^{1/2}\mathrm{e}^{r/4m}\sinh t/4m \quad , \end{aligned} \tag{27.153}$$

Remarks:

1. In these coordinates, the original "Schwarzschild patch" $r > 2m$, the region of validity of the Schwarzschild coordinates, corresponds to the region $-\infty < U < 0$, $0 < V < \infty$, or, in terms of X and T, to the region $X > 0$ and $X^2 - T^2 > 0$, or $|T| < X$. As Figure 19 shows, this 'Schwarzschild patch' is mapped to the first quadrant of the Kruskal-Szekeres metric, bounded by the lines $X = \pm T$.

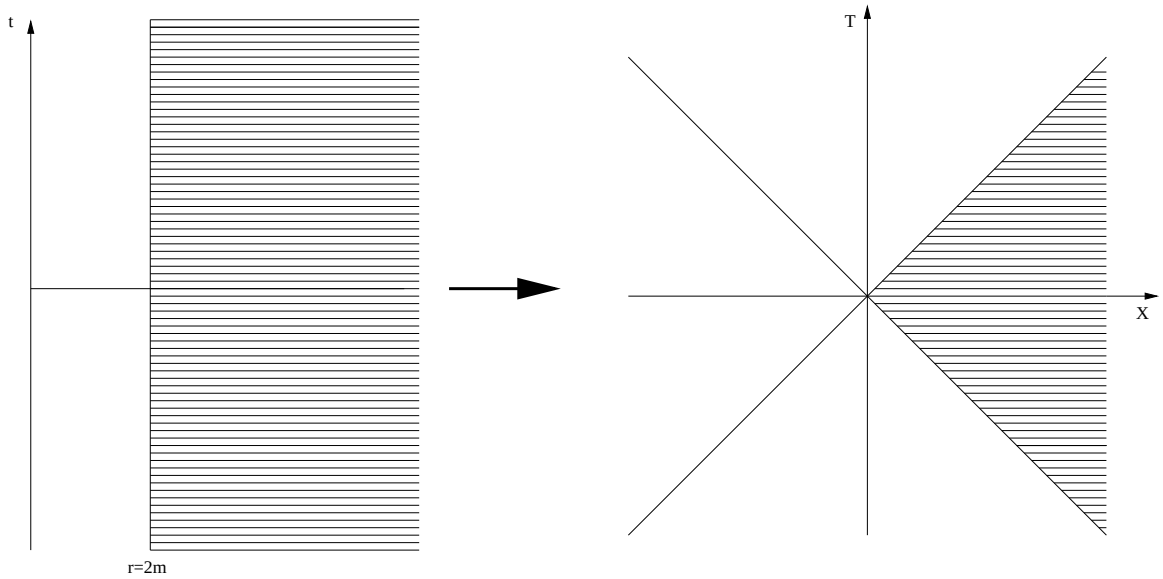


Figure 19: Schwarzschild patch in the Kruskal-Szekeres metric: the half-plane $r > 2m$ is mapped to the quadrant between the lines $X = \pm T$ in the Kruskal-Szekeres metric.

2. As in Minkowski space, null lines are given by $X = \pm T + const.$. In particular, therefore, lightcones have the standard Minkowskian form (slope ± 1) everywhere.

3. Surfaces of constant r are given by the lines (hyperboloids) $X^2 - T^2 = const.$

4. In particular, the boundary of the Schwarzschild patch is the horizon $r = 2m$. In terms of (U, V) this is mapped to $UV = 0$ which is the union of the two lines (null surfaces) $U = 0$ and $V = 0$, and in terms of (T, X) one has

$$r = 2m \quad \Rightarrow \quad X = \pm T \ , \tag{27.154}$$

5. In the region $r \gtrsim 2m$, the above coordinate transformation (27.153) reduces approximately (and up to constant factors) to the transformation (26.80) between the Rindler-like coordinates in that region and Minkowski coordinates,

$$T \sim t_M \sim (r - 2m)^{1/2} \sinh t/4m \quad , \quad X \sim x_M \sim (r - 2m)^{1/2} \cosh t/4m \tag{27.155}$$

6. In these coordinates the metric is now manifestly completely non-singular and regular not only at the horizons but everywhere except possibly at $r = 0$. We will discuss this extension of the Schwarzschild space-time in more detail in the next section.

7. Minor cosmetic improvements can be obtained by introducing, instead of U and V (27.142), the rescaled coordinates (u_K, v_K) through

$$\begin{aligned} du_K = \mathrm{e}^{-u/4m} du \quad &\Rightarrow \quad u_K = -4m\mathrm{e}^{-u/4m} = 4m\, U \\ dv_K = \mathrm{e}^{v/4m} dv \quad &\Rightarrow \quad v_K = +4m\mathrm{e}^{+v/4m} = 4m\, V \ . \end{aligned} \tag{27.156}$$

In particular, the coordinates (u_K, v_K) are now dimensionful (with dimension of length) while the coordinates (U, V) are dimensionless. In terms of (u_k, v_K) the metric takes the form

$$ds^2 = -\frac{2m}{r}\mathrm{e}^{-r/2m} du_K dv_K + r^2 d\Omega^2 \quad , \tag{27.157}$$

but then factors of $(16m^2)$ will reappear in other places. We will mostly work with the dimensionless coordinates (U, V) in the following.

8. It is also possible to further scale u_K and v_K so that the exponential prefactor has the form $\exp(-(r-2m)/2m)$ (e.g. by defining r^* with a different integration constant, $r^* \to r^* - 2m$), so that for $r \to 2m$ the metric tends to $ds^2 \to -du_k dv_k$ without additional numerical factors like e^{-1} but, as I said, this is pure cosmetics. In any case, as we will learn later on, U and V are coordinates that are naturally defined only up to affine transformations, so one choice is as good as any other.

9. Writing the implicit relation (27.147) between r and (U, V) as

$$-UV/e = (r/2m - 1)\mathrm{e}^{(r/2m-1)} \quad , \tag{27.158}$$

one sees that r can be expressed in terms of the *Lambert function* W,

$$r(U, V) = 2m\,(W(-UV/e) + 1) \tag{27.159}$$

defined by

$$x = W(x)\mathrm{e}^{W(x)} \tag{27.160}$$

($W(x) \geq W(-1/e) = -1$, $W(0) = 0$ is the horizon).[86]

10. Note that the transformation (27.156) is strictly identical to the transformation (3.97) between Minkowski and Rindler null coordinates, with the identification $a = 1/4m$. This is much more profound than it sounds at first. In particular, we will learn in section 27.10 that $\kappa = 1/4m$ is the *surface gravity* of a black hole, i.e. a measure of the gravitational acceleration at the horizon. It is therefore this acceleration that plays the same role in the black hole context as the acceleration called a in the Rindler context. Moreover, this identity is at the heart of the deep relation between the so-called *Unruh Effect* (thermal nature of a Minkowski QFT vacuum state when seen by an accelerating observer, already mentioned in section 7.8) and the *Hawking Effect* (quantum radiation of black holes).[87]

[86] For some applications of the Lambert function in this context see e.g. K. Lake, *Some notes on the Kruskal - Szekeres completion*, `arXiv:1002.3600 [gr-qc]`, *Some further notes on the Kruskal - Szekeres completion*, `arXiv:1202.0860 [gr-qc]`.

[87] For an introduction to QFT in curved spacce-time and a discussion of these effects, see e.g. the monographs N. Birrell, P. Davies, *Quantum Fields in Curved Space*, V. Mukhanov, S. Winitzki, *Quantum Effects in Gravity*, or the on-line lecture notes J. Traschen, *An Introduction to Black Hole Evaporation*, `arXiv:gr-qc/0010055`, T. Jacobson, *Introduction to Quantum Fields in Curved Spacetime and the Hawking Effect*, `arXiv:gr-qc/0308048`, C. Krishnan, *Quantum Field Theory, Black Holes and Holography*, `arXiv:1011.5875`.

11. Further insight into the nature of Kruskal-Szekeres coordinates will be provided in section 27.10 (where we elucidate the physical significance of the coordinates (U, V)) and in section 31.9, where we outline the construction of such coordinates for more general black hole metrics.

27.8 Maximal Extension of Schwarzschild: the Kruskal Diagram

Now that we have the coordinates X and T, we can let them range over all the values for which the metric is non-singular. The only remaining singularity is at $r = 0$, which corresponds to the two sheets of the hyperboloid

$$r = 0 \quad \Leftrightarrow \quad T^2 - X^2 = 1 \quad \Leftrightarrow \quad T = \pm\sqrt{1 + X^2} \ . \tag{27.161}$$

That $r = 0$ is indeed a real singularity that cannot be removed by a coordinate transformation can be shown by calculating some invariant of the curvature tensor, like the Kretschmann scalar $K = R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma}$ (8.50). On purely dimensional grounds, from

$$\left(\partial_r^2(1 - 2m/r)\right)^2 \sim m^2/r^6 \ , \tag{27.162}$$

say, one would expect K to be proportional to m^2/r^6, the crucial feature being that the constant of proportionality is not zero, explicit calculations (this is a doable but thoroughly unenlightning exercise) showing that

$$R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma} = 48\frac{m^2}{r^6} \ . \tag{27.163}$$

Thus the geometry is genuinely singular at $r = 0$. Nevertheless, since the metric is non-singular for all values of (X, T) subject to the constraint $r > 0$ or $T^2 - X^2 < 1$, there is no physical reason to exclude the regions in the other quadrants also satisfying this condition.

By including them, we obtain the *Kruskal-Szekeres Extension* of the Schwarzschild metric, displayed in the *Kruskal diagram* in Figure 20. This extension of the Schwarzschild metric was discovered independently by Kruskal and Szekeres in 1960 and presents us with an amazingly rich and complex picture of what originally appeared to be a rather simple (and perhaps even dull) solution to the Einstein equations. It can be shown that this represents the maximal analytic extension of the Schwarzschild metric in the sense that every affinely parametrised geodesic can either be continued to infinite values of its parameter or runs into the singularity at $r = 0$ at some finite value of the affine parameter.[88]

In addition to the Schwarzschild patch, quadrant I, we have three other regions, living in the quadrants II, III, and IV, each of them having its own peculiarities. Note that

[88] One can also show that any spherically symmetric solution of the vacuum Einstein equations is locally isometric to some domain of the Schwarzschild-Kruskal solution. For a proof of this variant of a generalised Birkhoff theorem see e.g. N. Straumann, *General Relativity*.

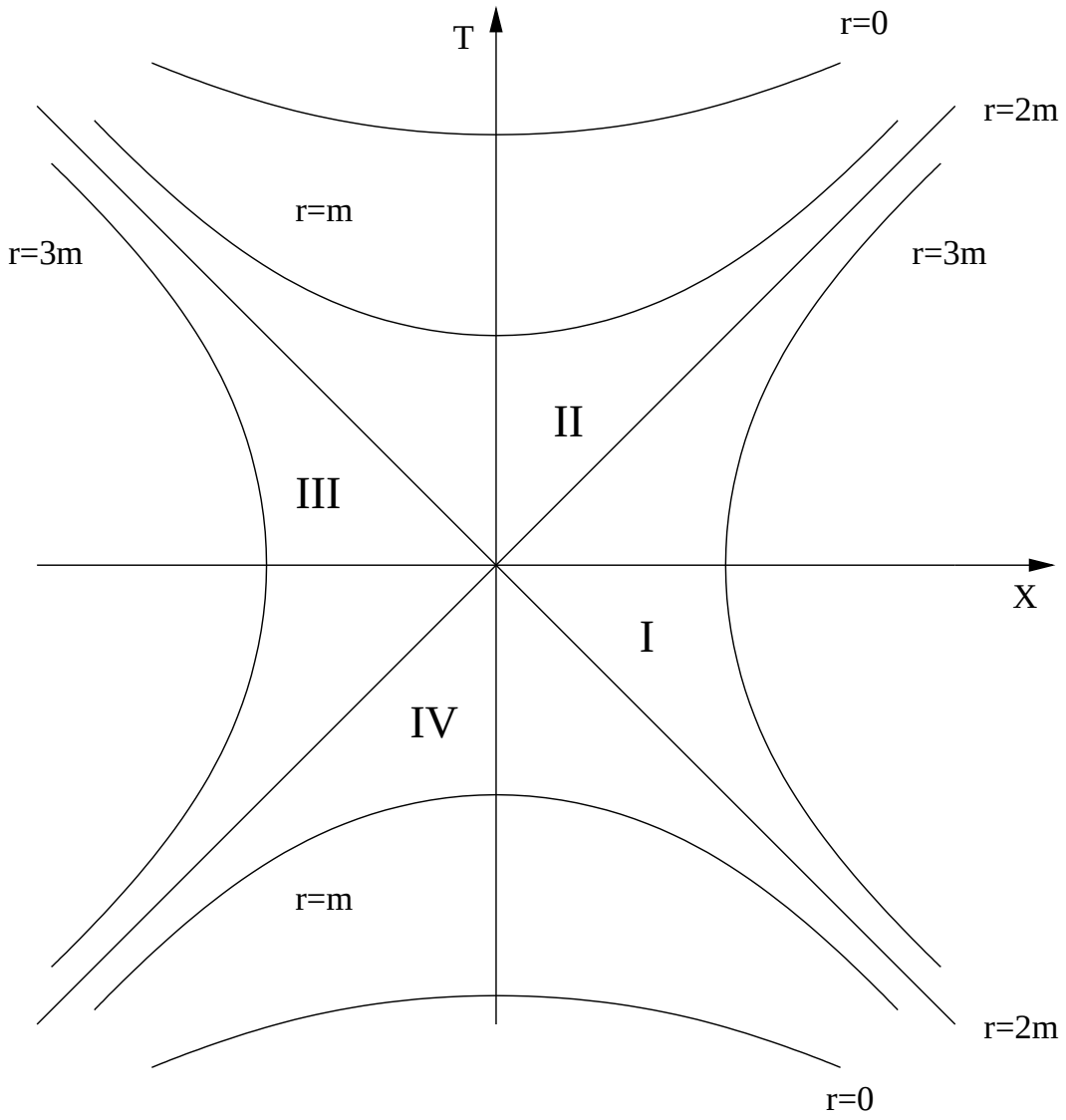


Figure 20: Complete Kruskal-Szekeres universe. Diagonal lines are null, lines of constant r are hyperbolas. Region I is the Schwarzschild patch, separated by the horizon from regions II and IV. The Eddington-Finkelstein coordinates (v, r) cover regions I and II, (u, r) cover regions I and IV. Regions I and III are filled with lines of constant $r > 2m$. They are causally disconnected. Observers in regions I and III can receive signals from region IV and send signals to region II. An observer in region IV can send signals into both regions I and III (and therefore also to region II) and must have emerged from the singularity at $r = 0$ at a finite proper time in the past. Any observer entering region II will be able to receive signals from regions I and III (and therefore also from IV) and will reach the singularity at $r = 0$ in finite time. Events occuring in region II cannot be observed in any of the other regions.

obviously the conversion formulae from $(r,t) \to (X,T)$ in the quadrants II, III and IV differ from those given above for quadrant I. E.g. in region II one can use Schwarzschild(-like) coordinates in which the metric reads

$$ds^2 = \left(\frac{2m}{r} - 1\right) dt^2 - \left(\frac{2m}{r} - 1\right)^{-1} dr^2 + r^2 d\Omega^2 \tag{27.164}$$

(these are not the same coordinates as those in patch I, as we have seen we cannot continue the Schwarzschild coordinates across the horizon), and in this quadrant (where r is a time coordinate etc.) the relation between Schwarzschild and Kruskal coordinates is

$$\begin{aligned} X &= (1 - r/2m)^{1/2} \mathrm{e}^{r/4m} \sinh t/4m \\ T &= (1 - r/2m)^{1/2} \mathrm{e}^{r/4m} \cosh t/4m \ . \end{aligned} \tag{27.165}$$

To get acquainted with the Kruskal diagram, let us note the following basic facts (some of which we had already noted for region I in the previous section).

1. Null lines are diagonals $X = \pm T + const.$, just as in Minkowski space. This greatly facilitates the exploration of the causal structure of the Kruskal-Szekeres metric.

2. In particular, the horizon corresponds to the two lines $X = \pm T$.

3. Lines of constant r are hyperbolas. For $r > 2m$ they fill the quadrants I and III, for $r < 2m$ the other regions II and IV.

4. In particular, the singularity at $r = 0$ is given by the two sheets of the hyperbola $T^2 - X^2 = 1$.

5. Notice in particular also that in regions II and IV worldlines with $r = const.$ are no longer timelike but spacelike. Including the transverse 2-sphere, this should be rephrased as either "lines of constant (r, θ, ϕ) are spacelike for $r < 2m$" or "the surfaces of constant r are spacelike for $r < 2m$".

6. Lines of constant Schwarzschild time t are straight lines through the origin. E.g. in region I one has $X = (\coth(t/4m))T$, with the future horizon $X = T$ corresponding, as expected, to $t \to \infty$.

7. The Eddington-Finkelstein coordinates (v, r) cover the regions I and II, the coordinates (u, r) the regions I and IV.

8. Quadrant III is completely new and is separated from region I by a spacelike distance. That is, regions I and III are causally disconnected. This is the region that already prematurely made a brief appearance when we analysed the coordinate transformation between Schwarzschild and isotropic coordinates at the end of section 24.5. Thus isotropic coordinates actually cover the regions I and III.

Now let us see what all this tells us about the physics of the Kruskal-Szekeres metric.

- An observer in region I (the familiar patch) can send signals into region II and receive signals from region IV.
- The same is true for an observer in the causally disconnected region III.
- Once an observer enters region II from, say, region I, he cannot escape from it anymore and he will run into the catastrophic region $r = 0$ in finite proper time.
- As a reward for his or her foolishness, between having crossed the horizon and being crushed to death, our observer will for the first time be able to receive signals and meet observers emerging from the mirror world in region III.
- Events occurring in region II cannot be observed anywhere outside that region (black hole).
- Finally, an observer in region IV must have emerged from the (past) singularity at $r = 0$ a finite proper time in the past and can send signals and enter into either of the regions I or III.

An even better visualisation of the causal *and global* structure of the maximally extended Schwarzschild solution is provided by its *Carter-Penrose Conformal Diagram* (or *Penrose diagram* for short), given in Figure 21. Its construction (and the notation used here) will be explained in section 28.

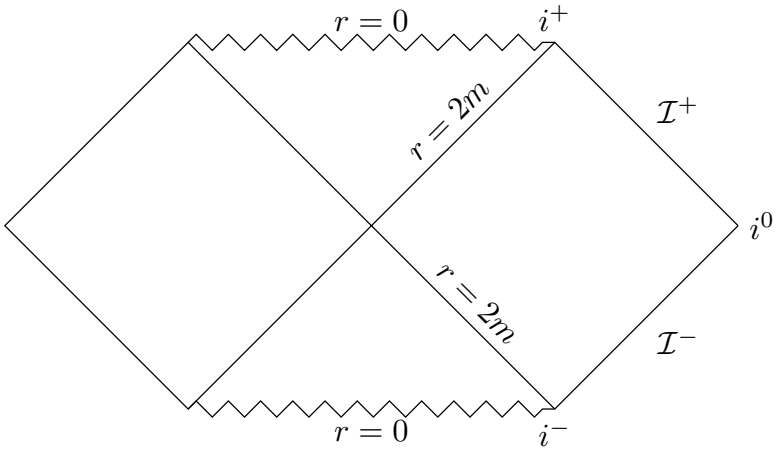


Figure 21: Carter-Penrose Conformal Diagram of the maximal Kruskal-Szekeres extension of the Schwarzschild space-time. For more details, see section 28.

All in all, the picture that we have uncovered of the complete maximally extended Schwarzschild space-time is quite intricate and rich, but also somewhat peculiar (to say the least). In particular, the mirror region III and the white hole region IV appear to

be quite unphysical. As we will see in section 29.3, reassuringly the existence of these regions is an artefact of an eternal black hole solution and these regions do not actually exist for astrophysical black hole solutions that are formed by gravitational collapse.

27.9 Properties of the Asymptotically Timelike Killing Vector

An interesting aspect of the Kruskal-Szekeres geometry is its dynamical character. This may appear to be a strange thing to say since we explicitly started off with a static metric, but this statement applies only to region I (and its mirror III). An investigation of the behaviour of spacelike slices analogous to that we performed for region I (see Figure 11) reveals a dynamical picture of continuing gravitational collapse in region II. In simple terms, the loss of staticity can be understood by noting that the timelike Killing vector field $\xi = \partial_t$ of region I, when expressed in terms of Kruskal coordinates, becomes null on the horizon and spacelike in region II.

Indeed it is easy to check from (27.147) and (27.148) that the time-translation symmetry $(t, r) \to (t + c, r)$ of the Schwarzschild patch corresponds to the transformation

$$U \to \mathrm{e}^{-c/4m} U \qquad V \to \mathrm{e}^{c/4m} V \ . \tag{27.166}$$

This is a boost in the (U, V) or (T, X)-plane which leaves the entire Kruskal metric invariant since

$$dU dV = dT^2 - dX^2 \tag{27.167}$$

is invariant and $r = r(U, V)$ depends on U and V only via the boost-invariant (time-independent) quantity UV (27.147),

$$UV = T^2 - X^2 = -(r/2m - 1)\mathrm{e}^{r/2m} \quad \Rightarrow \quad r = r(T^2 - X^2) \ . \tag{27.168}$$

Thus this symmetry is generated by the Killing vector

$$\xi = (V\partial_V - U\partial_U)/4m = (X\partial_T + T\partial_X)/4m \ . \tag{27.169}$$

It follows that ξ has norm proportional to $T^2 - X^2$,

$$||\xi||^2 = \frac{2m}{r}\mathrm{e}^{-r/2m}(T^2 - X^2) \ . \tag{27.170}$$

There are thus three different cases to consider, each of them interesting in its own right:

- ξ timelike

  ξ is timelike in the original region I (and in the mirror region III), with (27.151)

  $$I: \quad ||K||^2 = -\frac{2m}{r}\mathrm{e}^{-r/2m}(\frac{r}{2m} - 1)\mathrm{e}^{r/2m} = -(1 - \frac{2m}{r}) \ , \tag{27.171}$$

  confirming that $\xi = \partial_t$ in region I. We thus recover the statement that the Schwarzschild metric in the Schwarzschild patch is static.

- ξ spacelike

  ξ is spacelike in region II. Thus region II has no timelike Killing vector field, therefore cannot possibly be static, but has instead an additional spacelike Killing vector field - cf. the Remark 1 in section 24.6 in connection with Birkhoff's theorem.

  Related to this is the fact, already mentioned above, that in regions II and IV the slices of constant r are no longer timelike but spacelike surfaces. Thus they are analogous to, say, constant t or T slices for $r > 2m$. Just as it does not make sense to ask "where is the slice $t = 1$?" (say), only "when is $t = 1$?", or "where is $r = 3m$?", in these regions it makes no sense to ask "where is $r = m$?", only "when is $r = m$?".

- ξ null

  ξ is null on the horizon. This turns out to be the most interesting case, and turns out to be one of the characteristic features of static black holes in general, and therefore I will elaborate on this a bit in the following.

27.10 First Encounter with Killing Horizons and Surface Gravity

In this section, we will perform a rather pedestrian analysis of some properties of the horizon which are related to the fact that the Killing vector ξ becomes null on this null surface. These properties can be derived and understood without a general knowledge of the geometry of null hypersurfaces, as discussed in section 17, but it is useful to keep in mind that they are just special cases of the properties of null hypersurfaces, and of more general Killing horizons (to be discussed later on in section 32.5).

In particular, from section 17.2 one knows on general grounds that the integral curves of the normal vector ξ are (possibly non-affinely parametrised) null geodesics, but it is instructive to rederive this here from scratch, and from a slightly different perspective.

ξ is null on (and tangent to) the horizon $T = \pm X$ or $UV = 0$. In fact it reduces to $\xi = (V/4m)\partial_V = \partial_v$ on the horizon $U = 0$ (and to $\xi = \partial_u$ on $V = 0$). As a consequence, due to the characteristic peculiarities of null hypersurfaces (section 17), ξ is also normal to the horizon. In this context (or from this perspective) the horizon is known as a *Killing horizon* (the locus where a Killing vector becomes null and orthogonal to a null surface).

Therefore ξ generates translations along the horizon, $v \to v + c$, and is called the *null generator* of (this branch of) the horizon. Thus v can naturally be used as a coordinate there.

Evidently, ξ vanishes at the "point" $U = V = 0$ where the other horizon $V = 0$ branches off. Actually this is of course not a point but a perfectly respectable 2-sphere

of radius $r = 2m$ known as the *bifurcation surface* of the Killing horizon of the Schwarzschild geometry, and ξ vanishing means that this 2-sphere is invariant under the time-translation generated by ξ, something that is also evident from (27.166). On $U = 0$ this branching point lies at $V \to 0 \Rightarrow v \to -\infty$, so the Eddington-Finkelstein coordinate $v \in (-\infty, +\infty)$ only covers the half-line $U = 0, V > 0$ of the horizon.

On the other hand the line $U = 0$ is itself an "outgoing" (radial) null geodesic, but in light of the above v cannot possibly be an affine parameter along that geodesic (the affine parameter should not reach infinite values half-way along the geodesic). The failure of v to be an affine parameter on the horizon can be quantified by calculating the acceleration $\nabla_\xi \xi = \nabla_{\partial t} \partial_t$ (5.98) of the (integral curves of the) Killing vector and then taking the limit $r \to 2m$. This calculation is quite painless in Eddington-Finkelstein coordinates (v, r) in which $\xi = \partial_v$ everywhere,

$$\xi = \partial_t = (\partial_t v)\partial_v + (\partial_t r)\partial_r = \partial_v \tag{27.172}$$

($\xi = \partial_v$ is evidently a Killing vector because the components of the metric do not depend on v in Eddington-Finkelstein coordinates). Then the acceleration is

$$\nabla_\xi \xi \equiv (\nabla_\xi \xi)^\alpha \partial_\alpha = \Gamma^\alpha_{vv} \partial_\alpha = (f'/2)(f\partial_r + \partial_v) \tag{27.173}$$

where $f(r) = 1 - 2m/r$. Thus for $r \to 2m$ one finds

$$\lim_{r\to 2m} \nabla_\xi \xi = \left(\tfrac{1}{2} f'(r)|_{r=2m}\right) \xi = \left(\frac{m}{r^2}|_{r=2m}\right) \xi = \tfrac{1}{4m} \xi \ . \tag{27.174}$$

Remarks:

1. Since we have $\nabla_\xi \xi \sim \xi$ on the horizon, this shows first of all that there it generates a non-affinely parametrised geodesic (see (2.136) and (5.100)). Moreover, we see that one interpretation of the ubiquitous factor

   $$\kappa = \frac{1}{4m} \tag{27.175}$$

   is that it measures the *inaffinity*, i.e. the failure of v to be an affine parameter on the horizon,

   $$\lim_{r\to 2m} \nabla_\xi \xi = \kappa \xi \ . \tag{27.176}$$

2. Another interpretation brought out by the same calculation (27.174) is that $\kappa = 1/4m$ is the *surface gravity* which measures the strength of the gravitational force (acceleration) $\mathsf{a}(r)$ (26.11) acting on a static observer at the horizon, but as measured at infinity (by taking into account the redshift factor $f(r)^{1/2}$),

   $$\kappa = \lim_{r\to 2m} f(r)^{1/2} \mathsf{a}(r) = \tfrac{1}{2} f'(r)|_{r=2m} = 1/4m \ . \tag{27.177}$$

Anyway, to return to the beginning of this story, we have seen that v is not an affine parameter along the horizon. It turns out, however, that the Kruskal coordinate V is an affine parameter there (and this is one way of understanding why Kruskal coordinates are so natural for exploring the causal structure of the metric), meaning that the null curve

$$x^\alpha(\lambda) = (U(\lambda), V(\lambda), \theta(\lambda), \phi(\lambda)) = (0, \lambda, \theta_0, \phi_0) \tag{27.178}$$

is an affinely parametrised null geodesic. This follows on the nose from (27.174) and the result (2.138) of section 2.7. Noting that κ is constant (actually not just along the geodesic but on the entire horizon, but the former is all we need), (2.138) with $\tau \to \lambda$ and $\sigma \to v$ becomes (dropping integration constants)

$$\frac{d\lambda}{dv} \sim \mathrm{e}^{\kappa v} \quad \to \quad \lambda(v) \sim \kappa^{-1}\mathrm{e}^{\kappa v} \sim V \ . \tag{27.179}$$

Another way to see this, which provides some more insight, is to analyse the geodesic Lagrangian and the conserved quantity associated to ξ in ingoing Eddington-Finkelstein coordinates (one could also work in Kruskal coordinates, but nothing is gained by that). The elementary steps in the calculation are the following:

- From the metric (27.82) we deduce that for a radial null-geodesics one has

  $$-f(r)\dot{v}^2 + 2\dot{v}\dot{r} = 0 \tag{27.180}$$

  where a dot denotes a derivative with respect to the affine parameter λ. The geodesics with $v = \text{const}$ describe ingoing null-geodesics and we are not interested in these, so we have

  $$-f(r)\dot{v} + 2\dot{r} = 0 \ , \tag{27.181}$$

  and since $u = t - r^* = v - 2r^*$ and $dr^*/dr = f(r)^{-1}$, this is equivalent to $\dot{u} = 0$ and, as anticipated, describes outgoing geodesics.

- The conserved quantity E associated to the Killing vector $\xi = \partial_v$ is

  $$f(r)\dot{v} - \dot{r} = E \ . \tag{27.182}$$

  From the two preceding equations we immediately deduce (with a convenient parametrisation of the integration constant)

  $$\dot{r} = E \quad \Rightarrow \quad r(\lambda) = E(\lambda - \lambda_0) + r_s \ . \tag{27.183}$$

  For the null geodesic along the horizon we are ultimately interested in, we have $r(\lambda) = r_s$, i.e. $E = 0$, but we need to approach this with some care, so we keep the general solution for now. Analogously, for v we find the equation

  $$f(r)\dot{v} = 2E \quad \Rightarrow \quad \dot{v} = \frac{2Er(\lambda)}{r(\lambda) - r_s} = 2E + \frac{2r_s}{\lambda - \lambda_0} \ . \tag{27.184}$$

We can now take the limit $E \to 0$ with impunity, and are left with

$$\dot{v} = \frac{2r_s}{\lambda - \lambda_0} \quad \Rightarrow \quad v(\lambda) = 2r_s \ln(\lambda - \lambda_0) + \text{const.} \ . \tag{27.185}$$

- The prefactor $2r_s = 4m$ is now precisely such that it cancels the factor $1/4m$ in the definition of the Kruskal coordinate V (27.142), so that

$$V(\lambda) = \mathrm{e}^{v(\lambda)/4m} = a(\lambda - \lambda_0) \ , \tag{27.186}$$

  which is precisely the statement that V is related to λ by an affine transformation. Thus V is an affine parameter along the horizon $U = 0$, as claimed.

- We also see explicitly that for other outgoing null geodesics, i.e. those with $E \neq 0$, and with solution

$$v(\lambda) = 2E\lambda + 4m \ln(\lambda - \lambda_0) + \text{const.} \ , \tag{27.187}$$

  neither v nor V is an affine parameter. However, it is apparent from the explicit solution $r(\lambda) = E\lambda + b$ given above that, for $E \neq 0$, $r(\lambda)$ is related to λ by an affine transformation, and hence for these geodesics r is an affine parameter.

Remarks:

1. We now have two natural coordinates on the future horizon $U = 0, V > 0$ which we can for instance use to measure the frequency of incoming waves. $\xi = \partial_t = \partial_v$ measures what is commonly called Killing frequency (this requires no further explanation since it is associated to the Killing vector which generates time-translations in the Schwarzschild patch), and is the natural notion of frequency to be used by static observers with 4-velocvity $u^\alpha \sim \xi^\alpha$.

   ∂_V, on the other hand, measures the so-called free fall frequency since, more or less by construction, a freely falling observer in the Schwarzschild geometry near $r = 2m$ will see approximately the Minkowski space-time metric $ds^2 \sim -dUdV$ (recall that the transformation (3.97) between Rindler and Minkowski coordinates is strictly analogous to the transformation (27.156) between Schwarzschild and Kruskal coordinates). The exponential relation (27.142) between them reflects the exponential blue- or redshift we first encountered in section 26.5.

2. As a concluding remark to this section, I cannot resist mentioning that the notion of surface gravity κ plays a crucial role in the analysis of the classical dynamics of general black holes, and even more so in the semi-classical context, since it is directly proportional to the temperature of the famous Hawking radiation of an evaporating black hole,

$$T_H = \frac{\kappa}{2\pi} = \frac{1}{8\pi G_N M} \ . \tag{27.188}$$

For more details on this and other related advanced topics I am not able to cover or do justice to here, I refer you to the references given in footnote 87 of section 27.7, as well as to the superb Cambridge lecture notes on Black Holes.[89]

27.11 From Eddington-Finkelstein to Israel(-Klösch-Strobl) Coordinates

There is one more coordinate system for the Schwarzschild geometry that I want to mention because it is quite remarkable and, equally remarkably, apparently not widely known or used. It was discovered by W. Israel in 1966, and rediscovered several times since, most recently by T. Klösch and T. Strobl in a different particularly insightful way.[90] I will introduce these coordinates in a way that is complementary to those in these articles (and that is also motivated by certain generalisations, that I will however not discuss here).

Recall that the Kruskal coordinates were based on suitably combining the outgoing and ingoing Eddington-Finkelstein coordinates. Now, more generally one frequently encounters the situation that one knows a solution either in coordinates adapted to ingoing null geodesics, or in coordinates adapted to outgoing null geodesics, but usually not both (and given one constructing the other is usually a hard problem that may have no simple analytical solution).

Thus, let us assume that we are given (or have found) the Schwarzschild metric in outgoing Eddington-Finkelstein coordinates,

$$ds^2 = -f(r)du^2 - 2du\,dr + r^2 d\Omega^2 \quad , \quad f(r) = 1 - 2m/r \ . \tag{27.189}$$

We are happy and proud of this, but we quickly realise that this cannot be the end of the story because the above coordinates provide an incomplete covering of the space-time. The simplest way to discover this is to study radial lightrays or null geodesics, governed by the two equations

$$f(r)\dot{u}^2 + 2\dot{u}\dot{r} = 0 \tag{27.190}$$

(the null condition), and

$$f(r)\dot{u} + \dot{r} = E \tag{27.191}$$

(due to the u-translation invariance). One set of null geodesics is simply given by $\dot{u} = 0$, i.e. $u = u_0$ constant, and for these one has

$$u(\tau) = u_0 \quad \Rightarrow \quad \dot{r} = E \quad \Rightarrow \quad r(\tau) = E\tau + r_0 \ . \tag{27.192}$$

[89] P. Townsend, *Black Holes*, `arXiv:gr-qc/9707012v1`.

[90] W. Israel, *New Interpretation of the extended Schwarzschild Manifold*, Phys. Rev. 143 (1966) 1016-1021; T. Klösch, T. Strobl, *Explicit Global Coordinates for Schwarzschild and Reissner-Nordstroem*, `arXiv:gr-qc/9507011`. It is also mentioned in section 15.4.4 of the directory of exact solutions, H. Stephani, D. Kramer, M. MacCallum, C. Hoenslaers, E. Herlt, *Exact Solutions to Einstein's Field Equations* (2nd Edition). See also K. Lake, *An explicit global covering of the Schwarzschild-Tangherlini black holes*, `arXiv:gr-qc/0306073`, and *Maximally extended, explicit and regular coverings of the Schwarzschild - de Sitter vacua in arbitrary dimension*, `arXiv:gr-qc/0507031` for some generalisations.

For future-oriented null geodesics one needs $E > 0$, and therefore one has $\dot{r} > 0$. These are the outgoing null geodesics to which the outgoing Eddington-Finkelstein coordinate system is adapted. Here r_0 is an integration constant which, by an affine transformation (actually just a shift) of τ, we could e.g. without loss of generality choose to be $r_0 = 2m$. Then the solution describes the outgoing null geodesics that emerge from the past event horizon at $r = 2m$ for $\tau = 0$.

The other set of (thus ingoing) null geodesics has $\dot{u} \neq 0$ and is therefore governed by the equations

$$f(r)\dot{u} + 2\dot{r} = 0 \quad \text{and} \quad f(r)\dot{u} + \dot{r} = E \ . \tag{27.193}$$

Subtracting the two one finds $\dot{r} = -E$, and therefore (again with a convenient choice of integration constant or origin of τ)

$$r(\tau) = 2m - E\tau \quad , \quad u(\tau) = -4m \log|\tau| + 2E\tau + c \qquad (-\infty < \tau < 0) \tag{27.194}$$

The restriction on τ arises because for $\tau \to 0_-$ one has

$$\tau \to 0_- \quad \Rightarrow \quad r \to 2m \quad , \quad u \to \infty \ . \tag{27.195}$$

Thus we discover that we reach $u = +\infty$ in finite (affine) time, we run out of coordinate space as the ingoing null geodesics approach $r = 2m$ (and the coordinate u is evidently not suitable for describing what happens beyond $r = 2m, u = \infty$).

Note that this locus is most certainly not the past event horizon ($r = 2m$, $u = u_0$ finite), as we know that we can only cross that horizon along future-directed paths from smaller to larger values of r (the white hole). Thus we have discovered a new barrier (which also turns out to be a horizon, and which, with hindsight, we know is the future event horizon), and we realise that the outgoing Eddington-Finkelstein coordinates do not cover the whole space-time.

Now let us assume that we do not have the luxury of being able to appeal to ingoing Eddington-Finkelstein coordinates to construct a future extension (and subsequently the maximal Kruskal-Szekeres extension) of this space-time. How could we proceed?

Since τ is finite as $u = +\infty$ is reached, it is natural to introduce something like τ as a new coordinate. Looking at the leading term of $u(\tau)$ for $\tau \to 0$, one is then led to introduce a new coordinate x through

$$u(\tau) \approx -4m \log|\tau| \quad \Rightarrow \quad u(x) = -4m \log|x| \ , \tag{27.196}$$

so that the complete range of u, $u \in (-\infty, +\infty)$, is covered as x runs over the interval $x \in (-\infty, 0)$. This clearly now permits to continue the ingoing null geodesics beyond $u = +\infty$. However, if one just replaces $u \to x$, the metric appears to be singular at $x = 0$. This can be rectified by introducing a new coordinate y through the relation

$$r - 2m = xy \ , \tag{27.197}$$

its range subject to the condition $r > 0$. To better understand the rationale for this change of variables, note that as $u \to \infty$ one has $r - 2m \sim \tau \sim x$ so that $(r-2m)/x$ remains finite in the limit - and can therefore be used as a new coordinate, the one that we have called y.

It is straightforward to see that in these coordinates the metric takes the form

$$ds^2 = \frac{8my^2}{xy+2m}dx^2 + 8m\,dx\,dy + (xy+2m)^2 d\Omega^2 \quad . \tag{27.198}$$

This is the Schwarzschild metric in *Israel coordinates*. Before discussing some of it most important properties, note that by the simple scaling $u_I = 4mx$, so that

$$u_I = 4mx = -4me^{-u/4m} \equiv u_K \tag{27.199}$$

is just the "canonically normalised" Kruskal coordinate introduced in (27.156), and the relabelling $y \to v_I$, one can put the metric into the somewhat more common "canoncially normalised" form

$$ds^2 = \frac{v_I^2}{2m((u_I/4m)v_I + 2m)}du_I^2 + 2\,du_I\,dv_I + ((u_I/4m)v_I + 2m)^2 d\Omega^2 \quad . \tag{27.200}$$

Here are some of the key-properties of this metric (to describe these I will continue to use the coordinates (x, y), i.e. the form (27.198) of the metric):

1. First of all, one sees that the metric is completely non-singular as long as $xy > -2m$, and one can therefore let x and y run over all the values for which this condition is satisfied.

2. The horizon is given by the degenerate hyperboloid

$$r = 2m \quad \Leftrightarrow \quad xy = 0: \quad \{x = 0\} \cup \{y = 0\} \tag{27.201}$$

3. Then one sees immediately that this space-time covers four distinct "patches":

   - For $x < 0$ and $y < 0$ one has $r > 2m$: this is the original Schwarzschild patch.
   - There are two regions for which x and y have opposite sign, subject to the condition $xy > -2m$: these are the white hole region behind the past horizon $(x < 0, y > 0)$, covered by the original outgoing Eddington-Finkelstein coordinates, and the new region beyond the future event horizon with $x > 0, y < 0$.
   - Finally, for $x > 0$ and $y > 0$ one also has $r > 2m$: this is a distinct region isometric to the Schwarzschild patch, the mirror region.

4. Thus the Israel coordinates provide not only a future extension of the Schwarzschild metric in Schwarzschild or outgoing Eddington-Finkelstein coordinates but actually a complete covering of the maximal Kruskal-Szekeres extension of the Schwarzschild space-time.

5. The usual Kruskal-Szekeres coordinates (U, V) are related to the Israel coordinates by

$$U \sim x \quad , \quad V \sim y\, \mathrm{e}^{xy/2m+1} \quad . \tag{27.202}$$

6. The main difference to (and advantage compared with) Kruskal-Szekeres coordinates is that for the Israel coordinates the coordinate transformation to Schwarzschild coordinates and its inverse are completely explicit (whereas the radial coordinate r is only given implicitly in terms of the Kruskal-Szekeres coordinates). E.g. for $x < 0$ one has

$$\left.\begin{array}{ll} u & = -4m\log(-x) \\ r & = xy + 2m \end{array}\right\} \quad \Leftrightarrow \quad \left\{\begin{array}{ll} x & = -\mathrm{e}^{-u/4m} \\ y & = -(r-2m)\mathrm{e}^{u/4m} \end{array}\right. \tag{27.203}$$

7. By undoing the coordinate transformation $u = t - r^*$ from Schwarzschild to Eddington-Finkelstein coordinates, one can also explicitly write the transformation from Schwarzschild coordinates (t, r) to the Israel coordinates,

$$r(x,y) = xy + 2m \quad , \quad t(x,y) = xy + 2m + \log|y/2mx| \quad . \tag{27.204}$$

8. By inspection, the metric depends only on products like xy, $dxdy$ etc. Thus it has the isometry

$$(x, y) \to (\lambda x, \lambda^{-1} y) \tag{27.205}$$

which corresponds (and reduces) to the time-translation symmetry in the Schwarzschild patch, generated by

$$\partial_t = (y\partial_y - x\partial_x)/4m \quad . \tag{27.206}$$

9. Radial null geodesics are thus goverened by the two equations

$$\frac{y^2}{xy+2m}\dot{x}^2 + \dot{x}\dot{y} = 0 \tag{27.207}$$

(the null condition), and

$$\frac{2y^2}{xy+2m}x\dot{x} + \dot{y}x - \dot{x}y = c \quad . \tag{27.208}$$

(from time-translation invariance). They are therefore of two kinds:

- One has the curves with x (as well as, of course, (θ, ϕ)) constant, so that these are straight lines in an (x, y)-diagram. For these curves, (27.208) implies that $\dot{y}$ is constant, so that y an affine parameter along these null geodesics.
- The other null geodesics are given by the relation

$$\frac{y^2}{xy+2m}\dot{x} + \dot{y} = 0 \quad . \tag{27.209}$$

Multiplying by x and subtracting twice that from (27.208), one finds

$$\dot{x}y + x\dot{y} = c \quad \Rightarrow \quad x(\tau)y(\tau) = c\tau + d \ . \tag{27.210}$$

Using this to eliminate x and $\dot{x}$ from (27.209), the equation of motion for y reduces to

$$\dot{y} = -(c/2m)y \quad \Rightarrow \quad y(\tau) = y_0 \mathrm{e}^{-(c/2m)\tau} \ , \tag{27.211}$$

or $c\tau = -2m \log(y(\tau)/y_0)$. Substituting this in (27.210), one finds that these radial null geodesics are given by the curves

$$xy = -2m \log(Cy) \tag{27.212}$$

in the (x, y)-plane, with C an arbitrary constant.

Armed with this information, it is straightforward to draw an Israel analogue of the Kruskal diagram, containing an equivalent amount of information. All in all, Israel coordinates are an attractive and easy to construct and understand alternative to Kruskal-Szekeres coordinates.

27.12 Appendix: Summary of Schwarzschild Coordinate Systems

Here is, to wrap up this section, a list of the coordinate systems that we have used to gain insight into the properties of the Schwarzschild metric. [As region III is isometric to region I, this doubling of possibilities (e.g. Schwarzschild coordinates cover I *or* III, etc.) has not been indicated in the last column. The perfectly valid but somewhat un-insightful option to use Schwarzschild coordinates only in region II, say, and variants thereof, have also not been indicated in the last column.]

Name	Line Element	Adapted to	Covers
Schwarzschild	$-f(r)dt^2 + f(r)^{-1}dr^2 + r^2 d\Omega^2$	Stationary Observers	I
Tortoise	$f(r)(-dt^2 + (dr^*)^2) + r(r^*)^2 d\Omega^2$	In & Out Null Geodesics	I
Isotropic	$-(g_-(\rho)/g_+(\rho))dt^2 + g_+(\rho)^2(d\rho^2 + \rho^2 d\Omega^2)$	Stationary Observers	I,III
Painlevé - Gullstrand	$-f(r)d\tau^2 \pm 2\sqrt{\frac{2m}{r}}dr d\tau + (dr^2 + r^2 d\Omega^2)$	In/Out Timelike Geodesics	I,II / I,IV
Lemaître	$-d\tau^2 + (2m/r(\tau,\rho))d\rho^2 + r(\tau,\rho)^2 d\Omega^2$	In Timelike Geodesics	I,II
Novikov	$-d\tau^2 + \frac{R^2+1}{R^2}(r')^2 dR^2 + r(\tau,R)^2 d\Omega^2$	In Timelike Geodesics	I-IV
Eddington - Finkelstein	$-f(r)d(v/u)^2 \pm 2d(v/u)dr + r^2 d\Omega^2$	In/Out Null Geodesics	I,II / I,IV
Kerr-Schild	$-d\tilde{t}^2 + dr^2 + r^2 d\Omega^2 + \frac{2m}{r}(d(\tilde{t} \pm r))^2$	In/Out Null Geodesics	I,II / I,IV
Kruskal - Szekeres	$F(r(T,X))(-dT^2 + dX^2) + r(T,X)^2 d\Omega^2$	In & Out Null Geodesics	I-IV
Israel	$F_I(x,y)dx^2 + 8m\, dxdy + (xy+2m)^2 d\Omega^2$	In/Out Null Geodesics	I-IV

Abbreviations:

$$
\begin{aligned}
f(r) = 1 - 2m/r \quad &, \quad g_\pm(\rho) = (1 \pm m/2\rho)^2 \\
F(r) = 32m^3 r^{-1} \mathrm{e}^{-r/2m} \quad &, \quad r(\tau,\rho) = [3\sqrt{2m}(\rho - \tau)/2]^{2/3} \\
F_I(x,y) = 8my^2/(xy+2m) \quad &, \quad r(r^*) \text{ from } r^* = r + 2m\ln(r/2m - 1) \qquad (27.213) \\
r(\tau,R) \quad \text{and} \quad r' = \partial r/\partial R \quad &\text{from} \quad (27.59) \\
r(T,X) \quad &\text{from} \quad X^2 - T^2 = (r/2m - 1)\mathrm{e}^{r/2m}
\end{aligned}
$$

33 Cosmology I: Basics

33.1 Preliminary Remarks

We now turn away from considering isolated systems (stars) to some (admittedly very idealised) description of the universe as a whole. This subject is known as Cosmology. It is certainly one of the most fascinating subjects of theoretical physics, dealing with such issues as the origin and ultimate fate and the large-scale structure of the universe.

Due to the difficulty of performing cosmological experiments and making precise measurements at large distances, many of the most basic questions about the universe are still unanswered today:

1. PAST:

   What actually happened at (or even before) what is usually called the Big Bang?

2. PRESENT:

   is our universe spatially finite or infinite?

3. FUTURE:

   Will our universe keep expanding forever or will it recollapse?

4. MATTER:

   What is Dark Matter?

5. ENERGY

   - Is Dark Energy, responsible for what appears to be a current phase of accelerated expansion of the universe, a cosmological constant?
   - If Dark Energy is indeed a cosmological constant, why is the cosmological constant so small and what determines its value?

While recent precision data, e.g. from supernovae surveys and detailed analysis of the cosmic microwave background radiation, suggest answers to at least some of these questions, these answers leave less wiggle-room for philosophical prejudices or esthetic preferences and actually just make the universe more mysterious than ever.

Of course, we cannot study any of these questions in detail, in particular because an important role in studying these questions is played by the interaction of cosmology with astronomy, astrophysics and elementary particle physics, each of these subjects deserving at least a course of its own.

Fortunately, however, many of the important features any realistic cosmological model should display are already present in some very simple models, the so-called *Friedmann-Lemaître-Robertson-Walker Models* (FLRW models) already studied in the 20's and

30's of the last century. They are based on the simplest possible ansatz for the metric compatible with the assumption that on large scales the universe is roughly homogeneous and isotropic (cf. the next section for a more detailed discussion of this *Cosmological Principle*) and have become the 'standard model' of cosmology.

We will see that they already display all the essential features such as

1. a Big Bang
2. an expanding universes (Hubble expansion) and a cosmological redshift.
3. different long-term behaviour (eternal expansion versus recollapse)

Our first aim will be to make maximal use of the symmetries that simple cosmological models should have to find a simple ansatz for the metric. Our guiding principle will be the *Cosmological Principle*.

33.2 Fundamental Assumption: The Cosmological Principle

At first, it may sound impossibly difficult to find solutions of the Einstein equations describing the universe as a whole. However: if one looks at the universe at large (very large) scales, in that process averaging over galaxies and even clusters of galaxies, then the situation simplifies a lot in several respects:

1. First of all, at those scales non-gravitational interactions can be completely ignored because they are either short-range (the nuclear forces) or compensate each other at large distances (electro-magnetism).
2. Furthermore we assume that the earth, and our solar system, or even our galaxy, have no privileged position in the universe (this is occasionally referred to as the *Copernican Principle*). This means that at large scales the universe should look the same from any point in the universe. Mathematically this means that there should be translational symmetries from any point of space to any other, in other words, space should be homogeneous.
3. Also, we assume that, at large scales, the universe looks the same in all directions. Thus there should be rotational symmetries and hence space should be isotropic.

Together, the second and third assumptions form the *Cosmological Principle*, which is the starting point for our discussion of cosmology and on which much of the work in cosmology is based. It is plausible (and true) that the assumption of isotropy (around

us) can be tested experimentally / observationally, while testing the assumption of homogeneity is evidently going to be more tricky.[138]

Making the above assumptions, it follows from our discussion in section 14, that the n-dimensional space (of course $n = 3$ for us) has n translational and $n(n-1)/2$ rotational Killing vectors, i.e. that the spatial metric is *maximally symmetric*. For $n = 3$, we will thus have six Killing vectors, two more than for the Schwarzschild metric, and the ansatz for the metric will simplify accordingly.

Note that, since we know from observation that the universe expands, we do not require a priori a maximally symmetric space-*time* as this would imply that there is also a timelike Killing vector.

What simplifies life considerably is the fact that (cf. the discussion in section 14) there are only three species of maximally symmetric spaces (for any n), namely

- flat space $\mathbb{R}^n$ (with its standard Euclidean metric),
- the sphere S^n (with tis standard "round" metric),
- and its negatively curved counterpart, the n-dimensional pseudosphere or hyperboloid we will call H^n.

Thus, for a space-time metric with maximally symmetric spacelike 'slices', the only unknown is the time-dependence of the overall size of the metric. More concretely, the metric can (now fixing the number of spatial dimensions to be $n = 3$) be chosen to be

$$ds^2 = -dt^2 + a^2(t)\left(\frac{dr^2}{1-kr^2} + r^2 d\Omega^2\right) \ , \tag{33.1}$$

where $k = 0, \pm 1$ corresponds to the three possibilities mentioned above. Thus the metric contains only one unknown function, the 'radius' or cosmic scale factor $a(t)$. This function will be determined by the Einstein equations via the matter content of the universe (we will of course be dealing with a non-vanishing energy-momentum tensor), modelled by a perfect fluid.

33.3 Fundamental Observations I: Olbers' Paradox

One paradox, popularised by Olbers (1826) but noticed before by others is the following. He asked the seemingly innocuous question "Why is the sky dark at night?". According to his calculation, reproduced below, the sky should instead be infinitely bright.

[138] The extent to which these assumptions, in particular homogeneity, can be observationally tested is discussed e.g. in C. Clarkson, R. Maartens, *Inhomogeneity and the foundations of concordance cosmology*, `arXiv:1005.2165 [astro-ph.CO]`, G. Ellis, *Inhomogeneity effects in Cosmology*, `arXiv:1103.2335 [astro-ph.CO]`, R. Maartens, *Is the Universe homogeneous?*, `arXiv:1104.1300 [astro-ph.CO]`.

The simplest assumption one could make in cosmology (prior to the discovery of the Hubble expansion) is that the universe is static, infinite and homogeneously filled with stars. In fact, this is probably the naive picture one has in mind when looking at the stars at night, and certainly for a long time astronomers had no reason to believe otherwise.

However, these simple assumptions immediately lead to a paradox, namely the conclusion that the night-sky should be infinitely bright (or at least very bright) whereas, as we know, the sky is actually quite dark at night. This is a nice example of how very simple observations can actually tell us something deep about nature (in this case, the nature of the universe). The argument runs as follows.

1. Assume that there is a star of brightness (luminosity) L at distance r. Then, since the star sends out light into all directions, the apparent luminosity A (neglecting absorption) will be
$$A(r) = L/4\pi r^2 \quad . \tag{33.2}$$

2. If the number density ν of stars is constant, then the number of stars at distances between r and $r + dr$ is
$$dN(r) = 4\pi\nu r^2 dr \quad . \tag{33.3}$$
Hence the total energy density due to the radiation of all the stars is
$$E = \int_0^\infty A(r)dN(r) = L\nu \int_0^\infty dr = \infty \quad . \tag{33.4}$$

3. Therefore the sky should be infinitely bright.

Now what is one to make of this? Clearly some of the assumptions in the above are much too naive. The way out suggested by Olbers is to take into account absorption effects and to postulate some absorbing interstellar medium, but this is also too naive because in an eternal universe we should now be in a stage of thermal equilibrium. Hence the postulated interstellar medium should emit as much energy as it absorbs, so this will not reduce the radiant energy density either.

Of course, the stars themselves are not transparent, so they could block out light completely from distant sources, but if this is to rescue the situation, one would need to postulate so many stars that every line of sight ends on a star, but then the night sky would be bright (though not infinitely bright) and not dark.

Modern cosmological models can resolve this problem in a variety of ways. For instance, the universe could be static but finite (there are such solutions, but this is nevertheless an unlikely scenario) or the universe is not eternal since there was a 'Big Bang' (and this is a more likely scenario).

33.4 Fundamental Observations II: The Hubble(-Lemaître) Expansion

We have already discussed one of the fundamental inputs of simple cosmological models, namely the cosmological principle. This led us to consider space-times with maximally-symmetric spacelike slices. One of the few other things that is definitely known about the universe, and that tells us something about the time-dependence of the universe, is that it expands or, at least that it appears to be expanding.

In fact, in the 1920's and 1930's, the astronomer Edwin Hubble made a remarkable discovery regarding the motion of galaxies. He found that light from distant galaxies is systematically redshifted (increased in wave-length λ), the increase being proportional to the distance d of the galaxy,

$$z := \frac{\Delta\lambda}{\lambda} \propto d \ . \tag{33.5}$$

Hubble interpreted this redshift as due to a Doppler effect and therefore ascribed a *recessional velocity* $v = cz$ to the galaxy. While, as we will see, this pure Doppler shift explanation is not tenable or at least not always the most useful way of phrasing things, the terminology has stuck, and Hubble's law can be written in the form

$$v = Hd \ , \tag{33.6}$$

where H is *Hubble's constant*. To set the historical record straight: credit for this fundamental discovery should perhaps (also) go to G. Lemaître.[139]

We will see later that in most cosmological models H is actually a function of time, so the H in the above equation should then be interpreted as the value H_0 of H today. It is one of the main goals of observational cosmology to determine H_0 and H as precisely as possible, and the main problem here is naturally a precise determination of the distances of distant galaxies. This is a complex and fascinating issue in its own right, but one that we will not go into here (safe for a brief mention of the *luminosity distance* at the end of section 34.9).[140] I will just conclude this section with one comment on the units usually employed to express galactic distances and the Hubble constant H_0.

Galactic distances are frequently measured in *mega-parsecs* (Mpc). A parsec is the distance from which a star subtends an angle of 2 arc-seconds at the two diametrically opposite ends of the earth's orbit. This unit arose because of the old trigonometric

[139] See e.g. M. Way, H. Nussbaumer, *The linear redshift-distance relationship: Lemaître beats Hubble by two years*, `arXiv:1104.3031v1 [physics.hist-ph]`; J.-P. Luminet, *Editorial note to "The beginning of the world from the point of view of quantum theory"*, `arXiv:1105.6271v1 [physics.hist-ph]`, and in particular also M. Livio, *The Expanding Universe: Lost (in Translation) and Found*, `http://hubblesite.org/pubinfo/pdf/2011/36/pdf.pdf` for an important addition to this debate.

[140] For a beautiful introduction to this subject of the *Cosmic Distance Ladder*, see the pdf-slides of a public talk by Fields Medalist (the mathematics counterpart of a Nobel Laureate!) Terry Tao at `http://terrytao.files.wordpress.com/2010/10/cosmic-distance-ladder.pdf` (you can also find a video of the talk on youtube).

method of measuring stellar distances (a triangle is determined by the length of one side and the two adjacent angles). 1 parsec is approximately 3×10^{18} cm, a little over 3 light-years. The Hubble constant is therefore often expressed in units of km s^{-1} $(\mathrm{Mpc})^{-1}$. The best currently available estimates point to a value of H_0 in the range (using a standard parametrisation)

$$\begin{aligned} H_0 &= 100h\mathrm{km/s/Mpc} \ , \\ h &= 0.71 \pm 0.06 \ . \end{aligned} \tag{33.7}$$

We will usually prefer to express it just in terms of inverse units of time. The above result leads to an order of magnitude range of

$$H_0^{-1} \approx 10^{10} \text{ years} \tag{33.8}$$

(whereas Hubble's original estimate was more in the 10^9 year range).

34 Cosmology II: Geometry and Physics of Robertson-Walker Metrics

34.1 Mathematical Model: the Robertson-Walker Metric

Having determined that the metric of a maximally symmetric space is of the simple form (14.27), we can now deduce that a space-time metric satisfying the Cosmological Principle can be chosen to be of the form (33.1),

$$ds^2 = -dt^2 + a^2(t)\left[\frac{dr^2}{1-kr^2} + r^2 d\Omega^2\right] \quad . \tag{34.1}$$

Here we have used the fact that (as in the ansatz for a spherically symmetric metric) non-trival g_{tt} and g_{tr} can be removed by a coordinate transformation. This metric is known as the *Friedmann-Robertson-Walker metric* or just the Robertson-Walker metric, and spatial coordinates in which the metric takes this form are called *comoving coordinates*, for reasons that will become apparent below. The function $a(t)$, the "radius of the universe", is known as the *cosmic (or cosmological) scale factor*.

It is perhaps useful at this point to recall the symmetries of this metric (the isometries). See sections 14.4 and 39.1 for a more detailed and general discussion:

1. For $k = 0$, the spatial part of the metric is just (up to the overall factor $a(t)^2$), the Euclidean metric on $\mathbb{R}^3$. The isometry group of the Euclidean metric is the 6-parameter Euclidean group $\mathsf{E}(3)$, consisting of spatial translations and spatial rotations,

   $$k = 0: \quad \text{Isometry Group} = \mathsf{E}(3) = \mathsf{SO}(3) \ltimes \mathbb{R}^3 \quad . \tag{34.2}$$

   These are also symmetries of the space-time metric (34.1) for $k = 0$. For generic choices of $a(t)$, there are no further isometries of the metric.

2. For $k = +1$, the spatial part of the metric is just (up to the overall factor $a(t)^2$), the standard metric on the 3-sphere S^3, the one induced on S^3 from the Euclidean metric

   $$ds^2 = dx^2 + dy^2 + dz^2 + dw^2 \tag{34.3}$$

   on $\mathbb{R}^4$ via the embedding

   $$S^3 = \{(x, y, z, w) \in \mathbb{R}^4 : \; x^2 + y^2 + z^2 + w^2 = 1\} \quad . \tag{34.4}$$

   The isometry group of this metric on S^3 is therefore the group $\mathsf{SO}(4)$ of 4-dimensional rotations leaving the 4-dimensional Euclidean metric (and norm) invariant,

   $$k = +1: \quad \text{Isometry Group} = \mathsf{SO}(4) \quad . \tag{34.5}$$

This group is also 6-dimensional. These are also symmetries of the space-time metric (34.1) for $k = +1$. For generic choices of $a(t)$, there are no further isometries of the metric.

3. For $k = -1$, the spatial part of the metric is just (up to the overall factor $a(t)^2$), the standard metric on the 3-hyperboloid H^3, the one induced on H^3 from the Lorentzian metric

$$ds^2 = dx^2 + dy^2 + dz^2 - dw^2 \tag{34.6}$$

on $\mathbb{R}^4$ via the embedding

$$H^3 = \{(x, y, z, w) \in \mathbb{R}^4 : \; x^2 + y^2 + z^2 - w^2 = -1\} \; . \tag{34.7}$$

The isometry group of this metric on H^3 is therefore the group $\mathsf{SO}(3, 1)$ of 4-dimensional rotations leaving the 4-dimensional Lorentzian metric (and norm) invariant,

$$k = -1 : \quad \text{Isometry Group} = \mathsf{SO}(3, 1) \; . \tag{34.8}$$

This group is also 6-dimensional. These are also symmetries of the space-time metric (34.1) for $k = -1$. For generic choices of $a(t)$, there are no further isometries of the metric.

REMARKS:

1. The isometry group for $k = -1$ happens to be isomorphic to the Lorentz group of 4-dimensional Minkowski space. However, here it is realised purely on the spatial part of the metric, while the time coordinate t is not transformed at all. We will later on write the $k = -1$ metric in such a way, via a suitable coordinate transformation, that the $\mathsf{SO}(3, 1)$ symmetries are realised precisely by standard Lorentz transformations on the coordinates (cf. section 34.6).

2. As mentioned in the discussion around (14.29), a rescaling of k by a positive constant, $k \to k/L^2$, is equivalent to an overall constant rescaling of the spatial metric,

$$ds^2 = \frac{dr^2}{1 - kr^2/L^2} + r^2 d\Omega^2 = L^2 \left(\frac{d\tilde{r}^2}{1 - k\tilde{r}^2} + \tilde{r}^2 d\Omega^2 \right) \; , \tag{34.9}$$

where $\tilde{r} = r/L$. In the context of the Robertson-Walker metric (34.1), a scaling of k is thus equivalent to a scaling of the cosmic scale factor $a(t)$. This scaling freedom can be used

   - either to fix $k = 0, \pm 1$ once and for all,
   - or to normalise $a(t)$ in a convenient way, e.g. by the condition $a(t_0) = 1$ for t_0 the time "today".

In the latter case, the 3 different possibilities for the spatial geometry, are distinguished by $k < 0, k = 0, k > 0$. For the most part we work with the first option, but for certain questions (like "why is space so close to being flat today?") it is convenient to rephrase and express this in terms of k ("why is k so close to zero?"), which is evidently only meaningful if one does not restrict k to the 3 discrete values $k = 0, \pm 1$.

3. Another convenient way of representing this metric, that we will occasionally make use of below, is as

$$ds^2 = -dt^2 + a(t)^2(d\psi^2 + g_k(\psi)^2 d\Omega^2) \quad . \tag{34.10}$$

where (cf. (14.34) and (14.35))

$$g_k(\psi) = \begin{cases} \psi & k = 0 \\ \sin\psi & k = +1 \\ \sinh\psi & k = -1 \end{cases} \tag{34.11}$$

4. The metric of the three-space at constant t is

$$g_{ij} = a^2(t)\tilde{g}_{ij} \quad , \tag{34.12}$$

where $\tilde{g}_{ij}$ is the maximally symmetric spatial metric. Thus for $k = +1$, $a(t)$ directly gives the size (radius) of the universe. For $k = -1$, space is infinite, so no such interpretation is possible, but nevertheless $a(t)$ still sets the scale for the geometry of the universe, e.g. in the sense that the curvature scalar $R^{(3)}$ of the metric g_{ij} is related to the curvature scalar $\tilde{R}^{(3)}$ of $\tilde{g}_{ij}$ by

$$R^{(3)}(t) = \frac{1}{a^2(t)}\tilde{R}^{(3)} \quad . \tag{34.13}$$

Finally, for $k = 0$, three-space is flat and also infinite, but one could replace $\mathbb{R}^3$ by a three-torus T^3 (still flat but now compact) and then $a(t)$ would once again be related directly to the size of the universe at constant t.

5. Through the dependence of $a(t)$ on t, proper length scales and distances in the constant time surfaces depend on time. Thus $a(t)$ changes or sets the scale, i.e. $a(t)$ plays the role of a cosmological scale factor.

6. Note that the case $k = +1$ opened up for the very first time the possibility of considering, even conceiving, an unbounded but finite universe! These and other generalisations made possible by a general relativistic approach to cosmology are important as more naive (Newtonian) models of the universe immediately lead to paradoxes or contradictions (as we have seen e.g. in the discussion of Olbers' paradox in section 33.3).

34.2 Timelike Geodesics and Comoving Observers

We now look at timelike geodesics in the Robertson-Walker geometry. We will analyse the general case momentarily, but we can already identify one privileged class of geodesic observers by inspection. Indeed, as already noted (way back) in the discussion around (3.11), in a space-time of the Robertson-Walker form there is a particularly simple class of timelike geodesics, given by fixing once and for all the spatial coordinates of the observer.

Let us quickly rederive this result here. Note that, since $g_{tt} = -1$ is a constant and the off-diagonal time - space components of the metric are zero, $g_{tk} = 0$, one has

$$\Gamma_{\mu tt} = 0 \ . \tag{34.14}$$

Therefore the vector field ∂_t is geodesic, which can be expressed as the statement that

$$\nabla_t \partial_t := \Gamma^{\mu}_{tt} \partial_\mu = 0 \ . \tag{34.15}$$

In simpler terms this means that the curves $\vec{x} = const.$ ($\vec{x}$ referring to the spatial coordinates),

$$\tau \to (t(\tau), \vec{x}(\tau)) = (\tau, \vec{x}_0) \tag{34.16}$$

are geodesics.

Hence, in this coordinate system, observers remaining at fixed values of the spatial coordinates are in free fall. In other words, the coordinate system is falling with them or *comoving*, and the proper time τ along such geodesics coincides with the coordinate time or cosmic time t, $d\tau = dt$. It is these observers of constant $\vec{x}$ or constant (r, θ, ϕ) who all see the same isotropic universe at a given value of t.

Remarks:

1. This may sound a bit strange but a good way to visualise such a coordinate system is, as in Figure 44, as a mesh of coordinate lines drawn on a balloon that is being inflated or deflated (according to the behaviour of $a(t)$). Draw some dots on that balloon (that will eventually represent galaxies or clusters of galaxies). As the balloon is being inflated or deflated, the dots will move but the coordinate lines will move with them and the dots remain at fixed spatial coordinate values. Thus, as we now know, regardless of the behaviour of $a(t)$, these dots follow a geodesic, and we will thus think of galaxies in this description as being in free fall.

2. Recall that we had already encountered analogous comoving coordinates in our discussion of Lemaître coordinates for the Schwarzschild metric in section 27.3 and subsequently in equation (29.26) of section 29.3, when we had introduced proper time of the freely falling particles on the surface of the star to describe the metric induced on the surface of the star.

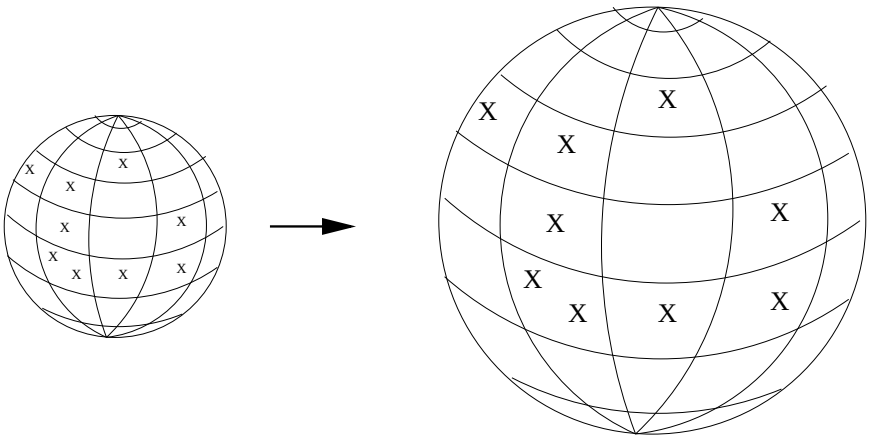


Figure 44: Illustration of a comoving coordinate system: Even though the sphere (universe) expands, the X's (galaxies) remain at the same spatial coordinates. These trajectories are geodesics and hence the X's (galaxies) can be considered to be in free fall. The figure also shows (cf. the discussion in section 6.8) that it is the number density per unit coordinate volume that is conserved, *not* the density per unit proper volume.

3. Another advantage of the comoving coordinate system is that the six-parameter family of isometries just acts on the spatial part of the metric. Indeed, let $K^i\partial_i$ be a Killing vector of the maximally symmetric spatial metric. Then $K^i\partial_i$ is also a Killing vector of the Robertson-Walker metric. This would not be the case if one had e.g. made an x-dependent coordinate transformation of t or a t-dependent coordinate transformation of the x^i. In those cases there would of course still be six Killing vectors, but they would have a more complicated form.

The worldlines of comoving observers discussed above are special timelike geodesics. To discuss the general case, it will be convenient to write the metric in the form (34.10)

$$ds^2 = -dt^2 + a(t)^2(d\psi^2 + g_k(\psi)^2 d\Omega^2) \ . \tag{34.17}$$

By spatial maximal symmetry and the associated conserved (angular) momenta, we can without loss of generality consider motion in the (t,ψ)-direction, so that $\dot{t}$ and $\dot{\psi}$ are related by

$$\dot{t}^2 - a(t)^2\dot{\psi}^2 = 1 \ , \tag{34.18}$$

where, as usual, an overdot refers to differentiation with respect to proper time.

Even though we do not have a timelike Killing vector (and its associated conserved energy) to further simplify this, in the case at hand we have plenty of spacelike Killing vectors $V^\alpha\partial_\alpha$ with $V^t = 0$. Among them there will be ψ-translational Killing vectors which have the form

$$V = f(\theta,\phi)\partial_\psi + \dots \tag{34.19}$$

(the 3-dimensional counterparts of the Killing vectors $V_{(1)}, V_{(2)}$ (9.55) of the 2-sphere, say). Associated to any such Killing vector and the timelike geodesic there is the conserved momentum

$$P = \dot{x}^\alpha V_\alpha = a(t)^2 \dot{\psi} f(\theta, \phi) \ . \tag{34.20}$$

Since θ and ϕ are constant along the lightray, we can absorb $f(\theta, \phi)$ into the definition of P, and thus we have

$$\dot{\psi} = P/a^2 \ . \tag{34.21}$$

Therefore (34.18) can be written as

$$\dot{t} = \sqrt{1 + P^2/a^2} \ . \tag{34.22}$$

In particular, we see that comoving observers are characterised by $P = 0$,

$$P = 0 \quad \Rightarrow \quad \dot{\psi} = 0 \quad \text{(comoving)} \ , \tag{34.23}$$

and that precisely for these observers the cosmic time t coincides with their proper time,

$$P = 0 \quad \Leftrightarrow \quad \dot{t} = 1 \quad \Leftrightarrow \quad dt = d\tau \ . \tag{34.24}$$

Nevertheless, even in the general case it is useful to combine the two previous equations to obtain an equation for ψ as a function of t, namely (assuming $P \neq 0$)

$$\frac{d\psi(t)}{dt} = \frac{\dot{\psi}}{\dot{t}} = \frac{P}{\sqrt{a(t)^4 + P^2 a(t)^2}} = \frac{1}{a(t)\sqrt{1 + a(t)^2/P^2}} \ . \tag{34.25}$$

In general, even for simple power-law behaviours for $a(t)$ (which we will typically find as solutions to the Einstein equations in the spatially flat case $k = 0$), this equation cannot be solved in closed form (but can be approximated by a tractable, even elementary, integral when $a(t) \gg |P|$ or when $a(t) \ll |P|$).

34.3 Velocity - Distance Relation, Recessional Velocities and the Hubble Sphere

The Robertson-Walker metric immediately, and in complete generality, implies a crude distance - velocity relation reminiscent of Hubble's law. Namely, let us ego- or geocentrically place ourselves at the origin $r = 0$ (remember that because of maximal symmetry this point is as good as any other and in no way privileged). Consider another galaxy following the comoving geodesic at the fixed value $r = r_1$. Its "instantaneous" proper distance $R_p(t)$ at time t can be calculated from

$$dR_p = a(t) \frac{dr}{(1 - kr^2)^{1/2}} \quad \Rightarrow \quad R_p(t) = a(t) f_k(r_1) \ , \tag{34.26}$$

where

$$f_k(r) = \begin{cases} r & k = 0 \\ \arcsin r & k = +1 \\ (\sinh)^{-1} r & k = -1 \end{cases} \tag{34.27}$$

Note that the $f_k(r)$ are the inverses of the functions $g_k(\psi)$ defined in (34.11) (the precise form of $f_k(r)$ will however be irrelevant for this argument). If we use the coordinates (t, ψ) instead of (t, r), the instantaneous proper distance to a point with coordinate ψ_1 simply has the form

$$R_p(t) = a(t)\psi_1 \quad . \tag{34.28}$$

It follows that its proper velocity is

$$V_p(t) \equiv \frac{d}{dt} R_p(t) = \dot{a}(t)\psi_1 = H(t) R_p(t) \quad . \tag{34.29}$$

Here we have introduced the *Hubble parameter*

$$H(t) = \frac{\dot{a}(t)}{a(t)} \quad , \tag{34.30}$$

which plays a pivotal role in cosmology.

The relation (34.29) clearly expresses something like Hubble's law $v = Hd$ (33.6): all objects run away from each other with velocities proportional to their distance. And the above derivation shows that such an (at first sight perhaps surprising) relationship between (some kind of) velocity and (some kind of) distance can arise as a general purely geometric statement about the geometry of space-time. We will have much more to say about $H(t)$, and about the relation between distance and redshift z (which is what is actually observed), below.

Calculating the second time-derivative of $R_p(t)$, one finds

$$\frac{d^2}{dt^2} R_p(t) = (\frac{d}{dt} H(t)) R_p(t) + H(t) \frac{d}{dt} R_p(t) = \frac{\ddot{a}(t)}{a(t)} R_p(t) \tag{34.31}$$

since

$$\frac{d}{dt} H(t) = \frac{\ddot{a}(t)}{a(t)} - \frac{\dot{a}(t)^2}{a(t)^2} = \frac{\ddot{a}(t)}{a(t)} - H(t)^2 \quad . \tag{34.32}$$

Thus the cosmological expansion or contraction can be visualised as acting like a linear harmonic oscillator force on the separation of comoving objects, with (in general time-dependent) real or imaginary frequency $\omega(t)$,

$$\frac{d^2}{dt^2} R_p(t) + \omega(t)^2 R_p(t) = 0 \quad , \quad \omega(t)^2 = -\frac{\ddot{a}(t)}{a(t)} \quad . \tag{34.33}$$

Universes with an accelerating expansion thus lead to imaginary frequencies, and hence to an exponential-like rather than harmonic motion (over periods of time during which the time-dependence of the frequency can be neglected).

If the object is not comoving, i.e. is not sitting at a fixed value of ψ, say, one has

$$R_p(t) = a(t)\psi(t) \quad , \tag{34.34}$$

and then its proper velocity is

$$V_p(t) = \frac{d}{dt}R_p(t) = H(t)R_p(t) + a(t)\frac{d\psi(t)}{dt} \quad . \tag{34.35}$$

Here "proper" refers to the fact that it is the t-derivative of the instantaneous proper distance, but it should be kept in mind that t is not the proper time for non-comoving observers. This proper velocity can be decomposed into what are known as its *recessional velocity* (equal to the proper velocity for a comoving object) and its *peculiar velocity* (relative to the recessional velocity or *Hubble flow*),

$$V_p(t) = H(t)R_p(t) + a(t)\frac{d\psi(t)}{dt} \equiv V_{rec}(t) + V_{pec}(t) \quad . \tag{34.36}$$

As a consequence of (34.18), one has

$$V_{pec}(t)^2 = a(t)^2 \left(\frac{d\psi(t)}{dt}\right)^2 < 1 \quad . \tag{34.37}$$

Alternatively this follows from (34.25), which allows us to express the peculiar velocity as

$$V_{pec}(t) = a(t)\frac{d\psi(t)}{dt} = \frac{1}{\sqrt{1 + a(t)^2/P^2}} \quad . \tag{34.38}$$

Either way we see that the peculiar velocity is always "subluminal".

However, the recessional velocity is not restricted in this way and in most cosmological models superluminal recessional velocities will occur for objects which are suffficiently far away (in the sense of having a sufficiently large proper distance $R_p(t)$).

REMARKS:

1. There is absolutely nothing illegal or pathological about this because V_{rec} measures the rate of an increase in distance between two objects, not any locally measurable velocity. For example, even in Special Relativity, if in your rest-frame you send off two objects in opposite directions at speeds $> c/2$ each, then the distance between them grows at a rate (measured with respect to your proper time) larger than c, but clearly you have not violated or disproven Special Relativity by doing this.

2. Moreover, in a curved space-time a comparison of vectors (say velocity vectors) at different points requires some care. Indeed, as explained in section 5.8, this requires some notion of parallel transport in order to have two vectors at the same point that one can meaningfully compare.[141]

[141] See e.g. A. Kaya, *Hubble's law and faster than light expansion speeds*, `arXiv:1107.5168 [physics.gen-ph]` for an analysis of this issue from the parallel transport point of view.

3. For a further illustration of the harmlessness of superluminal recessional velocities in the present context, it suffices to consider the special case $a(t) = t$, in which case one has

$$a(t) = t \quad \Rightarrow \quad H(t) = 1/t \quad \Rightarrow \quad V_{rec}(t) = R_p(t)/t \ . \tag{34.39}$$

At any given time t, objects further away than t appear to have superluminal recession velocities. On the other hand, as shown in section 37.1, the space-time with $a(t) = t$ and $k = -1$ is just a part of Minkowski space. Thus even Minkowski space can be foliated in such a way (by hyperboloids in the future lightcone) that events appear to have superluminal recession velocities, but evidently there is nothing here that violates any of the postulates of Special Relativity.

4. Cosmologists frequently refer to the sphere beyond which the recessional velocity exceeds the speed of light as the *Hubble sphere*. Its radius $R_H(t)$ at time t, the *Hubble radius*, is (restoring for once and temporarily the speed of light c)

$$V_{rec}(t) = c \quad \Leftrightarrow \quad R_H(t) = c/H(t) \ . \tag{34.40}$$

Misleadingly, this surface is also often referred to as the *Hubble horizon*, the reason for this apparently being the idea or belief that we can never observe objects outside the Hubble sphere, but this is in general not correct. In particular, it is not correct to say (perhaps based on a mistaken analogy with special relativistic reasoning) that objects with recessional velocities $V_{rec} > c$ are infinitely redshifted and therefore invisible to us. There is indeed a cosmological redshift, worked out in section 34.8, and there is also an ensuing Hubble-like redshift - distance relation in Robertson-Walker geometries, derived in section 34.9. However, this cannot be written as a standard special relativistic recessional velocity - redshift relation involving the recessional velocity.[142]

5. It is true that there are limits to how much of the universe one can observe at any given time, and it is also true that in certain situations the Hubble radius $R_H(t)$ provides one with an order of magnitude estimate of the size of the visible universe.

However, it is certainly misleading (even though some people appear to be obsessed with this) to think of the visible universe (or the inside of the Hubble sphere) as somehow being like the inside of a Schwarzschild black hole or some such nonsense. In fact, with any standard definition of a black hole going beyond pop-sci culture wisdom this statement is so obviously wrong or misleading in so many respects that I don't even know where to start (so don't get me started). Nevertheless I will briefly revisit this claim in section 36.9. For the time you may enjoy poking holes into this statement yourself ...

[142] See also T. Davis, C. Lineweaver, *Superluminal Recession Velocities*, `arXiv:astro-ph/0011070` for a nice discussion of this.

We will return to some of these issues in a (slightly) more quantitative way later on, in sections 36.7 – 36.9, with the Friedmann equations (i.e. the Einstein equations for the standard model of cosmology we are in the process of developing) at our disposal.

34.4 Painlevé-Gullstrand-like Coordinates for Comoving Observers

In section 27.2 we introduce coordinates for the Schwarzschild metric that are adapted to radial geodesic observers, known as Painlevé-Gullstrand coordinates. We can do something analogous for the comoving observers of the Robertson-Walker metrics.

For $k = 0$ the Robertson-Walker metric is

$$ds^2 = -dt^2 + a(t)^2(dr^2 + r^2 d\Omega^2) \ . \tag{34.41}$$

To put this into PG-like form, we keep t (which is, after all, already the proper time of comoving observers) but introduce, instead of the comoving coordinate r the area radius

$$\tilde{r}(t, r) = a(t)r \ . \tag{34.42}$$

In terms of this the Robertson-Walker metric takes the PG-like form

$$\begin{aligned} ds^2 &= -(1 - \tilde{r}^2 H(t)^2)dt^2 - 2\tilde{r}H(t)dtd\tilde{r} + (d\tilde{r}^2 + \tilde{r}^2 d\Omega^2) \\ &= -dt^2 + (d\tilde{r} - \tilde{r}H(t)dt)^2 + \tilde{r}^2 d\Omega^2 \ , \end{aligned} \tag{34.43}$$

where, as above, $H(t) = \dot{a}(t)/a(t)$ is the Hubble parameter.

Remarks:

1. This is clearly analogous to the Schwarzschild metric in PG coordinates (27.12)

$$\begin{aligned} ds^2 &= -(1 - 2m/r)dT^2 + 2\sqrt{2m/r}dTdr + (dr^2 + r^2 d\Omega^2) \\ &= -dT^2 + \left(dr + \sqrt{2m/r}dT\right)^2 + r^2 d\Omega^2 \ . \end{aligned} \tag{34.44}$$

2. Just as the Schwarzschild metric in PG coordinates is adapted to observers with $dr = -\sqrt{2m/r}dT$ (for which T is proper time, $\dot{r} = -\sqrt{2m/r}$ describing geodesic radial free fall with $E = 1$), the metric (34.43) is adapted to observers with

$$\frac{d}{dt}\tilde{r} = H(t)\tilde{r} \tag{34.45}$$

which are precisely the comoving observers $\tilde{r}(t) = a(t)r$ with r fixed obeying the Hubble law (34.29).

3. Performing the same coordinate transformation to the area radius $\tilde{r} = a(t)r$ for $k \neq 0$, one finds the metric

$$ds^2 = -dt^2 + \frac{1}{1 - k\tilde{r}^2/a^2}\left(d\tilde{r} - \tilde{r}H(t)dt\right)^2 + \tilde{r}^2 d\Omega^2 \ . \tag{34.46}$$

4. While these PG-like coordinates are not widely used in the cosmological context, we make use of them in section 29.5, in the description of the interior geometry of a collapsing star (because this PG-like form of the metric makes it particularly easy to match the interior metric to the exterior Schwarzschild metric).

34.5 Conformal Time η

Writing the Robertson-Walker metric as

$$ds^2 = -dt^2 + a(t)^2 d\tilde{s}^2 = a(t)^2[-dt^2/a(t)^2 + d\tilde{s}^2] \ , \tag{34.47}$$

where a tilde refers to the maximally symmetric spatial metric, we see that it is natural to introduce a new time-coordinate η through

$$d\eta = dt/a(t) \ , \tag{34.48}$$

in terms of which the Robertson-Walker metric takes the simple form

$$d\eta = dt/a(t) \quad \Rightarrow \quad ds^2 = a^2(\eta)(-d\eta^2 + d\tilde{s}^2) \tag{34.49}$$

($a(\eta)$ is short (and sloppy) for $a(t(\eta))$). In terms of polar coordinates (34.10), this becomes

$$ds^2 = a^2(\eta)(-d\eta^2 + d\psi^2 + g_k(\psi)^2 d\Omega^2) \ . \tag{34.50}$$

Remarks:

1. In particular, "radial" null lines are determined by $d\eta = \pm d\psi$, as in flat space, and η is also known as *conformal time*. This coordinate is very convenient for discussing the causal structure of the Friedmann-Lemaître-Robertson-Walker universes.

2. Since the η-dependence resides exclusively in the overall conformal factor of the metric, the vector ∂_η is a *conformal Killing vector* of the Robertson-Walker metric in the sense of (10.5),

$$C = \partial_\eta : \quad \nabla_\alpha C_\beta + \nabla_\beta C_\alpha = 2\frac{a'(\eta)}{a(\eta)}\, g_{\alpha\beta} = 2\dot{a}(t)\, g_{\alpha\beta} \ . \tag{34.51}$$

3. We will use conformal time immediately below to establish the conformal flatness of the Robertson-Walker metrics, in section 37.3 to solve the cosmological Einstein equations in a particular case, and we will use the above conformal Killing vector and the associated conserved charge for null geodesics (10.9) in the discussion of the cosmological redshift in section 34.8.

In the spatially flat case $k = 0$, (34.50) shows that Robertson-Walker metrics are conformally flat, here written in radial polar coordinates with $\psi = r$,

$$k = 0 \quad \Rightarrow \quad ds^2 = a^2(\eta)(-d\eta^2 + dr^2 + r^2 d\Omega^2) = a(\eta)^2 \eta_{\alpha\beta} dx^\alpha dx^\beta \ . \tag{34.52}$$

This is actually true for all Robertson-Walker metrics, i.e. also for $k \neq 0$, but the coordinate transformation required to exhibit this as explicitly as in the $k = 0$ case is somewhat more involved (and, as we will see below, the conformal factor does not depend just on the Minkowski time coordinate).

As a first step it will be convenient to introduce null coordinates

$$u = \eta - \psi \quad , \quad v = \eta + \psi \ , \tag{34.53}$$

in terms of which the metric (34.50) takes the form

$$ds^2 = a^2(\eta) \left[-du\, dv + g_k((v-u)/2)^2 \, d\Omega^2 \right] \ . \tag{34.54}$$

This metric is conformally flat if the line element in brackets, which does not depend on the cosmic scale factor $a(t)$ or $a(\eta)$, is conformally flat, so let us focus on

$$d\bar{s}^2 = ds^2/a^2(\eta) = -du\, dv + g_k((v-u)/2)^2 \, d\Omega^2 \ , \tag{34.55}$$

where, recall, $g_k(\psi) = \sin(\mathrm{h})\psi$ for $k = \pm 1$ respectively.

Given a metric of this form, it is natural to consider transformations of the form

$$U = U(u) \quad , \quad V = V(v) \tag{34.56}$$

because under such transformations $dudv$ and $dUdV$ are just related by an overall conformal factor. It is now straightforward to check, using some basic trigonometric identities (or their hyperbolic counterparts) that the specific coordinate transformation

$$U = 2\tan(\mathrm{h})u/2 \quad , \quad V = 2\tan(\mathrm{h})v/2 \tag{34.57}$$

(we will consider an alternative transformation for $k = -1$ below) is such that it transforms $d\bar{s}^2$ into

$$d\bar{s}^2 = \left(\cos(\mathrm{h})^2 u/2\right) \left(\cos(\mathrm{h})^2 v/2\right) \left[-dU\, dV + \tfrac{1}{4}(V-U)^2 d\Omega^2 \right] \ . \tag{34.58}$$

Now the metric in brackets is just the Minkowski metric, written in radial null coordinates, as can be seen by undoing the transformation to null coordinates through

$$U = T - R \quad , \quad V = T + R \ , \tag{34.59}$$

which results in

$$-dU\, dV + \tfrac{1}{4}(V-U)^2 d\Omega^2 = -dT^2 + dR^2 + R^2 d\Omega^2 \ , \tag{34.60}$$

as claimed. This establishes explicitly the conformal flatness of the Robertson-Walker metrics even for $k \neq 0$. In particular, therefore, the Weyl tensor of a Robertson-Walker metric is zero (section 11.4), and the Riemann tensor can be expressed in terms of its traces, the Ricci tensor and the Ricci scalar.

Note that for $k \neq 0$ the conformal factor relating the Robertson-Walker metric to the Minkowski metric is not just a function of the Minkowski time T alone, but a particular function of T and R,

$$k \neq 0: \qquad ds^2_{FRW} = \Omega^2(T,R)\left(-dT^2 + dR^2 + R^2 d\Omega^2\right) \quad . \tag{34.61}$$

This could hardly be otherwise because (recall the discussion in section 34.1) for $k \neq 0$ the isometry group is not the Euclidean group, the symmetry group of the Euclidean metric $dR^2 + R^2 d\Omega^2$, but rather it is $\mathsf{SO}(4)$ for $k = +1$ and $\mathsf{SO}(3,1)$ for $k = -1$.

In the latter case, we can make this symmetry manifest by slightly modifying the above procedure. To that end consider, instead of (34.57), the transformation

$$k = -1: \quad U = T - R = \mathrm{e}^{\,u} \quad , \quad V = T + R = \mathrm{e}^{\,v} \quad , \tag{34.62}$$

leading to

$$T = \mathrm{e}^{\,\eta}\cosh\psi \quad , \quad R = \mathrm{e}^{\,\eta}\sinh\psi \quad . \tag{34.63}$$

Then a straightforward calculation, using

$$R^2 = \mathrm{e}^{\,2\eta}\sinh^2\psi = UV\sinh^2\psi \quad , \tag{34.64}$$

shows that

$$\begin{aligned} ds^2 &= -dt^2 + a(t)^2(d\psi^2 + \sinh^2\psi\, d\Omega^2) \\ &= a(\eta)^2\left[-du\, dv + \sinh^2\psi\, d\Omega^2\right] \\ &= \frac{a(\eta)^2}{UV}\left[-dU\, dV + UV\sinh^2\psi\, d\Omega^2\right] \\ &= \mathrm{e}^{\,-2\eta}a(\eta)^2\left[-dT^2 + dR^2 + R^2 d\Omega^2\right] \quad . \end{aligned} \tag{34.65}$$

Here the term in brackets is just the 4-dimensional Minkowski metric,

$$-dT^2 + dR^2 + R^2 d\Omega^2 = -dT^2 + d\vec{X}^2 \quad , \tag{34.66}$$

and the conformal prefactor is only a function of η (or t) which, via

$$T^2 - R^2 = \mathrm{e}^{\,2\eta} \quad , \tag{34.67}$$

is itself only a function of the Lorentz invariant quantity

$$T^2 - R^2 = T^2 - \vec{X}^2 \quad . \tag{34.68}$$

Written in this way, the complete $\mathsf{SO}(3,1)$ symmetry of the $k = -1$ metrics is manifest. In the original comoving coordinates, it was realised as the spatial isometry group of the hyperboloid H^3, with t not transformed. In the new form of the metric, it is explicitly realised as the Lorentz transformations of (an auxiliary) 4-dimensional Minkowski space-time with coordinates $(T, \vec{X})$, but with the original time coordinate t or η transforming as a Lorentz scalar under these transformations, as it should.

34.7 Area Measurements and Number Counts

The aim of this and the subsequent sections is to learn as much as possible about the general properties of Robertson-Walker geometries (without using the Einstein equations) with the aim of looking for observational means of distinguishing e.g. among the models with $k = 0, \pm 1$.

To get a feeling for the geometry of the Schwarzschild metric, we studied the properties of areas and lengths in the Schwarzschild geometry. Spatial length measurements are rather obvious in the Robertson-Walker geometry, so here we focus on the properties of areas.

We write the spatial part of the Robertson-Walker metric in polar coordinates as (34.10)

$$ds^2 = a(t)^2[d\psi^2 + g_k^2(\psi)d\Omega^2] \ , \tag{34.69}$$

where $g_k(\psi) = \psi, \sin\psi, \sinh\psi$ for $k = 0, +1, -1$ (see (14.35)). Now the radius of a surface $\psi = \psi_0$ around the point $\psi = 0$ (or any other point, our space is isotropic and homogeneous) is given by

$$\rho = a\int_0^{\psi_0} d\psi = a\psi_0 \ . \tag{34.70}$$

On the other hand, the area of this surface is determined by the induced metric $a^2 g_k^2(\psi_0)d\Omega^2$ and is

$$A(\rho) = a^2 g_k^2(\psi_0)\int_0^{2\pi} d\phi \int_0^{\pi} d\theta \sin\theta = 4\pi a^2 g_k^2(\rho/a) \ . \tag{34.71}$$

For $k = 0$, this is just the standard behaviour

$$A(\rho) = 4\pi\rho^2 \ , \tag{34.72}$$

but for $k = \pm 1$ the geometry looks quite different. For $k = +1$, we have

$$A(\rho) = 4\pi a^2 \sin^2(\rho/a) \ . \tag{34.73}$$

Thus the area reaches a maximum for $\rho = \pi a/2$ (or $\psi = \pi/2$), then *decreases* again for larger values of ρ and goes to zero as $\rho \to \pi a$. Already the maximal area, $A_{max} = 4\pi a^2$ is much smaller than the area of a sphere of the same radius in Euclidean space, which would be $4\pi\rho^2 = \pi^3 a^2$.

This behaviour is best visualised by replacing the three-sphere by the two-sphere and looking at the circumference of circles as a function of their distance from the origin (see Figure 45).

For $k = -1$, we have

$$A(\rho) = 4\pi a^2 \sinh^2(\rho/a) \ , \tag{34.74}$$

so in this case the area grows much more rapidly with the radius than in flat space.

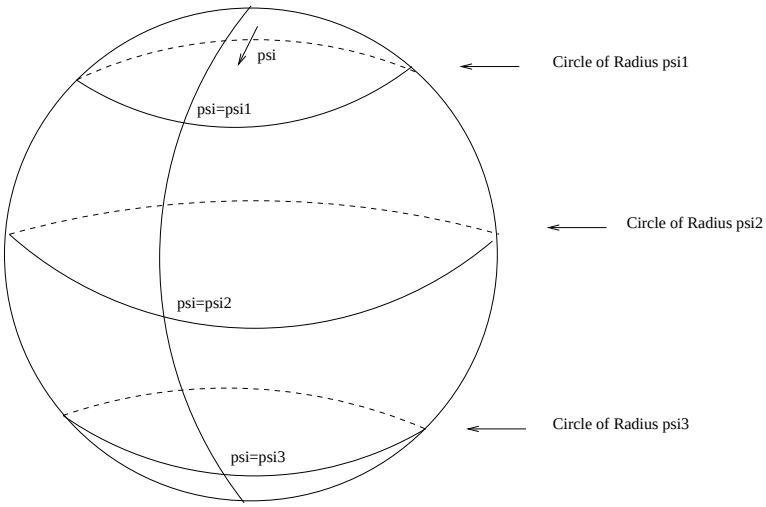


Figure 45: Visualisation of the $k = +1$ Robertson-Walker geometry via a two-sphere of unit radius: Circles of radius ψ, measured along the two-sphere, have an area which grows at first, reaches a maximum at $\psi = \pi/2$ and goes to zero when $\psi \to \pi$. E.g. the maximum value of the circumference, at $\psi = \pi/2$, namely 2π, is much smaller than the circumference of a circle with the same radius $\pi/2$ in a flat geometry, namely $2\pi \times \pi/2 = \pi^2$. Only for ψ very small does one approximately see a standard Euclidean geometry.

In principle, this distinct behaviour of areas in the models with $k = 0, \pm 1$ might allow for an empirical determination of k. For instance, one might make the assumption that there is a homogeneous distribution of the number and brightness of galaxies, and one could try to determine observationally the number of galaxies as a function of their apparent luminosity. As in the discussion of Olbers' paradox, the radiation flux would be proportional to $F \propto 1/\rho^2$. In Euclidean space ($k = 0$), one would expect the number $N(F)$ of galaxies with flux greater than F, i.e. distances less than ρ to behave like ρ^3, so that the expected Euclidean behaviour would be

$$N(F) \propto F^{-3/2} \quad . \tag{34.75}$$

Any empirical departure from this behaviour could thus be an indication of a universe with $k \neq 0$, but clearly, to decide this, many other factors (redshift, evolution of stars, etc.) would have to be taken into account. This illustrates as a matter of principle how the geometry of the spatial slices influences, and can be encoded in, observable quantities. In practice, however,

> [...] the statistical uncertainties, together with source evolution (which affects the detection probabilities), prevent this from being a useful test of k.[143]

[143] G. Ellis, R. Maartens, M. MacCallum, *Relativistic Cosmology*, section 7.6.

34.8 Cosmological Redshift

The most important information about the cosmic scale factor $a(t)$ comes from the observation of shifts in the frequency of light emitted by distant sources.

To calculate the expected shift in a Robertson-Walker geometry, let us again place ourselves at the origin $r = 0$. We consider a radially travelling electro-magnetic wave (a lightray) and consider the equation $d\tau^2 = 0$ or

$$dt^2 = a^2(t)\frac{dr^2}{1-kr^2} \quad . \tag{34.76}$$

Since the cosmological scale factor $a(t)$ sets the length scale, one may expect that wave lengths at different times are related by

$$\frac{\lambda(t_1)}{\lambda(t_0)} = \frac{a(t_1)}{a(t_0)} \quad , \tag{34.77}$$

leading to the relation

$$\frac{\omega(t_0)}{\omega(t_1)} = \frac{a(t_1)}{a(t_0)} \quad , \tag{34.78}$$

among the frequencies. This is indeed the correct result.

As in our discussion of the gravitational redshift in section 3.5, I will analyse this situation in two ways, in a geometric optics approach, where we trace the lightrays in the above geometry, and in a slightly more covariant language using the geodesic equation and the symmetries and associated conserved charges. I will also give a third, essentially one-line (but perhaps at first somewhat obscure looking), derivation based on the conformal Killing vector (34.51) and its associated conserved charge.

1. Let us assume that the wave leaves a galaxy located at $r = r_1$ at the time t_1. Then it will reach us at $r = 0$ at a time t_0 given by

$$f_k(r_1) = \int_{r_1}^{0} \frac{dr}{\sqrt{1-kr^2}} = \int_{t_1}^{t_0} \frac{dt}{a(t)} \quad . \tag{34.79}$$

   Note that there will only be a solution to this equation if the light from the galaxy at $r = r_1$ actually reaches us at a time t_0. In this sense galaxies whose light has not yet reached us (or may perhaps never reach us) are implicitly (and now, having said this, explicitly) excluded from the analysis - after all, such galaxies are not particularly useful for analysing redshifts.

   As typical galaxies will be comoving, i.e. have have constant spatial coordinates, $f_k(r_1)$ (34.27) is time-independent. If the next wave crest leaves the galaxy at r_1 at time $t_1 + \delta t_1$, it will arrive at a time $t_0 + \delta t_0$ determined by

$$f_k(r_1) = \int_{t_1+\delta t_1}^{t_0+\delta t_0} \frac{dt}{a(t)} \quad . \tag{34.80}$$

Subtracting these two equations and making the (eminently reasonable) assumption that the cosmic scale factor $a(t)$ does not vary significantly over the period δt given by the frequency of light, we obtain

$$\frac{\delta t_0}{a(t_0)} = \frac{\delta t_1}{a(t_1)} \quad . \tag{34.81}$$

Indeed, say that $b(t)$ is the integral of $1/a(t)$. Then we have

$$b(t_0 + \delta t_0) - b(t_1 + \delta t_1) = b(t_0) - b(t_1) \quad , \tag{34.82}$$

and Taylor expanding to first order, we obtain

$$b'(t_0)\delta t_0 = b'(t_1)\delta t_1 \quad , \tag{34.83}$$

which is the same as (34.81). Therefore the observed frequency ω_0 is related to the emitted frequency ω_1 by

$$\frac{\omega_0}{\omega_1} = \frac{a(t_1)}{a(t_0)} \quad , \tag{34.84}$$

precisely as anticipated in (34.78). I will comment further on this result below.

2. As in derivation 2 of section 3.5, we describe a lightray by the null wave vector $k^\mu = (\omega, \vec{k})$. The frequency measured by an observer with velocity u^μ is then $\omega = -u^\mu k_\mu$ (3.117).

   Adapting the discussion of timelike geodesics in the Robertson-Walker geometry in section 34.2 to null rays, we can choose the wave vector to be of the form

$$k^\mu = (\dot{t}, \dot{\psi}, 0, 0) \tag{34.85}$$

   with

$$-\dot{t}^2 + a(t)^2\dot{\psi}^2 = 0 \quad . \tag{34.86}$$

   In section 3.5 we used the timelike Killing vector of the static spherically symmetric metric, and its associated conserved energy, to relate the measured frequencies for static observers at different radial positions and to determine the gravitational redshift. Here we use one of the spatial Killing vectors to deduce, as in section 34.2, that

$$\dot{\psi} = P/a^2 \tag{34.87}$$

   for some constant P. Thus

$$\dot{t} = |P|/a \quad \Rightarrow \quad k^\alpha = (|P|/a)\ (1, \pm 1/a, 0, 0) \quad . \tag{34.88}$$

   The observers we are interested in are the comoving observers at fixed values of the spatial coordinates, i.e. with $u^\alpha = (1, 0, 0, 0)$. Thus these measure the frequency

$$\omega(t) = -k^\alpha u_\alpha = |P|/a(t) \quad . \tag{34.89}$$

In particular, for the ratio of frequencies at times t_0 and t_1 one has

$$\frac{\omega(t_0)}{\omega(t_1)} = \frac{a(t_1)}{a(t_0)} \quad , \tag{34.90}$$

in complete agreement with the result (34.78) found previously in (34.84).

3. Alternatively, and even more quickly, one can use the *conformal Killing vector* (34.51)

$$C = \partial_\eta = a(t)\partial_t \quad \text{or} \quad C^\alpha = a(t)u^\alpha \tag{34.91}$$

of the Robertson-Walker metric, and the associated conserved quantity $C_\alpha \dot{x}^\alpha$ (10.9) for null geodesics, to deduce

$$C_\alpha k^\alpha = -a(t)\omega(t) = \text{const.} \quad , \tag{34.92}$$

leading imediately to the same conclusion (34.90).

Astronomers like to express this result in terms of the redshift parameter (see the discussion of Hubble's law above)

$$z = \frac{\lambda_0 - \lambda_1}{\lambda_1} \quad , \tag{34.93}$$

which in view of the above result we can write as

$$z = \frac{a(t_0)}{a(t_1)} - 1 \quad . \tag{34.94}$$

Thus if the universe expands one has $z > 0$ and there is a redshift while in a contracting universe with $a(t_0) < a(t_1)$ the light of distant glaxies would be blueshifted.

Remarks:

1. This cosmological redshift has nothing to do with the star's own gravitational field - that contribution to the redshift is completely negligible compared to the effect of the cosmological redshift.

2. Unlike the gravitational redshift we discussed before, this cosmological redshift is symmetric between receiver and emitter, i.e. light sent from the earth to the distant galaxy would likewise be redshifted if we observe a redshift of the distant galaxy.

3. However, like the gravitational redshift, the final result depends only on the position (time) of emission and arrival of the lightray, not on the intermediate gravitational field (cosmic scale factor). This illustrates that fundamentally the redshift is due to the different reference frames used by emitter and observer, not due to the fact that something happens to the lightray along the way. We will briefly return to this matter from a slightly different perspective in section 34.11.

4. While the previous remark suggests a purely Doppler-like explanation of the cosmological redshift it is best to think of the redshift as a combined effect of gravitational and Doppler redshifts. Without additional choices (like preferred families of intermediate observers) it is not very meaningful to separate this into the two and/or to interpret this only in terms of one of them.[144]

5. Nowadays, astronomers tend to express the distance of a galaxy not in terms of light-years or megaparsecs, but directly in terms of the observed redshift factor z, the conversion to distance then following from some version of Hubble's law. It is good to keep in mind that when cosmologists talk about small distances, i.e. small redshifts $z \approx 0.1$, this corresponds to a distance of approximately 1 billion light-years! The largest observed redshift of a galaxy is currently $z \approx 10$, corresponding to a distance of the order of 13 billion light-years.

6. The cosmic microwave background radiation (CMBR), which originated just a couple of 100.000 years after the Big Bang ($\approx$ 370.000 years), has $z \gtrsim 1000$. This was the time when atoms were formed, and the CMBR photons were decoupled and emitted. This happened at a temperature of $T_{dec} \approx 3000K$. Comparing with the fact the temperature of the CMBR today is $T_{cmbr} \approx 3K$ and using $T \propto a^{-1}$ (this is essentially a reformulation of our above result for the redshift, since - up to conversion factors $\hbar$ and k - frequency = energy = temperature), one then finds the above-quoted estimate for z.[145]

34.9 Redshift - Distance Relation (Hubble's Law)

We have seen that there is a cosmological redshift in Robertson-Walker geometries. Our aim will now be to see if and how these geometries are capable of explaining Hubble's law that the redshift is approximately proportional to the distance and how the Hubble constant is related to the cosmic scale factor $a(t)$.

For a long time, reliable data for cosmological redshifts as well as for distance measurements were only available for small values of z, and thus it was common to consider the case where $t_0 - t_1$ and r_1 are small, i.e. small on cosmological scales. This allows one to find a redshift-distance relation which can e.g. be written as a power-series in z. Such a formula is not quite good enough for modern purposes, however, and I will come back to this below.

[144] For a lucid discussion of this issue, actually in the end arguing in favour of the pure Doppler interpretation, see E. Bunn, D. Hogg, *The kinematic origin of the cosmological redshift*, `arXiv:0808.1081`.

[145] For an introduction to the rich physics of the CMBR see e.g. the review articles D. Samtleben, S. Staggs, B. Winstein, *The Cosmic Microwave Background for Pedestrians: A Review for Particle and Nuclear Physicists*, `arXiv:0803.0834 [astro-ph]`, or A. Jones, A. Lasenby, *The Cosmic Microwave Background*, Living Rev. Relativity 1, (1998), 11; `http://www.livingreviews.org/lrr-1998-11`.

Assuming the validity of such an expansion, this allows us in particular to expand $a(t)$ in a Taylor series,

$$a(t) = a(t_0) + (t-t_0)\dot{a}(t_0) + \tfrac{1}{2}(t-t_0)^2\ddot{a}(t_0) + \dots \tag{34.95}$$

Let us introduce the *Hubble parameter* $H(t)$ (which already made a brief appearance in (34.30) of section 34.3) and the *deceleration parameter* $q(t)$ by

$$\begin{aligned} H(t) &= \frac{\dot{a}(t)}{a(t)} \\ q(t) &= -\frac{a(t)\ddot{a}(t)}{\dot{a}(t)^2} \quad , \end{aligned} \tag{34.96}$$

and denote their present day values by a subscript zero, i.e. $H_0 = H(t_0)$ and $q_0 = q(t_0)$. $H(t)$ measures the expansion velocity as a function of time while $q(t)$ measures whether the expansion velocity is increasing or decreasing. We will also denote $a_0 = a(t_0)$ and $a(t_1) = a_1$. In terms of these parameters, the Taylor expansion can be written as

$$a(t) = a_0(1 + H_0(t-t_0) - \tfrac{1}{2}q_0H_0^2(t-t_0)^2 + \dots) \quad . \tag{34.97}$$

Higher order terms in this expansion are known as *jerk* (3rd derivative) and *snap* (4th derivative).[146]

We can use the expansion (34.97) to express z as a function of $(t_0 - t_1)$, and we can in principle use (34.79) to express r_1 as a function of $(t_0 - t_1)$. Combining the two results and eliminating $(t_0 - t_1)$, one therefore obtains the sought-for relation between the redshift z and the (coordinate-) distance r_1. The result is given in (34.104). The derivation is primarily an exercise in inverting series expansions and not *per se* particularly enlightning.

From (34.97) one finds that the redshift parameter z, as a power series in time, is

$$\frac{1}{1+z} = \frac{a_1}{a_0} = 1 + (t_1 - t_0)H_0 - \tfrac{1}{2}q_0H_0^2(t_1-t_0)^2 + \dots \tag{34.98}$$

or

$$z = (t_0 - t_1)H_0 + (1 + \tfrac{1}{2}q_0)H_0^2(t_0-t_1)^2 + \dots \tag{34.99}$$

For small $H_0(t_0 - t_1)$ this can be inverted,

$$t_0 - t_1 = \frac{1}{H_0}[z - (1 + \tfrac{1}{2}q_0)z^2 + \dots] \quad . \tag{34.100}$$

We can also use (34.79) to express $(t_0 - t_1)$ in terms of r_1. On the one hand we have

$$\int_{t_1}^{t_0} \frac{dt}{a(t)} = r_1 + \mathcal{O}(r_1^3) \quad , \tag{34.101}$$

[146] If you want to know why the 5th and 6th order derivative terms are known as *crackle* and *pop* respectively, look at M. Dunajski, G. Gibbons, *Cosmic Jerk, Snap and Beyond*, `arXiv:0807.0207 [gr-qc]`.

while expanding $a(t)$ in the denominator we get

$$\begin{aligned}\int_{t_1}^{t_0}\frac{dt}{a(t)} &= \frac{1}{a_0}\int_{t_1}^{t_0}\frac{dt}{(1+(t-t_0)H_0+\ldots)}\\ &= \frac{1}{a_0}\int_{t_1}^{t_0}dt\,[1+(t_0-t)H_0+\ldots]\\ &= \frac{1}{a_0}[(t_0-t_1)+t_0(t_0-t_1)H_0-\tfrac{1}{2}(t_0^2-t_1^2)H_0+\ldots]\\ &= \frac{1}{a_0}[(t_0-t_1)+\tfrac{1}{2}(t_0-t_1)^2H_0+\ldots]\ .\end{aligned}\tag{34.102}$$

Therefore we get

$$r_1=\frac{1}{a_0}[(t_0-t_1)+\tfrac{1}{2}(t_0-t_1)^2H_0+\ldots]\ .\tag{34.103}$$

Using (34.100), we finally obtain

$$a_0r_1=\frac{1}{H_0}[z-\tfrac{1}{2}(1+q_0)z^2+\ldots]\ .\tag{34.104}$$

This clearly indicates to first order a linear dependence of the redshift on the distance of the galaxy and identifies H_0, the present day value of the Hubble parameter, as playing the role of the Hubble constant introduced in (33.6).

REMARKS:

1. Note that the linear relation (34.29) between recessional velocity and distance of comoving objects is exact while (34.104) shows that the relation between redshift and distance is only approximately linear for small z.

2. Nowadays, cosmologists routinely deal with objects with redshifts $z>1$. For such objects, the relation (34.104), a power-series exansion in z, is evidently not appropriate. In section 38.1 we will derive a "non-perturbative" formula for $H=H(z)$ (the value of the Hubble parameter at the time an object emitted the light that we now observe with redshift z), namely (38.8)

$$H(z)=H_0(1+z)\left[1+(\Omega_M)_0z+(\Omega_\Lambda)_0\left(\frac{1}{(1+z)^2}-1\right)\right]^{1/2}\ .\tag{34.105}$$

   Here Ω_M and Ω_Λ are the so-called *density parameters* associated to matter and a cosmological constant, the subscript 0 denoting their value today (so that $H(z)$ is expressed in terms of quantities that are in principle directly or indirectly observable).

3. Returning to the case of small z, even in that case (34.104) is not yet a very useful way of expressing Hubble's law even in that case. First of all, the distance a_0r_1 that appears in this expression is not the proper distance (unless $k=0$), but is at least equal to it in our approximation. Note that a_0r_1 is the present distance to the galaxy, not the distance at the time the light was emitted.

4. Even proper distance is not directly measurable or observable and thus, to compare this formula with experiment, one needs to relate r_1 to the measures of distance used by astronomers.

One practical way of doing this is based on the so-called *luminosity distance* d_L. If for some reasons one knows the absolute luminosity of a distant star (for instance because it shows a certain characteristic behaviour known from other stars nearby whose distances can be measured by direct means - such objects are known as *standard candles*), then one can compare this absolute luminosity L with the apparent luminosity A. Then one can define the luminosity distance d_L by (cf. (33.2))

$$d_L^2 = \frac{L}{4\pi A} \quad . \tag{34.106}$$

We thus need to relate d_L to the coordinate distance r_1. The key relation is

$$\frac{A}{L} = \frac{1}{4\pi a_0^2 r_1^2} \frac{1}{1+z} \frac{a_1}{a_0} = \frac{1}{4\pi a_0^2 r_1^2 (1+z)^2} \quad . \tag{34.107}$$

Here the first factor arises from dividing by the area of the sphere at distance $a_0 r_1$ and would be the only term in a flat geometry (see the discssion of Olbers' paradox). In a Robertson-Walker geometry, however, the photon flux will be diluted. The second factor is due to the fact that each individual photon is being redshifted. And the third factor (identical to the second) is due to the fact that as a consequence of the expansion of the universe, photons emitted a time δt apart will be measured a time $(1+z)\delta t$ apart. Hence the relation between r_1 and d_L is

$$d_L = (L/4\pi A)^{1/2} = r_1 a(t_0)(1+z) \quad . \tag{34.108}$$

Intuitively, the fact that for z positive d_L is larger than the actual (proper) distance of the galaxy can be understood by noting that the redshift makes an object look darker (further away) than it actually is.

This can be inserted into (34.104) to give an expression for the redshift in terms of d_L, *Hubble's law*

$$d_L = H_0^{-1}[z + \tfrac{1}{2}(1-q_0)z^2 + \ldots] \quad . \tag{34.109}$$

The program would then be to collect as much astronomical information as possible on the relation between d_L and z in order to determine the parameters q_0 and H_0.

34.10 Klein-Gordon Scalar Field in a Cosmological Background

The study of fields propagating in a cosmological background space-time plays an important role in cosmology, in particular when these fields arise as perturbations of the metric itself (cosmological perturbations). Here we will be content with simply deriving

the equations of motion of a free massive Klein-Gordon scalar field Ψ in a spatially flat ($k = 0$) Robertson-Walker background

$$ds^2 = -dt^2 + a(t)^2 d\vec{x}^2 \quad , \tag{34.110}$$

described by the standard action (6.11)

$$S[\Psi] = \int \sqrt{g} d^4x \left[-\tfrac{1}{2} g^{\alpha\beta} \partial_\alpha \Psi \partial_\beta \Psi - \tfrac{1}{2} m^2 \Psi^2 \right] \quad , \tag{34.111}$$

and very briefly qualitatively discussing some of the properties of these equations. This a continuation and variation of the theme begun in section 6.3 (general formalism for scalar fields in a gravitational field), section 7.8 (scalar field in Rindler coordinates), and section 26.8 (scalar field in the Schwarzschild space-time).

In the original coordinates $(t, \vec{x})$, the action explicitly takes the form

$$S[\Psi] = \tfrac{1}{2} \int dt \; d^3x \; a(t)^3 \left(\dot{\Psi}^2 - (\vec{\nabla}\Psi)^2 / a(t)^2 - m^2 \Psi^2 \right) \quad , \tag{34.112}$$

leading to the equation of motion

$$\ddot{\Psi} + 3(\dot{a}/a)\Psi - (\Delta\Psi)/a^2 + m^2 \Psi = 0 \quad . \tag{34.113}$$

This exhibits a characteristic drag/friction term proportional to the Hubble parameter $H(t)$.

However, in order to make the setting as close as possible to that of a scalar field in Minkowski space (which is useful e.g. if one is intent on quantising the scalar field afterwards), it is useful to employ the conformal time coordinate η, already introduced in (34.48) and defined by

$$\begin{aligned} ds^2 &= -dt^2 + a(t)^2 d\vec{x}^2 = a(t)^2 (-dt^2/a(t)^2 + d\vec{x}^2) \\ &\equiv A(\eta)^2 (-d\eta^2 + d\vec{x}^2) \quad , \end{aligned} \tag{34.114}$$

with $A(\eta(t)) = a(t)$ and $d/d\eta = a(d/dt)$. Denoting an η-derivative by a prime, $d\Psi/d\eta = \Psi'$, the action can then more explicitly be written as

$$S[\Psi] = \tfrac{1}{2} \int d^3x \; d\eta \; A(\eta)^2 \left[(\Psi')^2 - (\vec{\nabla}\Psi)^2 - m^2 A(\eta)^2 \Psi^2 \right] \quad . \tag{34.115}$$

We see that the field Ψ has a non-canonical kinetic term $\sim A(\eta)^2 (\Psi')^2$, leading to Euler-Lagrange equations of motion containing "friction" terms,

$$\tfrac{d}{d\eta}(A^2 \Psi') = A^2 (\Psi'' + 2(A'/A)\Psi') \quad , \tag{34.116}$$

which are awkward (in the classical theory, but even more so for quantisation). Happily, these non-canonical terms can be eliminated by the field redefinition

$$\phi(\eta, \vec{x}) = A(\eta) \Psi(\eta, \vec{x}) \quad . \tag{34.117}$$

Indeed, up to a total derivative term one then finds

$$S[\Psi] \to S[\phi] = \tfrac{1}{2}\int d^3x\, d\eta\, \left[(\phi')^2 - (\vec{\nabla}\phi)^2 - (m^2A^2 - A''/A)\phi^2\right] \ . \tag{34.118}$$

Remarks:

1. Observe that $S[\phi]$ is the standard action for a Klein-Gordon field in Minkowski space, its only mildly exotic feature being the time-dependent mass term with an effective mass

$$m^2_{\text{eff}}(\eta) = m^2A(\eta)^2 - A''(\eta)/A(\eta) \ . \tag{34.119}$$

Thus the interaction of the scalar field ϕ with the gravitational background is entirely encoded in this time-dependent mass term. Note that this effective mass term is even present when the original field is massless, $m^2 = 0$.

2. This purely geometric contribution to the mass term can be interpreted as an induced non-minimal coupling to the scalar curvature R of the space-time. Indeed, we have

$$A''/A = (1/a)(a\frac{d}{dt})(a\frac{d}{dt})a = \frac{d}{dt}(a\dot{a}) = a\ddot{a} + \dot{a}^2 \ . \tag{34.120}$$

Comparison with the result (35.7) for the Ricci scalar of the Robertson-Walker metric for $k = 0$ shows that

$$A''/A = Ra^2/6 = RA^2/6 \ , \tag{34.121}$$

so that the effective mass can also be written as

$$m^2_{\text{eff}}(\eta) = A(\eta)^2(m^2 - R(\eta)/6) \ . \tag{34.122}$$

3. This non-minimal coupling to the scalar curvature, and the factor 1/6, are (or should be) reminiscent of the *conformal coupling* $\xi R\phi^2$ of a scalar field discussed in section 22.3. If instead of with the action (34.111) we start off with the non-minimally coupled action (22.102)

$$S_\xi[\Psi] = -\tfrac{1}{2}\int \sqrt{g}d^4x\, \left(g^{\alpha\beta}\partial_\alpha\Psi\partial_\beta\Psi + \xi R\Psi^2\right) \ , \tag{34.123}$$

then instead of (34.118) we will find the action

$$S_\xi[\phi] = \tfrac{1}{2}\int d^3x\, d\eta\, \left[(\phi')^2 - (\vec{\nabla}\phi)^2 + A(\eta)^2(\xi - 1/6)R(\eta)\phi^2\right] \ . \tag{34.124}$$

In particular, when ξ takes the value (22.103)

$$\xi = \frac{D-2}{4(D-1)} = +1/6 \ , \tag{34.125}$$

for a conformal coupling, the action

$$S_{\xi=1/6}[\phi] = \tfrac{1}{2}\int d^3x\, d\eta\, \left[(\phi')^2 - (\vec{\nabla}\phi)^2\right] \tag{34.126}$$

is simply the action of a free massless scalar field in Minkowski space. This is as it should be: the action with $\xi = 1/6$ (and $m^2 = 0$) is conformally invariant, and the Robertson-Walker metric is conformally flat (this is manifest in (34.114)). Thus the action with this value of ξ must reduce to the free action in Minkowski space, the rescaling (34.117) of the scalar field reflecting the non-trivial conformal weight of a scalar field in $D = 4$.

When one introduces the non-mimimal ξ-coupling in the action together with a non-zero explicit mass term, then everything goes through as above, the only difference being that the effective mass is now ξ-dependent,

$$m^2_{\text{eff}}(\eta) = A(\eta)^2\left(m^2 + (\xi - 1/6)R(\eta)\right) \quad . \tag{34.127}$$

4. The equations of motion are

$$\phi''(\eta,\vec{x}) - \Delta\phi(\eta,\vec{x}) + m^2_{\text{eff}}(\eta)\phi(\eta,\vec{x}) = 0 \quad . \tag{34.128}$$

Spatial flatness $k = 0$ brings with it the simplifying feature that we can expand the spatial dependence of the fields in standard Fourier modes. Upon spatial Fourier expansion,

$$\phi(\eta,\vec{x}) \sim \int d^3k\, \phi_{\vec{k}}(\eta)\, \mathrm{e}^{i\vec{k}.\vec{x}} \quad , \tag{34.129}$$

one sees that each mode $\phi_{\vec{k}}$ satisfies the equation

$$\phi''_{\vec{k}}(\eta) + \omega^2_k(\eta)\phi_{\vec{k}}(\eta) = 0 \tag{34.130}$$

of a time-dependent harmonic oscillator, with time-dependent frequency

$$\omega^2_k(\eta) = m^2_{\text{eff}}(\eta) + k^2 \quad . \tag{34.131}$$

5. The crucial feature of this action and the mode equations are their explicit time-dependence which means that the energy of ϕ is not conserved. This in turn will lead to the important phenomena of particle or mode production in a cosmological background.

6. For example, one can consider the (evidently highly idealised) situation where the cosmic scale factor is asymptotically constant in the remote past and in the remote future. During these early and late periods the metric is essentially the Minkowski metric (possibly up to a rescaling of the coordinates), and one thus has a preferred notion of particles during those eras, uniquely determined by the asymptotic Poincaré symmetry. However, these definitions of particles need

not (and will amost invariably not) agree when there is an intermediate time-dependent phase. For instance, the early time vacuum Heisenberg state would not be interpreted or seen as a vacuum by the late time observer, and this disagreement about the particle content is then interpreted as a particle production due to the time-dependent gravitational field. See the references in footnote 87 (section 27.7) for a detailed discussion of these and other related fascinating issues.

34.11 Comments on Cosmic Expansion as "Expansion of Space"

As an aside, and as a conclusion to this section, let me make some comments on (and issue a *caveat* regarding) the seductive picture of an inflating balloon as the model for an expanding universe, as depicted e.g. in Figure 44 of section 34.2.

1. For many purposes, this picture certainly provides an instructive and illuminating analogy:

   - it illustrates how an expanding universe can look the same to all comoving observers;
   - in particular, it illustrates how it is possible for everything to move away from any given (comoving) observer without that observer actually being singled out as special;
   - it shows that a spatially homogeneous expansion naturally gives rise to a universal velocity-distance relation;
   - it illustrates (or is at least meant to illustrate) that expansion of the universe is something intrinsic and does not mean expansion in and into some space into which the universe has somehow been embedded.

2. The above picture of the inflating balloon is often used to describe the expansion of the universe as "an expansion of space itself", and it is then, in view of the success of this picture, tempting to *ascribe* the behaviour of light and test particles in the universe (which are mathematically described by the geodesic equations) to such an expansion of space.

   This manner of speaking and this imagery may provide some further useful intuitive understanding for some effects in the general relativistic description of cosmology. However, unless one defines precisely what one means by this, the notion of expanding space is not without its pitfalls and as with all analogies here one runs the risk of pushing this analogy too far. In particular, the danger hides in the above word "ascribe", i.e. in the risk of confusing cause and effect, or cause and effective description. Fundamentally, there is no (new?) force that (somehow) acts on space to (somehow) make it expand, and that can therefore be invoked

to "explain" (in a Newtonian way) the behaviour of particles and light in such a space-time.[147]

3. The danger of thinking of the cosmological expansion in terms of the expansion of space and a corresponding agent responsible for this is well-illustrated by the cosmological redshift discussed in section 34.8, and derived there on the basis of the null geodesic equations.

   If (somehow) the expansion of space were fundamentally responsible for this effect, one would expect this effect to be cumulative and to depend on the gravitational field (or cosmic scale factor) $a(t)$ during the entire period of propagation. However, as we have seen, and as stressed at the end of section 34.8, the final result (34.94) for the redshift, $z = a(t_0)/a(t_1) - 1$, depends only on the values of the cosmological scale factor at the times t_0 and t_1, indicating that the redshift is *not* a cumulative effect due to the expansion of space while it was traversed by the lightray, but that it can (equally intuitively and perhaps more correctly) be ascribed to the fact that emitter and observer do not share the same inertial frame.

4. The issue also, and in particular, arises when it comes to frequently asked questions such as "which objects participate in the cosmic expansion?" (do you expand with the universe? does a hydrogen atom? does our solar system?) or, to use cosmologists' jargon, "which objects join the Hubble flow?", which have generated a lot of confusion over the decades. Here again the "expanding space" image may lead to a misleading intuition, in particular when space is then viewed as some kind of viscous fluid which will invariably drag other objects along with it when it expands.[148]

5. These kinds of questions have a long history in general relativity, dating back at least to an article by Einstein and Straus in 1945 entitled *The Influence of the Expansion of Space on the Gravitational Fields Surrounding the Individual Stars*. The Einstein-Straus solution, known as the *Einstein-Straus vacuole* is a space-time that is obtained by a cut-and-paste procedure from a suitable cosmological solution, removing a ball of mass M and replacing it by a Schwarzschild solution of the same mass. This is essentially an inside-out version of the Oppenheimer-Snyder collapse solution (removing a ball from Schwarzschild and replacing it

[147] I strongly recommend the lucid and engaging discussion of these issues in M. Francis, L. Barnes, J. Berian James, G. Lewis, *Expanding Space: the Root of all Evil?*, `arXiv:0707.0380 [astro-ph]`. See also A. Whiting, *The Expansion of Space: Free Particle Motion and the Cosmological Redshift*, `arXiv:astro-ph/0404095`, and J. Peacock, *A diatribe on expanding space*, `arXiv:0809.4573 [astro-ph]` for related discussions.

[148] For a clear discussion of these issues for geodesic but non-comoving particles, highlighting the necessity to define precisely and carefully what one means by "joining the Hubble flow", see L. Barnes, M. Francis, J. Berian James, G. Lewis, *Joining the Hubble Flow: Implications for Expanding Space*, `arXiv:astro-ph/0609271`.

by a contracting cosmological solution modelling the collapsing star) discussed in section 29. The Einstein-Straus procedure can also be applied multiple times around various "centers" and can then be used to model inhomogeneities in an otherwise homogeneous universe (and in this context the model is then known to cosmologists as the *Swiss cheese model*).

6. Since the work of Einstein and Straus, a lot of work has gone into finding exact solutions of the Einstein equations that describe gravitational objects like stars or black holes somehow "embedded" into cosmological backgrounds. On the basis of such exact solutions one can then (try to) answer the question if a given bound object takes part in the cosmic expansion or not, and try to to develop some intuition for this issue that complements intuition coming from more Newtonian considerations.

   There is a common folklore statement or rule of thumb to the effect that "gravitationally (or otherwise) bound systems do not expand with the universe", and while this statement undoubtedly has a certain validity it requires a more precise formulation to decide if or when such a statement is not only true but also has some non-trivial content.

7. The most prominent class of solutions among these hybrid star-cosmology solutions (apart from black holes in (anti-)de Sitter space, the Schwarzschild (anti-)de Sitter metrics (30.7)) is the so-called McVittie solution, found already in 1933. It consists of a crude superposition of a (in the simplest case $k = 0$) Robertson-Walker metric,

$$ds^2 = -dt^2 + a(t)^2 d\vec{x}^2 \quad , \tag{34.132}$$

   with the Schwarzschild metric in isotropic coordinates (24.46)

$$ds^2 = -\frac{(1 - \frac{m}{2|\vec{x}|})^2}{(1 + \frac{m}{2|\vec{x}|})^2} dt^2 + (1 + \frac{m}{2|\vec{x}|})^4 \, d\vec{x}^2 \quad , \tag{34.133}$$

   and thus has the form

$$ds^2 = -\frac{(1 - \frac{m}{2a(t)|\vec{x}|})^2}{(1 + \frac{m}{2a(t)|\vec{x}|})^2} dt^2 + (1 + \frac{m}{2a(t)|\vec{x}|})^4 \, a(t)^2 \, d\vec{x}^2 \quad . \tag{34.134}$$

   While this metric is easy to write down, it leads to a somewhat peculiar energy-momentum tensor, and therefore its physical interpretation and significance are somewhat obscure. These issues, as well as aspects of the global structure of the McVittie space-time, continue to be debated in the literature to this day.[149]

[149] For a pedagogical review of these topics and a discussion of the literature see e.g. M. Carrera, D. Giulini, *On the influence of the global cosmological expansion on the local dynamics in the Solar System*, `arXiv:gr-qc/0602098`, *Influence of global cosmological expansion on local dynamics and kinematics*, `arXiv:0810.2712 [gr-qc]`. For more on the on-going McVittie debate, see also N. Kaloper, M. Kleban, D. Martin, *McVittie's Legacy: Black Holes in an Expanding Universe* and references therein and thereto.

35 Cosmology III: Friedmann-Lemaître-Robertson-Walker Cosmology

So far, we have only used the kinematical framework provided by the Robertson-Walker metrics and we never used the Einstein equations. The benefit of this is that it allows one to deduce relations betweens observed quantities and assumptions about the universe which are valid even if the Einstein equations are not entirely correct, perhaps because of higher derivative or other quantum corrections in the early universe.

Now, on the other hand we will have to be more specific, specify the matter content and solve the Einstein equations for $a(t)$. We will see that a lot about the solutions of the Einstein equations can already be deduced from a purely qualitative analysis of these equations, without having to resort to explicit solutions (section 36). Exact solutions will then be the subject of section 37.

35.1 Curvature and Einstein Tensor of the Robertson-Walker Metric

Of course, the first thing we need to discuss solutions of the Einstein equations is the Ricci tensor of the Robertson-Walker metric. Since we already know the curvature tensor of the maximally symmetric spatial metric entering the Robertson-Walker metric (and its contractions), this is not difficult.

1. First of all, we write the Robertson-Walker metric as

$$ds^2 = -dt^2 + a(t)^2\, d\tilde{s}^2 = -dt^2 + a(t)^2\, \tilde{g}_{ij} dx^i dx^j \quad . \tag{35.1}$$

   In this section all objects with a tilde, $\tilde{}$, will refer to 3-dimensional quantities calculated with respect to the maximally symmetric metric $\tilde{g}_{ij}$.

2. One can then calculate the Christoffel symbols in terms of $a(t)$ and $\tilde{\Gamma}^i{}_{jk}$. The non-vanishing components are (we had already established that $\Gamma^\mu_{00} = 0$)

$$\Gamma^i{}_{jk} = \tilde{\Gamma}^i{}_{jk} \quad , \quad \Gamma^i{}_{j0} = \frac{\dot{a}}{a}\delta^i_j \quad , \quad \Gamma^0{}_{ij} = \dot{a} a \tilde{g}_{ij} \quad . \tag{35.2}$$

3. The non-zero components of the Riemann tensor are

$$\begin{aligned} R^i{}_{0j0} &= -\frac{\ddot{a}}{a}\delta^i_j \quad \Leftrightarrow \quad R^0{}_{i0j} = a\ddot{a}\tilde{g}_{ij} \\ R^k{}_{ilj} &= \tilde{R}^k{}_{ilj} + \dot{a}^2(\delta^k_l \tilde{g}_{ij} - \delta^k_j \tilde{g}_{il}) \quad . \end{aligned} \tag{35.3}$$

   where, by maximal symmetry, $\tilde{R}_{kilj}$ has the form (14.10)

$$\tilde{R}^k{}_{ilj} = k(\delta^k_l \tilde{g}_{ij} - \delta^k_j \tilde{g}_{il}) \quad . \tag{35.4}$$

4. The partial contraction of the purely spatial components of the Riemann tensor over the spatial indices is thus

$$\begin{aligned} R^k_{ikj} &= \tilde{R}_{ij} + 2\dot{a}^2\tilde{g}_{ij} \\ &= 2k\tilde{g}_{ij} + 2\dot{a}^2\tilde{g}_{ij} \ . \end{aligned} \tag{35.5}$$

5. Therefore the non-zero components of the space-time Ricci tensor are

$$\begin{aligned} R_{00} &= -3\frac{\ddot{a}}{a} \\ R_{ij} &= (a\ddot{a} + 2\dot{a}^2 + 2k)\tilde{g}_{ij} \\ &= (\frac{\ddot{a}}{a} + 2\frac{\dot{a}^2}{a^2} + \frac{2k}{a^2})g_{ij} \ . \end{aligned} \tag{35.6}$$

6. Thus the Ricci scalar is

$$R = \frac{6}{a^2}(a\ddot{a} + \dot{a}^2 + k) \ , \tag{35.7}$$

7. Finally, therefore, putting everything together, we find that the Einstein tensor has the components

$$\begin{aligned} G_{00} &= 3(\frac{\dot{a}^2}{a^2} + \frac{k}{a^2}) \\ G_{0i} &= 0 \\ G_{ij} &= -(\frac{2\ddot{a}}{a} + \frac{\dot{a}^2}{a^2} + \frac{k}{a^2})g_{ij} \ . \end{aligned} \tag{35.8}$$

REMARKS:

1. It follows on symmetry grounds (spatial maximal symmetry) alone

   - that the only potentially non-vanishing components of the Einstein tensor are G_{00} and G_{ij};
   - that $G_{ij} \sim g_{ij}$
   - and that the coefficients are only functions of t, not functions of the spatial coordinates.

   A formal proof of this is given in section 35.3. It is phrased there as a statement about the energy-momentum tensor in a Robertson-Walker metric, but the result is a general statement about the structure of spatially maximally symmetric space-time tensors. Thus in a sense the only non-trivial content of the above calculation is in the precise form of the t-dependent coefficients of G_{00} and G_{ij}.

2. We already know that in a maximally symmetric space not only can we express the Ricci tensor in terms of the Riemann tensor (namely as a contraction thereof) but we can also write the Riemann tensor algebraically in terms of the Ricci tensor (and even just in terms of the Ricci scalar), as is obvious from (35.4).

Even though the Robertson-Walker metrics are not space-time maximally symmetric, it is nevertheless true that even in this case the Riemann tensor can be expressed algebraically in terms of the Ricci tensor. Indeed, it is easy to see that the components of the Riemann tensor given in (35.3) can be written in terms of the components of the Ricci tensor in (35.6) simply as

$$\begin{aligned} R^i{}_{0j0} &= \tfrac{1}{3} R_{00} \delta^i{}_j \\ R^k{}_{ilj} &= \tfrac{1}{2}(\delta^k{}_l R_{ij} - \delta^k{}_j R_{il}) \quad . \end{aligned} \tag{35.9}$$

On general grounds this follows from the fact, established in section 34.6, that the Robertson-Walker metrics are conformally flat so that the Weyl tensor vanishes.

3. The significance of this statement lies in the fact that it shows that a vacuum solution of the Einstein equations with spatial maximal symmetry is necessarily flat Minkowski space. This is perhaps as it should be, and at least vaguely Machian, but it is still good to have established this here once and for all since by just solving the vacuum equations one may (and will) find a solution that at first sight appears to be non-trivial, namely the *Milne universe* to be discussed in section 37.1, but which can then be shown to be just Minkowski space written in some non-inertial coordinates. The above result (35.9) shows that this had to be true.

4. Occasionally, in particular for the canonical analysis (i.e. developing the Hamiltonian formalism), it is useful to know the Ricci scalar (i.e. the Einstein-Hilbert Lagrangian) for the slightly more general metric

$$ds^2 = -N^2(t')(dt')^2 + a^2(t') d\tilde{s}^2 \quad , \tag{35.10}$$

where the function $N(t')$ is known as the *lapse function* (cf. (21.18)). Instead of redoing the calculation of the scalar curvature in this case, one can simply use the change of variable

$$dt = N(t')dt' \quad \Rightarrow \quad \frac{d}{dt} = \frac{1}{N}\frac{d}{dt'} \tag{35.11}$$

to rewrite the final result (35.7) as (a prime on a or N denoting a derivative with respect to t')

$$R = \frac{6}{a^2}(a\ddot{a} + \dot{a}^2 + k) = \frac{6}{a^2 N^3}(N(aa'' + (a')^2) - aa'N' + kN^3) \quad . \tag{35.12}$$

We will come back to and make use of this result in section 35.8.

5. The results for the Christoffel symbols (35.2) and the Riemann tensor (35.3) are true in any dimension, i.e. for a general n-dimensional maximally symmetric space, and the first time that a dimension-dependence enters is in the factors of 2 and 3 in equations (35.5) and (35.6), which arise from taking traces. If one replaces the

spatial dimension $3 \to n$, equations (35.6) - (35.8) take the form

$$\begin{aligned} R_{00} &= -n\frac{\ddot{a}}{a} \\ R_{ij} &= \left(\frac{\ddot{a}}{a} + (n-1)\frac{k+\dot{a}^2}{a^2}\right) g_{ij} \\ R &= 2n\frac{\ddot{a}}{a} + n(n-1)\frac{k+\dot{a}^2}{a^2} \\ G_{00} &= \frac{n(n-1)}{2}\frac{k+\dot{a}^2}{a^2} \\ G_{ij} &= -(n-1)\left(\frac{\ddot{a}}{a} + \frac{(n-2)}{2}\frac{k+\dot{a}^2}{a^2}\right) g_{ij} \quad . \end{aligned} \tag{35.13}$$

35.2 Matter Content: A Perfect Fluid

Next we need to specify the matter content. On physical grounds one might perhaps like to argue that in the approximation underlying the cosmological principle galaxies (or clusters) should be treated as non-interacting particles or a perfect fluid (first discussed in section 7.2). As it turns out, we do not need to do this as either the symmetries of the metric or comparison with the Einstein tensor determined above fix the energy-momentum tensor to be that of a perfect fluid anyway.

In section 35.3 I will give a formal argument for this using Killing vectors. Informally we can already deduce this from the structure of the Einstein tensor obtained above. Comparing (35.8) with the Einstein equation $G_{\alpha\beta} = 8\pi G_N T_{\alpha\beta}$, we deduce that the Einstein equations can only have a solution with a Robertson-Walker metric if the energy-momentum tensor is of the form

$$\begin{aligned} T_{00} &= \rho(t) \\ T_{0i} &= 0 \\ T_{ij} &= p(t) g_{ij} \quad , \end{aligned} \tag{35.14}$$

where $p(t)$ and $\rho(t)$ are some functions of time. A covariant way of writing this tensor is as

$$T_{\alpha\beta} = (p+\rho) u_\alpha u_\beta + p g_{\alpha\beta} \quad , \tag{35.15}$$

where $u^\alpha = (1,0,0,0)$ in a comoving coordinate system. This is precisely the energy-momentum tensor of a *perfect fluid* (cf. sections 7.2 and 7.5). In this context u^α is known as the *velocity field* of the fluid, and the comoving coordinates are those with respect to which the fluid is at rest. ρ is the energy-density of the perfect fluid and p is the pressure.

In general, this matter content has to be supplemented by an equation of state. This is usually assumed to be that of a *barytropic fluid*, i.e. one whose pressure depends only

on its density, $p = p(\rho)$. The most useful toy-models of cosmological fluids arise from considering a linear relationship between p and ρ, of the type

$$p = w\rho \ , \tag{35.16}$$

where w is known as the equation of state parameter. Occasionally also more exotic equations of state are considered, but the above covers a wide variety of commonly considered fluids and gases and other simple thermodynamic systems.

Consider e.g. a system whose entropy S is some function of the (internal) energy E and the (spatial) volume V,

$$S = S(E, V) \ . \tag{35.17}$$

Then the 1st law of thermodynamics

$$TdS = dE + pdV \tag{35.18}$$

implies

$$T = \left(\frac{\partial S}{\partial E}\right)^{-1}_{V} \quad , \quad p = T\left(\frac{\partial S}{\partial V}\right)_{E} \quad , \tag{35.19}$$

and thus

$$p = w\rho = wE/V \quad \Leftrightarrow \quad V\partial_V S = wE\partial_E S \ . \tag{35.20}$$

Thus the condition $p = w\rho$ is simply the statement that S is a function of $V^w E$,

$$p = w\rho \quad \Rightarrow \quad S = S(V^w E) \ . \tag{35.21}$$

Here are the most common and useful special cases of the equation of state $p = w\rho$.

1. Dust

   For non-interacting particles, there is no pressure, $p = 0$, i.e. $w = 0$, the energy-momentum tensor has the simple form

   $$T_{\alpha\beta} = \rho u_\alpha u_\beta \tag{35.22}$$

   and such matter is usually referred to as *dust*,

   $$\text{dust:} \qquad p = 0 \quad \Rightarrow \quad w = 0 \ . \tag{35.23}$$

   This is generally considered to be a good description of baryonic matter (and cold dark matter) today.

2. Radiation

   This corresponds to $w = 1/3$ (in 1+3 dimensions). One way to see this is to note that the trace of a perfect fluid energy-momentum tensor is

   $$T^\alpha_{\ \alpha} = -\rho + 3p \ . \tag{35.24}$$

For electro-magnetic radiation, for example, the energy-momentum tensor is that of Maxwell theory and hence traceless (7.121). Therefore electromagnetic *radiation* in an FLRW universe (in particular compatibility with the symmetries implies neglecting all anisotropies) has the equation of state

$$\text{radiation:} \qquad p = \rho/3 \quad \Rightarrow \quad w = 1/3 \ . \tag{35.25}$$

Alternatively, this can be deduced from familiar statements about the thermodynamics of electromagnetic radiation (i.e. the photon gas). E.g. $S \sim E/T$ and $\rho \sim T^4$ imply

$$S \sim E/T \sim E(E/V)^{-1/4} = E^{3/4}V^{1/4} = (V^{1/3}E)^{3/4} \ , \tag{35.26}$$

which (by (35.21)) also implies $w = 1/3$.

As an aside, note that one generalisation of this equation of state for radiation in $d = 3$ spatial dimensions to general spatial dimension d is a perfect fluid with a traceless energy-momentum tensor (describing what one might call a "Weyl invariant" or "conformal" fluid - cf. the discussion in section 7.7). Thus the energy-momentum tensor has to satisfy

$$g^{\alpha\beta}T_{\alpha\beta} = T^\alpha_\alpha = -(\rho + p) + (d+1)p = 0 \ , \tag{35.27}$$

leading to the equation of state

$$\text{conformal fluid:} \qquad p = \rho/d \quad \Rightarrow \quad w = 1/d \ . \tag{35.28}$$

and

$$T_{\alpha\beta} = \frac{\rho}{d}(g_{\alpha\beta} + (d+1)u_\alpha u_\beta) \ . \tag{35.29}$$

3. Cosmological Constant

A cosmological constant Λ, on the other hand, corresponds, as we will see, to a matter contribution with $p = -\rho$, i.e. $w = -1$,

$$\text{cosmological constant:} \qquad p = -\rho \quad \Rightarrow \quad w = -1 \ . \tag{35.30}$$

Thus either ρ is negative or p is negative.

To see the relation between a $w = -1$ perfect fluid and a cosmological constant, note that the Einstein equations with a cosmological constant (19.46) give a contribution to the energy-momentum tensor proportional to $g_{\mu\nu}$. Comparing this with the covariant form (35.15) of the energy-momentum tensor, one deduces that a cosmological constant Λ is tantamount to adding matter with $p = -\rho$. Specifically, one has

$$\begin{aligned} G_{\mu\nu} + \Lambda g_{\mu\nu} = 8\pi G_N T_{\mu\nu} \quad &\Leftrightarrow \quad G_{\mu\nu} = 8\pi G_N \left(T_{\mu\nu} - (\Lambda/8\pi G_N) g_{\mu\nu}\right) \\ &\Rightarrow \quad T^\Lambda_{\mu\nu} = -\frac{\Lambda}{8\pi G_N} g_{\mu\nu} \\ &\Rightarrow \quad T^\Lambda_{00} \equiv \rho_\Lambda = \frac{\Lambda}{8\pi G_N} = -p_\Lambda \ . \end{aligned} \tag{35.31}$$

This is the identification anticipated in (19.50).

REMARKS:

1. In section 22.1 we had introduced various *energy conditions*, the *null energy condition* (NEC), the *weak energy condition* (WEC), the *dominant energy condition* (DEC), and the *strong energy condition* (SEC), and had also analysed their implications for a perfect fluid energy-momentum tensor. In particular, the conditions that we found were

$$\begin{aligned} \text{NEC:} \quad & \rho + p \geq 0 \\ \text{WEC:} \quad & \rho \geq 0 \quad , \quad \rho + p \geq 0 \\ \text{DEC:} \quad & \rho \geq |p| \\ \text{SEC:} \quad & \rho + p \geq 0 \quad , \quad \rho + 3p \geq 0 \ . \end{aligned} \tag{35.32}$$

With the equation of state $p = w\rho$ these energy conditions can now be written as conditions on w. For physical (gravitating instead of anti-gravitating) matter one usually requires at least the condition $\rho > 0$ (positive energy density). With this condition, the NEC and the WEC are equivalent and require

$$\rho > 0 \quad \text{and NEC/WEC} \quad \Rightarrow \quad w \geq -1 \ , \tag{35.33}$$

while the SEC requires (the 2nd condition is then stronger than the 1st)

$$\rho > 0 \quad \text{and SEC} \quad \Rightarrow \quad w \geq -1/3 \ . \tag{35.34}$$

Finally, the DEC implies $\rho > 0$ as well as

$$\text{DEC} \quad \Rightarrow \quad |w| \leq 1 \ . \tag{35.35}$$

Some of the conclusions about the qualitative behaviour of the solutions to the Einstein equations in section 36 rely on the strict validity of the SEC, i.e. on the assumption that (at least in the era of interest) the matter content of the universe is dominated by stuff with $w > -1/3$.

On the other hand, as we had seen, a cosmological constant has $w = -1$, and thus either ρ is negative or p is negative. Therefore this violates either $\rho > 0$ (the WEC and the DEC) or $\rho + 3p > 0$ (the SEC).

2. The equation of state parameter need not necessarily be a constant. Consider for instance a scalar field ϕ. Such a field will respect the symmetries of a Robertson-Walker metric (and hence can potentially give rise to a solution of the Einstein equations of the Robertson-Walker form) if it depends only on t and not on the spatial coordinates, $\phi = \phi(t)$. For such a scalar field, the energy-momentum tensor (7.75),

$$T_{\alpha\beta} = \partial_\alpha\phi\partial_\beta\phi - \tfrac{1}{2} g_{\alpha\beta}\, g^{\mu\nu}\partial_\mu\phi\partial_\nu\phi - g_{\alpha\beta}V(\phi) \ , \tag{35.36}$$

reduces to

$$T_{00} = \tfrac{1}{2}\dot{\phi}^2 + V(\phi) \quad , \quad T_{ij} = \left(\tfrac{1}{2}\dot{\phi}^2 - V(\phi)\right) \, g_{ij} \quad , \tag{35.37}$$

which is thus indeed of the general perfect fluid form (35.14). The energy and pressure density are

$$\rho(t) = \tfrac{1}{2}\dot{\phi}^2 + V(\phi) \quad , \quad p(t) = \tfrac{1}{2}\dot{\phi}^2 - V(\phi) \quad , \tag{35.38}$$

leading to the time-dependent equation of state parameter

$$w(t) = \frac{p(t)}{\rho(t)} = \frac{\frac{1}{2}\dot{\phi}^2 - V(\phi)}{\frac{1}{2}\dot{\phi}^2 + V(\phi)} \quad . \tag{35.39}$$

Note that here we are treating the scalar field as a source term for the Einstein equations. This should be contrasted with (and should not be confused with) what what we did in section 6.3, where we developed the general formalism for scalar fields in a fixed gravitational field, neglecting the backreaction of the matter fields on the gravitational field / metric.

Note also that (35.39) mimics a cosmological constant, i.e. $w = -1$, during periods (of the scalar field slowly rolling down a very flat potential) where the kinetic energy term is negligible compared with the potential energy term. This can also be seen directly from the action (6.14), a constant potential leading to a constant contribution to the matter or gravity Lagrangian, a.k.a. as a cosmological constant, and plays an important role in models of *inflation*.

3. Occasionally, more exotic equations of state are also considered in cosmology. For example, extended objects like (cosmic) strings, or membranes (domain walls) or their higher-dimensional generalisations typically have a positive tension, and therefore a negative (contribution to the) pressure.[150] In particular, the equation of state of a gas of non-relativistic strings in d spatial dimensions is

$$\text{non-relativistic string gas:} \qquad p = -\rho/d \quad \Rightarrow \quad w = -1/d \quad . \tag{35.40}$$

4. Another exotic object that appears occasionally in the context of cosmology is the so-called *Chaplygin gas*, with equation of state

$$\text{Chaplygin gas:} \qquad p = -A/\rho \qquad (A > 0) \quad . \tag{35.41}$$

We will briefly return to the properties of the Chaplygin gas at the end of section 35.6 - cf. (35.99) and (35.100), but for the most part we will concentrate on the linear equations of state $p = w\rho$ in the following.

[150] The equation of state for such extended objects, affectionately and collectively known as "p-branes" or just "branes", is derived e.g. in T. Boehm, R. Brandenberger, *On T-Duality in Brane Gas Cosmology*, `arXiv:hep-th/0208188`.

5. A null variant of the dust energy-momentum tensor (35.22), with u^α replaced by a null vector k^α (*null dust*) will appear as the source term for a toy model of a radiating star, the *Vaidya metric*,

$$T_{\alpha\beta} = \rho k_\alpha k_\beta \tag{35.42}$$

to be discussed in sections 40 - 42 (cf. in particular section 40.4). Since this energy-momentum tensor is manifestly traceless,

$$T^\alpha_{\ \alpha} = \rho k^\alpha k_\alpha = 0 \ , \tag{35.43}$$

it can alternatively be regarded as a variant of the energy-momentum tensor for radiation (also with $\rho = 3p$).

35.3 Appendix: Space-Time Tensors with Maximal Spatial Symmetry

Here is the formal argument that the energy-momentum tensor necessarily has the form given in (35.14).[151]

It is of course a consequence of the Einstein equations that any symmetries of the Ricci (or Einstein) tensor also have to be symmetries of the energy-momentum tensor. Now we know that the metric $\tilde{g}_{ij}$ has six Killing vectors $K^{(a)}$ and that (in the comoving coordinate system) these are also Killing vectors of the Robertson-Walker metric,

$$L_{K^{(a)}} \tilde{g}_{ij} = 0 \quad \Rightarrow \quad L_{K^{(a)}} g_{\mu\nu} = 0 \ . \tag{35.44}$$

Therefore also the Ricci and Einstein tensors have these symmetries,

$$L_{K^{(a)}} g_{\mu\nu} = 0 \quad \Rightarrow \quad L_{K^{(a)}} R_{\mu\nu} = 0 \ , \ \ L_{K^{(a)}} G_{\mu\nu} = 0 \ . \tag{35.45}$$

To prove this one can either (non-covariantly) choose, for each Killing vector, an adapted coordinate system, or one generalises the argument given in section 13.4 for the Ricci scalar, $L_K R = 0$, to the Ricci tensor.

The Einstein equations then imply that $T_{\mu\nu}$ should have these symmetries,

$$L_{K^{(a)}} G_{\mu\nu} = 0 \quad \Rightarrow \quad L_{K^{(a)}} T_{\mu\nu} = 0 \ . \tag{35.46}$$

Moreover, since the $L_{K^{(a)}}$ act like three-dimensional coordinate transformations, in order to see what these conditions mean we can make a $(3+1)$-decomposition of the energy-momentum tensor. From the three-dimensional point of view, T_{00} transforms like a scalar under coordinate transformations (and Lie derivatives), T_{0i} like a vector, and T_{ij} like a symmetric tensor. Thus we need to determine what are the three-dimensional

[151] This elementary argument, which requires no "higher knowledge" about Lie groups and invariants, is adapted from the discussion in section 13.4 of S. Weinberg, *Gravitation and Cosmology*.

scalars, vectors and symmetric tensors that are invariant under the full six-parameter group of the three-dimensional isometries.

For scalars ϕ we thus require (calling K now any one of the Killing vectors of $\tilde{g}_{ij}$),

$$L_K\phi = K^i\partial_i\phi = 0 \ . \tag{35.47}$$

Since $K^i(x)$ can take any value in a maximally symmetric space (homogeneity), this implies that ϕ has to be constant (as a function on the three-dimensional space) and therefore T_{00} can only be a function of time,

$$T_{00} = \rho(t) \ . \tag{35.48}$$

For vectors, it is almost obvious that no invariant vectors can exist because any vector would single out a particular direction and therefore spoil isotropy. The formal argument (as a warm up for the argument for tensors) is the following. We have

$$L_K V^i = K^j\tilde{\nabla}_j V^i + V^j\tilde{\nabla}_j K^i \ . \tag{35.49}$$

We now choose the Killing vectors such that $K^i(x) = 0$ but $\tilde{\nabla}_i K_j \equiv K_{ij}$ is an arbitrary anti-symmetric matrix. Then the first term disappears and we have

$$L_K V^i = 0 \ \ \Rightarrow \ \ K_{ij}V^j = 0 \ . \tag{35.50}$$

To make the anti-symmetry manifest, we rewrite this as

$$K_{ij}V^j = K_{kj}\delta^k_i V^j = \tfrac{1}{2}K_{kj}(\delta^k_i V^j - \delta^j_i V^k) = 0 \ . \tag{35.51}$$

If this is to hold for all anti-symmetric matrices, we must have

$$\delta^k_i V^j = \delta^j_i V^k \ \ , \tag{35.52}$$

and by contraction one obtains $nV^j = V^j$, and hence $V_j = 0$. Therefore, as expected, there is no invariant vector field and

$$T_{0i} = 0 \ . \tag{35.53}$$

We now come to symmetric tensors. Once again we choose our Killing vectors to vanish at a given point x and such that K_{ij} is an arbitrary anti-symmetric matrix. Then the condition

$$L_K T_{ij} = K^k\tilde{\nabla}_k T_{ij} + \tilde{\nabla}_i K^k T_{kj} + \tilde{\nabla}_j K^k T_{ik} = 0 \tag{35.54}$$

reduces to

$$K_{mn}(\tilde{g}^{mk}\delta^n_i T_{kj} + \tilde{g}^{mk}\delta^n_j T_{ik}) = 0 \ . \tag{35.55}$$

If this is to hold for all anti-symmetric matrices K_{mn}, the anti-symmetric part of the term in brackets must be zero or, in other words, it must be symmetric in the indices m and n, i.e.

$$\tilde{g}^{mk}\delta^n_i T_{kj} + \tilde{g}^{mk}\delta^n_j T_{ik} = \tilde{g}^{nk}\delta^m_i T_{kj} + \tilde{g}^{nk}\delta^m_j T_{ik} \ . \tag{35.56}$$

Contracting over the indices n and i, one obtains

$$n\tilde{g}^{mk}T_{kj} + \tilde{g}^{mk}T_{jk} = \tilde{g}^{mk}T_{kj} + \delta^m_j T^k_k \quad . \tag{35.57}$$

Therefore

$$T_{ij} = \frac{\tilde{g}_{ij}}{n} T^k_k \quad . \tag{35.58}$$

Now we already know that the scalar T^k_k has to be a constant. Thus we conclude that the only invariant tensor is the metric itself, and therefore the T_{ij}-components of the energy-momentum tensor can only be a function of t times $\tilde{g}_{ij}$. Writing this function as $p(t)a^2(t)$, we arrive at

$$T_{ij} = p(t) g_{ij} \quad . \tag{35.59}$$

We thus see that the energy-momentum tensor is determined by two functions, $\rho(t)$ and $p(t)$, precisely as in (35.14).

35.4 Conservation Laws for Perfect Fluids

The same arguments as above show that a current J^μ in a Robertson-Walker metric has to be of the form $J^\mu = (n(t), 0, 0, 0)$ in comoving coordinates, or

$$J^\mu = n(t) u^\mu \tag{35.60}$$

in covariant form. Here $n(t)$ could be a number density like a galaxy number density. It gives the number density per unit proper volume. The conservation law $\nabla_\mu J^\mu = 0$ is equivalent to

$$\nabla_\mu J^\mu = 0 \quad \Leftrightarrow \quad \partial_t(\sqrt{g} n(t)) = 0 \quad . \tag{35.61}$$

Thus we see that $n(t)$ is not constant, but the number density per unit coordinate volume is (as we had already anticipated in the picture of the balloon, Figure 44). For a Robertson-Walker metric, the time-dependent part of $\sqrt{g}$ is $a(t)^3$, and thus the conservation law says

$$n(t) a(t)^3 = const. \tag{35.62}$$

Let us now turn to the conservation laws associated with the energy-momentum tensor,

$$\nabla_\mu T^{\mu\nu} = 0 \quad . \tag{35.63}$$

The spatial components of this conservation law,

$$\nabla_\mu T^{\mu i} = 0 \quad , \tag{35.64}$$

turn out to be identically satisfied, by virtue of the fact that the u^μ are geodesic and that the functions ρ and p are only functions of time.

This could hardly be otherwise because $\nabla_\mu T^{\mu i}$ transforms as a vector under spatial rotations, and rotational invariance (isotropy) now implies that this vector has to be zero identically. Nevertheless it is of course instructive to check this explicitly.

The only interesting conservation law is thus the zero-component

$$\nabla_\mu T^{\mu 0} = \partial_\mu T^{\mu 0} + \Gamma^\mu_{\mu\nu} T^{\nu 0} + \Gamma^0_{\mu\nu} T^{\mu\nu} = 0 \quad , \tag{35.65}$$

which for a perfect fluid with $T_{00} = \rho(t)$ and $T_{ij} = p(t) g_{ij}$ becomes

$$\partial_t \rho(t) + \Gamma^\mu_{\mu 0} \rho(t) + \Gamma^0_{00} \rho(t) + \Gamma^0_{ij} T^{ij} = 0 \quad . \tag{35.66}$$

Inserting the explicit expressions (35.2) for the Christoffel symbols, one finds

$$\dot\rho = -3(\rho + p)\frac{\dot a}{a} \quad . \tag{35.67}$$

Introducing some fixed comoving volume $\sim v$ and its associated proper volume

$$V(t) = a(t)^3 v \quad , \tag{35.68}$$

and noting that

$$\frac{dV(t)}{dt} = 3a(t)^2 \dot a(t) v = 3H(t) V \quad , \tag{35.69}$$

the conservation equation (35.67) can be written in a perhaps more suggestive and familiar (mechanical or thermodynamical) form as

$$\frac{dE}{dt} = -p \frac{dV}{dt} \tag{35.70}$$

where

$$E(t) = \rho(t) V(t) \tag{35.71}$$

is the total energy in the volume V. Comparing with the 1st law of thermodynamics (35.18), this equation thus encodes the statement that the time evolution of the perfect fluid is *adiabatic* in the sense that its entropy remains constant,

$$\frac{dS(t)}{dt} = 0 \quad . \tag{35.72}$$

35.5 Conservation Laws and Comoving Congruences

Before discussing some special cases of the solutions of (35.67) in section 35.6 below, it is instructive to rederive the above results in a somewhat more general and covariant manner.[152] Thus we consider a general velocity field $u^\mu(x)$ with $u^\mu u_\mu = -1$, and the perfect fluid energy-momentum tensor (35.15),

$$T_{\mu\nu} = (p + \rho) u_\mu u_\nu + p g_{\mu\nu} \quad , \tag{35.73}$$

[152] For a detailed exposition of cosmology in this covariant framework see e.g. G. Ellis, H. van Elst, *Cosmological Models* (Cargèse Lectures 1998), `arXiv:gr-qc/9812046`, or the even more detailed monograph by G. Ellis, R. Maartens, M. MacCallum, *Relativistic Cosmology*.

with for the time being ρ and p arbitrary functions of the space-time coordinates. Note that $u^\mu\nabla_\mu$ is the covariant derivative along the integral curves of u^μ, the object we denoted D_τ or $D/D\tau$ in section 5.7. Acting on scalars we will simply denote it, as usual, by an overdot, i.e.

$$u^\mu\nabla_\mu\rho \equiv \dot\rho \tag{35.74}$$

etc. Let us now see what the conditon $\nabla_\mu T^{\mu\nu} = 0$ (which has to hold if this energy-momentum tensor is to give us a solution to the Einstein equations) tells us.

1. We first consider the case $p = 0$, so this corresponds to a pressure free perfect fluid (and is used in cosmology to e.g. model cold dark or baryonic matter). Then one has

$$T_{\mu\nu} = \rho u_\mu u_\nu \quad \Rightarrow \quad \nabla_\mu T^{\mu\nu} = (\dot\rho + \rho\nabla_\mu u^\mu)u^\nu + \rho u^\mu\nabla_\mu u^\nu \ . \tag{35.75}$$

   Here $\nabla_\mu u^\mu \equiv \theta$ is (and measures) the *expansion* of the velocity field u^μ (and was introduced previously, in the context of the Raychaudhuri equation, in (12.34)), and the last term $u^\mu\nabla_\mu u^\nu \equiv a^\nu$ is its acceleration (5.98), so that we can also write this equation as

$$(\dot\rho + \theta\rho)u^\nu + \rho a^\nu = 0 \ . \tag{35.76}$$

   Since u^ν and a^ν are orthogonal to each other,

$$u_\mu u^\mu = -1 \quad \Rightarrow \quad u_\mu a^\mu = 0 \ , \tag{35.77}$$

   this equation breaks up into two independent pieces,

$$\nabla_\mu T^{\mu\nu} = 0 \quad \Leftrightarrow \quad \dot\rho + \theta\rho = 0 \quad \text{and} \quad a^\nu = u^\mu\nabla_\mu u^\nu = 0 \ . \tag{35.78}$$

   Its time (energy flow) component is a continuity equation, while its space (momentum flow) part tells us that the particles have to move on geodesics.

2. Now what happens if we include pressure p? This corresponds to adding $p(g^{\mu\nu} + u^\mu u^\nu) \equiv ph^{\mu\nu}$, but this tensor is orthogonal to u^μ (cf. again the discussion in section 12.2 on the Raychaudhuri equation),

$$u^\mu h_{\mu\nu} \equiv u^\mu(g_{\mu\nu} + u_\mu u_\nu) = u_\nu - u_\nu = 0 \ . \tag{35.79}$$

   Therefore the equation $\nabla_\mu T^{\mu\nu} = 0$, with

$$T_{\mu\nu} = (\rho+p)u_\mu u_\nu + pg_{\mu\nu} \quad \Rightarrow \quad \nabla_\mu T^{\mu\nu} = (\dot\rho+\theta(\rho+p))u^\nu + (\rho+p)a^\nu + (\nabla_\mu p)h^{\mu\nu} \tag{35.80}$$

   again breaks up nicely into two orthogonal pieces. The part tangent to u^μ tells us that

$$\dot\rho + \theta(\rho + p) = 0 \ , \tag{35.81}$$

so this is a conservation law, and the part orthogonal to u^μ gives

$$(\rho + p)a^\nu + (\nabla_\mu p)h^{\mu\nu} = 0 \ , \tag{35.82}$$

which is a curved-space relativstic generalisation of the Euler equations of a perfect fluid.

In particular, now the velocity field is not composed of geodesics unless the derivative of p in the directions orthogonal to u^μ (i.e. the spatial derivative) is zero, $h^{\nu\mu}\nabla_\mu p = 0$. This is precisely the situation we are considering in cosmology, where $\rho = \rho(t)$ and $p = p(t)$ depend only on t, which is the proper time of the comoving observers described by the velocity field u^μ.

Returning thus to the cosmological setting, where we have (correctly, and uniquely as we now know) chosen the matter to move along geodesics, we are left with the continuity equation (35.81), which is now the same as (35.67) because for $u^\mu = (1, 0, 0, 0)$ in comoving coordinates one has

$$\theta = \nabla_\mu u^\mu = \frac{1}{\sqrt{g}}\partial_\mu(\sqrt{g}u^\mu) = a(t)^{-3}\partial_t(a(t)^3) = 3\dot{a}(t)/a(t) \ . \tag{35.83}$$

As an aside, note that this equation implies

$$\frac{d}{dt}\theta = 3(\ddot{a}/a - \dot{a}^2/a^2) \ , \tag{35.84}$$

which in terms of the Hubble parameter $H = \dot{a}/a$ and the deceleration parameter $q = -\ddot{a}a/\dot{a}^2$ (34.96) can be written as

$$\frac{d}{dt}\theta = -3H^2(q + 1) \ . \tag{35.85}$$

This equation is a special case of the Raychaudhuri equation (12.36) for timelike geodesic congruences,

$$\frac{d}{d\tau}\theta = -\tfrac{1}{3}\theta^2 - \sigma^{\mu\nu}\sigma_{\mu\nu} + \omega^{\mu\nu}\omega_{\mu\nu} - R_{\mu\nu}u^\mu u^\nu \ . \tag{35.86}$$

Indeed, specialising (35.86) to the family of comoving observers in a Robertson-Walker geometry and noting that

- the proper time τ is the cosmological time t
- the rotation is zero (either by explicit calculation or, more ot the point, because the geodesics are orthogonal to the hypersurfaces $t = const.$, or on symmetry grounds)
- the shear is zero (either by explicit calculation or on symmetry grounds)
- the relevant component of the Ricci tensor is $R_{00} = -3\ddot{a}/a$,

oner sees that (35.86) reduces to

$$\frac{d}{dt}\theta = -\tfrac{1}{3}\theta^2 - R_{00} = -3\dot{a}^2/a^2 + 3\ddot{a}/a \tag{35.87}$$

which is identical to (35.84). Thus we see how the parameters $H(t)$ and $q(t)$, originally introduced to characterise the first terms in a Taylor expansion of the cosmic scale factor also govern the local behaviour of freely falling observers like (clusters of) galaxies. In particular, in an expanding universe the congruences of comoving observers diverge / expand ($\theta > 0$) but the rate of expansion decreases ($\dot{\theta} < 0$) provided that $q > -1$.

35.6 Conservation Laws for Specific Equations of State

We now look at the solutions of (35.67),

$$\dot{\rho} = -3(\rho + p)\frac{\dot{a}}{a} \ , \tag{35.88}$$

for some specific equations of state.

1. For instance, when the pressure of the cosmic matter is negligible, like in the universe today, and we can treat the galaxies (without disrespect) as dust, then one has

$$w = 0 \quad \Rightarrow \quad \frac{\dot{\rho}}{\rho} = -3\frac{\dot{a}}{a} \ , \tag{35.89}$$

and this equation can trivially be integrated to

$$\rho(t)a(t)^3 = \text{const.} \tag{35.90}$$

Thus the (proper) density is proportional to the inverse (proper) spatial volume, an unsurprising (and reassuring) result. This of course also follows on the nose from the conservation law in the form (35.70):

$$p = 0 \quad \Rightarrow \quad \frac{d}{dt}E = 0 \ . \tag{35.91}$$

2. On the other hand, if the universe is dominated by, say, radiation, then one has the equation of state $p = \rho/3$, and the conservation equation reduces to

$$w = 1/3 \quad \Rightarrow \quad \frac{\dot{\rho}}{\rho} = -4\frac{\dot{a}}{a} \ , \tag{35.92}$$

and therefore

$$\rho(t)a(t)^4 = \text{const.} \tag{35.93}$$

The reason why the energy density of photons decreases faster with $a(t)$ than that of dust is of course ... the redshift:

$$\frac{1}{a(t)^4} = \frac{1}{a(t)^3}(\text{number density of photons}) \times \frac{1}{a(t)}(\text{redshift of individual photons}) \ . \tag{35.94}$$

3. More generally, for matter with equation of state parameter w one finds

$$p = w\rho \quad \Rightarrow \quad \rho(t)a(t)^{3(1+w)} = \text{const.} \tag{35.95}$$

The interpretation of this equation becomes somewhat more transparent in terms of the thermodynamic variant (35.70) of the conservation law. For the linear equation of state $p = w\rho$, it reduces to

$$\frac{dE}{dt} = -wE\frac{dV}{dt} \quad \Leftrightarrow \quad \frac{d}{dt}(V^w E) = 0 \ . \tag{35.96}$$

By (35.21), this is simply the statement that the entropy, which is a function of $V^w E$, is constant (cf. (35.72)).

4. In particular, for $w = -1$, $\rho = E/V$ itself is constant,

$$w = -1 \quad \Rightarrow \quad \rho(t) = \text{const.} \tag{35.97}$$

as it should be, in view of its identification (35.31) with a cosmological constant.

5. Quite generally, we see from (35.88) that in an expanding universe (i.e. $\dot{a} > 0$), the energy density of matter satisfying the null energy condition (NEC) $\rho + p \geq 0$ cannot increase (and will necessarily decrease for matter satisfying the strict NEC, with $\rho + p > 0$),

$$\dot{a} > 0 \quad , \quad \rho + p \geq 0 \quad \text{(NEC)} \quad \Rightarrow \quad \dot{\rho} \leq 0 \ . \tag{35.98}$$

6. As the final example, consider the peculiar Chaplygin gas with equation of state $p = -A/\rho$ (35.41) with A constant. In this case (35.88) reads

$$p = -A/\rho \quad \Rightarrow \quad \dot{\rho} = -3\left(\rho - \frac{A}{\rho}\right)\frac{\dot{a}}{a} \ , \tag{35.99}$$

which has the exact solution[153]

$$\rho(t) = \sqrt{A + B/a(t)^6} \tag{35.100}$$

(with integration constant B). In the context of a cosmology of an expanding universe, this has the remarkable property of interpolating between what appears to be a dust-filled universe at early times (small $a(t)$),

$$a(t) \quad \text{small} \quad \Rightarrow \quad \rho(t) \sim \sqrt{B}/a(t)^3 \ , \tag{35.101}$$

and a universe filled with a cosmological constant at late times (large $a(t)$),

$$a(t) \quad \text{large} \quad \Rightarrow \quad \rho(t) \sim \sqrt{A} \quad \Rightarrow \quad p(t) \sim -\sqrt{A} \ . \tag{35.102}$$

Thus this mimics the evolution of an ordinary (matter-filled) universe to one dominated by *dark energy*, resembling the evolution of our universe (cf. section 38.1).

[153] A. Kamenshchik, U. Moschella, V. Pasquier, *An alternative to quintessence*, `arXiv:gr-qc/0103004`.

After these preliminaries, we are now prepared to tackle (hence first to determine) the Einstein equations in this setting.

Allowing for the presence of a cosmological constant, we thus consider the equations

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = 8\pi G_N T_{\mu\nu} \ . \tag{35.103}$$

Alternatively we can write these equations as

$$R_{\mu\nu} = 8\pi G_N (T_{\mu\nu} - \tfrac{1}{2} g_{\mu\nu} T^{\lambda}_{\ \lambda}) + \Lambda g_{\mu\nu} \ . \tag{35.104}$$

Because of isotropy, there are only two independent equations, e.g. the (00)-component and any one of the non-zero (ij)-components. Using (35.6), we find

$$\begin{aligned} -3\frac{\ddot{a}}{a} &= 4\pi G_N(\rho + 3p) - \Lambda \\ \frac{\ddot{a}}{a} + 2\frac{\dot{a}^2}{a^2} + 2\frac{k}{a^2} &= 4\pi G_N(\rho - p) + \Lambda \ . \end{aligned} \tag{35.105}$$

Alternatively, instead of the spatial components of (35.104) one could have used the (00)-component of (35.103) and the expression in (35.8) for G_{00} to deduce

$$G_{00} - \Lambda = 8\pi G_N \, T_{00} \quad \Rightarrow \quad \frac{\dot{a}^2}{a^2} + \frac{k}{a^2} = \frac{8\pi G_N}{3}\rho + \frac{\Lambda}{3} \tag{35.106}$$

(and this equation could have also been obtained by appropriate subtraction of the previous 2 equations). Either way, we supplement these by the conservation equation

$$\dot{\rho} = -3(\rho + p)\frac{\dot{a}}{a} \ . \tag{35.107}$$

and thus end up with the set of equations

$$\boxed{\begin{array}{lrcl} (F1) & \frac{\dot{a}^2}{a^2} + \frac{k}{a^2} &=& \frac{8\pi G_N}{3}\rho + \frac{\Lambda}{3} \\ (F2) & -3\frac{\ddot{a}}{a} &=& 4\pi G_N(\rho + 3p) - \Lambda \\ (F3) & \dot{\rho} &=& -3(\rho + p)\frac{\dot{a}}{a} \end{array}} \tag{35.108}$$

Together, these are known as the *Friedmann-Lemaître equations* or *Friedmann equations*. These equations form the basis, and govern every aspect, of the standard Friedmann-Lemaître-Robertson-Walker models of cosmology.

Remarks:

1. I will usually refer to the collection of these equations, without disrespect to Lemaître but just out of habit, to the Friedmann equations. When referring to individual equations, rather than quoting equation numbers I will simply refer to them as the equations (F1), (F2), and (F3) respectively.

2. In terms of the Hubble parameter $H(t)$ and the deceleration parameter $q(t)$, these equations can also be written as

$$\begin{array}{lrcl} (F1') & H^2 & = & \frac{8\pi G_N}{3}\rho - \frac{k}{a^2} + \frac{\Lambda}{3} \\ (F2') & q & = & \frac{1}{3H^2}[4\pi G_N(\rho + 3p) - \Lambda] \\ (F3') & \frac{d}{dt}(\rho a^3) & = & -3Hpa^3 \ . \end{array}$$

3. Commonly, (F1) is referred to as *the Friedmann equation*. As mentioned above, this is precisely the (00)-component of the Einstein equations.

4. (F2), on the other hand, can be interpreted as the Raychaudhuri equation (35.87). Indeed, with $\theta = 3\dot{a}/a$ and

$$R_{00} = 8\pi G_N(T_{00} + \tfrac{1}{2}T^\alpha_\alpha) - \Lambda = 4\pi G_N\ (\rho + 3p) - \Lambda \tag{35.109}$$

one finds that

$$\frac{d}{dt}\theta = -\tfrac{1}{3}\theta^2 - R_{00} \quad \Leftrightarrow \quad -3\ddot{a}/a = 4\pi G_N(\rho + 3p) - \Lambda \tag{35.110}$$

which is precisely (F2).

5. In writing the above equations, I have separated out the cosmological constant Λ from the remaining matter contributions. Of course, using (35.31), it could have just been treated as one other perfect fluid contribution (with $w = -1$). Occasionally either one or the other way of writing these equations is (marginally) more convenient.

6. Note that because of the Bianchi identities, the Einstein equations and the conservation equations should not be independent, and indeed they are not:

   (a) It is easy to see that (F1) and (F3) imply the second order equation (F2) so that, a pleasant simplification, in practice one only has to deal with the two first order equations (F1) and (F3). Sometimes, however, (F2) is easier to solve than (F1), because it is linear in $\ddot{a}(t)$, and then (F1) is just used to fix one constant of integration.

   (b) It is also easy to see that (F1) and (F2) imply (F3), i.e. that the gravity equations of motion imply the matter equations of motion, a general and fundamental feature of general relativity.

   (c) Finally, formally (F2) and (F3) also imply (F1), with k (which only appears in (F1)) arising as an integration constant.

7. Note also that the parameter k appears in the Friedmann equations only in the combination k/a^2. Thus a scaling of k can be compensated by a corresponding scaling of $a(t)$. This reflects the fact, mentioned several times already, that only this particular combination has individual meaning, and that one can exploit the scaling freedom to e.g. set $a_0 = 1$ or $k = 0, \pm 1$.

8. One can use (35.13) to determine the analogue of the Friedmann equations in any space-time dimension $D = n + 1$.

   - The indepedent components of the Einstein equation can then be chosen to be, from $G_{00} = \kappa T_{00}$ (for the reason for now calling the coupling constant κ and not just $8\pi G_N$, as in four dimensions, see the discussion around (19.44)),

   $$\frac{k + \dot{a}^2}{a^2} = \frac{2\kappa}{n(n-1)}\rho \ , \tag{35.111}$$

   and from R_{00}, say, using (19.45) and the fact that the trace of a perfect fluid energy-momentum tensor is now

   $$T^\lambda_\lambda = -\rho + np \ , \tag{35.112}$$

   one finds

   $$\frac{\ddot{a}}{a} = -\frac{\kappa}{n(n-1)}\left[(n-2)\rho + np\right] \ . \tag{35.113}$$

   - Finally the continuity equation takes the form

   $$\dot{\rho} = -n(\rho + p)\frac{\dot{a}}{a} \ . \tag{35.114}$$

9. As an aside note that, as the Robertson-Walker metrics are, in particular, spherically symmetric, and written in the manifestly spherically-symmetric form (24.80), we have the notion of the Misner-Sharp mass (24.82) for spherical symmetry at our disposal. In terms of the area radius (34.42),

   $$\tilde{r}(t, r) = a(t)r \ , \tag{35.115}$$

   the Misner-Sharp mass function is given by (24.82)

   $$M_{MS}(t, r)m(t, r) = \tfrac{1}{2}\tilde{r}\left(1 - g^{ab}\partial_a\tilde{r}\partial_b\tilde{r}\right) \ . \tag{35.116}$$

   With $g^{tt} = -1$ and $g^{rr} = (1 - kr^2)/a(t)^2$, this reduces to

   $$m(t, r) = \tfrac{1}{2}a(t)r^3(\dot{a}(t)^2 + k) \ . \tag{35.117}$$

   Using the Friedmann equation (F1), this can be written in a more informative way as

   $$\begin{aligned} m(t, r) &= \tfrac{1}{2}a(t)^3r^3\left(\frac{8\pi G_N}{3}\rho + \frac{\Lambda}{3}\right) \\ &= \frac{4\pi}{3}\tilde{r}^3 G_N\left(\rho + \rho_\Lambda\right) \ , \end{aligned} \tag{35.118}$$

   with $\rho_\Lambda = \Lambda/8\pi G_N$ the energy density (35.31) associated with the cosmological constant. Note that this result, which again, as in section 24.6, has the interpretation as "mass = coordinate volume (not proper volume) × density", does not depend explicitly on either the pressure p or the curvature k. In particular it is independent of the equation of state relating p and ρ.

35.8 Lagrangian Formulation of the Friedmann Equations

For many purposes it is useful to cast the above set of Friedmann-Lemaître equations into a Lagrangian or Hamiltonian form. In particular (and this is the motivation for doing this here), this system of equations is sufficiently simple to provide a concrete illustration of some of the general features of the canonical Hamiltonian formulation of general relativity discussed in a somewhat cursory way in section 21.

Rather than specialising the general results of that section to the case at hand, we will adopt a more pedestrian approach here and derive these results from scratch. This will make contact with and hopefully shed some light on a variety of different issues that have arisen in various parts of these notes, e.g.

- the characteristic constraints of general relativity (sections 19.7 and 21), in particular the *Hamiltonian constraint*
- the *Gibbons-Hawking-York boundary term* discussed in section 20.5
- the *ADM action* discussed in section 23.4
- the role and significance of the *lapse function* (cf. (21.18)) introduced into the Robertson-Walker metric (for what appeared to be no good reason at that point) at the end of section 35.1.

In the previous section, in order to arrive at the Friedmann equations, we made an ansatz for the metric based on symmetry considerations (spatial maximal symmetry), namely

$$ds^2 = -dt^2 + a(t)^2 d\tilde{s}^2 \tag{35.119}$$

and then plugged this (together with a compatible ansatz for the energy-momentum tensor) into the full Einstein equations. Such a reduction procedure is always possible at the level of the equations of motion. In order to develop the Hamiltonian formulation of the Friedmann equations, it is convenient to start with the Lagrangian formulation. However, such a reduction procedure (plugging an ansatz for the solution into the action and then solving the resulting Euler-Lagrange equations) is not necessarily consistent at the level of the Lagrangian or action, i.e. the Euler-Lagrange equations of the reduced Langrangian will not necessarily agree with the reduced equations of motion and will not necessarily give rise to solutions of the original unreduced equations.

Indeed, in the present context it is pretty obvious that, with the single gravitational degree of freedom $a(t)$, associated with the size of the spatial metric, it is impossible to derive both the (one independent) spatial component of the Einstein equations (the Friedmann equation F2, say) and the time-time component of the Einstein equations (the Friedmann equation F1) from a Lagrangian depending just on $a(t)$ (and the matter variables).

It turns out, however, that just by introducing as an additional variable a non-trivial time-time component of the metric (whose variation can then be used to impose the Friedmann equation F1), the reduction is consistent also at the Lagrangian level. We will introduce this additional variable in the form of the lapse function $N(t)$ of (35.10), but we will now simply write t for the time-coordinate and not t', so that the ansatz for the metric is

$$ds^2 = -N^2(t)dt^2 + a^2(t)d\tilde{s}^2 \ , \tag{35.120}$$

and we can always choose the $N(t) = 1$ "t is comoving proper time" gauge at the end of the calculations.

In section 35.1 we had already determined the Ricci scalar of this metric, namely (now again, consistent with $t' \to t$ denoting time-derivatives by overdots rather than primes) (35.12)

$$R = \frac{6}{a^2N^3}(N(a\ddot{a} + \dot{a}^2) - a\dot{a}\dot{N} + kN^3) \ . \tag{35.121}$$

The Einstein-Hilbert Lagangian density is therefore

$$\sqrt{g}R = Na^3\sqrt{\tilde{g}}R = \sqrt{\tilde{g}}\frac{6a}{N^2}(N(a\ddot{a} + \dot{a}^2) - a\dot{a}\dot{N} + kN^3) \ . \tag{35.122}$$

The only dependence on the spatial coordinates is in the spatial volume element $\sqrt{\tilde{g}}$. Therefore, integrating this Lagrangian density over the space-time, one obtains a (potentially infinite) volume factor from the integration over the spatial coordinates which we will simply drop. Thus in the infinite-volume case ($k = -1$ or $k = 0$ without periodic toroidal identifications) this is not really a reduction in the strict technical sense. However, this is not our main concern here. Our aim is simply to obtain a Lagrangian formulation of the Friedmann equations and (as we will see) this can be accomplished by just dropping the integration over the spatial coordinates.

We are thus left with the 1-dimensional (mechanics) gravitational Lagrangian

$$L_{EH} = \frac{6a}{N^2}(N(a\ddot{a} + \dot{a}^2) - a\dot{a}\dot{N} + kN^3) \ . \tag{35.123}$$

Note that this depends not only on $a(t)$ and $\dot{a}(t)$, and $N(t)$ and $\dot{N}(t)$, but also (linearly) on $\ddot{a}(t)$. This of course reflects the general property of the Einstein-Hilbert Lagrangian that it depends on 2nd derivatives of the metric, thus necessitating the introduction of boundary terms etc., as discussed in sections 20.4 and 20.5.

Naively, to eliminate this $\ddot{a}$-term, we can integrate the first term of the Lagrangian by parts,

$$6a^2\ddot{a}N^{-1} = \frac{d}{dt}(6a^2\dot{a}N^{-1}) + 6aN^{-2}(-2\dot{a}^2N + a\dot{a}\dot{N}) \ . \tag{35.124}$$

We see that this has the effect of changing the sign of the 2nd term of (35.123) and cancelling the 3rd term $\sim \dot{N}$, so that we have

$$L_{EH} = \frac{6a}{N^2}(-N\dot{a}^2 + kN^3) + \frac{d}{dt}(6a^2\dot{a}N^{-1}) \ . \tag{35.125}$$

It turns out that the total derivative term is cancelled precisely by the Gibbons-Hawking-York boundary term discussed in section 20.5. This should not come as a surprise: after all, that was its purpose. To see this explicitly, note that with any of the definitions or characterisations of the extrinsic curvature tensor given in section 18, one finds that the extrinsic curvature of the constant time t hypersurfaces (with unit normal vector $(1/N)\partial_t$) in the ambient space-time is

$$K_{ij} = \frac{1}{2N}\partial_t g_{ij} = \frac{\dot{a}}{aN} g_{ij} \tag{35.126}$$

so that the trace is

$$K = g^{ij}K_{ij} = \frac{3\dot{a}}{aN} \ . \tag{35.127}$$

Thus the Gibbons-Hawking-York boundary term (20.61) reduces to (again dropping the spatial volume element $\sqrt{\bar{g}}$ and using $\epsilon = -1$ for spacelike hypersurfaces)

$$2\epsilon\sqrt{h}\, K \to L_{GHY} = -2a^3(3\dot{a}/aN) = -6a^2\dot{a}N^{-1} \ . \tag{35.128}$$

The complete gravitational Lagrangian is now, in analogy with the complete standard gravitational action (20.67),

$$S_g[g_{\alpha\beta}] = S_{EH}[g_{\alpha\beta}] + S_{GHY}[g_{\alpha\beta}] \ , \tag{35.129}$$

the Lagrangian

$$\begin{aligned} L_g &= L_{EH} + \frac{d}{dt}L_{GHY} \\ &= \frac{6a}{N^2}(-N\dot{a}^2 + kN^3) \\ &= 6N(-a(\dot{a}/N)^2 + ka) \ . \end{aligned} \tag{35.130}$$

Inclusion of the cosmological constant Λ,

$$\sqrt{g}R \to \sqrt{g}(R - 2\Lambda) \ , \tag{35.131}$$

leads to

$$L_g = 6N(-a(\dot{a}/N)^2 + ka - \Lambda a^3/3) \ . \tag{35.132}$$

Remarks:

1. Note that, as desired, the Lagrangian now only depends on the "fields" $a(t)$ and $N(t)$ and (at most) their 1st derivatives.

2. Actually, we see that the Lagrangian depends only on $N(t)$, not its time-derivative $\dot{N}(t)$. This will also turn out to be the case for the matter action (which typically does not depend on any derivatives of the metric at all). Thus the role of $N(t)$ is just that of a Lagrange multiplier, and as we will see the constraint it imposes is simply the Friedmann equation F1.

3. One can also obtain (35.130) directly from the Gauss-Codazzi (21.6) or ADM form (21.42) of the action,

$$S_{ADM}[g_{\alpha\beta}] = \int \sqrt{g} d^4x \, (\bar{R} + K^{ij}K_{ij} - K^2) \ . \tag{35.133}$$

Indeed, the 3-dimensional scalar curvature is simply (this follows e.g. from (14.19))

$$\bar{R} = 6k/a^2 \ , \tag{35.134}$$

while from (35.126) one has

$$K^{ij}K_{ij} - K^2 = 3(\dot{a}/aN)^2 - 9(\dot{a}/aN)^2 = -6(\dot{a}/aN)^2 \tag{35.135}$$

so that (again dropping the spatial volume element $\sqrt{\bar{g}}$), one finds

$$L_{ADM} = a^3 N(-6(\dot{a}/aN)^2 + 6k/a^2) = L_g \ . \tag{35.136}$$

We now need to add matter. The phenomenological description of matter in terms of the energy and pressure densities $\rho(t)$ and $p(t)$ is not sufficient for a Lagrangian formulation. The simplest matter model with a "microscopic" Lagrangian description is that of a scalar field in the gravitational field described by the class of metrics (35.120), and this is the model we will consider.

The starting point is thus the action (6.14) for a scalar field with a potential $V(\phi)$,

$$S[\phi, g_{\alpha\beta}] = \int \sqrt{g} d^4x \left[-\tfrac{1}{2} g^{\alpha\beta} \partial_\alpha \phi \partial_\beta \phi - V(\phi) \right] \ . \tag{35.137}$$

Compatibility with the symmetries of the gravitational field, in particular the spatial homogeneity, requires that the scalar field is spatially constant and is thus only a function of time t. With this assumption (and again dropping $\sqrt{\bar{g}}$), the matter Lagrangian reduces to

$$L_m = Na^3(\dot{\phi}^2/2N^2 - V(\phi)) = N(a^3\dot{\phi}^2/2N^2 - a^3V(\phi)) \ . \tag{35.138}$$

Therefore the total gravitational + matter Lagrangian and action are (reinstating the gravitational coupling constant)

$$\begin{aligned} L_{tot} &= \frac{1}{16\pi G_N} L_g + L_m \\ &= N\left(-\frac{3}{4\pi G_N} \frac{a\dot{a}^2}{2N^2} + \frac{a^3\dot{\phi}^2}{2N^2} + \frac{3ka}{8\pi G_N} - \frac{\Lambda a^3}{8\pi G_N} - a^3 V(\phi) \right) \end{aligned} \tag{35.139}$$

and

$$\begin{aligned} S_{tot}[a, \phi, N] &= \int dt \, L_{tot} \\ &= \int dt \, N\left(-\frac{3}{4\pi G_N} \frac{a\dot{a}^2}{2N^2} + \frac{a^3\dot{\phi}^2}{2N^2} + \frac{3ka}{8\pi G_N} - \frac{\Lambda a^3}{8\pi G_N} - a^3 V(\phi) \right) \end{aligned} \tag{35.140}$$

Before analysing the equations of motion and the Hamiltonian arising from this Lagrangian and action, let us note the following points:

1. From this form of the action it is evident that one role of $N(t)$ is to ensure time reparametrisation invariance of the action. Indeed, dt and $N(t)$ only appear in the combinations $N(t)dt$ or $(1/N(t))(d/dt)$. Hence the action remains unchanged under time reparametrisations $t \to \tilde{t}(t)$ if one simultaneously transforms $N(t)$ to a new lapse function $\tilde{N}(\tilde{t})$, say, according to

$$N(t)dt = \tilde{N}(\tilde{t})d\tilde{t} \quad \Leftrightarrow \quad \tilde{N} = N(dt/d\tilde{t}) \ . \tag{35.141}$$

   This is the remnant of the general covariance of the original action, and the mechanism here is the same as that which rendered the "parent" geodesic action (2.111) reparametrisation invariant.

2. We see that this action does not involve the time-derivative of the lapse function $N(t)$. Hence $N(t)$ acts as a Lagrange multiplier enforcing a constraint. This constraint can be regarded as the constraint associated to this reparametrisation invariance, and such constraints are thus a characteristic feature of any reparametrisation invariant or generally covariant system.

3. In this combined gravity plus matter action, we now see very clearly that the gravitational kinetic term has the opposite sign of the matter kinetic term. This is a particular (and particularly obvious) manifestation of the general fact (mentioned in connection with the DeWitt metric at the end of section 21.2) that the gravitational kinetic term is not positive definite and that the negative "direction" in field space is associated with overall spatial volume deformations.

4. Finally, this form of the action makes it particularly obvious that the cosmological constant term can also be regarded as leading to (or arising from) a constant shift of the potential for the matter fields,

$$\frac{\Lambda a^3}{8\pi G_N} + a^3 V(\phi) = a^3(V(\phi) + \rho_\Lambda) \ . \tag{35.142}$$

   We will therefore absorb the cosmological constant term into the scalar potential in the following and not carry it around explicitly.

Now let us look at the Euler-Lagrange equations arising from this action.

- The Euler-Lagrange Equation for $N(t)$

  As mentioned before, $N(t)$ acts as a Lagrange multiplier enforcing the constraint

$$\frac{\delta S_{tot}}{\delta N} = 0 \quad \Rightarrow \quad \frac{\partial L}{\partial N(t)} = 0 \ . \tag{35.143}$$

  Explicitly this is the condition

$$-\frac{3}{4\pi G_N}\frac{a\dot{a}^2}{2N^2} + \frac{a^3\dot{\phi}^2}{2N^2} - \frac{3ka}{8\pi G_N} + a^3 V(\phi) = 0 \ . \tag{35.144}$$

This really is a constraint rather than an equation of motion, because it only depends on the fields and their 1st derivatives, not their 2nd derivatives, and thus constitutes a condition on initial data on some constant time initial hypersurface.

In the gauge $N(t) = 1$ (which we can now choose, after having determined the equation of motion arising from varying N in the action), this constraint becomes

$$\frac{3}{8\pi G_N}(a\dot{a}^2 + ak) = \frac{a^3\dot{\phi}^2}{2} + a^3 V(\phi) \quad , \tag{35.145}$$

or

$$\frac{\dot{a}^2 + k}{a^2} = \frac{8\pi G_N}{3}(\frac{1}{2}\dot{\phi}^2 + V(\phi)) \quad . \tag{35.146}$$

Now recall that for such a scalar field the energy density and pressure are given by (35.38)

$$\rho(t) = \tfrac{1}{2}\dot{\phi}^2 + V(\phi) \quad , \quad p(t) = \tfrac{1}{2}\dot{\phi}^2 - V(\phi) \quad . \tag{35.147}$$

Hence we can write the constraint as

$$\frac{\dot{a}^2 + k}{a^2} = \frac{8\pi G_N}{3}\rho \quad , \tag{35.148}$$

and this we now recognise as precisely the Friedmann equation F1.

Having derived the constraint arising from the variation of the lapse function N, we can now simplify our life by using the reparametrisation invariance to set $N(t) = 1$. Thus we can work with the simpler action

$$S_{tot}[a, \phi] = \int dt \left(-\frac{3}{4\pi G_N}\frac{a\dot{a}^2}{2} + \frac{a^3\dot{\phi}^2}{2} + \frac{3ka}{8\pi G_N} - a^3 V(\phi) \right) \quad . \tag{35.149}$$

- The Euler-Lagrange equation for $a(t)$

  This is the equation

$$\frac{d}{dt}\left(-\frac{3a\dot{a}}{4\pi G_N} \right) = 3\left(-\frac{1}{4\pi G_N}\frac{\dot{a}^2}{2} + \frac{a^2\dot{\phi}^2}{2} + \frac{k}{8\pi G_N} - a^2 V(\phi) \right) \quad , \tag{35.150}$$

  or

$$\frac{\ddot{a}}{a} = -4\pi G_N p - \frac{\dot{a}^2 + k}{2a^2} \quad , \tag{35.151}$$

  where $p = p(t)$ is the pressure (35.147). Using the constraint (the Friedmann equation F1) derived above, this becomes

$$\frac{\ddot{a}}{a} = -\frac{4\pi G_N}{3}(\rho + 3p) \quad , \tag{35.152}$$

  which is precisely the Friedmann equation F2.

- The Euler-Lagrange equation for $\phi(t)$

  Finally, the equation of motion for the scalar field is

$$\frac{d}{dt}(a^3\dot{\phi}) = -a^3 V'(\phi) \quad \Leftrightarrow \quad \ddot{\phi} + 3(\dot{a}/a)\dot{\phi} + V'(\phi) = 0 \ , \tag{35.153}$$

  exhibiting the same friction term coupling to the Hubble parameter as (34.113). Multiplying by $\dot{\phi}$ and noting that

$$\rho + p = \dot{\phi}^2 \tag{35.154}$$

  and

$$\dot{\rho} = \dot{\phi}\ddot{\phi} + V'(\phi)\dot{\phi} \ , \tag{35.155}$$

  it is evident that this equation can equivalently be written as

$$\ddot{\phi} + 3(\dot{a}/a)\dot{\phi} + V'(\phi) = 0 \quad \Leftrightarrow \quad \dot{\rho} + 3(\rho + p)(\dot{a}/a) = 0 \ , \tag{35.156}$$

  which is the Friedmann equation F3.

We have thus verified that we have indeed obtained a Lagrangian description of the complete set of Friedmann equations.

35.9 Hamiltonian Formulation of the Friedmann Equations

Having obtained the Lagrangian formulation, it is now quite straightforward to pass to a Hamiltonian formulation, and this despite the presence of constraints which, in general, can significantly complicate the Hamiltonian formulation (and subsequent canonical quantisation, say) of such systems.[154]

Our starting point is the Lagrangian (35.139) (with Λ absorbed into $V(\phi)$), i.e.

$$L_{tot} = N\left(-\frac{3}{4\pi G_N}\frac{a\dot{a}^2}{2N^2} + \frac{a^3\dot{\phi}^2}{2N^2} + \frac{3ka}{8\pi G_N} - a^3 V(\phi)\right) \ . \tag{35.157}$$

In order to streamline the following discussion, it is convenient to consider the fields $a(t)$ and $\phi(t)$ as the two coordinates

$$Q^A = (a, \phi) \tag{35.158}$$

of a 2-dimensional dynamical system, with metric

$$G_{AB} = \mathrm{diag}(-3a/4\pi G_N, a^3) \ . \tag{35.159}$$

[154] For detailed textbook discussions of this subject, see e.g. M. Henneaux, C. Teitelboim, *Quantisation of Gauge Systems*, or H. Rothe, K. Rothe, *Classical and Quantum Dynamics of Constrained Hamiltonian Systems*.

With this notation, the Lagrangian can be written as

$$L_{tot} = N\left(\frac{1}{2N^2}G_{AB}\dot{Q}^A\dot{Q}^B - V(Q)\right) \tag{35.160}$$

where the potential is

$$V(Q) = a^3V(\phi) - \frac{3ka}{8\pi G_N} \ . \tag{35.161}$$

Were it not for the presence of the lapse function N as an addditional dynamical variable, this would be a completely standard classical mechanics Lagrangian, with an obvious corresponding Hamiltonian. In particular, the momenta conjugate to the variables Q^A take the almost standard form

$$P_A = \frac{\partial L_{tot}}{\partial \dot{Q}^A} = G_{AB}\dot{Q}^B/N \ , \tag{35.162}$$

i.e.

$$P_a = -\frac{3a\dot{a}}{4\pi G_N N} \quad , \quad P_\phi = \frac{a^3\dot{\phi}}{N} \ . \tag{35.163}$$

These relations can, as usual, be used to eliminate the velocities in favour of the momenta. The major novelty is thus the presence of N, whose conjugate momentum vanishes,

$$P_N = \frac{\partial L_{tot}}{\partial \dot{N}} = 0 \ . \tag{35.164}$$

This is our first constraint (and a *primary constraint* in the terminology of constrained systems). While this relation does not allow us to eliminate $\dot{N}$ in terms of the momentum P_N, this is not an issue here since the Lagrangian does not depend on $\dot{N}$ in the first place.

We can now follow the standard procedure to construct the Hamiltonian, via (in the case at hand, whether or not we include the $P_N\dot{N}$-term evidently makes no difference)

$$H_{tot} = P_N\dot{N} + P_A\dot{Q}^A - L_{tot} \ . \tag{35.165}$$

Then one finds the Hamiltonian.

$$H_{tot}(N, Q, P) = N\left(\frac{1}{2}G^{AB}P_AP_B + V(Q)\right) \equiv N\mathcal{H}(Q, P) \ . \tag{35.166}$$

As far as its dependence on (Q, P) is concerned, this presents no surprises: time evolution is given by the Hamilton equations

$$\begin{aligned} \dot{Q}^A &= +\frac{\partial H_{tot}}{\partial P_A} = \{Q^A, H_{tot}\} \\ \dot{P}_A &= -\frac{\partial H_{tot}}{\partial Q_A} = \{P_A, H_{tot}\} \ , \end{aligned} \tag{35.167}$$

and the 1st of these just reproduces the definition of the momenta (35.162), while the 2nd then reproduces the Euler-Lagrange equations for the fields $Q^A = (a, \phi)$ discussed in the previous section.

Turning to N, consistency of the primary constraint $P_N = 0$ (i.e. the condition that it be preserved in time) requires

$$\dot{P}_N = \{P_N, H_{tot}\} = -\mathcal{H} = 0 \ . \tag{35.168}$$

Thus the primary constraint $p_N = 0$ gives rise to the *secondary constraint*

$$\mathcal{H}(Q, P) = 0 \tag{35.169}$$

(and there are no further constraints in this class of examples). This *Hamiltonian constraint* is precisely the Friedmann equation F1, i.e. the condition that was imposed in the Lagrangian formulation by the Lagrange multipler N,

$$\frac{\partial L_{tot}}{\partial N} = 0 \quad \Leftrightarrow \quad \mathcal{H} = 0 \ . \tag{35.170}$$

A painless way to see this is to note that the N-dependence in the Lagrangian (35.160),

$$L_{tot} = \frac{1}{2N} G_{AB} \dot{Q}^A \dot{Q}^B - N V(Q) \ , \tag{35.171}$$

is precisely such that differentiation with respect to N changes the relative sign between the 2 terms and thus essentially implements the Legendre transformation from the Lagrangian to the Hamiltonian (expressed as a function of the velocities),

$$-\frac{\partial L_{tot}}{\partial N} = \frac{1}{2N^2} G_{AB} \dot{Q}^A \dot{Q}^B + V(Q) = \mathcal{H} \ . \tag{35.172}$$

This structure

$$H = N\mathcal{H} \tag{35.173}$$

of the Hamiltonian, with the constraint $\mathcal{H} = 0$, is again characteristic of parametrisation invariant or generally covariant systems. In particular, this provides a concrete illustration of the general features of the gravitational Hamiltonian mentioned in section 21.7.

36 Cosmology IV: Qualitative Analysis

A lot can be deduced about the solutions of the Friedmann-Lemaître equations, i.e. the evolution of the universe in the Friedmann-Lemaître-Robertson-Walker cosmologies, without solving the equations directly and even without specifying a precise equation of state, i.e. a relation between p and ρ. In the following we will, in turn, discuss the Big Bang, the age of the universe, and its long term behaviour, from this qualitative point of view. I will then introduce the notions of critical density and density parameters, and discuss some global and causal aspects of these cosmological models (Penrose diagrams, horizons, ...).

36.1 The Past I: Big Bang (Existence of an Initial Singularity)

One amazing thing about the Friedmann-Lemaître-Robertson-Walker models is that all of them (provided that the matter content is reasonably physical - I will be more precise about this below) predict an initial singularity, commonly known as a Big Bang. This is very easy to see.

The Friedmann equation (F2),

$$-3\frac{\ddot{a}}{a} = 4\pi G_N(\rho + 3p) \quad \Leftrightarrow \quad q = \frac{4\pi G_N}{3H^2}(\rho + 3p) \ , \tag{36.1}$$

shows that, as long as the right-hand side is positive, one has $q > 0$, i.e. $\ddot{a} < 0$ so that the universe is decelerating due to gravitational attraction. This is the case for standard matter ($\rho > 0$) when it satisfies the *strong energy condition* (SEC) strictly (cf. the discussion in section 35.2, in particular (35.34)),

$$\text{strict SEC} \quad \Rightarrow \quad \rho + 3p > 0 \quad \Rightarrow \quad \ddot{a} < 0 \ . \tag{36.2}$$

It is also true for a negative cosmological constant (its negative energy density being outweighed by 3 times its positive pressure). It need not be true, however, in the presence of a positive cosmological constant which provides an accelerating contribution to the expansion of the universe. We will, for the time being, continue with the assumption that Λ is zero or, at least, non-positive, even though, as we will discuss later, recent evidence (strongly) suggests the presence of a non-negligible positive cosmological constant in our universe today (which is, however, totally irrelevant for the energy budget of the early universe).

Since $a > 0$ by definition, $\dot{a}(t_0) > 0$ because we observe a redshift, and $\ddot{a} < 0$ because $\rho + 3p > 0$, it follows that there cannot have been a turning point in the past and $a(t)$ must be concave downwards. Therefore $a(t)$ must have reached $a = 0$ at some *finite* time in the past. We will call this time $t = 0$, $a(0) = 0$.

As ρa^4 is constant for radiation (an apppropriate description of earlier periods of the universe), this shows that the energy density grows like $1/a^4$ as $a \to 0$ so this leads to quite a singular situation.

Once again, as in our discussion of black holes, it is natural to wonder at this point if the singularities predicted by General Relativity in the case of cosmological models are generic or only artefacts of the highly symmetric situations we were considering. And again there are singularity theorems applicable to these situations which state that, under reasonable assumptions about the matter content, singularities will occur independently of assumptions about symmetries.

36.2 The Past II: Age of the Universe

With the normalisation $a(0) = 0$, it is fair to call t_0 the age of the universe. If $\ddot{a}$ had been zero in the past for all $t \leq t_0$, then we would have

$$\ddot{a} = 0 \quad \Rightarrow \quad a(t) = a_0 t/t_0 \ , \tag{36.3}$$

and

$$\dot{a}(t) = a_0/t_0 = \dot{a}_0 \ . \tag{36.4}$$

This would determine the age of the universe to be

$$\ddot{a} = 0 \quad \Rightarrow \quad t_0 = \frac{a_0}{\dot{a}_0} = H_0^{-1} \ , \tag{36.5}$$

where H_0^{-1} is the *Hubble time*. However, provided that $\ddot{a} < 0$ for $t \leq t_0$ (as discussed above, this holds under suitable conditions on the matter content - which may or may not be realised in our universe), the actual age of the universe must be smaller than this,

$$\ddot{a} < 0 \quad \Rightarrow \quad t_0 < H_0^{-1} \ . \tag{36.6}$$

Thus the Hubble time H_0 sets an upper bound on the age of the universe. See Figure 46 for an illustration of this. In particular, this means that one can obtain an upper limit on the age of the universe by determining the leading (linear) term in the Hubble relation (34.104) from observations of redshifts of galaxies!

36.3 The Future: Long Term Behaviour

Let us now try to take a look into the future of the universe. Again we will see that it is remarkably simple to extract relevant information from the Friedmann equations without ever having to solve an equation.

We will assume that $\Lambda = 0$ and that we are dealing with matter with $\rho > 0$ and $w > -1/3$ (the SEC). The Friedmann equation (F1) can be written as

$$\dot{a}^2 = \frac{8\pi G_N}{3}\rho a^2 - k \ . \tag{36.7}$$

The left-hand side is manifestly non-negative. Let us see what this tells us about the right-hand side. Focus on the first term $\sim \rho a^2$. This term is strictly positive and, according to (35.95), behaves as

$$\rho a^2 \sim a^{-3(1+w)+2} = a^{-(1+3w)} \quad . \tag{36.8}$$

Thus for $w > -1/3$ the exponent is negative, so that if and when the cosmic scale factor $a(t)$ goes to infinity, one has

$$\lim_{a\to\infty} \rho a^2 = 0 \quad . \tag{36.9}$$

Now let us look at the second term on the right-hand side of (36.7), and analyse the 3 choices for k. For $k = -1$ or $k = 0$, the right hand side of (36.7) is strictly positive. Therefore $\dot{a}$ is never zero and since $\dot{a}_0 > 0$, we must have

$$\dot{a}(t) > 0 \ \ \forall\, t \quad . \tag{36.10}$$

Thus we can immediately conclude that open and flat universes must expand forever, i.e. they are open in space *and* time.

By taking into account (36.9), we can even be somewhat more precise about the long term behaviour. For $k = 0$, we learn that

$$k = 0: \qquad \lim_{a\to\infty} \dot{a}^2 = 0 \quad . \tag{36.11}$$

Thus the universe keeps expanding but more and more slowly as time goes on. By the same reasoning we see that for $k = -1$ we have

$$k = -1: \qquad \lim_{a\to\infty} \dot{a}^2 = 1 \quad . \tag{36.12}$$

Thus the universe keeps expanding, reaching a constant limiting velocity.

For $k = +1$, validity of (36.9) would lead us to conclude that $\dot{a}^2 \to -1$, but this is obviously a contradiction. Therefore we learn that the $k = +1$ universes never reach $a \to \infty$ and that there is therefore a maximal radius a_{max}. This maximal radius occurs for $\dot{a} = 0$ and therefore

$$k = +1: \qquad a_{max}^2 = \frac{3}{8\pi G_N \rho} \quad . \tag{36.13}$$

Note that intuitively this makes sense. For larger ρ or larger G_N the gravitational attraction is stronger, and therefore the maximal radius of the universe will be smaller. Since we have $\ddot{a} < 0$ also at a_{max}, again there is no turning point and the universe recontracts back to zero size leading to a *Big Crunch*. Therefore, spatially closed universes ($k = +1$) with physical matter are also closed in time. All of these findings are summarised in Figure 46.

If the cosmological constant Λ is not zero, this poetic correspondence "(open/closed) in space $\Leftrightarrow$ (open/closed) in time" is no longer necessarily true since the cosmological constant (with $\rho_\Lambda a^2 \sim a^2$) will dominate over other forms of matter for sufficiently large $a(t)$ - the asymptotic solutions in this case are given in section 37.5.

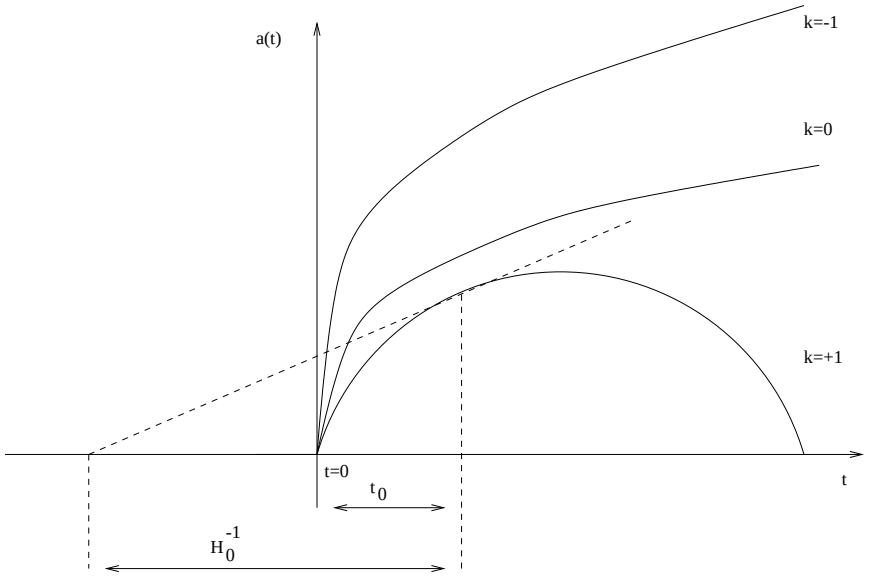


Figure 46: Qualitative behaviour of the Friedmann-Lemaître-Robertson-Walker models for $\Lambda = 0$. All models start with a Big Bang. For $k = +1$ the universe reaches a maximum radius and recollapses after a finite time. For $k = 0$, the universe keeps expanding but the expansion velocity tends to zero for $t \to \infty$ or $a \to \infty$. For $k = -1$, the expansion velocity approaches a non-zero constant value. Also shown is the significance of the Hubble time for the $k = +1$ universe showing clearly that H_0^{-1} gives an upper bound on the age of the universe.

In order to make the dependence of the Friedmann equation (F1) on the equation of state parameters w_b and on $a(t)$ more manifest, it is useful to use the conservation law (35.95,36.8) to write

$$\frac{8\pi G_N}{3}\rho_b(t)a(t)^2 = C_b a(t)^{-(1+3w_b)} \quad \Leftrightarrow \quad \Omega_b \dot{a}(t)^2 = C_b a(t)^{-(1+3w_b)} \tag{36.14}$$

for some constant C_b. Then the Friedmann equation takes the more explicit (in the sense that all the dependence on the cosmic scale factor $a(t)$ is explicit) form

$$\dot{a}^2 = \sum_b C_b a^{-(1+3w_b)} - k + \frac{\Lambda}{3}a^2 \ . \tag{36.15}$$

In addition to the vacuum energy (and pressure) provided by Λ, there are typically two other kinds of matter which are relevant in our approximation, namely matter in the form of dust ($w = 0$) and radiation ($w = 1/3$). Denoting the corresponding constants by C_m and C_r respectively, the Friedmann equation that we will be dealing with takes the form

$$(F1'') \quad \dot{a}^2 = \frac{C_m}{a} + \frac{C_r}{a^2} - k + \frac{\Lambda}{3}a^2 \ , \tag{36.16}$$

illustrating the qualitatively different contributions to the time-evolution.

One can then characterise the different eras in the evolution of the universe by which of the above terms dominates, i.e. gives the leading contribution to the equation of motion for a. This already gives some insight into the physics of the situation. We will call a universe

1. *matter dominated* if C_m/a dominates
2. *radiation dominated* if C_r/a^2 dominates
3. *curvature dominated* if k dominates
4. *cosmological constant dominated* if Λa^2 dominates

Here are some immediate consequences of the Friedmann equation (F1"):

1. No matter how small C_r is, provided that it is non-zero, for sufficiently small values of a that term will dominate and one is in the radiation dominated era. In that case, one finds the characteristic behaviour

$$\dot{a}^2 = \frac{C_r}{a^2} \quad \Rightarrow \quad a(t) = (4C_r)^{1/4} t^{1/2} \ . \tag{36.17}$$

It is more informative to trade the constant C_r for the condition $a(t_0) = a_0$, which leads to

$$a(t) = a_0 (t/t_0)^{1/2} \ . \tag{36.18}$$

2. On the other hand, if C_m dominates, one has the characteristic behaviour

$$\dot{a}^2 = \frac{C_m}{a} \quad \Rightarrow \quad a(t) = (9C_m/4)^{1/3} t^{2/3} \tag{36.19}$$

or

$$a(t) = a_0 (t/t_0)^{2/3} \tag{36.20}$$

Both this and the previous example illustrate the characteristic late-time behaviour $a(t) \to \infty$ with $\dot{a}(t) \to 0$ of the $k = 0$ models.

3. For general equation of state parameter $w \neq -1$, one similarly has

$$a(t) = a_0 (t/t_0)^h \quad , \quad h = \frac{2}{3(1+w)} \quad . \tag{36.21}$$

This describes a decelerating universe ($h(h-1) < 0 \Rightarrow 0 < h < 1$) for $w > -1/3$ and an accelerating universe ($h > 1$) for $-1 < w < -1/3$. This evidently reflects, and agrees with, the fact that from (F2) one has $\ddot{a} < 0$ for $\rho + 3p > 0$ and $\ddot{a} > 0$ for $\rho + 3p < 0$.

4. For the special case $w = -1/3$, one has $h = 1$ and thus the linear evolution $a(t) \sim t$. Since, as noted in (36.38) below, one can formally attribute an equation of state parameter $w_k = -1/3$ to the curvature contribution to the Friedmann equation, this solution arises not only for an exotic matter component with $w = -1/3$ and $k = 0$, but also for an empty universe with $k = -1$. We will look at the latter (the *Milne universe*) in more detail in section 37.1 (evidently an empty universe with $k = +1$, governed by $\dot{a}^2 = -1$, is not possible).

In spite of sharing the same Friedmann equation (F1') and the same solution $a(t)$, these two universes with $w = -1/3$ are decidedly not identical for the obvious reason that one is empty and the other one is not, and thus they solve the Einstein equations with very different energy-momentum tensors (alternatively, e.g. their Misner-Sharp masses (35.118) are different). More dramatically, the scalar curvature $R(t)$ for $a(t) = t$ is

$$R(t) = \frac{6}{a^2}(a\ddot{a} + \dot{a}^2 + k) = \frac{6(1+k)}{t^2} \quad . \tag{36.22}$$

Thus there is a singularity at $t = 0$ for $k = 0$ while $R(t) = 0$ for the Milne metric with $k = -1$ (which turns out to be just Minkowski space in disguise).

Note, as an side, that one cannot conclude from the fact alone that one has $k = -1$ and the other one has $k = 0$ that they are different, since it is possible that a given universe can be foliated in different ways by spatial hypersurfaces with different curvatures. An example of this is provided by the de Sitter universe, the solution to the Friedmann equations with a positive cosmological constant (and no other matter) - see section 37.5.

5. For sufficiently large a, the cosmological constant Λ, if not identically zero, will always dominate, no matter how small the cosmological constant may be, as all the other energy-content of the universe gets more and more diluted. In particular, for $k = 0$, the Friedmann equation for a positive cosmological constant reduces to

$$\dot{a}^2 = (\Lambda/3)a^2 \quad \Rightarrow \quad a(t) = a(t_0)\mathrm{e}^{\pm\sqrt{\Lambda/3}(t-t_0)} \ , \tag{36.23}$$

with Hubble parameter

$$H(t) = \frac{\dot{a}}{a} = \pm\sqrt{\Lambda/3} = H_0 \ . \tag{36.24}$$

This gives the $k = 0$ metric of de Sitter space,

$$ds^2 = -dt^2 + \mathrm{e}^{2H_0 t}d\vec{x}^2 \ . \tag{36.25}$$

6. Only for $\Lambda = 0$ does k dominate for large a and one obtains, as we saw before, a constant expansion velocity (for $k = 0, -1$).

7. We will find and discuss various other exact solutions in section 37.

36.5 Density Parameters and the Critical Density

The primary purpose of this section is to introduce some convenient and commonly used notation and terminology in cosmology associated with the Friedmann equation (F1'). We will now include the cosmological constant in our analysis. For starters, however, let us again consider the case $\Lambda = 0$ (or include ρ_Λ as one contribution to ρ). (F1') can be written as

$$\frac{8\pi G_N \rho}{3H^2} - 1 = \frac{k}{a^2 H^2} \ . \tag{36.26}$$

If one defines the *critical density* ρ_{cr} by

$$\rho_{cr} = \frac{3H^2}{8\pi G_N} \ , \tag{36.27}$$

and the *density parameter* Ω by

$$\Omega = \frac{\rho}{\rho_{cr}} = \frac{8\pi G_N \ \rho}{3H^2} \ , \tag{36.28}$$

then (F1') becomes

$$\Omega - 1 = \frac{k}{a^2 H^2} \tag{36.29}$$

Thus the sign of k is determined by whether the actual energy density ρ in the universe is greater than, equal to, or less than the critical density,

$$\begin{array}{ccccccc} \Omega < 1 & \Leftrightarrow & \rho < \rho_{cr} & \Leftrightarrow & k = -1 & \Leftrightarrow & \text{open} \\ \Omega = 1 & \Leftrightarrow & \rho = \rho_{cr} & \Leftrightarrow & k = 0 & \Leftrightarrow & \text{flat} \\ \Omega > 1 & \Leftrightarrow & \rho > \rho_{cr} & \Leftrightarrow & k = +1 & \Leftrightarrow & \text{closed} \end{array}$$

Note that, in particular, $\rho_{cr} = 0$ at a turning point (maximal radius, $H = 0$), and correspondingly $\Omega \to \infty$ there.

This can be generalised to several species of (not mutually interacting) matter, characterised by equation of state parameters w_b, subject to the condition $w_b > 0$ or $w_b > -1/3$, with density parameters

$$\Omega_b = \frac{\rho_b}{\rho_{cr}} \ . \tag{36.30}$$

The total matter contribution Ω_M is then

$$\Omega_M = \sum_b \Omega_b \ . \tag{36.31}$$

Along the same lines we can also include the cosmological constant Λ. Inspection of the Friedmann equations reveals that the presence of a cosmological constant is equivalent to adding matter $(\rho_\Lambda, p_\Lambda)$ with

$$\Lambda \quad \Leftrightarrow \quad \rho_\Lambda = -p_\Lambda = \frac{\Lambda}{8\pi G_N} \quad w_\Lambda = -1 \ . \tag{36.32}$$

in agreement with what we had already deduced in (35.31). Note that this identification is consistent with the conservation law (F3), since Λ is constant.

Then the Friedmann equation (F1') with a cosmological constant can be written as

$$(F1') \quad \Leftrightarrow \quad \Omega_M + \Omega_\Lambda = 1 + \frac{k}{a^2 H^2} \ , \tag{36.33}$$

where

$$\Omega_\Lambda = \frac{\rho_\Lambda}{\rho_{cr}} = \frac{\Lambda}{3H^2} \ . \tag{36.34}$$

One can also formally attribute a density parameter

$$\Omega_k = -\frac{k}{a^2 H^2} \tag{36.35}$$

to the curvature, so that the Friedmann equation (F1) can now succinctly (if somewhat obscurely) be written as the condition that the sum of all density parameters be equal to 1,

$$(F1') \quad \Leftrightarrow \quad \Omega_M + \Omega_\Lambda + \Omega_k = 1 \ . \tag{36.36}$$

Note that the Friedmann equation can thus be regarded as a kind of balance equation among the contributions of matter, dark energy and (negative) curvature: if you have more of one, you need less of the other and vice versa.

The corresponding curvature energy density ρ_k can now be determined from

$$\Omega_k = \frac{8\pi G_N}{3H^2}\rho_k \stackrel{!}{=} -\frac{k}{a^2 H^2} \ , \tag{36.37}$$

and from (F2), which does not depend on k, one deduces that the corrsponding pressure is $p_k = w_k \rho_k$ with $w_k = -1/3$, so that $\rho_k + 3p_k = 0$. Thus the curvature contribution can be described as

$$k \quad \Leftrightarrow \quad \rho_k = -3p_k = \frac{-3k}{8\pi G_N a^2} \quad w_k = -1/3 \ . \tag{36.38}$$

The continuity equation (F3) is identically satisfied in this case (or, if you prefer, requires that k is constant).

Finally, the 2nd order equation (F2') can also be written in terms of the density parameters,

$$q = \tfrac{1}{2} \sum_b (1 + 3w_b)\Omega_b - \Omega_\Lambda \ . \tag{36.39}$$

Denoting the values of the parameters today, at time $t = t_0$, by a subscript 0, the two key equations relating the cosmological parameters are

$$\begin{aligned} (\Omega_M)_0 + (\Omega_\Lambda)_0 + (\Omega_k)_0 &= 1 \\ \tfrac{1}{2}(1 + 3w_0)(\Omega_M)_0 - (\Omega_\Lambda)_0 &= q_0 \end{aligned} \tag{36.40}$$

Clearly, it is of utmost importance to determine the various contributions ρ_a to the matter density ρ of the universe (and to determine ρ_{cr} e.g. by measurements of the Hubble parameter $H(t)$, i.e. of H_0, q_0 etc.). From H_0 and q_0 and $(\Omega_M)_0$ one can then in principle determine $(\Omega_\Lambda)_0$ and $(\Omega_k)_0$.

36.6 Causal Structure and Penrose Diagrams for $k = 0$

In order to gain a better understanding of a space-time, it is always useful to study its null geodesics, the behaviour of lightcones and thus its causal structure. In this section, as a first step towards this, we will shed light on the global structure of (spatially flat, $k = 0$) cosmological space-times by constructing Penrose diagrams for them. The reason for the restriction to $k = 0$ is that we already know the exact solution to the Friedmann equations in this case, and with a single matter component characterised by the equation of state parameter w, namely

$$ds^2 = -dt^2 + a(t)^2 d\vec{x}^2 \tag{36.41}$$

with (36.21)

$$k = 0 \quad , \quad p = w\rho \quad \Rightarrow \quad a(t) = a_0 (t/t_0)^h \quad , \quad h = \frac{2}{3(1+w)} \tag{36.42}$$

(for $w \neq -1$). The case of a cosmological constant ($w = -1$), i.e. (anti-)de Sitter space, needs to be treated separately. Penrose diagrams for some other ($k = +1$) solutions will be given in section 37.

One crucial feature that will emerge from this analysis is the possible presence of *cosmological horizons* which delimit the regions of space-time that can be in causal contact at a given time or that are visible at a given time or in principle in the infinite future. This will be discussed further in section 36.7 below. This will also allow us to then, in sections 36.8 and 36.9, better understand the significance (or lack thereof) of the Hubble sphere or Hubble radius $R_H(t)$ (34.40) introduced in section 34.3.

For later reference, we note that the acceleration is

$$\ddot{a}(t) = h(h-1)\frac{a(t)}{t^2} \quad , \quad h(h-1) = -\frac{2(w+1/3)}{3(1+w)^2} \quad , \tag{36.43}$$

so that

$$h(h-1) \qquad \begin{cases} < 0 \ \text{(decelerating)} & \text{for } \ 0 < h < 1 \quad w > -1/3 \\ > 0 \ \text{(accelerating)} & \text{for } \ h > 1 \qquad w < -1/3 \end{cases} \tag{36.44}$$

Now the 1st step will be the introduction of the conformal time coordinate η (cf. section 34.5) defined by

$$d\eta(t) = dt/a(t) \tag{36.45}$$

in terms of which the metric can be written as

$$ds^2 = a(\eta)^2(-d\eta^2 + d\vec{x}^2) \ . \tag{36.46}$$

This makes it manifest that the metric is conformally flat, and thus its causal structure is particularly easy to analyse. All we need to pay attention to is the range of η.

In the case at hand, we have

$$a(t) = a_0 t_0^{-h} t^h \quad \Rightarrow \quad d\eta = a_0^{-1} t_0^h t^{-h} dt \tag{36.47}$$

and therefore

$$\eta(t) = a_0^{-1} t_0^h t^{1-h}/(1-h) \tag{36.48}$$

for $h \neq 1$ and

$$\eta(t) = a_0^{-1} t_0^h \log|t| \tag{36.49}$$

for $h = 1, w = -1/3$. Since the range of t is $(0, \infty)$, the range of η is

$$t \in (0, +\infty) \quad \Rightarrow \quad \begin{cases} \eta \in (0, +\infty) & \text{for } \ 0 < h < 1 \\ \eta \in (-\infty, +\infty) & \text{for } \ h = 1 \\ \eta \in (-\infty, 0) & \text{for } \ h > 1 \ . \end{cases} \tag{36.50}$$

This shows that the causal structure and Penrose diagram of the space-time for $h = 1, w = -1/3$ are identical to that of Minkowski space in Figure 23, with the only crucial difference that there is now a singularity at i^-, i.e. at $t \to 0, \eta \to -\infty$ (recall that the scalar curvature is $R(t) \sim t^{-2}$ (36.22)).

For $h \neq 1$, on the other hand, as indicated in Figure 48, the Penrose diagram is given by the upper (lower) half of the Minkowski Penrose diagram respectively, with the addition of the initial spacelike singularity. This has drastic implications for the global causal structure (and the existence of cosmological horizons) to be explored below.

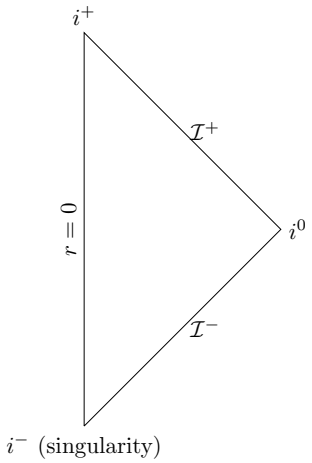


Figure 47: Penrose Diagram for the $k = 0, w = -1/3$ solution.

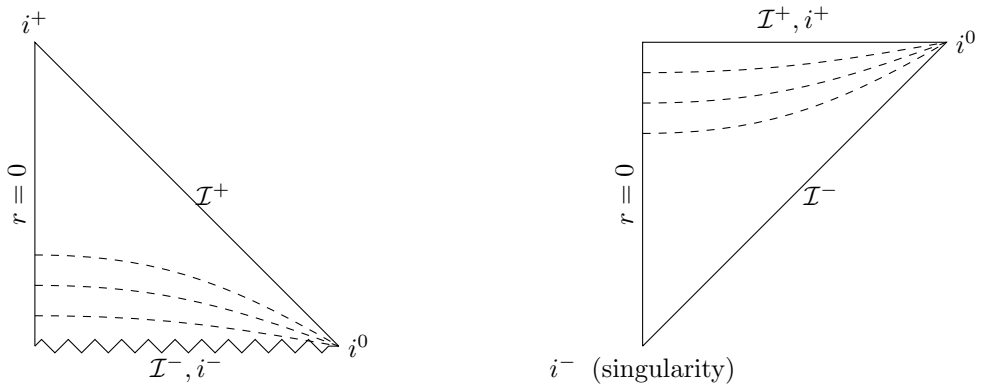


Figure 48: Penrose Diagram for the $k = 0, w \neq -1/3, -1$ solutions. On the left, the diagram for decelerating solutions with $0 < h < 1$, $w > -1/3$, on the right that for acelerating solutions with $h > 1$. Also schematically indicated (as dashed lines) are some surfaces of constant time t or η.

36.7 Lightrays and Cosmological Particle and Event Horizons

We now turn to one particularly relevant implication of the preceding analysis, namely the existence of (observer-dependent) *horizons* in cosmological space-times. Thus we will mainly consider the spatially flat case $k = 0$ in the following, but in principle everything can easily and immediately be extended to $k \neq 0$, e.g. by working in the polar coordinates (34.10), with $r \to \psi$. In the following, the term "comoving distance" will refer to the coordinate distance as measured by the comoving radial coordinate r.

Let us first (re-)analyse the evolution of lightrays in the Robertson-Walker geometry. We have already studied one aspect of light propagation in Robertson-Walker geometries in section 34.8, namely the cosmological redshift. Here we will look at the propagation

of light from a more global perspective (or at least with more global intentions).

By spatial isotropy, we can without loss of generality choose the lightrays to be purely radial. Then they are governed by the equation

$$dt^2 = a(t)^2 dr^2 \quad \Leftrightarrow \quad \frac{dr_\gamma(t)}{dt} = \pm\frac{1}{a(t)} \ . \tag{36.51}$$

Here, in order to avoid a proliferation of objects with the same anonymous name r, we call the comoving photon path $r_\gamma(t)$. Since $a(t)$ is non-negative, lightrays propagate either in the direction of increasing comoving radial coordinate distance r, or in the direction of decreasing r. In particular, lightrays coming towards us (at $r = 0$) and reaching $r = 0$ at $t = t_0$ are governed by the equation

$$\frac{dr_\gamma(t)}{dt} = -\frac{1}{a(t)} \quad \Rightarrow \quad r_\gamma(t) = \int_t^{t_0} \frac{dt}{a(t)} \ . \tag{36.52}$$

We will occasionally also denote this solution by

$$r_\gamma(t) \equiv r_\gamma(t; t_0) \quad \text{with} \quad r_\gamma(t_0, t_0) = 0 \ , \tag{36.53}$$

when we want to make the dependence of the solution on t_0 more explicit. We see that this solution can be directly expressed in terms of conformal time η as

$$r_\gamma(t; t_0) = \eta(t_0) - \eta(t) \ . \tag{36.54}$$

One can also, if one wishes, express light propagation in terms of the (instantaneous) proper spacelike distance

$$R_\gamma(t) = a(t) r_\gamma(t) \ , \tag{36.55}$$

i.e. the spatial proper distance to the photon as measured by the maximally symmetric spatial geometry on the slice of constant time t but, as we will see in sections 36.8 and 36.9, this is not without its pitfalls.

As an example, for matter with the equation of state parameter $w \neq -1$ and cosmic scale factor $a(t) = a_0(t/t_0)^h$ (36.21), the equation for lightrays is

$$dr_\gamma(t)/dt = -t_0^h t^{-h}/a_0 \quad , \quad h = \frac{2}{3(1+w)} \ . \tag{36.56}$$

Either from this, or from the result for $\eta(t)$ from (36.48), one finds that for $h \neq 1$

$$r_\gamma(t) = \frac{t_0}{a_0(1-h)}(1 - (t/t_0)^{1-h}) \ . \tag{36.57}$$

and

$$R_\gamma(t) = \frac{1}{1-h}(t_0^{1-h} t^h - t) \tag{36.58}$$

(while for $h = 1$ one finds a logarithmic behaviour). We now look at 2 limiting cases of these lightrays.

1. Cosmological Particle Horizon

Backtracking lightrays that reach us today, the largest comoving distance of an object about which we can have received any (causal) information by today or that can have had any causal influence on us is given by

$$r_{ph}(t_0) = \lim_{t\to 0_+} r_\gamma(t;t_0) = \eta(t_0) - \eta(0) \quad . \tag{36.59}$$

(with $t = 0$ the time of the initial singularity). If this integral converges, this means that there is a region of the universe, namely that at comoving distance $r > r_{ph}(t_0)$, with which we could not have had any causal contact until today. In that case $r_{ph}(t_0)$ defines the *cosmological particle horizon* (or just *particle horizon*).

Expressed in terms of conformal time, whether or not there is a particle horizon is equivalent to the question if $\eta(t)$ is finite or diverges as $t \to 0$. From the analysis of the previous section for the single-component models with equation of state $p = w\rho$, $w \neq -1$, we can thus conclude that

- for the non-decelerating models, $h \geq 1$, one has $\eta(t) \to -\infty$, and therefore the space-time is causally equivalent to Minkowski space in the past, and there is no particle horizon;
- for the decelerating models with $0 < h < 1$, $w > -1/3$, or those that are dominated by ordinary matter or radiation at early times, say, $\eta(0)$ is finite, the causal structure is as in the left-hand diagram of Figure 48 and there is a particle horizon.

Remarks:

(a) It is also common to express the particle horizon in terms of

$$R_{ph}(t_0) = a(t_0) r_{ph}(t_0) = a(t_0) \int_0^{t_0} \frac{dt}{a(t)} \quad , \tag{36.60}$$

giving the present instantaneous proper distance to the particle horizon. At any time t_0, the particle horizon is a spatial 2-sphere with radius $R_{ph}(t_0)$ around an observer, no information being available about objects outside that 2-sphere. This particle horizon is observer-dependent, i.e. other comoving observers at other values of r will also have a particle horizon, but it will be a 2-sphere of radius $R_{ph}(t_0)$ around them.

(b) As (36.59) shows, $r_{ph}(t_0)$ grows monotonically with t_0 (so that more and more of the universe becomes visible as time goes on). A good way of visualising the particle horizon is to draw $r_{ph}(t)$ as a function of t in a (t,r)- or (η,r)-diagram.[155] For a comoving observer at $r = r_0$, this is simply the future

[155] Note that often the particle horizon is depicted in a different and somewhat less informative, or at

lightcone of the point $(0, r_0)$, and the particle horizon size at any given time t_0 is given by the intersection of this lightcone null surface with the spacelike constant $t = t_0$ surface.

(c) This is indicated in a Penrose diagram in Figure 49. It is evident from this diagram that the past lightcone of an event at some finite time $\eta > \eta(0)$ cannot cover the entire space, and that at early times suffficiently spatially separated events have non-intersecting past lightcones, i.e. have never been in causal contact in the past. We will come back to one problematic aspect and consequence of this in section 38.3 in our brief discussion of the (aptly) so-called horizon problem in cosmology.

In interpreting this diagram, it is also good to keep in mind the highly distorted nature of constant time hypersurfaces at late times, which all end up at i^0. Thus the particle horizon really reaches infinity $\mathcal{I}^+$ at $t \to \infty$ or $\eta \to +\infty$.

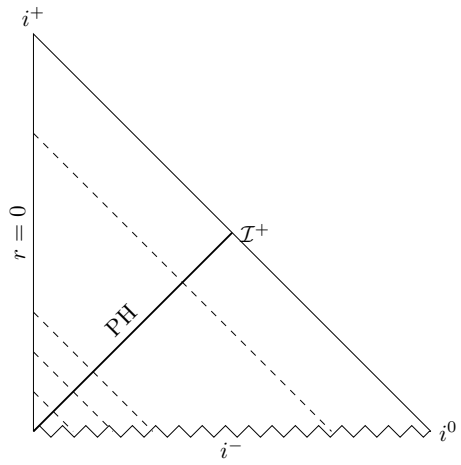


Figure 49: Cosmological Particle Horizon: indicated are the past lightcones of a comoving observer at $r = 0$ (dashed lines) as well as that observer's particle horizon (thick solid line).

This diagram also shows that the particle horizon can be regarded as the boundary of the region that can be influenced by the observer.

(d) One can of course also define these horizons for the models with $k \neq 0$. The case $k = +1$ is different in the sense that space is compact. E.g. the range of ψ in

$$ds^2 = a(\eta)^2(-d\eta^2 + d\psi^2 + \sin^2\psi d\Omega^2) \tag{36.61}$$

least potentially more confusing, way - see Figures 1 and 3 and section 3.4 of T. Davis, C. Lineweaver, *Expanding Confusion: common misconceptions of cosmological horizons and the superluminal expansion of the Universe*, `arXiv:astro-ph/0310808` for a nice discussion and illustration of this.

is finite, so that even if there is a particle horizon initially, it may be possible to see all of space (and later on all around the space at the back of one's head) at a later time, namely when the η-life-time of the universe is $\geq \pi$ (respectively $\geq 2\pi$). An example of this is provided by the matter dominated $k = +1$ universe, whose conformal diagram is given in Figure 51 of section 37.3.

(e) The particle horizon is a horizon in the sense that it is null and acts as a one-way membrane: any object that is inside the particle horizon or has crossed the particle horizon towards us will remain inside the particle horizon at all future times. This is also evident from the diagram.

2. Cosmological Event Horizon

From the above discussion it is also clear that the cosmological space-time is causally inequivalent to Minkowski space in the far future if η is bounded from above as $t \to \infty$. If it is bounded, then the past lightcone of an observer will never be able to cover all of space-time even as $t \to \infty$, i.e. there will be regions of space-time from which that observer can never ever receive any information, and in this case the past lightcone is known as a *cosmological event horizon*.

The radial coordinate size of the past lightcone from $t_0 = \infty$ at some time t is given by the future partner of (36.59), namely

$$r_{eh}(t) = \lim_{t_0 \to \infty} r_\gamma(t; t_0) = \eta(\infty) - \eta(t) \quad , \tag{36.62}$$

and there is an event horizon iff this integral is finite, i.e. iff $\eta(\infty)$ is finite.

Going back to our $k = 0$ single-component matter example, it is now precisely the accelerating cosmologies with $w < -1/3$ that exhibit an event horizon. This is iluustrated in Figure 50.

REMARKS:

(a) Since current observations suggest that our universe will be dominated by a positive cosmological constant in the future, i.e. that it will be asymptotically de Sitter in the future, while it was dominated by standard types of matter in the past, the standard hot big bang / Λ-CDM model of our universe has both a particle horizon and an event horizon.

(b) This cosmological event horizon (or the issue whether or not one exists) is not particularly relevant for observational cosmology today (or in the foreseeable future), but it is of great theoretical interest since, at least in the (asymptotically) de Sitter case, in some respects this (again observer-dependent) horizon appears to have more in common with a true black hole horizon

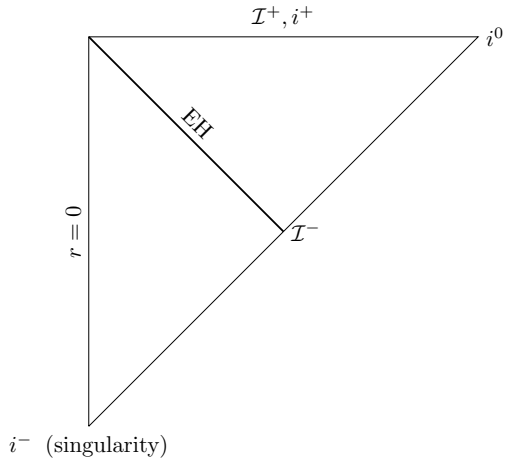


Figure 50: Cosmological Event Horizon: the region outside the event horizon (above the line EH) is invisible to a comoving observer at $r = 0$.

(like a temperature and an entropy) than one would have had the right to expect.[156]

(c) The event horizon is a horizon in the sense that it is null and acts as a one-way membrane: any object that is outside the event horizon or has crossed the event horizon away from us will remain outside the event horizon at all future times. Given that the event horizon is the past lightcone at a point at future infinity, these statements are almost a tautology, and are also evident from the diagram.

36.8 Lightrays and the Hubble Sphere

Recall that the Hubble radius $R_H(t)$ was introduced in section 34.3 as the (instantaneous) proper distance at which the recessional velocity (34.29)

$$V_{rec}(t) \equiv \frac{d}{dt} R_p(t) = H(t) R_p(t) \tag{36.63}$$

of a comoving object with $R_p(t) = a(t) r_1$ equals the speed of light,

$$V_{rec}(t) = c \quad \Leftrightarrow \quad R_H(t) = c/H(t) = ca(t)/\dot{a}(t) \tag{36.64}$$

For a contracting universe this should of course be replaced by $R_H = -c/H = c/|H|$, but until further notice we will consider the expanding case in the following (and we will from now on set $c = 1$ again).

[156] See e.g. M. Spradlin, A. Strominger, A. Volovich, *Les Houches Lectures on de Sitter Space*, `arXiv:hep-th/0110007` and references therein.

I also mentioned in section 34.3 that the Hubble sphere is occasionally also referred to as the Hubble horizon. Let us see if, or to which extent, it has such a role and how it is related to the cosmological particle and event horizons discussed in the previous section.

The most direct connection between the Hubble radius and the propgation of light arises from describing a photon path via its proper distance $R_\gamma(t)$ (36.55). This proper distance of an (out- or ingoing) lightray changes with time according to

$$\frac{d}{dt}R_\gamma(t) = \dot{a}(t)r_\gamma(t) \pm 1 = \frac{R_\gamma(t)}{R_H(t)} \pm 1 \ . \tag{36.65}$$

Thus expressed (or plotted) in terms of proper distance, an ingoing lightray exhibits (or appears to exhibit) quite a different behaviour from the strictly monotonous $\dot{r}_\gamma < 0$ evolution of $r_\gamma(t)$. In particular, for such ingoing lightrays the proper distance reaches a maximum at a time $t = t_m$ when

$$r_\gamma(t_m) = 1/\dot{a}(t_m) = r_H(t_m) \quad \Leftrightarrow \quad R_\gamma(t_m) = R_H(t_m) \ , \tag{36.66}$$

i.e. when the lightray crosses the Hubble sphere, and then decreases towards $r = 0$. Here

$$r_H(t) = R_H(t)/a(t) = 1/\dot{a}(t) \ . \tag{36.67}$$

is the time-dependent comoving radial coordinate of the Hubble radius. This behaviour can intuitively (but with caution) be attributed to the fact that at the time when the lightray crosses the Hubble radius $R_H(t)$ it has recessional velocity (i.e. away from us) equal to c and is thus momentarily at rest with respect to us.

Before discussing how (not) to interpret this result, let us determine how the Hubble radius evolves with time. The evolution equation for $R_H(t)$ is

$$\frac{d}{dt}R_H(t) = \frac{d}{dt}(a(t)/\dot{a}(t)) = 1 + q(t) \ , \tag{36.68}$$

while that for $r_H(t)$ is

$$\frac{d}{dt}r_H(t) = \frac{d}{dt}(1/\dot{a}(t)) = q(t)/a(t) \ . \tag{36.69}$$

Using (36.39) in the schematic form

$$q(t) = \tfrac{1}{2}(1 + 3w)\Omega_M - \Omega_\Lambda \tag{36.70}$$

and

$$k = 0 \quad \Rightarrow \quad \Omega_M + \Omega_\Lambda = 1 \ , \tag{36.71}$$

these equations can be written as

$$\begin{aligned} \frac{d}{dt}R_H(t) &= \tfrac{3}{2}(1 + w)\Omega_M \\ a(t)\frac{d}{dt}r_H(t) &= \tfrac{3}{2}(1 + w)\Omega_M - 1 \ . \end{aligned} \tag{36.72}$$

In particular, if one has a single species of matter (or just a cosmological constant), i.e. $\Omega_M = 1$ (or $\Omega_M = 0$) one has

$$q = \tfrac{1}{2}(1+3w) \ , \tag{36.73}$$

which is constant, and

$$\begin{aligned} \frac{d}{dt}R_H(t) &= \tfrac{3}{2}(1+w) \\ a(t)\frac{d}{dt}r_H(t) &= \tfrac{1}{2}(1+3w) \ . \end{aligned} \tag{36.74}$$

This implies that the Hubble radius increases monotonically in time for

$$\begin{aligned} w > -1 \quad &\Rightarrow \quad \frac{d}{dt}R_H(t) > 0 \\ w > -\frac{1}{3} \quad &\Rightarrow \quad \frac{d}{dt}r_H(t) > 0 \ . \end{aligned} \tag{36.75}$$

Evidently for $w > -1$ and $a(t) \sim t^h$ the solution for $R_H(t)$ reproduces the definition $R_H = a/\dot{a}$,

$$w > -1 \quad \Rightarrow \quad R_H(t) = \tfrac{3}{2}(1+w)t = t/h \ , \tag{36.76}$$

while for $w = -1$,

$$w = -1 \quad \Rightarrow \quad R_H(t) = R_H(t_0) = 1/H_0 \ . \tag{36.77}$$

With these results at our disposal, we can now make the following elementary observations:

1. Since $R_H(t)$ is non-decreasing for $w \geq -1$, it is a triviality that the maximal proper distance $R_\gamma(t_m) = R_H(t_m)$ a lightray reached in the past that arrives today is bounded from above by the value of the Hubble radius today,

   $$R_\gamma(t_m) = R_H(t_m) < R_H(t_0) \ . \tag{36.78}$$

   (It is also easy to sharpen this upper bound somewhat for $w > -1/3$, but we will not need this).

2. Thus we see that the objects whose light we receive today at $r = 0$ cannot be further away from us today than at a proper distance $R_H(t_0)$ so that the Hubble radius $R_H(t_0)$ provides some kind of (rough) upper bound on the distance of such objects. This may suggest (to some) that therefore the Hubble radius provides a limit to what we can see (or can have seen) of the universe at time t_0.[157] However, this appears to me to be at best an extremely misleading way of phrasing things.

[157] See e.g. F. Melia, *The Cosmic Horizon*, `arXiv:0711.4181 [astro-ph]`, or the plots and numerical calculations performed in O. Bikwa, F. Melia, A. Shevchuk, *Photon Geodesics in FRW Cosmologies*, `arXiv:1112.4774 [astro-ph.CO]` in order to confirm (36.78) (which requires no numerical confirmation) in some examples.

In fact, even though $R_\gamma(t)$ reaches a maximum at $t = t_m$, this is not the case for the comoving radial coordinate $r_\gamma(t)$ along the photon path. The monotonous behaviour $\dot{r}_\gamma < 0$ for ingoing lightrays implies that

$$t < t_m \quad \Rightarrow \quad r_\gamma(t) > r_\gamma(t_m) \tag{36.79}$$

so that information about objects at larger distances $r > r_H(t_m)$ than the Hubble radius can easily reach us. This distance is bounded only by the *particle horizon* $r_{ph}(t)$ which is what one gets when one tracks $r_\gamma(t)$ back to $t = 0$. Therefore it is precisely the particle horizon which tells us about which points / comoving objects in the constant time spatial surface we can already have obtained information, and not the Hubble radius.

3. The metric induced on the Hubble surface Σ_H defined by $r = r_H(t)$ is

$$\begin{aligned} ds^2|_{\Sigma_H} &= -dt^2 + a(t)^2(dr_H(t)^2 + r_H(t)^2 d\Omega^2) \\ &= -(1 - q(t)^2)dt^2 + R_H(t)^2 d\Omega^2 \ . \end{aligned} \tag{36.80}$$

Therefore this surface Σ_H is null iff $q = \pm 1$. In these two cases,

$$q = -1 \quad \Rightarrow \quad w = -1 \quad , \quad q = +1 \quad \Rightarrow \quad w = +\frac{1}{3} \ , \tag{36.81}$$

(i.e. a pure positive cosmological constant or radiation) the Hubble radius agrees with the event horizon or the particle horizon respectively,

$$\begin{aligned} w = -1 \quad &\Rightarrow \quad R_H(t) = R_{eh}(t) = R_H(t_0) \\ w = +\frac{1}{3} \quad &\Rightarrow \quad R_H(t) = R_{ph}(t) \ . \end{aligned} \tag{36.82}$$

In a universe which is dominated by radiation in the distant past and a cosmological constant in the far future, the Hubble radius will interpolate between the particle hoirzon in the past and the event horizon in the future, but it has no particular significance inbetween.

4. Indeed, more generally one has

$$\Sigma_H \quad \text{is} \quad \begin{cases} \text{spacelike} & \text{for} & w > 1/3 \\ \text{null} & \text{for} & w = 1/3 \\ \text{timelike} & \text{for} & -1 < w < 1/3 \\ \text{null} & \text{for} & w = -1 \\ \text{spacelike} & \text{for} & w < -1 \end{cases} \tag{36.83}$$

Thus for the physically relevant intermediate range $-1 < w < 1/3$ the Hubble surface is timelike, causally there is nothing strange or interesting going on there, and nothing prevents one from crossing it mulitple times in both directions. In particular it cannot and will not act like a one-way membrane.

36.9 Hubble "Horizon": Is our Universe (like) a Black Hole?

Hinchcliff's rule:
Whenever the title of a paper (section) is a question with a yes/no answer, the answer is "no".

In the previous section we collected some elementary properties of the Hubble radius (Hubble sphere). In particular, we saw that the Hubble sphere is only null in 2 special cases, in which it coincides with the particle or event horizon respectively.

Nevertheless, in spite of these elementary and well-known facts, strangely there is some debate in the current literature about the significance of the Hubble "horizon" (or lack thereof), and even the bizzarre idea that the "visible universe" (defined by $R_H(t)$) is somehow like the inside of a black hole.

As far as I can tell one of the contributing factors to this is the fact that in cosmology, distances to distant objects are commonly expressed in terms of their (somewhat fictitious) instantaneous proper spacelike distance $R(t_0) = a(t_0)r$ from us today, and not (for instance) in terms of their (approximately constant) comoving coordinate distance, or some measure of distance at the time the objects emitted the light that we receive today. In principle, this is perfectly fine, of course, and with due care myths about the Hubble radius can also be exorcised from this point of view.[158]

In practice, however, this use of $R(t)$ leads to a somewhat allegorical (and therefore potentially misleading) way of talking about perfectly mundane things:

- For example, as we have seen above in our discussion of lightrays, one can say (referring to $R_\gamma(t)$) that lightrays reach a maximum distance and then turn around to come towards us, but viewed in terms of comoving coordinates the lightrays just continue in the direction of decreasing r. Thus care needs to be taken to separate the physical motion of objects through space from artefacts arising from describing them in terms of their instantaneous proper spacelike distance.

- Let us consider the spatially flat decelerating case $k = 0$ and $w > -1/3$, so that one has $a(t) \sim t^h$ for some $0 < h < 1$. Then

$$a(t) \sim t^h \quad \Rightarrow \quad R_H(t) = t/h \ . \tag{36.84}$$

  In this setting one frequently encounters the following kind of argument to "explain" from this point of view why light can reach us from outside the Hubble sphere: for $0 < h < 1$ the Hubble sphere expands faster than the universe; thus

[158] See e.g. P. van Oirschot, J. Kwan, G. Lewis, *Through the Looking Glass: Why the "Cosmic Horizon" is not a horizon*, `arXiv:1001.4795 [astro-ph.CO]`, G. Lewis, P. van Oirschot, *How does the Hubble Sphere limit our view of the Universe?*, `arXiv:1203.0032` for illuminating discussions and enjoyable dissections of these issues from this perspective.

light emitted from a receding galaxy initially outside the Hubble sphere can eventually be overtaken by the Hubble radius and can then become visible to us. True, and very figurative, in terms of some radius overtaking some other radius, but this totally obscures the fact that the Hubble sphere has nothing to do with this and that the only thing that matters for what is visible to us today is if the object is inside the particle horizon or not, i.e. has comoving coordinate $r < r_{ph}(t_0)$ or not.

The above argument also seems to suggest that somehow things change when one has an accelerating universe with $h > 1$, and this is again true, but the thing that changes is that (as we have seen) for $h > 1$ there is simply no particle horizon and therefore no obstruction to seeing objects at any time. This simple fact is again obscured by the above argument.

Nevertheless, as I mentioned already in section 34.3, some people (and not just laymen who understand neither cosmology nor black holes) appear to be obsessed with the idea that our visible universe, defined (counterfactually, as we saw above) as the interior of the Hubble sphere, is somehow like the inside of a Schwarzschild black hole, the Hubble sphere playing the role of its horizon.

This is usually based on some variant of one of the following arguments.[159]

1. Friedmann Equation and the Schwarzschild Radius

   The Friedmann equation
   $$H^2 = \frac{8\pi G_N}{3}\rho - \frac{k}{a^2} \tag{36.85}$$
   can (by dividing by H^3) be written as
   $$R_H = 2G_N \frac{4\pi R_H^3}{3}\rho - \frac{k}{a^2 H^3} \quad . \tag{36.86}$$
   Therefore the statement that $k = 0$ is equivalent to the statement that the Hubble radius R_H is equal to the (would-be) Schwarzschild radius $R_s = 2G_N M$ associated with the total mass
   $$M(R_H) = \frac{4\pi R_H^3}{3}\rho \tag{36.87}$$
   contained in the Hubble sphere,
   $$k = 0 \quad \Leftrightarrow \quad R_H = R_s = 2G_N\, M(R_H) \quad . \tag{36.88}$$
   There may be something profound in this, I don't know, but just saying "hey, it's the Schwarzschild radius, hence I have a black hole" is not!

[159] See e.g. F. Melia (footnote 157) or F. Melia, M. Abdelquader, *The Cosmological Spacetime*, `arXiv:0907.5394 [astro-ph.CO]`.

- In particular, as we have seen above, for reasonable equations of state the "horizon" $r = r_H$ is timelike and we (and lightrays) can simply leave this would-be interior of the black hole. The region we cannot in principle escape from is that bounded by the particle horizon, but the above numerology does not work in that case (unless $w = 1/3$, so that $R_{ph} = R_H$).
- Moreover, as discussed at length in section 32, a black hole is not defined by "mass m enclosed in a sphere of radius $r < 2m$" but by reference to the outside of the black hole and the existence of an *event horizon* and the fact that there is a region of space-time causally sealed off from the asymptotic region.
- Other quasi-local characterisations of black hole-like objects also do not refer directly to mass and radius but to trapped surfaces and the behaviour of lightrays (and more on this below).
- In particular, we will see below that for $k = 0$ (but only for $k = 0$), the Hubble "horizon" is an apparent horizon, but one with properties that are quite different to those of the apparent horizon of a black hole.[160]
- Last but not least, the interior of a black hole exhibits a future spacelike singularity while our universe appears to have emerged from an initial (past) spacelike singularity (and, no, saying "I meant *white* hole, not black hole" will not help, see the discussion below).

2. Writing the FRW Metric in Schwarzschild-like Form

   In section 34.4 it was shown that the metric can be written in a form which resembles that of the Schwarzschild metric in Painlevé-Gullstrand-coordinates. Using the current notation ($R = a(t)r$ was called $\tilde{r}$ in section 34.4 and we use $R_H = 1/H$) we can write the general Robertson-Walker metric as (34.43)

$$\begin{aligned} ds^2 &= -(1 - R^2H(t)^2)dt^2 - 2RH(t)dtdR + (dR^2 + R^2d\Omega^2) \\ &= -(1 - (R/R_H)^2)dt^2 - 2(R/R_H)dtdR + (dR^2 + R^2d\Omega^2) \\ &= -\Phi(dt + (R/\Phi R_H)dR)^2 + \Phi^{-1}dR^2 + R^2d\Omega^2 \quad , \end{aligned} \tag{36.89}$$

   where

$$\Phi(t, R) = 1 - (R/R_H(t))^2 \quad . \tag{36.90}$$

   This form of the metric shows that there is an infinite time-dilation between the proper time of an observer at constant $R = R_0$ as $R_0 \to R_H$ and the cosmological time t, vaguely suggestive of the same phenomenon in the Schwarzschild metric. However, this is clearly a problem of the observers moving at constant $R(t) =$

[160] Moreover, there is nothing new about the observation (36.88) regarding the relation between the Hubble radius and the Schwarzschild-like radius R_s. It is e.g. already pointed out in section 10.1 of S. Hawking and G. Ellis, *The large scale structure of space-time*, that the numerology leading to (36.88) provides an "intuitive way of viewing this result".

$a(t)r$, not of the space-time itself. These observers at constant R are not the standard comoving observers (at constant r). Their 4-velocity has the form

$$u^t = \Phi(t,R_0)^{-1/2} \quad , \quad u^R = 0 \tag{36.91}$$

in the above (t,R)-coordinates, and

$$u^t = \Phi(t,R_0)^{-1/2} \quad , \quad u^r = \Phi(t,R_0)^{-1/2}(-R_0H(t)/a(t)) \tag{36.92}$$

in the standard comoving (t,r) coordinates, with

$$\Phi(t,R_0) = 1 - R_0^2H(t)^2 \ . \tag{36.93}$$

so that $u^t \to \infty$ for $R_0 \to R_H = a(t)/\dot{a}(t)$, while comoving observers have $u^t = 1$. Calculating the acceleration of these observers (in comoving coordinates, say), one finds

$$a^\alpha = u^\beta\nabla_\beta u^\alpha = \Phi^{-2}(q + H^2R_0^2)(R_0H^2)(-R_0H, 1/a) \tag{36.94}$$

with norm

$$g_{\alpha\beta}a^\alpha a^\beta = \Phi^{-3}(q + H^2R_0^2)^2(R_0H^2)^2 \ . \tag{36.95}$$

Clearly this diverges as $R_0 \to R_H = 1/H$, and therefore this is exactly like the space-time seen by ∞ acceleration observers in a Rindler space-time who detect a fictitious Rindler horizon.

Thus R_H is not like an event horizon and there is nothing like a black hole in sight. In fact, the best and most informative way of saying what $R_H(t)$ is and what its significance is (or is not) is that (for $k = 0$) $R_H(t)$ is a spherically symmetric marginally trapped tube (MTT) foliated by spherically symmetric marginally trapped surfaces (MTSs). In other words, with respect to a spherically symmetric foliation of space-time, for $k = 0$ (but only for $k = 0$) the surface $r = r_H(t)$ is an *apparent (3-)horizon* (cf. the discussion in section 32.9). Moreover, as the hypersurface $r = r_H(t)$ is timelike in the range $-1 < w < 1/3$, the horizon-terminology is really not very appropriate and, as mentioned in section 32.9, in current terminology such a hypersurface is referred to as a *timelike membrane*.

Concretely, and in elementary terms, this means the following. Consider (for any k) the equation

$$a(t)\frac{dr}{\sqrt{1-kr^2}} = \pm dt \tag{36.96}$$

for out- respectively ingoing lightrays, and write this in terms of the area radius $R(t) = a(t)r$,

$$dR_\pm = \sqrt{1-kR^2/a^2}\left(\pm 1 + RH/\sqrt{1-kR^2/a^2}\right)dt \ . \tag{36.97}$$

This can of course also be deduced directly from the PG-like metric (36.89) and its $k \neq 0$ counterpart (34.46).

For sufficiently small values of R,

$$R|H|/\sqrt{1-kR^2/a^2} < 1 \quad \Leftrightarrow \quad R(t) < \frac{1}{\sqrt{H(t)^2 + k/a(t)^2}} \equiv R_{ah}(t) \ , \tag{36.98}$$

this exhibits the standard behaviour

$$R(t) < R_{ah}(t) \quad \Rightarrow \quad \begin{cases} dR_+/dt > 0 \\ dR_-/dt < 0 \end{cases} \tag{36.99}$$

that one solution, R_+, propagates to larger values of R, while the other, R_-, propagates to smaller values of R. Evidently something special happens at $R = R_{ah}$ and for larger values one has

$$R(t) > R_{ah}(t) \quad \Rightarrow \quad \begin{cases} H(t) > 0: & dR_+/dt > 0 \quad \text{and} \quad dR_-(t)/dt > 0 \\ H(t) < 0: & dR_+/dt < 0 \quad \text{and} \quad dR_-(t)/dt < 0 \end{cases} \tag{36.100}$$

Thus, for a contracting universe this means that the spheres of constant R and t are *trapped* for $R > R_{ah}(t)$, like the 2-spheres inside the event horizon of the Schwarzschild black hole, with both ingoing and (would-be) outgoing radial lightrays moving towards smaller radii. However, far from indicating the presence of a black hole in this case, these trapped spheres and the apparent horizon $R_{ah}(t)$ indicate (together with an energy condition) a future cosmological singularity.

In an expanding universe, on the other hand, for $R > R_{ah}$ both in- and outgoing lightrays move to larger values of R. The spheres with $R > R_{ah}$ are thus the opposite of trapped surfaces, i.e. anti-trapped or trapped towards the past (and reflect the existence of a big bang singularity in the past).

The fact that both in- and outgoing expansions are positive is also a characteristic feature of the (unphysical) "white hole" region of a black hole in the region before the past event horizon. However, in that case it is the region around $r = 0$, and with $r < 2m$, that contains the anti-trapped surfaces while sufficiently large spheres show normal behaviour. Here, on the contrary, it is sufficiently large spheres around the comoving observer that are anti-trapped while in a sufficiently small region around any cmoving observer there are no (anti-)trapped surfaces.

Noting that for $k = 0$ the apparent horizon is equal to the Hubble radius,

$$k = 0 \quad \Rightarrow \quad R_{ah}(t) = R_H(t) \ , \tag{36.101}$$

all of this is simply a restatement of the discussion about the behaviour of lightrays in the previous section, and nothing seems to be gained by coining a new name for this well-established concept of an apparent horizon. Moreover, for $k \neq 0$ the Hubble sphere is not even an apparent horizon or some other marginally trapped tube, so it seems best not to associate the word "Hubble" with the word "horizon" at all.

37 Cosmology V: Some Exact Solutions

We have seen that a lot can be learnt about the Friedmann-Lemaître-Robertson-Walker models without ever having to solve a differential equation. On the other hand, more precise information can be obtained by specifying an equation of state for the matter content and solving explicitly the Friedmann equations. We have already seen some special exact solutions ($k = \Lambda = 0$ and only one species of matter) in section 36.4. Solutions with $k \neq 0$ but $\Lambda = 0$ are also easy to obtain. Together with the exact solutions for just $k \neq 0$ and $\Lambda \neq 0$, but with no other matter, these are the most useful solutions in practice, but it is also possible to solve the equations explicitly in some other cases.[161]

37.1 Milne Universe

Let us start our excursion into exact solutions by looking at the totally unphysical case of a completely empty universe, i.e. $\rho = p = \Lambda = 0$, and only k possibly not zero. As trivial as this may be, it has its pedagogical value, which is why I am including this case here.

In this case (F1) reduces to

$$\dot{a}^2 + k = 0 \quad \Rightarrow \quad k \leq 0 \ . \tag{37.1}$$

For $k = 0$, one has $\dot{a} = 0$ and thus $a(t) = a_0$, and the metric is

$$ds^2 = -dt^2 + a_0^2 d\vec{x}^2 \ . \tag{37.2}$$

It should not come as a suprise that the metric we have found is just that of Minkowski space (the constant a_0 can be absorbed into a rescaling of the spatial coordinates).

The case $k = -1$ is a bit more interesting. In that case the equation to solve is (I will call the time-coordinate τ instead of t for reasons that will perhaps become apparent below)

$$\dot{a}^2 = +1 \quad \Rightarrow \quad a(\tau) = \pm(\tau - \tau_0) \ . \tag{37.3}$$

and the resulting space-time metric is (choosing without loss of generality $\tau_0 = 0$)

$$ds^2 = -d\tau^2 + \tau^2 \, d\tilde{\Omega}_3^2 \tag{37.4}$$

where $d\tilde{\Omega}_3^2$ is the line-element on the unit 3-hyperboloid (14.33).

This appears to describe a non-trivial universe (known as the *Milne universe*) with either (for $a(\tau) = +\tau$) a big bang at $\tau = 0$ and with the universe subsequently expanding

[161] For explicit solutions of the Friedmann equations for $\Lambda \neq 0$ and non-trivial matter content (generalising the solution (38.26)) see e.g. B. Aldrovandi, R. Cuzinatto, L. Medeiros, *Analytic solutions for the Λ-FRW Model*, `arXiv:gr-qc/0508073`.

linearly with $\tau > 0$, or (for $a(\tau) = -\tau$) with a linearly contracting universe for $\tau < 0$, ending in a "big crunch" at $\tau = 0$, and all this in spite of the fact that the universe is empty.

This is deceptive, however (not that it is empty, but that it is non-trivial). As always, we need to be careful to disentangle coordinate artefacts from genuine geometrical statements. Indeed, this space-time is again nothing other than (a part of) Minkowski space-time. We had already anticipated this in section 35.1, in connection with (35.9) which expresses the Riemann tensor of a Robertson-Walker metric in terms of its Ricci tensor (and which therefore implies that a vacuum solution will necessarily be flat).

Nevertheless, it will be instructive to see this explicitly. To that end we start with the Minkowski metric

$$ds^2 = -dt^2 + d\vec{x}^2 \tag{37.5}$$

and introduce coordinates that are adapted to the family of space-like hyperboloids

$$t^2 - \vec{x}^2 = \tau^2 > 0 \ . \tag{37.6}$$

For $t > 0$ these hyperboloids fill (foliate) the interior of the future lightcone at the origin (and the interior of the past lightcone for $t < 0$), and if you draw these hyperboloids you can perceive what appears to be a non-trivial dynamical evolution of these surfaces. To show that this "fake" dynamics is precisely the dynamics exhibited by the Milne metric, introduce new coordinates $(\tau, \rho, \text{angles})$ via

$$t = \tau \cosh \rho \quad , \quad \vec{x} = \vec{n} \tau \sinh \rho \tag{37.7}$$

where $\vec{n}$ is a unit vector on S^2, so that

$$\vec{n}.\vec{n} = 1 \quad , \quad \vec{n}.d\vec{n} = 0 \quad , \quad d\vec{n}.d\vec{n} = d\Omega_2^2 \ . \tag{37.8}$$

Then the Minkowski metric becomes precisely the Milne metric (37.4),

$$ds^2 = -d\tau^2 + \tau^2 d\rho^2 + \tau^2 \sinh^2 \rho \, d\Omega_2^2 \ . \tag{37.9}$$

Remarks:

1. These coordinates only cover a part of Minkowski space-time, namely the interior of the future (and past) lightcone of the origin. They are adapted to the comoving (and thus geodesic) observers of the Milne metric, i.e. to families of observers with constant values $\rho = \rho_0$, $\vec{n} = \vec{n}_0$. Thus in terms of Minkowski coordinates, the worldlines of these observers are described by

$$t(\tau) = \tau \cosh \rho_0 \quad , \quad \vec{x}(\tau) = \vec{n}_0 \tau \sinh \rho_0 \tag{37.10}$$

or

$$\vec{x}(t) = (\vec{n}_0 \tanh \rho_0) t \equiv \vec{v}_0 t \tag{37.11}$$

with

$$(v_0)^2 = \tanh^2 \rho_0 \leq 1 \quad . \tag{37.12}$$

Thus these are geodesic observers emanating radially with different initial directions and velocities from the origin, the constant-time hyperboloids of the Milne metric being the surfaces which these observers reach at their proper time τ.

2. The coordinate singularity at $\tau = 0$ is simpy due to the fact that the worldlines of these observers all intersect at the origin (and thus do not provide good coordinates there).

3. These coordinates are, as may have occurred to you by now, the future and past relatives of the Rindler coordinates discussed (way back) in sections 1.3 and 3.4 (and then again in the context of the near-horizon geometry of the Schwarzschild metric in section 26.6), in particular of the spherical Rindler space discussed at the end of section 3.4:

   Introducing Rindler-like coordinates $(\tau, \rho, \text{angles})$ adapted to the timelike hyperboloids

   $$\vec{x}^2 - t^2 = \rho^2 \tag{37.13}$$

   via

   $$t = \rho \sinh \tau \quad , \quad \vec{x} = \vec{n} \rho \cosh \tau \quad , \tag{37.14}$$

   with $\vec{n}$ a spatial unit vector, $\vec{n}.\vec{n} = 1$, the Minkowski metric takes the form (3.100)

   $$\begin{aligned} ds^2 &= -\rho^2 d\tau^2 + d\rho^2 + \rho^2 \cosh^2 \tau \, d\Omega_2^2 \\ &= d\rho^2 + \rho^2(-d\tau^2 + \cosh^2 \tau \, d\Omega_2^2) \quad . \end{aligned} \tag{37.15}$$

   This is the spherical Rindler metric. The metric in brackets in the 2nd line will reappear below, in the guise of the (2+1)-dimensional de Sitter metric.

4. Thus, we see that we can foliate Minkowski space in different ways:

   - in the standard inertial coordinates, slices of constant Minkowski time t are Euclidean spaces $\mathbb{R}^3$, and slices of constant x^3, say, are Minkowski spaces $\mathbb{R}^{1,2}$;
   - in terms of Minkowski time and spatial spherical coordinates,

     $$ds^2 = -dt^2 + dr^2 + r^2 \, d\Omega_2^2 \tag{37.16}$$

     slices of constant r are of the form $S^2 \times \mathbb{R}$ (and we will recognise these below, in more fancy terminology, as (2+1)-dimensional versions of the Einstein static universe);
   - in Milne coordinates, slices of constant τ are hyperboloids H^3;
   - in the Rindler-like coordinates (37.15), slices of constant ρ are are de Sitter spaces dS_3.

5. The "singularity" of the Milne metric is of course just a coordinate singularity, but with some discrete identifications on the hyperboloid this becomes an honest conical singularity at $\tau = 0$. This is a nice (and much studied) toy-model of a big bang (everywhere flat except at $\tau = 0$).

37.2 Einstein Static Universe

This particular solution is mainly of historical interest. Before the discovery and understanding of the Hubble expansion of the universe, it was natural to assume that the universe can be described by a static solution compatible with the cosmological principle (homogeneity and isotropy). This led Einstein to introduce the (in-)famous cosmological constant. Let us see how this comes about.

Static means that $a(t) = a_0, \dot{a}(t) = \ddot{a}(t) = 0$. (F2) with $\ddot{a} = 0$ and a zero cosmological constant leads to

$$\text{(F2)}(\Lambda = 0) \quad \Rightarrow \quad 4\pi G_N(\rho + 3p) = 0 \ . \tag{37.17}$$

This equation is simply not satisfied for ordinary matter, and therefore no static solution of the Friedmann equations with $\Lambda = 0$ exists. (Re-)introducing Λ, however, (F2) then tells us that

$$\text{(F2)} \quad \Rightarrow \quad 4\pi G_N(\rho + 3p) = \Lambda \ . \tag{37.18}$$

Moreover, the conservation law (F3) with $\dot{a} = 0$ implies

$$\text{(F3)} \quad \Rightarrow \quad \dot{\rho} = 0 \ , \tag{37.19}$$

which now implies that also $\dot{p} = 0$. (37.18) then simply fixes the constant Λ in terms of the constants ρ and p. For a standard matter content of the universe, say $\rho = \rho_m + \rho_r$, this requires that $\Lambda > 0$. Finally, the first Friedmann equation (F1) now becomes an algebraic equation for $a(t) = a_0$, namely

$$\begin{aligned} \text{(F1)} \quad &\Rightarrow \quad \frac{k}{a^2} = \frac{8\pi G_N}{3}\rho + \frac{\Lambda}{3} = 4\pi G_N(\rho + p) \\ &\Rightarrow \quad k = +1 \quad \text{and} \quad a_0^2 = (4\pi G_N(\rho + p))^{-1} \ . \end{aligned} \tag{37.20}$$

This is thus a static solution of the Friedmann equations,

$$ds^2 = -dt^2 + a_0^2 \, d\Omega_3^2 \ , \tag{37.21}$$

the *Einstein Static Universe*.

Remarks:

1. The topology of the solution is $\mathbb{R} \times S^3$, the radius a_0 of the S^3 being smaller for larger energy and pressure density (bigger gravitational attraction) and vice-versa.

2. In this solution, the gravitational attraction is precisely balanced by the positive cosmological constant. Note that even though a positive cosmological constant has a positive energy density $\rho_\Lambda > 0$, it has a negative pressure, and the net effect of a positive cosmological constant in (F2) is that of gravitational repulsion rather than attraction,
$$\rho_\Lambda + 3p_\Lambda = -2\rho_\Lambda < 0 \ . \tag{37.22}$$

3. The precise balance between matter and cosmological constant required by this solution also implies that it is unstable to perturbations of either Λ or ρ, p: a slight increase in Λ relative to $\rho + p$, no matter how small, will make the universe expand, and a slight decrease will make it collapse. This alone is enough to make this particular solution unphysical and shows that, even with inclusion of a cosmological constant, an expanding or collapsing universe is practically inevitable, thus undermining the original motivation for introducing the cosmological constant in the first place.

37.3 Matter Dominated Era

We now return to somewhat more realistic solutions of the Friedmann equations. In a matter dominated era (but retaining the curvature term) we have to solve the equation
$$\dot{a}^2 = \frac{C_m}{a} - k \ . \tag{37.23}$$
We will now look in turn at the cases $k = 0, +1, -1$.

1. For $k = 0$, this is the equation we already discussed above, leading to the solution (36.20),
$$a(t) = a_0 (t/t_0)^{2/3} \tag{37.24}$$
This solution is also known as the Einstein - de Sitter universe. It describes a universe expanding from an initial singularity at $t = 0$. Another solution, arising from taking the negative square root of the Friedmann equation $\dot{a}^2 = C_m/a$,
$$a^{1/2}\dot{a} = -(C_m)^{1/2} \quad \Rightarrow \quad a(t) = a_0 (t_f - t)^{2/3}/(t_f - t_0)^{2/3} \ , \tag{37.25}$$
describes a universe collapsing to a big crunch at $t = t_f$.

2. For $k = +1$, the equation is
$$\dot{a}^2 = \frac{C_m}{a} - 1 \ . \tag{37.26}$$
We recall that in this case we will have a recollapsing universe, with $a_{max} = C_m$ attained for $\dot{a} = 0$,
$$\dot{a} = 0 \quad \Rightarrow \quad a = a_{max} = C_m \ . \tag{37.27}$$

The equation (37.26) can be solved in closed form for t as a function of a, and the solution to

$$\frac{dt}{da} = \left(\frac{a}{a_{max} - a}\right)^{1/2} \tag{37.28}$$

is

$$t(a) = \frac{a_{max}}{2} \arccos(1 - 2a/a_{max}) - \sqrt{a a_{max} - a^2} \ , \tag{37.29}$$

as can easily be verified. The universe starts at $t = 0$ with $a(0) = 0$, reaches its maximum $a = a_{max}$ at

$$t_{max} = a_{max} \arccos(-1)/2 = a_{max}\pi/2 \ , \tag{37.30}$$

and ends in a Big Crunch at $t = 2t_{max}$.

The curve $a(t)$ is a cycloid, as is most readily seen by writing the solution in parametrised form. For this it is convenient to use the conformal time coordinate η introduced in (34.48) through

$$d\eta = dt/a(t) \quad \Rightarrow \quad \partial_t = a(\eta)^{-1}\partial_\eta \quad , \quad \partial_\eta = a(t)\partial_t \ . \tag{37.31}$$

Denoting a derivative with respect to η by a prime, and noting that $\dot{a} = a'/a$, one then finds that for $k \neq 0$ the Friedman equation (37.23) can be written as

$$\begin{aligned} \dot{a}^2 + k = \frac{C_m}{a} \quad &\Leftrightarrow \quad (a')^2 + ka^2 = C_m a \\ &\Leftrightarrow \quad ((a - C_m/2k)')^2 + k(a - C_m/2k)^2 = kC_m^2/4 \ . \end{aligned} \tag{37.32}$$

Thus for $k = +1$, the solution to the Friedmann equation can be written as

$$a(\eta) - C_m/2 = (C_m/2)\cos(\eta - \eta_0) \ . \tag{37.33}$$

Choosing $\eta_0 = \pi$ (so that $a(\eta = 0) = 0$), and integrating the relation $dt/d\eta = a(\eta)$ to find $t(\eta)$, one then finds the solution

$$\begin{aligned} a(\eta) &= \frac{a_{max}}{2}(1 - \cos\eta) \\ t(\eta) &= \frac{a_{max}}{2}(\eta - \sin\eta) \ , \end{aligned} \tag{37.34}$$

which makes it transparent that the curve is indeed a cycloid, roughly as indicated in Figure 46.

The maximal radius is reached at

$$t_{max} = t(a = a_{max}) = t(\eta = \pi) = a_{max}\pi/2 \tag{37.35}$$

(with $a_{max} = C_m$), as before, and the total lifetime of the universe is $2t_{max}$.

3. Analogously, for $k = -1$ the Friedmann equation in parametrised form (37.32) can be solved in terms of hyperbolic (rather than trigonometric) functions,

$$\begin{aligned} a(\eta) &= \frac{C_m}{2}(\cosh\eta - 1) \\ t(\eta) &= \frac{C_m}{2}(\sinh\eta - \eta) \ . \end{aligned} \tag{37.36}$$

REMARKS:

1. We see that for small times (for which matter dominates over curvature) the solutions for $k \neq 0$ reduce to $t \sim \eta^3, a \sim \eta^2$ and therefore $a \sim t^{2/3}$ which is indeed the exact solution for $k = 0$.

2. Analogously, for late times in the $k = -1$ model one finds that $a(\eta) \sim t(\eta)$, reproducing the expected late-time behaviour $\dot{a} \to 1$ of section 36.3.

3. In section 29 we will use the exact solutions of this matter dominated phase to describe the interior geometry of collapsing stars.

It is instructive to display the causal structure of the $k = +1$ metric in a conformal diagram. Recall that in terms of conformal time the metric takes the form

$$ds^2 = a(\eta)^2(-d\eta^2 + d\Omega_3^2) = a(\eta)^2(-d\eta^2 + d\psi^2 + \sin^2\psi d\Omega_2^2) \ , \tag{37.37}$$

Suppressing the transverse 2-sphere and eliminating the conformal factor $a(\eta)^2$, we are thus led to consider the metric

$$d\tilde{s}^2 = -d\eta^2 + d\psi^2 \ . \tag{37.38}$$

Noting that in the case at hand the range of both η and ψ is finite,

$$\eta \in [0, 2\pi] \quad , \quad \psi \in [0, \pi] \ , \tag{37.39}$$

we see that no further redefinition of the coordinates is required to obtain a conformal diagram of this solution (Figure 51).

Here are some of the characteristic features of this diagram:

1. Since this solution is spatially compact, there is no analogue here of spatial infinity i^0. As a consequence, this diagram a priori looks very different from Penrose diagrams for asymptotically flat space-times, in particular also from those for spatially flat cosmologies in section 36.6.

2. All timelike and null geodesics begin at the initial (spacelike) singularity at $\eta = 0$ and end at the final singualrity at $\eta = 2\pi$.

3. A lightray sent out at the Big Bang will reach the antipodal point of the sphere exactly at the time $\eta_{max} = \pi$ the universe reaches its maximal radius, and will have circled around the universe exactly once precisely at the time $2\eta_{max}$ of the final big crunch.

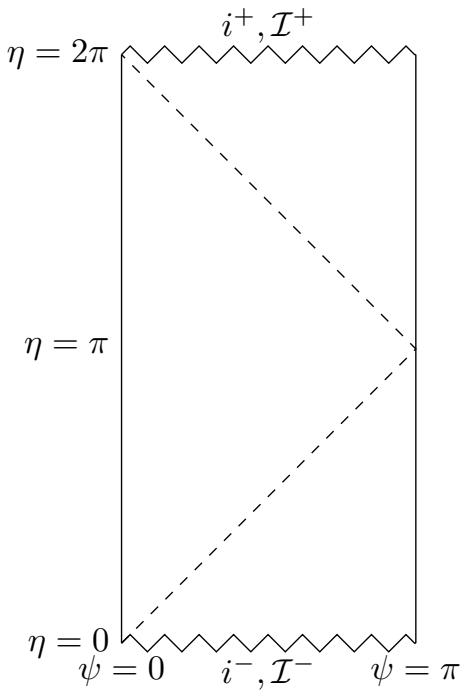


Figure 51: Conformal Diagram of the $k = +1$ matter dominated universe.

37.4 Radiation Dominated Era

We now consider the situation when radiation is dominant (as is expected during some time in the very early universe). In this case we need to solve

$$a^2\dot{a}^2 = C_r - ka^2 \quad . \tag{37.40}$$

For $k = 0$ we had already seen the solution in (36.18),

$$a(t) = a_0(t/t_0)^{1/2} \tag{37.41}$$

(and there is also evidently a corresponding collapsing solution). For $k = +1$, on the other hand, there will be a maximal radius a_{max} at

$$k = +1: \quad \dot{a} = 0 \quad \Rightarrow \quad a_{max} = C_r^{1/2} \quad . \tag{37.42}$$

Here are 2 ways to solve the Friedmann equation for $k \neq 0$:

1. Because $a(t)$ appears only quadratically, it is convenient to make the change of variables $b = a^2$. Then one obtains

$$\frac{\dot{b}^2}{4} + kb = C_r \quad . \tag{37.43}$$

For $k = \pm 1$, one necessarily has $b(t) = b_0 + b_1 t + b_2 t^2$. Fixing $b(0) = 0$, one easily finds the solution

$$a(t) = [2C_r^{1/2}t - kt^2]^{1/2} \quad . \tag{37.44}$$

As expected this reduces to $a(t) \sim t^{1/2}$ for small times where the curvature term is irrelevant.

- For $k = +1$ one has
$$a(0) = a(2C_r^{1/2}) = 0 \ , \tag{37.45}$$
and $a = a_{max}$ at the time
$$t_{max} = C_r^{1/2} \quad \Rightarrow \quad a(t_{max}) = a_{max} = C_r^{1/2} \ . \tag{37.46}$$
- For $k = -1$, on the other hand, the universe expands forever, the late-time behaviour being given by
$$a(t) \to (-kt^2)^{1/2} = t \ , \tag{37.47}$$
again as expected.

All this is of course in agreement with the results of the qualitative discussion given earlier.

2. It is also instructive to solve these equations in terms of conformal time η. Using $\dot{a} = a'/a$, the Friedmann equation becomes
$$(a')^2 + ka^2 = C_r \ . \tag{37.48}$$
Thus the solution for $k = +1$ is obviously (with the choice $a(\eta = 0) = 0$)
$$k = +1 \quad \Rightarrow \quad a(\eta) = C_r^{1/2} \sin \eta \tag{37.49}$$
and for $k = -1$ one has
$$k = -1 \quad \Rightarrow \quad a(\eta) = C_r^{1/2} \sinh \eta \ . \tag{37.50}$$
Using $dt/d\eta = a(\eta)$ one can also find $t(\eta)$. E.g. for $k = +1$ one has
$$t(\eta) = C_r^{1/2}(1 - \cos \eta) \ , \tag{37.51}$$
where the integration constant has been chosen such that
$$\eta = 0 \quad \Rightarrow \quad a(\eta = 0) = 0 \quad , \quad t(\eta = 0) = 0 \ . \tag{37.52}$$
One also has $\eta_{max} = \pi/2$ and the recollapse to $a(\eta) = 0$ at $\eta = \pi$ corresponding to $t = 2C_r^{1/2}$, as in the 1st derivation.

Again it is instructive to display the $k = +1$ solution in a conformal diagram (Figure 52) and to compare it with that of the matter dominated solution (Figure 51). The main difference is that here the range of η is equal to the range of ψ,
$$\eta \in [0, \pi] \quad , \quad \psi \in [0, \pi] \ . \tag{37.53}$$
As a consequence, the conformal diagram is a square, and any lightray sent out at the Big Bang can only travel half-way around the universe during the lifetime of the universe.

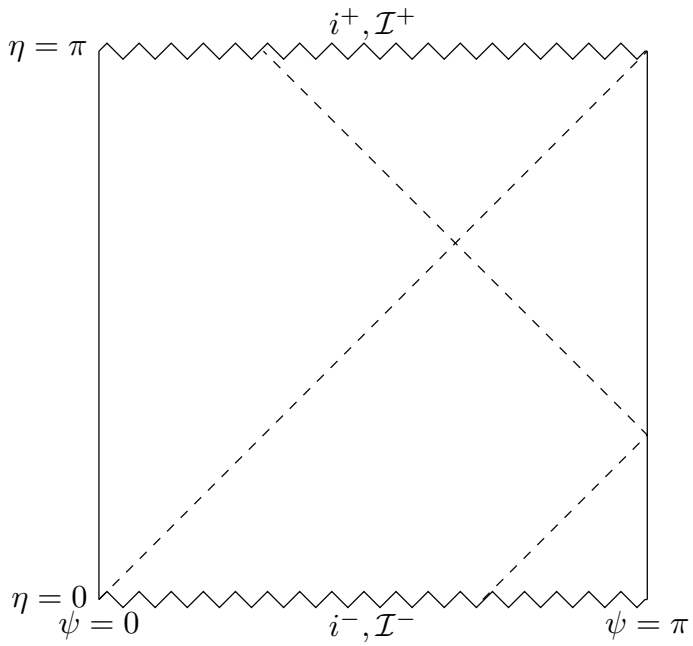


Figure 52: Conformal Diagram of the $k = +1$ radiation dominated universe.

37.5 Cosmological Constant Dominated Era: (Anti-) de Sitter Space

This case is of considerable interest for at least two reasons. On the one hand, as we know, Λ is the dominant driving force for $a(t)$ very large, and may therefore, if current observations are to be believed (see section 38.1), dominate the late-time behaviour of our universe.

On the other hand, the currently most popular cosmological models trying to also address and solve the so-called *horizon problem* and *flatness problem* (cf. the discussion in section 38.3) of the standard FLRW model of cosmology (as well as a number of other issues) use a mechanism called *inflation* based on an era of exponential expansion during some time in the very early universe. This is typically generated by something that acts effectively like a cosmological constant.

Thus the equations to solve are, setting $\rho = 0$ and $p = 0$ but retaining Λ and k,

$$\dot{a}^2 = -k + \frac{\Lambda}{3} a^2 \quad . \tag{37.54}$$

We see immediately that Λ has to be positive for $k = +1$ or $k = 0$, whereas for $k = -1$ both positive and negative Λ are possible,

$$\dot{a}^2 = -k + \frac{\Lambda}{3} a^2 \quad \Rightarrow \quad \begin{cases} \text{if } \Lambda > 0 & k = 0, \pm 1 \text{ possible} \\ \text{if } \Lambda < 0 & k = -1 \end{cases} \tag{37.55}$$

This is one instance where the solution to the second order equation (F2),

$$\ddot{a} = \frac{\Lambda}{3} a \quad , \tag{37.56}$$

is more immediate, namely trigonometric functions for $\Lambda < 0$ (only possible for $k = -1$) and hyperbolic functions for $\Lambda > 0$. The first order equation then fixes the constants of integration according to the value of k.

It will be convenient to express Λ in terms of a length-scale ℓ (which will turn out to be the curvature radius of the space-time) through

$$|\Lambda|/3 = \ell^{-2} \tag{37.57}$$

so that the two Friedmann equations read

$$\dot{a}^2 = -k \pm a/\ell^2 \quad , \quad \ddot{a} = \pm a/\ell^2 \quad , \tag{37.58}$$

the plus/minus referring to $\Lambda > 0$ and $\Lambda < 0$ respectively.

Solutions for $\Lambda > 0$:

1. $k = 0$: The solutions are evidently (cf. (36.23))

$$a_\pm(t) \sim \mathrm{e}^{\pm t/\ell} \tag{37.59}$$

leading to the two metrics (related by $t \to -t$)

$$ds^2 = -dt^2 + \mathrm{e}^{\pm 2t/\ell} d\vec{x}^2 \tag{37.60}$$

2. $k = +1$: The solution is

$$ds^2 = -dt^2 + \ell^2 \cosh^2 t/\ell \; d\Omega_3^2 \quad . \tag{37.61}$$

3. $k = -1$: The solution is

$$ds^2 = -dt^2 + \ell^2 \sinh^2 t/\ell \; d\tilde{\Omega}_3^2 \quad . \tag{37.62}$$

Remarks:

1. It turns out that all 3 metrics actually represent the same space-time metric, just written in different coordinates. This space-time is known as the *de Sitter space-time*. Thus these 3 metrics exhibit different slicings of the de Sitter space-time (or, henceforth, de Sitter space for short, or just dS space), with the t = const. slices being $\mathbb{R}^3$, S^3 and H^3 respectively.

2. One way to establish this would be to directly exhibit the coordinate transformations that map one metric to the other, but this is messy and does not provide any additional insight. We will proceed in a different way in section 39 below. Indeed, it turns out that the de Sitter solution is the unique maximally symmetric spacetime with positive curvature (cf. the discussion in sections 14 and 39), and this perspective will provide us with a more efficient and insightful way of constructing different coordinate systems and exploring the relations among them.

3. All 3 metrics exhibit an exponential expansion or collapse at early and/or late times. The $k = -1$ metric also appears to exhibit a big bang singularity as $t \to 0$, but this is an illusion. Indeed, as $t \to 0$ the metric approaches the metric (37.4) of the Milne universe, and we already know that the singularity at $\tau = 0$ of the Milne metric is just a coordinate singularity.

4. The coordinates appearing in the $k = +1$ metric turn out to cover de Sitter space globally. Thus the global picture of de Sitter space is that of a 3-sphere that

   - started out with infinite radius in the infinite past $t \to -\infty$,
   - undergoes an exponential contraction to a sphere of radius ℓ at time $t = 0$
   - and then again undergoes an exponential expansion to infinite size as $t \to +\infty$.

   It is clear from this description that e.g. the 2 metrics for $k = 0$ only cover the contracting (-) or expanding (+) period of the de Sitter universe.

5. For later purposes it is useful to write the expanding $k = 0$ metric in terms of conformal time η (cf. sections 34.5 and 36.6) as

$$ds^2 = \mathrm{e}^{2t/\ell}(-dt^2\mathrm{e}^{-2t/\ell} + d\vec{x}^2) = \frac{\ell^2}{\eta^2}(-d\eta^2 + d\vec{x}^2) \tag{37.63}$$

   where

$$\eta = -\ell\mathrm{e}^{-t/\ell} \ , \tag{37.64}$$

   with $\eta \to -\infty$ for $a(t(\eta)) \to 0$. This clearly exhibits the simple Minkowskian causal structure of de Sitter space in the expanding phase / patch.

6. The (2+1)-dimensional counterpart of the $k = +1$ metric is the metric that appeared in the Rindler-like coordinate system (37.15). Thus we can say that these Rindler-like coordinates provide a (radial) foliation of the left- and right Rindler wedges by de Sitter spaces.

7. A detailed discussion of many different coordinate systems for de Sitter space is given in section 39.2 below.

Solution for $\Lambda < 0$:

In this case, the only possibility (within the Robertson-Walker ansatz for the metric) is $k = -1$, and the solution is

$$ds^2 = -dt^2 + \ell^2 \sin^2 t/\ell \, d\tilde{\Omega}_3^2 \ . \tag{37.65}$$

Remarks:

1. This solution is nowadays known as the *anti-de Sitter space-time* (or *anti-de Sitter space* or AdS space for short).

2. As its $\Lambda > 0, k = -1$ counterpart, it appears to exhibit a big bang singularity as $t \to 0$, but this is again an illusion since this metric also approaches the non-singular Milne metric (37.4) as $t \to 0$.

3. AdS space turns out to be the unique maximally symmetric space-time with negative curvature.

4. The above cosmological coordinates do not provide a complete covering of anti-de Sitter space. In (one choice of) global coordinates the metric turns out to take the form (39.75)

$$ds^2 = -\cosh^2 \rho d\tau^2 + d\rho^2 + \sinh^2 \rho d\Omega_2^2 \tag{37.66}$$

   with $0 \leq \rho < \infty$ and $-\infty < \tau < \infty$. In these coordinates the metric is time-independent and therefore (unlike dS) AdS has a global timelike Killing vector, namely ∂_τ.

5. A detailed discussion of many different coordinate systems for anti-de Sitter space is given in section 39.3 below.

38 Cosmology VI: The Universe Today - Insights and Puzzles

38.1 Λ-CDM Cold Dark Matter + Dark Energy Model

Let us recall the key equations governing the evolution of the universe which, in the notation introduced in section 36.5, take the compact (albeit somewhat obscure) form

$$\begin{aligned} \Omega_M + \Omega_\Lambda + \Omega_k &= 1 \\ \tfrac{1}{2}(1+3w)\Omega_M - \Omega_\Lambda &= q \end{aligned} \tag{38.1}$$

Their version today (at $t = t_0$) is (36.40)

$$\begin{aligned} (\Omega_M)_0 + (\Omega_\Lambda)_0 + (\Omega_k)_0 &= 1 \\ \tfrac{1}{2}(1+3w_0)(\Omega_M)_0 - (\Omega_\Lambda)_0 &= q_0 \end{aligned} \tag{38.2}$$

In the universe today, the radiation contribution to the matter content is negligible and the only non-negligible matter content appears to be that of $w = 0$ pressureless matter, and thus

$$q_0 = \tfrac{1}{2}(\Omega_M)_0 - (\Omega_\Lambda)_0 \quad . \tag{38.3}$$

In that case Ω_M is related to $(\Omega_M)_0$ by

$$\Omega_M = \frac{\Omega_M}{(\Omega_M)_0}(\Omega_M)_0 = \frac{\rho_M}{(\rho_M)_0}\frac{(\rho_{cr})_0}{\rho_{cr}}(\Omega_M)_0 = \frac{a_0^3}{a^3}\frac{H_0^2}{H^2}(\Omega_M)_0 \quad . \tag{38.4}$$

Likewise, for the cosmological constant and curvature contributions one has

$$\Omega_\Lambda = \frac{H_0^2}{H^2}(\Omega_\Lambda)_0 \quad , \quad \Omega_k = \frac{a_0^2 H_0^2}{a^2 H^2}(\Omega_k)_0 \quad . \tag{38.5}$$

Thus the Friedmann equation can be written as

$$H^2 = H_0^2\left(\frac{a_0^3}{a^3}(\Omega_M)_0 + \frac{a_0^2}{a^2}(\Omega_k)_0 + (\Omega_\Lambda)_0\right) \quad . \tag{38.6}$$

Using the constraint $(\Omega_M)_0 + (\Omega_\Lambda)_0 + (\Omega_k)_0 = 1$ to eliminate $(\Omega_k)_0$ from this equation, one finds

$$H^2 = \frac{H_0^2 a_0^2}{a^2}\left(1 + (\frac{a_0}{a} - 1)(\Omega_M)_0 + (\frac{a^2}{a_0^2} - 1)(\Omega_\Lambda)_0\right) \quad . \tag{38.7}$$

This equation is also frequently written as an equation for H as a function of the redshift z, $H = H(z)$, which can be obtained by the substitution $a_0/a = 1 + z$,

$$H(z) = H_0(1+z)\left[1 + (\Omega_M)_0 z + (\Omega_\Lambda)_0\left(\frac{1}{(1+z)^2} - 1\right)\right]^{1/2} \quad . \tag{38.8}$$

This expression is useful because it is expressed directly in terms of observable quantities, with $H(z)$ the value of the Hubble parameter at the time an object emitted the light that we now observe with redshift z. Using

$$1 + z = \frac{a_0}{a(t)} \quad \Rightarrow \quad \frac{dz}{H(z)} = -\frac{a_0 dt}{a(t)} \tag{38.9}$$

it is also possible to convert other evolution equations in t (or integrals over t) into evolution equations in z (or integrals over z).

For a long time it was believed that $\Omega_\Lambda = 0$ (or at least negligibly small) today. In that case one would have

$$(\Omega_M)_0 + (\Omega_k)_0 = 1 \tag{38.10}$$

and

$$q_0 = \tfrac{1}{2}(\Omega_M)_0 > 0 \ . \tag{38.11}$$

Therefore the curvature parameter k would be directly related to the value q_0 of the deceleration parameter today:

$$\begin{array}{lllll} q_0 > 1/2 & \Rightarrow & \rho_0 > (\rho_{cr})_0 & \Rightarrow & k = +1 \\ q_0 < 1/2 & \Rightarrow & \rho_0 < (\rho_{cr})_0 & \Rightarrow & k = -1 \end{array} \tag{38.12}$$

While there were large uncertainties about q_0, observations indicated a value of ρ_0 much smaller than the critical density $(\rho_{cr})_0$ (even taking into account not just visible matter but also some invisible ("dark") matter whose existence had been deduced indirectly from astrophysical observations like the rotations of galaxies). Thus this strongly suggested a decelerating open $k = -1$ universe. While perhaps not the most hospitable place in the long run, at least this scenario had the virtue of simplicity.

However, exciting recent developments and observations in cosmology and astrophysics have provided strong evidence for a very different and extremely intriguing and puzzling picture of the universe today. Since I am not an expert on these matters, I will just summarise the results here.[162]

1. Estimates for the current matter contribution $\Omega_M = \sum_b \Omega_b$ are

$$(\Omega_M)_0 \approx 0.3 \ . \tag{38.13}$$

2. Ordinary (visible, baryonic) matter only accounts for a small fraction of this, namely

$$(\Omega_{M,\text{visible}})_0 \approx 0.04 \ . \tag{38.14}$$

   Most of the matter density of the universe must therefore be due to some form of (as yet ill-understood, non-relativistic, weakly interacting) *Dark Matter* or *Cold Dark Matter* (CDM).

3. A detailed analysis of the fine structure of the anisotropies of the *Cosmic Microwave Background* suggests that the universe is spatially flat, $k = 0$, i.e. that the total density of whatever fills the universe is very close to the critical density.

[162] For excellent recent introductions to cosmology which explain the significance of these results, how they were obtained and much more, see e.g. M. Trodden and S. Carroll, *TASI Lectures: Introduction to Cosmology*, `arXiv:astro-ph/0401547`, and J. Frieman, M. Turner, D. Huterer, *Dark Energy and the Accelerating Universe*, `arXiv:0803.0982v1 [astro-ph]`.

4. Observations of redshifts of particular types of *Supernovae* in distant galaxies lead to the conclusion that the universe must currently be in a phase of *accelerated expansion*, with the data well explained by assuming that the *Dark Energy* responsible for this acceleration (thus with equation of state parameter $w < -1/3$) is due to the existence of a positive cosmological constant ($w = -1$) with

$$q_0 = \tfrac{1}{2}(\Omega_M)_0 - (\Omega_\Lambda)_0 < 0 \tag{38.15}$$

and, more specifically,

$$(\Omega_\Lambda)_0 \approx 0.7 \ . \tag{38.16}$$

Since then

$$(\Omega_M)_0 + (\Omega_\Lambda)_0 \approx 1 \quad \Leftrightarrow \quad (\Omega_k)_0 \approx 0 \ , \tag{38.17}$$

these last two (completely independent) observations are evidently compatible with each other.

Remarks:

1. Approximate numerical values for the parameters Λ and H_0 characterising the universe today are

$$H_0 \approx 2.3 \times 10^{-18}\ \mathsf{s}^{-1} \quad , \quad \Lambda \approx 10^{-35}\ \mathsf{s}^{-2} \ . \tag{38.18}$$

2. Somewhat more informative are perhaps the current estimates for the age of the universe,

$$t_0 \approx 13.8 \times 10^9 \text{years} \tag{38.19}$$

and the energy density of dark energy, approximately

$$\rho_\Lambda \approx (10^{-3}\ \mathsf{eV})^4 \ . \tag{38.20}$$

3. The equation of state parameter for dark energy is known to very close to $w = -1$,

$$w \approx -0.98 \pm 0.05 \ , \tag{38.21}$$

so a cosmological constant indeed appears to be a very plausible candidate for dark energy.

4. The fact that there is something like a cosmological constant is perhaps not particularly puzzling as such (in the absence of a good reason why it should not have been there in the first place), but nevertheless there are a number of puzzling issues related to the value of the cosmological constant - see section 38.4 for a brief discussion.

38.2 Exact Λ-CDM Solution

The exact solution for a spatially flat universe dominated by non-interacting (cold dark) matter was already given in (36.20),

$$a(t) = a_0 (t/t_0)^{2/3} \quad . \tag{38.22}$$

and that for a spatially flat universe dominated by a cosmological constant was given in (36.23),

$$a(t) = a_0 \mathrm{e}^{\sqrt{\Lambda/3}(t-t_0)} \quad . \tag{38.23}$$

An exact solution of the Friedmann equations can also be written down when both types of energy/matter are present simultaneously (as in the universe today). In this case the Friedmann equation is just

$$\Omega_M + \Omega_\Lambda = 1 \quad , \tag{38.24}$$

so that the Friedmann equation of the Λ-*CDM Model* (cold dark matter with a cosmological constant) can be written as

$$H^2 = H_0^2 \left(\frac{a_0^3}{a^3} (\Omega_M)_0 + (\Omega_\Lambda)_0 \right) \quad . \tag{38.25}$$

An exact solution of this equation is

$$\begin{aligned} a(t) &= a_0 \left(\frac{(\Omega_M)_0}{(\Omega_\Lambda)_0} \right)^{1/3} \left(\sinh((3/2)\sqrt{(\Omega_\Lambda)_0} H_0 t) \right)^{2/3} \\ &\equiv a_0 \left(\frac{(\Omega_M)_0}{(\Omega_\Lambda)_0} \right)^{1/3} (\sinh(t/t_\Omega))^{2/3} \quad . \end{aligned} \tag{38.26}$$

This evidently reproduces the above power-law (exponential) behaviour at early (late) times. The transition between the decelerating matter dominated and accelerating Λ-dominated phases occurs at the time $t = t_\Lambda$ at which

$$\ddot{a}(t_\Lambda) = 0 \quad . \tag{38.27}$$

Calculating

$$\frac{d^2}{dt^2} \sinh^{2/3}(t/t_\Omega) = (2/3) t_\Omega^{-2} \left(-(1/3) \sinh^{-4/3}(t/t_\Omega) \cosh^2(t/t_\Omega) + \sinh^{2/3}(t/t_\Omega) \right) \tag{38.28}$$

one finds that at $t = t_\Lambda$

$$\sinh^2(t_\Lambda/t_\Omega) = 1/2 \quad , \tag{38.29}$$

and plugging this back into (38.26), one sees that

$$\frac{a(t_\Lambda)}{a_0} = \left(\frac{(\Omega_M)_0}{2(\Omega_\Lambda)_0} \right)^{1/3} \quad . \tag{38.30}$$

With the values for the density parameters given above, $(\Omega_M)_0 = 0.3$, $(\Omega_\Lambda)_0 = 0.7$, this is roughly

$$a(t_\Lambda) \approx 0.6\, a_0 \quad . \tag{38.31}$$

Thus the transition occurred at a time when the universe was roughly speaking "half as big" as today, and at a corresponding value

$$z = \frac{a_0}{a(t_\Lambda)} - 1 \approx 0.66 \tag{38.32}$$

of the redshift parameter.

38.3 Flatness and Horizon Problems

The currently favoured Λ-CDM scenario with $(\Omega_M)_0 = 0.3$, $(\Omega_\Lambda)_0 = 0.7$ and $k = 0$ has been tested and confirmed in various independent ways. Nevertheless, it is a mystery and raises all kinds of questions and puzzles, in particular because the emergence of this particular universe appears to be somewhat unnatural (although it may perhaps be difficult to quantify this sentiment) and to require an incredible amount of fine-tuning.

I will not say anything about the dark matter component, since an explanation presumably needs to be found in the realm of particle physics (every model of physics beyond the standard model worth its salt has its own dark matter candidates), and since the story is in any case sufficiently strange and interesting even without worrying about, or having to put up with, things like WIMPS (weakly interacting massive particles), or MACHOS (massive compact halo objects), or *binos*, *winos* and other *neutralinos*.

Here, in a nutshell, are some of these problems or puzzles (and for more in-depth and far-reaching discussions of these issues please consult the extensive literature on cosmology). The first two (discussed in this section) are independent of the recent discovery of dark energy, and appear to require some fine-tuning or other mechanism to intervene in the very early universe. The third, on the other hand, has to do with dark energy / the cosmological constant today, and with the relatively recent period of accelerated expansion of the universe and is discussed in section 38.4.

I should stress that, while I generally make an attempt in these notes to present just the (well-established) facts, this is not entirely possible in this and the subsequent section, since there is no general consensus on how to resolve these issues (in particular those arising in connection with dark energy). As a consequence, these sections contain not just facts but certainly and unavoidably also some opinions (whereas there was no point in sharing with you my opinion about the Riemann curvature tensor, say - I don't even have one).

1. The Flatness Problem

The flatness problem is the problem that a universe with a value of the density parameter $\Omega = \Omega_M + \Omega_\Lambda \approx 1$ today, equivalently $(\Omega_k)_0 \approx 0$, appears to be "unnatural".

To see in simple terms what the issue is, note that from the Friedmann equation we can read off that

$$H^2 = \frac{8\pi G_N}{3}\rho - \frac{k}{a^2} \quad \Rightarrow \quad \Omega_k = -\frac{k}{a^2 H^2} \ . \tag{38.33}$$

Now for a matter or radiation dominated universe we have $a(t) \sim t^{2/3}$ or $a(t) \sim t^{1/2}$ (with subleading corrections when $k \neq 0$), and therefore

$$a^2 H^2 = \dot{a}^2 \sim \begin{cases} t^{-2/3} & \text{for } \ w = 0 \\ t^{-1} & \text{for } \ w = 1/3 \end{cases} \tag{38.34}$$

so that

$$\Omega_k(t) \sim \begin{cases} t^{+2/3} & \text{for } \ w = 0 \\ t & \text{for } \ w = 1/3 \end{cases} \tag{38.35}$$

Thus a small value of $(\Omega_k)_0 \approx 0$ today (whether to within a few percent, as observations suggest, or even give or take a couple of orders of magnitude) seems to require $\Omega_k = 0$ with an enormous accuracy at some given time in the early universe.

It is more common to phrase this issue in terms of the critical density ρ_{cr} and the density parameter $\Omega = \Omega_M + \Omega_\Lambda$ for matter / radiation plus dark energy (but the contribution of the latter is negligible in the early universe). To that end we rewrite the Friedmann equation as

$$H^2 = \frac{8\pi G_N}{3}\rho - \frac{k}{a^2} \quad \Leftrightarrow \quad (\rho_{cr} - \rho)a^2 = -\frac{3k}{8\pi G_N} \ , \tag{38.36}$$

the point of rewriting it in this way being that what now appears on the right-hand side is a constant. In terms of the density parameter Ω we have

$$(\Omega^{-1} - 1)\rho a^2 = -\frac{3k}{8\pi G_N} \ . \tag{38.37}$$

Since the right-hand side is a constant, the product of the two terms on the left-hand side is a constant. However, as we have already seen (and used) in section 36.3, for ordinary matter ($w > -1/3$) the factor ρa^2 decreases (enormously) as the universe expands. This has to be (and will be, by the above equation) compensated and accompanied by a huge increase in the first factor. Thus, to have $(\Omega)_0 \approx 1$ today appears to require the density $\rho(t)$ of the universe to be equal to the critical density $\rho_{cr}(t)$ with an enormous accuracy at some given time in the early universe.

For example, in the matter dominated phase one has $\rho a^2 \sim a^{-1}$. In this case one can write (38.37) as

$$(\Omega^{-1} - 1)/a = (\Omega_0^{-1} - 1)/a_0 \quad \Leftrightarrow \quad \Omega = \frac{\Omega_0}{\Omega_0 + (a/a_0)(1 - \Omega_0)} \ , \tag{38.38}$$

or

$$\Delta\Omega \equiv \frac{\Omega - 1}{\Omega} = \frac{a}{a_0}\Delta\Omega_0 \ . \tag{38.39}$$

Going back to the beginning of the matter-dominated phase when radiation decoupled from matter, $z = a_0/a \approx 10^3$, one sees that

$$\Delta\Omega \approx 10^{-3}\Delta\Omega_0 \ , \tag{38.40}$$

so that at that time the deviation from flatness (as measured by $\Delta\Omega$) was smaller by a factor of 10^{-3} than it is today.

However, this may begin to look like a more serious fine-tuning issue when one compares with the very early universe and specifies a particular ("earliest") time in the past. Concretely, let us (for the sake of argument) assume that

- the relevant early time is at the Planck scale, $t_P \approx 10^{-43}\mathtt{s}$
- and that matter consists of dust and radiation, with radiation dominating at early times, so that

$$\rho_r(t)a(t)^2 \sim a(t)^{-2} \sim t^{-1} \ . \tag{38.41}$$

With the age of the universe t_0 taken to be (only orders of magnitude will be relevant for this argument) $t_0 \approx 10^{17}\mathtt{s}$, one finds that

$$(\Omega^{-1} - 1)_{t=t_0} \approx 10^{-2} \quad \Rightarrow \quad (\Omega^{-1} - 1)_{t=t_p} \approx 10^{-62} \ . \tag{38.42}$$

In particular, any miniscule deviation from this tiny value at $t = t_p$, i.e. a miniscule difference between the actual and the critical density of the universe at that time, would have led to a universe completely incompatible with obervations. The universe would have most likely either recollapsed after a very short time ($\Omega \to \infty$), or expanded so quickly as to prevent the formation of any structure in the universe ($\Omega \to 0$). In fact, $\Omega = 0$ and $\Omega = \infty$ are the only attractors (attractive fixed points) of the theory in the terminology of dynamical systems, while $\Omega = 1$ is a repeller (unstable fixed point).

What is one supposed to make of this?

Now it should be borne in mind that the fact that Ω was very close to 1 in the past is not in itself a fine-tuning issue: after all, $\Omega \to 1$ in the past is *implied* by the Friedmann equations anyway, no matter what the size of Ω today. So one option is to declare that the above argument regarding finetuning is bogus, in particular since extrapolating all the way back to the Planck time (or the GUT era or ...), and maintaining the assumptions of isotropy and homogeneity that underlie the Friedmann equations requires some enormous leap of faith.[163]

[163] For various critical examinations of this flatness problem, see e.g. P. Helbig, *Is there a flatness problem in classical cosmology?*, `arXiv:1112.1666 [astro-ph.CO]` and M. Holman, *How Problematic is the Near-Euclidean Spatial Geometry of the Large-Scale Universe?*, `arXiv:1803.05148 [gr-qc]`, and references therein and thereto.

But if the flatness problem is perceived as a real problem, then no explanation for this can be found in the standard model of cosmology we have been discussing here so far (FLRW with standard matter content at early times). Occasionally, therefore, weakly anthropic arguments of the kind "if Ω had not been fine-tuned to this value by some (unknown) mechanism/deity, we couldn't be here in the first place to ask the question why Ω takes the value it does today" have been advocated.

While this cannot be ruled out (this is the whole problem with this entire anthropic reasoning business), there is actually a mechanism which naturally leads to the required tiny value of $\Omega - 1$ at early times, namely *inflation*.[164] As this mechanism is also invoked to solve other problems of the standard model of cosmology (such as the horizon problem, see below) which are (a) possibly more serious and (b) not obviously of anthropic significance, one could perhaps consider the anthropophile value of Ω_0 as an unintentional (but serendipitous) side-effect of inflation.

What inflation does is to postulate, for a number of reasons, a brief but highly significant period of exponential expansion in the very early universe, as could be triggered by the presence of a cosmological constant (36.23),

$$H^2 = (\Lambda/3) \quad \Rightarrow \quad a(t) \sim \mathrm{e}^{Ht} \ . \tag{38.43}$$

or more precisely by something that acts like a cosmological constant during a certain period, e.g. an almost constant scalar field (35.39). During such a period of exponential expansion, the factor ρa^2 grown enormously, and correspondingly $\Omega^{-1} - 1$ can be driven arbitrarily close to zero. When, after the end of inflation, the universe resumes its normal (radiation-dominated, say) decelerating expansion (this requires a process called *reheating*, since the universe has cooled down significantly during the inflationary period), the initial value of $\Omega^{-1} - 1$ for this phase can easily be small enough to account for $\Omega \approx 1$ to within a few percent today.

2. The Horizon Problem

   The key issue here is the high degree of isotropy of the universe that we observe, in particular as manifested in the cosmic microwave background radiation (CMBR, see the references in section 34.8), which is isotropic to better than one part in 10^5. This is a puzzle because, as we will see below, according to the standard model of cosmology, the different regions of the spatial surface at $t = t_{ls}$ (the time of

[164] For introductions to inflation, see e.g. A. Linde, *Particle Physics and Inflationary Cosmology*, `arXiv:hep-th/0503203`; D. Baumann, *TASI Lectures on Inflation*, `arXiv:0907.5424 [hep-th]`; S. Tsujikawa, *Introductory review of cosmic inflation*, `arXiv:hep-ph/0304257`; A. Liddle, *An introduction to cosmological inflation*, `arXiv:astro-ph/9901124`; A. Linde, *Inflationary Cosmology*, `arXiv:0705.0164v2 [hep-th]`; R. Brandenberger, *Inflationary Cosmology: Progress and Problems*, `arXiv:hep-ph/9910410`.

"last scattering") from which the CMBR photons that we observe today originated were not ever in causal contact since the beginning of the universe. Thus standard causal microphysics provides no explanation for why these different regions were so precisely at the same temperature. This is schematically illustrated in Figure 53.

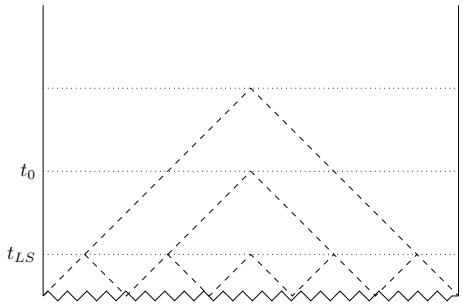


Figure 53: Horizon Problem ($k = +1$ cartoon): causally disconnected patches at the time t_{LS} of last scattering become simultaneously visible at later times.

To quantify this somewhat, note that the maximal spatial coordinate distance photons could have travelled between the big bang at $t = 0$ and $t = t_{ls}$ is (setting $k = 0$ for simplicity, the k-dependence is insignificant for this argument)

$$dt^2 = a(t)^2 dr^2 \quad \Rightarrow \quad r(t_{ls}) = r_{ph}(t_{ls}) = \int_0^{t_{ls}} \frac{dt}{a(t)} \ , \tag{38.44}$$

the particle horizon at time $t = t_{ls}$. Converting this to proper distance at time $t = t_{ls}$, one obtains the quantity

$$R_{ph}(t_{ls}) = a(t_{ls}) \int_0^{t_{ls}} \frac{dt}{a(t)} \ , \tag{38.45}$$

and for a radiation-dominated universe with $a(t) \sim t^{1/2}$ one finds

$$R_{ph}(t_{ls}) = t_{ls}^{1/2} \int_0^{t_{ls}} dt \ t^{-1/2} = 2t_{ls} \ . \tag{38.46}$$

Its significance in the present context is that the past light cone of events that are further than $2R_{ph}(t_{ls})$ apart on the surface $t = t_{ls}$ do not intersect, so that they are so far apart that they were never before in causal contact. In particular this means that no causal interaction can be responsible for the temperature being the same at the two events.

In the (matter dominated) meantime (i.e. between t_{ls} and t_0, today) the size of such a causal patch of size $\sim R_{ph}(t_{ls})$ on the last scattering surface has expanded to proper size

$$\frac{a(t_0)}{a(t_{ls})} R_{ph}(t_{ls}) \sim (t_0/t_{ls})^{2/3} t_{ls} = t_0^{2/3} t_{ls}^{1/3} \ . \tag{38.47}$$

On the other hand, the distance over which the CMBR photons could have travelled since $t = t_{ls}$ is

$$a(t_0) \int_{t_{ls}}^{t_0} \frac{dt}{a(t)} = 3t_0^{2/3}(t_0^{1/3} - t_{ls}^{1/3}) \sim t_0 \ , \tag{38.48}$$

where we have dropped the second term since $t_0 \sim 10^{10}$ years while $t_{ls} \sim 10^5$ years (once again all numbers here and below are just meant to be order of magnitude estimates). Thus the region of the last scattering surface from which we receive the CMBR photons today is much larger than a causal patch, their ratio being (one can of course calculate this either at $t = t_{ls}$ or at $t = t_0$, here we have chosen the latter option)

$$t_0/(t_0^{2/3} t_{ls}^{1/3}) = (t_0/t_{ls})^{1/3} \approx 10^{5/3} \ . \tag{38.49}$$

What this means is that the sky splits into roughly $4\pi 10^{10/3} \gtrsim 10^4$ disconnected patches, that were never in communication before sending light to us. In view of this the observed isotropy of the CMBR is not only astounding but utterly implausible.

Again, inflation solves this in an extremely natural way. Inflation operates at a time $t_i \ll t_{ls}$ (perhaps some time between 10^{-36}s and 10^{-32}s after the big bang) and it can easily inflate a tiny causally connected patch at that time $t = t_i$ to such a size that at time $t = t_{ls}$ it is (more than) large enough to explain the isotropy of the CMBR.

This is easy to understand by looking at the expanding de Sitter metric in conformal time (37.63),

$$ds^2 = \mathrm{e}^{2t/\ell}(-dt^2 \mathrm{e}^{-2t/\ell} + d\vec{x}^2) = \frac{\ell^2}{\eta^2}(-d\eta^2 + d\vec{x}^2) \tag{38.50}$$

with

$$\eta = -\ell \mathrm{e}^{-t/\ell} \ . \tag{38.51}$$

This shows that in a conformal diagram whatever came before inflation can now be pushed to arbitrarily large negative values of η, and therefore the horizon problem as depicted in Figure 53 does not arise.

The majority view among cosmologists appears to be that for these and other reasons inflation should be considered to be part of the standard model of cosmology. However, science is not decided by opinion polls and it is good scientific practice to keep an open mind. In particular, one should keep in mind the possibility that for instance the two (flatness and horizon) problems (or perhaps better: puzzles) may indicate problems with the approximation of the early universe by a FLRW cosmology and are thus not

something that should (or needs to, or perhaps even can) be solved by (inflationary) modifications within the class of FLRW models.[165]

The most significant success of inflation, however, is that it not only "explains" the high degree of isotropy of the CMBR but that it also provides a mechanism for, and a precise quantitative account of, density perturbations and the small inhomogeneities exhibited by the CMBR (namely as arising from quantum vacuum fluctuations). Simple models of inflation appear to be able to account for the latest (2013) precision measurements and data from the Planck satellite[166], but the debate over this and other aspects of inflation continues.[167] I will not try to constantly update these notes as this discussion evolves.[168]

Ultimately, in order for inflation to be considered a natural solution to the above problems (and others I have not mentioned - see the literature cited above), one needs to be able to show that inflation arises fairly naturally (in some precise sense) and does not itself require a comparably huge amount of fine-tuning in order to resolve these issues. This is a complicated and intensely debated issue, and one I don't feel sufficiently competent and knowledgeable about to have an informed opinion, let alone utter an opinion in public. It will perhaps only be settled once one has a better understanding of the physics in the very early, pre-inflationary, era that is supposed to be responsible for setting the initial conditions for inflation.

38.4 Aspects of the Cosmological Constant Problem(s)

This is a complex, confusing and multi-faceted problem that has been around for a long time, and I will not remotely be able to do justice to it here.[169] It has been sharpened and brought to the forefront again by the discovery of dark energy.[170]

We have remarked before that the cosmological constant looks like a vacuum energy contribution to the energy-momentum tensor. It is perhaps better to turn this around and

[165] For example, there are claims that in certain inhomogeneous Lemaître-Tolman cosmological models the horizon problem can be avoided without invoking inflation - see e.g. chapter 18.17 of J. Plebanski, A. Krasinski, *An Introduction to General Relativity and Cosmology*.

[166] A. Linde, *Inflationary Cosmology after Planck 2013*, arXiv:1402.0526 [hep-th].

[167] A. Ijjas, P. Steinhardt, A. Loeb, *Inflationary schism after Planck2013*, arXiv:1402.6980 [astro-ph.CO].

[168] For a recent (at the time of writing this) assessment of inflation see J. Martin, *Cosmic Inflation: Trick or Treat?*, arXiv:1902.05286 [astro-ph.CO].

[169] The classic reference for this is the authoritative and influential article by S. Weinberg, *The cosmological constant problem*, Rev. Mod. Phys. 61 (1989) 1-23.

[170] For recent reviews see e.g. S. Carroll, *The Cosmological Constant*, Living Rev. Relativity 4 (2001) 1, http://www.livingreviews.org/lrr-2001-1; J. Polchinski, *The Cosmological Constant and the String Landscape*, arXiv:hep-th/0603249; T. Padmanabhan, *Cosmological Constant - the Weight of the Vacuum*, arXiv:hep-th/0212290; R. Bousso, *TASI Lectures on the Cosmological Constant*, arXiv:0708.4231 [hep-th], J. Martin, *Everything You Always Wanted To Know About The Cosmological Constant Problem (But Were Afraid To Ask)*, arXiv:1205.3365 [astro-ph.CO].

to say that vacuum energy is one potential contribution to the cosmological constant, in the sense that the associated energy density ρ_Λ can be written as

$$\rho_\Lambda = (\rho_\Lambda)_{\text{bare}} + (\rho_\Lambda)_{\text{vac}} = \rho_{\Lambda_0} + \rho_{\text{vac}} \quad . \tag{38.52}$$

Here

$$\rho_{\Lambda_0} = \frac{\Lambda_0}{8\pi G_N} \tag{38.53}$$

is associated with a "bare" cosmological constant Λ_0, some parameter in the action, while ρ_{vac} is a quantum contribution arising from the energy in the ground state or vacuum, i.e. it is the vacuum expectation value of the Hamiltonian density operator $\hat{T}_{00}$,

$$\rho_{\text{vac}} = < \text{vac}|\hat{T}_{00}|\text{vac} > \quad . \tag{38.54}$$

In a Poincaré-invariant field theory in Minkowski space, with a Poincaré-invariant ground state, this will give rise to a contribution

$$< \text{vac}|\hat{T}_{ab}|\text{vac} > = -\rho_{\text{vac}}\eta_{ab} \quad , \tag{38.55}$$

so the equation of state

$$p_{\text{vac}} = -\rho_{\text{vac}} \tag{38.56}$$

is implied by Lorentz invariance.

This suggests that this extends to quantum field theory in a gravitational background (as described e.g. in the references in footnote 87 in section 27.7) in the form

$$< \text{vac}|\hat{T}_{\alpha\beta}|\text{vac} > = -\rho_{\text{vac}} g_{\alpha\beta} \quad . \tag{38.57}$$

In discussions of the cosmological constant problem, it is usually assumed that (38.57) holds, but it is worth bearing in mind that this is a non-trivial (and possibly incorrect) assumption:

1. First of all, the semi-classical approach of treating quantised matter fields in a classical gravitational background is not something that is completely internally consistent, but it is generally believed to be a valid approximation for sufficiently weak gravitational fields (for which one expects whatever quantum gravitational effects may be present in principle to be completely negligible).

2. Even in Minkowski space an expression like (38.55) requires some sort of regularisation procedure, and the procedures that are common or privileged in the case of Poincaré-invariant field theories may either not be available or may not be in any way privileged when one considers a general curved background. Thus there are ambiguities in the calculation of $< \hat{T}_{\alpha\beta} >$ which may effect the validity of (38.57).

One (relatively harmless) ambiguity of this kind would be the addition of a term proportional to the Einstein tensor $G_{\alpha\beta}$ to the right-hand side of (38.57),

$$< \text{vac}|\hat{T}_{\alpha\beta}|\text{vac} >= -\rho_{\text{vac}} g_{\alpha\beta} + \frac{\gamma}{8\pi G_N} G_{\alpha\beta} \tag{38.58}$$

for some constant γ. This would be compatible with $\nabla^\alpha T_{\alpha\beta} = 0$ and with the Minkowskian limit, and would amount to a renormalisation of Newton's constant G_N.

3. Finally, and perhaps most crucially, while reasonable Poincaré-invariant quantum field theories in Minkowski space typically have a preferred and unique Poincaré-invariant ground state, the vacuum, the key feature of quantum field theory in a curved background is that quantum field theory can be extended to this case but that this uniqueness is lost (and it is precisely this non-uniqueness that is at the heart of the characteristic phenomena of quantum field theory in a gravitational field like particle production and Hawking radiation). As a consequence, the state unceremoniously called $|\text{vac} >$ in (38.57) is not unique, and there is no *a priori* reason to believe that (38.57) will hold for any putative vacuum state.

In spite of all this it is usually assumed that something like (38.57) is at least approximately true for sufficiently weak fields, and we will proceed with this assumption.

With the vacuum-energy contributing to the cosmological constant, it is natural to imagine that contributions to the cosmological constant can arise from many quantum field theory processes such as phase-transitions etc, and are of the order of the momentum cut-off, say. Now in field theory in a fixed background, only energy-differences matter, not absolute energies, but the inclusion of gravity and its presumed universal coupling to all forms of energy change this. It is then not clear if one can simply drop finite (let alone infinite) energy-differences from one's equations.

Depending on the physics or physical process one is looking at, natural estimates for the energy scale of a cosmological constant produced in this way via some phase transition, say, might be in the MeV or GeV range, or perhaps (via some wild extrapolation) even at the (ultimate Planck cut-off) scale of

$$E_P = \sqrt{\frac{\hbar c^5}{8\pi G_N}} \approx 10^{18}\ \text{GeV} \tag{38.59}$$

(with the inclusion of the factor of 8π in the definition this is known as the reduced Planck energy). But one does not really have to go that far to see that there is an issue: after all, any of these values are many many orders of magnitude above what was and is compatible with observation, or even with the very existence of our universe.

One cannot simply "solve" this problem by using the bare value Λ_0 of the cosmological constant to cancel the vacuum contribution by hand, because

- this would require an enormous (and unexplained) fine-tuning of this bare value
- contributions to the vacuum energy are expected to arise at various instances during the evolution of the universe while this cancellation could at best be achieved at one point in time during the evolution of the universe (if this cancellation is chosen to take place too early, it will be incompatible with observations today while if it happens too late it will be incompatible with the well-established thermal history of the universe).

It therefore seemed natural to seek some mechanism that would simply make the net cosmological constant identically zero, and a lot of effort went into finding or inventing some mechanism responsible for this. Nowadays, with the strong evidence for dark energy, this cannot be the solution and the original question/problem has morphed into at least 3 distinct questions, namely

1. Why is the cosmological constant Λ not huge?
2. What is the origin of Dark Energy, with $\rho_{DE} \sim (10^{-3}\ \mathsf{eV})^4$?
3. Why is ρ_Λ (constant in time) of the same order of magnitude as ρ_M (today)?

and we will briefly address these in turn.

1. Why is Λ not huge?

   This is the same problem (described above) as before the discovery of dark energy, and is essentially the question "why the vacuum does not gravitate" (Polchinski; see his article in footnote 170 for a very illuminating discussion of this issue).

   This whole business points to some loop-hole in the reasoning leading to (38.57) and the estimates for the size of this quantity.[171] It may even indicate some fundamental lack of (or perhaps mis-) understanding about how gravity couples to quantum fields.[172]

   One on the face of it rather radical, but actually quite conservative, possibility that has been explored (and that I mention here not because I think it is the most plausible explanation but because it is easy to explain and also quite interesting in its own right) is to modify the Einstein equations in such a way that the

[171] See e.g. Q. Wang, Z. Zhu, W. Unruh, *How the huge energy of quantum vacuum gravitates to drive the slow accelerating expansion of the Universe*, `arXiv:1703.00543 [gr-qc]` where it is argued that, because of quantum fluctuations, the quantum vacuum actually gravitates substantially differently from a cosmological constant.

[172] See S. Hollands, R. Wald, *Quantum Field Theory Is Not Merely Quantum Mechanics Applied to Low Energy Effective Degrees of Freedom*, `arXiv:gr-qc/0405082` for some reflections on this issue.

gravitational field does not couple directly to any contribution to the energy-momentum tensor $\sim g_{\alpha\beta}$, i.e. such that they are invariant under shifts

$$T_{\alpha\beta} \to T_{\alpha\beta} + c g_{\alpha\beta} \tag{38.60}$$

for constant c. To that end one postulates the so-called *trace-free Einstein equations*

$$R_{\alpha\beta} - \tfrac{1}{4} g_{\alpha\beta} R = 8\pi G_N \left(T_{\alpha\beta} - \tfrac{1}{4} g_{\alpha\beta} T\right) \ , \tag{38.61}$$

supplemented by the usual covariant energy-momentum tensor conservation law

$$\nabla_\alpha T^{\alpha\beta} = 0 \tag{38.62}$$

(which is not implied by the modified Einstein equations (38.61)).[173] Note that both sides of (38.61) are manifestly traceless. In particular, therefore, any contribution to the energy-momentum tensor $\sim g_{\alpha\beta}$ (like a cosmological constant) does not contribute to (38.61),

$$T_{\alpha\beta} \sim g_{\alpha\beta} \quad \Rightarrow \quad T_{\alpha\beta} - \tfrac{1}{4} g_{\alpha\beta} T = 0 \ , \tag{38.63}$$

and therefore does not couple directly to gravity (said differently, the source term is invariant under the shift (38.60), as required).

While this looks like a major departure from the usual (and well-tested) Einstein equations, and this might make one believe that the equations (38.61) can easily be ruled out experimentally, what actually happens is more subtle and somewhat surprising. Namely, writing (38.61) as

$$R_{\alpha\beta} - \tfrac{1}{2} g_{\alpha\beta} R = 8\pi G_N \left(T_{\alpha\beta} - \tfrac{1}{4} g_{\alpha\beta} T\right) - \tfrac{1}{4} g_{\alpha\beta} R \tag{38.64}$$

and using the contracted Bianchi idenities $\nabla_\alpha G^{\alpha\beta} = 0$ as well as the postulated $\nabla_\alpha T^{\alpha\beta} = 0$, one deduces that

$$\nabla_\alpha (8\pi G_N \, T + R) = 0 \quad \Rightarrow \quad 8\pi G_N \, T = -R + 4\Lambda \tag{38.65}$$

for some integration constant Λ. Plugging this result back into (38.64), one finds

$$\begin{aligned} R_{\alpha\beta} - \tfrac{1}{2} g_{\alpha\beta} R &= 8\pi G_N \, T_{\alpha\beta} - \tfrac{1}{4} g_{\alpha\beta} [8\pi G_N \, T + R] \\ &= 8\pi G_N \, T_{\alpha\beta} - g_{\alpha\beta} \Lambda \ . \end{aligned} \tag{38.66}$$

This is nothing other than the usual Einstein equations with a cosmological constant,

$$R_{\alpha\beta} - \tfrac{1}{2} g_{\alpha\beta} R + \Lambda g_{\alpha\beta} = 8\pi G_N \, T_{\alpha\beta} \ , \tag{38.67}$$

[173] See e.g. G. Ellis, H. van Elst, J. Murugan, J.-P. Uzan, *On the Trace-Free Einstein Equations as a Viable Alternative to General Relativity*, `arXiv:1008.1196v2 [gr-qc]` and references therein for a more detailed discussion and the history of this and other variants of *unimodular gravity*, originally even considered by Einstein himself in 1919.

the crucial difference being that here Λ is *not* determined by the matter content and its vacuum energy but arises solely as an integration constant. While this does not explain the observed tiny value of the cosmological constant, it separates this issue from that of the vacuum fluctuations. Note also that this scenario does not rule out a gravitational coupling to general quantum corrections to states of some physical system (Lamb shift, Casimir energies, ...) which are not Lorentz invariant, as required e.g. by precision tests of the equivalence principle (see the discussion regarding "do vacuum fluctuations gravitate?" in the reviews by Polchinski and Martin in footnote 170).

2. What is the origin of Dark Energy, with $\rho_{DE} \sim (10^{-3}\ \mathsf{eV})^4$?

   Perhaps this is the wrong question altogether. After all, it could just be that Λ happens to be a new constant of nature, like G_N, that has now been found to be non-zero and that has been experimentally determined, and whose precise value cannot currently (or perhaps even in principle) be derived from some underlying theory.

   But a priori it is also possible that the value can be derived, and/or that dark energy is due to something other than a cosmological constant. Again anthropic reasoning (shudder!) can be invoked to argue that the above is a plausible value for the cosmological constant. It would however of course be very desirable to find alternative explanations.

   Numerous models have been proposed that do give rise to a late-time acceleration of the universe for one reason or another (one of the buzzwords here is *quintessence*). Most of them, however, assume (explicitly or implicitly) that somehow the first problem has been solved and that there is no (bare or combined) cosmological constant source term in the Einstein equations.

   It is not clear at all, however, what kind of mechanism could produce on the nose an effect of the desired size. One curious observation, which may provide us with a clue, is that the energy scale E_Λ associated with the cosmological constant,

$$E_\Lambda \approx 10^{-3}\ \mathsf{eV} = 10^{-12}\ \mathsf{GeV}\ , \tag{38.68}$$

   which is much much smaller than the "natural" Planck energy scale,

$$E_\Lambda \approx 10^{-30} E_P\ , \tag{38.69}$$

   is however of the order of the geometric mean of the (current-day) Hubble scale and the Planck scale (I do not know who (if anybody) should be credited with this observation which has probably been made independently multiple times). In energy units this is the curious relation

$$H_0 \approx 10^{-42}\ \mathsf{GeV}\ ,\ E_P \approx 10^{18}\mathsf{GeV} \quad \Rightarrow \quad E_\Lambda \approx (E_P H_0)^{1/2}\ , \tag{38.70}$$

and in time-units, expressed in terms of the (reduced) *Planck time*

$$t_P = \sqrt{\frac{\hbar 8\pi G_N}{c^5}} \approx 10^{-43}\ \mathtt{s}\ , \tag{38.71}$$

this is the equally baffling statement that

$$(H_0)^{-1} \approx 10^{60} t_P \quad , \quad \Lambda \approx 10^{-120} (t_P)^{-2} \approx (H_0)^2\ . \tag{38.72}$$

This is certainly intriguing, as it appears to relate the cosmological constant to both a fundamental UV cut-off (the Planck scale) and a current IR cut-off (the Hubble scale), which has led to some "holographic" ruminations.[174] This curious relation is also one aspect of the so-called "coincidence problem", and we will briefly come back to this below.

A more conservative realisation of the scenario described by (38.70) is also possible.[175] For concreteness, consider the scalar field in a cosmological background discussed in section 34.10, with a UV momentum cutoff at a momentum $k = k_{UV}$. Namely if, for one reason or another, one can argue that the leading quartic divergence in the vacuum energy

$$\rho_{\rm vac} \sim \tfrac{1}{2} \int \frac{d^3k}{(2\pi)^3} \sqrt{k^2 + m_{eff}^2} = \frac{(k_{UV})^4}{16\pi^2} + \dots \tag{38.73}$$

(which would also be present in, and hence destroy, Minkowski space) should be subtracted, then the subleading term is quite generically of the form

$$\rho_{\rm vac} \sim (k_{UV})^2 (H_0)^2 \tag{38.74}$$

(in the model in section 34.10 this H_0^2-term would arise from the scalar curvature term in the effective mass). With a cut-off at the Planck scale, this gives a vacuum energy contribution of the order of the observed dark energy density,

$$k_{UV} \sim k_P \quad \Rightarrow \quad \rho_{\rm vac} \sim (E_P)^2 (H_0)^2 \sim (\rho_\Lambda)_{\rm observed}\ . \tag{38.75}$$

In designing such scenarios, care should be taken that the vacuum contribution thus determined really has an equation of state parameter equal (or very close to) $w = -1$, so that $\rho_{\rm vac}$ calculated today is really (approximately) constant (and does not behave like $\rho_{\rm vac}(t) \sim H(t)^2$, say, which would be in conflict with many cosmological observations). Exploring scenarios of this kind is clearly a constructive alternative to anthropic incantations.

[174] See e.g. A. Cohen, D. Kaplan, A. Nelson, *Effective Field Theory, Black Holes, and the Cosmological Constant*, `arXiv:hep-th/9803132`, M. Li, *A Model of Holographic Dark Energy*, `arXiv:hep-th/0403127`, T. Padmanabhan, *Vacuum Fluctuations of Energy Density can lead to the observed Cosmological Constant*, `arXiv:hep-th/0406060`, S. Hsu, A. Zee, *A speculative relation between the cosmological constant and the Planck mass*, `arXiv:hep-th/0406142`, and references thereto for theoretical speculations along these lines.

[175] See e.g. M. Maggiore, *Zero-point quantum fluctuations and dark energy*, `arXiv:1004.1782 [astro-ph.CO]`, and D. Bernard, A. LeClair, *Scrutinizing the Cosmological Constant Problem and a possible resolution*, `arXiv:1211.4848 [hep-th]`.

3. Why is ρ_Λ (constant in time) of the same order of magnitude as ρ_M (today)?

   This is weird, and one version of what is known as the *Coincidence Problem* in cosmology. To see why this is peculiar, note that ρ_Λ is constant while $\rho_M \sim a^{-3}$ so that the ratio of the two behaves as

$$\frac{\rho_\Lambda(t)}{\rho_M(t)} \sim a(t)^3 \ . \tag{38.76}$$

   It is therefore, one might reason, extremely unlikely to find oneself at a time when the two happen to be comparable in size, but nevertheless this is precisely the time we find ourselves at (how precisely depends among other things on what one means by "comparable in size"). However, any such statement has to also rely on some probability distribution for something (the probability that observers exist at some time t_0?) that allows one to quantify this statement somewhat. But it is not clear if this is an issue that really requires an explanation.

   If one thinks that this coincidence is indeed a problem that requires a solution, then one can try to come up with models in which dark energy does not behave like a cosmological constant at all times. However:

   > "Why obfuscate? If a poet sees something that walks like a duck and swims like a duck and quacks like a duck, we will forgive him for entertaining more fanciful possibilities, It could be a unicorn in a duck suit - who's to say! But we know that more likely, it's a duck."[176]

   Fair enough.[177]

   Note by the way that this coincidence issue is equivalent to the empirical observation (38.70) mentioned above. Namely, with $\rho_\Lambda \sim \rho_M$, and both of order of the critival density because $\Omega_M + \Omega_\Lambda = 1$, one has

$$\rho_\Lambda = \frac{\Lambda}{8\pi G_N} \sim (\rho_c)_0 = \frac{3H_0^2}{8\pi G_N} \sim H_0^2 E_P^2 \tag{38.77}$$

   which is just (38.70). One can also eliminate any reference to Planck units and the Planck scale and write this as (38.72)

$$\Lambda \approx (H_0)^2 \ . \tag{38.78}$$

   Thus another (and perhaps more objective and constructive) way of stating the coincidence issue is as the curious fact that the two a priori completely unrelated time-scales, one set by $(H_0)^{-1}$ (which turns out to be remarkably close to the current best estimates for the age of the universe) and the other set by the energy density or curvature radius of dark energy, are approximately equal. Why should

[176] R. Bousso, *TASI Lectures on the Cosmological Constant*, `arXiv:0708.4231 [hep-th]`.

[177] See also E. Bianchi, C. Rovelli, *Why all these prejudices against a constant?*, `arXiv:1002.3966 [astro-ph.CO]` for further reflections on this and related issues.

the cosmological constant today be related to the age of the universe today??? Perhaps this is indeed not a coincidence and is trying to tell us something.

It is not clear if a fundamental physics (non-multiverse, non-anthropic) explanation for dark energy can be found that is capable of resolving all 3 of the cosmological constant problems at once, but perhaps one should at least be humble and continue to entertain the possibility, alluded to above, that this issue is telling us that we are missing something fundamental about the coupling of gravity and quantum fields.

This is not to say that there is not a strong case to be made for what appears to be an extreme fine-tuning of other (standard model) parameters to support everything from primordial nucleosynthesis to chemistry and life as we know it.[178] However, among other things precisely because of the proviso "as we know it", and also because of the (as far as I know) limited understanding of chemistry based on other than standard model gauge theories, this begs the question and the significance of these findings is far from clear and difficult to assess.

[178] See e.g. L. Barnes, *The Fine-Tuning of the Universe for Intelligent Life*, `arXiv:1112.4647 [physics.hist-ph]` for a recent assessment of the situation.